```
 LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    2 ************************************************************************
                                                    3 *
                                                    4 *        PLO-Extension Facility function tests
                                                    5 *
                                                    6 *        James Wekel November 2026
                                                    7 ************************************************************************


                                                    9 ************************************************************************
                                                   10 *
                                                   11 *        basic instruction tests
                                                   12 *
                                                   13 ************************************************************
                                                   14 *  This program tests proper functioning of the PLO-Extension
                                                   15 *  facility functions. Specification exceptions and access-register
                                                   16 *  mode are not tested.
                                                   17 *
                                                   18 *  PLEASE NOTE that the tests are very SIMPLE TESTS designed to catch
                                                   19 *  obvious coding errors.  None of the tests are thorough.  They are
                                                   20 *  NOT designed to test all aspects of any of the instructions.
                                                   21 *
                                                   22 *------------------------------------------------------------------
                                                   23 * Programmer's Note:
                                                   24 *
                                                   25 * The following tests have a LOT of duplicate (copy/paste) code
                                                   26 * to allow for minor modifications specific to each test.
                                                   27 *------------------------------------------------------------------
                                                   28 *
                                                   29 ************************************************************************
                                                   30 *
                                                   31 *  Example Hercules Testcase:
                                                   32 *
                                                   33 *    *Testcase plo-01-extension
                                                   34 *    *
                                                   35 *    *   PLO-Extension Functions Facility tests
                                                   36 *    *
                                                   37 *    *   # ---------------------------------------------------------
                                                   38 *    *   #  This tests only the basic function of the instruction.
                                                   39 *    *   #  Exceptions are NOT tested.
                                                   40 *    *   # ---------------------------------------------------------
                                                   41 *    *
                                                   42 *    mainsize    2
                                                   43 *    numcpu      1
                                                   44 *    sysclear
                                                   45 *    archlvl     z/Arch
                                                   46 *
                                                   47 *    loadcore    "$(testpath)/plo-01-extension.core" 0x0
                                                   48 *
                                                   49 *    diag8cmd    enable    # (needed for messages to Hercules console)
                                                   50 *    runtest     5         #
                                                   51 *    diag8cmd    disable   # (reset back to default)
                                                   52 *
                                                   53 *    *Done
                                                   54 *
                                                   55 ************************************************************************
```

```
 LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    57 ************************************************************************
                                                    58 *        FCHECK Macro - Is a Facility Bit set?
                                                    59 *
                                                    60 *        If the facility bit is NOT set, an message is issued and
                                                    61 *        the test is skipped.
                                                    62 *
                                                    63 *        Fcheck uses R0, R1 and R2
                                                    64 *
                                                    65 * eg.    FCHECK 134,'vector-packed-decimal'
                                                    66 ************************************************************************
                                                    67          MACRO
                                                    68          FCHECK &BITNO,&NOTSETMSG
                                                    69 .*                      &BITNO : facility bit number to check
                                                    70 .*                      &NOTSETMSG : 'facility name'
                                                    71          LCLA  &FBBYTE             Facility bit in Byte
                                                    72          LCLA  &FBBIT              Facility bit within Byte
                                                    73
                                                    74          LCLA  &L(8)
                                                    75 &L(1)    SetA  128,64,32,16,8,4,2,1  bit positions within byte
                                                    76
                                                    77 &FBBYTE  SETA  &BITNO/8
                                                    78 &FBBIT   SETA  &L((&BITNO-(&FBBYTE*8))+1)
                                                    79 .*       MNOTE 0,'checking Bit=&BITNO: FBBYTE=&FBBYTE, FBBIT=&FBBIT'
                                                    80
                                                    81          B     X&SYSNDX
                                                    82 *                                  Fcheck data area
                                                    83 *                                  skip messgae
                                                    84 SKT&SYSNDX DC  C'          Skipping tests: '
                                                    85          DC    C&NOTSETMSG
                                                    86          DC    C' facility (bit &BITNO) is not installed.'
                                                    87 SKL&SYSNDX EQU *-SKT&SYSNDX
                                                    88 *                                  facility bits
                                                    89          DS    FD                  gap
                                                    90 FB&SYSNDX DS   4FD
                                                    91          DS    FD                  gap
                                                    92 *
                                                    93 X&SYSNDX EQU *
                                                    94          LA    R0,((X&SYSNDX-FB&SYSNDX)/8)-1
                                                    95          STFLE FB&SYSNDX           get facility bits
                                                    96
                                                    97          XGR   R0,R0
                                                    98          IC    R0,FB&SYSNDX+&FBBYTE     get fbit byte
                                                    99          N     R0,=F'&FBBIT'            is bit set?
                                                   100          BNZ   XC&SYSNDX
                                                   101 *
                                                   102 * facility bit not set, issue message and exit
                                                   103 *
                                                   104          LA    R0,SKL&SYSNDX       message length
                                                   105          LA    R1,SKT&SYSNDX       message address
                                                   106          BAL   R2,MSG
                                                   107
                                                   108          B     EOJ
                                                   109 XC&SYSNDX EQU *
                                                   110          MEND
                                                   111
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    113 * ================================================================
                                                    114 * LOW CORE
                                                    115 * ================================================================
                                                    116 *
                               00000000  0004F280   117 PLOETST  START 0
                               00000000  00000001   118 A000LABL EQU   *
00000000                       00000000             119          USING PLOETST,R0          Low core addressability
                                                    120
                               00000140  00000000   121 SVOLDPSW EQU   PLOETST+X'140'      z/Arch Supervisor call old PSW


00000000                       00000000  000001A0   123          ORG   PLOETST+X'1A0'      z/Architecure RESTART PSW
000001A0  00000001 80000000                         124          DC    X'0000000180000000'
000001A8  00000000 00002000                         125          DC    AD(BEGIN)


000001B0                       000001B0  000001D0   127          ORG   PLOETST+X'1D0'      z/Architecure PROGRAM CHECK PSW
000001D0  00020001 80000000                         128          DC    X'0002000180000000'
000001D8  00000000 0000DEAD                         129          DC    AD(X'DEAD')
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    131 ************************************************************************
                                                    132 *               The actual "PLOETST" program itself...
                                                    133 ************************************************************************
                                                    134 *
                                                    135 *  Architecture Mode: z/Arch
                                                    136 *  Register Usage:
                                                    137 *
                                                    138 *   R0-R4    (work)
                                                    139 *   R5       Testing control table - base current entry
                                                    140 *   R6-R7    (work)
                                                    141 *   R8       First base register
                                                    142 *   R9       Second base register
                                                    143 *   R10      Third base register
                                                    144 *   R11-R13  (work)
                                                    145 *   R14      Subroutine call return
                                                    146 *   R15      Secondary Subroutine call return or work
                                                    147 *
                                                    148 ************************************************************************

000001E0                       000001E0  00002000   150          ORG   PLOETST+X'2000'      Start of actual test program...

00002000                       00002000             152          USING  BEGIN,R8           FIRST Base Register
00002000                       00003000             153          USING  BEGIN+4096,R9      SECOND Base Register
00002000                       00004000             154          USING  BEGIN+8192,R10     THIRD Base Register

00002000  0580                                       156 BEGIN    BALR  R8,0               Initalize FIRST base register
00002002  0680                                       157          BCTR  R8,0               Initalize FIRST base register
00002004  0680                                       158          BCTR  R8,0               Initalize FIRST base register

00002006  4190 8800                    00000800     160          LA    R9,2048(,R8)       Initalize SECOND base register
0000200A  4190 9800                    00000800     161          LA    R9,2048(,R9)       Initalize SECOND base register
                                                    162
0000200E  41A0 9800                    00000800     163          LA    R10,2048(,R9)      Initalize THIRD base register
00002012  41A0 A800                    00000800     164          LA    R10,2048(,R10)     Initalize THIRD base register
                                                    165
00002016  B600 870C                    0000270C     166          STCTL R0,R0,CTLR0        Store CR0 to enable AFP
0000201A  9604 870D                    0000270D     167          OI    CTLR0+1,X'04'      Turn on AFP bit
0000201E  9602 870D                    0000270D     168          OI    CTLR0+1,X'02'      Turn on Vector bit
00002022  B700 870C                    0000270C     169          LCTL  R0,R0,CTLR0        Reload updated CR0
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    171 ********************************************************************************
                                                    172 * Is PLO-extension facility installed  (bit 87)
                                                    173 ********************************************************************************
                                                    174
                                                    175          FCHECK 87,'PLO-extension'
00002026  47F0 80A8                       000020A8  176+         B     X0001
                                                    177+*                                Fcheck data area
                                                    178+*                                skip messgae
0000202A  40404040 40404040                         179+SKT0001  DC    C'          Skipping tests: '
00002044  D7D3D660 85A7A385                         180+         DC    C'PLO-extension'
00002051  40868183 899389A3                         181+         DC    C' facility (bit 87) is not installed.'
                            0000004B  00000001    182+SKL0001  EQU   *-SKT0001
                                                    183+*                                facility bits
00002078  00000000 00000000                         184+         DS    FD                gap
00002080  00000000 00000000                         185+FB0001   DS    4FD
000020A0  00000000 00000000                         186+         DS    FD                gap
                                                    187+*
                            000020A8  00000001    188+X0001    EQU   *
000020A8  4100 0004                 00000004      189+         LA    R0,((X0001-FB0001)/8)-1
000020AC  B2B0 8080                 00002080      190+         STFLE FB0001            get facility bits
000020B0  B982 0000                               191+         XGR   R0,R0
000020B4  4300 808A                 0000208A      192+         IC    R0,FB0001+10            get fbit byte
000020B8  5400 867C                 0000267C      193+         N     R0,=F'1'                is bit set?
000020BC  4770 80D0                 000020D0      194+         BNZ   XC0001
                                                    195+*
                                                    196+* facility bit not set, issue message and exit
                                                    197+*
000020C0  4100 004B                 0000004B      198+         LA    R0,SKL0001        message length
000020C4  4110 802A                 0000202A      199+         LA    R1,SKT0001        message address
000020C8  4520 8598                 00002598      200+         BAL   R2,MSG
000020CC  47F0 8660                 00002660      201+         B     EOJ
                            000020D0  00000001    202+XC0001   EQU   *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     204 ********************************************************************************
                                                     205 *    Do tests in the PLOETST table
                                                     206 ********************************************************************************
                                                     207
000020D0  58C0 8680                        00002680  208          L     R12,=A(PLOTESTS)    get table of test address
                                                     209
                                 000020D4  00000001  210 NEXTTST  EQU   *
000020D4  5850 C000                        00000000  211          L     R5,0(0,R12)         get test address
000020D8  1255                                       212          LTR   R5,R5                have a test?
000020DA  4780 8188                        00002188  213          BZ    ENDTEST                 done?
                                                     214
000020DE                         00000000            215          USING PLOETEST,R5
                                                     216
                                                     217 * save before a test
000020DE  EB0F 8108 0024                   00002108  218          STMG  R0,R15,SVTST        Save regs before test
                                                     219
                                                     220 * start a test
000020E4  E300 5004 0015                   00000004  221          LGH   R0,TNUM             Save current test number
000020EA  5000 A004                        00004004  222          ST    R0,TESTING
                                                     223
000020EE  58B0 5000                        00000000  224          L     R11,TSUB            get address of test routine
000020F2  05BB                                       225          BALR  R11,R11             do test
                                                     226
                                                     227 * restore after a test
000020F4  EB0F 8108 0004                   00002108  228          LMG   R0,R15,SVTST        restore regs after test
                                                     229
000020FA  41C0 C004                        00000004  230          LA    R12,4(0,R12)        next test address
000020FE  47F0 80D4                        000020D4  231          B     NEXTTST
                                                     232
00002108  00000000 00000000                          233 SVTST    DS    16D                 save area


                                                     235 ********************************************************************************
                                                     236 * end of testing; set ending psw
                                                     237 ********************************************************************************
                                 00002188  00000001  238 ENDTEST  EQU   *
00002188  5810 A000                        00004000  239          L     R1,FAILED           did a test fail?
0000218C  1211                                       240          LTR   R1,R1
0000218E  4780 8660                        00002660  241          BZ    EOJ                 No, exit
00002192  47F0 8678                        00002678  242          B     FAILTEST            Yes, exit with BAD PSW
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    244 ************************************************************************
                                                    245 * CCMSG:       cc was not as expected
                                                    246 *              R5  = address of test control table
                                                    247 *              R15 = return address
                                                    248 ************************************************************************
                             00002196  00000001     249 CCMSG    EQU   *
                                                    250 *
                                                    251 ** check for ignore cc as PLO did not update CC
                                                    252 *
00002196  9563 500A                    0000000A     253          CLI   CC,99               99 -->  cc not updated
0000219A  478F 0000                    00000000     254          BE    0(R15)
                                                    255
0000219E  5800 867C                    0000267C     256          L     R0,=F'1'            set GLOBAL failed test indicator
000021A2  5000 A000                    00004000     257          ST    R0,FAILED
                                                    258 *
                                                    259 * extract CC extracted PSW
                                                    260 *
000021A6  5810 5010                    00000010     261          L     R1,CCPSW
000021AA  8810 000C                    0000000C     262          SRL   R1,12
000021AE  5410 8684                    00002684     263          N     R1,=XL4'3'
000021B2  4210 500C                    0000000C     264          STC   R1,CCFOUND          save cc
                                                    265 *
                                                    266 * FILL IN MESSAGE
                                                    267 *
000021B6  4820 5004                    00000004     268          LH    R2,TNUM             get test number and convert
000021BA  4E20 A1A1                    000041A1     269          CVD   R2,DECNUM
000021BE  D211 A18B A175       0000418B  00004175     270          MVC   PRT3,EDIT
000021C4  DE11 A18B A1A1       0000418B  000041A1     271          ED    PRT3,DECNUM
000021CA  D202 A091 A198       00004091  00004198     272          MVC   CCPRTNUM(3),PRT3+13 fill in message with test #
                                                    273
000021D0  B982 0022                                 274          XGR   R2,R2               get FUN as U8
000021D4  4320 5007                    00000007     275          IC    R2,FUN
000021D8  4E20 A1A1                    000041A1     276          CVD   R2,DECNUM           and convert
000021DC  D211 A18B A175       0000418B  00004175     277          MVC   PRT3,EDIT
000021E2  DE11 A18B A1A1       0000418B  000041A1     278          ED    PRT3,DECNUM
000021E8  D201 A0BD A199       000040BD  00004199     279          MVC   CCPRTFUN(2),PRT3+14 fill in message with CC field
                                                    280
000021EE  B982 0022                                 281          XGR   R2,R2               get CC as U8
000021F2  4320 500A                    0000000A     282          IC    R2,CC
000021F6  4E20 A1A1                    000041A1     283          CVD   R2,DECNUM           and convert
000021FA  D211 A18B A175       0000418B  00004175     284          MVC   PRT3,EDIT
00002200  DE11 A18B A1A1       0000418B  000041A1     285          ED    PRT3,DECNUM
00002206  D200 A0CE A19A       000040CE  0000419A     286          MVC   CCPRTEXP(1),PRT3+15 fill in message with CC field
                                                    287
0000220C  B982 0022                                 288          XGR   R2,R2               get CCFOUND as U8
00002210  4320 500C                    0000000C     289          IC    R2,CCFOUND
00002214  4E20 A1A1                    000041A1     290          CVD   R2,DECNUM           and convert
00002218  D211 A18B A175       0000418B  00004175     291          MVC   PRT3,EDIT
0000221E  DE11 A18B A1A1       0000418B  000041A1     292          ED    PRT3,DECNUM
00002224  D200 A0DE A19A       000040DE  0000419A     293          MVC   CCPRTGOT(1),PRT3+15 fill in message with ccfound
                                                    294 *
                                                    295 *        Use Hercules Diagnose for Message to console
                                                    296 *
0000222A  4100 005F                    0000005F     297          LA    R0,CCPRTLNG         message length
0000222E  4110 A081                    00004081     298          LA    R1,CCPRTLINE        message address
                                                    299
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00002232  EBF5 8248 0024                 00002248   300          STMG  R15,R5,FCCSAV       save regs r15 to R5
                                                    301
00002238  4520 8598                      00002598   302          BAL   R2,MSG
                                                    303
0000223C  EBF5 8248 0004                 00002248   304          LMG   R15,R5,FCCSAV       restore regs
00002242  07FF                                      305          BR    R15                 Return to caller

00002248  00000000 00000000                         307 FCCSAV   DC     16D'0'             save area for MSG call
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    309 ************************************************************************
                                                    310 * URMSG:       UR was not as expected
                                                    311 *              R5  = address of test control table
                                                    312 *              R15 = return address
                                                    313 *
                                                    314 *              note: UR 99 - UR was not updated
                                                    315 ************************************************************************
                               000022C8  00000001   316 URMSG    EQU   *
                                                    317 *
                                                    318 ** check for ignore UR as PLO did not update UR
                                                    319 *
000022C8  9563 5008                     00000008    320          CLI   UR,99               99 -->  UR not updated
000022CC  478F 0000                     00000000    321          BE    0(R15)
                                                    322
000022D0  5800 867C                     0000267C    323          L     R0,=F'1'            set GLOBAL failed test indicator
000022D4  5000 A000                     00004000    324          ST    R0,FAILED
                                                    325 *
                                                    326 * FILL IN MESSAGE
                                                    327 *
000022D8  4820 5004                     00000004    328          LH    R2,TNUM             get test number and convert
000022DC  4E20 A1A1                     000041A1    329          CVD   R2,DECNUM
000022E0  D211 A18B A175        0000418B  00004175    330          MVC   PRT3,EDIT
000022E6  DE11 A18B A1A1        0000418B  000041A1    331          ED    PRT3,DECNUM
000022EC  D202 A0F0 A198        000040F0  00004198    332          MVC   URPRTNUM(3),PRT3+13 fill in message with test #
                                                    333
000022F2  B982 0022                                 334          XGR   R2,R2               get FUN as U8
000022F6  4320 5007                     00000007    335          IC    R2,FUN
000022FA  4E20 A1A1                     000041A1    336          CVD   R2,DECNUM            and convert
000022FE  D211 A18B A175        0000418B  00004175    337          MVC   PRT3,EDIT
00002304  DE11 A18B A1A1        0000418B  000041A1    338          ED    PRT3,DECNUM
0000230A  D201 A11C A199        0000411C  00004199    339          MVC   URPRTFUN(2),PRT3+14 fill in message with Fun field
                                                    340
00002310  B982 0022                                 341          XGR   R2,R2               get UR as U8
00002314  4320 5008                     00000008    342          IC    R2,UR
00002318  4E20 A1A1                     000041A1    343          CVD   R2,DECNUM            and convert
0000231C  D211 A18B A175        0000418B  00004175    344          MVC   PRT3,EDIT
00002322  DE11 A18B A1A1        0000418B  000041A1    345          ED    PRT3,DECNUM
00002328  D200 A12D A19A        0000412D  0000419A    346          MVC   URPRTEXP(1),PRT3+15 fill in message with UR field
                                                    347
0000232E  B982 0022                                 348          XGR   R2,R2               get URFOUND as U8
00002332  4320 5009                     00000009    349          IC    R2,URFOUND
00002336  4E20 A1A1                     000041A1    350          CVD   R2,DECNUM            and convert
0000233A  D211 A18B A175        0000418B  00004175    351          MVC   PRT3,EDIT
00002340  DE11 A18B A1A1        0000418B  000041A1    352          ED    PRT3,DECNUM
00002346  D200 A13D A19A        0000413D  0000419A    353          MVC   URPRTGOT(1),PRT3+15 fill in message with urfound
                                                    354 *
                                                    355 *        Use Hercules Diagnose for Message to console
                                                    356 *
0000234C  4100 005F                     0000005F    357          LA    R0,URPRTLNG         message length
00002350  4110 A0E0                     000040E0    358          LA    R1,URPRTLINE        messagfe address
                                                    359
00002354  EBF5 8368 0024                00002368    360          STMG  R15,R5,URSAV        save regs r15 to R5
                                                    361
0000235A  4520 8598                     00002598    362          BAL   R2,MSG
                                                    363
0000235E  EBF5 8368 0004                00002368    364          LMG   R15,R5,URSAV       restore regs
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00002364  07FF                                      365          BR    R15                Return to caller
00002368  00000000 00000000                         367 URSAV    DC     16D'0'            save area for MSG call
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     369 ************************************************************************
                                                     370 * Fail Message to console
                                                     371 *              R5  = address of test control table
                                                     372 *              R15 = return address
                                                     373 ************************************************************************
                               000023E8  00000001    374 FAILMSG  EQU   *
000023E8  5800 867C                      0000267C    375          L     R0,=F'1'            set GLOBAL failed test indicator
000023EC  5000 A000                      00004000    376          ST    R0,FAILED
                                                     377
000023F0  4820 5004                      00000004    378          LH    R2,TNUM             get test number and convert
000023F4  4E20 A1A1                      000041A1    379          CVD   R2,DECNUM
000023F8  D211 A18B A175       0000418B  00004175    380          MVC   PRT3,EDIT
000023FE  DE11 A18B A1A1       0000418B  000041A1    381          ED    PRT3,DECNUM
00002404  D202 A018 A198       00004018  00004198    382          MVC   FAILTNUM(3),PRT3+13 fill in message with test #
                                                     383
0000240A  D263 A01D 868C       0000401D  0000268C    384          MVC   FAILTXT(L'FAILTXT),=CL128' '  set to blanks
00002410  B982 0022                                  385          XGR   R2,R2
00002414  5820 5018                      00000018    386          L     R2,FAILTXTA
00002418  D263 A01D 2000       0000401D  00000000    387          MVC   FAILTXT(L'FAILTXT),0(R2)      load message text
                                                     388
                                                     389 *
                                                     390 *        Use Hercules Diagnose for Message to console
                                                     391 *
0000241E  4100 0079                      00000079    392          LA    R0,FAILLNG          message length
00002422  4110 A008                      00004008    393          LA    R1,FAILLINE         messagfe address
                                                     394
00002426  EBF5 8438 0024                 00002438    395          STMG  R15,R5,FLSAV        save regs r15 to R5
                                                     396
0000242C  4520 8598                      00002598    397          BAL   R2,MSG
                                                     398
00002430  EBF5 8438 0004                 00002438    399          LMG   R15,R5,FLSAV        restore regs
00002436  07FF                                       400          BR    R15                 Return to caller

00002438  00000000 00000000                          402 FLSAV    DC     16D'0'             save area for MSG call
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    404 ***************************************************************************
                                                    405 * Test: Not Available Message
                                                    406 *              R5  = address of test control table
                                                    407 *              R15 = return address
                                                    408 ***************************************************************************
                               000024B8  00000001   409 NAMSG    EQU   *
000024B8  5800 867C                      0000267C   410          L     R0,=F'1'            set GLOBAL failed test indicator
000024BC  5000 A000                      00004000   411          ST    R0,FAILED
                                                    412
000024C0                       00000000             413          USING PLOAVAL,R5
                                                    414
000024C0  4820 5004                      00000004   415          LH    R2,ATNUM            get test number and convert
000024C4  4E20 A1A1                      000041A1   416          CVD   R2,DECNUM
000024C8  D211 A18B A175       0000418B  00004175   417          MVC   PRT3,EDIT
000024CE  DE11 A18B A1A1       0000418B  000041A1   418          ED    PRT3,DECNUM
000024D4  D202 A14F A198       0000414F  00004198   419          MVC   NAPRTNUM(3),PRT3+13 fill in message with test #
                                                    420
000024DA  B982 0022                                 421          XGR   R2,R2               get FUN as U8
000024DE  4320 5007                      00000007   422          IC    R2,AFUN
000024E2  4E20 A1A1                      000041A1   423          CVD   R2,DECNUM           and convert
000024E6  D211 A18B A175       0000418B  00004175   424          MVC   PRT3,EDIT
000024EC  DE11 A18B A1A1       0000418B  000041A1   425          ED    PRT3,DECNUM
000024F2  D201 A161 A199       00004161  00004199   426          MVC   NAPRTFUN(2),PRT3+14 fill in message with Fun field
                                                    427
000024F8                                            428          DROP R5
                                                    429 *
                                                    430 *        Use Hercules Diagnose for Message to console
                                                    431 *
000024F8  4100 0036                      00000036   432          LA    R0,NAPRTLNG         message length
000024FC  4110 A13F                      0000413F   433          LA    R1,NAPRTLINE        messagfe address
                                                    434
00002500  EBF5 8518 0024                 00002518   435          STMG  R15,R5,NACCSAV      save regs r15 to R5
                                                    436
00002506  4520 8598                      00002598   437          BAL   R2,MSG
                                                    438
0000250A  EBF5 8518 0004                 00002518   439          LMG   R15,R5,NACCSAV      restore regs
00002510  07FF                                      440          BR    R15                 Return to caller

00002518  00000000 00000000                         442 NACCSAV  DC     16D'0'             save area for MSG call
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    444 ************************************************************************
                                                    445 *        Issue HERCULES MESSAGE
                                                    446 *              R0 = length of message
                                                    447 *              R1 = message address
                                                    448 *              R2 = return address
                                                    449 ************************************************************************

00002598  4900 8688                         00002688  451 MSG      CH    R0,=H'0'            Do we even HAVE a message?
0000259C  07D2                                        452          BNHR  R2                  No, ignore

0000259E  9002 85D4                         000025D4  454          STM   R0,R2,MSGSAVE       Save registers

000025A2  4900 868A                         0000268A  456          CH    R0,=AL2(L'MSGMSG)   Message length within limits?
000025A6  47D0 85AE                         000025AE  457          BNH   MSGOK               Yes, continue
000025AA  4100 005F                         0000005F  458          LA    R0,L'MSGMSG         No, set to maximum

000025AE  1820                                        460 MSGOK    LR    R2,R0               Copy length to work register
000025B0  0620                                        461          BCTR  R2,0                Minus-1 for execute
000025B2  4420 85E0                         000025E0  462          EX    R2,MSGMVC           Copy message to O/P buffer

000025B6  4120 200A                         0000000A  464          LA    R2,1+L'MSGCMD(,R2)  Calculate true command length
000025BA  4110 85E6                         000025E6  465          LA    R1,MSGCMD           Point to true command

000025BE  83120008                                    467          DC    X'83',X'12',X'0008' Issue Hercules Diagnose X'008'
000025C2  4780 85CE                         000025CE  468          BZ    MSGRET              Return if successful
                                                    469
000025C6  1222                                        470          LTR   R2,R2               Is Diag8 Ry (R2) 0?
000025C8  4780 85CE                         000025CE  471          BZ    MSGRET                an error occurred but coninue
                                                    472
000025CC  0000                                        473          DC    H'0'                CRASH for debugging purposes

000025CE  9802 85D4                         000025D4  475 MSGRET   LM    R0,R2,MSGSAVE       Restore registers
000025D2  07F2                                        476          BR    R2                  Return to caller




000025D4  00000000 00000000                           478 MSGSAVE  DC    3F'0'               Registers save area
                                                    479
000025E0  D200 85EF 1000      000025EF  00000000  480 MSGMVC   MVC   MSGMSG(0),0(R1)      Executed instruction


000025E6  D4E2C7D5 D6C8405C                           482 MSGCMD   DC    C'MSGNOH * '        *** HERCULES MESSAGE COMMAND ***
000025EF  40404040 40404040                           483 MSGMSG   DC    CL95' '             The message text to be displayed
                                                    484
00002650                                              485          DS    0D
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     487 ************************************************************************
                                                     488 *        Normal completion or Abnormal termination PSWs
                                                     489 ************************************************************************

00002650  00020001 80000000                          491 EOJPSW   DC    0D'0',X'0002000180000000',AD(0)
00002660  B2B2 8650                    00002650      493 EOJ      LPSWE EOJPSW               Normal completion

00002668  00020001 80000000                          495 FAILPSW  DC    0D'0',X'0002000180000000',AD(X'BAD')
00002678  B2B2 8668                    00002668      497 FAILTEST LPSWE FAILPSW              Abnormal termination

                                                     499 ************************************************************************
                                                     500 *        Working Storage
                                                     501 ************************************************************************

0000267C                                             503          LTORG ,                    Literals pool
0000267C  00000001                                   504                =F'1'
00002680  000045B8                                   505                =A(PLOTESTS)
00002684  00000003                                   506                =XL4'3'
00002688  0000                                       507                =H'0'
0000268A  005F                                       508                =AL2(L'MSGMSG)
0000268C  40404040 40404040                          509                =CL128' '
                                                     510
0000270C  00000000                                   511 CTLR0    DS    F
00002710  00000000                                   512          DS    F
                                                     513
                                                     514 * Constants
                                                     515
                           00000400  00000001        516 K        EQU   1024               One KB
                           00001000  00000001        517 PAGE     EQU   (4*K)              Size of one page
                           00010000  00000001        518 K64      EQU   (64*K)             64 KB
                           00100000  00000001        519 MB       EQU   (K*K)              1 MB
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     521 *=====================================================================
                                                     522 *
                                                     523 *  NOTE: start data on an address that is easy to display
                                                     524 *        within Hercules
                                                     525 *
                                                     526 *=====================================================================
                                                     527
00002714                       00002714  00004000    528          ORG   PLOETST+X'4000'
00004000  00000000                                   529 FAILED   DC    F'0'                 some test failed?
00004004  00000000                                   530 TESTING  DC    F'0'                 current test number
                                                     532 *************************************************************************
                                                     533 *        TEST failed : instruction data
                                                     534 *************************************************************************
                                                     535 *
                                                     536 *        failed message and associated editting
                                                     537 *
00004008  40404040 40404040                          538 FAILLINE DC    C'         Test # '
00004018  A7A7A7                                     539 FAILTNUM DC    C'xxx'
0000401B  7A40                                       540          DC    C': '
                               00000015  00000001    541 FAILHLNG EQU   *-FAILLINE           fail line header length
                                                     542
0000401D  40404040 40404040                          543 FAILTXT  DC    CL100' '             Test fail message
                               00000079  00000001    544 FAILLNG  EQU   *-FAILLINE
                                                     545
                                                     547 *************************************************************************
                                                     548 *        TEST failed : CC message
                                                     549 *************************************************************************
                                                     550 *
                                                     551 *        failed message and associated editting
                                                     552 *
00004081  40404040 40404040                          553 CCPRTLINE DC   C'         Test # '
00004091  A7A7A7                                     554 CCPRTNUM DC    C'xxx'
00004094  7A40A699 96958740                          555          DC    c': wrong cc for PLO instruction'
000040B2  404D86A4 9583A389                          556          DC    c' (function '
000040BD  A7A7                                       557 CCPRTFUN DC    CL2'xx'
000040BF  5D                                         558          DC    c')'
000040C0  4085A797 8583A385                          559          DC    C' expected: cc='
000040CE  A7                                         560 CCPRTEXP DC    C'x'
000040CF  6B                                         561          DC    C','
000040D0  40998583 8589A585                          562          DC    C' received: cc='
000040DE  A7                                         563 CCPRTGOT DC    C'x'
000040DF  4B                                         564          DC    C'.'
                               0000005F  00000001    565 CCPRTLNG   EQU   *-CCPRTLINE
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    567 ************************************************************************
                                                    568 *        TEST failed : unequal results (UR) field message
                                                    569 ************************************************************************
                                                    570 *
                                                    571 *        failed message and associated editting
                                                    572 *
000040E0  40404040 40404040                         573 URPRTLINE DC   C'         Test # '
000040F0  A7A7A7                                    574 URPRTNUM DC    C'xxx'
000040F3  7A40A699 96958740                         575          DC    c': wrong UR for PLO instruction'
00004111  404D86A4 9583A389                         576          DC    c' (function '
0000411C  A7A7                                      577 URPRTFUN DC    CL2'xx'
0000411E  5D                                        578          DC    c')'
0000411F  4085A797 8583A385                         579          DC    C' expected: UR='
0000412D  A7                                        580 URPRTEXP DC    C'x'
0000412E  6B                                        581          DC    C','
0000412F  40998583 8589A585                         582          DC    C' received: UR='
0000413D  A7                                        583 URPRTGOT DC    C'x'
0000413E  4B                                        584          DC    C'.'
                            0000005F  00000001      585 URPRTLNG   EQU   *-URPRTLINE


                                                    587 ************************************************************************
                                                    588 *        TEST failed : PLO function is Not Available
                                                    589 ************************************************************************
                                                    590 *
                                                    591 *        failed message and associated editting
                                                    592 *
0000413F  40404040 40404040                         593 NAPRTLINE DC   C'         Test # '
0000414F  A7A7A7                                    594 NAPRTNUM DC    C'xxx'
00004152  7A40D7D3 D64086A4                         595          DC    C': PLO function '
00004161  A7A7                                      596 NAPRTFUN DC    CL2'xx'
00004163  4089A240 D5D6E340                         597          DC    C' is NOT installed.'
                                                    598
                            00000036  00000001      599 NAPRTLNG EQU   *-NAPRTLINE
                                                    600
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     602 ************************************************************************
                                                     603 *        TEST failed : message working storge
                                                     604 ************************************************************************
00004175  40212020 20202020                          605 EDIT     DC    XL18'402120202020202020202020202020202020'
                                                     606
00004187  7E7E7E6E                                   607          DC    C'===>'
0000418B  40404040 40404040                          608 PRT3     DC    CL18' '
0000419D  4C7E7E7E                                   609          DC    C'<==='
000041A1  00000000 00000000                          610 DECNUM   DS    CL16
                                                     611
                                                     612 * ------------------------------------------------------------------
                                                     613 * Random Bytes for Test Data (1 KB)
                                                     614 * ------------------------------------------------------------------
000041B8                                             615          DS    0D
000041B8  7A1FC308 9DE246B1                          616 RANDX    DC    XL16'7A 1F C3 08 9D E2 46 B1 5C F7 20 8B 64 DA 03 91'
000041C8  BE550CA7 3ED87219                          617          DC    XL16'BE 55 0C A7 3E D8 72 19 F4 6B 80 2D C9 14 EF 5A'
000041D8  31AC77E0 4B960FD3                          618          DC    XL16'31 AC 77 E0 4B 96 0F D3 68 B9 25 FE 52 8D 17 C6'
000041E8  EB40A16E 09D5832A                          619          DC    XL16'EB 40 A1 6E 09 D5 83 2A F8 5F BC 12 74 C1 9A 36'
000041F8  0DE759B2 43FC861B                          620          DC    XL16'0D E7 59 B2 43 FC 86 1B D0 6C 28 A5 93 4E F1 7D'
00004208  C81165DE 3B90FA27                          621          DC    XL16'C8 11 65 DE 3B 90 FA 27 54 AD 02 79 E4 B7 4C 88'
00004218  16D970A3 5ECB34F5                          622          DC    XL16'16 D9 70 A3 5E CB 34 F5 81 0A BD 62 E9 47 9F 23'
00004228  D65801AF 75CA3DE1                          623          DC    XL16'D6 58 01 AF 75 CA 3D E1 92 6F 18 B4 50 FD 87 2C'
00004238  A844DB0E 69C235F0                          624          DC    XL16'A8 44 DB 0E 69 C2 35 F0 9B 56 ED 21 7F B8 04 D4'
00004248  63AA1DE6 4895CF30                          625          DC    XL16'63 AA 1D E6 48 95 CF 30 F9 82 5B 13 DC 76 A0 2F'
00004258  8E41F3B5 0B67CE24                          626          DC    XL16'8E 41 F3 B5 0B 67 CE 24 99 DA 53 E8 1A 7C C7 32'
00004268  F68549B0 1ED26A94                          627          DC    XL16'F6 85 49 B0 1E D2 6A 94 3F EB 07 5D A9 70 C4 26'
00004278  DD1084F2 38AB61C5                          628          DC    XL16'DD 10 84 F2 38 AB 61 C5 0C 97 4A E3 7B B6 29 D1'
00004288  5F8A15CC 73F800A6                          629          DC    XL16'5F 8A 15 CC 73 F8 00 A6 42 9C E5 37 BD 68 02 D7'
00004298  904DEA22 7EC059F1                          630          DC    XL16'90 4D EA 22 7E C0 59 F1 0F A4 6D DB 35 88 B3 1C'
000042A8  E0712BC9 469EF460                          631          DC    XL16'E0 71 2B C9 46 9E F4 60 AD 18 D5 83 3A BF 54 07'
000042B8  A73CF108 6DB249DE                          632          DC    XL16'A7 3C F1 08 6D B2 49 DE 15 83 5A C6 2F 91 E4 70'
000042C8  D30B58AC E9347F12                          633          DC    XL16'D3 0B 58 AC E9 34 7F 12 C1 66 9D 42 F8 25 B7 5E'
000042D8  81DA3704 9AE26BC5                          634          DC    XL16'81 DA 37 04 9A E2 6B C5 18 F3 4D 76 A0 2C BF 53'
000042E8  E7901D68 C83FA572                          635          DC    XL16'E7 90 1D 68 C8 3F A5 72 0E DB 46 B1 59 F6 23 8C'
000042F8  35CE7A11 E0945FB8                          636          DC    XL16'35 CE 7A 11 E0 94 5F B8 27 D4 63 0A F9 41 AD 86'
00004308  C31A75EC 509F28D7                          637          DC    XL16'C3 1A 75 EC 50 9F 28 D7 62 B4 0D E8 39 87 FA 14'
00004318  6FC03295 4ABE07D1                          638          DC    XL16'6F C0 32 95 4A BE 07 D1 7C E5 21 89 56 F2 A9 3D'
00004328  B648DF10 649C2AF5                          639          DC    XL16'B6 48 DF 10 64 9C 2A F5 73 C7 0F 82 ED 31 A4 5B'
00004338  D926904F B36AE10C                          640          DC    XL16'D9 26 90 4F B3 6A E1 0C 78 C2 55 AF 13 F7 84 3E'
00004348  69D5209B 47EA7106                          641          DC    XL16'69 D5 20 9B 47 EA 71 06 BD 52 F0 38 8E C9 16 A2'
00004358  5DE62B97 40BC037E                          642          DC    XL16'5D E6 2B 97 40 BC 03 7E D0 19 A8 65 F4 33 8A C4'
00004368  1F74EB57 920AD843                          643          DC    XL16'1F 74 EB 57 92 0A D8 43 B5 6C F1 2E 80 CD 39 A6'
00004378  E3519E24 7BC80DB0                          644          DC    XL16'E3 51 9E 24 7B C8 0D B0 67 F5 32 8F DA 14 46 AC'
00004388  79E90593 2DC658B1                          645          DC    XL16'79 E9 05 93 2D C6 58 B1 7F 30 ED 84 12 A5 6E D2'
00004398  4CF8219A 63BE07D6                          646          DC    XL16'4C F8 21 9A 63 BE 07 D6 50 C3 88 1B F0 35 A9 72'
000043A8  DB448D17 E269B72C                          647          DC    XL16'DB 44 8D 17 E2 69 B7 2C 95 0F C1 5A F6 38 AE 73'
000043B8  4FA8D21C 9B673EF0                          648          DC    XL16'4F A8 D2 1C 9B 67 3E F0 85 14 C9 72 AD 5F E1 38'
000043C8  B72D94FE 610AC357                          649          DC    XL16'B7 2D 94 FE 61 0A C3 57 88 D1 4B AE 23 F9 76 10'
000043D8  D8459A3C E78116BF                          650          DC    XL16'D8 45 9A 3C E7 81 16 BF 52 CC 09 74 A1 ED 68 35'
000043E8  FA20B35D 8EC7419C                          651          DC    XL16'FA 20 B3 5D 8E C7 41 9C 6B F4 17 DA 80 2E 53 BD'
000043F8  31CE7A05 E8924FB1                          652          DC    XL16'31 CE 7A 05 E8 92 4F B1 6D 23 FC 58 A7 10 D9 84'
00004408  65B8E31F 904AC67D                          653          DC    XL16'65 B8 E3 1F 90 4A C6 7D 2B F5 18 A2 DE 73 0C 49'
00004418  87D43FB9 21EC5608                          654          DC    XL16'87 D4 3F B9 21 EC 56 08 C1 7A AD 34 F2 6E 95 10'
00004428  CB42F81D 73A60ED5                          655          DC    XL16'CB 42 F8 1D 73 A6 0E D5 89 2C B7 50 EF 14 68 93'
00004438  3AD186FC 275BE09D                          656          DC    XL16'3A D1 86 FC 27 5B E0 9D 41 B8 12 C7 6F A4 35 EE'
00004448  904CDB18 75A2F63E                          657          DC    XL16'90 4C DB 18 75 A2 F6 3E C9 07 B1 5D E8 24 7A CF'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00004458  16E3588B D032FA6D                          658          DC    XL16'16 E3 58 8B D0 32 FA 6D 97 41 BC 0F 74 A5 E1 2C'
00004468  F78913CE 5AD420B6                          659          DC    XL16'F7 89 13 CE 5A D4 20 B6 7F E2 49 9C 35 A8 01 DD'
00004478  68F12B95 C07E14BA                          660          DC    XL16'68 F1 2B 95 C0 7E 14 BA 53 EF 86 39 D7 42 AC 10'
00004488  B46DE821 F9570CC3                          661          DC    XL16'B4 6D E8 21 F9 57 0C C3 7A DE 45 98 12 FB 64 3E'
00004498  8D30C75B E1A419F2                          662          DC    XL16'8D 30 C7 5B E1 A4 19 F2 76 0F D8 43 BE 95 28 6A'
000044A8  F35E80CA 176DB924                          663          DC    XL16'F3 5E 80 CA 17 6D B9 24 E7 41 0A DC 58 92 AF 35'
000044B8  34C17DE9 52AB06F8                          664          DC    XL16'34 C1 7D E9 52 AB 06 F8 1E 93 6C D4 27 B0 5F EA'
000044C8  7914CE63 A82DF145                          665          DC    XL16'79 14 CE 63 A8 2D F1 45 9B 30 E7 5A 8C D6 12 BF'
000044D8  E2580FB4 71DA2983                          666          DC    XL16'E2 58 0F B4 71 DA 29 83 C5 1C F7 6E 90 3D A1 4B'
000044E8  0DFE6738 C2951AB8                          667          DC    XL16'0D FE 67 38 C2 95 1A B8 4F E3 72 DC 06 A9 54 21'
000044F8  BC47D18E 23F5609A                          668          DC    XL16'BC 47 D1 8E 23 F5 60 9A 14 C8 7D E0 35 B2 6F 91'
00004508  A619FB42 D70C853E                          669          DC    XL16'A6 19 FB 42 D7 0C 85 3E E1 74 28 CA 5D 90 B3 67'
00004518  12E853AC 39F60BD1                          670          DC    XL16'12 E8 53 AC 39 F6 0B D1 7C A5 24 EF 68 93 4D B7'
00004528  F02A9E65 C318DB47                          671          DC    XL16'F0 2A 9E 65 C3 18 DB 47 81 5C EA 36 72 0F B9 D4'
00004538  58A13CE7 942FD06B                          672          DC    XL16'58 A1 3C E7 94 2F D0 6B C5 19 FE 43 87 B2 0D 7A'
00004548  CD7016F9 4AB528E3                          673          DC    XL16'CD 70 16 F9 4A B5 28 E3 8D 51 AC 37 F0 64 9B 12'
00004558  26DE815B C4F7309D                          674          DC    XL16'26 DE 81 5B C4 F7 30 9D EA 14 68 A2 3F D1 75 BC'
00004568  910EF467 2BC853AD                          675          DC    XL16'91 0E F4 67 2B C8 53 AD 18 E7 45 9A D0 36 FB 72'
00004578  AF3DC184 59E217B6                          676          DC    XL16'AF 3D C1 84 59 E2 17 B6 F8 20 D7 6A 93 4C 0F E1'
00004588  73B924DA 058EC741                          677          DC    XL16'73 B9 24 DA 05 8E C7 41 F2 6D 10 AC 57 E8 3B 96'
00004598  1F62D5A8 34EB790C                          678          DC    XL16'1F 62 D5 A8 34 EB 79 0C C0 4A 97 F3 2E BD 61 85'
000045A8  E4378AD2 5CF019B7                          679          DC    XL16'E4 37 8A D2 5C F0 19 B7 46 CE 23 9D 70 1B A5 F8'
                                                     680
                                                     681 * ------------------------------------------------------------------
                                                     682 * Test Table is Located here .....
                                                     683 * ------------------------------------------------------------------
000045B8                                             684 PLOTESTS DS    0D
                                                     685 *
                                                     686 *         PTTABLE will be located here
                                                     687 *
000045B8                  000045B8  000049C0         688          ORG   *+(256*4)+8          1K for pointers for up to 256 tests
                                                     689 *                                        plus 8 bytes for table end
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    691 ********************************************************************************
                                                    692 *        PLOETEST DSECT
                                                    693 ********************************************************************************


                                                    695 PLOETEST DSECT ,
00000000  00000000                                  696 TSUB     DC    A(0)               pointer  to test
00000004  0000                                      697 TNUM     DC    H'00'              Test Number
00000006  00                                        698          DC    XL1'00'
00000007  00                                        699 FUN      DC    HL1'00'            Function
00000008  00                                        700 UR       DC    HL1'00'            expected unequal results (UR)
00000009  00                                        701 URFOUND  DS    HL1'00'            extracted UR from PL
                                                    702
0000000A  00                                        703 CC       DC    HL1'00'            cc
0000000B  00                                        704 CCMASK   DC    HL1'00'            not expected CC mask
                                                    705
0000000C  00                                        706 CCFOUND  DS    X                  extracted cc
00000010  00000000 00000000                         707 CCPSW    DS    2F                 extract PSW after test (has CC)
                                                    708
00000018  00000000                                  709 FAILTXTA DC    A(0)               address of test fail message
0000001C  00000000                                  710 OPSRCA   DC    A(0)               address of operand source
                           00000020  00000001       711 OPREPA   EQU   *                  address of operand replacement
00000020  00000000                                  712 OPDSTA   DC    A(0)               address of operand destination
00000024  00000000                                  713 PLA      DC    A(0)               address of parameter list
00000028  00000000                                  714 RANDXA   DC    A(0)               address of random source data


                                                    716 ********************************************************************************
                                                    717 *        PLOEAVL DSECT
                                                    718 ********************************************************************************


                                                    720 PLOAVAL  DSECT ,
00000000  00000000                                  721 ATSUB    DC    A(0)               pointer  to test
00000004  0000                                      722 ATNUM    DC    H'00'              Test Number
00000006  00                                        723          DC    XL1'00'
00000007  00                                        724 AFUN     DC    HL1'00'            Function
00000008                                            725          DS    0F
```

```
 LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   727 ********************************************************************************
                                                   728 *     Macros to help build test tables
                                                   729 ********************************************************************************


                                                   731 *-------------------------------------------------------------------------
                                                   732 * TSTPLO : test whether a PLO function is available/installed.
                                                   733 *          If a functionm is installed, CC=0. Not installed, CC=3.
                                                   734 *          - &fun: the function number
                                                   735 *-------------------------------------------------------------------------
                                                   736          MACRO
                                                   737          TSTPLO  &FUN
                                                   738          GBLA  &TNUM
                                                   739 &TNUM    SETA  &TNUM+1
                                                   740 *
                                                   741 * test if PLO function is available
                                                   742 *
                                                   743          USING *,R5                base for test data and test routine
                                                   744          USING PLOAVAL,R5
                                                   745
                                                   746 T&TNUM   DC    A(CODE&TNUM)           address of test routine
                                                   747          DC    H'&TNUM'              test number
                                                   748          DC    X'00'
                                                   749          DC    HL1'&FUN'             Function Number
                                                   750 *
                                                   751 CODE&TNUM DS    0F
                                                   752          LA    R1,(256+&FUN)         T bit 55  (+256)
                                                   753          LR    R0,R1
                                                   754          PLO   0,0,0,0
                                                   755          BZ    0(R11)                Function is installed, return
                                                   756
                                                   757          BAL   R15,NAMSG             Function not available message
                                                   758          BR    R11                   return
                                                   759          DROP  R5
                                                   760 *
                                                   761          MEND
                                                   762
                                                   763 *-------------------------------------------------------------------------
                                                   764 * TSTLBL : insert a label with a test number suffix
                                                   765 *          - &name: the label name
                                                   766 *-------------------------------------------------------------------------
                                                   767          MACRO
                                                   768          TSTLBL  &NAME
                                                   769          GBLA  &TNUM
                                                   770          LCLC  &X
                                                   771 &X       SETC  '&NAME&TNUM'
                                                   772          DC    CL32'>>>&X<<<'        eye catcher
                                                   773 &X       DS  0C
                                                   774          MEND
```

```
 LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   776 *-------------------------------------------------------------------
                                                   777 * TSTEND : end a test
                                                   778 *-------------------------------------------------------------------
                                                   779          MACRO
                                                   780          TSTEND  &NAME
                                                   781          GBLA  &TNUM
                                                   782          DC    CL32'END of test: &TNUM <<<'
                                                   783          MEND
                                                   784
                                                   785 *-------------------------------------------------------------------
                                                   786 * TSTCODE : The code for the test follows the label
                                                   787 *-------------------------------------------------------------------
                                                   788          MACRO
                                                   789          TSTCODE  &NAME
                                                   790          GBLA  &TNUM
                                                   791          LCLC  &X
                                                   792 &X       SETC  'CODE&TNUM'
                                                   793 &X       DS    0F
                                                   794          MEND
                                                   795
                                                   796 *-------------------------------------------------------------------
                                                   797 * TSTDATA: The data for the test follows.
                                                   798 *          Data is aligned to the 1.5K boundary after the TSTHDR
                                                   799 *-------------------------------------------------------------------
                                                   800          MACRO
                                                   801          TSTDATA  &NAME
                                                   802          GBLA  &THDRNUM
                                                   803          ORG   PLOETST+(128*K)+&THDRNUM*(3*K)+(K+512)
                                                   804          MEND
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     806 *-------------------------------------------------------------------
                                                     807 * TSTHDR: The start of a PLOE function test
                                                     808 *         Test is aligned to 3K boundary starting at 128K
                                                     809 *              (1.5K data/code; 1.5K instruction test data)
                                                     810 *
                                                     811 *        Note: PLOETEST DSECT needs to be in-sync with TSTHDR
                                                     812 *-------------------------------------------------------------------
                                                     813          MACRO
                                                     814          TSTHDR  &CC,&UR,&FUN
                                                     815 .*                      &CC    - expected CC
                                                     816 .*                      &UR    - expected unequal results
                                                     817 .*                      &FUN   - function
                                                     818
                                                     819          LCLA  &XCC(4)  &CC has mask values for FAILED condition codes
                                                     820 &XCC(1)  SETA  7                   CC != 0
                                                     821 &XCC(2)  SETA  11                  CC != 1
                                                     822 &XCC(3)  SETA  13                  CC != 2
                                                     823 &XCC(4)  SETA  14                  CC != 3
                                                     824
                                                     825          GBLA  &TNUM
                                                     826 &TNUM    SETA  &TNUM+1
                                                     827
                                                     828          GBLA  &THDRNUM
                                                     829 &THDRNUM SETA  &THDRNUM+1
                                                     830
                                                     831          ORG   PLOETST+(128*K)+&THDRNUM*(3*K)
                                                     832          DC    CL32'START of test: &TNUM'
                                                     833
                                                     834          USING *,R5                base for test data and test routine
                                                     835          USING PLOETEST,R5
                                                     836
                                                     837 T&TNUM   DC    A(CODE&TNUM)        address of test code
                                                     838          DC    H'&TNUM'            test number
                                                     839          DC    X'00'
                                                     840          DC    HL1'&FUN'           Function
                                                     841          DC    HL1'&UR'            expected unequal results (UR)
                                                     842          DC    HL1'00'             found UR
                                                     843
                                                     844          DC    HL1'&CC'            cc
                                                     845          AIF   ('&CC' EQ '99').NOCC
                                                     846          DC    HL1'&XCC(&CC+1)'    not expected CC mask
                                                     847          AGO   .CCCONT
                                                     848 .NOCC    ANOP
                                                     849          DC    HL1'00'             cc not checked
                                                     850 .CCCONT  ANOP
                                                     851
                                                     852          DS    X                   extracted cc
                                                     853          DS    2F                  extract PSW after test (has CC)
                                                     854          DC    A(FMSG&TNUM)        address of test fail message
                                                     855          DC    A(OPSRC&TNUM)       address of operand source
                                                     856          DC    A(OPDST&TNUM)       address of operand destination
                                                     857          DC    A(PL&TNUM)          address of parameter list
                                                     858          DC    A(RANDX+&TNUM)      address of random source data
                                                     859 .*
                                                     860          MEND
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

                                                      862 *-------------------------------------------------------------------
                                                      863 * PTTABLE: generate table of pointers to individual tests
                                                      864 *-------------------------------------------------------------------
                                                      865
                                                      866          MACRO
                                                      867          PTTABLE
                                                      868          GBLA  &TNUM
                                                      869          LCLA  &CUR
                                                      870 &CUR     SETA  1
                                                      871 .*
                                                      872 TTABLE   DS    0F
                                                      873 .LOOP    ANOP
                                                      874 .*
                                                      875          DC    A(T&CUR)
                                                      876 .*
                                                      877 &CUR     SETA  &CUR+1
                                                      878          AIF   (&CUR LE &TNUM).LOOP
                                                      879 *
                                                      880          DC    A(0)                END OF TABLE
                                                      881          DC    A(0)
                                                      882 .*
                                                      883          MEND
                                                      884
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     886 ********************************************************************************
                                                     887 *        PLO-Ext tests
                                                     888 ********************************************************************************
                               00000000  0004F280    889 PLOETST  CSECT
                                                     890          PRINT DATA
                                                     891 * ------------------------------------------------------------------------
                                                     892 * Test PLO_Extension functions are available/installed
                                                     893 * ------------------------------------------------------------------------
                                                     894          TSTPLO 24
                                                     895+*
                                                     896+* test if PLO function is available
                                                     897+*
000049C0                       000049C0              898+         USING *,R5                base for test data and test routine
000049C0                       00000000              899+         USING PLOAVAL,R5
000049C0  000049C8                                   900+T1       DC    A(CODE1)               address of test routine
000049C4  0001                                       901+         DC    H'1'              test number
000049C6  00                                         902+         DC    X'00'
000049C7  18                                         903+         DC    HL1'24'           Function Number
                                                     904+*
000049C8                                             905+CODE1    DS    0F
000049C8  4110 0118                    00000118      906+         LA    R1,(256+24)       T bit 55  (+256)
000049CC  1801                                       907+         LR    R0,R1
000049CE  EE00 0000 0000       00000000  00000000    908+         PLO   0,0,0,0
000049D4  478B 0000                    00000000      909+         BZ    0(R11)            Function is installed, return
000049D8  45F0 84B8                    000024B8      910+         BAL   R15,NAMSG         Function not available message
000049DC  07FB                                       911+         BR    R11               return
000049DE                                             912+         DROP  R5
                                                     913+*
                                                     914          TSTPLO 25
                                                     915+*
                                                     916+* test if PLO function is available
                                                     917+*
000049DE                       000049DE              918+         USING *,R5                base for test data and test routine
000049DE                       00000000              919+         USING PLOAVAL,R5
000049E0  000049E8                                   920+T2       DC    A(CODE2)               address of test routine
000049E4  0002                                       921+         DC    H'2'              test number
000049E6  00                                         922+         DC    X'00'
000049E7  19                                         923+         DC    HL1'25'           Function Number
                                                     924+*
000049E8                                             925+CODE2    DS    0F
000049E8  4110 0119                    00000119      926+         LA    R1,(256+25)       T bit 55  (+256)
000049EC  1801                                       927+         LR    R0,R1
000049EE  EE00 0000 0000       00000000  00000000    928+         PLO   0,0,0,0
000049F4  478B 0000                    00000000      929+         BZ    0(R11)            Function is installed, return
000049F8  45F0 84B8                    000024B8      930+         BAL   R15,NAMSG         Function not available message
000049FC  07FB                                       931+         BR    R11               return
000049FE                                             932+         DROP  R5
                                                     933+*
                                                     934          TSTPLO 26
                                                     935+*
                                                     936+* test if PLO function is available
                                                     937+*
000049FE                       000049FE              938+         USING *,R5                base for test data and test routine
000049FE                       00000000              939+         USING PLOAVAL,R5
00004A00  00004A08                                   940+T3       DC    A(CODE3)               address of test routine
00004A04  0003                                       941+         DC    H'3'              test number
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00004A06  00                                       942+          DC    X'00'
00004A07  1A                                       943+          DC    HL1'26'             Function Number
                                                   944+*
00004A08                                           945+CODE3     DS    0F
00004A08  4110 011A                      0000011A  946+          LA    R1,(256+26)         T bit 55  (+256)
00004A0C  1801                                     947+          LR    R0,R1
00004A0E  EE00 0000 0000      00000000   00000000  948+          PLO   0,0,0,0
00004A14  478B 0000                      00000000  949+          BZ    0(R11)              Function is installed, return
00004A18  45F0 84B8                      000024B8  950+          BAL   R15,NAMSG           Function not available message
00004A1C  07FB                                     951+          BR    R11                 return
00004A1E                                           952+          DROP  R5
                                                   953+*
                                                   954           TSTPLO 27
                                                   955+*
                                                   956+* test if PLO function is available
                                                   957+*
00004A1E                  00004A1E                 958+          USING *,R5                base for test data and test routine
00004A1E                  00000000                 959+          USING PLOAVAL,R5
00004A20  00004A28                                 960+T4        DC    A(CODE4)               address of test routine
00004A24  0004                                     961+          DC    H'4'                test number
00004A26  00                                       962+          DC    X'00'
00004A27  1B                                       963+          DC    HL1'27'             Function Number
                                                   964+*
00004A28                                           965+CODE4     DS    0F
00004A28  4110 011B                      0000011B  966+          LA    R1,(256+27)         T bit 55  (+256)
00004A2C  1801                                     967+          LR    R0,R1
00004A2E  EE00 0000 0000      00000000   00000000  968+          PLO   0,0,0,0
00004A34  478B 0000                      00000000  969+          BZ    0(R11)              Function is installed, return
00004A38  45F0 84B8                      000024B8  970+          BAL   R15,NAMSG           Function not available message
00004A3C  07FB                                     971+          BR    R11                 return
00004A3E                                           972+          DROP  R5
                                                   973+*
                                                   974           TSTPLO 28
                                                   975+*
                                                   976+* test if PLO function is available
                                                   977+*
00004A3E                  00004A3E                 978+          USING *,R5                base for test data and test routine
00004A3E                  00000000                 979+          USING PLOAVAL,R5
00004A40  00004A48                                 980+T5        DC    A(CODE5)               address of test routine
00004A44  0005                                     981+          DC    H'5'                test number
00004A46  00                                       982+          DC    X'00'
00004A47  1C                                       983+          DC    HL1'28'             Function Number
                                                   984+*
00004A48                                           985+CODE5     DS    0F
00004A48  4110 011C                      0000011C  986+          LA    R1,(256+28)         T bit 55  (+256)
00004A4C  1801                                     987+          LR    R0,R1
00004A4E  EE00 0000 0000      00000000   00000000  988+          PLO   0,0,0,0
00004A54  478B 0000                      00000000  989+          BZ    0(R11)              Function is installed, return
00004A58  45F0 84B8                      000024B8  990+          BAL   R15,NAMSG           Function not available message
00004A5C  07FB                                     991+          BR    R11                 return
00004A5E                                           992+          DROP  R5
                                                   993+*
                                                   994           TSTPLO 29
                                                   995+*
                                                   996+* test if PLO function is available
                                                   997+*
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00004A5E                     00004A5E               998+          USING *,R5                base for test data and test routine
00004A5E                     00000000               999+          USING PLOAVAL,R5
00004A60  00004A68                                 1000+T6        DC    A(CODE6)               address of test routine
00004A64  0006                                     1001+          DC    H'6'                   test number
00004A66  00                                       1002+          DC    X'00'
00004A67  1D                                       1003+          DC    HL1'29'                Function Number
                                                   1004+*
00004A68                                           1005+CODE6     DS    0F
00004A68  4110 011D                     0000011D   1006+          LA    R1,(256+29)            T bit 55  (+256)
00004A6C  1801                                     1007+          LR    R0,R1
00004A6E  EE00 0000 0000     00000000  00000000   1008+          PLO   0,0,0,0
00004A74  478B 0000                     00000000   1009+          BZ    0(R11)                 Function is installed, return
00004A78  45F0 84B8                     000024B8   1010+          BAL   R15,NAMSG              Function not available message
00004A7C  07FB                                     1011+          BR    R11                    return
00004A7E                                           1012+          DROP  R5
                                                   1013+*
                                                   1014           TSTPLO 30
                                                   1015+*
                                                   1016+* test if PLO function is available
                                                   1017+*
00004A7E                     00004A7E              1018+          USING *,R5                base for test data and test routine
00004A7E                     00000000              1019+          USING PLOAVAL,R5
00004A80  00004A88                                 1020+T7        DC    A(CODE7)               address of test routine
00004A84  0007                                     1021+          DC    H'7'                   test number
00004A86  00                                       1022+          DC    X'00'
00004A87  1E                                       1023+          DC    HL1'30'                Function Number
                                                   1024+*
00004A88                                           1025+CODE7     DS    0F
00004A88  4110 011E                     0000011E   1026+          LA    R1,(256+30)            T bit 55  (+256)
00004A8C  1801                                     1027+          LR    R0,R1
00004A8E  EE00 0000 0000     00000000  00000000   1028+          PLO   0,0,0,0
00004A94  478B 0000                     00000000   1029+          BZ    0(R11)                 Function is installed, return
00004A98  45F0 84B8                     000024B8   1030+          BAL   R15,NAMSG              Function not available message
00004A9C  07FB                                     1031+          BR    R11                    return
00004A9E                                           1032+          DROP  R5
                                                   1033+*
                                                   1034           TSTPLO 31
                                                   1035+*
                                                   1036+* test if PLO function is available
                                                   1037+*
00004A9E                     00004A9E              1038+          USING *,R5                base for test data and test routine
00004A9E                     00000000              1039+          USING PLOAVAL,R5
00004AA0  00004AA8                                 1040+T8        DC    A(CODE8)               address of test routine
00004AA4  0008                                     1041+          DC    H'8'                   test number
00004AA6  00                                       1042+          DC    X'00'
00004AA7  1F                                       1043+          DC    HL1'31'                Function Number
                                                   1044+*
00004AA8                                           1045+CODE8     DS    0F
00004AA8  4110 011F                     0000011F   1046+          LA    R1,(256+31)            T bit 55  (+256)
00004AAC  1801                                     1047+          LR    R0,R1
00004AAE  EE00 0000 0000     00000000  00000000   1048+          PLO   0,0,0,0
00004AB4  478B 0000                     00000000   1049+          BZ    0(R11)                 Function is installed, return
00004AB8  45F0 84B8                     000024B8   1050+          BAL   R15,NAMSG              Function not available message
00004ABC  07FB                                     1051+          BR    R11                    return
00004ABE                                           1052+          DROP  R5
                                                   1053+*
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    1054          TSTPLO 32
                                                    1055+*
                                                    1056+* test if PLO function is available
                                                    1057+*
00004ABE                      00004ABE              1058+         USING *,R5                base for test data and test routine
00004ABE                      00000000              1059+         USING PLOAVAL,R5
00004AC0  00004AC8                                  1060+T9       DC    A(CODE9)               address of test routine
00004AC4  0009                                      1061+         DC    H'9'              test number
00004AC6  00                                        1062+         DC    X'00'
00004AC7  20                                        1063+         DC    HL1'32'           Function Number
                                                    1064+*
00004AC8                                            1065+CODE9    DS    0F
00004AC8  4110 0120                       00000120  1066+         LA    R1,(256+32)       T bit 55  (+256)
00004ACC  1801                                      1067+         LR    R0,R1
00004ACE  EE00 0000 0000      00000000    00000000  1068+         PLO   0,0,0,0
00004AD4  478B 0000                       00000000  1069+         BZ    0(R11)            Function is installed, return
00004AD8  45F0 84B8                       000024B8  1070+         BAL   R15,NAMSG         Function not available message
00004ADC  07FB                                      1071+         BR    R11               return
00004ADE                                            1072+         DROP  R5
                                                    1073+*
                                                    1074          TSTPLO 33
                                                    1075+*
                                                    1076+* test if PLO function is available
                                                    1077+*
00004ADE                      00004ADE              1078+         USING *,R5                base for test data and test routine
00004ADE                      00000000              1079+         USING PLOAVAL,R5
00004AE0  00004AE8                                  1080+T10      DC    A(CODE10)              address of test routine
00004AE4  000A                                      1081+         DC    H'10'             test number
00004AE6  00                                        1082+         DC    X'00'
00004AE7  21                                        1083+         DC    HL1'33'           Function Number
                                                    1084+*
00004AE8                                            1085+CODE10   DS    0F
00004AE8  4110 0121                       00000121  1086+         LA    R1,(256+33)       T bit 55  (+256)
00004AEC  1801                                      1087+         LR    R0,R1
00004AEE  EE00 0000 0000      00000000    00000000  1088+         PLO   0,0,0,0
00004AF4  478B 0000                       00000000  1089+         BZ    0(R11)            Function is installed, return
00004AF8  45F0 84B8                       000024B8  1090+         BAL   R15,NAMSG         Function not available message
00004AFC  07FB                                      1091+         BR    R11               return
00004AFE                                            1092+         DROP  R5
                                                    1093+*
                                                    1094          TSTPLO 34
                                                    1095+*
                                                    1096+* test if PLO function is available
                                                    1097+*
00004AFE                      00004AFE              1098+         USING *,R5                base for test data and test routine
00004AFE                      00000000              1099+         USING PLOAVAL,R5
00004B00  00004B08                                  1100+T11      DC    A(CODE11)              address of test routine
00004B04  000B                                      1101+         DC    H'11'             test number
00004B06  00                                        1102+         DC    X'00'
00004B07  22                                        1103+         DC    HL1'34'           Function Number
                                                    1104+*
00004B08                                            1105+CODE11   DS    0F
00004B08  4110 0122                       00000122  1106+         LA    R1,(256+34)       T bit 55  (+256)
00004B0C  1801                                      1107+         LR    R0,R1
00004B0E  EE00 0000 0000      00000000    00000000  1108+         PLO   0,0,0,0
00004B14  478B 0000                       00000000  1109+         BZ    0(R11)            Function is installed, return
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00004B18  45F0 84B8                    000024B8  1110+          BAL   R15,NAMSG           Function not available message
00004B1C  07FB                                   1111+          BR    R11                 return
00004B1E                                         1112+          DROP  R5
                                                 1113+*
                                                 1114           TSTPLO 35
                                                 1115+*
                                                 1116+* test if PLO function is available
                                                 1117+*
00004B1E                  00004B1E               1118+          USING *,R5                base for test data and test routine
00004B1E                  00000000               1119+          USING PLOAVAL,R5
00004B20  00004B28                               1120+T12       DC    A(CODE12)              address of test routine
00004B24  000C                                   1121+          DC    H'12'               test number
00004B26  00                                     1122+          DC    X'00'
00004B27  23                                     1123+          DC    HL1'35'             Function Number
                                                 1124+*
00004B28                                         1125+CODE12    DS    0F
00004B28  4110 0123                    00000123  1126+          LA    R1,(256+35)         T bit 55  (+256)
00004B2C  1801                                   1127+          LR    R0,R1
00004B2E  EE00 0000 0000  00000000     00000000  1128+          PLO   0,0,0,0
00004B34  478B 0000                    00000000  1129+          BZ    0(R11)              Function is installed, return
00004B38  45F0 84B8                    000024B8  1130+          BAL   R15,NAMSG           Function not available message
00004B3C  07FB                                   1131+          BR    R11                 return
00004B3E                                         1132+          DROP  R5
                                                 1133+*
                                                 1134           TSTPLO 36
                                                 1135+*
                                                 1136+* test if PLO function is available
                                                 1137+*
00004B3E                  00004B3E               1138+          USING *,R5                base for test data and test routine
00004B3E                  00000000               1139+          USING PLOAVAL,R5
00004B40  00004B48                               1140+T13       DC    A(CODE13)              address of test routine
00004B44  000D                                   1141+          DC    H'13'               test number
00004B46  00                                     1142+          DC    X'00'
00004B47  24                                     1143+          DC    HL1'36'             Function Number
                                                 1144+*
00004B48                                         1145+CODE13    DS    0F
00004B48  4110 0124                    00000124  1146+          LA    R1,(256+36)         T bit 55  (+256)
00004B4C  1801                                   1147+          LR    R0,R1
00004B4E  EE00 0000 0000  00000000     00000000  1148+          PLO   0,0,0,0
00004B54  478B 0000                    00000000  1149+          BZ    0(R11)              Function is installed, return
00004B58  45F0 84B8                    000024B8  1150+          BAL   R15,NAMSG           Function not available message
00004B5C  07FB                                   1151+          BR    R11                 return
00004B5E                                         1152+          DROP  R5
                                                 1153+*
                                                 1154           TSTPLO 37
                                                 1155+*
                                                 1156+* test if PLO function is available
                                                 1157+*
00004B5E                  00004B5E               1158+          USING *,R5                base for test data and test routine
00004B5E                  00000000               1159+          USING PLOAVAL,R5
00004B60  00004B68                               1160+T14       DC    A(CODE14)              address of test routine
00004B64  000E                                   1161+          DC    H'14'               test number
00004B66  00                                     1162+          DC    X'00'
00004B67  25                                     1163+          DC    HL1'37'             Function Number
                                                 1164+*
00004B68                                         1165+CODE14    DS    0F
```

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  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00004B68  4110 0125                      00000125  1166+          LA    R1,(256+37)          T bit 55  (+256)
00004B6C  1801                                     1167+          LR    R0,R1
00004B6E  EE00 0000 0000      00000000  00000000  1168+          PLO   0,0,0,0
00004B74  478B 0000                      00000000  1169+          BZ    0(R11)               Function is installed, return
00004B78  45F0 84B8                      000024B8  1170+          BAL   R15,NAMSG            Function not available message
00004B7C  07FB                                     1171+          BR    R11                  return
00004B7E                                           1172+          DROP  R5
                                                   1173+*
                                                   1174           TSTPLO 38
                                                   1175+*
                                                   1176+* test if PLO function is available
                                                   1177+*
00004B7E                  00004B7E                 1178+          USING *,R5                 base for test data and test routine
00004B7E                  00000000                 1179+          USING PLOAVAL,R5
00004B80  00004B88                                 1180+T15       DC    A(CODE15)               address of test routine
00004B84  000F                                     1181+          DC    H'15'                test number
00004B86  00                                       1182+          DC    X'00'
00004B87  26                                       1183+          DC    HL1'38'              Function Number
                                                   1184+*
00004B88                                           1185+CODE15    DS    0F
00004B88  4110 0126                      00000126  1186+          LA    R1,(256+38)          T bit 55  (+256)
00004B8C  1801                                     1187+          LR    R0,R1
00004B8E  EE00 0000 0000      00000000  00000000  1188+          PLO   0,0,0,0
00004B94  478B 0000                      00000000  1189+          BZ    0(R11)               Function is installed, return
00004B98  45F0 84B8                      000024B8  1190+          BAL   R15,NAMSG            Function not available message
00004B9C  07FB                                     1191+          BR    R11                  return
00004B9E                                           1192+          DROP  R5
                                                   1193+*
                                                   1194           TSTPLO 39
                                                   1195+*
                                                   1196+* test if PLO function is available
                                                   1197+*
00004B9E                  00004B9E                 1198+          USING *,R5                 base for test data and test routine
00004B9E                  00000000                 1199+          USING PLOAVAL,R5
00004BA0  00004BA8                                 1200+T16       DC    A(CODE16)               address of test routine
00004BA4  0010                                     1201+          DC    H'16'                test number
00004BA6  00                                       1202+          DC    X'00'
00004BA7  27                                       1203+          DC    HL1'39'              Function Number
                                                   1204+*
00004BA8                                           1205+CODE16    DS    0F
00004BA8  4110 0127                      00000127  1206+          LA    R1,(256+39)          T bit 55  (+256)
00004BAC  1801                                     1207+          LR    R0,R1
00004BAE  EE00 0000 0000      00000000  00000000  1208+          PLO   0,0,0,0
00004BB4  478B 0000                      00000000  1209+          BZ    0(R11)               Function is installed, return
00004BB8  45F0 84B8                      000024B8  1210+          BAL   R15,NAMSG            Function not available message
00004BBC  07FB                                     1211+          BR    R11                  return
00004BBE                                           1212+          DROP  R5
                                                   1213+*
                                                   1214           TSTPLO 40
                                                   1215+*
                                                   1216+* test if PLO function is available
                                                   1217+*
00004BBE                  00004BBE                 1218+          USING *,R5                 base for test data and test routine
00004BBE                  00000000                 1219+          USING PLOAVAL,R5
00004BC0  00004BC8                                 1220+T17       DC    A(CODE17)               address of test routine
00004BC4  0011                                     1221+          DC    H'17'                test number
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00004BC6  00                                       1222+          DC    X'00'
00004BC7  28                                       1223+          DC    HL1'40'              Function Number
                                                   1224+*
00004BC8                                           1225+CODE17    DS    0F
00004BC8  4110 0128                      00000128  1226+          LA    R1,(256+40)          T bit 55  (+256)
00004BCC  1801                                     1227+          LR    R0,R1
00004BCE  EE00 0000 0000      00000000   00000000  1228+          PLO   0,0,0,0
00004BD4  478B 0000                      00000000  1229+          BZ    0(R11)               Function is installed, return
00004BD8  45F0 84B8                      000024B8  1230+          BAL   R15,NAMSG            Function not available message
00004BDC  07FB                                     1231+          BR    R11                  return
00004BDE                                           1232+          DROP  R5
                                                   1233+*
                                                   1234           TSTPLO 41
                                                   1235+*
                                                   1236+* test if PLO function is available
                                                   1237+*
00004BDE                  00004BDE                 1238+          USING *,R5                 base for test data and test routine
00004BDE                  00000000                 1239+          USING PLOAVAL,R5
00004BE0  00004BE8                                 1240+T18       DC    A(CODE18)               address of test routine
00004BE4  0012                                     1241+          DC    H'18'                test number
00004BE6  00                                       1242+          DC    X'00'
00004BE7  29                                       1243+          DC    HL1'41'              Function Number
                                                   1244+*
00004BE8                                           1245+CODE18    DS    0F
00004BE8  4110 0129                      00000129  1246+          LA    R1,(256+41)          T bit 55  (+256)
00004BEC  1801                                     1247+          LR    R0,R1
00004BEE  EE00 0000 0000      00000000   00000000  1248+          PLO   0,0,0,0
00004BF4  478B 0000                      00000000  1249+          BZ    0(R11)               Function is installed, return
00004BF8  45F0 84B8                      000024B8  1250+          BAL   R15,NAMSG            Function not available message
00004BFC  07FB                                     1251+          BR    R11                  return
00004BFE                                           1252+          DROP  R5
                                                   1253+*
                                                   1254           TSTPLO 42
                                                   1255+*
                                                   1256+* test if PLO function is available
                                                   1257+*
00004BFE                  00004BFE                 1258+          USING *,R5                 base for test data and test routine
00004BFE                  00000000                 1259+          USING PLOAVAL,R5
00004C00  00004C08                                 1260+T19       DC    A(CODE19)               address of test routine
00004C04  0013                                     1261+          DC    H'19'                test number
00004C06  00                                       1262+          DC    X'00'
00004C07  2A                                       1263+          DC    HL1'42'              Function Number
                                                   1264+*
00004C08                                           1265+CODE19    DS    0F
00004C08  4110 012A                      0000012A  1266+          LA    R1,(256+42)          T bit 55  (+256)
00004C0C  1801                                     1267+          LR    R0,R1
00004C0E  EE00 0000 0000      00000000   00000000  1268+          PLO   0,0,0,0
00004C14  478B 0000                      00000000  1269+          BZ    0(R11)               Function is installed, return
00004C18  45F0 84B8                      000024B8  1270+          BAL   R15,NAMSG            Function not available message
00004C1C  07FB                                     1271+          BR    R11                  return
00004C1E                                           1272+          DROP  R5
                                                   1273+*
                                                   1274           TSTPLO 43
                                                   1275+*
                                                   1276+* test if PLO function is available
                                                   1277+*
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00004C1E                     00004C1E              1278+          USING *,R5                base for test data and test routine
00004C1E                     00000000              1279+          USING PLOAVAL,R5
00004C20  00004C28                                 1280+T20       DC    A(CODE20)              address of test routine
00004C24  0014                                     1281+          DC    H'20'                  test number
00004C26  00                                       1282+          DC    X'00'
00004C27  2B                                       1283+          DC    HL1'43'                Function Number
                                                   1284+*
00004C28                                           1285+CODE20    DS    0F
00004C28  4110 012B                     0000012B   1286+          LA    R1,(256+43)            T bit 55  (+256)
00004C2C  1801                                     1287+          LR    R0,R1
00004C2E  EE00 0000 0000      00000000  00000000   1288+          PLO   0,0,0,0
00004C34  478B 0000                     00000000   1289+          BZ    0(R11)                 Function is installed, return
00004C38  45F0 84B8                     000024B8   1290+          BAL   R15,NAMSG              Function not available message
00004C3C  07FB                                     1291+          BR    R11                    return
00004C3E                                           1292+          DROP  R5
                                                   1293+*
                                                   1294           TSTPLO 44
                                                   1295+*
                                                   1296+* test if PLO function is available
                                                   1297+*
00004C3E                     00004C3E              1298+          USING *,R5                base for test data and test routine
00004C3E                     00000000              1299+          USING PLOAVAL,R5
00004C40  00004C48                                 1300+T21       DC    A(CODE21)              address of test routine
00004C44  0015                                     1301+          DC    H'21'                  test number
00004C46  00                                       1302+          DC    X'00'
00004C47  2C                                       1303+          DC    HL1'44'                Function Number
                                                   1304+*
00004C48                                           1305+CODE21    DS    0F
00004C48  4110 012C                     0000012C   1306+          LA    R1,(256+44)            T bit 55  (+256)
00004C4C  1801                                     1307+          LR    R0,R1
00004C4E  EE00 0000 0000      00000000  00000000   1308+          PLO   0,0,0,0
00004C54  478B 0000                     00000000   1309+          BZ    0(R11)                 Function is installed, return
00004C58  45F0 84B8                     000024B8   1310+          BAL   R15,NAMSG              Function not available message
00004C5C  07FB                                     1311+          BR    R11                    return
00004C5E                                           1312+          DROP  R5
                                                   1313+*
                                                   1314           TSTPLO 45
                                                   1315+*
                                                   1316+* test if PLO function is available
                                                   1317+*
00004C5E                     00004C5E              1318+          USING *,R5                base for test data and test routine
00004C5E                     00000000              1319+          USING PLOAVAL,R5
00004C60  00004C68                                 1320+T22       DC    A(CODE22)              address of test routine
00004C64  0016                                     1321+          DC    H'22'                  test number
00004C66  00                                       1322+          DC    X'00'
00004C67  2D                                       1323+          DC    HL1'45'                Function Number
                                                   1324+*
00004C68                                           1325+CODE22    DS    0F
00004C68  4110 012D                     0000012D   1326+          LA    R1,(256+45)            T bit 55  (+256)
00004C6C  1801                                     1327+          LR    R0,R1
00004C6E  EE00 0000 0000      00000000  00000000   1328+          PLO   0,0,0,0
00004C74  478B 0000                     00000000   1329+          BZ    0(R11)                 Function is installed, return
00004C78  45F0 84B8                     000024B8   1330+          BAL   R15,NAMSG              Function not available message
00004C7C  07FB                                     1331+          BR    R11                    return
00004C7E                                           1332+          DROP  R5
                                                   1333+*
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   1334          TSTPLO 46
                                                   1335+*
                                                   1336+* test if PLO function is available
                                                   1337+*
00004C7E                       00004C7E            1338+         USING *,R5                base for test data and test routine
00004C7E                       00000000            1339+         USING PLOAVAL,R5
00004C80  00004C88                                 1340+T23      DC    A(CODE23)              address of test routine
00004C84  0017                                     1341+         DC    H'23'                  test number
00004C86  00                                       1342+         DC    X'00'
00004C87  2E                                       1343+         DC    HL1'46'                Function Number
                                                   1344+*
00004C88                                           1345+CODE23   DS    0F
00004C88  4110 012E                    0000012E    1346+         LA    R1,(256+46)            T bit 55  (+256)
00004C8C  1801                                     1347+         LR    R0,R1
00004C8E  EE00 0000 0000       00000000  00000000  1348+         PLO   0,0,0,0
00004C94  478B 0000                    00000000    1349+         BZ    0(R11)                 Function is installed, return
00004C98  45F0 84B8                    000024B8    1350+         BAL   R15,NAMSG              Function not available message
00004C9C  07FB                                     1351+         BR    R11                    return
00004C9E                                           1352+         DROP  R5
                                                   1353+*
                                                   1354          TSTPLO 47
                                                   1355+*
                                                   1356+* test if PLO function is available
                                                   1357+*
00004C9E                       00004C9E            1358+         USING *,R5                base for test data and test routine
00004C9E                       00000000            1359+         USING PLOAVAL,R5
00004CA0  00004CA8                                 1360+T24      DC    A(CODE24)              address of test routine
00004CA4  0018                                     1361+         DC    H'24'                  test number
00004CA6  00                                       1362+         DC    X'00'
00004CA7  2F                                       1363+         DC    HL1'47'                Function Number
                                                   1364+*
00004CA8                                           1365+CODE24   DS    0F
00004CA8  4110 012F                    0000012F    1366+         LA    R1,(256+47)            T bit 55  (+256)
00004CAC  1801                                     1367+         LR    R0,R1
00004CAE  EE00 0000 0000       00000000  00000000  1368+         PLO   0,0,0,0
00004CB4  478B 0000                    00000000    1369+         BZ    0(R11)                 Function is installed, return
00004CB8  45F0 84B8                    000024B8    1370+         BAL   R15,NAMSG              Function not available message
00004CBC  07FB                                     1371+         BR    R11                    return
00004CBE                                           1372+         DROP  R5
                                                   1373+*
                                                   1374          TSTPLO 48
                                                   1375+*
                                                   1376+* test if PLO function is available
                                                   1377+*
00004CBE                       00004CBE            1378+         USING *,R5                base for test data and test routine
00004CBE                       00000000            1379+         USING PLOAVAL,R5
00004CC0  00004CC8                                 1380+T25      DC    A(CODE25)              address of test routine
00004CC4  0019                                     1381+         DC    H'25'                  test number
00004CC6  00                                       1382+         DC    X'00'
00004CC7  30                                       1383+         DC    HL1'48'                Function Number
                                                   1384+*
00004CC8                                           1385+CODE25   DS    0F
00004CC8  4110 0130                    00000130    1386+         LA    R1,(256+48)            T bit 55  (+256)
00004CCC  1801                                     1387+         LR    R0,R1
00004CCE  EE00 0000 0000       00000000  00000000  1388+         PLO   0,0,0,0
00004CD4  478B 0000                    00000000    1389+         BZ    0(R11)                 Function is installed, return
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00004CD8  45F0 84B8                    000024B8  1390+          BAL   R15,NAMSG           Function not available message
00004CDC  07FB                                   1391+          BR    R11                 return
00004CDE                                         1392+          DROP  R5
                                                 1393+*
                                                 1394           TSTPLO 49
                                                 1395+*
                                                 1396+* test if PLO function is available
                                                 1397+*
00004CDE                  00004CDE               1398+          USING *,R5                base for test data and test routine
00004CDE                  00000000               1399+          USING PLOAVAL,R5
00004CE0  00004CE8                               1400+T26       DC    A(CODE26)              address of test routine
00004CE4  001A                                   1401+          DC    H'26'               test number
00004CE6  00                                     1402+          DC    X'00'
00004CE7  31                                     1403+          DC    HL1'49'             Function Number
                                                 1404+*
00004CE8                                         1405+CODE26    DS    0F
00004CE8  4110 0131                    00000131  1406+          LA    R1,(256+49)         T bit 55  (+256)
00004CEC  1801                                   1407+          LR    R0,R1
00004CEE  EE00 0000 0000  00000000     00000000  1408+          PLO   0,0,0,0
00004CF4  478B 0000                    00000000  1409+          BZ    0(R11)              Function is installed, return
00004CF8  45F0 84B8                    000024B8  1410+          BAL   R15,NAMSG           Function not available message
00004CFC  07FB                                   1411+          BR    R11                 return
00004CFE                                         1412+          DROP  R5
                                                 1413+*
                                                 1414           TSTPLO 50
                                                 1415+*
                                                 1416+* test if PLO function is available
                                                 1417+*
00004CFE                  00004CFE               1418+          USING *,R5                base for test data and test routine
00004CFE                  00000000               1419+          USING PLOAVAL,R5
00004D00  00004D08                               1420+T27       DC    A(CODE27)              address of test routine
00004D04  001B                                   1421+          DC    H'27'               test number
00004D06  00                                     1422+          DC    X'00'
00004D07  32                                     1423+          DC    HL1'50'             Function Number
                                                 1424+*
00004D08                                         1425+CODE27    DS    0F
00004D08  4110 0132                    00000132  1426+          LA    R1,(256+50)         T bit 55  (+256)
00004D0C  1801                                   1427+          LR    R0,R1
00004D0E  EE00 0000 0000  00000000     00000000  1428+          PLO   0,0,0,0
00004D14  478B 0000                    00000000  1429+          BZ    0(R11)              Function is installed, return
00004D18  45F0 84B8                    000024B8  1430+          BAL   R15,NAMSG           Function not available message
00004D1C  07FB                                   1431+          BR    R11                 return
00004D1E                                         1432+          DROP  R5
                                                 1433+*
                                                 1434           TSTPLO 51
                                                 1435+*
                                                 1436+* test if PLO function is available
                                                 1437+*
00004D1E                  00004D1E               1438+          USING *,R5                base for test data and test routine
00004D1E                  00000000               1439+          USING PLOAVAL,R5
00004D20  00004D28                               1440+T28       DC    A(CODE28)              address of test routine
00004D24  001C                                   1441+          DC    H'28'               test number
00004D26  00                                     1442+          DC    X'00'
00004D27  33                                     1443+          DC    HL1'51'             Function Number
                                                 1444+*
00004D28                                         1445+CODE28    DS    0F
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00004D28  4110 0133                    00000133  1446+          LA    R1,(256+51)          T bit 55  (+256)
00004D2C  1801                                   1447+          LR    R0,R1
00004D2E  EE00 0000 0000      00000000  00000000  1448+          PLO   0,0,0,0
00004D34  478B 0000                    00000000  1449+          BZ    0(R11)               Function is installed, return
00004D38  45F0 84B8                    000024B8  1450+          BAL   R15,NAMSG            Function not available message
00004D3C  07FB                                   1451+          BR    R11                  return
00004D3E                                         1452+          DROP  R5
                                                 1453+*
                                                 1454           TSTPLO 52
                                                 1455+*
                                                 1456+* test if PLO function is available
                                                 1457+*
00004D3E                       00004D3E          1458+          USING *,R5                 base for test data and test routine
00004D3E                       00000000          1459+          USING PLOAVAL,R5
00004D40  00004D48                               1460+T29       DC    A(CODE29)               address of test routine
00004D44  001D                                   1461+          DC    H'29'                test number
00004D46  00                                     1462+          DC    X'00'
00004D47  34                                     1463+          DC    HL1'52'              Function Number
                                                 1464+*
00004D48                                         1465+CODE29    DS    0F
00004D48  4110 0134                    00000134  1466+          LA    R1,(256+52)          T bit 55  (+256)
00004D4C  1801                                   1467+          LR    R0,R1
00004D4E  EE00 0000 0000      00000000  00000000  1468+          PLO   0,0,0,0
00004D54  478B 0000                    00000000  1469+          BZ    0(R11)               Function is installed, return
00004D58  45F0 84B8                    000024B8  1470+          BAL   R15,NAMSG            Function not available message
00004D5C  07FB                                   1471+          BR    R11                  return
00004D5E                                         1472+          DROP  R5
                                                 1473+*
                                                 1474           TSTPLO 53
                                                 1475+*
                                                 1476+* test if PLO function is available
                                                 1477+*
00004D5E                       00004D5E          1478+          USING *,R5                 base for test data and test routine
00004D5E                       00000000          1479+          USING PLOAVAL,R5
00004D60  00004D68                               1480+T30       DC    A(CODE30)               address of test routine
00004D64  001E                                   1481+          DC    H'30'                test number
00004D66  00                                     1482+          DC    X'00'
00004D67  35                                     1483+          DC    HL1'53'              Function Number
                                                 1484+*
00004D68                                         1485+CODE30    DS    0F
00004D68  4110 0135                    00000135  1486+          LA    R1,(256+53)          T bit 55  (+256)
00004D6C  1801                                   1487+          LR    R0,R1
00004D6E  EE00 0000 0000      00000000  00000000  1488+          PLO   0,0,0,0
00004D74  478B 0000                    00000000  1489+          BZ    0(R11)               Function is installed, return
00004D78  45F0 84B8                    000024B8  1490+          BAL   R15,NAMSG            Function not available message
00004D7C  07FB                                   1491+          BR    R11                  return
00004D7E                                         1492+          DROP  R5
                                                 1493+*
                                                 1494           TSTPLO 54
                                                 1495+*
                                                 1496+* test if PLO function is available
                                                 1497+*
00004D7E                       00004D7E          1498+          USING *,R5                 base for test data and test routine
00004D7E                       00000000          1499+          USING PLOAVAL,R5
00004D80  00004D88                               1500+T31       DC    A(CODE31)               address of test routine
00004D84  001F                                   1501+          DC    H'31'                test number
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00004D86  00                                       1502+          DC    X'00'
00004D87  36                                       1503+          DC    HL1'54'              Function Number
                                                   1504+*
00004D88                                           1505+CODE31    DS    0F
00004D88  4110 0136                      00000136  1506+          LA    R1,(256+54)          T bit 55  (+256)
00004D8C  1801                                     1507+          LR    R0,R1
00004D8E  EE00 0000 0000      00000000   00000000  1508+          PLO   0,0,0,0
00004D94  478B 0000                      00000000  1509+          BZ    0(R11)               Function is installed, return
00004D98  45F0 84B8                      000024B8  1510+          BAL   R15,NAMSG            Function not available message
00004D9C  07FB                                     1511+          BR    R11                  return
00004D9E                                           1512+          DROP  R5
                                                   1513+*
                                                   1514           TSTPLO 55
                                                   1515+*
                                                   1516+* test if PLO function is available
                                                   1517+*
00004D9E                       00004D9E            1518+          USING *,R5                 base for test data and test routine
00004D9E                       00000000            1519+          USING PLOAVAL,R5
00004DA0  00004DA8                                 1520+T32       DC    A(CODE32)               address of test routine
00004DA4  0020                                     1521+          DC    H'32'                test number
00004DA6  00                                       1522+          DC    X'00'
00004DA7  37                                       1523+          DC    HL1'55'              Function Number
                                                   1524+*
00004DA8                                           1525+CODE32    DS    0F
00004DA8  4110 0137                      00000137  1526+          LA    R1,(256+55)          T bit 55  (+256)
00004DAC  1801                                     1527+          LR    R0,R1
00004DAE  EE00 0000 0000      00000000   00000000  1528+          PLO   0,0,0,0
00004DB4  478B 0000                      00000000  1529+          BZ    0(R11)               Function is installed, return
00004DB8  45F0 84B8                      000024B8  1530+          BAL   R15,NAMSG            Function not available message
00004DBC  07FB                                     1531+          BR    R11                  return
00004DBE                                           1532+          DROP  R5
                                                   1533+*
                                                   1534           TSTPLO 56
                                                   1535+*
                                                   1536+* test if PLO function is available
                                                   1537+*
00004DBE                       00004DBE            1538+          USING *,R5                 base for test data and test routine
00004DBE                       00000000            1539+          USING PLOAVAL,R5
00004DC0  00004DC8                                 1540+T33       DC    A(CODE33)               address of test routine
00004DC4  0021                                     1541+          DC    H'33'                test number
00004DC6  00                                       1542+          DC    X'00'
00004DC7  38                                       1543+          DC    HL1'56'              Function Number
                                                   1544+*
00004DC8                                           1545+CODE33    DS    0F
00004DC8  4110 0138                      00000138  1546+          LA    R1,(256+56)          T bit 55  (+256)
00004DCC  1801                                     1547+          LR    R0,R1
00004DCE  EE00 0000 0000      00000000   00000000  1548+          PLO   0,0,0,0
00004DD4  478B 0000                      00000000  1549+          BZ    0(R11)               Function is installed, return
00004DD8  45F0 84B8                      000024B8  1550+          BAL   R15,NAMSG            Function not available message
00004DDC  07FB                                     1551+          BR    R11                  return
00004DDE                                           1552+          DROP  R5
                                                   1553+*
                                                   1554           TSTPLO 57
                                                   1555+*
                                                   1556+* test if PLO function is available
                                                   1557+*
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00004DDE                     00004DDE              1558+          USING *,R5                base for test data and test routine
00004DDE                     00000000              1559+          USING PLOAVAL,R5
00004DE0  00004DE8                                 1560+T34       DC    A(CODE34)             address of test routine
00004DE4  0022                                     1561+          DC    H'34'             test number
00004DE6  00                                       1562+          DC    X'00'
00004DE7  39                                       1563+          DC    HL1'57'           Function Number
                                                   1564+*
00004DE8                                           1565+CODE34    DS    0F
00004DE8  4110 0139                    00000139    1566+          LA    R1,(256+57)       T bit 55  (+256)
00004DEC  1801                                     1567+          LR    R0,R1
00004DEE  EE00 0000 0000     00000000  00000000    1568+          PLO   0,0,0,0
00004DF4  478B 0000                    00000000    1569+          BZ    0(R11)            Function is installed, return
00004DF8  45F0 84B8                    000024B8    1570+          BAL   R15,NAMSG         Function not available message
00004DFC  07FB                                     1571+          BR    R11               return
00004DFE                                           1572+          DROP  R5
                                                   1573+*
                                                   1574           TSTPLO 58
                                                   1575+*
                                                   1576+* test if PLO function is available
                                                   1577+*
00004DFE                     00004DFE              1578+          USING *,R5                base for test data and test routine
00004DFE                     00000000              1579+          USING PLOAVAL,R5
00004E00  00004E08                                 1580+T35       DC    A(CODE35)             address of test routine
00004E04  0023                                     1581+          DC    H'35'             test number
00004E06  00                                       1582+          DC    X'00'
00004E07  3A                                       1583+          DC    HL1'58'           Function Number
                                                   1584+*
00004E08                                           1585+CODE35    DS    0F
00004E08  4110 013A                    0000013A    1586+          LA    R1,(256+58)       T bit 55  (+256)
00004E0C  1801                                     1587+          LR    R0,R1
00004E0E  EE00 0000 0000     00000000  00000000    1588+          PLO   0,0,0,0
00004E14  478B 0000                    00000000    1589+          BZ    0(R11)            Function is installed, return
00004E18  45F0 84B8                    000024B8    1590+          BAL   R15,NAMSG         Function not available message
00004E1C  07FB                                     1591+          BR    R11               return
00004E1E                                           1592+          DROP  R5
                                                   1593+*
                                                   1594           TSTPLO 59
                                                   1595+*
                                                   1596+* test if PLO function is available
                                                   1597+*
00004E1E                     00004E1E              1598+          USING *,R5                base for test data and test routine
00004E1E                     00000000              1599+          USING PLOAVAL,R5
00004E20  00004E28                                 1600+T36       DC    A(CODE36)             address of test routine
00004E24  0024                                     1601+          DC    H'36'             test number
00004E26  00                                       1602+          DC    X'00'
00004E27  3B                                       1603+          DC    HL1'59'           Function Number
                                                   1604+*
00004E28                                           1605+CODE36    DS    0F
00004E28  4110 013B                    0000013B    1606+          LA    R1,(256+59)       T bit 55  (+256)
00004E2C  1801                                     1607+          LR    R0,R1
00004E2E  EE00 0000 0000     00000000  00000000    1608+          PLO   0,0,0,0
00004E34  478B 0000                    00000000    1609+          BZ    0(R11)            Function is installed, return
00004E38  45F0 84B8                    000024B8    1610+          BAL   R15,NAMSG         Function not available message
00004E3C  07FB                                     1611+          BR    R11               return
00004E3E                                           1612+          DROP  R5
                                                   1613+*
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   1614          TSTPLO 60
                                                   1615+*
                                                   1616+* test if PLO function is available
                                                   1617+*
00004E3E                       00004E3E            1618+         USING *,R5                base for test data and test routine
00004E3E                       00000000            1619+         USING PLOAVAL,R5
00004E40  00004E48                                 1620+T37      DC    A(CODE37)              address of test routine
00004E44  0025                                     1621+         DC    H'37'                  test number
00004E46  00                                       1622+         DC    X'00'
00004E47  3C                                       1623+         DC    HL1'60'                Function Number
                                                   1624+*
00004E48                                           1625+CODE37   DS    0F
00004E48  4110 013C                     0000013C   1626+         LA    R1,(256+60)            T bit 55  (+256)
00004E4C  1801                                     1627+         LR    R0,R1
00004E4E  EE00 0000 0000       00000000 00000000   1628+         PLO   0,0,0,0
00004E54  478B 0000                     00000000   1629+         BZ    0(R11)                 Function is installed, return
00004E58  45F0 84B8                     000024B8   1630+         BAL   R15,NAMSG              Function not available message
00004E5C  07FB                                     1631+         BR    R11                    return
00004E5E                                           1632+         DROP  R5
                                                   1633+*
                                                   1634          TSTPLO 61
                                                   1635+*
                                                   1636+* test if PLO function is available
                                                   1637+*
00004E5E                       00004E5E            1638+         USING *,R5                base for test data and test routine
00004E5E                       00000000            1639+         USING PLOAVAL,R5
00004E60  00004E68                                 1640+T38      DC    A(CODE38)              address of test routine
00004E64  0026                                     1641+         DC    H'38'                  test number
00004E66  00                                       1642+         DC    X'00'
00004E67  3D                                       1643+         DC    HL1'61'                Function Number
                                                   1644+*
00004E68                                           1645+CODE38   DS    0F
00004E68  4110 013D                     0000013D   1646+         LA    R1,(256+61)            T bit 55  (+256)
00004E6C  1801                                     1647+         LR    R0,R1
00004E6E  EE00 0000 0000       00000000 00000000   1648+         PLO   0,0,0,0
00004E74  478B 0000                     00000000   1649+         BZ    0(R11)                 Function is installed, return
00004E78  45F0 84B8                     000024B8   1650+         BAL   R15,NAMSG              Function not available message
00004E7C  07FB                                     1651+         BR    R11                    return
00004E7E                                           1652+         DROP  R5
                                                   1653+*
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
                                                    1655 * ------------------------------------------------------------------------
                                                    1656 * NOTE:  A test is structured as follows:
                                                    1657 *
                                                    1658 *        TSTHDR
                                                    1659 *        TSTCODE
                                                    1660 *
                                                    1661 *        ... code to test the function
                                                    1662 *
                                                    1663 *        TSTDATA                  test data is located here
                                                    1664 *        TSTLBL OPDST             Operand Destination
                                                    1665 *        DC    XL128'0'
                                                    1666 *
                                                    1667 *        TSTLBL OPSRC             Operand Source
                                                    1668 *        DC    XL128'0'
                                                    1669 *
                                                    1670 *        TSTLBL PL                Parameter List
                                                    1671 *        DC    XL256'00'
                                                    1672 *
                                                    1673 *        TSTLBL FMSG              fail message
                                                    1674 *        DC    C'failed PLO-Extension '
                                                    1675 *        DC    CLxx'.......'
                                                    1676 *
                                                    1677 *        TSTEND
                                                    1678 * ------------------------------------------------------------------------
                                                    1679
                                                    1680 *========================================================================
                                                    1681 *       Quadruple Load / Store Tests
                                                    1682 *========================================================================
                                                    1683
                                                    1684 * ------------------------------------------------------------------------
                                                    1685 * Test Quadruple Load(QLO: 48) / Store(QSTO: 61) instructions
                                                    1686 * ------------------------------------------------------------------------
                                                    1687          TSTHDR 99,99,99
00004E7E                          00004E7E  00020C00 1688+         ORG   PLOETST+(128*K)+1*(3*K)
00020C00  E2E3C1D9 E3409686                         1689+         DC    CL32'START of test: 39'
00020C08  40A385A2 A37A40F3
00020C10  F9404040 40404040
00020C18  40404040 40404040
00020C20                          00020C20          1690+         USING *,R5               base for test data and test routine
00020C20                          00000000          1691+         USING PLOETEST,R5
00020C20  00020C4C                                  1692+T39      DC    A(CODE39)          address of test code
00020C24  0027                                      1693+         DC    H'39'              test number
00020C26  00                                        1694+         DC    X'00'
00020C27  63                                        1695+         DC    HL1'99'            Function
00020C28  63                                        1696+         DC    HL1'99'            expected unequal results (UR)
00020C29  00                                        1697+         DC    HL1'00'            found UR
00020C2A  63                                        1698+         DC    HL1'99'            cc
00020C2B  00                                        1699+         DC    HL1'00'            cc not checked
00020C2C  00                                        1700+         DS    X                  extracted cc
00020C30  00000000 00000000                         1701+         DS    2F                 extract PSW after test (has CC)
00020C38  00021480                                  1702+         DC    A(FMSG39)          address of test fail message
00020C3C  000212C0                                  1703+         DC    A(OPSRC39)         address of operand source
00020C40  00021220                                  1704+         DC    A(OPDST39)         address of operand destination
00020C44  00021360                                  1705+         DC    A(PL39)            address of parameter list
00020C48  000041DF                                  1706+         DC    A(RANDX+39)        address of random source data
                                                    1707
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   1708          TSTCODE
00020C4C                                           1709+CODE39   DS    0F
                                                   1710 *                                load random source
00020C4C  E310 501C 0014                0000001C   1711          LGF   R1,OPSRCA
00020C52  E320 5028 0014                00000028   1712          LGF   R2,RANDXA
00020C58  E330 5024 0014                00000024   1713          LGF   R3,PLA
                                                   1714
00020C5E  D27F 1000 2000      00000000  00000000   1715          MVC   0(128,R1),0(R2)
                                                   1716 *
                                                   1717 * Quadruple Load(QLO: 48)
                                                   1718 *
00020C64  E313 0000 0024                00000000   1719          STG   R1,0(R3)            load OP1 address in PL
                                                   1720
00020C6A  4111 0020                     00000020   1721          LA    R1,32(R1)
00020C6E  E313 0008 0024                00000008   1722          STG   R1,8(R3)            load OP2 address in PL
                                                   1723
00020C74  4111 0020                     00000020   1724          LA    R1,32(R1)
00020C78  E313 0010 0024                00000010   1725          STG   R1,16(R3)           load OP3 address in PL
                                                   1726
00020C7E  4111 0020                     00000020   1727          LA    R1,32(R1)
00020C82  E313 0018 0024                00000018   1728          STG   R1,24(R3)           load OP4 address in PL
                                                   1729
00020C88  B982 0000                                1730          XGR   R0,R0
00020C8C  4100 0030                     00000030   1731          LA    R0,48
00020C90  EE10 0000 3000      00000000  00000000   1732          PLO   R1,0,0,0(R3)        Quadruple Load(QLO: 48)
                                                   1733 *
                                                   1734 * Quadruple Store(QSTO: 61)
                                                   1735 *
00020C96  E310 5020 0014                00000020   1736          LGF   R1,OPDSTA           Destination address
00020C9C  E330 5024 0014                00000024   1737          LGF   R3,PLA
                                                   1738
00020CA2  E313 0000 0024                00000000   1739          STG   R1,0(R3)            load OP1 address in PL
                                                   1740
00020CA8  4111 0020                     00000020   1741          LA    R1,32(R1)
00020CAC  E313 0008 0024                00000008   1742          STG   R1,8(R3)            load OP2 address in PL
                                                   1743
00020CB2  4111 0020                     00000020   1744          LA    R1,32(R1)
00020CB6  E313 0010 0024                00000010   1745          STG   R1,16(R3)           load OP3 address in PL
                                                   1746
00020CBC  4111 0020                     00000020   1747          LA    R1,32(R1)
00020CC0  E313 0018 0024                00000018   1748          STG   R1,24(R3)           load OP4 address in PL
                                                   1749
00020CC6  B982 0000                                1750          XGR   R0,R0
00020CCA  4100 003D                     0000003D   1751          LA    R0,61
00020CCE  EE10 0000 3000      00000000  00000000   1752          PLO   R1,0,0,0(R3)        Quadruple Store(QSTO: 61)
                                                   1753 *
                                                   1754 * compare results
                                                   1755 *
00020CD4  E310 501C 0014                0000001C   1756          LGF   R1,OPSRCA           source
00020CDA  E330 5020 0014                00000020   1757          LGF   R3,OPDSTA           Destination
                                                   1758
00020CE0  D51F 1000 3000      00000000  00000000   1759          CLC   0(32,R1),0(R3)
00020CE6  4770 50E8                     00020D08   1760          BNE   F4861               OP1 Different, error
00020CEA  D51F 1020 3020      00000020  00000020   1761          CLC   32(32,R1),32(R3)
00020CF0  4770 50E8                     00020D08   1762          BNE   F4861               OP2 Different, error
00020CF4  D51F 1040 3040      00000040  00000040   1763          CLC   64(32,R1),64(R3)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00020CFA  4770 50E8                      00020D08  1764           BNE   F4861               OP3 Different, error
00020CFE  D51F 1060 3060      00000060   00000060  1765           CLC   96(32,R1),96(R3)
00020D04  478B 0000                      00000000  1766           BE    0(R11)              OP4 same, return
                                                   1767
                              00020D08   00000001  1768 F4861     EQU   *
00020D08  45F0 83E8                      000023E8  1769           BAL   R15,FAILMSG
00020D0C  07FB                                     1770           BR    R11
                                                   1771
                                                   1772           TSTDATA
00020D0E                      00020D0E   00021200  1773+          ORG   PLOETST+(128*K)+1*(3*K)+(K+512)
                                                   1774           TSTLBL OPDST              Operand Destination (Results)
00021200  6E6E6ED6 D7C4E2E3                       1775+          DC    CL32'>>>OPDST39<<<' eye catcher
00021208  F3F94C4C 4C404040
00021210  40404040 40404040
00021218  40404040 40404040
00021220                                          1776+OPDST39   DS    0C
00021220  00000000 00000000                       1777           DC    XL128'0'
00021228  00000000 00000000
00021230  00000000 00000000
00021238  00000000 00000000
00021240  00000000 00000000
00021248  00000000 00000000
00021250  00000000 00000000
00021258  00000000 00000000
00021260  00000000 00000000
00021268  00000000 00000000
00021270  00000000 00000000
00021278  00000000 00000000
00021280  00000000 00000000
00021288  00000000 00000000
00021290  00000000 00000000
00021298  00000000 00000000
                                                   1778
                                                   1779           TSTLBL OPSRC              Operand Source
000212A0  6E6E6ED6 D7E2D9C3                       1780+          DC    CL32'>>>OPSRC39<<<' eye catcher
000212A8  F3F94C4C 4C404040
000212B0  40404040 40404040
000212B8  40404040 40404040
000212C0                                          1781+OPSRC39   DS    0C
000212C0  00000000 00000000                       1782           DC    XL128'0'
000212C8  00000000 00000000
000212D0  00000000 00000000
000212D8  00000000 00000000
000212E0  00000000 00000000
000212E8  00000000 00000000
000212F0  00000000 00000000
000212F8  00000000 00000000
00021300  00000000 00000000
00021308  00000000 00000000
00021310  00000000 00000000
00021318  00000000 00000000
00021320  00000000 00000000
00021328  00000000 00000000
00021330  00000000 00000000
00021338  00000000 00000000
                                                   1783
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   1784          TSTLBL PL                  Parameter List
00021340  6E6E6ED7 D3F3F94C                        1785+         DC    CL32'>>>PL39<<<'     eye catcher
00021348  4C4C4040 40404040
00021350  40404040 40404040
00021358  40404040 40404040
00021360                                           1786+PL39     DS    0C
00021360  00000000 00000000                        1787          DC    XL256'00'
00021368  00000000 00000000
00021370  00000000 00000000
00021378  00000000 00000000
00021380  00000000 00000000
00021388  00000000 00000000
00021390  00000000 00000000
00021398  00000000 00000000
000213A0  00000000 00000000
000213A8  00000000 00000000
000213B0  00000000 00000000
000213B8  00000000 00000000
000213C0  00000000 00000000
000213C8  00000000 00000000
000213D0  00000000 00000000
000213D8  00000000 00000000
000213E0  00000000 00000000
000213E8  00000000 00000000
000213F0  00000000 00000000
000213F8  00000000 00000000
00021400  00000000 00000000
00021408  00000000 00000000
00021410  00000000 00000000
00021418  00000000 00000000
00021420  00000000 00000000
00021428  00000000 00000000
00021430  00000000 00000000
00021438  00000000 00000000
00021440  00000000 00000000
00021448  00000000 00000000
00021450  00000000 00000000
00021458  00000000 00000000
                                                   1788
                                                   1789          TSTLBL FMSG                fail message
00021460  6E6E6EC6 D4E2C7F3                        1790+         DC    CL32'>>>FMSG39<<<'   eye catcher
00021468  F94C4C4C 40404040
00021470  40404040 40404040
00021478  40404040 40404040
00021480                                           1791+FMSG39   DS    0C
00021480  86818993 858440D7                        1792          DC    C'failed PLO-Extension '
00021488  D3D660C5 A7A38595
00021490  A2899695 40
00021495  D8A48184 99A49793                        1793          DC    CL96'Quadruple Load(QLO: 48) / Store(QSTO: 61)        '
0002149D  8540D396 81844DD8
000214A5  D3D67A40 F4F85D40
000214AD  6140E2A3 9699854D
000214B5  D8E2E3D6 7A40F6F1
000214BD  5D404040 40404040
000214C5  40404040 40404040
000214CD  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000214D5  40404040 40404040                                1794
000214DD  40404040 40404040                                1795          TSTEND
000214E5  40404040 40404040                                1796+         DC    CL32'END of test: 39 <<<'
000214ED  40404040 40404040                                1794
                                                           1795          TSTEND
000214F5  C5D5C440 968640A3                                1796+         DC    CL32'END of test: 39 <<<'
000214FD  85A2A37A 40F3F940
00021505  4C4C4C40 40404040
0002150D  40404040 40404040                                1797
                                                           1798 * ------------------------------------------------------------------
                                                           1799 * Test Quadruple Load(QLX: 47) / Store(QSTX: 60) instructions
                                                           1800 * ------------------------------------------------------------------
                                                           1801          TSTHDR 99,99,99
00021515                    00021515  00021800             1802+         ORG   PLOETST+(128*K)+2*(3*K)
00021800  E2E3C1D9 E3409686                                1803+         DC    CL32'START of test: 40'
00021808  40A385A2 A37A40F4
00021810  F0404040 40404040
00021818  40404040 40404040
00021820                    00021820                       1804+         USING *,R5                base for test data and test routine
00021820                    00000000                       1805+         USING PLOETEST,R5
00021820  0002184C                                         1806+T40      DC    A(CODE40)           address of test code
00021824  0028                                             1807+         DC    H'40'               test number
00021826  00                                               1808+         DC    X'00'
00021827  63                                               1809+         DC    HL1'99'             Function
00021828  63                                               1810+         DC    HL1'99'             expected unequal results (UR)
00021829  00                                               1811+         DC    HL1'00'             found UR
0002182A  63                                               1812+         DC    HL1'99'             cc
0002182B  00                                               1813+         DC    HL1'00'             cc not checked
0002182C  00                                               1814+         DS    X                   extracted cc
00021830  00000000 00000000                                1815+         DS    2F                  extract PSW after test (has CC)
00021838  00022080                                         1816+         DC    A(FMSG40)           address of test fail message
0002183C  00021EC0                                         1817+         DC    A(OPSRC40)          address of operand source
00021840  00021E20                                         1818+         DC    A(OPDST40)          address of operand destination
00021844  00021F60                                         1819+         DC    A(PL40)             address of parameter list
00021848  000041E0                                         1820+         DC    A(RANDX+40)         address of random source data
                                                           1821
                                                           1822          TSTCODE
0002184C                                                   1823+CODE40   DS    0F
                                                           1824 *                                    load random source
0002184C  E310 501C 0014                0000001C           1825          LGF   R1,OPSRCA
00021852  E320 5028 0014                00000028           1826          LGF   R2,RANDXA
00021858  E330 5024 0014                00000024           1827          LGF   R3,PLA
                                                           1828
0002185E  D27F 1000 2000      00000000  00000000           1829          MVC   0(128,R1),0(R2)
                                                           1830 *
                                                           1831 * Quadruple Load(QLX: 47)
                                                           1832 *
00021864  E313 0000 0024                00000000           1833          STG   R1,0(R3)            load OP1 address in PL
                                                           1834
0002186A  4111 0020                     00000020           1835          LA    R1,32(R1)
0002186E  E313 0008 0024                00000008           1836          STG   R1,8(R3)            load OP2 address in PL
                                                           1837
00021874  4111 0020                     00000020           1838          LA    R1,32(R1)
00021878  E313 0010 0024                00000010           1839          STG   R1,16(R3)           load OP3 address in PL
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   1840
0002187E  4111 0020                     00000020  1841          LA    R1,32(R1)
00021882  E313 0018 0024                00000018  1842          STG   R1,24(R3)          load OP4 address in PL
                                                   1843
00021888  B982 0000                               1844          XGR   R0,R0
0002188C  4100 002F                     0000002F  1845          LA    R0,47
00021890  EE10 0000 3000      00000000  00000000  1846          PLO   R1,0,0,0(R3)       Quadruple Load(QLX: 47)
                                                   1847 *
                                                   1848 * Quadruple Store(QSTX: 60)
                                                   1849 *
00021896  E310 5020 0014                00000020  1850          LGF   R1,OPDSTA          Destination address
0002189C  E330 5024 0014                00000024  1851          LGF   R3,PLA
                                                   1852
000218A2  E313 0000 0024                00000000  1853          STG   R1,0(R3)           load OP1 address in PL
                                                   1854
000218A8  4111 0020                     00000020  1855          LA    R1,32(R1)
000218AC  E313 0008 0024                00000008  1856          STG   R1,8(R3)           load OP2 address in PL
                                                   1857
000218B2  4111 0020                     00000020  1858          LA    R1,32(R1)
000218B6  E313 0010 0024                00000010  1859          STG   R1,16(R3)          load OP3 address in PL
                                                   1860
000218BC  4111 0020                     00000020  1861          LA    R1,32(R1)
000218C0  E313 0018 0024                00000018  1862          STG   R1,24(R3)          load OP4 address in PL
                                                   1863
000218C6  B982 0000                               1864          XGR   R0,R0
000218CA  4100 003C                     0000003C  1865          LA    R0,60
000218CE  EE10 0000 3000      00000000  00000000  1866          PLO   R1,0,0,0(R3)       Quadruple Store(QSTX: 60)
                                                   1867 *
                                                   1868 * compare results
                                                   1869 *
000218D4  E310 501C 0014                0000001C  1870          LGF   R1,OPSRCA          source
000218DA  E330 5020 0014                00000020  1871          LGF   R3,OPDSTA          Destination
                                                   1872
000218E0  D50F 1000 3000      00000000  00000000  1873          CLC   0(16,R1),0(R3)
000218E6  4770 50E8                     00021908  1874          BNE   F4760              OP1 Different, error
000218EA  D50F 1020 3020      00000020  00000020  1875          CLC   32(16,R1),32(R3)
000218F0  4770 50E8                     00021908  1876          BNE   F4760              OP2 Different, error
000218F4  D50F 1040 3040      00000040  00000040  1877          CLC   64(16,R1),64(R3)
000218FA  4770 50E8                     00021908  1878          BNE   F4760              OP3 Different, error
000218FE  D50F 1060 3060      00000060  00000060  1879          CLC   96(16,R1),96(R3)
00021904  478B 0000                     00000000  1880          BE    0(R11)             OP4 same, return
                                                   1881
                              00021908  00000001  1882 F4760    EQU   *
00021908  45F0 83E8                     000023E8  1883          BAL   R15,FAILMSG
0002190C  07FB                                    1884          BR    R11
                                                   1885
                                                   1886          TSTDATA
0002190E                      0002190E  00021E00  1887+         ORG   PLOETST+(128*K)+2*(3*K)+(K+512)
                                                   1888          TSTLBL OPDST             Operand Destination (Results)
00021E00  6E6E6ED6 D7C4E2E3                       1889+         DC    CL32'>>>OPDST40<<<' eye catcher
00021E08  F4F04C4C 4C404040
00021E10  40404040 40404040
00021E18  40404040 40404040
00021E20                                          1890+OPDST40  DS    0C
00021E20  00000000 00000000                       1891          DC    XL128'0'
00021E28  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00021E30  00000000 00000000
00021E38  00000000 00000000
00021E40  00000000 00000000
00021E48  00000000 00000000
00021E50  00000000 00000000
00021E58  00000000 00000000
00021E60  00000000 00000000
00021E68  00000000 00000000
00021E70  00000000 00000000
00021E78  00000000 00000000
00021E80  00000000 00000000
00021E88  00000000 00000000
00021E90  00000000 00000000
00021E98  00000000 00000000
                                          1892
                                          1893          TSTLBL OPSRC               Operand Source
00021EA0  6E6E6ED6 D7E2D9C3               1894+         DC    CL32'>>>OPSRC40<<<'  eye catcher
00021EA8  F4F04C4C 4C404040
00021EB0  40404040 40404040
00021EB8  40404040 40404040
00021EC0                                  1895+OPSRC40  DS    0C
00021EC0  00000000 00000000               1896          DC    XL128'0'
00021EC8  00000000 00000000
00021ED0  00000000 00000000
00021ED8  00000000 00000000
00021EE0  00000000 00000000
00021EE8  00000000 00000000
00021EF0  00000000 00000000
00021EF8  00000000 00000000
00021F00  00000000 00000000
00021F08  00000000 00000000
00021F10  00000000 00000000
00021F18  00000000 00000000
00021F20  00000000 00000000
00021F28  00000000 00000000
00021F30  00000000 00000000
00021F38  00000000 00000000
                                          1897
                                          1898          TSTLBL PL                  Parameter List
00021F40  6E6E6ED7 D3F4F04C               1899+         DC    CL32'>>>PL40<<<'     eye catcher
00021F48  4C4C4040 40404040
00021F50  40404040 40404040
00021F58  40404040 40404040
00021F60                                  1900+PL40     DS    0C
00021F60  00000000 00000000               1901          DC    XL256'00'
00021F68  00000000 00000000
00021F70  00000000 00000000
00021F78  00000000 00000000
00021F80  00000000 00000000
00021F88  00000000 00000000
00021F90  00000000 00000000
00021F98  00000000 00000000
00021FA0  00000000 00000000
00021FA8  00000000 00000000
00021FB0  00000000 00000000
00021FB8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00021FC0  00000000 00000000
00021FC8  00000000 00000000
00021FD0  00000000 00000000
00021FD8  00000000 00000000
00021FE0  00000000 00000000
00021FE8  00000000 00000000
00021FF0  00000000 00000000
00021FF8  00000000 00000000
00022000  00000000 00000000
00022008  00000000 00000000
00022010  00000000 00000000
00022018  00000000 00000000
00022020  00000000 00000000
00022028  00000000 00000000
00022030  00000000 00000000
00022038  00000000 00000000
00022040  00000000 00000000
00022048  00000000 00000000
00022050  00000000 00000000
00022058  00000000 00000000
                                      1902
                                      1903          TSTLBL FMSG                fail message
00022060  6E6E6EC6 D4E2C7F4           1904+         DC    CL32'>>>FMSG40<<<'   eye catcher
00022068  F04C4C4C 40404040
00022070  40404040 40404040
00022078  40404040 40404040
00022080                              1905+FMSG40   DS    0C
00022080  86818993 858440D7           1906          DC    C'failed PLO-Extension '
00022088  D3D660C5 A7A38595
00022090  A2899695 40
00022095  D8A48184 99A49793           1907          DC    CL96'Quadruple Load(QLX: 47) / Store(QSTX: 60)'
0002209D  8540D396 81844DD8
000220A5  D3E77A40 F4F75D40
000220AD  6140E2A3 9699854D
000220B5  D8E2E3E7 7A40F6F0
000220BD  5D404040 40404040
000220C5  40404040 40404040
000220CD  40404040 40404040
000220D5  40404040 40404040
000220DD  40404040 40404040
000220E5  40404040 40404040
000220ED  40404040 40404040
                                      1908
                                      1909          TSTEND
000220F5  C5D5C440 968640A3           1910+         DC    CL32'END of test: 40 <<<'
000220FD  85A2A37A 40F4F040
00022105  4C4C4C40 40404040
0002210D  40404040 40404040
                                      1911
                                      1912 * ------------------------------------------------------------------
                                      1913 * Test Quadruple Load(QLG: 46) / Store(QSTG: 59) instructions
                                      1914 * ------------------------------------------------------------------
                                      1915          TSTHDR 99,99,99
00022115                  00022115  00022400  1916+         ORG   PLOETST+(128*K)+3*(3*K)
00022400  E2E3C1D9 E3409686           1917+         DC    CL32'START of test: 41'
00022408  40A385A2 A37A40F4
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00022410  F1404040 40404040
00022418  40404040 40404040
00022420                    00022420            1918+          USING *,R5              base for test data and test routine
00022420                    00000000            1919+          USING PLOETEST,R5
00022420  0002244C                              1920+T41       DC    A(CODE41)         address of test code
00022424  0029                                  1921+          DC    H'41'             test number
00022426  00                                    1922+          DC    X'00'
00022427  63                                    1923+          DC    HL1'99'           Function
00022428  63                                    1924+          DC    HL1'99'           expected unequal results (UR)
00022429  00                                    1925+          DC    HL1'00'           found UR
0002242A  63                                    1926+          DC    HL1'99'           cc
0002242B  00                                    1927+          DC    HL1'00'           cc not checked
0002242C  00                                    1928+          DS    X                 extracted cc
00022430  00000000 00000000                    1929+          DS    2F                extract PSW after test (has CC)
00022438  00022C80                              1930+          DC    A(FMSG41)         address of test fail message
0002243C  00022AC0                              1931+          DC    A(OPSRC41)        address of operand source
00022440  00022A20                              1932+          DC    A(OPDST41)        address of operand destination
00022444  00022B60                              1933+          DC    A(PL41)           address of parameter list
00022448  000041E1                              1934+          DC    A(RANDX+41)       address of random source data
                                                1935
                                                1936           TSTCODE
0002244C                                        1937+CODE41    DS    0F
                                                1938 *                                 load random source
0002244C  E310 501C 0014            0000001C    1939           LGF   R1,OPSRCA
00022452  E320 5028 0014            00000028    1940           LGF   R2,RANDXA
00022458  E330 5024 0014            00000024    1941           LGF   R3,PLA
                                                1942
0002245E  D27F 1000 2000  00000000  00000000    1943           MVC   0(128,R1),0(R2)
                                                1944 *
                                                1945 * Quadruple Load(QLX: 46)
                                                1946 *
00022464  E313 0000 0024            00000000    1947           STG   R1,0(R3)          load OP1 address in PL
                                                1948
0002246A  4111 0020                 00000020    1949           LA    R1,32(R1)
0002246E  E313 0008 0024            00000008    1950           STG   R1,8(R3)          load OP2 address in PL
                                                1951
00022474  4111 0020                 00000020    1952           LA    R1,32(R1)
00022478  E313 0010 0024            00000010    1953           STG   R1,16(R3)         load OP3 address in PL
                                                1954
0002247E  4111 0020                 00000020    1955           LA    R1,32(R1)
00022482  E313 0018 0024            00000018    1956           STG   R1,24(R3)         load OP4 address in PL
                                                1957
00022488  B982 0000                             1958           XGR   R0,R0
0002248C  4100 002E                 0000002E    1959           LA    R0,46
00022490  EE10 0000 3000  00000000  00000000    1960           PLO   R1,0,0,0(R3)      Quadruple Load(QLG: 46)
                                                1961 *
                                                1962 * Quadruple Store(QSTX: 59)
                                                1963 *
00022496  E310 5020 0014            00000020    1964           LGF   R1,OPDSTA         Destination address
0002249C  E330 5024 0014            00000024    1965           LGF   R3,PLA
                                                1966
000224A2  E313 0000 0024            00000000    1967           STG   R1,0(R3)          load OP1 address in PL
                                                1968
000224A8  4111 0020                 00000020    1969           LA    R1,32(R1)
000224AC  E313 0008 0024            00000008    1970           STG   R1,8(R3)          load OP2 address in PL
                                                1971
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000224B2  4111 0020                    00000020  1972          LA    R1,32(R1)
000224B6  E313 0010 0024               00000010  1973          STG   R1,16(R3)           load OP3 address in PL
                                                 1974
000224BC  4111 0020                    00000020  1975          LA    R1,32(R1)
000224C0  E313 0018 0024               00000018  1976          STG   R1,24(R3)           load OP4 address in PL
                                                 1977
000224C6  B982 0000                              1978          XGR   R0,R0
000224CA  4100 003B                    0000003B  1979          LA    R0,59
000224CE  EE10 0000 3000  00000000     00000000  1980          PLO   R1,0,0,0(R3)        Quadruple Store(QSTG: 59)
                                                 1981 *
                                                 1982 * compare results
                                                 1983 *
000224D4  E310 501C 0014               0000001C  1984          LGF   R1,OPSRCA           source
000224DA  E330 5020 0014               00000020  1985          LGF   R3,OPDSTA           Destination
                                                 1986
000224E0  D507 1000 3000  00000000     00000000  1987          CLC   0(8,R1),0(R3)
000224E6  4770 50E8                    00022508  1988          BNE   F4659               OP1 Different, error
000224EA  D507 1020 3020  00000020     00000020  1989          CLC   32(8,R1),32(R3)
000224F0  4770 50E8                    00022508  1990          BNE   F4659               OP2 Different, error
000224F4  D507 1040 3040  00000040     00000040  1991          CLC   64(8,R1),64(R3)
000224FA  4770 50E8                    00022508  1992          BNE   F4659               OP3 Different, error
000224FE  D507 1060 3060  00000060     00000060  1993          CLC   96(8,R1),96(R3)
00022504  478B 0000                    00000000  1994          BE    0(R11)              OP4 same, return
                                                 1995
                          00022508     00000001  1996 F4659    EQU   *
00022508  45F0 83E8                    000023E8  1997          BAL   R15,FAILMSG
0002250C  07FB                                   1998          BR    R11
                                                 1999
                                                 2000          TSTDATA
0002250E                  0002250E     00022A00  2001+         ORG   PLOETST+(128*K)+3*(3*K)+(K+512)
                                                 2002          TSTLBL OPDST              Operand Destination (Results)
00022A00  6E6E6ED6 D7C4E2E3                      2003+         DC    CL32'>>>OPDST41<<<' eye catcher
00022A08  F4F14C4C 4C404040
00022A10  40404040 40404040
00022A18  40404040 40404040
00022A20                                         2004+OPDST41  DS    0C
00022A20  00000000 00000000                      2005          DC    XL128'0'
00022A28  00000000 00000000
00022A30  00000000 00000000
00022A38  00000000 00000000
00022A40  00000000 00000000
00022A48  00000000 00000000
00022A50  00000000 00000000
00022A58  00000000 00000000
00022A60  00000000 00000000
00022A68  00000000 00000000
00022A70  00000000 00000000
00022A78  00000000 00000000
00022A80  00000000 00000000
00022A88  00000000 00000000
00022A90  00000000 00000000
00022A98  00000000 00000000
                                                 2006
                                                 2007          TSTLBL OPSRC              Operand Source
00022AA0  6E6E6ED6 D7E2D9C3                      2008+         DC    CL32'>>>OPSRC41<<<' eye catcher
00022AA8  F4F14C4C 4C404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00022AB0  40404040 40404040
00022AB8  40404040 40404040
00022AC0                                         2009+OPSRC41  DS    0C
00022AC0  00000000 00000000                      2010          DC    XL128'0'
00022AC8  00000000 00000000
00022AD0  00000000 00000000
00022AD8  00000000 00000000
00022AE0  00000000 00000000
00022AE8  00000000 00000000
00022AF0  00000000 00000000
00022AF8  00000000 00000000
00022B00  00000000 00000000
00022B08  00000000 00000000
00022B10  00000000 00000000
00022B18  00000000 00000000
00022B20  00000000 00000000
00022B28  00000000 00000000
00022B30  00000000 00000000
00022B38  00000000 00000000
                                                 2011
                                                 2012          TSTLBL PL                   Parameter List
00022B40  6E6E6ED7 D3F4F14C                      2013+         DC    CL32'>>>PL41<<<'      eye catcher
00022B48  4C4C4040 40404040
00022B50  40404040 40404040
00022B58  40404040 40404040
00022B60                                         2014+PL41     DS    0C
00022B60  00000000 00000000                      2015          DC    XL256'00'
00022B68  00000000 00000000
00022B70  00000000 00000000
00022B78  00000000 00000000
00022B80  00000000 00000000
00022B88  00000000 00000000
00022B90  00000000 00000000
00022B98  00000000 00000000
00022BA0  00000000 00000000
00022BA8  00000000 00000000
00022BB0  00000000 00000000
00022BB8  00000000 00000000
00022BC0  00000000 00000000
00022BC8  00000000 00000000
00022BD0  00000000 00000000
00022BD8  00000000 00000000
00022BE0  00000000 00000000
00022BE8  00000000 00000000
00022BF0  00000000 00000000
00022BF8  00000000 00000000
00022C00  00000000 00000000
00022C08  00000000 00000000
00022C10  00000000 00000000
00022C18  00000000 00000000
00022C20  00000000 00000000
00022C28  00000000 00000000
00022C30  00000000 00000000
00022C38  00000000 00000000
00022C40  00000000 00000000
00022C48  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00022C50  00000000 00000000
00022C58  00000000 00000000
                                          2016
                                          2017          TSTLBL FMSG                fail message
00022C60  6E6E6EC6 D4E2C7F4               2018+         DC     CL32'>>>FMSG41<<<'  eye catcher
00022C68  F14C4C4C 40404040
00022C70  40404040 40404040
00022C78  40404040 40404040
00022C80                                  2019+FMSG41   DS     0C
00022C80  86818993 858440D7               2020          DC     C'failed PLO-Extension '
00022C88  D3D660C5 A7A38595
00022C90  A2899695 40
00022C95  D8A48184 99A49793               2021          DC     CL96'Quadruple Load(QLG: 46) / Store(QSTG: 59)'
00022C9D  8540D396 81844DD8
00022CA5  D3C77A40 F4F65D40
00022CAD  6140E2A3 9699854D
00022CB5  D8E2E3C7 7A40F5F9
00022CBD  5D404040 40404040
00022CC5  40404040 40404040
00022CCD  40404040 40404040
00022CD5  40404040 40404040
00022CDD  40404040 40404040
00022CE5  40404040 40404040
00022CED  40404040 40404040
                                          2022
                                          2023          TSTEND
00022CF5  C5D5C440 968640A3               2024+         DC     CL32'END of test: 41 <<<'
00022CFD  85A2A37A 40F4F140
00022D05  4C4C4C40 40404040
00022D0D  40404040 40404040
                                          2025
                                          2026 * ------------------------------------------------------------------
                                          2027 * Test Quadruple Load(QL: 45) / Store(QST: 58) instructions
                                          2028 * ------------------------------------------------------------------
                                          2029          TSTHDR 99,99,99
00022D15                        00022D15  00023000      2030+         ORG    PLOETST+(128*K)+4*(3*K)
00023000  E2E3C1D9 E3409686               2031+         DC     CL32'START of test: 42'
00023008  40A385A2 A37A40F4
00023010  F2404040 40404040
00023018  40404040 40404040
00023020                        00023020               2032+         USING  *,R5                base for test data and test routine
00023020                        00000000               2033+         USING  PLOETEST,R5
00023020  0002304C                                     2034+T42      DC     A(CODE42)           address of test code
00023024  002A                                         2035+         DC     H'42'               test number
00023026  00                                           2036+         DC     X'00'
00023027  63                                           2037+         DC     HL1'99'             Function
00023028  63                                           2038+         DC     HL1'99'             expected unequal results (UR)
00023029  00                                           2039+         DC     HL1'00'             found UR
0002302A  63                                           2040+         DC     HL1'99'             cc
0002302B  00                                           2041+         DC     HL1'00'             cc not checked
0002302C  00                                           2042+         DS     X                   extracted cc
00023030  00000000 00000000                            2043+         DS     2F                  extract PSW after test (has CC)
00023038  00023880                                     2044+         DC     A(FMSG42)           address of test fail message
0002303C  000236C0                                     2045+         DC     A(OPSRC42)          address of operand source
00023040  00023620                                     2046+         DC     A(OPDST42)          address of operand destination
00023044  00023760                                     2047+         DC     A(PL42)             address of parameter list
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00023048  000041E2                        2048+          DC    A(RANDX+42)          address of random source data
                                          2049
                                          2050           TSTCODE
0002304C                                  2051+CODE42    DS    0F
                                          2052 *                                    load random source
0002304C  E310 501C 0014                  0000001C  2053           LGF   R1,OPSRCA
00023052  E320 5028 0014                  00000028  2054           LGF   R2,RANDXA
00023058  E330 5024 0014                  00000024  2055           LGF   R3,PLA
                                          2056
0002305E  D27F 1000 2000      00000000    00000000  2057           MVC   0(128,R1),0(R2)
                                          2058 *
                                          2059 * Quadruple Load(QLX: 45`
                                          2060 *
00023064  E313 0000 0024                  00000000  2061           STG   R1,0(R3)       load OP1 address in PL
                                          2062
0002306A  4111 0020                       00000020  2063           LA    R1,32(R1)
0002306E  E313 0008 0024                  00000008  2064           STG   R1,8(R3)       load OP2 address in PL
                                          2065
00023074  4111 0020                       00000020  2066           LA    R1,32(R1)
00023078  E313 0010 0024                  00000010  2067           STG   R1,16(R3)      load OP3 address in PL
                                          2068
0002307E  4111 0020                       00000020  2069           LA    R1,32(R1)
00023082  E313 0018 0024                  00000018  2070           STG   R1,24(R3)      load OP4 address in PL
                                          2071
00023088  B982 0000                       00000000  2072           XGR   R0,R0
0002308C  4100 002D                       0000002D  2073           LA    R0,45
00023090  EE10 0000 3000      00000000    00000000  2074           PLO   R1,0,0,0(R3)   Quadruple Load(QL: 45)
                                          2075 *
                                          2076 * Quadruple Store(QST: 58)
                                          2077 *
00023096  E310 5020 0014                  00000020  2078           LGF   R1,OPDSTA      Destination address
0002309C  E330 5024 0014                  00000024  2079           LGF   R3,PLA
                                          2080
000230A2  E313 0000 0024                  00000000  2081           STG   R1,0(R3)       load OP1 address in PL
                                          2082
000230A8  4111 0020                       00000020  2083           LA    R1,32(R1)
000230AC  E313 0008 0024                  00000008  2084           STG   R1,8(R3)       load OP2 address in PL
                                          2085
000230B2  4111 0020                       00000020  2086           LA    R1,32(R1)
000230B6  E313 0010 0024                  00000010  2087           STG   R1,16(R3)      load OP3 address in PL
                                          2088
000230BC  4111 0020                       00000020  2089           LA    R1,32(R1)
000230C0  E313 0018 0024                  00000018  2090           STG   R1,24(R3)      load OP4 address in PL
                                          2091
000230C6  B982 0000                       00000000  2092           XGR   R0,R0
000230CA  4100 003A                       0000003A  2093           LA    R0,58
000230CE  EE10 0000 3000      00000000    00000000  2094           PLO   R1,0,0,0(R3)   Quadruple Store(QST: 58)
                                          2095 *
                                          2096 * compare results
                                          2097 *
000230D4  E310 501C 0014                  0000001C  2098           LGF   R1,OPSRCA      source
000230DA  E330 5020 0014                  00000020  2099           LGF   R3,OPDSTA      Destination
                                          2100
000230E0  D503 1000 3000      00000000    00000000  2101           CLC   0(4,R1),0(R3)
000230E6  4770 50E8                       00023108  2102           BNE   F4558          OP1 Different, error
000230EA  D503 1020 3020      00000020    00000020  2103           CLC   32(4,R1),32(R3)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000230F0  4770 50E8                      00023108  2104           BNE   F4558                OP2 Different, error
000230F4  D503 1040 3040      00000040  00000040  2105           CLC   64(4,R1),64(R3)
000230FA  4770 50E8                      00023108  2106           BNE   F4558                OP3 Different, error
000230FE  D503 1060 3060      00000060  00000060  2107           CLC   96(4,R1),96(R3)
00023104  478B 0000                      00000000  2108           BE    0(R11)               OP4 same, return
                                                   2109
                              00023108  00000001  2110 F4558     EQU   *
00023108  45F0 83E8                      000023E8  2111           BAL   R15,FAILMSG
0002310C  07FB                                     2112           BR    R11
                                                   2113
                                                   2114           TSTDATA
0002310E                      0002310E  00023600  2115+          ORG   PLOETST+(128*K)+4*(3*K)+(K+512)
                                                   2116           TSTLBL OPDST               Operand Destination (Results)
00023600  6E6E6ED6 D7C4E2E3                       2117+          DC    CL32'>>>OPDST42<<<' eye catcher
00023608  F4F24C4C 4C404040
00023610  40404040 40404040
00023618  40404040 40404040
00023620                                          2118+OPDST42   DS    0C
00023620  00000000 00000000                       2119           DC    XL128'0'
00023628  00000000 00000000
00023630  00000000 00000000
00023638  00000000 00000000
00023640  00000000 00000000
00023648  00000000 00000000
00023650  00000000 00000000
00023658  00000000 00000000
00023660  00000000 00000000
00023668  00000000 00000000
00023670  00000000 00000000
00023678  00000000 00000000
00023680  00000000 00000000
00023688  00000000 00000000
00023690  00000000 00000000
00023698  00000000 00000000
                                                   2120
                                                   2121           TSTLBL OPSRC               Operand Source
000236A0  6E6E6ED6 D7E2D9C3                       2122+          DC    CL32'>>>OPSRC42<<<' eye catcher
000236A8  F4F24C4C 4C404040
000236B0  40404040 40404040
000236B8  40404040 40404040
000236C0                                          2123+OPSRC42   DS    0C
000236C0  00000000 00000000                       2124           DC    XL128'0'
000236C8  00000000 00000000
000236D0  00000000 00000000
000236D8  00000000 00000000
000236E0  00000000 00000000
000236E8  00000000 00000000
000236F0  00000000 00000000
000236F8  00000000 00000000
00023700  00000000 00000000
00023708  00000000 00000000
00023710  00000000 00000000
00023718  00000000 00000000
00023720  00000000 00000000
00023728  00000000 00000000
00023730  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00023738  00000000 00000000
                                          2125
                                          2126          TSTLBL PL                 Parameter List
00023740  6E6E6ED7 D3F4F24C               2127+         DC    CL32'>>>PL42<<<'    eye catcher
00023748  4C4C4040 40404040
00023750  40404040 40404040
00023758  40404040 40404040
00023760                                  2128+PL42     DS    0C
00023760  00000000 00000000               2129          DC    XL256'00'
00023768  00000000 00000000
00023770  00000000 00000000
00023778  00000000 00000000
00023780  00000000 00000000
00023788  00000000 00000000
00023790  00000000 00000000
00023798  00000000 00000000
000237A0  00000000 00000000
000237A8  00000000 00000000
000237B0  00000000 00000000
000237B8  00000000 00000000
000237C0  00000000 00000000
000237C8  00000000 00000000
000237D0  00000000 00000000
000237D8  00000000 00000000
000237E0  00000000 00000000
000237E8  00000000 00000000
000237F0  00000000 00000000
000237F8  00000000 00000000
00023800  00000000 00000000
00023808  00000000 00000000
00023810  00000000 00000000
00023818  00000000 00000000
00023820  00000000 00000000
00023828  00000000 00000000
00023830  00000000 00000000
00023838  00000000 00000000
00023840  00000000 00000000
00023848  00000000 00000000
00023850  00000000 00000000
00023858  00000000 00000000
                                          2130
                                          2131          TSTLBL FMSG               fail message
00023860  6E6E6EC6 D4E2C7F4               2132+         DC    CL32'>>>FMSG42<<<'  eye catcher
00023868  F24C4C4C 40404040
00023870  40404040 40404040
00023878  40404040 40404040
00023880                                  2133+FMSG42   DS    0C
00023880  86818993 858440D7               2134          DC    C'failed PLO-Extension '
00023888  D3D660C5 A7A38595
00023890  A2899695 40
00023895  D8A48184 99A49793               2135          DC    CL96'Quadruple Load(QL: 45) / Store(QST: 58)'
0002389D  8540D396 81844DD8
000238A5  D37A40F4 F55D4061
000238AD  40E2A396 99854DD8
000238B5  E2E37A40 F5F85D40
000238BD  40404040 40404040
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
000238C5  40404040 40404040
000238CD  40404040 40404040
000238D5  40404040 40404040
000238DD  40404040 40404040
000238E5  40404040 40404040
000238ED  40404040 40404040
                                        2136
                                        2137          TSTEND
000238F5  C5D5C440 968640A3             2138+         DC    CL32'END of test: 42 <<<'
000238FD  85A2A37A 40F4F240
00023905  4C4C4C40 40404040
0002390D  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    2140 *=============================================================================
                                                    2141 *       Triple Load / Store Tests
                                                    2142 *=============================================================================
                                                    2143
                                                    2144 * -----------------------------------------------------------------------
                                                    2145 * Test Triple Load(TLO: 44) / Store(TSTO: 57) instructions
                                                    2146 * -----------------------------------------------------------------------
                                                    2147          TSTHDR 99,99,99
00023915                          00023915  00023C00  2148+         ORG   PLOETST+(128*K)+5*(3*K)
00023C00  E2E3C1D9 E3409686                         2149+         DC    CL32'START of test: 43'
00023C08  40A385A2 A37A40F4
00023C10  F3404040 40404040
00023C18  40404040 40404040
00023C20                          00023C20          2150+         USING *,R5                base for test data and test routine
00023C20                          00000000          2151+         USING PLOETEST,R5
00023C20  00023C4C                                  2152+T43      DC    A(CODE43)           address of test code
00023C24  002B                                      2153+         DC    H'43'               test number
00023C26  00                                        2154+         DC    X'00'
00023C27  63                                        2155+         DC    HL1'99'             Function
00023C28  63                                        2156+         DC    HL1'99'             expected unequal results (UR)
00023C29  00                                        2157+         DC    HL1'00'             found UR
00023C2A  63                                        2158+         DC    HL1'99'             cc
00023C2B  00                                        2159+         DC    HL1'00'             cc not checked
00023C2C  00                                        2160+         DS    X                   extracted cc
00023C30  00000000 00000000                         2161+         DS    2F                  extract PSW after test (has CC)
00023C38  00024480                                  2162+         DC    A(FMSG43)           address of test fail message
00023C3C  000242C0                                  2163+         DC    A(OPSRC43)          address of operand source
00023C40  00024220                                  2164+         DC    A(OPDST43)          address of operand destination
00023C44  00024360                                  2165+         DC    A(PL43)             address of parameter list
00023C48  000041E3                                  2166+         DC    A(RANDX+43)         address of random source data
                                                    2167
                                                    2168          TSTCODE
00023C4C                                            2169+CODE43   DS    0F
                                                    2170 *                                  load random source
00023C4C  E310 501C 0014                 0000001C   2171          LGF   R1,OPSRCA
00023C52  E320 5028 0014                 00000028   2172          LGF   R2,RANDXA
00023C58  E330 5024 0014                 00000024   2173          LGF   R3,PLA
                                                    2174
00023C5E  D27F 1000 2000       00000000  00000000   2175          MVC   0(128,R1),0(R2)
                                                    2176 *
                                                    2177 * Triple Load(TLO: 44)
                                                    2178 *
00023C64  E313 0000 0024                 00000000   2179          STG   R1,0(R3)            load OP1 address in PL
                                                    2180
00023C6A  4111 0020                      00000020   2181          LA    R1,32(R1)
00023C6E  E313 0008 0024                 00000008   2182          STG   R1,8(R3)            load OP2 address in PL
                                                    2183
00023C74  4111 0020                      00000020   2184          LA    R1,32(R1)
00023C78  E313 0010 0024                 00000010   2185          STG   R1,16(R3)           load OP3 address in PL
                                                    2186
00023C7E  B982 0000                                 2187          XGR   R0,R0
00023C82  4100 002C                      0000002C   2188          LA    R0,44
00023C86  EE10 0000 3000       00000000  00000000   2189          PLO   R1,0,0,0(R3)        Triple Load(TLO: 44)
                                                    2190 *
                                                    2191 * Triple Store(TSTO: 57)
                                                    2192 *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00023C8C  E310 5020 0014                 00000020  2193          LGF   R1,OPDSTA           Destination address
00023C92  E330 5024 0014                 00000024  2194          LGF   R3,PLA
                                                   2195
00023C98  E313 0000 0024                 00000000  2196          STG   R1,0(R3)            load OP1 address in PL
                                                   2197
00023C9E  4111 0020                      00000020  2198          LA    R1,32(R1)
00023CA2  E313 0008 0024                 00000008  2199          STG   R1,8(R3)            load OP2 address in PL
                                                   2200
00023CA8  4111 0020                      00000020  2201          LA    R1,32(R1)
00023CAC  E313 0010 0024                 00000010  2202          STG   R1,16(R3)           load OP3 address in PL
                                                   2203
00023CB2  B982 0000                                2204          XGR   R0,R0
00023CB6  4100 0039                      00000039  2205          LA    R0,57
00023CBA  EE10 0000 3000       00000000  00000000  2206          PLO   R1,0,0,0(R3)        Triple Store(TSTO: 57)
                                                   2207 *
                                                   2208 * compare results
                                                   2209 *
00023CC0  E310 501C 0014                 0000001C  2210          LGF   R1,OPSRCA           source
00023CC6  E330 5020 0014                 00000020  2211          LGF   R3,OPDSTA           Destination
                                                   2212
00023CCC  D51F 1000 3000       00000000  00000000  2213          CLC   0(32,R1),0(R3)
00023CD2  4770 50CA                      00023CEA  2214          BNE   F4457               OP1 Different, error
00023CD6  D51F 1020 3020       00000020  00000020  2215          CLC   32(32,R1),32(R3)
00023CDC  4770 50CA                      00023CEA  2216          BNE   F4457               OP2 Different, error
00023CE0  D51F 1040 3040       00000040  00000040  2217          CLC   64(32,R1),64(R3)
00023CE6  478B 0000                      00000000  2218          BE    0(R11)              OP3 same, return
                                                   2219
                          00023CEA       00000001  2220 F4457    EQU   *
00023CEA  45F0 83E8                      000023E8  2221          BAL   R15,FAILMSG
00023CEE  07FB                                     2222          BR    R11
                                                   2223
                                                   2224          TSTDATA
00023CF0                  00023CF0       00024200  2225+         ORG   PLOETST+(128*K)+5*(3*K)+(K+512)
                                                   2226          TSTLBL OPDST              Operand Destination (Results)
00024200  6E6E6ED6 D7C4E2E3                       2227+         DC    CL32'>>>OPDST43<<<' eye catcher
00024208  F4F34C4C 4C404040
00024210  40404040 40404040
00024218  40404040 40404040
00024220                                           2228+OPDST43  DS    0C
00024220  00000000 00000000                        2229          DC    XL128'0'
00024228  00000000 00000000
00024230  00000000 00000000
00024238  00000000 00000000
00024240  00000000 00000000
00024248  00000000 00000000
00024250  00000000 00000000
00024258  00000000 00000000
00024260  00000000 00000000
00024268  00000000 00000000
00024270  00000000 00000000
00024278  00000000 00000000
00024280  00000000 00000000
00024288  00000000 00000000
00024290  00000000 00000000
00024298  00000000 00000000
                                                   2230
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    2231           TSTLBL OPSRC               Operand Source
000242A0  6E6E6ED6 D7E2D9C3                         2232+          DC    CL32'>>>OPSRC43<<<'  eye catcher
000242A8  F4F34C4C 4C404040
000242B0  40404040 40404040
000242B8  40404040 40404040
000242C0                                            2233+OPSRC43   DS    0C
000242C0  00000000 00000000                         2234           DC    XL128'0'
000242C8  00000000 00000000
000242D0  00000000 00000000
000242D8  00000000 00000000
000242E0  00000000 00000000
000242E8  00000000 00000000
000242F0  00000000 00000000
000242F8  00000000 00000000
00024300  00000000 00000000
00024308  00000000 00000000
00024310  00000000 00000000
00024318  00000000 00000000
00024320  00000000 00000000
00024328  00000000 00000000
00024330  00000000 00000000
00024338  00000000 00000000
                                                    2235
                                                    2236           TSTLBL PL                  Parameter List
00024340  6E6E6ED7 D3F4F34C                         2237+          DC    CL32'>>>PL43<<<'     eye catcher
00024348  4C4C4040 40404040
00024350  40404040 40404040
00024358  40404040 40404040
00024360                                            2238+PL43      DS    0C
00024360  00000000 00000000                         2239           DC    XL256'00'
00024368  00000000 00000000
00024370  00000000 00000000
00024378  00000000 00000000
00024380  00000000 00000000
00024388  00000000 00000000
00024390  00000000 00000000
00024398  00000000 00000000
000243A0  00000000 00000000
000243A8  00000000 00000000
000243B0  00000000 00000000
000243B8  00000000 00000000
000243C0  00000000 00000000
000243C8  00000000 00000000
000243D0  00000000 00000000
000243D8  00000000 00000000
000243E0  00000000 00000000
000243E8  00000000 00000000
000243F0  00000000 00000000
000243F8  00000000 00000000
00024400  00000000 00000000
00024408  00000000 00000000
00024410  00000000 00000000
00024418  00000000 00000000
00024420  00000000 00000000
00024428  00000000 00000000
00024430  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00024438  00000000 00000000
00024440  00000000 00000000
00024448  00000000 00000000
00024450  00000000 00000000
00024458  00000000 00000000
                                        2240
                                        2241          TSTLBL FMSG               fail message
00024460  6E6E6EC6 D4E2C7F4             2242+         DC    CL32'>>>FMSG43<<<'  eye catcher
00024468  F34C4C4C 40404040
00024470  40404040 40404040
00024478  40404040 40404040
00024480                                2243+FMSG43   DS    0C
00024480  86818993 858440D7             2244          DC    C'failed PLO-Extension '
00024488  D3D660C5 A7A38595
00024490  A2899695 40
00024495  E3998997 938540D3             2245          DC    CL96'Triple Load(TLO: 44) / Store(TSTO: 57)'
0002449D  9681844D E3D3D67A
000244A5  40F4F45D 406140E2
000244AD  A3969985 4DE3E2E3
000244B5  D67A40F5 F75D4040
000244BD  40404040 40404040
000244C5  40404040 40404040
000244CD  40404040 40404040
000244D5  40404040 40404040
000244DD  40404040 40404040
000244E5  40404040 40404040
000244ED  40404040 40404040
                                        2246
                                        2247          TSTEND
000244F5  C5D5C440 968640A3             2248+         DC    CL32'END of test: 43 <<<'
000244FD  85A2A37A 40F4F340
00024505  4C4C4C40 40404040
0002450D  40404040 40404040
                                        2249
                                        2250 * ------------------------------------------------------------------
                                        2251 * Test Triple Load(TLX: 43) / Store(TSTX: 56) instructions
                                        2252 * ------------------------------------------------------------------
                                        2253          TSTHDR 99,99,99
00024515                  00024515  00024800  2254+         ORG   PLOETST+(128*K)+6*(3*K)
00024800  E2E3C1D9 E3409686             2255+         DC    CL32'START of test: 44'
00024808  40A385A2 A37A40F4
00024810  F4404040 40404040
00024818  40404040 40404040
00024820                  00024820          2256+         USING *,R5                base for test data and test routine
00024820                  00000000          2257+         USING PLOETEST,R5
00024820  0002484C                          2258+T44      DC    A(CODE44)           address of test code
00024824  002C                              2259+         DC    H'44'               test number
00024826  00                                2260+         DC    X'00'
00024827  63                                2261+         DC    HL1'99'             Function
00024828  63                                2262+         DC    HL1'99'             expected unequal results (UR)
00024829  00                                2263+         DC    HL1'00'             found UR
0002482A  63                                2264+         DC    HL1'99'             cc
0002482B  00                                2265+         DC    HL1'00'             cc not checked
0002482C  00                                2266+         DS    X                   extracted cc
00024830  00000000 00000000                 2267+         DS    2F                  extract PSW after test (has CC)
00024838  00025080                          2268+         DC    A(FMSG44)           address of test fail message
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002483C  00024EC0                                2269+          DC    A(OPSRC44)          address of operand source
00024840  00024E20                                2270+          DC    A(OPDST44)          address of operand destination
00024844  00024F60                                2271+          DC    A(PL44)             address of parameter list
00024848  000041E4                                2272+          DC    A(RANDX+44)         address of random source data
                                                  2273
                                                  2274           TSTCODE
0002484C                                          2275+CODE44    DS    0F
                                                  2276 *                                    load random source
0002484C  E310 501C 0014                0000001C  2277           LGF   R1,OPSRCA
00024852  E320 5028 0014                00000028  2278           LGF   R2,RANDXA
00024858  E330 5024 0014                00000024  2279           LGF   R3,PLA
                                                  2280
0002485E  D27F 1000 2000      00000000  00000000  2281           MVC   0(128,R1),0(R2)
                                                  2282 *
                                                  2283 * Triple Load(TLX: 43)
                                                  2284 *
00024864  E313 0000 0024                00000000  2285           STG   R1,0(R3)            load OP1 address in PL
                                                  2286
0002486A  4111 0020                     00000020  2287           LA    R1,32(R1)
0002486E  E313 0008 0024                00000008  2288           STG   R1,8(R3)            load OP2 address in PL
                                                  2289
00024874  4111 0020                     00000020  2290           LA    R1,32(R1)
00024878  E313 0010 0024                00000010  2291           STG   R1,16(R3)           load OP3 address in PL
                                                  2292
0002487E  B982 0000                               2293           XGR   R0,R0
00024882  4100 002B                     0000002B  2294           LA    R0,43
00024886  EE10 0000 3000      00000000  00000000  2295           PLO   R1,0,0,0(R3)        Triple Load(TLX: 43)
                                                  2296 *
                                                  2297 * Triple Store(TSTX: 56)
                                                  2298 *
0002488C  E310 5020 0014                00000020  2299           LGF   R1,OPDSTA           Destination address
00024892  E330 5024 0014                00000024  2300           LGF   R3,PLA
                                                  2301
00024898  E313 0000 0024                00000000  2302           STG   R1,0(R3)            load OP1 address in PL
                                                  2303
0002489E  4111 0020                     00000020  2304           LA    R1,32(R1)
000248A2  E313 0008 0024                00000008  2305           STG   R1,8(R3)            load OP2 address in PL
                                                  2306
000248A8  4111 0020                     00000020  2307           LA    R1,32(R1)
000248AC  E313 0010 0024                00000010  2308           STG   R1,16(R3)           load OP3 address in PL
                                                  2309
000248B2  B982 0000                               2310           XGR   R0,R0
000248B6  4100 0038                     00000038  2311           LA    R0,56
000248BA  EE10 0000 3000      00000000  00000000  2312           PLO   R1,0,0,0(R3)        Triple Store(TSTX: 56)
                                                  2313 *
                                                  2314 * compare results
                                                  2315 *
000248C0  E310 501C 0014                0000001C  2316           LGF   R1,OPSRCA           source
000248C6  E330 5020 0014                00000020  2317           LGF   R3,OPDSTA           Destination
                                                  2318
000248CC  D50F 1000 3000      00000000  00000000  2319           CLC   0(16,R1),0(R3)
000248D2  4770 50CA                     000248EA  2320           BNE   F4356               OP1 Different, error
000248D6  D50F 1020 3020      00000020  00000020  2321           CLC   32(16,R1),32(R3)
000248DC  4770 50CA                     000248EA  2322           BNE   F4356               OP2 Different, error
000248E0  D50F 1040 3040      00000040  00000040  2323           CLC   64(16,R1),64(R3)
000248E6  478B 0000                     00000000  2324           BE    0(R11)              OP3 same, return
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    2325
                               000248EA  00000001  2326 F4356    EQU   *
000248EA  45F0 83E8                      000023E8  2327          BAL   R15,FAILMSG
000248EE  07FB                                     2328          BR    R11
                                                   2329
                                                   2330          TSTDATA
000248F0                       000248F0  00024E00  2331+         ORG   PLOETST+(128*K)+6*(3*K)+(K+512)
                                                   2332          TSTLBL OPDST              Operand Destination (Results)
00024E00  6E6E6ED6 D7C4E2E3                        2333+         DC    CL32'>>>OPDST44<<<' eye catcher
00024E08  F4F44C4C 4C404040
00024E10  40404040 40404040
00024E18  40404040 40404040
00024E20                                           2334+OPDST44  DS    0C
00024E20  00000000 00000000                        2335          DC    XL128'0'
00024E28  00000000 00000000
00024E30  00000000 00000000
00024E38  00000000 00000000
00024E40  00000000 00000000
00024E48  00000000 00000000
00024E50  00000000 00000000
00024E58  00000000 00000000
00024E60  00000000 00000000
00024E68  00000000 00000000
00024E70  00000000 00000000
00024E78  00000000 00000000
00024E80  00000000 00000000
00024E88  00000000 00000000
00024E90  00000000 00000000
00024E98  00000000 00000000
                                                   2336
                                                   2337          TSTLBL OPSRC              Operand Source
00024EA0  6E6E6ED6 D7E2D9C3                        2338+         DC    CL32'>>>OPSRC44<<<' eye catcher
00024EA8  F4F44C4C 4C404040
00024EB0  40404040 40404040
00024EB8  40404040 40404040
00024EC0                                           2339+OPSRC44  DS    0C
00024EC0  00000000 00000000                        2340          DC    XL128'0'
00024EC8  00000000 00000000
00024ED0  00000000 00000000
00024ED8  00000000 00000000
00024EE0  00000000 00000000
00024EE8  00000000 00000000
00024EF0  00000000 00000000
00024EF8  00000000 00000000
00024F00  00000000 00000000
00024F08  00000000 00000000
00024F10  00000000 00000000
00024F18  00000000 00000000
00024F20  00000000 00000000
00024F28  00000000 00000000
00024F30  00000000 00000000
00024F38  00000000 00000000
                                                   2341
                                                   2342          TSTLBL PL                 Parameter List
00024F40  6E6E6ED7 D3F4F44C                        2343+         DC    CL32'>>>PL44<<<'    eye catcher
00024F48  4C4C4040 40404040
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00024F50  40404040 40404040
00024F58  40404040 40404040
00024F60                                             2344+PL44     DS    0C
00024F60  00000000 00000000                          2345          DC    XL256'00'
00024F68  00000000 00000000
00024F70  00000000 00000000
00024F78  00000000 00000000
00024F80  00000000 00000000
00024F88  00000000 00000000
00024F90  00000000 00000000
00024F98  00000000 00000000
00024FA0  00000000 00000000
00024FA8  00000000 00000000
00024FB0  00000000 00000000
00024FB8  00000000 00000000
00024FC0  00000000 00000000
00024FC8  00000000 00000000
00024FD0  00000000 00000000
00024FD8  00000000 00000000
00024FE0  00000000 00000000
00024FE8  00000000 00000000
00024FF0  00000000 00000000
00024FF8  00000000 00000000
00025000  00000000 00000000
00025008  00000000 00000000
00025010  00000000 00000000
00025018  00000000 00000000
00025020  00000000 00000000
00025028  00000000 00000000
00025030  00000000 00000000
00025038  00000000 00000000
00025040  00000000 00000000
00025048  00000000 00000000
00025050  00000000 00000000
00025058  00000000 00000000
                                                     2346
                                                     2347          TSTLBL FMSG               fail message
00025060  6E6E6EC6 D4E2C7F4                          2348+         DC    CL32'>>>FMSG44<<<'  eye catcher
00025068  F44C4C4C 40404040
00025070  40404040 40404040
00025078  40404040 40404040
00025080                                             2349+FMSG44   DS    0C
00025080  86818993 858440D7                          2350          DC    C'failed PLO-Extension '
00025088  D3D660C5 A7A38595
00025090  A2899695 40
00025095  E3998997 938540D3                          2351          DC    CL96'Triple Load(TLX: 43) / Store(TSTX: 56)'
0002509D  9681844D E3D3E77A
000250A5  40F4F35D 406140E2
000250AD  A3969985 4DE3E2E3
000250B5  E77A40F5 F65D4040
000250BD  40404040 40404040
000250C5  40404040 40404040
000250CD  40404040 40404040
000250D5  40404040 40404040
000250DD  40404040 40404040
000250E5  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000250ED  40404040 40404040                                2352
                                                           2353          TSTEND
000250F5  C5D5C440 968640A3                                2354+         DC    CL32'END of test: 44 <<<'
000250FD  85A2A37A 40F4F440
00025105  4C4C4C40 40404040
0002510D  40404040 40404040                                2355
                                                           2356 * ------------------------------------------------------------------
                                                           2357 * Test Triple Load(TLG: 42) / Store(TSTG: 55) instructions
                                                           2358 * ------------------------------------------------------------------
                                                           2359          TSTHDR 99,99,99
00025115                     00025115  00025400            2360+         ORG   PLOETST+(128*K)+7*(3*K)
00025400  E2E3C1D9 E3409686                                2361+         DC    CL32'START of test: 45'
00025408  40A385A2 A37A40F4
00025410  F5404040 40404040
00025418  40404040 40404040
00025420                     00025420                      2362+         USING *,R5                base for test data and test routine
00025420                     00000000                      2363+         USING PLOETEST,R5
00025420  0002544C                                         2364+T45      DC    A(CODE45)           address of test code
00025424  002D                                             2365+         DC    H'45'               test number
00025426  00                                               2366+         DC    X'00'
00025427  63                                               2367+         DC    HL1'99'             Function
00025428  63                                               2368+         DC    HL1'99'             expected unequal results (UR)
00025429  00                                               2369+         DC    HL1'00'             found UR
0002542A  63                                               2370+         DC    HL1'99'             cc
0002542B  00                                               2371+         DC    HL1'00'             cc not checked
0002542C  00                                               2372+         DS    X                   extracted cc
00025430  00000000 00000000                                2373+         DS    2F                  extract PSW after test (has CC)
00025438  00025C80                                         2374+         DC    A(FMSG45)           address of test fail message
0002543C  00025AC0                                         2375+         DC    A(OPSRC45)          address of operand source
00025440  00025A20                                         2376+         DC    A(OPDST45)          address of operand destination
00025444  00025B60                                         2377+         DC    A(PL45)             address of parameter list
00025448  000041E5                                         2378+         DC    A(RANDX+45)         address of random source data
                                                           2379
                                                           2380          TSTCODE
0002544C                                                   2381+CODE45   DS    0F
                                                           2382 *                                    load random source
0002544C  E310 501C 0014                 0000001C          2383          LGF   R1,OPSRCA
00025452  E320 5028 0014                 00000028          2384          LGF   R2,RANDXA
00025458  E330 5024 0014                 00000024          2385          LGF   R3,PLA
                                                           2386
0002545E  D27F 1000 2000      00000000   00000000          2387          MVC   0(128,R1),0(R2)
                                                           2388 *
                                                           2389 * Triple Load(TLG: 42)
                                                           2390 *
00025464  E313 0000 0024                 00000000          2391          STG   R1,0(R3)            load OP1 address in PL
                                                           2392
0002546A  4111 0020                      00000020          2393          LA    R1,32(R1)
0002546E  E313 0008 0024                 00000008          2394          STG   R1,8(R3)            load OP2 address in PL
                                                           2395
00025474  4111 0020                      00000020          2396          LA    R1,32(R1)
00025478  E313 0010 0024                 00000010          2397          STG   R1,16(R3)           load OP3 address in PL
                                                           2398
0002547E  B982 0000                                        2399          XGR   R0,R0
00025482  4100 002A                      0000002A          2400          LA    R0,42
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00025486  EE10 0000 3000      00000000  00000000  2401          PLO   R1,0,0,0(R3)        Triple Load(TLG: 42)
                                                   2402 *
                                                   2403 * Triple Store(TSTG: 55)
                                                   2404 *
0002548C  E310 5020 0014                00000020  2405          LGF   R1,OPDSTA           Destination address
00025492  E330 5024 0014                00000024  2406          LGF   R3,PLA
                                                   2407
00025498  E313 0000 0024                00000000  2408          STG   R1,0(R3)            load OP1 address in PL
                                                   2409
0002549E  4111 0020                     00000020  2410          LA    R1,32(R1)
000254A2  E313 0008 0024                00000008  2411          STG   R1,8(R3)            load OP2 address in PL
                                                   2412
000254A8  4111 0020                     00000020  2413          LA    R1,32(R1)
000254AC  E313 0010 0024                00000010  2414          STG   R1,16(R3)           load OP3 address in PL
                                                   2415
000254B2  B982 0000                               2416          XGR   R0,R0
000254B6  4100 0037                     00000037  2417          LA    R0,55
000254BA  EE10 0000 3000      00000000  00000000  2418          PLO   R1,0,0,0(R3)        Triple Store(TSTG: 55)
                                                   2419 *
                                                   2420 * compare results
                                                   2421 *
000254C0  E310 501C 0014                0000001C  2422          LGF   R1,OPSRCA           source
000254C6  E330 5020 0014                00000020  2423          LGF   R3,OPDSTA           Destination
                                                   2424
000254CC  D507 1000 3000      00000000  00000000  2425          CLC   0(8,R1),0(R3)
000254D2  4770 50CA                     000254EA  2426          BNE   F4255               OP1 Different, error
000254D6  D507 1020 3020      00000020  00000020  2427          CLC   32(8,R1),32(R3)
000254DC  4770 50CA                     000254EA  2428          BNE   F4255               OP2 Different, error
000254E0  D507 1040 3040      00000040  00000040  2429          CLC   64(8,R1),64(R3)
000254E6  478B 0000                     00000000  2430          BE    0(R11)              OP3 same, return
                                                   2431
                                        000254EA  00000001  2432 F4255     EQU   *
000254EA  45F0 83E8                     000023E8  2433          BAL   R15,FAILMSG
000254EE  07FB                                    2434          BR    R11
                                                   2435
                                                   2436          TSTDATA
000254F0                                000254F0  00025A00  2437+         ORG   PLOETST+(128*K)+7*(3*K)+(K+512)
                                                   2438          TSTLBL OPDST              Operand Destination (Results)
00025A00  6E6E6ED6 D7C4E2E3                        2439+         DC    CL32'>>>OPDST45<<<' eye catcher
00025A08  F4F54C4C 4C404040
00025A10  40404040 40404040
00025A18  40404040 40404040
00025A20                                           2440+OPDST45  DS    0C
00025A20  00000000 00000000                        2441          DC    XL128'0'
00025A28  00000000 00000000
00025A30  00000000 00000000
00025A38  00000000 00000000
00025A40  00000000 00000000
00025A48  00000000 00000000
00025A50  00000000 00000000
00025A58  00000000 00000000
00025A60  00000000 00000000
00025A68  00000000 00000000
00025A70  00000000 00000000
00025A78  00000000 00000000
00025A80  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00025A88  00000000 00000000
00025A90  00000000 00000000
00025A98  00000000 00000000
                                          2442
                                          2443           TSTLBL OPSRC              Operand Source
00025AA0  6E6E6ED6 D7E2D9C3               2444+          DC    CL32'>>>OPSRC45<<<' eye catcher
00025AA8  F4F54C4C 4C404040
00025AB0  40404040 40404040
00025AB8  40404040 40404040
00025AC0                                  2445+OPSRC45   DS    0C
00025AC0  00000000 00000000               2446           DC    XL128'0'
00025AC8  00000000 00000000
00025AD0  00000000 00000000
00025AD8  00000000 00000000
00025AE0  00000000 00000000
00025AE8  00000000 00000000
00025AF0  00000000 00000000
00025AF8  00000000 00000000
00025B00  00000000 00000000
00025B08  00000000 00000000
00025B10  00000000 00000000
00025B18  00000000 00000000
00025B20  00000000 00000000
00025B28  00000000 00000000
00025B30  00000000 00000000
00025B38  00000000 00000000
                                          2447
                                          2448           TSTLBL PL                 Parameter List
00025B40  6E6E6ED7 D3F54C                 2449+          DC    CL32'>>>PL45<<<'    eye catcher
00025B48  4C4C4040 40404040
00025B50  40404040 40404040
00025B58  40404040 40404040
00025B60                                  2450+PL45      DS    0C
00025B60  00000000 00000000               2451           DC    XL256'00'
00025B68  00000000 00000000
00025B70  00000000 00000000
00025B78  00000000 00000000
00025B80  00000000 00000000
00025B88  00000000 00000000
00025B90  00000000 00000000
00025B98  00000000 00000000
00025BA0  00000000 00000000
00025BA8  00000000 00000000
00025BB0  00000000 00000000
00025BB8  00000000 00000000
00025BC0  00000000 00000000
00025BC8  00000000 00000000
00025BD0  00000000 00000000
00025BD8  00000000 00000000
00025BE0  00000000 00000000
00025BE8  00000000 00000000
00025BF0  00000000 00000000
00025BF8  00000000 00000000
00025C00  00000000 00000000
00025C08  00000000 00000000
00025C10  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00025C18  00000000 00000000
00025C20  00000000 00000000
00025C28  00000000 00000000
00025C30  00000000 00000000
00025C38  00000000 00000000
00025C40  00000000 00000000
00025C48  00000000 00000000
00025C50  00000000 00000000
00025C58  00000000 00000000
                                        2452
                                        2453          TSTLBL FMSG                fail message
00025C60  6E6E6EC6 D4E2C7F4             2454+         DC    CL32'>>>FMSG45<<<'   eye catcher
00025C68  F54C4C4C 40404040
00025C70  40404040 40404040
00025C78  40404040 40404040
00025C80                                2455+FMSG45   DS    0C
00025C80  86818993 858440D7             2456          DC    C'failed PLO-Extension '
00025C88  D3D660C5 A7A38595
00025C90  A2899695 40
00025C95  E3998997 938540D3             2457          DC    CL96'Triple Load(TLG: 42) / Store(TSTG: 55)'
00025C9D  9681844D E3D3C77A
00025CA5  40F4F25D 406140E2
00025CAD  A3969985 4DE3E2E3
00025CB5  C77A40F5 F55D4040
00025CBD  40404040 40404040
00025CC5  40404040 40404040
00025CCD  40404040 40404040
00025CD5  40404040 40404040
00025CDD  40404040 40404040
00025CE5  40404040 40404040
00025CED  40404040 40404040
                                        2458
                                        2459          TSTEND
00025CF5  C5D5C440 968640A3             2460+         DC    CL32'END of test: 45 <<<'
00025CFD  85A2A37A 40F4F540
00025D05  4C4C4C40 40404040
00025D0D  40404040 40404040
                                        2461
                                        2462 * ------------------------------------------------------------------
                                        2463 * Test Triple Load(TL: 41) / Store(TST: 54) instructions
                                        2464 * ------------------------------------------------------------------
                                        2465          TSTHDR 99,99,99
00025D15                  00025D15  00026000  2466+         ORG   PLOETST+(128*K)+8*(3*K)
00026000  E2E3C1D9 E3409686             2467+         DC    CL32'START of test: 46'
00026008  40A385A2 A37A40F4
00026010  F6404040 40404040
00026018  40404040 40404040
00026020                  00026020            2468+         USING *,R5                 base for test data and test routine
00026020                  00000000            2469+         USING PLOETEST,R5
00026020  0002604C                            2470+T46      DC    A(CODE46)            address of test code
00026024  002E                                2471+         DC    H'46'                test number
00026026  00                                  2472+         DC    X'00'
00026027  63                                  2473+         DC    HL1'99'              Function
00026028  63                                  2474+         DC    HL1'99'              expected unequal results (UR)
00026029  00                                  2475+         DC    HL1'00'              found UR
0002602A  63                                  2476+         DC    HL1'99'              cc
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002602B  00                                       2477+          DC    HL1'00'             cc not checked
0002602C  00                                       2478+          DS    X                   extracted cc
00026030  00000000 00000000                        2479+          DS    2F                  extract PSW after test (has CC)
00026038  00026880                                 2480+          DC    A(FMSG46)           address of test fail message
0002603C  000266C0                                 2481+          DC    A(OPSRC46)          address of operand source
00026040  00026620                                 2482+          DC    A(OPDST46)          address of operand destination
00026044  00026760                                 2483+          DC    A(PL46)             address of parameter list
00026048  000041E6                                 2484+          DC    A(RANDX+46)         address of random source data
                                                   2485
                                                   2486           TSTCODE
0002604C                                           2487+CODE46    DS    0F
                                                   2488 *                                   load random source
0002604C  E310 501C 0014                0000001C   2489           LGF   R1,OPSRCA
00026052  E320 5028 0014                00000028   2490           LGF   R2,RANDXA
00026058  E330 5024 0014                00000024   2491           LGF   R3,PLA
                                                   2492
0002605E  D27F 1000 2000      00000000  00000000   2493           MVC   0(128,R1),0(R2)
                                                   2494 *
                                                   2495 * Triple Load(TL: 41)
                                                   2496 *
00026064  E313 0000 0024                00000000   2497           STG   R1,0(R3)            load OP1 address in PL
                                                   2498
0002606A  4111 0020                     00000020   2499           LA    R1,32(R1)
0002606E  E313 0008 0024                00000008   2500           STG   R1,8(R3)            load OP2 address in PL
                                                   2501
00026074  4111 0020                     00000020   2502           LA    R1,32(R1)
00026078  E313 0010 0024                00000010   2503           STG   R1,16(R3)           load OP3 address in PL
                                                   2504
0002607E  B982 0000                                2505           XGR   R0,R0
00026082  4100 0029                     00000029   2506           LA    R0,41
00026086  EE10 0000 3000      00000000  00000000   2507           PLO   R1,0,0,0(R3)        Triple Load(TL: 41)
                                                   2508 *
                                                   2509 * Triple Store(TST: 54)
                                                   2510 *
0002608C  E310 5020 0014                00000020   2511           LGF   R1,OPDSTA           Destination address
00026092  E330 5024 0014                00000024   2512           LGF   R3,PLA
                                                   2513
00026098  E313 0000 0024                00000000   2514           STG   R1,0(R3)            load OP1 address in PL
                                                   2515
0002609E  4111 0020                     00000020   2516           LA    R1,32(R1)
000260A2  E313 0008 0024                00000008   2517           STG   R1,8(R3)            load OP2 address in PL
                                                   2518
000260A8  4111 0020                     00000020   2519           LA    R1,32(R1)
000260AC  E313 0010 0024                00000010   2520           STG   R1,16(R3)           load OP3 address in PL
                                                   2521
000260B2  B982 0000                                2522           XGR   R0,R0
000260B6  4100 0036                     00000036   2523           LA    R0,54
000260BA  EE10 0000 3000      00000000  00000000   2524           PLO   R1,0,0,0(R3)        Triple Store(TST: 54)
                                                   2525 *
                                                   2526 * compare results
                                                   2527 *
000260C0  E310 501C 0014                0000001C   2528           LGF   R1,OPSRCA           source
000260C6  E330 5020 0014                00000020   2529           LGF   R3,OPDSTA           Destination
                                                   2530
000260CC  D503 1000 3000      00000000  00000000   2531           CLC   0(4,R1),0(R3)
000260D2  4770 50CA                     000260EA   2532           BNE   F4154               OP1 Different, error
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000260D6  D503 1020 3020      00000020  00000020  2533           CLC   32(4,R1),32(R3)
000260DC  4770 50CA                     000260EA  2534           BNE   F4154                OP2 Different, error
000260E0  D503 1040 3040      00000040  00000040  2535           CLC   64(4,R1),64(R3)
000260E6  478B 0000                     00000000  2536           BE    0(R11)               OP3 same, return
                                                  2537
                              000260EA  00000001  2538 F4154     EQU   *
000260EA  45F0 83E8                     000023E8  2539           BAL   R15,FAILMSG
000260EE  07FB                                    2540           BR    R11
                                                  2541
                                                  2542           TSTDATA
000260F0                      000260F0  00026600  2543+          ORG   PLOETST+(128*K)+8*(3*K)+(K+512)
                                                  2544           TSTLBL OPDST               Operand Destination (Results)
00026600  6E6E6ED6 D7C4E2E3                       2545+          DC    CL32'>>>OPDST46<<<' eye catcher
00026608  F4F64C4C 4C404040
00026610  40404040 40404040
00026618  40404040 40404040
00026620                                          2546+OPDST46   DS    0C
00026620  00000000 00000000                       2547           DC    XL128'0'
00026628  00000000 00000000
00026630  00000000 00000000
00026638  00000000 00000000
00026640  00000000 00000000
00026648  00000000 00000000
00026650  00000000 00000000
00026658  00000000 00000000
00026660  00000000 00000000
00026668  00000000 00000000
00026670  00000000 00000000
00026678  00000000 00000000
00026680  00000000 00000000
00026688  00000000 00000000
00026690  00000000 00000000
00026698  00000000 00000000
                                                  2548
                                                  2549           TSTLBL OPSRC               Operand Source
000266A0  6E6E6ED6 D7E2D9C3                       2550+          DC    CL32'>>>OPSRC46<<<' eye catcher
000266A8  F4F64C4C 4C404040
000266B0  40404040 40404040
000266B8  40404040 40404040
000266C0                                          2551+OPSRC46   DS    0C
000266C0  00000000 00000000                       2552           DC    XL128'0'
000266C8  00000000 00000000
000266D0  00000000 00000000
000266D8  00000000 00000000
000266E0  00000000 00000000
000266E8  00000000 00000000
000266F0  00000000 00000000
000266F8  00000000 00000000
00026700  00000000 00000000
00026708  00000000 00000000
00026710  00000000 00000000
00026718  00000000 00000000
00026720  00000000 00000000
00026728  00000000 00000000
00026730  00000000 00000000
00026738  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   2553
                                                   2554          TSTLBL PL                 Parameter List
00026740  6E6E6ED7 D3F4F64C                        2555+         DC    CL32'>>>PL46<<<'    eye catcher
00026748  4C4C4040 40404040
00026750  40404040 40404040
00026758  40404040 40404040
00026760                                           2556+PL46     DS    0C
00026760  00000000 00000000                        2557          DC    XL256'00'
00026768  00000000 00000000
00026770  00000000 00000000
00026778  00000000 00000000
00026780  00000000 00000000
00026788  00000000 00000000
00026790  00000000 00000000
00026798  00000000 00000000
000267A0  00000000 00000000
000267A8  00000000 00000000
000267B0  00000000 00000000
000267B8  00000000 00000000
000267C0  00000000 00000000
000267C8  00000000 00000000
000267D0  00000000 00000000
000267D8  00000000 00000000
000267E0  00000000 00000000
000267E8  00000000 00000000
000267F0  00000000 00000000
000267F8  00000000 00000000
00026800  00000000 00000000
00026808  00000000 00000000
00026810  00000000 00000000
00026818  00000000 00000000
00026820  00000000 00000000
00026828  00000000 00000000
00026830  00000000 00000000
00026838  00000000 00000000
00026840  00000000 00000000
00026848  00000000 00000000
00026850  00000000 00000000
00026858  00000000 00000000
                                                   2558
                                                   2559          TSTLBL FMSG               fail message
00026860  6E6E6EC6 D4E2C7F4                        2560+         DC    CL32'>>>FMSG46<<<'  eye catcher
00026868  F64C4C4C 40404040
00026870  40404040 40404040
00026878  40404040 40404040
00026880                                           2561+FMSG46   DS    0C
00026880  86818993 858440D7                        2562          DC    C'failed PLO-Extension '
00026888  D3D660C5 A7A38595
00026890  A2899695 40
00026895  E3998997 938540D3                        2563          DC    CL96'Triple Load(TL: 41) / Store(TST: 54)'
0002689D  9681844D E3D37A40
000268A5  F4F15D40 6140E2A3
000268AD  9699854D E3E2E37A
000268B5  40F5F45D 40404040
000268BD  40404040 40404040
000268C5  40404040 40404040
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
000268CD  40404040 40404040
000268D5  40404040 40404040
000268DD  40404040 40404040
000268E5  40404040 40404040
000268ED  40404040 40404040
                                          2564
                                          2565          TSTEND
000268F5  C5D5C440 968640A3               2566+         DC    CL32'END of test: 46 <<<'
000268FD  85A2A37A 40F4F640
00026905  4C4C4C40 40404040
0002690D  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   2568 *=======================================================================
                                                   2569 *       Double Load / Store Tests
                                                   2570 *=======================================================================
                                                   2571
                                                   2572 * ---------------------------------------------------------------------
                                                   2573 * Test Double Load(DLO: 40) / Store(DSTO: 53) instructions
                                                   2574 * ---------------------------------------------------------------------
                                                   2575          TSTHDR 99,99,99
00026915                          00026915  00026C00 2576+         ORG   PLOETST+(128*K)+9*(3*K)
00026C00  E2E3C1D9 E3409686                        2577+         DC    CL32'START of test: 47'
00026C08  40A385A2 A37A40F4
00026C10  F7404040 40404040
00026C18  40404040 40404040
00026C20                          00026C20         2578+         USING *,R5                base for test data and test routine
00026C20                          00000000         2579+         USING PLOETEST,R5
00026C20  00026C4C                                 2580+T47      DC    A(CODE47)           address of test code
00026C24  002F                                     2581+         DC    H'47'               test number
00026C26  00                                       2582+         DC    X'00'
00026C27  63                                       2583+         DC    HL1'99'             Function
00026C28  63                                       2584+         DC    HL1'99'             expected unequal results (UR)
00026C29  00                                       2585+         DC    HL1'00'             found UR
00026C2A  63                                       2586+         DC    HL1'99'             cc
00026C2B  00                                       2587+         DC    HL1'00'             cc not checked
00026C2C  00                                       2588+         DS    X                   extracted cc
00026C30  00000000 00000000                        2589+         DS    2F                  extract PSW after test (has CC)
00026C38  00027480                                 2590+         DC    A(FMSG47)           address of test fail message
00026C3C  000272C0                                 2591+         DC    A(OPSRC47)          address of operand source
00026C40  00027220                                 2592+         DC    A(OPDST47)          address of operand destination
00026C44  00027360                                 2593+         DC    A(PL47)             address of parameter list
00026C48  000041E7                                 2594+         DC    A(RANDX+47)         address of random source data
                                                   2595
                                                   2596          TSTCODE
00026C4C                                           2597+CODE47   DS    0F
                                                   2598 *                                    load random source
00026C4C  E310 501C 0014                0000001C   2599          LGF   R1,OPSRCA
00026C52  E320 5028 0014                00000028   2600          LGF   R2,RANDXA
00026C58  E330 5024 0014                00000024   2601          LGF   R3,PLA
                                                   2602
00026C5E  D27F 1000 2000      00000000  00000000   2603          MVC   0(128,R1),0(R2)
                                                   2604 *
                                                   2605 * Double Load(DLO: 40)
                                                   2606 *
00026C64  E313 0000 0024                00000000   2607          STG   R1,0(R3)            load OP1 address in PL
                                                   2608
00026C6A  4111 0020                     00000020   2609          LA    R1,32(R1)
00026C6E  E313 0008 0024                00000008   2610          STG   R1,8(R3)            load OP2 address in PL
                                                   2611
00026C74  B982 0000                                2612          XGR   R0,R0
00026C78  4100 0028                     00000028   2613          LA    R0,40
00026C7C  EE10 0000 3000      00000000  00000000   2614          PLO   R1,0,0,0(R3)        Double Load(DLO: 40)
                                                   2615 *
                                                   2616 * Double Store(DSTO: 53)
                                                   2617 *
00026C82  E310 5020 0014                00000020   2618          LGF   R1,OPDSTA           Destination address
00026C88  E330 5024 0014                00000024   2619          LGF   R3,PLA
                                                   2620
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00026C8E  E313 0000 0024                 00000000  2621          STG   R1,0(R3)            load OP1 address in PL
                                                   2622
00026C94  4111 0020                      00000020  2623          LA    R1,32(R1)
00026C98  E313 0008 0024                 00000008  2624          STG   R1,8(R3)            load OP2 address in PL
                                                   2625
00026C9E  B982 0000                                2626          XGR   R0,R0
00026CA2  4100 0035                      00000035  2627          LA    R0,53
00026CA6  EE10 0000 3000      00000000   00000000  2628          PLO   R1,0,0,0(R3)        Double Store(DSTO: 53)
                                                   2629 *
                                                   2630 * compare results
                                                   2631 *
00026CAC  E310 501C 0014                 0000001C  2632          LGF   R1,OPSRCA           source
00026CB2  E330 5020 0014                 00000020  2633          LGF   R3,OPDSTA           Destination
                                                   2634
00026CB8  D51F 1000 3000      00000000   00000000  2635          CLC   0(32,R1),0(R3)
00026CBE  4770 50AC                      00026CCC  2636          BNE   F4053               OP1 Different, error
00026CC2  D51F 1020 3020      00000020   00000020  2637          CLC   32(32,R1),32(R3)
00026CC8  478B 0000                      00000000  2638          BE    0(R11)              OP2 same, return
                                                   2639
                              00026CCC   00000001  2640 F4053    EQU   *
00026CCC  45F0 83E8                      000023E8  2641          BAL   R15,FAILMSG
00026CD0  07FB                                     2642          BR    R11
                                                   2643
                                                   2644          TSTDATA
00026CD2                      00026CD2   00027200  2645+         ORG   PLOETST+(128*K)+9*(3*K)+(K+512)
                                                   2646          TSTLBL OPDST              Operand Destination (Results)
00027200  6E6E6ED6 D7C4E2E3                       2647+         DC    CL32'>>>OPDST47<<<' eye catcher
00027208  F4F74C4C 4C404040
00027210  40404040 40404040
00027218  40404040 40404040
00027220                                   2648+OPDST47  DS    0C
00027220  00000000 00000000                2649          DC    XL128'0'
00027228  00000000 00000000
00027230  00000000 00000000
00027238  00000000 00000000
00027240  00000000 00000000
00027248  00000000 00000000
00027250  00000000 00000000
00027258  00000000 00000000
00027260  00000000 00000000
00027268  00000000 00000000
00027270  00000000 00000000
00027278  00000000 00000000
00027280  00000000 00000000
00027288  00000000 00000000
00027290  00000000 00000000
00027298  00000000 00000000
                                           2650
                                           2651          TSTLBL OPSRC              Operand Source
000272A0  6E6E6ED6 D7E2D9C3                2652+         DC    CL32'>>>OPSRC47<<<' eye catcher
000272A8  F4F74C4C 4C404040
000272B0  40404040 40404040
000272B8  40404040 40404040
000272C0                                   2653+OPSRC47  DS    0C
000272C0  00000000 00000000                2654          DC    XL128'0'
000272C8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000272D0  00000000 00000000
000272D8  00000000 00000000
000272E0  00000000 00000000
000272E8  00000000 00000000
000272F0  00000000 00000000
000272F8  00000000 00000000
00027300  00000000 00000000
00027308  00000000 00000000
00027310  00000000 00000000
00027318  00000000 00000000
00027320  00000000 00000000
00027328  00000000 00000000
00027330  00000000 00000000
00027338  00000000 00000000
                                        2655
                                        2656          TSTLBL PL                   Parameter List
00027340  6E6E6ED7 D3F4F74C             2657+         DC    CL32'>>>PL47<<<'      eye catcher
00027348  4C4C4040 40404040
00027350  40404040 40404040
00027358  40404040 40404040
00027360                                2658+PL47     DS    0C
00027360  00000000 00000000             2659          DC    XL256'00'
00027368  00000000 00000000
00027370  00000000 00000000
00027378  00000000 00000000
00027380  00000000 00000000
00027388  00000000 00000000
00027390  00000000 00000000
00027398  00000000 00000000
000273A0  00000000 00000000
000273A8  00000000 00000000
000273B0  00000000 00000000
000273B8  00000000 00000000
000273C0  00000000 00000000
000273C8  00000000 00000000
000273D0  00000000 00000000
000273D8  00000000 00000000
000273E0  00000000 00000000
000273E8  00000000 00000000
000273F0  00000000 00000000
000273F8  00000000 00000000
00027400  00000000 00000000
00027408  00000000 00000000
00027410  00000000 00000000
00027418  00000000 00000000
00027420  00000000 00000000
00027428  00000000 00000000
00027430  00000000 00000000
00027438  00000000 00000000
00027440  00000000 00000000
00027448  00000000 00000000
00027450  00000000 00000000
00027458  00000000 00000000
                                        2660
                                        2661          TSTLBL FMSG                 fail message
00027460  6E6E6EC6 D4E2C7F4             2662+         DC    CL32'>>>FMSG47<<<'    eye catcher
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00027468  F74C4C4C 40404040
00027470  40404040 40404040
00027478  40404040 40404040
00027480                                    2663+FMSG47   DS    0C
00027480  86818993 858440D7                 2664          DC    C'failed PLO-Extension '
00027488  D3D660C5 A7A38595
00027490  A2899695 40
00027495  C496A482 938540D3                 2665          DC    CL96'Double Load(DLO: 40) / Store(DSTO: 53)'
0002749D  9681844D C4D3D67A
000274A5  40F4F05D 406140E2
000274AD  A3969985 4DC4E2E3
000274B5  D67A40F5 F35D4040
000274BD  40404040 40404040
000274C5  40404040 40404040
000274CD  40404040 40404040
000274D5  40404040 40404040
000274DD  40404040 40404040
000274E5  40404040 40404040
000274ED  40404040 40404040
                                            2666
                                            2667          TSTEND
000274F5  C5D5C440 968640A3                 2668+         DC    CL32'END of test: 47 <<<'
000274FD  85A2A37A 40F4F740
00027505  4C4C4C40 40404040
0002750D  40404040 40404040
                                            2669
                                            2670 * ------------------------------------------------------------------
                                            2671 * Test Double Load(DLX: 39) / Store(DSTX: 52) instructions
                                            2672 * ------------------------------------------------------------------
                                            2673          TSTHDR 99,99,99
00027515                    00027515  00027800  2674+         ORG   PLOETST+(128*K)+10*(3*K)
00027800  E2E3C1D9 E3409686                 2675+         DC    CL32'START of test: 48'
00027808  40A385A2 A37A40F4
00027810  F8404040 40404040
00027818  40404040 40404040
00027820                    00027820        2676+         USING *,R5                base for test data and test routine
00027820                    00000000        2677+         USING PLOETEST,R5
00027820  0002784C                          2678+T48      DC    A(CODE48)           address of test code
00027824  0030                              2679+         DC    H'48'               test number
00027826  00                                2680+         DC    X'00'
00027827  63                                2681+         DC    HL1'99'             Function
00027828  63                                2682+         DC    HL1'99'             expected unequal results (UR)
00027829  00                                2683+         DC    HL1'00'             found UR
0002782A  63                                2684+         DC    HL1'99'             cc
0002782B  00                                2685+         DC    HL1'00'             cc not checked
0002782C  00                                2686+         DS    X                   extracted cc
00027830  00000000 00000000                 2687+         DS    2F                  extract PSW after test (has CC)
00027838  00028080                          2688+         DC    A(FMSG48)           address of test fail message
0002783C  00027EC0                          2689+         DC    A(OPSRC48)          address of operand source
00027840  00027E20                          2690+         DC    A(OPDST48)          address of operand destination
00027844  00027F60                          2691+         DC    A(PL48)             address of parameter list
00027848  000041E8                          2692+         DC    A(RANDX+48)         address of random source data
                                            2693
                                            2694          TSTCODE
0002784C                                    2695+CODE48   DS    0F
                                            2696 *                                  load random source
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002784C  E310 501C 0014                 0000001C  2697          LGF   R1,OPSRCA
00027852  E320 5028 0014                 00000028  2698          LGF   R2,RANDXA
00027858  E330 5024 0014                 00000024  2699          LGF   R3,PLA
                                                   2700
0002785E  D27F 1000 2000      00000000   00000000  2701          MVC   0(128,R1),0(R2)
                                                   2702 *
                                                   2703 * Double Load(DLX: 39)
                                                   2704 *
00027864  E313 0000 0024                 00000000  2705          STG   R1,0(R3)            load OP1 address in PL
                                                   2706
0002786A  4111 0020                      00000020  2707          LA    R1,32(R1)
0002786E  E313 0008 0024                 00000008  2708          STG   R1,8(R3)            load OP2 address in PL
                                                   2709
00027874  B982 0000                                2710          XGR   R0,R0
00027878  4100 002B                      0000002B  2711          LA    R0,43
0002787C  EE10 0000 3000      00000000   00000000  2712          PLO   R1,0,0,0(R3)        Double Load(DLX: 39)
                                                   2713 *
                                                   2714 * Double Store(DSTX: 52)
                                                   2715 *
00027882  E310 5020 0014                 00000020  2716          LGF   R1,OPDSTA           Destination address
00027888  E330 5024 0014                 00000024  2717          LGF   R3,PLA
                                                   2718
0002788E  E313 0000 0024                 00000000  2719          STG   R1,0(R3)            load OP1 address in PL
                                                   2720
00027894  4111 0020                      00000020  2721          LA    R1,32(R1)
00027898  E313 0008 0024                 00000008  2722          STG   R1,8(R3)            load OP2 address in PL
                                                   2723
0002789E  B982 0000                                2724          XGR   R0,R0
000278A2  4100 0034                      00000034  2725          LA    R0,52
000278A6  EE10 0000 3000      00000000   00000000  2726          PLO   R1,0,0,0(R3)        Double Store(DSTX: 52)
                                                   2727 *
                                                   2728 * compare results
                                                   2729 *
000278AC  E310 501C 0014                 0000001C  2730          LGF   R1,OPSRCA           source
000278B2  E330 5020 0014                 00000020  2731          LGF   R3,OPDSTA           Destination
                                                   2732
000278B8  D50F 1000 3000      00000000   00000000  2733          CLC   0(16,R1),0(R3)
000278BE  4770 50AC                      000278CC  2734          BNE   F4352               OP1 Different, error
000278C2  D50F 1020 3020      00000020   00000020  2735          CLC   32(16,R1),32(R3)
000278C8  478B 0000                      00000000  2736          BE    0(R11)              OP2 same, return
                                                   2737
                              000278CC   00000001  2738 F4352    EQU   *
000278CC  45F0 83E8                      000023E8  2739          BAL   R15,FAILMSG
000278D0  07FB                                     2740          BR    R11
                                                   2741
                                                   2742          TSTDATA
000278D2                      000278D2   00027E00  2743+         ORG   PLOETST+(128*K)+10*(3*K)+(K+512)
                                                   2744          TSTLBL OPDST              Operand Destination (Results)
00027E00  6E6E6ED6 D7C4E2E3                        2745+         DC    CL32'>>>OPDST48<<<' eye catcher
00027E08  F4F84C4C 4C404040
00027E10  40404040 40404040
00027E18  40404040 40404040
00027E20                                           2746+OPDST48  DS    0C
00027E20  00000000 00000000                        2747          DC    XL128'0'
00027E28  00000000 00000000
00027E30  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00027E38  00000000 00000000
00027E40  00000000 00000000
00027E48  00000000 00000000
00027E50  00000000 00000000
00027E58  00000000 00000000
00027E60  00000000 00000000
00027E68  00000000 00000000
00027E70  00000000 00000000
00027E78  00000000 00000000
00027E80  00000000 00000000
00027E88  00000000 00000000
00027E90  00000000 00000000
00027E98  00000000 00000000
                                          2748
                                          2749          TSTLBL OPSRC               Operand Source
00027EA0  6E6E6ED6 D7E2D9C3               2750+         DC    CL32'>>>OPSRC48<<<'  eye catcher
00027EA8  F4F84C4C 4C404040
00027EB0  40404040 40404040
00027EB8  40404040 40404040
00027EC0                                  2751+OPSRC48  DS    0C
00027EC0  00000000 00000000               2752          DC    XL128'0'
00027EC8  00000000 00000000
00027ED0  00000000 00000000
00027ED8  00000000 00000000
00027EE0  00000000 00000000
00027EE8  00000000 00000000
00027EF0  00000000 00000000
00027EF8  00000000 00000000
00027F00  00000000 00000000
00027F08  00000000 00000000
00027F10  00000000 00000000
00027F18  00000000 00000000
00027F20  00000000 00000000
00027F28  00000000 00000000
00027F30  00000000 00000000
00027F38  00000000 00000000
                                       2753
                                       2754          TSTLBL PL                  Parameter List
00027F40  6E6E6ED7 D3F4F84C            2755+         DC    CL32'>>>PL48<<<'     eye catcher
00027F48  4C4C4040 40404040
00027F50  40404040 40404040
00027F58  40404040 40404040
00027F60                               2756+PL48     DS    0C
00027F60  00000000 00000000            2757          DC    XL256'00'
00027F68  00000000 00000000
00027F70  00000000 00000000
00027F78  00000000 00000000
00027F80  00000000 00000000
00027F88  00000000 00000000
00027F90  00000000 00000000
00027F98  00000000 00000000
00027FA0  00000000 00000000
00027FA8  00000000 00000000
00027FB0  00000000 00000000
00027FB8  00000000 00000000
00027FC0  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00027FC8  00000000 00000000
00027FD0  00000000 00000000
00027FD8  00000000 00000000
00027FE0  00000000 00000000
00027FE8  00000000 00000000
00027FF0  00000000 00000000
00027FF8  00000000 00000000
00028000  00000000 00000000
00028008  00000000 00000000
00028010  00000000 00000000
00028018  00000000 00000000
00028020  00000000 00000000
00028028  00000000 00000000
00028030  00000000 00000000
00028038  00000000 00000000
00028040  00000000 00000000
00028048  00000000 00000000
00028050  00000000 00000000
00028058  00000000 00000000
                                        2758
                                        2759          TSTLBL FMSG               fail message
00028060  6E6E6EC6 D4E2C7F4             2760+         DC    CL32'>>>FMSG48<<<'  eye catcher
00028068  F84C4C4C 40404040
00028070  40404040 40404040
00028078  40404040 40404040
00028080                                2761+FMSG48   DS    0C
00028080  86818993 858440D7             2762          DC    C'failed PLO-Extension '
00028088  D3D660C5 A7A38595
00028090  A2899695 40
00028095  C496A482 938540D3             2763          DC    CL96'Double Load(DLX: 39) / Store(DSTX: 52)'
0002809D  9681844D C4D3E77A
000280A5  40F3F95D 406140E2
000280AD  A3969985 4DC4E2E3
000280B5  E77A40F5 F25D4040
000280BD  40404040 40404040
000280C5  40404040 40404040
000280CD  40404040 40404040
000280D5  40404040 40404040
000280DD  40404040 40404040
000280E5  40404040 40404040
000280ED  40404040 40404040
                                        2764
                                        2765          TSTEND
000280F5  C5D5C440 968640A3             2766+         DC    CL32'END of test: 48 <<<'
000280FD  85A2A37A 40F4F840
00028105  4C4C4C40 40404040
0002810D  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   2768 *=====================================================================
                                                   2769 *       Double Store Tests
                                                   2770 *=====================================================================
                                                   2771
                                                   2772 * -------------------------------------------------------------------
                                                   2773 * Test Double Store(DSTG: 51) instruction
                                                   2774 * -------------------------------------------------------------------
                                                   2775          TSTHDR 99,99,51
00028115                          00028115  00028400 2776+         ORG   PLOETST+(128*K)+11*(3*K)
00028400  E2E3C1D9 E3409686                        2777+         DC    CL32'START of test: 49'
00028408  40A385A2 A37A40F4
00028410  F9404040 40404040
00028418  40404040 40404040
00028420                          00028420         2778+         USING *,R5                base for test data and test routine
00028420                          00000000         2779+         USING PLOETEST,R5
00028420  0002844C                                 2780+T49      DC    A(CODE49)           address of test code
00028424  0031                                     2781+         DC    H'49'               test number
00028426  00                                       2782+         DC    X'00'
00028427  33                                       2783+         DC    HL1'51'             Function
00028428  63                                       2784+         DC    HL1'99'             expected unequal results (UR)
00028429  00                                       2785+         DC    HL1'00'             found UR
0002842A  63                                       2786+         DC    HL1'99'             cc
0002842B  00                                       2787+         DC    HL1'00'             cc not checked
0002842C  00                                       2788+         DS    X                   extracted cc
00028430  00000000 00000000                        2789+         DS    2F                  extract PSW after test (has CC)
00028438  00028C80                                 2790+         DC    A(FMSG49)           address of test fail message
0002843C  00028AC0                                 2791+         DC    A(OPSRC49)          address of operand source
00028440  00028A20                                 2792+         DC    A(OPDST49)          address of operand destination
00028444  00028B60                                 2793+         DC    A(PL49)             address of parameter list
00028448  000041E9                                 2794+         DC    A(RANDX+49)         address of random source data
                                                   2795
                                                   2796          TSTCODE
0002844C                                           2797+CODE49   DS    0F
                                                   2798 *                                  load random source
0002844C  E310 501C 0014                0000001C   2799          LGF   R1,OPSRCA
00028452  E320 5028 0014                00000028   2800          LGF   R2,RANDXA
00028458  E330 5024 0014                00000024   2801          LGF   R3,PLA
                                                   2802
0002845E  D27F 1000 2000      00000000  00000000   2803          MVC   0(128,R1),0(R2)
                                                   2804 *
                                                   2805 * load of op1-data, op2-data
                                                   2806 *
00028464  E321 0000 0004                00000000   2807          LG    R2,0(R1)            Get OP1 from src
0002846A  E323 0040 0024                00000040   2808          STG   R2,64(R3)           load OP1-Data in PL
                                                   2809
00028470  E321 0008 0004                00000008   2810          LG    R2,8(R1)            Get OP2 from src
00028476  E323 0048 0024                00000048   2811          STG   R2,72(R3)           load OP2-Data in PL
                                                   2812 *
                                                   2813 * Double Store(DSTG: 51)
                                                   2814 *
0002847C  E310 5020 0014                00000020   2815          LGF   R1,OPDSTA           Destination address
00028482  E330 5024 0014                00000024   2816          LGF   R3,PLA
                                                   2817
00028488  E313 0000 0024                00000000   2818          STG   R1,0(R3)            load OP1 address in PL
                                                   2819
0002848E  4111 0020                     00000020   2820          LA    R1,32(R1)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00028492  E313 0008 0024                 00000008  2821          STG   R1,8(R3)            load OP2 address in PL
                                                   2822
00028498  B982 0000                              2823          XGR   R0,R0
0002849C  4100 0033                 00000033     2824          LA    R0,51
000284A0  EE10 0000 3000  00000000  00000000     2825          PLO   R1,0,0,0(R3)        Double Store(DSTG: 51)
                                                   2826 *
                                                   2827 * compare results
                                                   2828 *
000284A6  E310 501C 0014                 0000001C  2829          LGF   R1,OPSRCA           source
000284AC  E330 5020 0014                 00000020  2830          LGF   R3,OPDSTA           Destination
                                                   2831
000284B2  D507 1000 3000  00000000  00000000     2832          CLC   0(8,R1),0(R3)
000284B8  4770 50A6                 000284C6     2833          BNE   FXX51               OP1 Different, error
000284BC  D507 1008 3020  00000008  00000020     2834          CLC   8(8,R1),32(R3)
000284C2  478B 0000                 00000000     2835          BE    0(R11)              OP2 same, return
                                                   2836
                          000284C6  00000001     2837 FXX51    EQU   *
000284C6  45F0 83E8                 000023E8     2838          BAL   R15,FAILMSG
000284CA  07FB                                   2839          BR    R11
                                                   2840
                                                   2841          TSTDATA
000284CC                  000284CC  00028A00     2842+         ORG   PLOETST+(128*K)+11*(3*K)+(K+512)
                                                   2843          TSTLBL OPDST              Operand Destination (Results)
00028A00  6E6E6ED6 D7C4E2E3                      2844+         DC    CL32'>>>OPDST49<<<' eye catcher
00028A08  F4F94C4C 4C404040
00028A10  40404040 40404040
00028A18  40404040 40404040
00028A20                                         2845+OPDST49  DS    0C
00028A20  00000000 00000000                      2846          DC    XL128'0'
00028A28  00000000 00000000
00028A30  00000000 00000000
00028A38  00000000 00000000
00028A40  00000000 00000000
00028A48  00000000 00000000
00028A50  00000000 00000000
00028A58  00000000 00000000
00028A60  00000000 00000000
00028A68  00000000 00000000
00028A70  00000000 00000000
00028A78  00000000 00000000
00028A80  00000000 00000000
00028A88  00000000 00000000
00028A90  00000000 00000000
00028A98  00000000 00000000
                                                   2847
                                                   2848          TSTLBL OPSRC              Operand Source
00028AA0  6E6E6ED6 D7E2D9C3                      2849+         DC    CL32'>>>OPSRC49<<<' eye catcher
00028AA8  F4F94C4C 4C404040
00028AB0  40404040 40404040
00028AB8  40404040 40404040
00028AC0                                         2850+OPSRC49  DS    0C
00028AC0  00000000 00000000                      2851          DC    XL128'0'
00028AC8  00000000 00000000
00028AD0  00000000 00000000
00028AD8  00000000 00000000
00028AE0  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00028AE8  00000000 00000000
00028AF0  00000000 00000000
00028AF8  00000000 00000000
00028B00  00000000 00000000
00028B08  00000000 00000000
00028B10  00000000 00000000
00028B18  00000000 00000000
00028B20  00000000 00000000
00028B28  00000000 00000000
00028B30  00000000 00000000
00028B38  00000000 00000000
                                        2852
                                        2853          TSTLBL PL                  Parameter List
00028B40  6E6E6ED7 D3F4F94C             2854+         DC    CL32'>>>PL49<<<'     eye catcher
00028B48  4C4C4040 40404040
00028B50  40404040 40404040
00028B58  40404040 40404040
00028B60                                2855+PL49     DS    0C
00028B60  00000000 00000000             2856          DC    XL256'00'
00028B68  00000000 00000000
00028B70  00000000 00000000
00028B78  00000000 00000000
00028B80  00000000 00000000
00028B88  00000000 00000000
00028B90  00000000 00000000
00028B98  00000000 00000000
00028BA0  00000000 00000000
00028BA8  00000000 00000000
00028BB0  00000000 00000000
00028BB8  00000000 00000000
00028BC0  00000000 00000000
00028BC8  00000000 00000000
00028BD0  00000000 00000000
00028BD8  00000000 00000000
00028BE0  00000000 00000000
00028BE8  00000000 00000000
00028BF0  00000000 00000000
00028BF8  00000000 00000000
00028C00  00000000 00000000
00028C08  00000000 00000000
00028C10  00000000 00000000
00028C18  00000000 00000000
00028C20  00000000 00000000
00028C28  00000000 00000000
00028C30  00000000 00000000
00028C38  00000000 00000000
00028C40  00000000 00000000
00028C48  00000000 00000000
00028C50  00000000 00000000
00028C58  00000000 00000000
                                        2857
                                        2858          TSTLBL FMSG                fail message
00028C60  6E6E6EC6 D4E2C7F4             2859+         DC    CL32'>>>FMSG49<<<'   eye catcher
00028C68  F94C4C4C 40404040
00028C70  40404040 40404040
00028C78  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00028C80                                          2860+FMSG49   DS    0C
00028C80  86818993 858440D7                       2861          DC    C'failed PLO-Extension '
00028C88  D3D660C5 A7A38595
00028C90  A2899695 40
00028C95  C496A482 938540E2                       2862          DC    CL96'Double Store(DSTX: 51)'
00028C9D  A3969985 4DC4E2E3
00028CA5  E77A40F5 F15D4040
00028CAD  40404040 40404040
00028CB5  40404040 40404040
00028CBD  40404040 40404040
00028CC5  40404040 40404040
00028CCD  40404040 40404040
00028CD5  40404040 40404040
00028CDD  40404040 40404040
00028CE5  40404040 40404040
00028CED  40404040 40404040
                                                  2863
                                                  2864          TSTEND
00028CF5  C5D5C440 968640A3                       2865+         DC    CL32'END of test: 49 <<<'
00028CFD  85A2A37A 40F4F940
00028D05  4C4C4C40 40404040
00028D0D  40404040 40404040
                                                  2866
                                                  2867 * ------------------------------------------------------------------
                                                  2868 * Test Double Store(DSTG: 50) instruction
                                                  2869 * ------------------------------------------------------------------
                                                  2870          TSTHDR 99,99,50
00028D15                      00028D15  00029000  2871+         ORG   PLOETST+(128*K)+12*(3*K)
00029000  E2E3C1D9 E3409686                       2872+         DC    CL32'START of test: 50'
00029008  40A385A2 A37A40F5
00029010  F0404040 40404040
00029018  40404040 40404040
00029020                      00029020            2873+         USING *,R5                base for test data and test routine
00029020                      00000000            2874+         USING PLOETEST,R5
00029020  0002904C                                2875+T50      DC    A(CODE50)           address of test code
00029024  0032                                    2876+         DC    H'50'               test number
00029026  00                                      2877+         DC    X'00'
00029027  32                                      2878+         DC    HL1'50'             Function
00029028  63                                      2879+         DC    HL1'99'             expected unequal results (UR)
00029029  00                                      2880+         DC    HL1'00'             found UR
0002902A  63                                      2881+         DC    HL1'99'             cc
0002902B  00                                      2882+         DC    HL1'00'             cc not checked
0002902C  00                                      2883+         DS    X                   extracted cc
00029030  00000000 00000000                       2884+         DS    2F                  extract PSW after test (has CC)
00029038  00029880                                2885+         DC    A(FMSG50)           address of test fail message
0002903C  000296C0                                2886+         DC    A(OPSRC50)          address of operand source
00029040  00029620                                2887+         DC    A(OPDST50)          address of operand destination
00029044  00029760                                2888+         DC    A(PL50)             address of parameter list
00029048  000041EA                                2889+         DC    A(RANDX+50)         address of random source data
                                                  2890
                                                  2891          TSTCODE
0002904C                                          2892+CODE50   DS    0F
                                                  2893 *                                    load random source
0002904C  E310 501C 0014            0000001C      2894          LGF   R1,OPSRCA
00029052  E320 5028 0014            00000028      2895          LGF   R2,RANDXA
00029058  E330 5024 0014            00000024      2896          LGF   R3,PLA
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    2897
0002905E  D27F 1000 2000      00000000  00000000  2898          MVC   0(128,R1),0(R2)
                                                    2899 *
                                                    2900 * load of op1-data, op2-data
                                                    2901 *
00029064  5821 0000                     00000000  2902          L     R2,0(R1)            Get OP1 from src
00029068  5023 0040                     00000040  2903          ST    R2,64(R3)           load OP1-Data in PL
                                                    2904
0002906C  5821 0008                     00000008  2905          L     R2,8(R1)            Get OP2 from src
00029070  5023 0048                     00000048  2906          ST    R2,72(R3)           load OP2-Data in PL
                                                    2907 *
                                                    2908 * Double Store(DSTG: 50
                                                    2909 *
00029074  E310 5020 0014                00000020  2910          LGF   R1,OPDSTA           Destination address
0002907A  E330 5024 0014                00000024  2911          LGF   R3,PLA
                                                    2912
00029080  E313 0000 0024                00000000  2913          STG   R1,0(R3)            load OP1 address in PL
                                                    2914
00029086  4111 0020                     00000020  2915          LA    R1,32(R1)
0002908A  E313 0008 0024                00000008  2916          STG   R1,8(R3)            load OP2 address in PL
                                                    2917
00029090  B982 0000                               2918          XGR   R0,R0
00029094  4100 0032                     00000032  2919          LA    R0,50
00029098  EE10 0000 3000      00000000  00000000  2920          PLO   R1,0,0,0(R3)        Double Store(DSTG: 50)
                                                    2921 *
                                                    2922 * compare results
                                                    2923 *
0002909E  E310 501C 0014                0000001C  2924          LGF   R1,OPSRCA           source
000290A4  E330 5020 0014                00000020  2925          LGF   R3,OPDSTA           Destination
                                                    2926
000290AA  D503 1000 3000      00000000  00000000  2927          CLC   0(4,R1),0(R3)
000290B0  4770 509E                     000290BE  2928          BNE   FXX50               OP1 Different, error
000290B4  D503 1008 3020      00000008  00000020  2929          CLC   8(4,R1),32(R3)
000290BA  478B 0000                     00000000  2930          BE    0(R11)              OP2 same, return
                                                    2931
                              000290BE  00000001  2932 FXX50    EQU   *
000290BE  45F0 83E8                     000023E8  2933          BAL   R15,FAILMSG
000290C2  07FB                                    2934          BR    R11
                                                    2935
                                                    2936          TSTDATA
000290C4                      000290C4  00029600  2937+         ORG   PLOETST+(128*K)+12*(3*K)+(K+512)
                                                    2938          TSTLBL OPDST              Operand Destination (Results)
00029600  6E6E6ED6 D7C4E2E3               00000000  2939+         DC    CL32'>>>OPDST50<<<' eye catcher
00029608  F5F04C4C 4C404040               00000008
00029610  40404040 40404040               00000010
00029618  40404040 40404040               00000018
00029620                                  00000020  2940+OPDST50  DS    0C
00029620  00000000 00000000               00000020  2941          DC    XL128'0'
00029628  00000000 00000000               00000028
00029630  00000000 00000000               00000030
00029638  00000000 00000000               00000038
00029640  00000000 00000000               00000040
00029648  00000000 00000000               00000048
00029650  00000000 00000000               00000050
00029658  00000000 00000000               00000058
00029660  00000000 00000000               00000060
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00029668  00000000 00000000
00029670  00000000 00000000
00029678  00000000 00000000
00029680  00000000 00000000
00029688  00000000 00000000
00029690  00000000 00000000
00029698  00000000 00000000
                                        2942
                                        2943          TSTLBL OPSRC               Operand Source
000296A0  6E6E6ED6 D7E2D9C3             2944+         DC    CL32'>>>OPSRC50<<<'  eye catcher
000296A8  F5F04C4C 4C404040
000296B0  40404040 40404040
000296B8  40404040 40404040
000296C0                                2945+OPSRC50  DS    0C
000296C0  00000000 00000000             2946          DC    XL128'0'
000296C8  00000000 00000000
000296D0  00000000 00000000
000296D8  00000000 00000000
000296E0  00000000 00000000
000296E8  00000000 00000000
000296F0  00000000 00000000
000296F8  00000000 00000000
00029700  00000000 00000000
00029708  00000000 00000000
00029710  00000000 00000000
00029718  00000000 00000000
00029720  00000000 00000000
00029728  00000000 00000000
00029730  00000000 00000000
00029738  00000000 00000000
                                        2947
                                        2948          TSTLBL PL                  Parameter List
00029740  6E6E6ED7 D3F5F04C             2949+         DC    CL32'>>>PL50<<<'     eye catcher
00029748  4C4C4040 40404040
00029750  40404040 40404040
00029758  40404040 40404040
00029760                                2950+PL50     DS    0C
00029760  00000000 00000000             2951          DC    XL256'00'
00029768  00000000 00000000
00029770  00000000 00000000
00029778  00000000 00000000
00029780  00000000 00000000
00029788  00000000 00000000
00029790  00000000 00000000
00029798  00000000 00000000
000297A0  00000000 00000000
000297A8  00000000 00000000
000297B0  00000000 00000000
000297B8  00000000 00000000
000297C0  00000000 00000000
000297C8  00000000 00000000
000297D0  00000000 00000000
000297D8  00000000 00000000
000297E0  00000000 00000000
000297E8  00000000 00000000
000297F0  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000297F8  00000000 00000000
00029800  00000000 00000000
00029808  00000000 00000000
00029810  00000000 00000000
00029818  00000000 00000000
00029820  00000000 00000000
00029828  00000000 00000000
00029830  00000000 00000000
00029838  00000000 00000000
00029840  00000000 00000000
00029848  00000000 00000000
00029850  00000000 00000000
00029858  00000000 00000000
                                          2952
                                          2953          TSTLBL FMSG                fail message
00029860  6E6E6EC6 D4E2C7F5               2954+         DC    CL32'>>>FMSG50<<<'   eye catcher
00029868  F04C4C4C 40404040
00029870  40404040 40404040
00029878  40404040 40404040
00029880                                  2955+FMSG50   DS    0C
00029880  86818993 858440D7               2956          DC    C'failed PLO-Extension '
00029888  D3D660C5 A7A38595
00029890  A2899695 40
00029895  C496A482 938540E2               2957          DC    CL96'Double Store(DSTG: 50)'
0002989D  A3969985 4DC4E2E3
000298A5  C77A40F5 F05D4040
000298AD  40404040 40404040
000298B5  40404040 40404040
000298BD  40404040 40404040
000298C5  40404040 40404040
000298CD  40404040 40404040
000298D5  40404040 40404040
000298DD  40404040 40404040
000298E5  40404040 40404040
000298ED  40404040 40404040
                                          2958
                                          2959          TSTEND
000298F5  C5D5C440 968640A3               2960+         DC    CL32'END of test: 50 <<<'
000298FD  85A2A37A 40F5F040
00029905  4C4C4C40 40404040
0002990D  40404040 40404040
                                          2961
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   2963 *=======================================================================
                                                   2964 *       Load / Store Test
                                                   2965 *=======================================================================
                                                   2966
                                                   2967 * -----------------------------------------------------------------
                                                   2968 * Test Load(LO: 38) / Store(STO: 49) instructions
                                                   2969 * -----------------------------------------------------------------
                                                   2970          TSTHDR 99,99,99
00029915                          00029915  00029C00 2971+         ORG   PLOETST+(128*K)+13*(3*K)
00029C00  E2E3C1D9 E3409686                        2972+         DC    CL32'START of test: 51'
00029C08  40A385A2 A37A40F5
00029C10  F1404040 40404040
00029C18  40404040 40404040
00029C20                          00029C20         2973+         USING *,R5                base for test data and test routine
00029C20                          00000000         2974+         USING PLOETEST,R5
00029C20  00029C4C                                 2975+T51      DC    A(CODE51)           address of test code
00029C24  0033                                     2976+         DC    H'51'               test number
00029C26  00                                       2977+         DC    X'00'
00029C27  63                                       2978+         DC    HL1'99'             Function
00029C28  63                                       2979+         DC    HL1'99'             expected unequal results (UR)
00029C29  00                                       2980+         DC    HL1'00'             found UR
00029C2A  63                                       2981+         DC    HL1'99'             cc
00029C2B  00                                       2982+         DC    HL1'00'             cc not checked
00029C2C  00                                       2983+         DS    X                   extracted cc
00029C30  00000000 00000000                        2984+         DS    2F                  extract PSW after test (has CC)
00029C38  0002A480                                 2985+         DC    A(FMSG51)           address of test fail message
00029C3C  0002A2C0                                 2986+         DC    A(OPSRC51)          address of operand source
00029C40  0002A220                                 2987+         DC    A(OPDST51)          address of operand destination
00029C44  0002A360                                 2988+         DC    A(PL51)             address of parameter list
00029C48  000041EB                                 2989+         DC    A(RANDX+51)         address of random source data
                                                   2990
                                                   2991          TSTCODE
00029C4C                                           2992+CODE51   DS    0F
                                                   2993 *                                  load random source
00029C4C  E310 501C 0014                0000001C   2994          LGF   R1,OPSRCA
00029C52  E320 5028 0014                00000028   2995          LGF   R2,RANDXA
00029C58  E330 5024 0014                00000024   2996          LGF   R3,PLA
                                                   2997
00029C5E  D27F 1000 2000        00000000 00000000  2998          MVC   0(128,R1),0(R2)
                                                   2999 *
                                                   3000 * Load(LO: 38)
                                                   3001 *
00029C64  E313 0000 0024                00000000   3002          STG   R1,0(R3)            load OP1 address in PL
                                                   3003
00029C6A  B982 0000                                3004          XGR   R0,R0
00029C6E  4100 0026                     00000026   3005          LA    R0,38
00029C72  EE10 0000 3000        00000000 00000000  3006          PLO   R1,0,0,0(R3)        Load(LO: 38)
                                                   3007 *
                                                   3008 * Store(STO: 49)
                                                   3009 *
00029C78  E310 5020 0014                00000020   3010          LGF   R1,OPDSTA           Destination address
00029C7E  E330 5024 0014                00000024   3011          LGF   R3,PLA
                                                   3012
00029C84  E313 0000 0024                00000000   3013          STG   R1,0(R3)            load OP1 address in PL
                                                   3014
00029C8A  B982 0000                                3015          XGR   R0,R0
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00029C8E  4100 0031                    00000031  3016          LA    R0,49
00029C92  EE10 0000 3000      00000000  00000000  3017          PLO   R1,0,0,0(R3)        Store(STO: 49)
                                                  3018 *
                                                  3019 * compare results
                                                  3020 *
00029C98  E310 501C 0014               0000001C  3021          LGF   R1,OPSRCA           source
00029C9E  E330 5020 0014               00000020  3022          LGF   R3,OPDSTA           Destination
                                                  3023
00029CA4  D51F 1000 3000      00000000  00000000  3024          CLC   0(32,R1),0(R3)
00029CAA  478B 0000                    00000000  3025          BE    0(R11)              OP1 same, return
                                                  3026
00029CAE  45F0 83E8                    000023E8  3027          BAL   R15,FAILMSG
00029CB2  07FB                                   3028          BR    R11
                                                  3029
                                                  3030          TSTDATA
00029CB4                      00029CB4  0002A200  3031+         ORG   PLOETST+(128*K)+13*(3*K)+(K+512)
                                                  3032          TSTLBL OPDST              Operand Destination (Results)
0002A200  6E6E6ED6 D7C4E2E3                       3033+         DC    CL32'>>>OPDST51<<<' eye catcher
0002A208  F5F14C4C 4C404040
0002A210  40404040 40404040
0002A218  40404040 40404040
0002A220                                          3034+OPDST51  DS    0C
0002A220  00000000 00000000                       3035          DC    XL128'0'
0002A228  00000000 00000000
0002A230  00000000 00000000
0002A238  00000000 00000000
0002A240  00000000 00000000
0002A248  00000000 00000000
0002A250  00000000 00000000
0002A258  00000000 00000000
0002A260  00000000 00000000
0002A268  00000000 00000000
0002A270  00000000 00000000
0002A278  00000000 00000000
0002A280  00000000 00000000
0002A288  00000000 00000000
0002A290  00000000 00000000
0002A298  00000000 00000000
                                                  3036
                                                  3037          TSTLBL OPSRC              Operand Source
0002A2A0  6E6E6ED6 D7E2D9C3                       3038+         DC    CL32'>>>OPSRC51<<<' eye catcher
0002A2A8  F5F14C4C 4C404040
0002A2B0  40404040 40404040
0002A2B8  40404040 40404040
0002A2C0                                          3039+OPSRC51  DS    0C
0002A2C0  00000000 00000000                       3040          DC    XL128'0'
0002A2C8  00000000 00000000
0002A2D0  00000000 00000000
0002A2D8  00000000 00000000
0002A2E0  00000000 00000000
0002A2E8  00000000 00000000
0002A2F0  00000000 00000000
0002A2F8  00000000 00000000
0002A300  00000000 00000000
0002A308  00000000 00000000
0002A310  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0002A318  00000000 00000000
0002A320  00000000 00000000
0002A328  00000000 00000000
0002A330  00000000 00000000
0002A338  00000000 00000000
                                       3041
                                       3042          TSTLBL PL                  Parameter List
0002A340  6E6E6ED7 D3F5F14C            3043+         DC    CL32'>>>PL51<<<'     eye catcher
0002A348  4C4C4040 40404040
0002A350  40404040 40404040
0002A358  40404040 40404040
0002A360                               3044+PL51     DS    0C
0002A360  00000000 00000000            3045          DC    XL256'00'
0002A368  00000000 00000000
0002A370  00000000 00000000
0002A378  00000000 00000000
0002A380  00000000 00000000
0002A388  00000000 00000000
0002A390  00000000 00000000
0002A398  00000000 00000000
0002A3A0  00000000 00000000
0002A3A8  00000000 00000000
0002A3B0  00000000 00000000
0002A3B8  00000000 00000000
0002A3C0  00000000 00000000
0002A3C8  00000000 00000000
0002A3D0  00000000 00000000
0002A3D8  00000000 00000000
0002A3E0  00000000 00000000
0002A3E8  00000000 00000000
0002A3F0  00000000 00000000
0002A3F8  00000000 00000000
0002A400  00000000 00000000
0002A408  00000000 00000000
0002A410  00000000 00000000
0002A418  00000000 00000000
0002A420  00000000 00000000
0002A428  00000000 00000000
0002A430  00000000 00000000
0002A438  00000000 00000000
0002A440  00000000 00000000
0002A448  00000000 00000000
0002A450  00000000 00000000
0002A458  00000000 00000000
                                       3046
                                       3047          TSTLBL FMSG                fail message
0002A460  6E6E6EC6 D4E2C7F5            3048+         DC    CL32'>>>FMSG51<<<'   eye catcher
0002A468  F14C4C4C 40404040
0002A470  40404040 40404040
0002A478  40404040 40404040
0002A480                               3049+FMSG51   DS    0C
0002A480  86818993 858440D7            3050          DC    C'failed PLO-Extension '
0002A488  D3D660C5 A7A38595
0002A490  A2899695 40
0002A495  D3968184 4DD3D67A            3051          DC    CL96'Load(LO: 38) / Store(STO: 49)'
0002A49D  40F3F85D 406140E2
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0002A4A5  A3969985 4DE2E3D6
0002A4AD  7A40F4F9 5D404040
0002A4B5  40404040 40404040
0002A4BD  40404040 40404040
0002A4C5  40404040 40404040
0002A4CD  40404040 40404040
0002A4D5  40404040 40404040
0002A4DD  40404040 40404040
0002A4E5  40404040 40404040
0002A4ED  40404040 40404040
                                          3052
                                          3053          TSTEND
0002A4F5  C5D5C440 968640A3               3054+         DC    CL32'END of test: 51 <<<'
0002A4FD  85A2A37A 40F5F140
0002A505  4C4C4C40 40404040
0002A50D  40404040 40404040
                                          3055
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   3057 *============================================================================
                                                   3058 *       Quadruple Compare and Swap Tests
                                                   3059 *============================================================================
                                                   3060
                                                   3061 * ----------------------------------------------------------------------
                                                   3062 * Test Quadruple Compare and Swap (QCSO: 37) instruction : 1st NE
                                                   3063 * ----------------------------------------------------------------------
                                                   3064          TSTHDR 2,1,37             1st Operand NOT Equal (cc=2, UR=1)
0002A515                          0002A515  0002A800 3065+         ORG   PLOETST+(128*K)+14*(3*K)
0002A800  E2E3C1D9 E3409686                        3066+         DC    CL32'START of test: 52'
0002A808  40A385A2 A37A40F5
0002A810  F2404040 40404040
0002A818  40404040 40404040
0002A820                          0002A820         3067+         USING *,R5                base for test data and test routine
0002A820                          00000000         3068+         USING PLOETEST,R5
0002A820  0002A84C                                 3069+T52      DC    A(CODE52)           address of test code
0002A824  0034                                     3070+         DC    H'52'               test number
0002A826  00                                       3071+         DC    X'00'
0002A827  25                                       3072+         DC    HL1'37'             Function
0002A828  01                                       3073+         DC    HL1'1'              expected unequal results (UR)
0002A829  00                                       3074+         DC    HL1'00'             found UR
0002A82A  02                                       3075+         DC    HL1'2'              cc
0002A82B  0D                                       3076+         DC    HL1'13'             not expected CC mask
0002A82C  00                                       3077+         DS    X                   extracted cc
0002A830  00000000 00000000                        3078+         DS    2F                  extract PSW after test (has CC)
0002A838  0002B1E0                                 3079+         DC    A(FMSG52)           address of test fail message
0002A83C  0002AF40                                 3080+         DC    A(OPSRC52)          address of operand source
0002A840  0002AE20                                 3081+         DC    A(OPDST52)          address of operand destination
0002A844  0002B060                                 3082+         DC    A(PL52)             address of parameter list
0002A848  000041EC                                 3083+         DC    A(RANDX+52)         address of random source data
                                                   3084
                                                   3085          TSTCODE
0002A84C                                           3086+CODE52   DS    0F
                                                   3087 *                                  load random source
0002A84C  E310 501C 0014                0000001C   3088          LGF   R1,OPSRCA
0002A852  E320 5028 0014                00000028   3089          LGF   R2,RANDXA
0002A858  E330 5024 0014                00000024   3090          LGF   R3,PLA
0002A85E  E340 5020 0014                00000020   3091          LGF   R4,OPREPA
                                                   3092 *
                                                   3093 * Quadruple Compare and Swap (QCSO: 37)
                                                   3094 * Build Parambter List
                                                   3095 *
                                                   3096
0002A864  D27F 1000 2000  00000000      00000000   3097          MVC   0(128,R1),0(R2)     load Op1-Op4 data in source
0002A86A  D27F 3040 1000  00000040      00000000   3098          MVC   64(128,R3),0(R1)    copy to PL OP1-Op4 Compare
                                                   3099
0002A870  D27F 4000 2080  00000000      00000080   3100          MVC   0(128,R4),128(R2)   load initial Op1-Op4 data
0002A876  D27F 30C0 4000  000000C0      00000000   3101          MVC   192(128,R3),0(R4)   copy to PL Op1-Op4 Replacement
                                                   3102 ****
                                                   3103 * force a failure by changing the first operand
                                                   3104 ****
0002A87C  D207 1000 4000  00000000      00000000   3105          MVC   0(8,R1),0(R4)
                                                   3106 ****
                                                   3107
0002A882  E313 0000 0024                00000000   3108          STG   R1,0(R3)            load OP1 address in PL
                                                   3109
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002A888  4111 0020                    00000020  3110          LA    R1,32(R1)
0002A88C  E313 0008 0024               00000008  3111          STG   R1,8(R3)            load OP2 address in PL
                                                 3112
0002A892  4111 0020                    00000020  3113          LA    R1,32(R1)
0002A896  E313 0010 0024               00000010  3114          STG   R1,16(R3)           load OP3 address in PL
                                                 3115
0002A89C  4111 0020                    00000020  3116          LA    R1,32(R1)
0002A8A0  E313 0018 0024               00000018  3117          STG   R1,24(R3)           load OP4 address in PL
                                                 3118
0002A8A6  B982 0000                              3119          XGR   R0,R0
0002A8AA  4100 0025                    00000025  3120          LA    R0,37
0002A8AE  EE00 0000 3000  00000000     00000000  3121          PLO   0,0,0,0(R3)         Quadruple C/S (QCSO: 37)
                                                 3122
0002A8B4  B98D 0020                              3123          EPSW  R2,R0               extract psw
0002A8B8  5020 5010                    00000010  3124          ST    R2,CCPSW              to save CC
0002A8BC  E310 500B 0076               0000000B  3125          LB    R1,CCMASK           (failure CC mask)
0002A8C2  8910 0004                    00000004  3126          SLL   R1,4                (shift to BC instr CC position)
0002A8C6  4410 5104                    0002A924  3127          EX    R1,F371TSTCC            fail if...
                                                 3128
                          0002A8CA     00000001  3129 F371CKUR EQU   *
0002A8CA  D200 5009 3030  00000009     00000030  3130          MVC   URFOUND,48(R3)      Extract UR
0002A8D0  D500 5008 3030  00000008     00000030  3131          CLC   UR,48(R3)
0002A8D6  4780 50BE                    0002A8DE  3132          BE    F371CKRES
0002A8DA  45F0 82C8                    000022C8  3133          BAL   R15,URMSG
                                                 3134 *
                                                 3135 * compare results
                                                 3136 *
                          0002A8DE     00000001  3137 F371CKRES EQU   *
                                                 3138 *                                reset adresses
0002A8DE  E310 501C 0014               0000001C  3139          LGF   R1,OPSRCA
0002A8E4  E320 5028 0014               00000028  3140          LGF   R2,RANDXA
0002A8EA  E330 5024 0014               00000024  3141          LGF   R3,PLA
0002A8F0  E340 5020 0014               00000020  3142          LGF   R4,OPREPA
                                                 3143
0002A8F6  D51F 1000 3040  00000000     00000040  3144          CLC   0(32,R1),64(R3)
0002A8FC  4770 50FE                    0002A91E  3145          BNE   FXX137              OP1C Different, error
0002A900  D51F 1020 3060  00000020     00000060  3146          CLC   32(32,R1),96(R3)
0002A906  4770 50FE                    0002A91E  3147          BNE   FXX137              OP2C Different, error
0002A90A  D51F 1040 3080  00000040     00000080  3148          CLC   64(32,R1),128(R3)
0002A910  4770 50FE                    0002A91E  3149          BNE   FXX137              OP3C Different, error
0002A914  D51F 1060 30A0  00000060     000000A0  3150          CLC   96(32,R1),160(R3)
0002A91A  478B 0000                    00000000  3151          BE    0(R11)              OP4Csame, return
                                                 3152
                          0002A91E     00000001  3153 FXX137    EQU   *
0002A91E  45F0 83E8                    000023E8  3154          BAL   R15,FAILMSG
0002A922  07FB                                   3155          BR    R11
                                                 3156
0002A924  4700 5108                    0002A928  3157 F371TSTCC BC    0,F371CCMSG         fail if unexpected condition code
                          0002A928     00000001  3158 F371CCMSG EQU   *
0002A928  45F0 8196                    00002196  3159          BAL   R15,CCMSG
0002A92C  47F0 50AA                    0002A8CA  3160          B     F371CKUR
                                                 3161
                                                 3162          TSTDATA
0002A930                  0002A930     0002AE00  3163+         ORG   PLOETST+(128*K)+14*(3*K)+(K+512)
                                                 3164          TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
0002AE00  6E6E6ED6 D7C4E2E3                      3165+         DC    CL32'>>>OPDST52<<<' eye catcher
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0002AE08  F5F24C4C 4C404040
0002AE10  40404040 40404040
0002AE18  40404040 40404040
0002AE20                                           3166+OPDST52  DS    0C
0002AE20  00000000 00000000                        3167          DC    XL256'0'
0002AE28  00000000 00000000
0002AE30  00000000 00000000
0002AE38  00000000 00000000
0002AE40  00000000 00000000
0002AE48  00000000 00000000
0002AE50  00000000 00000000
0002AE58  00000000 00000000
0002AE60  00000000 00000000
0002AE68  00000000 00000000
0002AE70  00000000 00000000
0002AE78  00000000 00000000
0002AE80  00000000 00000000
0002AE88  00000000 00000000
0002AE90  00000000 00000000
0002AE98  00000000 00000000
0002AEA0  00000000 00000000
0002AEA8  00000000 00000000
0002AEB0  00000000 00000000
0002AEB8  00000000 00000000
0002AEC0  00000000 00000000
0002AEC8  00000000 00000000
0002AED0  00000000 00000000
0002AED8  00000000 00000000
0002AEE0  00000000 00000000
0002AEE8  00000000 00000000
0002AEF0  00000000 00000000
0002AEF8  00000000 00000000
0002AF00  00000000 00000000
0002AF08  00000000 00000000
0002AF10  00000000 00000000
0002AF18  00000000 00000000
                                                   3168
                                                   3169          TSTLBL OPSRC              Source: OP1-OP4
0002AF20  6E6E6ED6 D7E2D9C3                        3170+         DC    CL32'>>>OPSRC52<<<' eye catcher
0002AF28  F5F24C4C 4C404040
0002AF30  40404040 40404040
0002AF38  40404040 40404040
0002AF40                                           3171+OPSRC52  DS    0C
0002AF40  00000000 00000000                        3172          DC    XL256'0'
0002AF48  00000000 00000000
0002AF50  00000000 00000000
0002AF58  00000000 00000000
0002AF60  00000000 00000000
0002AF68  00000000 00000000
0002AF70  00000000 00000000
0002AF78  00000000 00000000
0002AF80  00000000 00000000
0002AF88  00000000 00000000
0002AF90  00000000 00000000
0002AF98  00000000 00000000
0002AFA0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0002AFA8  00000000 00000000
0002AFB0  00000000 00000000
0002AFB8  00000000 00000000
0002AFC0  00000000 00000000
0002AFC8  00000000 00000000
0002AFD0  00000000 00000000
0002AFD8  00000000 00000000
0002AFE0  00000000 00000000
0002AFE8  00000000 00000000
0002AFF0  00000000 00000000
0002AFF8  00000000 00000000
0002B000  00000000 00000000
0002B008  00000000 00000000
0002B010  00000000 00000000
0002B018  00000000 00000000
0002B020  00000000 00000000
0002B028  00000000 00000000
0002B030  00000000 00000000
0002B038  00000000 00000000
                                        3173
                                        3174          TSTLBL PL                   Parameter List
0002B040  6E6E6ED7 D3F5F24C             3175+         DC    CL32'>>>PL52<<<'      eye catcher
0002B048  4C4C4040 40404040
0002B050  40404040 40404040
0002B058  40404040 40404040
0002B060                                3176+PL52     DS    0C
0002B060  00000000 00000000             3177          DC    XL352'00'
0002B068  00000000 00000000
0002B070  00000000 00000000
0002B078  00000000 00000000
0002B080  00000000 00000000
0002B088  00000000 00000000
0002B090  00000000 00000000
0002B098  00000000 00000000
0002B0A0  00000000 00000000
0002B0A8  00000000 00000000
0002B0B0  00000000 00000000
0002B0B8  00000000 00000000
0002B0C0  00000000 00000000
0002B0C8  00000000 00000000
0002B0D0  00000000 00000000
0002B0D8  00000000 00000000
0002B0E0  00000000 00000000
0002B0E8  00000000 00000000
0002B0F0  00000000 00000000
0002B0F8  00000000 00000000
0002B100  00000000 00000000
0002B108  00000000 00000000
0002B110  00000000 00000000
0002B118  00000000 00000000
0002B120  00000000 00000000
0002B128  00000000 00000000
0002B130  00000000 00000000
0002B138  00000000 00000000
0002B140  00000000 00000000
0002B148  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002B150  00000000 00000000
0002B158  00000000 00000000
0002B160  00000000 00000000
0002B168  00000000 00000000
0002B170  00000000 00000000
0002B178  00000000 00000000
0002B180  00000000 00000000
0002B188  00000000 00000000
0002B190  00000000 00000000
0002B198  00000000 00000000
0002B1A0  00000000 00000000
0002B1A8  00000000 00000000
0002B1B0  00000000 00000000
0002B1B8  00000000 00000000
                                        3178
                                        3179          TSTLBL FMSG                fail message
0002B1C0  6E6E6EC6 D4E2C7F5             3180+         DC    CL32'>>>FMSG52<<<'   eye catcher
0002B1C8  F24C4C4C 40404040
0002B1D0  40404040 40404040
0002B1D8  40404040 40404040
0002B1E0                                3181+FMSG52   DS    0C
0002B1E0  86818993 858440D7             3182          DC    C'failed PLO-Extension '
0002B1E8  D3D660C5 A7A38595
0002B1F0  A2899695 40
0002B1F5  D8A48184 99A49793             3183          DC    CL96'Quadruple Compare and Swap (QCSO: 37): 1st NE'
0002B1FD  8540C396 94978199
0002B205  85408195 8440E2A6
0002B20D  8197404D D8C3E2D6
0002B215  7A40F3F7 5D7A40F1
0002B21D  A2A340D5 C5404040
0002B225  40404040 40404040
0002B22D  40404040 40404040
0002B235  40404040 40404040
0002B23D  40404040 40404040
0002B245  40404040 40404040
0002B24D  40404040 40404040
                                        3184
                                        3185          TSTEND
0002B255  C5D5C440 968640A3             3186+         DC    CL32'END of test: 52 <<<'
0002B25D  85A2A37A 40F5F240
0002B265  4C4C4C40 40404040
0002B26D  40404040 40404040
                                        3187
                                        3188 * ------------------------------------------------------------------
                                        3189 * Test Quadruple Compare and Swap (QCSO: 37) instruction : 2nd NE
                                        3190 * ------------------------------------------------------------------
                                        3191          TSTHDR 2,2,37              2nd Operand NOT Equal (cc=2, UR=2)
0002B275                  0002B275  0002B400  3192+         ORG   PLOETST+(128*K)+15*(3*K)
0002B400  E2E3C1D9 E3409686             3193+         DC    CL32'START of test: 53'
0002B408  40A385A2 A37A40F5
0002B410  F3404040 40404040
0002B418  40404040 40404040
0002B420                  0002B420      3194+         USING *,R5                 base for test data and test routine
0002B420                  00000000      3195+         USING PLOETEST,R5
0002B420  0002B44C                      3196+T53      DC    A(CODE53)            address of test code
0002B424  0035                          3197+         DC    H'53'                test number
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002B426  00                                       3198+          DC    X'00'
0002B427  25                                       3199+          DC    HL1'37'             Function
0002B428  02                                       3200+          DC    HL1'2'              expected unequal results (UR)
0002B429  00                                       3201+          DC    HL1'00'             found UR
0002B42A  02                                       3202+          DC    HL1'2'              cc
0002B42B  0D                                       3203+          DC    HL1'13'             not expected CC mask
0002B42C  00                                       3204+          DS    X                   extracted cc
0002B430  00000000 00000000                        3205+          DS    2F                  extract PSW after test (has CC)
0002B438  0002BDE0                                 3206+          DC    A(FMSG53)           address of test fail message
0002B43C  0002BB40                                 3207+          DC    A(OPSRC53)          address of operand source
0002B440  0002BA20                                 3208+          DC    A(OPDST53)          address of operand destination
0002B444  0002BC60                                 3209+          DC    A(PL53)             address of parameter list
0002B448  000041ED                                 3210+          DC    A(RANDX+53)         address of random source data
                                                   3211
                                                   3212           TSTCODE
0002B44C                                           3213+CODE53    DS    0F
                                                   3214 *                                   load random source
0002B44C  E310 501C 0014                0000001C   3215           LGF   R1,OPSRCA
0002B452  E320 5028 0014                00000028   3216           LGF   R2,RANDXA
0002B458  E330 5024 0014                00000024   3217           LGF   R3,PLA
0002B45E  E340 5020 0014                00000020   3218           LGF   R4,OPREPA
                                                   3219 *
                                                   3220 * Quadruple Compare and Swap (QCSO: 37)
                                                   3221 * Build Parambter List
                                                   3222 *
0002B464  D27F 1000 2000      00000000  00000000   3223           MVC   0(128,R1),0(R2)     load Op1-Op4 data in source
0002B46A  D27F 3040 1000      00000040  00000000   3224           MVC   64(128,R3),0(R1)    copy to PL OP1-Op4 Compare
                                                   3225
0002B470  D27F 4000 2080      00000000  00000080   3226           MVC   0(128,R4),128(R2)   load initial Op1-Op4 data
0002B476  D27F 30C0 4000      000000C0  00000000   3227           MVC   192(128,R3),0(R4)   copy to PL Op1-Op4 Replacement
                                                   3228 ****
                                                   3229 * force a failure by changing the 2nd operand
                                                   3230 ****
0002B47C  D207 1020 4000      00000020  00000000   3231           MVC   32(8,R1),0(R4)
                                                   3232 ****
                                                   3233
0002B482  E313 0000 0024                00000000   3234           STG   R1,0(R3)            load OP1 address in PL
                                                   3235
0002B488  4111 0020                     00000020   3236           LA    R1,32(R1)
0002B48C  E313 0008 0024                00000008   3237           STG   R1,8(R3)            load OP2 address in PL
                                                   3238
0002B492  4111 0020                     00000020   3239           LA    R1,32(R1)
0002B496  E313 0010 0024                00000010   3240           STG   R1,16(R3)           load OP3 address in PL
                                                   3241
0002B49C  4111 0020                     00000020   3242           LA    R1,32(R1)
0002B4A0  E313 0018 0024                00000018   3243           STG   R1,24(R3)           load OP4 address in PL
                                                   3244
0002B4A6  B982 0000                                3245           XGR   R0,R0
0002B4AA  4100 0025                     00000025   3246           LA    R0,37
0002B4AE  EE00 0000 3000      00000000  00000000   3247           PLO   0,0,0,0(R3)         Quadruple C/S (QCSO: 37)
                                                   3248
0002B4B4  B98D 0020                                3249           EPSW  R2,R0               extract psw
0002B4B8  5020 5010                     00000010   3250           ST    R2,CCPSW              to save CC
0002B4BC  E310 500B 0076                0000000B   3251           LB    R1,CCMASK           (failure CC mask)
0002B4C2  8910 0004                     00000004   3252           SLL   R1,4                (shift to BC instr CC position)
0002B4C6  4410 5104                     0002B524   3253           EX    R1,F372TSTCC            fail if...
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    3254
                               0002B4CA  00000001  3255 F372CKUR  EQU   *
0002B4CA  D200 5009 3030       00000009  00000030  3256           MVC   URFOUND,48(R3)      Extract UR
0002B4D0  D500 5008 3030       00000008  00000030  3257           CLC   UR,48(R3)
0002B4D6  4780 50BE                      0002B4DE  3258           BE    F372CKRES
0002B4DA  45F0 82C8                      000022C8  3259           BAL   R15,URMSG
                                                    3260 *
                                                    3261 * compare results
                                                    3262 *
                               0002B4DE  00000001  3263 F372CKRES EQU   *
                                                    3264 *                               reset adresses
0002B4DE  E310 501C 0014                 0000001C  3265           LGF   R1,OPSRCA
0002B4E4  E320 5028 0014                 00000028  3266           LGF   R2,RANDXA
0002B4EA  E330 5024 0014                 00000024  3267           LGF   R3,PLA
0002B4F0  E340 5020 0014                 00000020  3268           LGF   R4,OPREPA
                                                    3269
0002B4F6  D51F 1000 3040       00000000  00000040  3270           CLC   0(32,R1),64(R3)
0002B4FC  4770 50FE                      0002B51E  3271           BNE   FXX237                 OP1C Different, error
0002B500  D51F 1020 3060       00000020  00000060  3272           CLC   32(32,R1),96(R3)
0002B506  4770 50FE                      0002B51E  3273           BNE   FXX237                 OP2C Different, error
0002B50A  D51F 1040 3080       00000040  00000080  3274           CLC   64(32,R1),128(R3)
0002B510  4770 50FE                      0002B51E  3275           BNE   FXX237                 OP3C Different, error
0002B514  D51F 1060 30A0       00000060  000000A0  3276           CLC   96(32,R1),160(R3)
0002B51A  478B 0000                      00000000  3277           BE    0(R11)                 OP4Csame, return
                                                    3278
                               0002B51E  00000001  3279 FXX237    EQU   *
0002B51E  45F0 83E8                      000023E8  3280           BAL   R15,FAILMSG
0002B522  07FB                                     3281           BR    R11
                                                    3282
0002B524  4700 5108                      0002B528  3283 F372TSTCC BC     0,F372CCMSG          fail if unexpected condition code
                               0002B528  00000001  3284 F372CCMSG EQU   *
0002B528  45F0 8196                      00002196  3285           BAL   R15,CCMSG
0002B52C  47F0 50AA                      0002B4CA  3286           B     F372CKUR
                                                    3287
                                                    3288           TSTDATA
0002B530                       0002B530  0002BA00  3289+          ORG   PLOETST+(128*K)+15*(3*K)+(K+512)
                                                    3290           TSTLBL OPDST                 Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
0002BA00  6E6E6ED6 D7C4E2E3                         3291+          DC    CL32'>>>OPDST53<<<'    eye catcher
0002BA08  F5F34C4C 4C404040
0002BA10  40404040 40404040
0002BA18  40404040 40404040
0002BA20                                            3292+OPDST53   DS    0C
0002BA20  00000000 00000000                         3293           DC    XL256'0'
0002BA28  00000000 00000000
0002BA30  00000000 00000000
0002BA38  00000000 00000000
0002BA40  00000000 00000000
0002BA48  00000000 00000000
0002BA50  00000000 00000000
0002BA58  00000000 00000000
0002BA60  00000000 00000000
0002BA68  00000000 00000000
0002BA70  00000000 00000000
0002BA78  00000000 00000000
0002BA80  00000000 00000000
0002BA88  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0002BA90  00000000 00000000
0002BA98  00000000 00000000
0002BAA0  00000000 00000000
0002BAA8  00000000 00000000
0002BAB0  00000000 00000000
0002BAB8  00000000 00000000
0002BAC0  00000000 00000000
0002BAC8  00000000 00000000
0002BAD0  00000000 00000000
0002BAD8  00000000 00000000
0002BAE0  00000000 00000000
0002BAE8  00000000 00000000
0002BAF0  00000000 00000000
0002BAF8  00000000 00000000
0002BB00  00000000 00000000
0002BB08  00000000 00000000
0002BB10  00000000 00000000
0002BB18  00000000 00000000                          3294
                                                     3295          TSTLBL OPSRC              Source: OP1-OP4
0002BB20  6E6E6ED6 D7E2D9C3                          3296+         DC    CL32'>>>OPSRC53<<<' eye catcher
0002BB28  F5F34C4C 4C404040
0002BB30  40404040 40404040
0002BB38  40404040 40404040
0002BB40                                             3297+OPSRC53  DS    0C
0002BB40  00000000 00000000                          3298          DC    XL256'0'
0002BB48  00000000 00000000
0002BB50  00000000 00000000
0002BB58  00000000 00000000
0002BB60  00000000 00000000
0002BB68  00000000 00000000
0002BB70  00000000 00000000
0002BB78  00000000 00000000
0002BB80  00000000 00000000
0002BB88  00000000 00000000
0002BB90  00000000 00000000
0002BB98  00000000 00000000
0002BBA0  00000000 00000000
0002BBA8  00000000 00000000
0002BBB0  00000000 00000000
0002BBB8  00000000 00000000
0002BBC0  00000000 00000000
0002BBC8  00000000 00000000
0002BBD0  00000000 00000000
0002BBD8  00000000 00000000
0002BBE0  00000000 00000000
0002BBE8  00000000 00000000
0002BBF0  00000000 00000000
0002BBF8  00000000 00000000
0002BC00  00000000 00000000
0002BC08  00000000 00000000
0002BC10  00000000 00000000
0002BC18  00000000 00000000
0002BC20  00000000 00000000
0002BC28  00000000 00000000
0002BC30  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002BC38  00000000 00000000
                                                   3299
                                                   3300          TSTLBL PL                 Parameter List
0002BC40  6E6E6ED7 D3F5F34C                        3301+         DC    CL32'>>>PL53<<<'    eye catcher
0002BC48  4C4C4040 40404040
0002BC50  40404040 40404040
0002BC58  40404040 40404040
0002BC60                                           3302+PL53     DS    0C
0002BC60  00000000 00000000                        3303          DC    XL352'00'
0002BC68  00000000 00000000
0002BC70  00000000 00000000
0002BC78  00000000 00000000
0002BC80  00000000 00000000
0002BC88  00000000 00000000
0002BC90  00000000 00000000
0002BC98  00000000 00000000
0002BCA0  00000000 00000000
0002BCA8  00000000 00000000
0002BCB0  00000000 00000000
0002BCB8  00000000 00000000
0002BCC0  00000000 00000000
0002BCC8  00000000 00000000
0002BCD0  00000000 00000000
0002BCD8  00000000 00000000
0002BCE0  00000000 00000000
0002BCE8  00000000 00000000
0002BCF0  00000000 00000000
0002BCF8  00000000 00000000
0002BD00  00000000 00000000
0002BD08  00000000 00000000
0002BD10  00000000 00000000
0002BD18  00000000 00000000
0002BD20  00000000 00000000
0002BD28  00000000 00000000
0002BD30  00000000 00000000
0002BD38  00000000 00000000
0002BD40  00000000 00000000
0002BD48  00000000 00000000
0002BD50  00000000 00000000
0002BD58  00000000 00000000
0002BD60  00000000 00000000
0002BD68  00000000 00000000
0002BD70  00000000 00000000
0002BD78  00000000 00000000
0002BD80  00000000 00000000
0002BD88  00000000 00000000
0002BD90  00000000 00000000
0002BD98  00000000 00000000
0002BDA0  00000000 00000000
0002BDA8  00000000 00000000
0002BDB0  00000000 00000000
0002BDB8  00000000 00000000
                                                   3304
                                                   3305          TSTLBL FMSG               fail message
0002BDC0  6E6E6EC6 D4E2C7F5                        3306+         DC    CL32'>>>FMSG53<<<'  eye catcher
0002BDC8  F34C4C4C 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002BDD0  40404040 40404040
0002BDD8  40404040 40404040
0002BDE0                                         3307+FMSG53   DS    0C
0002BDE0  86818993 858440D7                      3308          DC    C'failed PLO-Extension '
0002BDE8  D3D660C5 A7A38595
0002BDF0  A2899695 40
0002BDF5  D8A48184 99A49793                      3309          DC    CL96'Quadruple Compare and Swap (QCSO: 37): 2nd NE'
0002BDFD  8540C396 94978199
0002BE05  85408195 8440E2A6
0002BE0D  8197404D D8C3E2D6
0002BE15  7A40F3F7 5D7A40F2
0002BE1D  958440D5 C5404040
0002BE25  40404040 40404040
0002BE2D  40404040 40404040
0002BE35  40404040 40404040
0002BE3D  40404040 40404040
0002BE45  40404040 40404040
0002BE4D  40404040 40404040
                                                 3310
                                                 3311          TSTEND
0002BE55  C5D5C440 968640A3                      3312+         DC    CL32'END of test: 53 <<<'
0002BE5D  85A2A37A 40F5F340
0002BE65  4C4C4C40 40404040
0002BE6D  40404040 40404040
                                                 3313
                                                 3314 * ------------------------------------------------------------------
                                                 3315 * Test Quadruple Compare and Swap (QCSO: 37) instruction : 3rd NE
                                                 3316 * ------------------------------------------------------------------
                                                 3317          TSTHDR 2,3,37             3rd Operand NOT Equal (cc=2, UR=3)
0002BE75                          0002BE75  0002C000 3318+         ORG   PLOETST+(128*K)+16*(3*K)
0002C000  E2E3C1D9 E3409686                      3319+         DC    CL32'START of test: 54'
0002C008  40A385A2 A37A40F5
0002C010  F4404040 40404040
0002C018  40404040 40404040
0002C020                          0002C020       3320+         USING *,R5                base for test data and test routine
0002C020                          00000000       3321+         USING PLOETEST,R5
0002C020  0002C04C                               3322+T54      DC    A(CODE54)           address of test code
0002C024  0036                                   3323+         DC    H'54'               test number
0002C026  00                                     3324+         DC    X'00'
0002C027  25                                     3325+         DC    HL1'37'             Function
0002C028  03                                     3326+         DC    HL1'3'              expected unequal results (UR)
0002C029  00                                     3327+         DC    HL1'00'             found UR
0002C02A  02                                     3328+         DC    HL1'2'              cc
0002C02B  0D                                     3329+         DC    HL1'13'             not expected CC mask
0002C02C  00                                     3330+         DS    X                   extracted cc
0002C030  00000000 00000000                      3331+         DS    2F                  extract PSW after test (has CC)
0002C038  0002C9E0                               3332+         DC    A(FMSG54)           address of test fail message
0002C03C  0002C740                               3333+         DC    A(OPSRC54)          address of operand source
0002C040  0002C620                               3334+         DC    A(OPDST54)          address of operand destination
0002C044  0002C860                               3335+         DC    A(PL54)             address of parameter list
0002C048  000041EE                               3336+         DC    A(RANDX+54)         address of random source data
                                                 3337
                                                 3338          TSTCODE
0002C04C                                         3339+CODE54   DS    0F
                                                 3340 *                                  load random source
0002C04C  E310 501C 0014                0000001C 3341          LGF   R1,OPSRCA
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002C052  E320 5028 0014                 00000028  3342          LGF   R2,RANDXA
0002C058  E330 5024 0014                 00000024  3343          LGF   R3,PLA
0002C05E  E340 5020 0014                 00000020  3344          LGF   R4,OPREPA
                                                   3345 *
                                                   3346 * Quadruple Compare and Swap (QCSO: 37)
                                                   3347 * Build Parambter List
                                                   3348 *
0002C064  D27F 1000 2000      00000000   00000000  3349          MVC   0(128,R1),0(R2)     load Op1-Op4 data in source
0002C06A  D27F 3040 1000      00000040   00000000  3350          MVC   64(128,R3),0(R1)    copy to PL OP1-Op4 Compare
                                                   3351
0002C070  D27F 4000 2080      00000000   00000080  3352          MVC   0(128,R4),128(R2)   load initial Op1-Op4 data
0002C076  D27F 30C0 4000      000000C0   00000000  3353          MVC   192(128,R3),0(R4)   copy to PL Op1-Op4 Replacement
                                                   3354 ****
                                                   3355 * force a failure by changing the 3nd operand
                                                   3356 ****
0002C07C  D207 1040 4000      00000040   00000000  3357          MVC   64(8,R1),0(R4)
                                                   3358 ****
                                                   3359
0002C082  E313 0000 0024                 00000000  3360          STG   R1,0(R3)            load OP1 address in PL
                                                   3361
0002C088  4111 0020                      00000020  3362          LA    R1,32(R1)
0002C08C  E313 0008 0024                 00000008  3363          STG   R1,8(R3)            load OP2 address in PL
                                                   3364
0002C092  4111 0020                      00000020  3365          LA    R1,32(R1)
0002C096  E313 0010 0024                 00000010  3366          STG   R1,16(R3)           load OP3 address in PL
                                                   3367
0002C09C  4111 0020                      00000020  3368          LA    R1,32(R1)
0002C0A0  E313 0018 0024                 00000018  3369          STG   R1,24(R3)           load OP4 address in PL
                                                   3370
0002C0A6  B982 0000                                3371          XGR   R0,R0
0002C0AA  4100 0025                      00000025  3372          LA    R0,37
0002C0AE  EE00 0000 3000      00000000   00000000  3373          PLO   0,0,0,0(R3)         Quadruple C/S (QCSO: 37)
                                                   3374
0002C0B4  B98D 0020                                3375          EPSW  R2,R0               extract psw
0002C0B8  5020 5010                      00000010  3376          ST    R2,CCPSW              to save CC
0002C0BC  E310 500B 0076                 0000000B  3377          LB    R1,CCMASK           (failure CC mask)
0002C0C2  8910 0004                      00000004  3378          SLL   R1,4                (shift to BC instr CC position)
0002C0C6  4410 5104                      0002C124  3379          EX    R1,F373TSTCC          fail if...
                                                   3380
                                0002C0CA   00000001  3381 F373CKUR  EQU   *
0002C0CA  D200 5009 3030      00000009   00000030  3382          MVC   URFOUND,48(R3)      Extract UR
0002C0D0  D500 5008 3030      00000008   00000030  3383          CLC   UR,48(R3)
0002C0D6  4780 50BE                      0002C0DE  3384          BE    F373CKRES
0002C0DA  45F0 82C8                      000022C8  3385          BAL   R15,URMSG
                                                   3386 *
                                                   3387 * compare results
                                                   3388 *
                                0002C0DE   00000001  3389 F373CKRES EQU   *
                                                   3390 *                            reset adresses
0002C0DE  E310 501C 0014                 0000001C  3391          LGF   R1,OPSRCA
0002C0E4  E320 5028 0014                 00000028  3392          LGF   R2,RANDXA
0002C0EA  E330 5024 0014                 00000024  3393          LGF   R3,PLA
0002C0F0  E340 5020 0014                 00000020  3394          LGF   R4,OPREPA
                                                   3395
0002C0F6  D51F 1000 3040      00000000   00000040  3396          CLC   0(32,R1),64(R3)
0002C0FC  4770 50FE                      0002C11E  3397          BNE   F373XX              OP1C Different, error
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002C100  D51F 1020 3060      00000020  00000060  3398           CLC   32(32,R1),96(R3)
0002C106  4770 50FE                     0002C11E  3399           BNE   F373XX               OP2C Different, error
0002C10A  D51F 1040 3080      00000040  00000080  3400           CLC   64(32,R1),128(R3)
0002C110  4770 50FE                     0002C11E  3401           BNE   F373XX               OP3C Different, error
0002C114  D51F 1060 30A0      00000060  000000A0  3402           CLC   96(32,R1),160(R3)
0002C11A  478B 0000                     00000000  3403           BE    0(R11)               OP4Csame, return
                                                  3404
                              0002C11E  00000001  3405 F373XX    EQU   *
0002C11E  45F0 83E8                     000023E8  3406           BAL   R15,FAILMSG
0002C122  07FB                                    3407           BR    R11
                                                  3408
0002C124  4700 5108                     0002C128  3409 F373TSTCC BC     0,F373CCMSG          fail if unexpected condition code
                              0002C128  00000001  3410 F373CCMSG EQU   *
0002C128  45F0 8196                     00002196  3411           BAL   R15,CCMSG
0002C12C  47F0 50AA                     0002C0CA  3412           B     F373CKUR
                                                  3413
                                                  3414           TSTDATA
0002C130                      0002C130  0002C600  3415+          ORG   PLOETST+(128*K)+16*(3*K)+(K+512)
                                                  3416           TSTLBL OPDST               Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
0002C600  6E6E6ED6 D7C4E2E3                       3417+          DC    CL32'>>>OPDST54<<<'  eye catcher
0002C608  F5F44C4C 4C404040
0002C610  40404040 40404040
0002C618  40404040 40404040
0002C620                                          3418+OPDST54   DS    0C
0002C620  00000000 00000000                       3419           DC    XL256'0'
0002C628  00000000 00000000
0002C630  00000000 00000000
0002C638  00000000 00000000
0002C640  00000000 00000000
0002C648  00000000 00000000
0002C650  00000000 00000000
0002C658  00000000 00000000
0002C660  00000000 00000000
0002C668  00000000 00000000
0002C670  00000000 00000000
0002C678  00000000 00000000
0002C680  00000000 00000000
0002C688  00000000 00000000
0002C690  00000000 00000000
0002C698  00000000 00000000
0002C6A0  00000000 00000000
0002C6A8  00000000 00000000
0002C6B0  00000000 00000000
0002C6B8  00000000 00000000
0002C6C0  00000000 00000000
0002C6C8  00000000 00000000
0002C6D0  00000000 00000000
0002C6D8  00000000 00000000
0002C6E0  00000000 00000000
0002C6E8  00000000 00000000
0002C6F0  00000000 00000000
0002C6F8  00000000 00000000
0002C700  00000000 00000000
0002C708  00000000 00000000
0002C710  00000000 00000000
0002C718  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    3420
                                                    3421           TSTLBL OPSRC               Source: OP1-OP4
0002C720  6E6E6ED6 D7E2D9C3                        3422+          DC    CL32'>>>OPSRC54<<<' eye catcher
0002C728  F5F44C4C 4C404040
0002C730  40404040 40404040
0002C738  40404040 40404040
0002C740                                           3423+OPSRC54   DS    0C
0002C740  00000000 00000000                        3424           DC    XL256'0'
0002C748  00000000 00000000
0002C750  00000000 00000000
0002C758  00000000 00000000
0002C760  00000000 00000000
0002C768  00000000 00000000
0002C770  00000000 00000000
0002C778  00000000 00000000
0002C780  00000000 00000000
0002C788  00000000 00000000
0002C790  00000000 00000000
0002C798  00000000 00000000
0002C7A0  00000000 00000000
0002C7A8  00000000 00000000
0002C7B0  00000000 00000000
0002C7B8  00000000 00000000
0002C7C0  00000000 00000000
0002C7C8  00000000 00000000
0002C7D0  00000000 00000000
0002C7D8  00000000 00000000
0002C7E0  00000000 00000000
0002C7E8  00000000 00000000
0002C7F0  00000000 00000000
0002C7F8  00000000 00000000
0002C800  00000000 00000000
0002C808  00000000 00000000
0002C810  00000000 00000000
0002C818  00000000 00000000
0002C820  00000000 00000000
0002C828  00000000 00000000
0002C830  00000000 00000000
0002C838  00000000 00000000
                                                    3425
                                                    3426           TSTLBL PL                  Parameter List
0002C840  6E6E6ED7 D3F5F44C                        3427+          DC    CL32'>>>PL54<<<'    eye catcher
0002C848  4C4C4040 40404040
0002C850  40404040 40404040
0002C858  40404040 40404040
0002C860                                           3428+PL54      DS    0C
0002C860  00000000 00000000                        3429           DC    XL352'00'
0002C868  00000000 00000000
0002C870  00000000 00000000
0002C878  00000000 00000000
0002C880  00000000 00000000
0002C888  00000000 00000000
0002C890  00000000 00000000
0002C898  00000000 00000000
0002C8A0  00000000 00000000
0002C8A8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0002C8B0  00000000 00000000
0002C8B8  00000000 00000000
0002C8C0  00000000 00000000
0002C8C8  00000000 00000000
0002C8D0  00000000 00000000
0002C8D8  00000000 00000000
0002C8E0  00000000 00000000
0002C8E8  00000000 00000000
0002C8F0  00000000 00000000
0002C8F8  00000000 00000000
0002C900  00000000 00000000
0002C908  00000000 00000000
0002C910  00000000 00000000
0002C918  00000000 00000000
0002C920  00000000 00000000
0002C928  00000000 00000000
0002C930  00000000 00000000
0002C938  00000000 00000000
0002C940  00000000 00000000
0002C948  00000000 00000000
0002C950  00000000 00000000
0002C958  00000000 00000000
0002C960  00000000 00000000
0002C968  00000000 00000000
0002C970  00000000 00000000
0002C978  00000000 00000000
0002C980  00000000 00000000
0002C988  00000000 00000000
0002C990  00000000 00000000
0002C998  00000000 00000000
0002C9A0  00000000 00000000
0002C9A8  00000000 00000000
0002C9B0  00000000 00000000
0002C9B8  00000000 00000000
                                        3430
                                        3431          TSTLBL FMSG                fail message
0002C9C0  6E6E6EC6 D4E2C7F5             3432+         DC    CL32'>>>FMSG54<<<'   eye catcher
0002C9C8  F44C4C4C 40404040
0002C9D0  40404040 40404040
0002C9D8  40404040 40404040
0002C9E0                                3433+FMSG54   DS    0C
0002C9E0  86818993 858440D7             3434          DC    C'failed PLO-Extension '
0002C9E8  D3D660C5 A7A38595
0002C9F0  A2899695 40
0002C9F5  D8A48184 99A49793             3435          DC    CL96'Quadruple Compare and Swap (QCSO: 37): 3rd NE'
0002C9FD  8540C396 94978199
0002CA05  85408195 8440E2A6
0002CA0D  8197404D D8C3E2D6
0002CA15  7A40F3F7 5D7A40F3
0002CA1D  998440D5 C5404040
0002CA25  40404040 40404040
0002CA2D  40404040 40404040
0002CA35  40404040 40404040
0002CA3D  40404040 40404040
0002CA45  40404040 40404040
0002CA4D  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    3436
                                                    3437          TSTEND
0002CA55  C5D5C440 968640A3                         3438+         DC    CL32'END of test: 54 <<<'
0002CA5D  85A2A37A 40F5F440
0002CA65  4C4C4C40 40404040
0002CA6D  40404040 40404040
                                                    3439
                                                    3440 * ------------------------------------------------------------------
                                                    3441 * Test Quadruple Compare and Swap (QCSO: 37) instruction : 4th NE
                                                    3442 * ------------------------------------------------------------------
                                                    3443          TSTHDR 2,4,37             4th Operand NOT Equal (cc=2, UR=4)
0002CA75                          0002CA75  0002CC00  3444+         ORG   PLOETST+(128*K)+17*(3*K)
0002CC00  E2E3C1D9 E3409686                         3445+         DC    CL32'START of test: 55'
0002CC08  40A385A2 A37A40F5
0002CC10  F5404040 40404040
0002CC18  40404040 40404040
0002CC20                          0002CC20          3446+         USING *,R5                base for test data and test routine
0002CC20                          00000000          3447+         USING PLOETEST,R5
0002CC20  0002CC4C                                  3448+T55      DC    A(CODE55)           address of test code
0002CC24  0037                                      3449+         DC    H'55'               test number
0002CC26  00                                        3450+         DC    X'00'
0002CC27  25                                        3451+         DC    HL1'37'             Function
0002CC28  04                                        3452+         DC    HL1'4'              expected unequal results (UR)
0002CC29  00                                        3453+         DC    HL1'00'             found UR
0002CC2A  02                                        3454+         DC    HL1'2'              cc
0002CC2B  0D                                        3455+         DC    HL1'13'             not expected CC mask
0002CC2C  00                                        3456+         DS    X                   extracted cc
0002CC30  00000000 00000000                         3457+         DS    2F                  extract PSW after test (has CC)
0002CC38  0002D5E0                                  3458+         DC    A(FMSG55)           address of test fail message
0002CC3C  0002D340                                  3459+         DC    A(OPSRC55)          address of operand source
0002CC40  0002D220                                  3460+         DC    A(OPDST55)          address of operand destination
0002CC44  0002D460                                  3461+         DC    A(PL55)             address of parameter list
0002CC48  000041EF                                  3462+         DC    A(RANDX+55)         address of random source data
                                                    3463
                                                    3464          TSTCODE
0002CC4C                                            3465+CODE55   DS    0F
                                                    3466 *                                  load random source
0002CC4C  E310 501C 0014                 0000001C   3467          LGF   R1,OPSRCA
0002CC52  E320 5028 0014                 00000028   3468          LGF   R2,RANDXA
0002CC58  E330 5024 0014                 00000024   3469          LGF   R3,PLA
0002CC5E  E340 5020 0014                 00000020   3470          LGF   R4,OPREPA
                                                    3471 *
                                                    3472 * Quadruple Compare and Swap (QCSO: 37)
                                                    3473 * Build Parambter List
                                                    3474 *
0002CC64  D27F 1000 2000       00000000  00000000   3475          MVC   0(128,R1),0(R2)     load Op1-Op4 data in source
0002CC6A  D27F 3040 1000       00000040  00000000   3476          MVC   64(128,R3),0(R1)    copy to PL OP1-Op4 Compare
                                                    3477
0002CC70  D27F 4000 2080       00000000  00000080   3478          MVC   0(128,R4),128(R2)   load initial Op1-Op4 data
0002CC76  D27F 30C0 4000       000000C0  00000000   3479          MVC   192(128,R3),0(R4)   copy to PL Op1-Op4 Replacement
                                                    3480 ****
                                                    3481 * force a failure by changing the 4th operand
                                                    3482 ****
0002CC7C  D207 1060 4000       00000060  00000000   3483          MVC   96(8,R1),0(R4)
                                                    3484 ****
                                                    3485
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002CC82  E313 0000 0024                 00000000  3486           STG   R1,0(R3)            load OP1 address in PL
                                                   3487
0002CC88  4111 0020                      00000020  3488           LA    R1,32(R1)
0002CC8C  E313 0008 0024                 00000008  3489           STG   R1,8(R3)            load OP2 address in PL
                                                   3490
0002CC92  4111 0020                      00000020  3491           LA    R1,32(R1)
0002CC96  E313 0010 0024                 00000010  3492           STG   R1,16(R3)           load OP3 address in PL
                                                   3493
0002CC9C  4111 0020                      00000020  3494           LA    R1,32(R1)
0002CCA0  E313 0018 0024                 00000018  3495           STG   R1,24(R3)           load OP4 address in PL
                                                   3496
0002CCA6  B982 0000                                3497           XGR   R0,R0
0002CCAA  4100 0025                      00000025  3498           LA    R0,37
0002CCAE  EE00 0000 3000       00000000  00000000  3499           PLO   0,0,0,0(R3)         Quadruple C/S (QCSO: 37)
                                                   3500
0002CCB4  B98D 0020                                3501           EPSW  R2,R0               extract psw
0002CCB8  5020 5010                      00000010  3502           ST    R2,CCPSW              to save CC
0002CCBC  E310 500B 0076                 0000000B  3503           LB    R1,CCMASK           (failure CC mask)
0002CCC2  8910 0004                      00000004  3504           SLL   R1,4                (shift to BC instr CC position)
0002CCC6  4410 5104                      0002CD24  3505           EX    R1,F374TSTCC          fail if...
                                                   3506
                                0002CCCA  00000001  3507 F374CKUR  EQU   *
0002CCCA  D200 5009 3030       00000009  00000030  3508           MVC   URFOUND,48(R3)      Extract UR
0002CCD0  D500 5008 3030       00000008  00000030  3509           CLC   UR,48(R3)
0002CCD6  4780 50BE                      0002CCDE  3510           BE    F374CKRES
0002CCDA  45F0 82C8                      000022C8  3511           BAL   R15,URMSG
                                                   3512 *
                                                   3513 * compare results
                                                   3514 *
                                0002CCDE  00000001  3515 F374CKRES EQU   *
                                                   3516 *                                  reset adresses
0002CCDE  E310 501C 0014                 0000001C  3517           LGF   R1,OPSRCA
0002CCE4  E320 5028 0014                 00000028  3518           LGF   R2,RANDXA
0002CCEA  E330 5024 0014                 00000024  3519           LGF   R3,PLA
0002CCF0  E340 5020 0014                 00000020  3520           LGF   R4,OPREPA
                                                   3521
0002CCF6  D51F 1000 3040       00000000  00000040  3522           CLC   0(32,R1),64(R3)
0002CCFC  4770 50FE                      0002CD1E  3523           BNE   F374XX              OP1C Different, error
0002CD00  D51F 1020 3060       00000020  00000060  3524           CLC   32(32,R1),96(R3)
0002CD06  4770 50FE                      0002CD1E  3525           BNE   F374XX              OP2C Different, error
0002CD0A  D51F 1040 3080       00000040  00000080  3526           CLC   64(32,R1),128(R3)
0002CD10  4770 50FE                      0002CD1E  3527           BNE   F374XX              OP3C Different, error
0002CD14  D51F 1060 30A0       00000060  000000A0  3528           CLC   96(32,R1),160(R3)
0002CD1A  478B 0000                      00000000  3529           BE    0(R11)              OP4Csame, return
                                                   3530
                                0002CD1E  00000001  3531 F374XX   EQU   *
0002CD1E  45F0 83E8                      000023E8  3532           BAL   R15,FAILMSG
0002CD22  07FB                                     3533           BR    R11
                                                   3534
0002CD24  4700 5108                      0002CD28  3535 F374TSTCC BC    0,F374CCMSG         fail if unexpected condition code
                                0002CD28  00000001  3536 F374CCMSG EQU   *
0002CD28  45F0 8196                      00002196  3537           BAL   R15,CCMSG
0002CD2C  47F0 50AA                      0002CCCA  3538           B     F374CKUR
                                                   3539
                                                   3540           TSTDATA
0002CD30                        0002CD30  0002D200  3541+          ORG   PLOETST+(128*K)+17*(3*K)+(K+512)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   3542           TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
0002D200  6E6E6ED6 D7C4E2E3                       3543+          DC    CL32'>>>OPDST55<<<' eye catcher
0002D208  F5F54C4C 4C404040
0002D210  40404040 40404040
0002D218  40404040 40404040
0002D220                                          3544+OPDST55   DS    0C
0002D220  00000000 00000000                       3545           DC    XL256'0'
0002D228  00000000 00000000
0002D230  00000000 00000000
0002D238  00000000 00000000
0002D240  00000000 00000000
0002D248  00000000 00000000
0002D250  00000000 00000000
0002D258  00000000 00000000
0002D260  00000000 00000000
0002D268  00000000 00000000
0002D270  00000000 00000000
0002D278  00000000 00000000
0002D280  00000000 00000000
0002D288  00000000 00000000
0002D290  00000000 00000000
0002D298  00000000 00000000
0002D2A0  00000000 00000000
0002D2A8  00000000 00000000
0002D2B0  00000000 00000000
0002D2B8  00000000 00000000
0002D2C0  00000000 00000000
0002D2C8  00000000 00000000
0002D2D0  00000000 00000000
0002D2D8  00000000 00000000
0002D2E0  00000000 00000000
0002D2E8  00000000 00000000
0002D2F0  00000000 00000000
0002D2F8  00000000 00000000
0002D300  00000000 00000000
0002D308  00000000 00000000
0002D310  00000000 00000000
0002D318  00000000 00000000
                                                   3546
                                                   3547           TSTLBL OPSRC              Source: OP1-OP4
0002D320  6E6E6ED6 D7E2D9C3                       3548+          DC    CL32'>>>OPSRC55<<<' eye catcher
0002D328  F5F54C4C 4C404040
0002D330  40404040 40404040
0002D338  40404040 40404040
0002D340                                          3549+OPSRC55   DS    0C
0002D340  00000000 00000000                       3550           DC    XL256'0'
0002D348  00000000 00000000
0002D350  00000000 00000000
0002D358  00000000 00000000
0002D360  00000000 00000000
0002D368  00000000 00000000
0002D370  00000000 00000000
0002D378  00000000 00000000
0002D380  00000000 00000000
0002D388  00000000 00000000
0002D390  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0002D398  00000000 00000000
0002D3A0  00000000 00000000
0002D3A8  00000000 00000000
0002D3B0  00000000 00000000
0002D3B8  00000000 00000000
0002D3C0  00000000 00000000
0002D3C8  00000000 00000000
0002D3D0  00000000 00000000
0002D3D8  00000000 00000000
0002D3E0  00000000 00000000
0002D3E8  00000000 00000000
0002D3F0  00000000 00000000
0002D3F8  00000000 00000000
0002D400  00000000 00000000
0002D408  00000000 00000000
0002D410  00000000 00000000
0002D418  00000000 00000000
0002D420  00000000 00000000
0002D428  00000000 00000000
0002D430  00000000 00000000
0002D438  00000000 00000000
                                      3551
                                      3552          TSTLBL PL                   Parameter List
0002D440  6E6E6ED7 D3F5F54C           3553+         DC    CL32'>>>PL55<<<'      eye catcher
0002D448  4C4C4040 40404040
0002D450  40404040 40404040
0002D458  40404040 40404040
0002D460                              3554+PL55     DS    0C
0002D460  00000000 00000000           3555          DC    XL352'00'
0002D468  00000000 00000000
0002D470  00000000 00000000
0002D478  00000000 00000000
0002D480  00000000 00000000
0002D488  00000000 00000000
0002D490  00000000 00000000
0002D498  00000000 00000000
0002D4A0  00000000 00000000
0002D4A8  00000000 00000000
0002D4B0  00000000 00000000
0002D4B8  00000000 00000000
0002D4C0  00000000 00000000
0002D4C8  00000000 00000000
0002D4D0  00000000 00000000
0002D4D8  00000000 00000000
0002D4E0  00000000 00000000
0002D4E8  00000000 00000000
0002D4F0  00000000 00000000
0002D4F8  00000000 00000000
0002D500  00000000 00000000
0002D508  00000000 00000000
0002D510  00000000 00000000
0002D518  00000000 00000000
0002D520  00000000 00000000
0002D528  00000000 00000000
0002D530  00000000 00000000
0002D538  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002D540  00000000 00000000
0002D548  00000000 00000000
0002D550  00000000 00000000
0002D558  00000000 00000000
0002D560  00000000 00000000
0002D568  00000000 00000000
0002D570  00000000 00000000
0002D578  00000000 00000000
0002D580  00000000 00000000
0002D588  00000000 00000000
0002D590  00000000 00000000
0002D598  00000000 00000000
0002D5A0  00000000 00000000
0002D5A8  00000000 00000000
0002D5B0  00000000 00000000
0002D5B8  00000000 00000000
                                        3556
                                        3557          TSTLBL FMSG                fail message
0002D5C0  6E6E6EC6 D4E2C7F5             3558+         DC    CL32'>>>FMSG55<<<'   eye catcher
0002D5C8  F54C4C4C 40404040
0002D5D0  40404040 40404040
0002D5D8  40404040 40404040
0002D5E0                                3559+FMSG55   DS    0C
0002D5E0  86818993 858440D7             3560          DC    C'failed PLO-Extension '
0002D5E8  D3D660C5 A7A38595
0002D5F0  A2899695 40
0002D5F5  D8A48184 99A49793             3561          DC    CL96'Quadruple Compare and Swap (QCSO: 37): 4th NE'
0002D5FD  8540C396 94978199
0002D605  85408195 8440E2A6
0002D60D  8197404D D8C3E2D6
0002D615  7A40F3F7 5D7A40F4
0002D61D  A38840D5 C5404040
0002D625  40404040 40404040
0002D62D  40404040 40404040
0002D635  40404040 40404040
0002D63D  40404040 40404040
0002D645  40404040 40404040
0002D64D  40404040 40404040
                                        3562
                                        3563          TSTEND
0002D655  C5D5C440 968640A3             3564+         DC    CL32'END of test: 55 <<<'
0002D65D  85A2A37A 40F5F540
0002D665  4C4C4C40 40404040
0002D66D  40404040 40404040
                                        3565
                                        3566
                                        3567 * ------------------------------------------------------------------
                                        3568 * Test Quadruple Compare and Swap (QCSO: 37) instruction : All EQ
                                        3569 * ------------------------------------------------------------------
                                        3570          TSTHDR 0,99,37             All Equal (cc=0, UR= unchanged)
0002D675                    0002D675  0002D800  3571+         ORG   PLOETST+(128*K)+18*(3*K)
0002D800  E2E3C1D9 E3409686             3572+         DC    CL32'START of test: 56'
0002D808  40A385A2 A37A40F5
0002D810  F6404040 40404040
0002D818  40404040 40404040
0002D820                    0002D820    3573+         USING *,R5                 base for test data and test routine
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0002D820                     00000000            3574+         USING PLOETEST,R5
0002D820  0002D84C                               3575+T56      DC    A(CODE56)           address of test code
0002D824  0038                                   3576+         DC    H'56'               test number
0002D826  00                                     3577+         DC    X'00'
0002D827  25                                     3578+         DC    HL1'37'             Function
0002D828  63                                     3579+         DC    HL1'99'             expected unequal results (UR)
0002D829  00                                     3580+         DC    HL1'00'             found UR
0002D82A  00                                     3581+         DC    HL1'0'              cc
0002D82B  07                                     3582+         DC    HL1'7'              not expected CC mask
0002D82C  00                                     3583+         DS    X                   extracted cc
0002D830  00000000 00000000                      3584+         DS    2F                  extract PSW after test (has CC)
0002D838  0002E1E0                               3585+         DC    A(FMSG56)           address of test fail message
0002D83C  0002DF40                               3586+         DC    A(OPSRC56)          address of operand source
0002D840  0002DE20                               3587+         DC    A(OPDST56)          address of operand destination
0002D844  0002E060                               3588+         DC    A(PL56)             address of parameter list
0002D848  000041F0                               3589+         DC    A(RANDX+56)         address of random source data
                                                 3590
                                                 3591          TSTCODE
0002D84C                                         3592+CODE56   DS    0F
                                                 3593 *                                  load random source
0002D84C  E310 501C 0014           0000001C      3594          LGF   R1,OPSRCA
0002D852  E320 5028 0014           00000028      3595          LGF   R2,RANDXA
0002D858  E330 5024 0014           00000024      3596          LGF   R3,PLA
0002D85E  E340 5020 0014           00000020      3597          LGF   R4,OPREPA
                                                 3598 *
                                                 3599 * Quadruple Compare and Swap (QCSO: 37)
                                                 3600 * Build Parambter List
                                                 3601 *
0002D864  D27F 1000 2000  00000000 00000000      3602          MVC   0(128,R1),0(R2)     load Op1-Op4 data in source
0002D86A  D27F 3040 1000  00000040 00000000      3603          MVC   64(128,R3),0(R1)    copy to PL OP1-Op4 Compare
                                                 3604
0002D870  D27F 4000 2080  00000000 00000080      3605          MVC   0(128,R4),128(R2)   load initial Op1R-Op4R data
0002D876  D27F 30C0 4000  000000C0 00000000      3606          MVC   192(128,R3),0(R4)   copy to PL Op1-Op4 Replacement
                                                 3607 ****
                                                 3608 * Do NOT force a failure by changing an operand
                                                 3609 ****
                                                 3610 *         MVC   96(8,R1),0(R4)
                                                 3611 ****
                                                 3612
0002D87C  E313 0000 0024           00000000      3613          STG   R1,0(R3)            load OP1 address in PL
                                                 3614
0002D882  4111 0020                00000020      3615          LA    R1,32(R1)
0002D886  E313 0008 0024           00000008      3616          STG   R1,8(R3)            load OP2 address in PL
                                                 3617
0002D88C  4111 0020                00000020      3618          LA    R1,32(R1)
0002D890  E313 0010 0024           00000010      3619          STG   R1,16(R3)           load OP3 address in PL
                                                 3620
0002D896  4111 0020                00000020      3621          LA    R1,32(R1)
0002D89A  E313 0018 0024           00000018      3622          STG   R1,24(R3)           load OP4 address in PL
                                                 3623
0002D8A0  B982 0000                              3624          XGR   R0,R0
0002D8A4  4100 0025                00000025      3625          LA    R0,37
0002D8A8  EE00 0000 3000  00000000 00000000      3626          PLO   0,0,0,0(R3)         Quadruple C/S (QCSO: 37)
                                                 3627
0002D8AE  B98D 0020                              3628          EPSW  R2,R0               extract psw
0002D8B2  5020 5010                00000010      3629          ST    R2,CCPSW              to save CC
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002D8B6  E310 500B 0076                 0000000B  3630          LB    R1,CCMASK           (failure CC mask)
0002D8BC  8910 0004                      00000004  3631          SLL   R1,4                (shift to BC instr CC position)
0002D8C0  4410 50FE                      0002D91E  3632          EX    R1,F375TSTCC            fail if...
                                                   3633
                               0002D8C4  00000001  3634 F375CKUR  EQU   *
0002D8C4  D200 5009 3030       00000009  00000030  3635          MVC   URFOUND,48(R3)      Extract UR
0002D8CA  D500 5008 3030       00000008  00000030  3636          CLC   UR,48(R3)
0002D8D0  4780 50B8                      0002D8D8  3637          BE    F375CKRES
0002D8D4  45F0 82C8                      000022C8  3638          BAL   R15,URMSG
                                                   3639 *
                                                   3640 * compare results
                                                   3641 *
                               0002D8D8  00000001  3642 F375CKRES EQU   *
                                                   3643 *                                  reset adresses
0002D8D8  E310 501C 0014                 0000001C  3644          LGF   R1,OPSRCA
0002D8DE  E320 5028 0014                 00000028  3645          LGF   R2,RANDXA
0002D8E4  E330 5024 0014                 00000024  3646          LGF   R3,PLA
0002D8EA  E340 5020 0014                 00000020  3647          LGF   R4,OPREPA
                                                   3648
                                                   3649 * All equal; replacement values copied to operand source
                                                   3650
0002D8F0  D51F 1000 30C0       00000000  000000C0  3651          CLC   0(32,R1),192(R3)
0002D8F6  4770 50F8                      0002D918  3652          BNE   F375XX              OP1R Different, error
0002D8FA  D51F 1020 30E0       00000020  000000E0  3653          CLC   32(32,R1),224(R3)
0002D900  4770 50F8                      0002D918  3654          BNE   F375XX              OP2R Different, error
0002D904  D51F 1040 3100       00000040  00000100  3655          CLC   64(32,R1),256(R3)
0002D90A  4770 50F8                      0002D918  3656          BNE   F375XX              OP3R Different, error
0002D90E  D51F 1060 3120       00000060  00000120  3657          CLC   96(32,R1),288(R3)
0002D914  478B 0000                      00000000  3658          BE    0(R11)              OP4R same, return
                                                   3659
                               0002D918  00000001  3660 F375XX   EQU   *
0002D918  45F0 83E8                      000023E8  3661          BAL   R15,FAILMSG
0002D91C  07FB                                     3662          BR    R11
                                                   3663
0002D91E  4700 5102                      0002D922  3664 F375TSTCC BC    0,F375CCMSG         fail if unexpected condition code
                               0002D922  00000001  3665 F375CCMSG EQU   *
0002D922  45F0 8196                      00002196  3666          BAL   R15,CCMSG
0002D926  47F0 50A4                      0002D8C4  3667          B     F375CKUR
                                                   3668
                                                   3669          TSTDATA
0002D92A                       0002D92A  0002DE00  3670+         ORG   PLOETST+(128*K)+18*(3*K)+(K+512)
                                                   3671          TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
0002DE00  6E6E6ED6 D7C4E2E3                        3672+         DC    CL32'>>>OPDST56<<<' eye catcher
0002DE08  F5F64C4C 4C404040
0002DE10  40404040 40404040
0002DE18  40404040 40404040
0002DE20                                           3673+OPDST56  DS    0C
0002DE20  00000000 00000000                        3674          DC    XL256'0'
0002DE28  00000000 00000000
0002DE30  00000000 00000000
0002DE38  00000000 00000000
0002DE40  00000000 00000000
0002DE48  00000000 00000000
0002DE50  00000000 00000000
0002DE58  00000000 00000000
0002DE60  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0002DE68  00000000 00000000
0002DE70  00000000 00000000
0002DE78  00000000 00000000
0002DE80  00000000 00000000
0002DE88  00000000 00000000
0002DE90  00000000 00000000
0002DE98  00000000 00000000
0002DEA0  00000000 00000000
0002DEA8  00000000 00000000
0002DEB0  00000000 00000000
0002DEB8  00000000 00000000
0002DEC0  00000000 00000000
0002DEC8  00000000 00000000
0002DED0  00000000 00000000
0002DED8  00000000 00000000
0002DEE0  00000000 00000000
0002DEE8  00000000 00000000
0002DEF0  00000000 00000000
0002DEF8  00000000 00000000
0002DF00  00000000 00000000
0002DF08  00000000 00000000
0002DF10  00000000 00000000
0002DF18  00000000 00000000
                                        3675
                                        3676          TSTLBL OPSRC              Source: OP1-OP4
0002DF20  6E6E6ED6 D7E2D9C3             3677+         DC    CL32'>>>OPSRC56<<<' eye catcher
0002DF28  F5F64C4C 4C404040
0002DF30  40404040 40404040
0002DF38  40404040 40404040
0002DF40                                3678+OPSRC56  DS    0C
0002DF40  00000000 00000000             3679          DC    XL256'0'
0002DF48  00000000 00000000
0002DF50  00000000 00000000
0002DF58  00000000 00000000
0002DF60  00000000 00000000
0002DF68  00000000 00000000
0002DF70  00000000 00000000
0002DF78  00000000 00000000
0002DF80  00000000 00000000
0002DF88  00000000 00000000
0002DF90  00000000 00000000
0002DF98  00000000 00000000
0002DFA0  00000000 00000000
0002DFA8  00000000 00000000
0002DFB0  00000000 00000000
0002DFB8  00000000 00000000
0002DFC0  00000000 00000000
0002DFC8  00000000 00000000
0002DFD0  00000000 00000000
0002DFD8  00000000 00000000
0002DFE0  00000000 00000000
0002DFE8  00000000 00000000
0002DFF0  00000000 00000000
0002DFF8  00000000 00000000
0002E000  00000000 00000000
0002E008  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0002E010  00000000 00000000
0002E018  00000000 00000000
0002E020  00000000 00000000
0002E028  00000000 00000000
0002E030  00000000 00000000
0002E038  00000000 00000000
                                       3680
                                       3681          TSTLBL PL                  Parameter List
0002E040  6E6E6ED7 D3F5F64C            3682+         DC    CL32'>>>PL56<<<'     eye catcher
0002E048  4C4C4040 40404040
0002E050  40404040 40404040
0002E058  40404040 40404040
0002E060                               3683+PL56     DS    0C
0002E060  00000000 00000000            3684          DC    XL352'00'
0002E068  00000000 00000000
0002E070  00000000 00000000
0002E078  00000000 00000000
0002E080  00000000 00000000
0002E088  00000000 00000000
0002E090  00000000 00000000
0002E098  00000000 00000000
0002E0A0  00000000 00000000
0002E0A8  00000000 00000000
0002E0B0  00000000 00000000
0002E0B8  00000000 00000000
0002E0C0  00000000 00000000
0002E0C8  00000000 00000000
0002E0D0  00000000 00000000
0002E0D8  00000000 00000000
0002E0E0  00000000 00000000
0002E0E8  00000000 00000000
0002E0F0  00000000 00000000
0002E0F8  00000000 00000000
0002E100  00000000 00000000
0002E108  00000000 00000000
0002E110  00000000 00000000
0002E118  00000000 00000000
0002E120  00000000 00000000
0002E128  00000000 00000000
0002E130  00000000 00000000
0002E138  00000000 00000000
0002E140  00000000 00000000
0002E148  00000000 00000000
0002E150  00000000 00000000
0002E158  00000000 00000000
0002E160  00000000 00000000
0002E168  00000000 00000000
0002E170  00000000 00000000
0002E178  00000000 00000000
0002E180  00000000 00000000
0002E188  00000000 00000000
0002E190  00000000 00000000
0002E198  00000000 00000000
0002E1A0  00000000 00000000
0002E1A8  00000000 00000000
0002E1B0  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002E1B8  00000000 00000000                          3685
                                                     3686          TSTLBL FMSG                fail message
0002E1C0  6E6E6EC6 D4E2C7F5                          3687+         DC    CL32'>>>FMSG56<<<'   eye catcher
0002E1C8  F64C4C4C 40404040
0002E1D0  40404040 40404040
0002E1D8  40404040 40404040
0002E1E0                                             3688+FMSG56   DS    0C
0002E1E0  86818993 858440D7                          3689          DC    C'failed PLO-Extension '
0002E1E8  D3D660C5 A7A38595
0002E1F0  A2899695 40
0002E1F5  D8A48184 99A49793                          3690          DC    CL96'Quadruple Compare and Swap (QCSO: 37): All EQ'
0002E1FD  8540C396 94978199
0002E205  85408195 8440E2A6
0002E20D  8197404D D8C3E2D6
0002E215  7A40F3F7 5D7A40C1
0002E21D  939340C5 D8404040
0002E225  40404040 40404040
0002E22D  40404040 40404040
0002E235  40404040 40404040
0002E23D  40404040 40404040
0002E245  40404040 40404040
0002E24D  40404040 40404040
                                                     3691
                                                     3692          TSTEND
0002E255  C5D5C440 968640A3                          3693+         DC    CL32'END of test: 56 <<<'
0002E25D  85A2A37A 40F5F640
0002E265  4C4C4C40 40404040
0002E26D  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    3695 *=============================================================================
                                                    3696 *       Quadruple Compare and Swap Tests
                                                    3697 *=============================================================================
                                                    3698
                                                    3699 * ----------------------------------------------------------------------
                                                    3700 * Test Quadruple Compare and Swap (QCSX: 36) instruction : 1st NE
                                                    3701 * ----------------------------------------------------------------------
                                                    3702          TSTHDR 2,1,36             1st Operand NOT Equal (cc=2, UR=1)
0002E275                          0002E275  0002E400 3703+         ORG   PLOETST+(128*K)+19*(3*K)
0002E400  E2E3C1D9 E3409686                         3704+         DC    CL32'START of test: 57'
0002E408  40A385A2 A37A40F5
0002E410  F7404040 40404040
0002E418  40404040 40404040
0002E420                          0002E420          3705+         USING *,R5                base for test data and test routine
0002E420                          00000000          3706+         USING PLOETEST,R5
0002E420  0002E44C                                  3707+T57      DC    A(CODE57)           address of test code
0002E424  0039                                      3708+         DC    H'57'               test number
0002E426  00                                        3709+         DC    X'00'
0002E427  24                                        3710+         DC    HL1'36'             Function
0002E428  01                                        3711+         DC    HL1'1'              expected unequal results (UR)
0002E429  00                                        3712+         DC    HL1'00'             found UR
0002E42A  02                                        3713+         DC    HL1'2'              cc
0002E42B  0D                                        3714+         DC    HL1'13'             not expected CC mask
0002E42C  00                                        3715+         DS    X                   extracted cc
0002E430  00000000 00000000                         3716+         DS    2F                  extract PSW after test (has CC)
0002E438  0002EDE0                                  3717+         DC    A(FMSG57)           address of test fail message
0002E43C  0002EB40                                  3718+         DC    A(OPSRC57)          address of operand source
0002E440  0002EA20                                  3719+         DC    A(OPDST57)          address of operand destination
0002E444  0002EC60                                  3720+         DC    A(PL57)             address of parameter list
0002E448  000041F1                                  3721+         DC    A(RANDX+57)         address of random source data
                                                    3722
                                                    3723          TSTCODE
0002E44C                                            3724+CODE57   DS    0F
                                                    3725 *                                  load random source
0002E44C  E310 501C 0014                 0000001C   3726          LGF   R1,OPSRCA
0002E452  E320 5028 0014                 00000028   3727          LGF   R2,RANDXA
0002E458  E330 5024 0014                 00000024   3728          LGF   R3,PLA
0002E45E  E340 5020 0014                 00000020   3729          LGF   R4,OPREPA
                                                    3730 *
                                                    3731 * Quadruple Compare and Swap (QCSX: 36)
                                                    3732 * Build Parambter List
                                                    3733 *
0002E464  D23F 1000 2000  00000000       00000000   3734          MVC   0(64,R1),0(R2)      load Op1-Op4 data in source
0002E46A  D23F 3040 1000  00000040       00000000   3735          MVC   64(64,R3),0(R1)     copy to PL OP1-Op4 Compare
                                                    3736
0002E470  D23F 4000 2080  00000000       00000080   3737          MVC   0(64,R4),128(R2)    load initial Op1-Op4 data
0002E476  D23F 30C0 4000  000000C0       00000000   3738          MVC   192(64,R3),0(R4)    copy to PL Op1-Op4 Replacement
                                                    3739 ****
                                                    3740 * force a failure by changing the first operand
                                                    3741 ****
0002E47C  D207 1000 4000  00000000       00000000   3742          MVC   0(8,R1),0(R4)
                                                    3743 ****
                                                    3744
0002E482  E313 0000 0024                 00000000   3745          STG   R1,0(R3)            load OP1 address in PL
                                                    3746
0002E488  4111 0010                      00000010   3747          LA    R1,16(R1)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002E48C  E313 0008 0024                 00000008  3748           STG   R1,8(R3)            load OP2 address in PL
                                                   3749
0002E492  4111 0010                      00000010  3750           LA    R1,16(R1)
0002E496  E313 0010 0024                 00000010  3751           STG   R1,16(R3)           load OP3 address in PL
                                                   3752
0002E49C  4111 0010                      00000010  3753           LA    R1,16(R1)
0002E4A0  E313 0018 0024                 00000018  3754           STG   R1,24(R3)           load OP4 address in PL
                                                   3755
0002E4A6  B982 0000                                3756           XGR   R0,R0
0002E4AA  4100 0024                      00000024  3757           LA    R0,36
0002E4AE  EE00 0000 3000       00000000  00000000  3758           PLO   0,0,0,0(R3)         Quadruple C/S (QCSX: 36)
                                                   3759
0002E4B4  B98D 0020                                3760           EPSW  R2,R0               extract psw
0002E4B8  5020 5010                      00000010  3761           ST    R2,CCPSW              to save CC
0002E4BC  E310 500B 0076                 0000000B  3762           LB    R1,CCMASK           (failure CC mask)
0002E4C2  8910 0004                      00000004  3763           SLL   R1,4                (shift to BC instr CC position)
0002E4C6  4410 5104                      0002E524  3764           EX    R1,F361TSTCC            fail if...
                                                   3765
                                0002E4CA  00000001  3766 F361CKUR EQU   *
0002E4CA  D200 5009 3030       00000009  00000030  3767           MVC   URFOUND,48(R3)      Extract UR
0002E4D0  D500 5008 3030       00000008  00000030  3768           CLC   UR,48(R3)
0002E4D6  4780 50BE                      0002E4DE  3769           BE    F361CKRES
0002E4DA  45F0 82C8                      000022C8  3770           BAL   R15,URMSG
                                                   3771 *
                                                   3772 * compare results
                                                   3773 *
                                0002E4DE  00000001  3774 F361CKRES EQU   *
                                                   3775 *                                  reset adresses
0002E4DE  E310 501C 0014                 0000001C  3776           LGF   R1,OPSRCA
0002E4E4  E320 5028 0014                 00000028  3777           LGF   R2,RANDXA
0002E4EA  E330 5024 0014                 00000024  3778           LGF   R3,PLA
0002E4F0  E340 5020 0014                 00000020  3779           LGF   R4,OPREPA
                                                   3780
0002E4F6  D50F 1000 3040       00000000  00000040  3781           CLC   0(16,R1),64(R3)
0002E4FC  4770 50FE                      0002E51E  3782           BNE   F361XX              OP1C Different, error
0002E500  D50F 1010 3050       00000010  00000050  3783           CLC   16(16,R1),80(R3)
0002E506  4770 50FE                      0002E51E  3784           BNE   F361XX              OP2C Different, error
0002E50A  D50F 1020 3060       00000020  00000060  3785           CLC   32(16,R1),96(R3)
0002E510  4770 50FE                      0002E51E  3786           BNE   F361XX              OP3C Different, error
0002E514  D50F 1030 3070       00000030  00000070  3787           CLC   48(16,R1),112(R3)
0002E51A  478B 0000                      00000000  3788           BE    0(R11)              OP4Csame, return
                                                   3789
                                0002E51E  00000001  3790 F361XX   EQU   *
0002E51E  45F0 83E8                      000023E8  3791           BAL   R15,FAILMSG
0002E522  07FB                                     3792           BR    R11
                                                   3793
0002E524  4700 5108                      0002E528  3794 F361TSTCC BC     0,F361CCMSG       fail if unexpected condition code
                                0002E528  00000001  3795 F361CCMSG EQU   *
0002E528  45F0 8196                      00002196  3796           BAL   R15,CCMSG
0002E52C  47F0 50AA                      0002E4CA  3797           B     F361CKUR
                                                   3798
                                                   3799           TSTDATA
0002E530                        0002E530  0002EA00  3800+          ORG   PLOETST+(128*K)+19*(3*K)+(K+512)
                                                   3801           TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
0002EA00  6E6E6ED6 D7C4E2E3                        3802+          DC    CL32'>>>OPDST57<<<' eye catcher
0002EA08  F5F74C4C 4C404040
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0002EA10  40404040 40404040
0002EA18  40404040 40404040
0002EA20                                             3803+OPDST57  DS    0C
0002EA20  00000000 00000000                          3804          DC    XL256'0'
0002EA28  00000000 00000000
0002EA30  00000000 00000000
0002EA38  00000000 00000000
0002EA40  00000000 00000000
0002EA48  00000000 00000000
0002EA50  00000000 00000000
0002EA58  00000000 00000000
0002EA60  00000000 00000000
0002EA68  00000000 00000000
0002EA70  00000000 00000000
0002EA78  00000000 00000000
0002EA80  00000000 00000000
0002EA88  00000000 00000000
0002EA90  00000000 00000000
0002EA98  00000000 00000000
0002EAA0  00000000 00000000
0002EAA8  00000000 00000000
0002EAB0  00000000 00000000
0002EAB8  00000000 00000000
0002EAC0  00000000 00000000
0002EAC8  00000000 00000000
0002EAD0  00000000 00000000
0002EAD8  00000000 00000000
0002EAE0  00000000 00000000
0002EAE8  00000000 00000000
0002EAF0  00000000 00000000
0002EAF8  00000000 00000000
0002EB00  00000000 00000000
0002EB08  00000000 00000000
0002EB10  00000000 00000000
0002EB18  00000000 00000000
                                                     3805
                                                     3806          TSTLBL OPSRC              Source: OP1-OP4
0002EB20  6E6E6ED6 D7E2D9C3                          3807+         DC    CL32'>>>OPSRC57<<<' eye catcher
0002EB28  F5F74C4C 4C404040
0002EB30  40404040 40404040
0002EB38  40404040 40404040
0002EB40                                             3808+OPSRC57  DS    0C
0002EB40  00000000 00000000                          3809          DC    XL256'0'
0002EB48  00000000 00000000
0002EB50  00000000 00000000
0002EB58  00000000 00000000
0002EB60  00000000 00000000
0002EB68  00000000 00000000
0002EB70  00000000 00000000
0002EB78  00000000 00000000
0002EB80  00000000 00000000
0002EB88  00000000 00000000
0002EB90  00000000 00000000
0002EB98  00000000 00000000
0002EBA0  00000000 00000000
0002EBA8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0002EBB0  00000000 00000000
0002EBB8  00000000 00000000
0002EBC0  00000000 00000000
0002EBC8  00000000 00000000
0002EBD0  00000000 00000000
0002EBD8  00000000 00000000
0002EBE0  00000000 00000000
0002EBE8  00000000 00000000
0002EBF0  00000000 00000000
0002EBF8  00000000 00000000
0002EC00  00000000 00000000
0002EC08  00000000 00000000
0002EC10  00000000 00000000
0002EC18  00000000 00000000
0002EC20  00000000 00000000
0002EC28  00000000 00000000
0002EC30  00000000 00000000
0002EC38  00000000 00000000
                                        3810
                                        3811          TSTLBL PL                 Parameter List
0002EC40  6E6E6ED7 D3F5F74C             3812+         DC    CL32'>>>PL57<<<'    eye catcher
0002EC48  4C4C4040 40404040
0002EC50  40404040 40404040
0002EC58  40404040 40404040
0002EC60                                3813+PL57     DS    0C
0002EC60  00000000 00000000             3814          DC    XL352'00'
0002EC68  00000000 00000000
0002EC70  00000000 00000000
0002EC78  00000000 00000000
0002EC80  00000000 00000000
0002EC88  00000000 00000000
0002EC90  00000000 00000000
0002EC98  00000000 00000000
0002ECA0  00000000 00000000
0002ECA8  00000000 00000000
0002ECB0  00000000 00000000
0002ECB8  00000000 00000000
0002ECC0  00000000 00000000
0002ECC8  00000000 00000000
0002ECD0  00000000 00000000
0002ECD8  00000000 00000000
0002ECE0  00000000 00000000
0002ECE8  00000000 00000000
0002ECF0  00000000 00000000
0002ECF8  00000000 00000000
0002ED00  00000000 00000000
0002ED08  00000000 00000000
0002ED10  00000000 00000000
0002ED18  00000000 00000000
0002ED20  00000000 00000000
0002ED28  00000000 00000000
0002ED30  00000000 00000000
0002ED38  00000000 00000000
0002ED40  00000000 00000000
0002ED48  00000000 00000000
0002ED50  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002ED58  00000000 00000000
0002ED60  00000000 00000000
0002ED68  00000000 00000000
0002ED70  00000000 00000000
0002ED78  00000000 00000000
0002ED80  00000000 00000000
0002ED88  00000000 00000000
0002ED90  00000000 00000000
0002ED98  00000000 00000000
0002EDA0  00000000 00000000
0002EDA8  00000000 00000000
0002EDB0  00000000 00000000
0002EDB8  00000000 00000000
                                        3815
                                        3816          TSTLBL FMSG                fail message
0002EDC0  6E6E6EC6 D4E2C7F5             3817+         DC    CL32'>>>FMSG57<<<'   eye catcher
0002EDC8  F74C4C4C 40404040
0002EDD0  40404040 40404040
0002EDD8  40404040 40404040
0002EDE0                                3818+FMSG57   DS    0C
0002EDE0  86818993 858440D7             3819          DC    C'failed PLO-Extension '
0002EDE8  D3D660C5 A7A38595
0002EDF0  A2899695 40
0002EDF5  D8A48184 99A49793             3820          DC    CL96'Quadruple Compare and Swap (QCSX: 36): 1st NE'
0002EDFD  8540C396 94978199
0002EE05  85408195 8440E2A6
0002EE0D  8197404D D8C3E2E7
0002EE15  7A40F3F6 5D7A40F1
0002EE1D  A2A340D5 C5404040
0002EE25  40404040 40404040
0002EE2D  40404040 40404040
0002EE35  40404040 40404040
0002EE3D  40404040 40404040
0002EE45  40404040 40404040
0002EE4D  40404040 40404040
                                        3821
                                        3822          TSTEND
0002EE55  C5D5C440 968640A3             3823+         DC    CL32'END of test: 57 <<<'
0002EE5D  85A2A37A 40F5F740
0002EE65  4C4C4C40 40404040
0002EE6D  40404040 40404040
                                        3824
                                        3825 * ------------------------------------------------------------------
                                        3826 * Test Quadruple Compare and Swap (QCSX: 36) instruction : 2nd NE
                                        3827 * ------------------------------------------------------------------
                                        3828          TSTHDR 2,2,36              2nd Operand NOT Equal (cc=2, UR=2)
0002EE75                  0002EE75  0002F000  3829+         ORG   PLOETST+(128*K)+20*(3*K)
0002F000  E2E3C1D9 E3409686             3830+         DC    CL32'START of test: 58'
0002F008  40A385A2 A37A40F5
0002F010  F8404040 40404040
0002F018  40404040 40404040
0002F020                  0002F020      3831+         USING *,R5                 base for test data and test routine
0002F020                  00000000      3832+         USING PLOETEST,R5
0002F020  0002F04C                      3833+T58      DC    A(CODE58)            address of test code
0002F024  003A                          3834+         DC    H'58'                test number
0002F026  00                            3835+         DC    X'00'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002F027  24                                       3836+          DC    HL1'36'             Function
0002F028  02                                       3837+          DC    HL1'2'              expected unequal results (UR)
0002F029  00                                       3838+          DC    HL1'00'             found UR
0002F02A  02                                       3839+          DC    HL1'2'              cc
0002F02B  0D                                       3840+          DC    HL1'13'             not expected CC mask
0002F02C  00                                       3841+          DS    X                   extracted cc
0002F030  00000000 00000000                        3842+          DS    2F                  extract PSW after test (has CC)
0002F038  0002F9E0                                 3843+          DC    A(FMSG58)           address of test fail message
0002F03C  0002F740                                 3844+          DC    A(OPSRC58)          address of operand source
0002F040  0002F620                                 3845+          DC    A(OPDST58)          address of operand destination
0002F044  0002F860                                 3846+          DC    A(PL58)             address of parameter list
0002F048  000041F2                                 3847+          DC    A(RANDX+58)         address of random source data
                                                   3848
                                                   3849           TSTCODE
0002F04C                                           3850+CODE58    DS    0F
                                                   3851 *                                   load random source
0002F04C  E310 501C 0014              0000001C     3852           LGF   R1,OPSRCA
0002F052  E320 5028 0014              00000028     3853           LGF   R2,RANDXA
0002F058  E330 5024 0014              00000024     3854           LGF   R3,PLA
0002F05E  E340 5020 0014              00000020     3855           LGF   R4,OPREPA
                                                   3856 *
                                                   3857 * Quadruple Compare and Swap (QCSX: 36)
                                                   3858 * Build Parambter List
                                                   3859 *
0002F064  D23F 1000 2000      00000000  00000000   3860           MVC   0(64,R1),0(R2)      load Op1-Op4 data in source
0002F06A  D23F 3040 1000      00000040  00000000   3861           MVC   64(64,R3),0(R1)     copy to PL OP1-Op4 Compare
                                                   3862
0002F070  D23F 4000 2080      00000000  00000080   3863           MVC   0(64,R4),128(R2)    load initial Op1-Op4 data
0002F076  D23F 30C0 4000      000000C0  00000000   3864           MVC   192(64,R3),0(R4)    copy to PL Op1-Op4 Replacement
                                                   3865 ****
                                                   3866 * force a failure by changing the 2nd operand
                                                   3867 ****
0002F07C  D207 1010 4000      00000010  00000000   3868           MVC   16(8,R1),0(R4)
                                                   3869 ****
                                                   3870
0002F082  E313 0000 0024                00000000   3871           STG   R1,0(R3)            load OP1 address in PL
                                                   3872
0002F088  4111 0010                     00000010   3873           LA    R1,16(R1)
0002F08C  E313 0008 0024                00000008   3874           STG   R1,8(R3)            load OP2 address in PL
                                                   3875
0002F092  4111 0010                     00000010   3876           LA    R1,16(R1)
0002F096  E313 0010 0024                00000010   3877           STG   R1,16(R3)           load OP3 address in PL
                                                   3878
0002F09C  4111 0010                     00000010   3879           LA    R1,16(R1)
0002F0A0  E313 0018 0024                00000018   3880           STG   R1,24(R3)           load OP4 address in PL
                                                   3881
0002F0A6  B982 0000                                3882           XGR   R0,R0
0002F0AA  4100 0024                     00000024   3883           LA    R0,36
0002F0AE  EE00 0000 3000      00000000  00000000   3884           PLO   0,0,0,0(R3)         Quadruple C/S (QCSX: 36)
                                                   3885
0002F0B4  B98D 0020                                3886           EPSW  R2,R0               extract psw
0002F0B8  5020 5010                     00000010   3887           ST    R2,CCPSW              to save CC
0002F0BC  E310 500B 0076                0000000B   3888           LB    R1,CCMASK           (failure CC mask)
0002F0C2  8910 0004                     00000004   3889           SLL   R1,4                (shift to BC instr CC position)
0002F0C6  4410 5104                     0002F124   3890           EX    R1,F362TSTCC            fail if...
                                                   3891
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                               0002F0CA  00000001  3892 F362CKUR  EQU   *
0002F0CA  D200 5009 3030       00000009  00000030  3893           MVC   URFOUND,48(R3)      Extract UR
0002F0D0  D500 5008 3030       00000008  00000030  3894           CLC   UR,48(R3)
0002F0D6  4780 50BE                      0002F0DE  3895           BE    F362CKRES
0002F0DA  45F0 82C8                      000022C8  3896           BAL   R15,URMSG
                                                   3897 *
                                                   3898 * compare results
                                                   3899 *
                               0002F0DE  00000001  3900 F362CKRES EQU   *
                                                   3901 *                               reset adresses
0002F0DE  E310 501C 0014                 0000001C  3902           LGF   R1,OPSRCA
0002F0E4  E320 5028 0014                 00000028  3903           LGF   R2,RANDXA
0002F0EA  E330 5024 0014                 00000024  3904           LGF   R3,PLA
0002F0F0  E340 5020 0014                 00000020  3905           LGF   R4,OPREPA
                                                   3906
0002F0F6  D50F 1000 3040       00000000  00000040  3907           CLC   0(16,R1),64(R3)
0002F0FC  4770 50FE                      0002F11E  3908           BNE   F362XX                 OP1C Different, error
0002F100  D50F 1010 3050       00000010  00000050  3909           CLC   16(16,R1),80(R3)
0002F106  4770 50FE                      0002F11E  3910           BNE   F362XX                 OP2C Different, error
0002F10A  D50F 1020 3060       00000020  00000060  3911           CLC   32(16,R1),96(R3)
0002F110  4770 50FE                      0002F11E  3912           BNE   F362XX                 OP3C Different, error
0002F114  D50F 1030 3070       00000030  00000070  3913           CLC   48(16,R1),112(R3)
0002F11A  478B 0000                      00000000  3914           BE    0(R11)                 OP4Csame, return
                                                   3915
                               0002F11E  00000001  3916 F362XX   EQU   *
0002F11E  45F0 83E8                      000023E8  3917           BAL   R15,FAILMSG
0002F122  07FB                                     3918           BR    R11
                                                   3919
0002F124  4700 5108                      0002F128  3920 F362TSTCC BC     0,F362CCMSG          fail if unexpected condition code
                               0002F128  00000001  3921 F362CCMSG EQU   *
0002F128  45F0 8196                      00002196  3922           BAL   R15,CCMSG
0002F12C  47F0 50AA                      0002F0CA  3923           B     F362CKUR
                                                   3924
                                                   3925           TSTDATA
0002F130                       0002F130  0002F600  3926+          ORG   PLOETST+(128*K)+20*(3*K)+(K+512)
                                                   3927           TSTLBL OPDST                 Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
0002F600  6E6E6ED6 D7C4E2E3                        3928+          DC    CL32'>>>OPDST58<<<'    eye catcher
0002F608  F5F84C4C 4C404040
0002F610  40404040 40404040
0002F618  40404040 40404040
0002F620                                           3929+OPDST58   DS    0C
0002F620  00000000 00000000                        3930           DC    XL256'0'
0002F628  00000000 00000000
0002F630  00000000 00000000
0002F638  00000000 00000000
0002F640  00000000 00000000
0002F648  00000000 00000000
0002F650  00000000 00000000
0002F658  00000000 00000000
0002F660  00000000 00000000
0002F668  00000000 00000000
0002F670  00000000 00000000
0002F678  00000000 00000000
0002F680  00000000 00000000
0002F688  00000000 00000000
0002F690  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002F698  00000000 00000000
0002F6A0  00000000 00000000
0002F6A8  00000000 00000000
0002F6B0  00000000 00000000
0002F6B8  00000000 00000000
0002F6C0  00000000 00000000
0002F6C8  00000000 00000000
0002F6D0  00000000 00000000
0002F6D8  00000000 00000000
0002F6E0  00000000 00000000
0002F6E8  00000000 00000000
0002F6F0  00000000 00000000
0002F6F8  00000000 00000000
0002F700  00000000 00000000
0002F708  00000000 00000000
0002F710  00000000 00000000
0002F718  00000000 00000000
                                      3931
                                      3932          TSTLBL OPSRC              Source: OP1-OP4
0002F720  6E6E6ED6 D7E2D9C3           3933+         DC    CL32'>>>OPSRC58<<<' eye catcher
0002F728  F5F84C4C 4C404040
0002F730  40404040 40404040
0002F738  40404040 40404040
0002F740                              3934+OPSRC58  DS    0C
0002F740  00000000 00000000           3935          DC    XL256'0'
0002F748  00000000 00000000
0002F750  00000000 00000000
0002F758  00000000 00000000
0002F760  00000000 00000000
0002F768  00000000 00000000
0002F770  00000000 00000000
0002F778  00000000 00000000
0002F780  00000000 00000000
0002F788  00000000 00000000
0002F790  00000000 00000000
0002F798  00000000 00000000
0002F7A0  00000000 00000000
0002F7A8  00000000 00000000
0002F7B0  00000000 00000000
0002F7B8  00000000 00000000
0002F7C0  00000000 00000000
0002F7C8  00000000 00000000
0002F7D0  00000000 00000000
0002F7D8  00000000 00000000
0002F7E0  00000000 00000000
0002F7E8  00000000 00000000
0002F7F0  00000000 00000000
0002F7F8  00000000 00000000
0002F800  00000000 00000000
0002F808  00000000 00000000
0002F810  00000000 00000000
0002F818  00000000 00000000
0002F820  00000000 00000000
0002F828  00000000 00000000
0002F830  00000000 00000000
0002F838  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   3936
                                                   3937          TSTLBL PL                 Parameter List
0002F840  6E6E6ED7 D3F5F84C                        3938+         DC    CL32'>>>PL58<<<'    eye catcher
0002F848  4C4C4040 40404040
0002F850  40404040 40404040
0002F858  40404040 40404040
0002F860                                           3939+PL58     DS    0C
0002F860  00000000 00000000                        3940          DC    XL352'00'
0002F868  00000000 00000000
0002F870  00000000 00000000
0002F878  00000000 00000000
0002F880  00000000 00000000
0002F888  00000000 00000000
0002F890  00000000 00000000
0002F898  00000000 00000000
0002F8A0  00000000 00000000
0002F8A8  00000000 00000000
0002F8B0  00000000 00000000
0002F8B8  00000000 00000000
0002F8C0  00000000 00000000
0002F8C8  00000000 00000000
0002F8D0  00000000 00000000
0002F8D8  00000000 00000000
0002F8E0  00000000 00000000
0002F8E8  00000000 00000000
0002F8F0  00000000 00000000
0002F8F8  00000000 00000000
0002F900  00000000 00000000
0002F908  00000000 00000000
0002F910  00000000 00000000
0002F918  00000000 00000000
0002F920  00000000 00000000
0002F928  00000000 00000000
0002F930  00000000 00000000
0002F938  00000000 00000000
0002F940  00000000 00000000
0002F948  00000000 00000000
0002F950  00000000 00000000
0002F958  00000000 00000000
0002F960  00000000 00000000
0002F968  00000000 00000000
0002F970  00000000 00000000
0002F978  00000000 00000000
0002F980  00000000 00000000
0002F988  00000000 00000000
0002F990  00000000 00000000
0002F998  00000000 00000000
0002F9A0  00000000 00000000
0002F9A8  00000000 00000000
0002F9B0  00000000 00000000
0002F9B8  00000000 00000000
                                                   3941
                                                   3942          TSTLBL FMSG               fail message
0002F9C0  6E6E6EC6 D4E2C7F5                        3943+         DC    CL32'>>>FMSG58<<<'  eye catcher
0002F9C8  F84C4C4C 40404040
0002F9D0  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002F9D8  40404040 40404040
0002F9E0                                         3944+FMSG58   DS    0C
0002F9E0  86818993 858440D7                      3945          DC    C'failed PLO-Extension '
0002F9E8  D3D660C5 A7A38595
0002F9F0  A2899695 40
0002F9F5  D8A48184 99A49793                      3946          DC    CL96'Quadruple Compare and Swap (QCSX: 36): 2nd NE'
0002F9FD  8540C396 94978199
0002FA05  85408195 8440E2A6
0002FA0D  8197404D D8C3E2E7
0002FA15  7A40F3F6 5D7A40F2
0002FA1D  958440D5 C5404040
0002FA25  40404040 40404040
0002FA2D  40404040 40404040
0002FA35  40404040 40404040
0002FA3D  40404040 40404040
0002FA45  40404040 40404040
0002FA4D  40404040 40404040
                                                 3947
                                                 3948          TSTEND
0002FA55  C5D5C440 968640A3                      3949+         DC    CL32'END of test: 58 <<<'
0002FA5D  85A2A37A 40F5F840
0002FA65  4C4C4C40 40404040
0002FA6D  40404040 40404040
                                                 3950
                                                 3951 * ------------------------------------------------------------------
                                                 3952 * Test Quadruple Compare and Swap (QCSX: 36) instruction : 3rd NE
                                                 3953 * ------------------------------------------------------------------
                                                 3954          TSTHDR 2,3,36             3rd Operand NOT Equal (cc=2, UR=3)
0002FA75                        0002FA75  0002FC00 3955+         ORG   PLOETST+(128*K)+21*(3*K)
0002FC00  E2E3C1D9 E3409686                      3956+         DC    CL32'START of test: 59'
0002FC08  40A385A2 A37A40F5
0002FC10  F9404040 40404040
0002FC18  40404040 40404040
0002FC20                        0002FC20         3957+         USING *,R5                base for test data and test routine
0002FC20                        00000000         3958+         USING PLOETEST,R5
0002FC20  0002FC4C                               3959+T59      DC    A(CODE59)           address of test code
0002FC24  003B                                   3960+         DC    H'59'               test number
0002FC26  00                                     3961+         DC    X'00'
0002FC27  24                                     3962+         DC    HL1'36'             Function
0002FC28  03                                     3963+         DC    HL1'3'              expected unequal results (UR)
0002FC29  00                                     3964+         DC    HL1'00'             found UR
0002FC2A  02                                     3965+         DC    HL1'2'              cc
0002FC2B  0D                                     3966+         DC    HL1'13'             not expected CC mask
0002FC2C  00                                     3967+         DS    X                   extracted cc
0002FC30  00000000 00000000                      3968+         DS    2F                  extract PSW after test (has CC)
0002FC38  000305E0                               3969+         DC    A(FMSG59)           address of test fail message
0002FC3C  00030340                               3970+         DC    A(OPSRC59)          address of operand source
0002FC40  00030220                               3971+         DC    A(OPDST59)          address of operand destination
0002FC44  00030460                               3972+         DC    A(PL59)             address of parameter list
0002FC48  000041F3                               3973+         DC    A(RANDX+59)         address of random source data
                                                 3974
                                                 3975          TSTCODE
0002FC4C                                         3976+CODE59   DS    0F
                                                 3977 *                                  load random source
0002FC4C  E310 501C 0014              0000001C   3978          LGF   R1,OPSRCA
0002FC52  E320 5028 0014              00000028   3979          LGF   R2,RANDXA
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002FC58  E330 5024 0014                 00000024  3980          LGF   R3,PLA
0002FC5E  E340 5020 0014                 00000020  3981          LGF   R4,OPREPA
                                                   3982 *
                                                   3983 * Quadruple Compare and Swap (QCSX: 36)
                                                   3984 * Build Parambter List
                                                   3985 *
0002FC64  D23F 1000 2000      00000000   00000000  3986          MVC   0(64,R1),0(R2)      load Op1-Op4 data in source
0002FC6A  D23F 3040 1000      00000040   00000000  3987          MVC   64(64,R3),0(R1)     copy to PL OP1-Op4 Compare
                                                   3988
0002FC70  D23F 4000 2080      00000000   00000080  3989          MVC   0(64,R4),128(R2)    load initial Op1-Op4 data
0002FC76  D23F 30C0 4000      000000C0   00000000  3990          MVC   192(64,R3),0(R4)    copy to PL Op1-Op4 Replacement
                                                   3991 ****
                                                   3992 * force a failure by changing the 3nd operand
                                                   3993 ****
0002FC7C  D207 1020 4000      00000020   00000000  3994          MVC   32(8,R1),0(R4)
                                                   3995 ****
                                                   3996
0002FC82  E313 0000 0024                 00000000  3997          STG   R1,0(R3)            load OP1 address in PL
                                                   3998
0002FC88  4111 0010                      00000010  3999          LA    R1,16(R1)
0002FC8C  E313 0008 0024                 00000008  4000          STG   R1,8(R3)            load OP2 address in PL
                                                   4001
0002FC92  4111 0010                      00000010  4002          LA    R1,16(R1)
0002FC96  E313 0010 0024                 00000010  4003          STG   R1,16(R3)           load OP3 address in PL
                                                   4004
0002FC9C  4111 0010                      00000010  4005          LA    R1,16(R1)
0002FCA0  E313 0018 0024                 00000018  4006          STG   R1,24(R3)           load OP4 address in PL
                                                   4007
0002FCA6  B982 0000                                4008          XGR   R0,R0
0002FCAA  4100 0024                      00000024  4009          LA    R0,36
0002FCAE  EE00 0000 3000      00000000   00000000  4010          PLO   0,0,0,0(R3)         Quadruple C/S (QCSX: 36)
                                                   4011
0002FCB4  B98D 0020                                4012          EPSW  R2,R0               extract psw
0002FCB8  5020 5010                      00000010  4013          ST    R2,CCPSW              to save CC
0002FCBC  E310 500B 0076                 0000000B  4014          LB    R1,CCMASK           (failure CC mask)
0002FCC2  8910 0004                      00000004  4015          SLL   R1,4                (shift to BC instr CC position)
0002FCC6  4410 5104                      0002FD24  4016          EX    R1,F363TSTCC            fail if...
                                                   4017
                                0002FCCA   00000001  4018 F363CKUR  EQU   *
0002FCCA  D200 5009 3030      00000009   00000030  4019          MVC   URFOUND,48(R3)      Extract UR
0002FCD0  D500 5008 3030      00000008   00000030  4020          CLC   UR,48(R3)
0002FCD6  4780 50BE                      0002FCDE  4021          BE    F363CKRES
0002FCDA  45F0 82C8                      000022C8  4022          BAL   R15,URMSG
                                                   4023 *
                                                   4024 * compare results
                                                   4025 *
                                0002FCDE   00000001  4026 F363CKRES EQU   *
                                                   4027 *                            reset adresses
0002FCDE  E310 501C 0014                 0000001C  4028          LGF   R1,OPSRCA
0002FCE4  E320 5028 0014                 00000028  4029          LGF   R2,RANDXA
0002FCEA  E330 5024 0014                 00000024  4030          LGF   R3,PLA
0002FCF0  E340 5020 0014                 00000020  4031          LGF   R4,OPREPA
                                                   4032
0002FCF6  D50F 1000 3040      00000000   00000040  4033          CLC   0(16,R1),64(R3)
0002FCFC  4770 50FE                      0002FD1E  4034          BNE   F363XX              OP1C Different, error
0002FD00  D50F 1010 3050      00000010   00000050  4035          CLC   16(16,R1),80(R3)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002FD06  4770 50FE                     0002FD1E  4036           BNE   F363XX              OP2C Different, error
0002FD0A  D50F 1020 3060      00000020  00000060  4037           CLC   32(16,R1),96(R3)
0002FD10  4770 50FE                     0002FD1E  4038           BNE   F363XX              OP3C Different, error
0002FD14  D50F 1030 3070      00000030  00000070  4039           CLC   48(16,R1),112(R3)
0002FD1A  478B 0000                     00000000  4040           BE    0(R11)              OP4Csame, return
                                                  4041
                              0002FD1E  00000001  4042 F363XX    EQU   *
0002FD1E  45F0 83E8                     000023E8  4043           BAL   R15,FAILMSG
0002FD22  07FB                                    4044           BR    R11
                                                  4045
0002FD24  4700 5108                     0002FD28  4046 F363TSTCC BC     0,F363CCMSG         fail if unexpected condition code
                              0002FD28  00000001  4047 F363CCMSG EQU   *
0002FD28  45F0 8196                     00002196  4048           BAL   R15,CCMSG
0002FD2C  47F0 50AA                     0002FCCA  4049           B     F363CKUR
                                                  4050
                                                  4051           TSTDATA
0002FD30                      0002FD30  00030200  4052+          ORG   PLOETST+(128*K)+21*(3*K)+(K+512)
                                                  4053           TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00030200  6E6E6ED6 D7C4E2E3                      4054+          DC    CL32'>>>OPDST59<<<' eye catcher
00030208  F5F94C4C 4C404040
00030210  40404040 40404040
00030218  40404040 40404040
00030220                                          4055+OPDST59   DS    0C
00030220  00000000 00000000                      4056           DC    XL256'0'
00030228  00000000 00000000
00030230  00000000 00000000
00030238  00000000 00000000
00030240  00000000 00000000
00030248  00000000 00000000
00030250  00000000 00000000
00030258  00000000 00000000
00030260  00000000 00000000
00030268  00000000 00000000
00030270  00000000 00000000
00030278  00000000 00000000
00030280  00000000 00000000
00030288  00000000 00000000
00030290  00000000 00000000
00030298  00000000 00000000
000302A0  00000000 00000000
000302A8  00000000 00000000
000302B0  00000000 00000000
000302B8  00000000 00000000
000302C0  00000000 00000000
000302C8  00000000 00000000
000302D0  00000000 00000000
000302D8  00000000 00000000
000302E0  00000000 00000000
000302E8  00000000 00000000
000302F0  00000000 00000000
000302F8  00000000 00000000
00030300  00000000 00000000
00030308  00000000 00000000
00030310  00000000 00000000
00030318  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   4058          TSTLBL OPSRC               Source: OP1-OP4
00030320  6E6E6ED6 D7E2D9C3                        4059+         DC    CL32'>>>OPSRC59<<<' eye catcher
00030328  F5F94C4C 4C404040
00030330  40404040 40404040
00030338  40404040 40404040
00030340                                           4060+OPSRC59  DS    0C
00030340  00000000 00000000                        4061          DC    XL256'0'
00030348  00000000 00000000
00030350  00000000 00000000
00030358  00000000 00000000
00030360  00000000 00000000
00030368  00000000 00000000
00030370  00000000 00000000
00030378  00000000 00000000
00030380  00000000 00000000
00030388  00000000 00000000
00030390  00000000 00000000
00030398  00000000 00000000
000303A0  00000000 00000000
000303A8  00000000 00000000
000303B0  00000000 00000000
000303B8  00000000 00000000
000303C0  00000000 00000000
000303C8  00000000 00000000
000303D0  00000000 00000000
000303D8  00000000 00000000
000303E0  00000000 00000000
000303E8  00000000 00000000
000303F0  00000000 00000000
000303F8  00000000 00000000
00030400  00000000 00000000
00030408  00000000 00000000
00030410  00000000 00000000
00030418  00000000 00000000
00030420  00000000 00000000
00030428  00000000 00000000
00030430  00000000 00000000
00030438  00000000 00000000
                                           4062
                                           4063          TSTLBL PL                  Parameter List
00030440  6E6E6ED7 D3F5F94C                4064+         DC    CL32'>>>PL59<<<'     eye catcher
00030448  4C4C4040 40404040
00030450  40404040 40404040
00030458  40404040 40404040
00030460                                   4065+PL59     DS    0C
00030460  00000000 00000000                4066          DC    XL352'00'
00030468  00000000 00000000
00030470  00000000 00000000
00030478  00000000 00000000
00030480  00000000 00000000
00030488  00000000 00000000
00030490  00000000 00000000
00030498  00000000 00000000
000304A0  00000000 00000000
000304A8  00000000 00000000
000304B0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000304B8  00000000 00000000
000304C0  00000000 00000000
000304C8  00000000 00000000
000304D0  00000000 00000000
000304D8  00000000 00000000
000304E0  00000000 00000000
000304E8  00000000 00000000
000304F0  00000000 00000000
000304F8  00000000 00000000
00030500  00000000 00000000
00030508  00000000 00000000
00030510  00000000 00000000
00030518  00000000 00000000
00030520  00000000 00000000
00030528  00000000 00000000
00030530  00000000 00000000
00030538  00000000 00000000
00030540  00000000 00000000
00030548  00000000 00000000
00030550  00000000 00000000
00030558  00000000 00000000
00030560  00000000 00000000
00030568  00000000 00000000
00030570  00000000 00000000
00030578  00000000 00000000
00030580  00000000 00000000
00030588  00000000 00000000
00030590  00000000 00000000
00030598  00000000 00000000
000305A0  00000000 00000000
000305A8  00000000 00000000
000305B0  00000000 00000000
000305B8  00000000 00000000
                                          4067
                                          4068          TSTLBL FMSG                fail message
000305C0  6E6E6EC6 D4E2C7F5               4069+         DC    CL32'>>>FMSG59<<<'   eye catcher
000305C8  F94C4C4C 40404040
000305D0  40404040 40404040
000305D8  40404040 40404040
000305E0                                  4070+FMSG59   DS    0C
000305E0  86818993 858440D7               4071          DC    C'failed PLO-Extension '
000305E8  D3D660C5 A7A38595
000305F0  A2899695 40
000305F5  D8A48184 99A49793               4072          DC    CL96'Quadruple Compare and Swap (QCSX: 36): 3rd NE'
000305FD  8540C396 94978199
00030605  85408195 8440E2A6
0003060D  8197404D D8C3E2E7
00030615  7A40F3F6 5D7A40F3
0003061D  998440D5 C5404040
00030625  40404040 40404040
0003062D  40404040 40404040
00030635  40404040 40404040
0003063D  40404040 40404040
00030645  40404040 40404040
0003064D  40404040 40404040
                                          4073
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   4074          TSTEND
00030655  C5D5C440 968640A3                        4075+         DC    CL32'END of test: 59 <<<'
0003065D  85A2A37A 40F5F940
00030665  4C4C4C40 40404040
0003066D  40404040 40404040
                                                   4076
                                                   4077 * -------------------------------------------------------------------
                                                   4078 * Test Quadruple Compare and Swap (QCSX: 36) instruction : 4th NE
                                                   4079 * -------------------------------------------------------------------
                                                   4080          TSTHDR 2,4,36             4th Operand NOT Equal (cc=2, UR=4)
00030675                    00030675  00030800     4081+         ORG   PLOETST+(128*K)+22*(3*K)
00030800  E2E3C1D9 E3409686                        4082+         DC    CL32'START of test: 60'
00030808  40A385A2 A37A40F6
00030810  F0404040 40404040
00030818  40404040 40404040
00030820                    00030820               4083+         USING *,R5                base for test data and test routine
00030820                    00000000               4084+         USING PLOETEST,R5
00030820  0003084C                                 4085+T60      DC    A(CODE60)           address of test code
00030824  003C                                     4086+         DC    H'60'               test number
00030826  00                                       4087+         DC    X'00'
00030827  24                                       4088+         DC    HL1'36'             Function
00030828  04                                       4089+         DC    HL1'4'              expected unequal results (UR)
00030829  00                                       4090+         DC    HL1'00'             found UR
0003082A  02                                       4091+         DC    HL1'2'              cc
0003082B  0D                                       4092+         DC    HL1'13'             not expected CC mask
0003082C  00                                       4093+         DS    X                   extracted cc
00030830  00000000 00000000                        4094+         DS    2F                  extract PSW after test (has CC)
00030838  000311E0                                 4095+         DC    A(FMSG60)           address of test fail message
0003083C  00030F40                                 4096+         DC    A(OPSRC60)          address of operand source
00030840  00030E20                                 4097+         DC    A(OPDST60)          address of operand destination
00030844  00031060                                 4098+         DC    A(PL60)             address of parameter list
00030848  000041F4                                 4099+         DC    A(RANDX+60)         address of random source data
                                                   4100
                                                   4101          TSTCODE
0003084C                                           4102+CODE60   DS    0F
                                                   4103 *                                  load random source
0003084C  E310 501C 0014              0000001C     4104          LGF   R1,OPSRCA
00030852  E320 5028 0014              00000028     4105          LGF   R2,RANDXA
00030858  E330 5024 0014              00000024     4106          LGF   R3,PLA
0003085E  E340 5020 0014              00000020     4107          LGF   R4,OPREPA
                                                   4108 *
                                                   4109 * Quadruple Compare and Swap (QCSX: 36)
                                                   4110 * Build Parambter List
                                                   4111 *
00030864  D23F 1000 2000  00000000    00000000     4112          MVC   0(64,R1),0(R2)      load Op1-Op4 data in source
0003086A  D23F 3040 1000  00000040    00000000     4113          MVC   64(64,R3),0(R1)     copy to PL OP1-Op4 Compare
                                                   4114
00030870  D23F 4000 2080  00000000    00000080     4115          MVC   0(64,R4),128(R2)    load initial Op1-Op4 data
00030876  D23F 30C0 4000  000000C0    00000000     4116          MVC   192(64,R3),0(R4)    copy to PL Op1-Op4 Replacement
                                                   4117 ****
                                                   4118 * force a failure by changing the 4th operand
                                                   4119 ****
0003087C  D207 1030 4000  00000030    00000000     4120          MVC   48(8,R1),0(R4)
                                                   4121 ****
                                                   4122
00030882  E313 0000 0024              00000000     4123          STG   R1,0(R3)            load OP1 address in PL
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    4124
00030888  4111 0010                     00000010  4125          LA    R1,16(R1)
0003088C  E313 0008 0024                00000008  4126          STG   R1,8(R3)           load OP2 address in PL
                                                    4127
00030892  4111 0010                     00000010  4128          LA    R1,16(R1)
00030896  E313 0010 0024                00000010  4129          STG   R1,16(R3)          load OP3 address in PL
                                                    4130
0003089C  4111 0010                     00000010  4131          LA    R1,16(R1)
000308A0  E313 0018 0024                00000018  4132          STG   R1,24(R3)          load OP4 address in PL
                                                    4133
000308A6  B982 0000                               4134          XGR   R0,R0
000308AA  4100 0024                     00000024  4135          LA    R0,36
000308AE  EE00 0000 3000  00000000      00000000  4136          PLO   0,0,0,0(R3)        Quadruple C/S (QCSX: 36)
                                                    4137
000308B4  B98D 0020                               4138          EPSW  R2,R0              extract psw
000308B8  5020 5010                     00000010  4139          ST    R2,CCPSW             to save CC
000308BC  E310 500B 0076                0000000B  4140          LB    R1,CCMASK          (failure CC mask)
000308C2  8910 0004                     00000004  4141          SLL   R1,4               (shift to BC instr CC position)
000308C6  4410 5104                     00030924  4142          EX    R1,F364TSTCC           fail if...
                                                    4143
                          000308CA      00000001  4144 F364CKUR  EQU   *
000308CA  D200 5009 3030  00000009      00000030  4145          MVC   URFOUND,48(R3)     Extract UR
000308D0  D500 5008 3030  00000008      00000030  4146          CLC   UR,48(R3)
000308D6  4780 50BE                     000308DE  4147          BE    F364CKRES
000308DA  45F0 82C8                     000022C8  4148          BAL   R15,URMSG
                                                    4149 *
                                                    4150 * compare results
                                                    4151 *
                          000308DE      00000001  4152 F364CKRES EQU   *
                                                    4153 *                             reset adresses
000308DE  E310 501C 0014                0000001C  4154          LGF   R1,OPSRCA
000308E4  E320 5028 0014                00000028  4155          LGF   R2,RANDXA
000308EA  E330 5024 0014                00000024  4156          LGF   R3,PLA
000308F0  E340 5020 0014                00000020  4157          LGF   R4,OPREPA
                                                    4158
000308F6  D50F 1000 3040  00000000      00000040  4159          CLC   0(16,R1),64(R3)
000308FC  4770 50FE                     0003091E  4160          BNE   FXX437             OP1C Different, error
00030900  D50F 1010 3050  00000010      00000050  4161          CLC   16(16,R1),80(R3)
00030906  4770 50FE                     0003091E  4162          BNE   FXX437             OP2C Different, error
0003090A  D50F 1020 3060  00000020      00000060  4163          CLC   32(16,R1),96(R3)
00030910  4770 50FE                     0003091E  4164          BNE   FXX437             OP3C Different, error
00030914  D50F 1030 3070  00000030      00000070  4165          CLC   48(16,R1),112(R3)
0003091A  478B 0000                     00000000  4166          BE    0(R11)             OP4Csame, return
                                                    4167
                          0003091E      00000001  4168 FXX437   EQU   *
0003091E  45F0 83E8                     000023E8  4169          BAL   R15,FAILMSG
00030922  07FB                                    4170          BR    R11
                                                    4171
00030924  4700 5108                     00030928  4172 F364TSTCC BC     0,F364CCMSG      fail if unexpected condition code
                          00030928      00000001  4173 F364CCMSG EQU   *
00030928  45F0 8196                     00002196  4174          BAL   R15,CCMSG
0003092C  47F0 50AA                     000308CA  4175          B     F364CKUR
                                                    4176
                                                    4177          TSTDATA
00030930                  00030930      00030E00  4178+         ORG   PLOETST+(128*K)+22*(3*K)+(K+512)
                                                    4179          TSTLBL OPDST             Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00030E00  6E6E6ED6 D7C4E2E3                         4180+         DC    CL32'>>>OPDST60<<<' eye catcher
00030E08  F6F04C4C 4C404040
00030E10  40404040 40404040
00030E18  40404040 40404040
00030E20                                            4181+OPDST60  DS    0C
00030E20  00000000 00000000                         4182          DC    XL256'0'
00030E28  00000000 00000000
00030E30  00000000 00000000
00030E38  00000000 00000000
00030E40  00000000 00000000
00030E48  00000000 00000000
00030E50  00000000 00000000
00030E58  00000000 00000000
00030E60  00000000 00000000
00030E68  00000000 00000000
00030E70  00000000 00000000
00030E78  00000000 00000000
00030E80  00000000 00000000
00030E88  00000000 00000000
00030E90  00000000 00000000
00030E98  00000000 00000000
00030EA0  00000000 00000000
00030EA8  00000000 00000000
00030EB0  00000000 00000000
00030EB8  00000000 00000000
00030EC0  00000000 00000000
00030EC8  00000000 00000000
00030ED0  00000000 00000000
00030ED8  00000000 00000000
00030EE0  00000000 00000000
00030EE8  00000000 00000000
00030EF0  00000000 00000000
00030EF8  00000000 00000000
00030F00  00000000 00000000
00030F08  00000000 00000000
00030F10  00000000 00000000
00030F18  00000000 00000000
                                                    4183
                                                    4184          TSTLBL OPSRC              Source: OP1-OP4
00030F20  6E6E6ED6 D7E2D9C3                         4185+         DC    CL32'>>>OPSRC60<<<' eye catcher
00030F28  F6F04C4C 4C404040
00030F30  40404040 40404040
00030F38  40404040 40404040
00030F40                                            4186+OPSRC60  DS    0C
00030F40  00000000 00000000                         4187          DC    XL256'0'
00030F48  00000000 00000000
00030F50  00000000 00000000
00030F58  00000000 00000000
00030F60  00000000 00000000
00030F68  00000000 00000000
00030F70  00000000 00000000
00030F78  00000000 00000000
00030F80  00000000 00000000
00030F88  00000000 00000000
00030F90  00000000 00000000
00030F98  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00030FA0  00000000 00000000
00030FA8  00000000 00000000
00030FB0  00000000 00000000
00030FB8  00000000 00000000
00030FC0  00000000 00000000
00030FC8  00000000 00000000
00030FD0  00000000 00000000
00030FD8  00000000 00000000
00030FE0  00000000 00000000
00030FE8  00000000 00000000
00030FF0  00000000 00000000
00030FF8  00000000 00000000
00031000  00000000 00000000
00031008  00000000 00000000
00031010  00000000 00000000
00031018  00000000 00000000
00031020  00000000 00000000
00031028  00000000 00000000
00031030  00000000 00000000
00031038  00000000 00000000
                                        4188
                                        4189          TSTLBL PL                   Parameter List
00031040  6E6E6ED7 D3F6F04C             4190+         DC    CL32'>>>PL60<<<'      eye catcher
00031048  4C4C4040 40404040
00031050  40404040 40404040
00031058  40404040 40404040
00031060                                4191+PL60     DS    0C
00031060  00000000 00000000             4192          DC    XL352'00'
00031068  00000000 00000000
00031070  00000000 00000000
00031078  00000000 00000000
00031080  00000000 00000000
00031088  00000000 00000000
00031090  00000000 00000000
00031098  00000000 00000000
000310A0  00000000 00000000
000310A8  00000000 00000000
000310B0  00000000 00000000
000310B8  00000000 00000000
000310C0  00000000 00000000
000310C8  00000000 00000000
000310D0  00000000 00000000
000310D8  00000000 00000000
000310E0  00000000 00000000
000310E8  00000000 00000000
000310F0  00000000 00000000
000310F8  00000000 00000000
00031100  00000000 00000000
00031108  00000000 00000000
00031110  00000000 00000000
00031118  00000000 00000000
00031120  00000000 00000000
00031128  00000000 00000000
00031130  00000000 00000000
00031138  00000000 00000000
00031140  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00031148  00000000 00000000
00031150  00000000 00000000
00031158  00000000 00000000
00031160  00000000 00000000
00031168  00000000 00000000
00031170  00000000 00000000
00031178  00000000 00000000
00031180  00000000 00000000
00031188  00000000 00000000
00031190  00000000 00000000
00031198  00000000 00000000
000311A0  00000000 00000000
000311A8  00000000 00000000
000311B0  00000000 00000000
000311B8  00000000 00000000
                                        4193
                                        4194          TSTLBL FMSG                fail message
000311C0  6E6E6EC6 D4E2C7F6             4195+         DC    CL32'>>>FMSG60<<<'   eye catcher
000311C8  F04C4C4C 40404040
000311D0  40404040 40404040
000311D8  40404040 40404040
000311E0                                4196+FMSG60   DS    0C
000311E0  86818993 858440D7             4197          DC    C'failed PLO-Extension '
000311E8  D3D660C5 A7A38595
000311F0  A2899695 40
000311F5  D8A48184 99A49793             4198          DC    CL96'Quadruple Compare and Swap (QCSX: 36): 4th NE'
000311FD  8540C396 94978199
00031205  85408195 8440E2A6
0003120D  8197404D D8C3E2E7
00031215  7A40F3F6 5D7A40F4
0003121D  A38840D5 C5404040
00031225  40404040 40404040
0003122D  40404040 40404040
00031235  40404040 40404040
0003123D  40404040 40404040
00031245  40404040 40404040
0003124D  40404040 40404040
                                        4199
                                        4200          TSTEND
00031255  C5D5C440 968640A3             4201+         DC    CL32'END of test: 60 <<<'
0003125D  85A2A37A 40F6F040
00031265  4C4C4C40 40404040
0003126D  40404040 40404040
                                        4202
                                        4203
                                        4204 * ------------------------------------------------------------------
                                        4205 * Test Quadruple Compare and Swap (QCSX: 36) instruction : All EQ
                                        4206 * ------------------------------------------------------------------
                                        4207          TSTHDR 0,99,36             All Equal (cc=0, UR= unchanged)
00031275                    00031275  00031400  4208+         ORG   PLOETST+(128*K)+23*(3*K)
00031400  E2E3C1D9 E3409686             4209+         DC    CL32'START of test: 61'
00031408  40A385A2 A37A40F6
00031410  F1404040 40404040
00031418  40404040 40404040
00031420                    00031420            4210+         USING *,R5                 base for test data and test routine
00031420                    00000000            4211+         USING PLOETEST,R5
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00031420  0003144C                                4212+T61      DC    A(CODE61)           address of test code
00031424  003D                                    4213+         DC    H'61'               test number
00031426  00                                      4214+         DC    X'00'
00031427  24                                      4215+         DC    HL1'36'             Function
00031428  63                                      4216+         DC    HL1'99'             expected unequal results (UR)
00031429  00                                      4217+         DC    HL1'00'             found UR
0003142A  00                                      4218+         DC    HL1'0'              cc
0003142B  07                                      4219+         DC    HL1'7'              not expected CC mask
0003142C  00                                      4220+         DS    X                   extracted cc
00031430  00000000 00000000                       4221+         DS    2F                  extract PSW after test (has CC)
00031438  00031DE0                                4222+         DC    A(FMSG61)           address of test fail message
0003143C  00031B40                                4223+         DC    A(OPSRC61)          address of operand source
00031440  00031A20                                4224+         DC    A(OPDST61)          address of operand destination
00031444  00031C60                                4225+         DC    A(PL61)             address of parameter list
00031448  000041F5                                4226+         DC    A(RANDX+61)         address of random source data
                                                  4227
                                                  4228          TSTCODE
0003144C                                          4229+CODE61   DS    0F
                                                  4230 *                                  load random source
0003144C  E310 501C 0014             0000001C     4231          LGF   R1,OPSRCA
00031452  E320 5028 0014             00000028     4232          LGF   R2,RANDXA
00031458  E330 5024 0014             00000024     4233          LGF   R3,PLA
0003145E  E340 5020 0014             00000020     4234          LGF   R4,OPREPA
                                                  4235 *
                                                  4236 * Quadruple Compare and Swap (QCSX: 36)
                                                  4237 * Build Parambter List
                                                  4238 *
00031464  D23F 1000 2000  00000000   00000000     4239          MVC   0(64,R1),0(R2)      load Op1-Op4 data in source
0003146A  D23F 3040 1000  00000040   00000000     4240          MVC   64(64,R3),0(R1)     copy to PL OP1-Op4 Compare
                                                  4241
00031470  D23F 4000 2080  00000000   00000080     4242          MVC   0(64,R4),128(R2)    load initial Op1R-Op4R data
00031476  D23F 30C0 4000  000000C0   00000000     4243          MVC   192(64,R3),0(R4)    copy to PL Op1-Op4 Replacement
                                                  4244 ****
                                                  4245 * Do NOT force a failure by changing an operand
                                                  4246 ****
                                                  4247 *         MVC   96(8,R1),0(R4)
                                                  4248 ****
                                                  4249
0003147C  E313 0000 0024                 00000000 4250          STG   R1,0(R3)            load OP1 address in PL
                                                  4251
00031482  4111 0010                      00000010 4252          LA    R1,16(R1)
00031486  E313 0008 0024                 00000008 4253          STG   R1,8(R3)            load OP2 address in PL
                                                  4254
0003148C  4111 0010                      00000010 4255          LA    R1,16(R1)
00031490  E313 0010 0024                 00000010 4256          STG   R1,16(R3)           load OP3 address in PL
                                                  4257
00031496  4111 0010                      00000010 4258          LA    R1,16(R1)
0003149A  E313 0018 0024                 00000018 4259          STG   R1,24(R3)           load OP4 address in PL
                                                  4260
000314A0  B982 0000                               4261          XGR   R0,R0
000314A4  4100 0024                      00000024 4262          LA    R0,36
000314A8  EE00 0000 3000  00000000   00000000     4263          PLO   0,0,0,0(R3)         Quadruple C/S (QCSX: 36)
                                                  4264
000314AE  B98D 0020                               4265          EPSW  R2,R0               extract psw
000314B2  5020 5010                      00000010 4266          ST    R2,CCPSW              to save CC
000314B6  E310 500B 0076                 0000000B 4267          LB    R1,CCMASK           (failure CC mask)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000314BC  8910 0004                    00000004  4268          SLL   R1,4                (shift to BC instr CC position)
000314C0  4410 50FE                    0003151E  4269          EX    R1,F365TSTCC          fail if...
                                                 4270
                               000314C4  00000001  4271 F365CKUR  EQU   *
000314C4  D200 5009 3030       00000009  00000030  4272          MVC   URFOUND,48(R3)      Extract UR
000314CA  D500 5008 3030       00000008  00000030  4273          CLC   UR,48(R3)
000314D0  4780 50B8                    000314D8  4274          BE    F365CKRES
000314D4  45F0 82C8                    000022C8  4275          BAL   R15,URMSG
                                                 4276 *
                                                 4277 * compare results
                                                 4278 *
                               000314D8  00000001  4279 F365CKRES EQU   *
                                                 4280 *                              reset adresses
000314D8  E310 501C 0014               0000001C  4281          LGF   R1,OPSRCA
000314DE  E320 5028 0014               00000028  4282          LGF   R2,RANDXA
000314E4  E330 5024 0014               00000024  4283          LGF   R3,PLA
000314EA  E340 5020 0014               00000020  4284          LGF   R4,OPREPA
                                                 4285
                                                 4286 * All equal; replacement values copied to operand source
                                                 4287
000314F0  D50F 1000 3080       00000000  00000080  4288          CLC   0(16,R1),128(R3)
000314F6  4770 50F8                    00031518  4289          BNE   FXX537              OP1R Different, error
000314FA  D50F 1010 3090       00000010  00000090  4290          CLC   16(16,R1),144(R3)
00031500  4770 50F8                    00031518  4291          BNE   FXX537              OP2R Different, error
00031504  D50F 1020 30A0       00000020  000000A0  4292          CLC   32(16,R1),160(R3)
0003150A  4770 50F8                    00031518  4293          BNE   FXX537              OP3R Different, error
0003150E  D50F 1030 30B0       00000030  000000B0  4294          CLC   48(16,R1),176(R3)
00031514  478B 0000                    00000000  4295          BE    0(R11)              OP4R same, return
                                                 4296
                               00031518  00000001  4297 FXX537   EQU   *
00031518  45F0 83E8                    000023E8  4298          BAL   R15,FAILMSG
0003151C  07FB                                   4299          BR    R11
                                                 4300
0003151E  4700 5102                    00031522  4301 F365TSTCC BC    0,F365CCMSG         fail if unexpected condition code
                               00031522  00000001  4302 F365CCMSG EQU   *
00031522  45F0 8196                    00002196  4303          BAL   R15,CCMSG
00031526  47F0 50A4                    000314C4  4304          B     F365CKUR
                                                 4305
                                                 4306          TSTDATA
0003152A                       0003152A  00031A00  4307+         ORG   PLOETST+(128*K)+23*(3*K)+(K+512)
                                                 4308          TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00031A00  6E6E6ED6 D7C4E2E3                      4309+         DC    CL32'>>>OPDST61<<<' eye catcher
00031A08  F6F14C4C 4C404040
00031A10  40404040 40404040
00031A18  40404040 40404040
00031A20                                         4310+OPDST61  DS    0C
00031A20  00000000 00000000                      4311          DC    XL256'0'
00031A28  00000000 00000000
00031A30  00000000 00000000
00031A38  00000000 00000000
00031A40  00000000 00000000
00031A48  00000000 00000000
00031A50  00000000 00000000
00031A58  00000000 00000000
00031A60  00000000 00000000
00031A68  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00031A70  00000000 00000000
00031A78  00000000 00000000
00031A80  00000000 00000000
00031A88  00000000 00000000
00031A90  00000000 00000000
00031A98  00000000 00000000
00031AA0  00000000 00000000
00031AA8  00000000 00000000
00031AB0  00000000 00000000
00031AB8  00000000 00000000
00031AC0  00000000 00000000
00031AC8  00000000 00000000
00031AD0  00000000 00000000
00031AD8  00000000 00000000
00031AE0  00000000 00000000
00031AE8  00000000 00000000
00031AF0  00000000 00000000
00031AF8  00000000 00000000
00031B00  00000000 00000000
00031B08  00000000 00000000
00031B10  00000000 00000000
00031B18  00000000 00000000
                                          4312
                                          4313          TSTLBL OPSRC              Source: OP1-OP4
00031B20  6E6E6ED6 D7E2D9C3               4314+         DC    CL32'>>>OPSRC61<<<' eye catcher
00031B28  F6F14C4C 4C404040
00031B30  40404040 40404040
00031B38  40404040 40404040
00031B40                                  4315+OPSRC61  DS    0C
00031B40  00000000 00000000               4316          DC    XL256'0'
00031B48  00000000 00000000
00031B50  00000000 00000000
00031B58  00000000 00000000
00031B60  00000000 00000000
00031B68  00000000 00000000
00031B70  00000000 00000000
00031B78  00000000 00000000
00031B80  00000000 00000000
00031B88  00000000 00000000
00031B90  00000000 00000000
00031B98  00000000 00000000
00031BA0  00000000 00000000
00031BA8  00000000 00000000
00031BB0  00000000 00000000
00031BB8  00000000 00000000
00031BC0  00000000 00000000
00031BC8  00000000 00000000
00031BD0  00000000 00000000
00031BD8  00000000 00000000
00031BE0  00000000 00000000
00031BE8  00000000 00000000
00031BF0  00000000 00000000
00031BF8  00000000 00000000
00031C00  00000000 00000000
00031C08  00000000 00000000
00031C10  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00031C18  00000000 00000000
00031C20  00000000 00000000
00031C28  00000000 00000000
00031C30  00000000 00000000
00031C38  00000000 00000000
                                       4317
                                       4318          TSTLBL PL                  Parameter List
00031C40  6E6E6ED7 D3F6F14C            4319+         DC    CL32'>>>PL61<<<'     eye catcher
00031C48  4C4C4040 40404040
00031C50  40404040 40404040
00031C58  40404040 40404040
00031C60                               4320+PL61     DS    0C
00031C60  00000000 00000000            4321          DC    XL352'00'
00031C68  00000000 00000000
00031C70  00000000 00000000
00031C78  00000000 00000000
00031C80  00000000 00000000
00031C88  00000000 00000000
00031C90  00000000 00000000
00031C98  00000000 00000000
00031CA0  00000000 00000000
00031CA8  00000000 00000000
00031CB0  00000000 00000000
00031CB8  00000000 00000000
00031CC0  00000000 00000000
00031CC8  00000000 00000000
00031CD0  00000000 00000000
00031CD8  00000000 00000000
00031CE0  00000000 00000000
00031CE8  00000000 00000000
00031CF0  00000000 00000000
00031CF8  00000000 00000000
00031D00  00000000 00000000
00031D08  00000000 00000000
00031D10  00000000 00000000
00031D18  00000000 00000000
00031D20  00000000 00000000
00031D28  00000000 00000000
00031D30  00000000 00000000
00031D38  00000000 00000000
00031D40  00000000 00000000
00031D48  00000000 00000000
00031D50  00000000 00000000
00031D58  00000000 00000000
00031D60  00000000 00000000
00031D68  00000000 00000000
00031D70  00000000 00000000
00031D78  00000000 00000000
00031D80  00000000 00000000
00031D88  00000000 00000000
00031D90  00000000 00000000
00031D98  00000000 00000000
00031DA0  00000000 00000000
00031DA8  00000000 00000000
00031DB0  00000000 00000000
00031DB8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    4322
                                                    4323          TSTLBL FMSG                fail message
00031DC0  6E6E6EC6 D4E2C7F6                         4324+         DC    CL32'>>>FMSG61<<<'   eye catcher
00031DC8  F14C4C4C 40404040
00031DD0  40404040 40404040
00031DD8  40404040 40404040
00031DE0                                            4325+FMSG61   DS    0C
00031DE0  86818993 858440D7                         4326          DC    C'failed PLO-Extension '
00031DE8  D3D660C5 A7A38595
00031DF0  A2899695 40
00031DF5  D8A48184 99A49793                         4327          DC    CL96'Quadruple Compare and Swap (QCSX: 36): All EQ'
00031DFD  8540C396 94978199
00031E05  85408195 8440E2A6
00031E0D  8197404D D8C3E2E7
00031E15  7A40F3F6 5D7A40C1
00031E1D  939340C5 D8404040
00031E25  40404040 40404040
00031E2D  40404040 40404040
00031E35  40404040 40404040
00031E3D  40404040 40404040
00031E45  40404040 40404040
00031E4D  40404040 40404040
                                                    4328
                                                    4329          TSTEND
00031E55  C5D5C440 968640A3                         4330+         DC    CL32'END of test: 61 <<<'
00031E5D  85A2A37A 40F6F140
00031E65  4C4C4C40 40404040
00031E6D  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   4332 *=============================================================================
                                                   4333 *       Quadruple Compare and Swap Tests
                                                   4334 *=============================================================================
                                                   4335
                                                   4336 * -----------------------------------------------------------------------
                                                   4337 * Test Quadruple Compare and Swap (QCSG: 35) instruction : 1st NE
                                                   4338 * -----------------------------------------------------------------------
                                                   4339          TSTHDR 2,1,35             1st Operand NOT Equal (cc=2, UR=1)
00031E75                          00031E75  00032000 4340+         ORG   PLOETST+(128*K)+24*(3*K)
00032000  E2E3C1D9 E3409686                        4341+         DC    CL32'START of test: 62'
00032008  40A385A2 A37A40F6
00032010  F2404040 40404040
00032018  40404040 40404040
00032020                          00032020         4342+         USING *,R5                base for test data and test routine
00032020                          00000000         4343+         USING PLOETEST,R5
00032020  0003204C                                 4344+T62      DC    A(CODE62)           address of test code
00032024  003E                                     4345+         DC    H'62'               test number
00032026  00                                       4346+         DC    X'00'
00032027  23                                       4347+         DC    HL1'35'             Function
00032028  01                                       4348+         DC    HL1'1'              expected unequal results (UR)
00032029  00                                       4349+         DC    HL1'00'             found UR
0003202A  02                                       4350+         DC    HL1'2'              cc
0003202B  0D                                       4351+         DC    HL1'13'             not expected CC mask
0003202C  00                                       4352+         DS    X                   extracted cc
00032030  00000000 00000000                        4353+         DS    2F                  extract PSW after test (has CC)
00032038  000329E0                                 4354+         DC    A(FMSG62)           address of test fail message
0003203C  00032740                                 4355+         DC    A(OPSRC62)          address of operand source
00032040  00032620                                 4356+         DC    A(OPDST62)          address of operand destination
00032044  00032860                                 4357+         DC    A(PL62)             address of parameter list
00032048  000041F6                                 4358+         DC    A(RANDX+62)         address of random source data
                                                   4359
                                                   4360          TSTCODE
0003204C                                           4361+CODE62   DS    0F
                                                   4362 *                                  load random source
0003204C  E310 501C 0014                0000001C   4363          LGF   R1,OPSRCA
00032052  E320 5028 0014                00000028   4364          LGF   R2,RANDXA
00032058  E330 5024 0014                00000024   4365          LGF   R3,PLA
0003205E  E340 5020 0014                00000020   4366          LGF   R4,OPREPA
                                                   4367 *
                                                   4368 * Quadruple Compare and Swap (QCSG: 35)
                                                   4369 * Build Parambter List
                                                   4370 *
00032064  D21F 1000 2000  00000000      00000000   4371          MVC   0(32,R1),0(R2)      load Op1-Op4 data in source
0003206A  D21F 3040 1000  00000040      00000000   4372          MVC   64(32,R3),0(R1)     copy to PL OP1-Op4 Compare
                                                   4373
00032070  D21F 4000 2080  00000000      00000080   4374          MVC   0(32,R4),128(R2)    load initial Op1-Op4 data
00032076  D21F 3080 4000  00000080      00000000   4375          MVC   128(32,R3),0(R4)    copy to PL Op1-Op4 Replacement
                                                   4376 ****
                                                   4377 * force a failure by changing the first operand
                                                   4378 ****
0003207C  D203 1000 4000  00000000      00000000   4379          MVC   0(4,R1),0(R4)
                                                   4380 ****
                                                   4381
00032082  E313 0000 0024                00000000   4382          STG   R1,0(R3)            load OP1 address in PL
                                                   4383
00032088  4111 0008                     00000008   4384          LA    R1,8(R1)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003208C  E313 0008 0024                 00000008  4385           STG   R1,8(R3)           load OP2 address in PL
                                                   4386
00032092  4111 0008                      00000008  4387           LA    R1,8(R1)
00032096  E313 0010 0024                 00000010  4388           STG   R1,16(R3)          load OP3 address in PL
                                                   4389
0003209C  4111 0008                      00000008  4390           LA    R1,8(R1)
000320A0  E313 0018 0024                 00000018  4391           STG   R1,24(R3)          load OP4 address in PL
                                                   4392
000320A6  B982 0000                                4393           XGR   R0,R0
000320AA  4100 0023                      00000023  4394           LA    R0,35
000320AE  EE00 0000 3000       00000000  00000000  4395           PLO   0,0,0,0(R3)        Quadruple C/S (QCSG: 35)
                                                   4396
000320B4  B98D 0020                                4397           EPSW  R2,R0              extract psw
000320B8  5020 5010                      00000010  4398           ST    R2,CCPSW             to save CC
000320BC  E310 500B 0076                 0000000B  4399           LB    R1,CCMASK          (failure CC mask)
000320C2  8910 0004                      00000004  4400           SLL   R1,4               (shift to BC instr CC position)
000320C6  4410 5104                      00032124  4401           EX    R1,F351TSTCC         fail if...
                                                   4402
                               000320CA  00000001  4403 F351CKUR EQU   *
000320CA  D200 5009 3030       00000009  00000030  4404           MVC   URFOUND,48(R3)     Extract UR
000320D0  D500 5008 3030       00000008  00000030  4405           CLC   UR,48(R3)
000320D6  4780 50BE                      000320DE  4406           BE    F351CKRES
000320DA  45F0 82C8                      000022C8  4407           BAL   R15,URMSG
                                                   4408 *
                                                   4409 * compare results
                                                   4410 *
                               000320DE  00000001  4411 F351CKRES EQU   *
                                                   4412 *                                reset adresses
000320DE  E310 501C 0014                 0000001C  4413           LGF   R1,OPSRCA
000320E4  E320 5028 0014                 00000028  4414           LGF   R2,RANDXA
000320EA  E330 5024 0014                 00000024  4415           LGF   R3,PLA
000320F0  E340 5020 0014                 00000020  4416           LGF   R4,OPREPA
                                                   4417
000320F6  D507 1000 3040       00000000  00000040  4418           CLC   0(8,R1),64(R3)
000320FC  4770 50FE                      0003211E  4419           BNE   F351XX             OP1C Different, error
00032100  D507 1008 3048       00000008  00000048  4420           CLC   8(8,R1),72(R3)
00032106  4770 50FE                      0003211E  4421           BNE   F351XX             OP2C Different, error
0003210A  D507 1010 3050       00000010  00000050  4422           CLC   16(8,R1),80(R3)
00032110  4770 50FE                      0003211E  4423           BNE   F351XX             OP3C Different, error
00032114  D507 1018 3058       00000018  00000058  4424           CLC   24(8,R1),88(R3)
0003211A  478B 0000                      00000000  4425           BE    0(R11)             OP4Csame, return
                                                   4426
                               0003211E  00000001  4427 F351XX   EQU   *
0003211E  45F0 83E8                      000023E8  4428           BAL   R15,FAILMSG
00032122  07FB                                     4429           BR    R11
                                                   4430
00032124  4700 5108                      00032128  4431 F351TSTCC BC    0,F351CCMSG        fail if unexpected condition code
                               00032128  00000001  4432 F351CCMSG EQU   *
00032128  45F0 8196                      00002196  4433           BAL   R15,CCMSG
0003212C  47F0 50AA                      000320CA  4434           B     F351CKUR
                                                   4435
                                                   4436           TSTDATA
00032130                       00032130  00032600  4437+          ORG   PLOETST+(128*K)+24*(3*K)+(K+512)
                                                   4438           TSTLBL OPDST             Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00032600  6E6E6ED6 D7C4E2E3                        4439+          DC    CL32'>>>OPDST62<<<' eye catcher
00032608  F6F24C4C 4C404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00032610  40404040 40404040
00032618  40404040 40404040
00032620                                           4440+OPDST62  DS    0C
00032620  00000000 00000000                        4441          DC    XL256'0'
00032628  00000000 00000000
00032630  00000000 00000000
00032638  00000000 00000000
00032640  00000000 00000000
00032648  00000000 00000000
00032650  00000000 00000000
00032658  00000000 00000000
00032660  00000000 00000000
00032668  00000000 00000000
00032670  00000000 00000000
00032678  00000000 00000000
00032680  00000000 00000000
00032688  00000000 00000000
00032690  00000000 00000000
00032698  00000000 00000000
000326A0  00000000 00000000
000326A8  00000000 00000000
000326B0  00000000 00000000
000326B8  00000000 00000000
000326C0  00000000 00000000
000326C8  00000000 00000000
000326D0  00000000 00000000
000326D8  00000000 00000000
000326E0  00000000 00000000
000326E8  00000000 00000000
000326F0  00000000 00000000
000326F8  00000000 00000000
00032700  00000000 00000000
00032708  00000000 00000000
00032710  00000000 00000000
00032718  00000000 00000000
                                           4442
                                           4443          TSTLBL OPSRC              Source: OP1-OP4
00032720  6E6E6ED6 D7E2D9C3                4444+         DC    CL32'>>>OPSRC62<<<' eye catcher
00032728  F6F24C4C 4C404040
00032730  40404040 40404040
00032738  40404040 40404040
00032740                                   4445+OPSRC62  DS    0C
00032740  00000000 00000000                4446          DC    XL256'0'
00032748  00000000 00000000
00032750  00000000 00000000
00032758  00000000 00000000
00032760  00000000 00000000
00032768  00000000 00000000
00032770  00000000 00000000
00032778  00000000 00000000
00032780  00000000 00000000
00032788  00000000 00000000
00032790  00000000 00000000
00032798  00000000 00000000
000327A0  00000000 00000000
000327A8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000327B0  00000000 00000000
000327B8  00000000 00000000
000327C0  00000000 00000000
000327C8  00000000 00000000
000327D0  00000000 00000000
000327D8  00000000 00000000
000327E0  00000000 00000000
000327E8  00000000 00000000
000327F0  00000000 00000000
000327F8  00000000 00000000
00032800  00000000 00000000
00032808  00000000 00000000
00032810  00000000 00000000
00032818  00000000 00000000
00032820  00000000 00000000
00032828  00000000 00000000
00032830  00000000 00000000
00032838  00000000 00000000
                                          4447
                                          4448          TSTLBL PL                 Parameter List
00032840  6E6E6ED7 D3F6F24C               4449+         DC    CL32'>>>PL62<<<'    eye catcher
00032848  4C4C4040 40404040
00032850  40404040 40404040
00032858  40404040 40404040
00032860                                  4450+PL62     DS    0C
00032860  00000000 00000000               4451          DC    XL352'00'
00032868  00000000 00000000
00032870  00000000 00000000
00032878  00000000 00000000
00032880  00000000 00000000
00032888  00000000 00000000
00032890  00000000 00000000
00032898  00000000 00000000
000328A0  00000000 00000000
000328A8  00000000 00000000
000328B0  00000000 00000000
000328B8  00000000 00000000
000328C0  00000000 00000000
000328C8  00000000 00000000
000328D0  00000000 00000000
000328D8  00000000 00000000
000328E0  00000000 00000000
000328E8  00000000 00000000
000328F0  00000000 00000000
000328F8  00000000 00000000
00032900  00000000 00000000
00032908  00000000 00000000
00032910  00000000 00000000
00032918  00000000 00000000
00032920  00000000 00000000
00032928  00000000 00000000
00032930  00000000 00000000
00032938  00000000 00000000
00032940  00000000 00000000
00032948  00000000 00000000
00032950  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00032958  00000000 00000000
00032960  00000000 00000000
00032968  00000000 00000000
00032970  00000000 00000000
00032978  00000000 00000000
00032980  00000000 00000000
00032988  00000000 00000000
00032990  00000000 00000000
00032998  00000000 00000000
000329A0  00000000 00000000
000329A8  00000000 00000000
000329B0  00000000 00000000
000329B8  00000000 00000000
                                        4452
                                        4453          TSTLBL FMSG                fail message
000329C0  6E6E6EC6 D4E2C7F6             4454+         DC    CL32'>>>FMSG62<<<'   eye catcher
000329C8  F24C4C4C 40404040
000329D0  40404040 40404040
000329D8  40404040 40404040
000329E0                                4455+FMSG62   DS    0C
000329E0  86818993 858440D7             4456          DC    C'failed PLO-Extension '
000329E8  D3D660C5 A7A38595
000329F0  A2899695 40
000329F5  D8A48184 99A49793             4457          DC    CL96'Quadruple Compare and Swap (QCSG: 35): 1st NE'
000329FD  8540C396 94978199
00032A05  85408195 8440E2A6
00032A0D  8197404D D8C3E2C7
00032A15  7A40F3F5 5D7A40F1
00032A1D  A2A340D5 C5404040
00032A25  40404040 40404040
00032A2D  40404040 40404040
00032A35  40404040 40404040
00032A3D  40404040 40404040
00032A45  40404040 40404040
00032A4D  40404040 40404040
                                        4458
                                        4459          TSTEND
00032A55  C5D5C440 968640A3             4460+         DC    CL32'END of test: 62 <<<'
00032A5D  85A2A37A 40F6F240
00032A65  4C4C4C40 40404040
00032A6D  40404040 40404040
                                        4461
                                        4462 * ------------------------------------------------------------------
                                        4463 * Test Quadruple Compare and Swap (QCSG: 35) instruction : 2nd NE
                                        4464 * ------------------------------------------------------------------
                                        4465          TSTHDR 2,2,35              2nd Operand NOT Equal (cc=2, UR=2)
00032A75                  00032A75  00032C00  4466+         ORG   PLOETST+(128*K)+25*(3*K)
00032C00  E2E3C1D9 E3409686             4467+         DC    CL32'START of test: 63'
00032C08  40A385A2 A37A40F6
00032C10  F3404040 40404040
00032C18  40404040 40404040
00032C20                  00032C20      4468+         USING *,R5                 base for test data and test routine
00032C20                  00000000      4469+         USING PLOETEST,R5
00032C20  00032C4C                      4470+T63      DC    A(CODE63)            address of test code
00032C24  003F                          4471+         DC    H'63'                test number
00032C26  00                            4472+         DC    X'00'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00032C27  23                                     4473+          DC    HL1'35'             Function
00032C28  02                                     4474+          DC    HL1'2'              expected unequal results (UR)
00032C29  00                                     4475+          DC    HL1'00'             found UR
00032C2A  02                                     4476+          DC    HL1'2'              cc
00032C2B  0D                                     4477+          DC    HL1'13'             not expected CC mask
00032C2C  00                                     4478+          DS    X                   extracted cc
00032C30  00000000 00000000                      4479+          DS    2F                  extract PSW after test (has CC)
00032C38  000335E0                               4480+          DC    A(FMSG63)           address of test fail message
00032C3C  00033340                               4481+          DC    A(OPSRC63)          address of operand source
00032C40  00033220                               4482+          DC    A(OPDST63)          address of operand destination
00032C44  00033460                               4483+          DC    A(PL63)             address of parameter list
00032C48  000041F7                               4484+          DC    A(RANDX+63)         address of random source data
                                                 4485
                                                 4486           TSTCODE
00032C4C                                         4487+CODE63    DS    0F
                                                 4488 *                                   load random source
00032C4C  E310 501C 0014             0000001C    4489           LGF   R1,OPSRCA
00032C52  E320 5028 0014             00000028    4490           LGF   R2,RANDXA
00032C58  E330 5024 0014             00000024    4491           LGF   R3,PLA
00032C5E  E340 5020 0014             00000020    4492           LGF   R4,OPREPA
                                                 4493 *
                                                 4494 * Quadruple Compare and Swap (QCSG: 35)
                                                 4495 * Build Parambter List
                                                 4496 *
00032C64  D21F 1000 2000       00000000  00000000 4497           MVC   0(32,R1),0(R2)      load Op1-Op4 data in source
00032C6A  D21F 3040 1000       00000040  00000000 4498           MVC   64(32,R3),0(R1)     copy to PL OP1-Op4 Compare
                                                 4499
00032C70  D21F 4000 2080       00000000  00000080 4500           MVC   0(32,R4),128(R2)    load initial Op1-Op4 data
00032C76  D21F 3080 4000       00000080  00000000 4501           MVC   128(32,R3),0(R4)    copy to PL Op1-Op4 Replacement
                                                 4502 ****
                                                 4503 * force a failure by changing the 2nd operand
                                                 4504 ****
00032C7C  D203 1008 4000       00000008  00000000 4505           MVC   8(4,R1),0(R4)
                                                 4506 ****
                                                 4507
00032C82  E313 0000 0024                 00000000 4508           STG   R1,0(R3)            load OP1 address in PL
                                                 4509
00032C88  4111 0008                      00000008 4510           LA    R1,8(R1)
00032C8C  E313 0008 0024                 00000008 4511           STG   R1,8(R3)            load OP2 address in PL
                                                 4512
00032C92  4111 0008                      00000008 4513           LA    R1,8(R1)
00032C96  E313 0010 0024                 00000010 4514           STG   R1,16(R3)           load OP3 address in PL
                                                 4515
00032C9C  4111 0008                      00000008 4516           LA    R1,8(R1)
00032CA0  E313 0018 0024                 00000018 4517           STG   R1,24(R3)           load OP4 address in PL
                                                 4518
00032CA6  B982 0000                              4519           XGR   R0,R0
00032CAA  4100 0023                      00000023 4520           LA    R0,35
00032CAE  EE00 0000 3000       00000000  00000000 4521           PLO   0,0,0,0(R3)         Quadruple C/S (QCSG: 35)
                                                 4522
00032CB4  B98D 0020                              4523           EPSW  R2,R0               extract psw
00032CB8  5020 5010                      00000010 4524           ST    R2,CCPSW              to save CC
00032CBC  E310 500B 0076                 0000000B 4525           LB    R1,CCMASK           (failure CC mask)
00032CC2  8910 0004                      00000004 4526           SLL   R1,4                (shift to BC instr CC position)
00032CC6  4410 5104                      00032D24 4527           EX    R1,F352TSTCC            fail if...
                                                 4528
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                00032CCA  00000001  4529 F352CKUR EQU   *
00032CCA  D200 5009 3030        00000009  00000030  4530          MVC   URFOUND,48(R3)      Extract UR
00032CD0  D500 5008 3030        00000008  00000030  4531          CLC   UR,48(R3)
00032CD6  4780 50BE                       00032CDE  4532          BE    F352CKRES
00032CDA  45F0 82C8                       000022C8  4533          BAL   R15,URMSG
                                                    4534 *
                                                    4535 * compare results
                                                    4536 *
                                00032CDE  00000001  4537 F352CKRES EQU   *
                                                    4538 *                              reset adresses
00032CDE  E310 501C 0014                  0000001C  4539          LGF   R1,OPSRCA
00032CE4  E320 5028 0014                  00000028  4540          LGF   R2,RANDXA
00032CEA  E330 5024 0014                  00000024  4541          LGF   R3,PLA
00032CF0  E340 5020 0014                  00000020  4542          LGF   R4,OPREPA
                                                    4543
00032CF6  D507 1000 3040        00000000  00000040  4544          CLC   0(8,R1),64(R3)
00032CFC  4770 50FE                       00032D1E  4545          BNE   F352XX              OP1C Different, error
00032D00  D507 1008 3048        00000008  00000048  4546          CLC   8(8,R1),72(R3)
00032D06  4770 50FE                       00032D1E  4547          BNE   F352XX              OP2C Different, error
00032D0A  D507 1010 3050        00000010  00000050  4548          CLC   16(8,R1),80(R3)
00032D10  4770 50FE                       00032D1E  4549          BNE   F352XX              OP3C Different, error
00032D14  D507 1018 3058        00000018  00000058  4550          CLC   24(8,R1),88(R3)
00032D1A  478B 0000                       00000000  4551          BE    0(R11)              OP4Csame, return
                                                    4552
                                00032D1E  00000001  4553 F352XX   EQU   *
00032D1E  45F0 83E8                       000023E8  4554          BAL   R15,FAILMSG
00032D22  07FB                                      4555          BR    R11
                                                    4556
00032D24  4700 5108                       00032D28  4557 F352TSTCC BC    0,F352CCMSG        fail if unexpected condition code
                                00032D28  00000001  4558 F352CCMSG EQU   *
00032D28  45F0 8196                       00002196  4559          BAL   R15,CCMSG
00032D2C  47F0 50AA                       00032CCA  4560          B     F352CKUR
                                                    4561
                                                    4562          TSTDATA
00032D30                        00032D30  00033200  4563+         ORG   PLOETST+(128*K)+25*(3*K)+(K+512)
                                                    4564          TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00033200  6E6E6ED6 D7C4E2E3                         4565+         DC    CL32'>>>OPDST63<<<' eye catcher
00033208  F6F34C4C 4C404040
00033210  40404040 40404040
00033218  40404040 40404040
00033220                                            4566+OPDST63  DS    0C
00033220  00000000 00000000                         4567          DC    XL256'0'
00033228  00000000 00000000
00033230  00000000 00000000
00033238  00000000 00000000
00033240  00000000 00000000
00033248  00000000 00000000
00033250  00000000 00000000
00033258  00000000 00000000
00033260  00000000 00000000
00033268  00000000 00000000
00033270  00000000 00000000
00033278  00000000 00000000
00033280  00000000 00000000
00033288  00000000 00000000
00033290  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00033298  00000000 00000000
000332A0  00000000 00000000
000332A8  00000000 00000000
000332B0  00000000 00000000
000332B8  00000000 00000000
000332C0  00000000 00000000
000332C8  00000000 00000000
000332D0  00000000 00000000
000332D8  00000000 00000000
000332E0  00000000 00000000
000332E8  00000000 00000000
000332F0  00000000 00000000
000332F8  00000000 00000000
00033300  00000000 00000000
00033308  00000000 00000000
00033310  00000000 00000000
00033318  00000000 00000000
                                        4568
                                        4569          TSTLBL OPSRC              Source: OP1-OP4
00033320  6E6E6ED6 D7E2D9C3             4570+         DC    CL32'>>>OPSRC63<<<' eye catcher
00033328  F6F34C4C 4C404040
00033330  40404040 40404040
00033338  40404040 40404040
00033340                                4571+OPSRC63  DS    0C
00033340  00000000 00000000             4572          DC    XL256'0'
00033348  00000000 00000000
00033350  00000000 00000000
00033358  00000000 00000000
00033360  00000000 00000000
00033368  00000000 00000000
00033370  00000000 00000000
00033378  00000000 00000000
00033380  00000000 00000000
00033388  00000000 00000000
00033390  00000000 00000000
00033398  00000000 00000000
000333A0  00000000 00000000
000333A8  00000000 00000000
000333B0  00000000 00000000
000333B8  00000000 00000000
000333C0  00000000 00000000
000333C8  00000000 00000000
000333D0  00000000 00000000
000333D8  00000000 00000000
000333E0  00000000 00000000
000333E8  00000000 00000000
000333F0  00000000 00000000
000333F8  00000000 00000000
00033400  00000000 00000000
00033408  00000000 00000000
00033410  00000000 00000000
00033418  00000000 00000000
00033420  00000000 00000000
00033428  00000000 00000000
00033430  00000000 00000000
00033438  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   4573
                                                   4574          TSTLBL PL                 Parameter List
00033440  6E6E6ED7 D3F6F34C                        4575+         DC    CL32'>>>PL63<<<'    eye catcher
00033448  4C4C4040 40404040
00033450  40404040 40404040
00033458  40404040 40404040
00033460                                           4576+PL63     DS    0C
00033460  00000000 00000000                        4577          DC    XL352'00'
00033468  00000000 00000000
00033470  00000000 00000000
00033478  00000000 00000000
00033480  00000000 00000000
00033488  00000000 00000000
00033490  00000000 00000000
00033498  00000000 00000000
000334A0  00000000 00000000
000334A8  00000000 00000000
000334B0  00000000 00000000
000334B8  00000000 00000000
000334C0  00000000 00000000
000334C8  00000000 00000000
000334D0  00000000 00000000
000334D8  00000000 00000000
000334E0  00000000 00000000
000334E8  00000000 00000000
000334F0  00000000 00000000
000334F8  00000000 00000000
00033500  00000000 00000000
00033508  00000000 00000000
00033510  00000000 00000000
00033518  00000000 00000000
00033520  00000000 00000000
00033528  00000000 00000000
00033530  00000000 00000000
00033538  00000000 00000000
00033540  00000000 00000000
00033548  00000000 00000000
00033550  00000000 00000000
00033558  00000000 00000000
00033560  00000000 00000000
00033568  00000000 00000000
00033570  00000000 00000000
00033578  00000000 00000000
00033580  00000000 00000000
00033588  00000000 00000000
00033590  00000000 00000000
00033598  00000000 00000000
000335A0  00000000 00000000
000335A8  00000000 00000000
000335B0  00000000 00000000
000335B8  00000000 00000000
                                                   4578
                                                   4579          TSTLBL FMSG               fail message
000335C0  6E6E6EC6 D4E2C7F6                        4580+         DC    CL32'>>>FMSG63<<<'  eye catcher
000335C8  F34C4C4C 40404040
000335D0  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000335D8  40404040 40404040
000335E0                                         4581+FMSG63   DS    0C
000335E0  86818993 858440D7                      4582          DC    C'failed PLO-Extension '
000335E8  D3D660C5 A7A38595
000335F0  A2899695 40
000335F5  D8A48184 99A49793                      4583          DC    CL96'Quadruple Compare and Swap (QCSG: 35): 2nd NE'
000335FD  8540C396 94978199
00033605  85408195 8440E2A6
0003360D  8197404D D8C3E2C7
00033615  7A40F3F5 5D7A40F2
0003361D  958440D5 C5404040
00033625  40404040 40404040
0003362D  40404040 40404040
00033635  40404040 40404040
0003363D  40404040 40404040
00033645  40404040 40404040
0003364D  40404040 40404040
                                                 4584
                                                 4585          TSTEND
00033655  C5D5C440 968640A3                      4586+         DC    CL32'END of test: 63 <<<'
0003365D  85A2A37A 40F6F340
00033665  4C4C4C40 40404040
0003366D  40404040 40404040
                                                 4587
                                                 4588 * ------------------------------------------------------------------
                                                 4589 * Test Quadruple Compare and Swap (QCSG: 35) instruction : 3rd NE
                                                 4590 * ------------------------------------------------------------------
                                                 4591          TSTHDR 2,3,35             3rd Operand NOT Equal (cc=2, UR=3)
00033675                        00033675  00033800 4592+         ORG   PLOETST+(128*K)+26*(3*K)
00033800  E2E3C1D9 E3409686                      4593+         DC    CL32'START of test: 64'
00033808  40A385A2 A37A40F6
00033810  F4404040 40404040
00033818  40404040 40404040
00033820                        00033820         4594+         USING *,R5                base for test data and test routine
00033820                        00000000         4595+         USING PLOETEST,R5
00033820  0003384C                               4596+T64      DC    A(CODE64)           address of test code
00033824  0040                                   4597+         DC    H'64'               test number
00033826  00                                     4598+         DC    X'00'
00033827  23                                     4599+         DC    HL1'35'             Function
00033828  03                                     4600+         DC    HL1'3'              expected unequal results (UR)
00033829  00                                     4601+         DC    HL1'00'             found UR
0003382A  02                                     4602+         DC    HL1'2'              cc
0003382B  0D                                     4603+         DC    HL1'13'             not expected CC mask
0003382C  00                                     4604+         DS    X                   extracted cc
00033830  00000000 00000000                      4605+         DS    2F                  extract PSW after test (has CC)
00033838  000341E0                               4606+         DC    A(FMSG64)           address of test fail message
0003383C  00033F40                               4607+         DC    A(OPSRC64)          address of operand source
00033840  00033E20                               4608+         DC    A(OPDST64)          address of operand destination
00033844  00034060                               4609+         DC    A(PL64)             address of parameter list
00033848  000041F8                               4610+         DC    A(RANDX+64)         address of random source data
                                                 4611
                                                 4612          TSTCODE
0003384C                                         4613+CODE64   DS    0F
                                                 4614 *                                  load random source
0003384C  E310 501C 0014              0000001C   4615          LGF   R1,OPSRCA
00033852  E320 5028 0014              00000028   4616          LGF   R2,RANDXA
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00033858  E330 5024 0014                 00000024  4617          LGF   R3,PLA
0003385E  E340 5020 0014                 00000020  4618          LGF   R4,OPREPA
                                                   4619 *
                                                   4620 * Quadruple Compare and Swap (QCSG: 35)
                                                   4621 * Build Parambter List
                                                   4622 *
00033864  D21F 1000 2000      00000000   00000000  4623          MVC   0(32,R1),0(R2)      load Op1-Op4 data in source
0003386A  D21F 3040 1000      00000040   00000000  4624          MVC   64(32,R3),0(R1)     copy to PL OP1-Op4 Compare
                                                   4625
00033870  D21F 4000 2080      00000000   00000080  4626          MVC   0(32,R4),128(R2)    load initial Op1-Op4 data
00033876  D21F 3080 4000      00000080   00000000  4627          MVC   128(32,R3),0(R4)    copy to PL Op1-Op4 Replacement
                                                   4628 ****
                                                   4629 * force a failure by changing the 3rd operand
                                                   4630 ****
0003387C  D203 1010 4000      00000010   00000000  4631          MVC   16(4,R1),0(R4)
                                                   4632 ****
                                                   4633
00033882  E313 0000 0024                 00000000  4634          STG   R1,0(R3)            load OP1 address in PL
                                                   4635
00033888  4111 0008                      00000008  4636          LA    R1,8(R1)
0003388C  E313 0008 0024                 00000008  4637          STG   R1,8(R3)            load OP2 address in PL
                                                   4638
00033892  4111 0008                      00000008  4639          LA    R1,8(R1)
00033896  E313 0010 0024                 00000010  4640          STG   R1,16(R3)           load OP3 address in PL
                                                   4641
0003389C  4111 0008                      00000008  4642          LA    R1,8(R1)
000338A0  E313 0018 0024                 00000018  4643          STG   R1,24(R3)           load OP4 address in PL
                                                   4644
000338A6  B982 0000                                4645          XGR   R0,R0
000338AA  4100 0023                      00000023  4646          LA    R0,35
000338AE  EE00 0000 3000      00000000   00000000  4647          PLO   0,0,0,0(R3)         Quadruple C/S (QCSG: 35)
                                                   4648
000338B4  B98D 0020                                4649          EPSW  R2,R0               extract psw
000338B8  5020 5010                      00000010  4650          ST    R2,CCPSW              to save CC
000338BC  E310 500B 0076                 0000000B  4651          LB    R1,CCMASK           (failure CC mask)
000338C2  8910 0004                      00000004  4652          SLL   R1,4                (shift to BC instr CC position)
000338C6  4410 5104                      00033924  4653          EX    R1,F353TSTCC            fail if...
                                                   4654
                                000338CA  00000001  4655 F353CKUR EQU   *
000338CA  D200 5009 3030      00000009   00000030  4656          MVC   URFOUND,48(R3)      Extract UR
000338D0  D500 5008 3030      00000008   00000030  4657          CLC   UR,48(R3)
000338D6  4780 50BE                      000338DE  4658          BE    F353CKRES
000338DA  45F0 82C8                      000022C8  4659          BAL   R15,URMSG
                                                   4660 *
                                                   4661 * compare results
                                                   4662 *
                                000338DE  00000001  4663 F353CKRES EQU   *
                                                   4664 *                            reset adresses
000338DE  E310 501C 0014                 0000001C  4665          LGF   R1,OPSRCA
000338E4  E320 5028 0014                 00000028  4666          LGF   R2,RANDXA
000338EA  E330 5024 0014                 00000024  4667          LGF   R3,PLA
000338F0  E340 5020 0014                 00000020  4668          LGF   R4,OPREPA
                                                   4669
000338F6  D507 1000 3040      00000000   00000040  4670          CLC   0(8,R1),64(R3)
000338FC  4770 50FE                      0003391E  4671          BNE   F353XX              OP1C Different, error
00033900  D507 1008 3048      00000008   00000048  4672          CLC   8(8,R1),72(R3)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT
00033906  4770 50FE                      0003391E  4673           BNE   F353XX               OP2C Different, error
0003390A  D507 1010 3050      00000010  00000050  4674           CLC   16(8,R1),80(R3)
00033910  4770 50FE                      0003391E  4675           BNE   F353XX               OP3C Different, error
00033914  D507 1018 3058      00000018  00000058  4676           CLC   24(8,R1),88(R3)
0003391A  478B 0000                      00000000  4677           BE    0(R11)               OP4Csame, return
                                                   4678
                              0003391E  00000001  4679 F353XX    EQU   *
0003391E  45F0 83E8                      000023E8  4680           BAL   R15,FAILMSG
00033922  07FB                                     4681           BR    R11
                                                   4682
00033924  4700 5108                      00033928  4683 F353TSTCC BC    0,F353CCMSG           fail if unexpected condition code
                              00033928  00000001  4684 F353CCMSG EQU    *
00033928  45F0 8196                      00002196  4685           BAL   R15,CCMSG
0003392C  47F0 50AA                      000338CA  4686           B     F353CKUR
                                                   4687
                                                   4688           TSTDATA
00033930                      00033930  00033E00  4689+          ORG   PLOETST+(128*K)+26*(3*K)+(K+512)
                                                   4690           TSTLBL OPDST               Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00033E00  6E6E6ED6 D7C4E2E3                       4691+          DC    CL32'>>>OPDST64<<<'  eye catcher
00033E08  F6F44C4C 4C404040
00033E10  40404040 40404040
00033E18  40404040 40404040
00033E20                                          4692+OPDST64   DS    0C
00033E20  00000000 00000000                       4693           DC    XL256'0'
00033E28  00000000 00000000
00033E30  00000000 00000000
00033E38  00000000 00000000
00033E40  00000000 00000000
00033E48  00000000 00000000
00033E50  00000000 00000000
00033E58  00000000 00000000
00033E60  00000000 00000000
00033E68  00000000 00000000
00033E70  00000000 00000000
00033E78  00000000 00000000
00033E80  00000000 00000000
00033E88  00000000 00000000
00033E90  00000000 00000000
00033E98  00000000 00000000
00033EA0  00000000 00000000
00033EA8  00000000 00000000
00033EB0  00000000 00000000
00033EB8  00000000 00000000
00033EC0  00000000 00000000
00033EC8  00000000 00000000
00033ED0  00000000 00000000
00033ED8  00000000 00000000
00033EE0  00000000 00000000
00033EE8  00000000 00000000
00033EF0  00000000 00000000
00033EF8  00000000 00000000
00033F00  00000000 00000000
00033F08  00000000 00000000
00033F10  00000000 00000000
00033F18  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   4695           TSTLBL OPSRC               Source: OP1-OP4
00033F20  6E6E6ED6 D7E2D9C3                       4696+          DC    CL32'>>>OPSRC64<<<' eye catcher
00033F28  F6F44C4C 4C404040
00033F30  40404040 40404040
00033F38  40404040 40404040
00033F40                                          4697+OPSRC64   DS    0C
00033F40  00000000 00000000                       4698           DC    XL256'0'
00033F48  00000000 00000000
00033F50  00000000 00000000
00033F58  00000000 00000000
00033F60  00000000 00000000
00033F68  00000000 00000000
00033F70  00000000 00000000
00033F78  00000000 00000000
00033F80  00000000 00000000
00033F88  00000000 00000000
00033F90  00000000 00000000
00033F98  00000000 00000000
00033FA0  00000000 00000000
00033FA8  00000000 00000000
00033FB0  00000000 00000000
00033FB8  00000000 00000000
00033FC0  00000000 00000000
00033FC8  00000000 00000000
00033FD0  00000000 00000000
00033FD8  00000000 00000000
00033FE0  00000000 00000000
00033FE8  00000000 00000000
00033FF0  00000000 00000000
00033FF8  00000000 00000000
00034000  00000000 00000000
00034008  00000000 00000000
00034010  00000000 00000000
00034018  00000000 00000000
00034020  00000000 00000000
00034028  00000000 00000000
00034030  00000000 00000000
00034038  00000000 00000000
                                                   4699
                                                   4700           TSTLBL PL                  Parameter List
00034040  6E6E6ED7 D3F6F44C                       4701+          DC    CL32'>>>PL64<<<'     eye catcher
00034048  4C4C4040 40404040
00034050  40404040 40404040
00034058  40404040 40404040
00034060                                          4702+PL64      DS    0C
00034060  00000000 00000000                       4703           DC    XL352'00'
00034068  00000000 00000000
00034070  00000000 00000000
00034078  00000000 00000000
00034080  00000000 00000000
00034088  00000000 00000000
00034090  00000000 00000000
00034098  00000000 00000000
000340A0  00000000 00000000
000340A8  00000000 00000000
000340B0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000340B8  00000000 00000000
000340C0  00000000 00000000
000340C8  00000000 00000000
000340D0  00000000 00000000
000340D8  00000000 00000000
000340E0  00000000 00000000
000340E8  00000000 00000000
000340F0  00000000 00000000
000340F8  00000000 00000000
00034100  00000000 00000000
00034108  00000000 00000000
00034110  00000000 00000000
00034118  00000000 00000000
00034120  00000000 00000000
00034128  00000000 00000000
00034130  00000000 00000000
00034138  00000000 00000000
00034140  00000000 00000000
00034148  00000000 00000000
00034150  00000000 00000000
00034158  00000000 00000000
00034160  00000000 00000000
00034168  00000000 00000000
00034170  00000000 00000000
00034178  00000000 00000000
00034180  00000000 00000000
00034188  00000000 00000000
00034190  00000000 00000000
00034198  00000000 00000000
000341A0  00000000 00000000
000341A8  00000000 00000000
000341B0  00000000 00000000
000341B8  00000000 00000000
                                        4704
                                        4705          TSTLBL FMSG                fail message
000341C0  6E6E6EC6 D4E2C7F6             4706+         DC    CL32'>>>FMSG64<<<'   eye catcher
000341C8  F44C4C4C 40404040
000341D0  40404040 40404040
000341D8  40404040 40404040
000341E0                                4707+FMSG64   DS    0C
000341E0  86818993 858440D7             4708          DC    C'failed PLO-Extension '
000341E8  D3D660C5 A7A38595
000341F0  A2899695 40
000341F5  D8A48184 99A49793             4709          DC    CL96'Quadruple Compare and Swap (QCSG: 35): 3rd NE'
000341FD  8540C396 94978199
00034205  85408195 8440E2A6
0003420D  8197404D D8C3E2C7
00034215  7A40F3F5 5D7A40F3
0003421D  998440D5 C5404040
00034225  40404040 40404040
0003422D  40404040 40404040
00034235  40404040 40404040
0003423D  40404040 40404040
00034245  40404040 40404040
0003424D  40404040 40404040
                                        4710
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   4711          TSTEND
00034255  C5D5C440 968640A3                       4712+         DC    CL32'END of test: 64 <<<'
0003425D  85A2A37A 40F6F440
00034265  4C4C4C40 40404040
0003426D  40404040 40404040
                                                   4713
                                                   4714 * ------------------------------------------------------------------
                                                   4715 * Test Quadruple Compare and Swap (QCSG: 35) instruction : 4th NE
                                                   4716 * ------------------------------------------------------------------
                                                   4717          TSTHDR 2,4,35             4th Operand NOT Equal (cc=2, UR=4)
00034275                    00034275  00034400     4718+         ORG   PLOETST+(128*K)+27*(3*K)
00034400  E2E3C1D9 E3409686                       4719+         DC    CL32'START of test: 65'
00034408  40A385A2 A37A40F6
00034410  F5404040 40404040
00034418  40404040 40404040
00034420                    00034420              4720+         USING *,R5                base for test data and test routine
00034420                    00000000              4721+         USING PLOETEST,R5
00034420  0003444C                                4722+T65      DC    A(CODE65)           address of test code
00034424  0041                                    4723+         DC    H'65'               test number
00034426  00                                      4724+         DC    X'00'
00034427  23                                      4725+         DC    HL1'35'             Function
00034428  04                                      4726+         DC    HL1'4'              expected unequal results (UR)
00034429  00                                      4727+         DC    HL1'00'             found UR
0003442A  02                                      4728+         DC    HL1'2'              cc
0003442B  0D                                      4729+         DC    HL1'13'             not expected CC mask
0003442C  00                                      4730+         DS    X                   extracted cc
00034430  00000000 00000000                       4731+         DS    2F                  extract PSW after test (has CC)
00034438  00034DE0                                4732+         DC    A(FMSG65)           address of test fail message
0003443C  00034B40                                4733+         DC    A(OPSRC65)          address of operand source
00034440  00034A20                                4734+         DC    A(OPDST65)          address of operand destination
00034444  00034C60                                4735+         DC    A(PL65)             address of parameter list
00034448  000041F9                                4736+         DC    A(RANDX+65)         address of random source data
                                                   4737
                                                   4738          TSTCODE
0003444C                                          4739+CODE65   DS    0F
                                                   4740 *                                  load random source
0003444C  E310 501C 0014              0000001C    4741          LGF   R1,OPSRCA
00034452  E320 5028 0014              00000028    4742          LGF   R2,RANDXA
00034458  E330 5024 0014              00000024    4743          LGF   R3,PLA
0003445E  E340 5020 0014              00000020    4744          LGF   R4,OPREPA
                                                   4745 *
                                                   4746 * Quadruple Compare and Swap (QCSG: 35)
                                                   4747 * Build Parambter List
                                                   4748 *
00034464  D21F 1000 2000  00000000    00000000    4749          MVC   0(32,R1),0(R2)      load Op1-Op4 data in source
0003446A  D21F 3040 1000  00000040    00000000    4750          MVC   64(32,R3),0(R1)     copy to PL OP1-Op4 Compare
                                                   4751
00034470  D21F 4000 2080  00000000    00000080    4752          MVC   0(32,R4),128(R2)    load initial Op1-Op4 data
00034476  D21F 3080 4000  00000080    00000000    4753          MVC   128(32,R3),0(R4)    copy to PL Op1-Op4 Replacement
                                                   4754 ****
                                                   4755 * force a failure by changing the 3rd operand
                                                   4756 ****
0003447C  D203 1018 4000  00000018    00000000    4757          MVC   24(4,R1),0(R4)
                                                   4758 ****
                                                   4759
00034482  E313 0000 0024              00000000    4760          STG   R1,0(R3)            load OP1 address in PL
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    4761
00034488  4111 0008                     00000008  4762          LA    R1,8(R1)
0003448C  E313 0008 0024                 00000008  4763          STG   R1,8(R3)          load OP2 address in PL
                                                    4764
00034492  4111 0008                     00000008  4765          LA    R1,8(R1)
00034496  E313 0010 0024                 00000010  4766          STG   R1,16(R3)         load OP3 address in PL
                                                    4767
0003449C  4111 0008                     00000008  4768          LA    R1,8(R1)
000344A0  E313 0018 0024                 00000018  4769          STG   R1,24(R3)         load OP4 address in PL
                                                    4770
000344A6  B982 0000                               4771          XGR   R0,R0
000344AA  4100 0023                     00000023  4772          LA    R0,35
000344AE  EE00 0000 3000  00000000      00000000  4773          PLO   0,0,0,0(R3)       Quadruple C/S (QCSG: 35)
                                                    4774
000344B4  B98D 0020                               4775          EPSW  R2,R0             extract psw
000344B8  5020 5010                     00000010  4776          ST    R2,CCPSW            to save CC
000344BC  E310 500B 0076                0000000B  4777          LB    R1,CCMASK         (failure CC mask)
000344C2  8910 0004                     00000004  4778          SLL   R1,4              (shift to BC instr CC position)
000344C6  4410 5104                     00034524  4779          EX    R1,F354TSTCC        fail if...
                                                    4780
                          000344CA      00000001  4781 F354CKUR EQU   *
000344CA  D200 5009 3030  00000009      00000030  4782          MVC   URFOUND,48(R3)    Extract UR
000344D0  D500 5008 3030  00000008      00000030  4783          CLC   UR,48(R3)
000344D6  4780 50BE                     000344DE  4784          BE    F354CKRES
000344DA  45F0 82C8                     000022C8  4785          BAL   R15,URMSG
                                                    4786 *
                                                    4787 * compare results
                                                    4788 *
                          000344DE      00000001  4789 F354CKRES EQU   *
                                                    4790 *                            reset adresses
000344DE  E310 501C 0014                0000001C  4791          LGF   R1,OPSRCA
000344E4  E320 5028 0014                00000028  4792          LGF   R2,RANDXA
000344EA  E330 5024 0014                00000024  4793          LGF   R3,PLA
000344F0  E340 5020 0014                00000020  4794          LGF   R4,OPREPA
                                                    4795
000344F6  D507 1000 3040  00000000      00000040  4796          CLC   0(8,R1),64(R3)
000344FC  4770 50FE                     0003451E  4797          BNE   F354XX            OP1C Different, error
00034500  D507 1008 3048  00000008      00000048  4798          CLC   8(8,R1),72(R3)
00034506  4770 50FE                     0003451E  4799          BNE   F354XX            OP2C Different, error
0003450A  D507 1010 3050  00000010      00000050  4800          CLC   16(8,R1),80(R3)
00034510  4770 50FE                     0003451E  4801          BNE   F354XX            OP3C Different, error
00034514  D507 1018 3058  00000018      00000058  4802          CLC   24(8,R1),88(R3)
0003451A  478B 0000                     00000000  4803          BE    0(R11)            OP4Csame, return
                                                    4804
                          0003451E      00000001  4805 F354XX   EQU   *
0003451E  45F0 83E8                     000023E8  4806          BAL   R15,FAILMSG
00034522  07FB                                    4807          BR    R11
                                                    4808
00034524  4700 5108                     00034528  4809 F354TSTCC BC   0,F354CCMSG       fail if unexpected condition code
                          00034528      00000001  4810 F354CCMSG EQU   *
00034528  45F0 8196                     00002196  4811          BAL   R15,CCMSG
0003452C  47F0 50AA                     000344CA  4812          B     F354CKUR
                                                    4813
                                                    4814          TSTDATA
00034530                  00034530      00034A00  4815+         ORG   PLOETST+(128*K)+27*(3*K)+(K+512)
                                                    4816          TSTLBL OPDST            Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00034A00  6E6E6ED6 D7C4E2E3                       4817+         DC    CL32'>>>OPDST65<<<' eye catcher
00034A08  F6F54C4C 4C404040
00034A10  40404040 40404040
00034A18  40404040 40404040
00034A20                                          4818+OPDST65  DS    0C
00034A20  00000000 00000000                       4819          DC    XL256'0'
00034A28  00000000 00000000
00034A30  00000000 00000000
00034A38  00000000 00000000
00034A40  00000000 00000000
00034A48  00000000 00000000
00034A50  00000000 00000000
00034A58  00000000 00000000
00034A60  00000000 00000000
00034A68  00000000 00000000
00034A70  00000000 00000000
00034A78  00000000 00000000
00034A80  00000000 00000000
00034A88  00000000 00000000
00034A90  00000000 00000000
00034A98  00000000 00000000
00034AA0  00000000 00000000
00034AA8  00000000 00000000
00034AB0  00000000 00000000
00034AB8  00000000 00000000
00034AC0  00000000 00000000
00034AC8  00000000 00000000
00034AD0  00000000 00000000
00034AD8  00000000 00000000
00034AE0  00000000 00000000
00034AE8  00000000 00000000
00034AF0  00000000 00000000
00034AF8  00000000 00000000
00034B00  00000000 00000000
00034B08  00000000 00000000
00034B10  00000000 00000000
00034B18  00000000 00000000
                                                  4820
                                                  4821          TSTLBL OPSRC              Source: OP1-OP4
00034B20  6E6E6ED6 D7E2D9C3                       4822+         DC    CL32'>>>OPSRC65<<<' eye catcher
00034B28  F6F54C4C 4C404040
00034B30  40404040 40404040
00034B38  40404040 40404040
00034B40                                          4823+OPSRC65  DS    0C
00034B40  00000000 00000000                       4824          DC    XL256'0'
00034B48  00000000 00000000
00034B50  00000000 00000000
00034B58  00000000 00000000
00034B60  00000000 00000000
00034B68  00000000 00000000
00034B70  00000000 00000000
00034B78  00000000 00000000
00034B80  00000000 00000000
00034B88  00000000 00000000
00034B90  00000000 00000000
00034B98  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00034BA0  00000000 00000000
00034BA8  00000000 00000000
00034BB0  00000000 00000000
00034BB8  00000000 00000000
00034BC0  00000000 00000000
00034BC8  00000000 00000000
00034BD0  00000000 00000000
00034BD8  00000000 00000000
00034BE0  00000000 00000000
00034BE8  00000000 00000000
00034BF0  00000000 00000000
00034BF8  00000000 00000000
00034C00  00000000 00000000
00034C08  00000000 00000000
00034C10  00000000 00000000
00034C18  00000000 00000000
00034C20  00000000 00000000
00034C28  00000000 00000000
00034C30  00000000 00000000
00034C38  00000000 00000000
                                        4825
                                        4826          TSTLBL PL                   Parameter List
00034C40  6E6E6ED7 D3F6F54C             4827+         DC    CL32'>>>PL65<<<'      eye catcher
00034C48  4C4C4040 40404040
00034C50  40404040 40404040
00034C58  40404040 40404040
00034C60                                4828+PL65     DS    0C
00034C60  00000000 00000000             4829          DC    XL352'00'
00034C68  00000000 00000000
00034C70  00000000 00000000
00034C78  00000000 00000000
00034C80  00000000 00000000
00034C88  00000000 00000000
00034C90  00000000 00000000
00034C98  00000000 00000000
00034CA0  00000000 00000000
00034CA8  00000000 00000000
00034CB0  00000000 00000000
00034CB8  00000000 00000000
00034CC0  00000000 00000000
00034CC8  00000000 00000000
00034CD0  00000000 00000000
00034CD8  00000000 00000000
00034CE0  00000000 00000000
00034CE8  00000000 00000000
00034CF0  00000000 00000000
00034CF8  00000000 00000000
00034D00  00000000 00000000
00034D08  00000000 00000000
00034D10  00000000 00000000
00034D18  00000000 00000000
00034D20  00000000 00000000
00034D28  00000000 00000000
00034D30  00000000 00000000
00034D38  00000000 00000000
00034D40  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00034D48  00000000 00000000
00034D50  00000000 00000000
00034D58  00000000 00000000
00034D60  00000000 00000000
00034D68  00000000 00000000
00034D70  00000000 00000000
00034D78  00000000 00000000
00034D80  00000000 00000000
00034D88  00000000 00000000
00034D90  00000000 00000000
00034D98  00000000 00000000
00034DA0  00000000 00000000
00034DA8  00000000 00000000
00034DB0  00000000 00000000
00034DB8  00000000 00000000
                                          4830
                                          4831          TSTLBL FMSG                fail message
00034DC0  6E6E6EC6 D4E2C7F6               4832+         DC    CL32'>>>FMSG65<<<'   eye catcher
00034DC8  F54C4C4C 40404040
00034DD0  40404040 40404040
00034DD8  40404040 40404040
00034DE0                                  4833+FMSG65   DS    0C
00034DE0  86818993 858440D7               4834          DC    C'failed PLO-Extension '
00034DE8  D3D660C5 A7A38595
00034DF0  A2899695 40
00034DF5  D8A48184 99A49793               4835          DC    CL96'Quadruple Compare and Swap (QCSG: 35): 4th NE'
00034DFD  8540C396 94978199
00034E05  85408195 8440E2A6
00034E0D  8197404D D8C3E2C7
00034E15  7A40F3F5 5D7A40F4
00034E1D  A38840D5 C5404040
00034E25  40404040 40404040
00034E2D  40404040 40404040
00034E35  40404040 40404040
00034E3D  40404040 40404040
00034E45  40404040 40404040
00034E4D  40404040 40404040
                                          4836
                                          4837          TSTEND
00034E55  C5D5C440 968640A3               4838+         DC    CL32'END of test: 65 <<<'
00034E5D  85A2A37A 40F6F540
00034E65  4C4C4C40 40404040
00034E6D  40404040 40404040
                                          4839
                                          4840 * ------------------------------------------------------------------
                                          4841 * Test Quadruple Compare and Swap (QCSG: 35) instruction : All EQ
                                          4842 * ------------------------------------------------------------------
                                          4843          TSTHDR 0,99,35              ALL EQ Equal (cc=0, UR=99
00034E75                        00034E75  00035000      4844+         ORG   PLOETST+(128*K)+28*(3*K)
00035000  E2E3C1D9 E3409686               4845+         DC    CL32'START of test: 66'
00035008  40A385A2 A37A40F6
00035010  F6404040 40404040
00035018  40404040 40404040
00035020                        00035020                4846+         USING *,R5                 base for test data and test routine
00035020                        00000000                4847+         USING PLOETEST,R5
00035020  0003504C                                      4848+T66      DC    A(CODE66)            address of test code
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00035024  0042                                    4849+          DC    H'66'              test number
00035026  00                                      4850+          DC    X'00'
00035027  23                                      4851+          DC    HL1'35'            Function
00035028  63                                      4852+          DC    HL1'99'            expected unequal results (UR)
00035029  00                                      4853+          DC    HL1'00'            found UR
0003502A  00                                      4854+          DC    HL1'0'             cc
0003502B  07                                      4855+          DC    HL1'7'             not expected CC mask
0003502C  00                                      4856+          DS    X                  extracted cc
00035030  00000000 00000000                       4857+          DS    2F                 extract PSW after test (has CC)
00035038  000359E0                                4858+          DC    A(FMSG66)          address of test fail message
0003503C  00035740                                4859+          DC    A(OPSRC66)         address of operand source
00035040  00035620                                4860+          DC    A(OPDST66)         address of operand destination
00035044  00035860                                4861+          DC    A(PL66)            address of parameter list
00035048  000041FA                                4862+          DC    A(RANDX+66)        address of random source data
                                                  4863
                                                  4864           TSTCODE
0003504C                                          4865+CODE66    DS    0F
                                                  4866 *                                  load random source
0003504C  E310 501C 0014              0000001C    4867           LGF   R1,OPSRCA
00035052  E320 5028 0014              00000028    4868           LGF   R2,RANDXA
00035058  E330 5024 0014              00000024    4869           LGF   R3,PLA
0003505E  E340 5020 0014              00000020    4870           LGF   R4,OPREPA
                                                  4871 *
                                                  4872 * Quadruple Compare and Swap (QCSG: 35)
                                                  4873 * Build Parambter List
                                                  4874 *
00035064  D21F 1000 2000  00000000    00000000    4875           MVC   0(32,R1),0(R2)     load Op1-Op4 data in source
0003506A  D21F 3040 1000  00000040    00000000    4876           MVC   64(32,R3),0(R1)    copy to PL OP1-Op4 Compare
                                                  4877
00035070  D21F 4000 2080  00000000    00000080    4878           MVC   0(32,R4),128(R2)   load initial Op1-Op4 data
00035076  D21F 3080 4000  00000080    00000000    4879           MVC   128(32,R3),0(R4)   copy to PL Op1-Op4 Replacement
                                                  4880 ****
                                                  4881 * force a failure by changing the 3rd operand
                                                  4882 ****
                                                  4883 *         MVC   24(4,R1),0(R4)
                                                  4884 ****
                                                  4885
0003507C  E313 0000 0024              00000000    4886           STG   R1,0(R3)           load OP1 address in PL
                                                  4887
00035082  4111 0008                   00000008    4888           LA    R1,8(R1)
00035086  E313 0008 0024              00000008    4889           STG   R1,8(R3)           load OP2 address in PL
                                                  4890
0003508C  4111 0008                   00000008    4891           LA    R1,8(R1)
00035090  E313 0010 0024              00000010    4892           STG   R1,16(R3)          load OP3 address in PL
                                                  4893
00035096  4111 0008                   00000008    4894           LA    R1,8(R1)
0003509A  E313 0018 0024              00000018    4895           STG   R1,24(R3)          load OP4 address in PL
                                                  4896
000350A0  B982 0000                               4897           XGR   R0,R0
000350A4  4100 0023                   00000023    4898           LA    R0,35
000350A8  EE00 0000 3000  00000000    00000000    4899           PLO   0,0,0,0(R3)        Quadruple C/S (QCSG: 35)
                                                  4900
000350AE  B98D 0020                               4901           EPSW  R2,R0              extract psw
000350B2  5020 5010                   00000010    4902           ST    R2,CCPSW             to save CC
000350B6  E310 500B 0076              0000000B    4903           LB    R1,CCMASK          (failure CC mask)
000350BC  8910 0004                   00000004    4904           SLL   R1,4              (shift to BC instr CC position)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000350C0  4410 50FE                     0003511E  4905          EX    R1,F355TSTCC            fail if...
                                                  4906
                            000350C4  00000001  4907 F355CKUR EQU   *
000350C4  D200 5009 3030    00000009  00000030  4908          MVC   URFOUND,48(R3)      Extract UR
000350CA  D500 5008 3030    00000008  00000030  4909          CLC   UR,48(R3)
000350D0  4780 50B8                     000350D8  4910          BE    F355CKRES
000350D4  45F0 82C8                     000022C8  4911          BAL   R15,URMSG
                                                  4912 *
                                                  4913 * compare results
                                                  4914 *
                            000350D8  00000001  4915 F355CKRES EQU   *
                                                  4916 *                                  reset adresses
000350D8  E310 501C 0014                0000001C  4917          LGF   R1,OPSRCA
000350DE  E320 5028 0014                00000028  4918          LGF   R2,RANDXA
000350E4  E330 5024 0014                00000024  4919          LGF   R3,PLA
000350EA  E340 5020 0014                00000020  4920          LGF   R4,OPREPA
                                                  4921
                                                  4922 * All equal; replacement values copied to operand source
                                                  4923
000350F0  D507 1000 3060    00000000  00000060  4924          CLC   0(8,R1),96(R3)
000350F6  4770 50F8                     00035118  4925          BNE   F355XX              OP1R Different, error
000350FA  D507 1008 3068    00000008  00000068  4926          CLC   8(8,R1),104(R3)
00035100  4770 50F8                     00035118  4927          BNE   F355XX              OP2R Different, error
00035104  D507 1010 3070    00000010  00000070  4928          CLC   16(8,R1),112(R3)
0003510A  4770 50F8                     00035118  4929          BNE   F355XX              OP3R Different, error
0003510E  D507 1018 3078    00000018  00000078  4930          CLC   24(8,R1),120(R3)
00035114  478B 0000                     00000000  4931          BE    0(R11)              OP4R same, return
                                                  4932
                            00035118  00000001  4933 F355XX   EQU   *
00035118  45F0 83E8                     000023E8  4934          BAL   R15,FAILMSG
0003511C  07FB                                    4935          BR    R11
                                                  4936
0003511E  4700 5102                     00035122  4937 F355TSTCC BC   0,F355CCMSG         fail if unexpected condition code
                            00035122  00000001  4938 F355CCMSG EQU   *
00035122  45F0 8196                     00002196  4939          BAL   R15,CCMSG
00035126  47F0 50A4                     000350C4  4940          B     F355CKUR
                                                  4941
                                                  4942          TSTDATA
0003512A                    0003512A  00035600  4943+          ORG   PLOETST+(128*K)+28*(3*K)+(K+512)
                                                  4944          TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00035600  6E6E6ED6 D7C4E2E3                      4945+          DC    CL32'>>>OPDST66<<<' eye catcher
00035608  F6F64C4C 4C404040
00035610  40404040 40404040
00035618  40404040 40404040
00035620                                          4946+OPDST66  DS    0C
00035620  00000000 00000000                       4947          DC    XL256'0'
00035628  00000000 00000000
00035630  00000000 00000000
00035638  00000000 00000000
00035640  00000000 00000000
00035648  00000000 00000000
00035650  00000000 00000000
00035658  00000000 00000000
00035660  00000000 00000000
00035668  00000000 00000000
00035670  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00035678  00000000 00000000
00035680  00000000 00000000
00035688  00000000 00000000
00035690  00000000 00000000
00035698  00000000 00000000
000356A0  00000000 00000000
000356A8  00000000 00000000
000356B0  00000000 00000000
000356B8  00000000 00000000
000356C0  00000000 00000000
000356C8  00000000 00000000
000356D0  00000000 00000000
000356D8  00000000 00000000
000356E0  00000000 00000000
000356E8  00000000 00000000
000356F0  00000000 00000000
000356F8  00000000 00000000
00035700  00000000 00000000
00035708  00000000 00000000
00035710  00000000 00000000
00035718  00000000 00000000
                                          4948
                                          4949          TSTLBL OPSRC              Source: OP1-OP4
00035720  6E6E6ED6 D7E2D9C3               4950+         DC    CL32'>>>OPSRC66<<<' eye catcher
00035728  F6F64C4C 4C404040
00035730  40404040 40404040
00035738  40404040 40404040
00035740                                  4951+OPSRC66  DS    0C
00035740  00000000 00000000               4952          DC    XL256'0'
00035748  00000000 00000000
00035750  00000000 00000000
00035758  00000000 00000000
00035760  00000000 00000000
00035768  00000000 00000000
00035770  00000000 00000000
00035778  00000000 00000000
00035780  00000000 00000000
00035788  00000000 00000000
00035790  00000000 00000000
00035798  00000000 00000000
000357A0  00000000 00000000
000357A8  00000000 00000000
000357B0  00000000 00000000
000357B8  00000000 00000000
000357C0  00000000 00000000
000357C8  00000000 00000000
000357D0  00000000 00000000
000357D8  00000000 00000000
000357E0  00000000 00000000
000357E8  00000000 00000000
000357F0  00000000 00000000
000357F8  00000000 00000000
00035800  00000000 00000000
00035808  00000000 00000000
00035810  00000000 00000000
00035818  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00035820  00000000 00000000
00035828  00000000 00000000
00035830  00000000 00000000
00035838  00000000 00000000
                                         4953
                                         4954          TSTLBL PL                   Parameter List
00035840  6E6E6ED7 D3F6F64C              4955+         DC    CL32'>>>PL66<<<'      eye catcher
00035848  4C4C4040 40404040
00035850  40404040 40404040
00035858  40404040 40404040
00035860                                 4956+PL66     DS    0C
00035860  00000000 00000000              4957          DC    XL352'00'
00035868  00000000 00000000
00035870  00000000 00000000
00035878  00000000 00000000
00035880  00000000 00000000
00035888  00000000 00000000
00035890  00000000 00000000
00035898  00000000 00000000
000358A0  00000000 00000000
000358A8  00000000 00000000
000358B0  00000000 00000000
000358B8  00000000 00000000
000358C0  00000000 00000000
000358C8  00000000 00000000
000358D0  00000000 00000000
000358D8  00000000 00000000
000358E0  00000000 00000000
000358E8  00000000 00000000
000358F0  00000000 00000000
000358F8  00000000 00000000
00035900  00000000 00000000
00035908  00000000 00000000
00035910  00000000 00000000
00035918  00000000 00000000
00035920  00000000 00000000
00035928  00000000 00000000
00035930  00000000 00000000
00035938  00000000 00000000
00035940  00000000 00000000
00035948  00000000 00000000
00035950  00000000 00000000
00035958  00000000 00000000
00035960  00000000 00000000
00035968  00000000 00000000
00035970  00000000 00000000
00035978  00000000 00000000
00035980  00000000 00000000
00035988  00000000 00000000
00035990  00000000 00000000
00035998  00000000 00000000
000359A0  00000000 00000000
000359A8  00000000 00000000
000359B0  00000000 00000000
000359B8  00000000 00000000
                                         4958
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    4959           TSTLBL FMSG                fail message
000359C0  6E6E6EC6 D4E2C7F6                         4960+          DC    CL32'>>>FMSG66<<<'   eye catcher
000359C8  F64C4C4C 40404040
000359D0  40404040 40404040
000359D8  40404040 40404040
000359E0                                            4961+FMSG66    DS    0C
000359E0  86818993 858440D7                         4962           DC    C'failed PLO-Extension '
000359E8  D3D660C5 A7A38595
000359F0  A2899695 40
000359F5  D8A48184 99A49793                         4963           DC    CL96'Quadruple Compare and Swap (QCSG: 35): All EQ'
000359FD  8540C396 94978199
00035A05  85408195 8440E2A6
00035A0D  8197404D D8C3E2C7
00035A15  7A40F3F5 5D7A40C1
00035A1D  939340C5 D8404040
00035A25  40404040 40404040
00035A2D  40404040 40404040
00035A35  40404040 40404040
00035A3D  40404040 40404040
00035A45  40404040 40404040
00035A4D  40404040 40404040
                                                    4964
                                                    4965           TSTEND
00035A55  C5D5C440 968640A3                         4966+          DC    CL32'END of test: 66 <<<'
00035A5D  85A2A37A 40F6F640
00035A65  4C4C4C40 40404040
00035A6D  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   4968 *=============================================================
                                                   4969 *       Quadruple Compare and Swap Tests
                                                   4970 *=============================================================
                                                   4971
                                                   4972 * -------------------------------------------------------------
                                                   4973 * Test Quadruple Compare and Swap (QCS: 34) instruction : 1st NE
                                                   4974 * -------------------------------------------------------------
                                                   4975          TSTHDR 2,1,34             1st Operand NOT Equal (cc=2, UR=1)
00035A75                          00035A75  00035C00 4976+         ORG   PLOETST+(128*K)+29*(3*K)
00035C00  E2E3C1D9 E3409686                        4977+         DC    CL32'START of test: 67'
00035C08  40A385A2 A37A40F6
00035C10  F7404040 40404040
00035C18  40404040 40404040
00035C20                          00035C20         4978+         USING *,R5                base for test data and test routine
00035C20                          00000000         4979+         USING PLOETEST,R5
00035C20  00035C4C                                 4980+T67      DC    A(CODE67)           address of test code
00035C24  0043                                     4981+         DC    H'67'               test number
00035C26  00                                       4982+         DC    X'00'
00035C27  22                                       4983+         DC    HL1'34'             Function
00035C28  01                                       4984+         DC    HL1'1'              expected unequal results (UR)
00035C29  00                                       4985+         DC    HL1'00'             found UR
00035C2A  02                                       4986+         DC    HL1'2'              cc
00035C2B  0D                                       4987+         DC    HL1'13'             not expected CC mask
00035C2C  00                                       4988+         DS    X                   extracted cc
00035C30  00000000 00000000                        4989+         DS    2F                  extract PSW after test (has CC)
00035C38  000365E0                                 4990+         DC    A(FMSG67)           address of test fail message
00035C3C  00036340                                 4991+         DC    A(OPSRC67)          address of operand source
00035C40  00036220                                 4992+         DC    A(OPDST67)          address of operand destination
00035C44  00036460                                 4993+         DC    A(PL67)             address of parameter list
00035C48  000041FB                                 4994+         DC    A(RANDX+67)         address of random source data
                                                   4995
                                                   4996          TSTCODE
00035C4C                                           4997+CODE67   DS    0F
                                                   4998 *                                  load random source
00035C4C  E310 501C 0014                0000001C   4999          LGF   R1,OPSRCA
00035C52  E320 5028 0014                00000028   5000          LGF   R2,RANDXA
00035C58  E330 5024 0014                00000024   5001          LGF   R3,PLA
00035C5E  E340 5020 0014                00000020   5002          LGF   R4,OPREPA
                                                   5003 *
                                                   5004 * Quadruple Compare and Swap (QCS: 34)
                                                   5005 * Build Parambter List
                                                   5006 *
00035C64  D20F 1000 2000       00000000 00000000   5007          MVC   0(16,R1),0(R2)      load Op1-Op4 data in source
00035C6A  D203 3040 1000       00000040 00000000   5008          MVC   64(4,R3),0(R1)      copy to PL OP1-Compare
00035C70  D203 3048 1004       00000048 00000004   5009          MVC   72(4,R3),4(R1)      copy to PL OP2-Compare
00035C76  D203 3050 1008       00000050 00000008   5010          MVC   80(4,R3),8(R1)      copy to PL OP3-Compare
00035C7C  D203 3058 100C       00000058 0000000C   5011          MVC   88(4,R3),12(R1)     copy to PL OP4-Compare
                                                   5012
00035C82  D21F 4000 2080       00000000 00000080   5013          MVC   0(32,R4),128(R2)    load initial Op1-Op4 data
00035C88  D203 3060 4000       00000060 00000000   5014          MVC   96(4,R3),0(R4)      copy to PL OP1-Replacement
00035C8E  D203 3068 4004       00000068 00000004   5015          MVC   104(4,R3),4(R4)     copy to PL OP2-Replacement
00035C94  D203 3070 4008       00000070 00000008   5016          MVC   112(4,R3),8(R4)     copy to PL OP3-Replacement
00035C9A  D203 3078 400C       00000078 0000000C   5017          MVC   120(4,R3),12(R4)    copy to PL OP4-Replacement
                                                   5018 ****
                                                   5019 * force a failure by changing the first operand
                                                   5020 ****
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00035CA0  D203 1000 4000      00000000  00000000  5021           MVC   0(4,R1),0(R4)
                                                  5022 ****
                                                  5023
00035CA6  E313 0000 0024                00000000  5024           STG   R1,0(R3)           load OP1 address in PL
                                                  5025
00035CAC  4111 0004                     00000004  5026           LA    R1,4(R1)
00035CB0  E313 0008 0024                00000008  5027           STG   R1,8(R3)           load OP2 address in PL
                                                  5028
00035CB6  4111 0004                     00000004  5029           LA    R1,4(R1)
00035CBA  E313 0010 0024                00000010  5030           STG   R1,16(R3)          load OP3 address in PL
                                                  5031
00035CC0  4111 0004                     00000004  5032           LA    R1,4(R1)
00035CC4  E313 0018 0024                00000018  5033           STG   R1,24(R3)          load OP4 address in PL
                                                  5034
00035CCA  B982 0000                               5035           XGR   R0,R0
00035CCE  4100 0022                     00000022  5036           LA    R0,34
00035CD2  EE00 0000 3000      00000000  00000000  5037           PLO   0,0,0,0(R3)        Quadruple C/S (QCS: 34)
                                                  5038
00035CD8  B98D 0020                               5039           EPSW  R2,R0              extract psw
00035CDC  5020 5010                     00000010  5040           ST    R2,CCPSW             to save CC
00035CE0  E310 500B 0076                0000000B  5041           LB    R1,CCMASK          (failure CC mask)
00035CE6  8910 0004                     00000004  5042           SLL   R1,4               (shift to BC instr CC position)
00035CEA  4410 5128                     00035D48  5043           EX    R1,F341TSTCC         fail if...
                                                  5044
                                00035CEE  00000001  5045 F341CKUR EQU   *
00035CEE  D200 5009 3030      00000009  00000030  5046           MVC   URFOUND,48(R3)     Extract UR
00035CF4  D500 5008 3030      00000008  00000030  5047           CLC   UR,48(R3)
00035CFA  4780 50E2                     00035D02  5048           BE    F341CKRES
00035CFE  45F0 82C8                     000022C8  5049           BAL   R15,URMSG
                                                  5050 *
                                                  5051 * compare results
                                                  5052 *
                                00035D02  00000001  5053 F341CKRES EQU   *
                                                  5054 *                                 reset adresses
00035D02  E310 501C 0014                0000001C  5055           LGF   R1,OPSRCA
00035D08  E320 5028 0014                00000028  5056           LGF   R2,RANDXA
00035D0E  E330 5024 0014                00000024  5057           LGF   R3,PLA
00035D14  E340 5020 0014                00000020  5058           LGF   R4,OPREPA
                                                  5059
00035D1A  D503 1000 3040      00000000  00000040  5060           CLC   0(4,R1),64(R3)
00035D20  4770 5122                     00035D42  5061           BNE   F341XX             OP1C Different, error
00035D24  D503 1004 3048      00000004  00000048  5062           CLC   4(4,R1),72(R3)
00035D2A  4770 5122                     00035D42  5063           BNE   F341XX             OP2C Different, error
00035D2E  D503 1008 3050      00000008  00000050  5064           CLC   8(4,R1),80(R3)
00035D34  4770 5122                     00035D42  5065           BNE   F341XX             OP3C Different, error
00035D38  D503 100C 3058      0000000C  00000058  5066           CLC   12(4,R1),88(R3)
00035D3E  478B 0000                     00000000  5067           BE    0(R11)             OP4Csame, return
                                                  5068
                                00035D42  00000001  5069 F341XX   EQU   *
00035D42  45F0 83E8                     000023E8  5070           BAL   R15,FAILMSG
00035D46  07FB                                    5071           BR    R11
                                                  5072
00035D48  4700 512C                     00035D4C  5073 F341TSTCC BC    0,F341CCMSG        fail if unexpected condition code
                                00035D4C  00000001  5074 F341CCMSG EQU   *
00035D4C  45F0 8196                     00002196  5075           BAL   R15,CCMSG
00035D50  47F0 50CE                     00035CEE  5076           B     F341CKUR
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    5077
                                                    5078          TSTDATA
00035D54                      00035D54  00036200  5079+          ORG   PLOETST+(128*K)+29*(3*K)+(K+512)
                                                    5080          TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00036200  6E6E6ED6 D7C4E2E3                        5081+          DC    CL32'>>>OPDST67<<<' eye catcher
00036208  F6F74C4C 4C404040
00036210  40404040 40404040
00036218  40404040 40404040
00036220                                          5082+OPDST67   DS    0C
00036220  00000000 00000000                       5083           DC    XL256'0'
00036228  00000000 00000000
00036230  00000000 00000000
00036238  00000000 00000000
00036240  00000000 00000000
00036248  00000000 00000000
00036250  00000000 00000000
00036258  00000000 00000000
00036260  00000000 00000000
00036268  00000000 00000000
00036270  00000000 00000000
00036278  00000000 00000000
00036280  00000000 00000000
00036288  00000000 00000000
00036290  00000000 00000000
00036298  00000000 00000000
000362A0  00000000 00000000
000362A8  00000000 00000000
000362B0  00000000 00000000
000362B8  00000000 00000000
000362C0  00000000 00000000
000362C8  00000000 00000000
000362D0  00000000 00000000
000362D8  00000000 00000000
000362E0  00000000 00000000
000362E8  00000000 00000000
000362F0  00000000 00000000
000362F8  00000000 00000000
00036300  00000000 00000000
00036308  00000000 00000000
00036310  00000000 00000000
00036318  00000000 00000000
                                                    5084
                                                    5085          TSTLBL OPSRC              Source: OP1-OP4
00036320  6E6E6ED6 D7E2D9C3                        5086+          DC    CL32'>>>OPSRC67<<<' eye catcher
00036328  F6F74C4C 4C404040
00036330  40404040 40404040
00036338  40404040 40404040
00036340                                          5087+OPSRC67   DS    0C
00036340  00000000 00000000                       5088           DC    XL256'0'
00036348  00000000 00000000
00036350  00000000 00000000
00036358  00000000 00000000
00036360  00000000 00000000
00036368  00000000 00000000
00036370  00000000 00000000
00036378  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00036380  00000000 00000000
00036388  00000000 00000000
00036390  00000000 00000000
00036398  00000000 00000000
000363A0  00000000 00000000
000363A8  00000000 00000000
000363B0  00000000 00000000
000363B8  00000000 00000000
000363C0  00000000 00000000
000363C8  00000000 00000000
000363D0  00000000 00000000
000363D8  00000000 00000000
000363E0  00000000 00000000
000363E8  00000000 00000000
000363F0  00000000 00000000
000363F8  00000000 00000000
00036400  00000000 00000000
00036408  00000000 00000000
00036410  00000000 00000000
00036418  00000000 00000000
00036420  00000000 00000000
00036428  00000000 00000000
00036430  00000000 00000000
00036438  00000000 00000000
                                          5089
                                          5090          TSTLBL PL                 Parameter List
00036440  6E6E6ED7 D3F6F74C               5091+         DC     CL32'>>>PL67<<<'   eye catcher
00036448  4C4C4040 40404040
00036450  40404040 40404040
00036458  40404040 40404040
00036460                                  5092+PL67     DS     0C
00036460  00000000 00000000               5093          DC     XL352'00'
00036468  00000000 00000000
00036470  00000000 00000000
00036478  00000000 00000000
00036480  00000000 00000000
00036488  00000000 00000000
00036490  00000000 00000000
00036498  00000000 00000000
000364A0  00000000 00000000
000364A8  00000000 00000000
000364B0  00000000 00000000
000364B8  00000000 00000000
000364C0  00000000 00000000
000364C8  00000000 00000000
000364D0  00000000 00000000
000364D8  00000000 00000000
000364E0  00000000 00000000
000364E8  00000000 00000000
000364F0  00000000 00000000
000364F8  00000000 00000000
00036500  00000000 00000000
00036508  00000000 00000000
00036510  00000000 00000000
00036518  00000000 00000000
00036520  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00036528  00000000 00000000
00036530  00000000 00000000
00036538  00000000 00000000
00036540  00000000 00000000
00036548  00000000 00000000
00036550  00000000 00000000
00036558  00000000 00000000
00036560  00000000 00000000
00036568  00000000 00000000
00036570  00000000 00000000
00036578  00000000 00000000
00036580  00000000 00000000
00036588  00000000 00000000
00036590  00000000 00000000
00036598  00000000 00000000
000365A0  00000000 00000000
000365A8  00000000 00000000
000365B0  00000000 00000000
000365B8  00000000 00000000
                                          5094
                                          5095          TSTLBL FMSG                fail message
000365C0  6E6E6EC6 D4E2C7F6               5096+         DC    CL32'>>>FMSG67<<<'   eye catcher
000365C8  F74C4C4C 40404040
000365D0  40404040 40404040
000365D8  40404040 40404040
000365E0                                  5097+FMSG67   DS    0C
000365E0  86818993 858440D7               5098          DC    C'failed PLO-Extension '
000365E8  D3D660C5 A7A38595
000365F0  A2899695 40
000365F5  D8A48184 99A49793               5099          DC    CL96'Quadruple Compare and Swap (QCS: 34): 1st NE'
000365FD  8540C396 94978199
00036605  85408195 8440E2A6
0003660D  8197404D D8C3E27A
00036615  40F3F45D 7A40F1A2
0003661D  A340D5C5 40404040
00036625  40404040 40404040
0003662D  40404040 40404040
00036635  40404040 40404040
0003663D  40404040 40404040
00036645  40404040 40404040
0003664D  40404040 40404040
                                          5100
                                          5101          TSTEND
00036655  C5D5C440 968640A3               5102+         DC    CL32'END of test: 67 <<<'
0003665D  85A2A37A 40F6F740
00036665  4C4C4C40 40404040
0003666D  40404040 40404040
                                          5103
                                          5104 * ------------------------------------------------------------------
                                          5105 * Test Quadruple Compare and Swap (QCS: 34) instruction : 2nd NE
                                          5106 * ------------------------------------------------------------------
                                          5107          TSTHDR 2,2,34              2nd Operand NOT Equal (cc=2, UR=2)
00036675                        00036675  00036800      5108+         ORG   PLOETST+(128*K)+30*(3*K)
00036800  E2E3C1D9 E3409686               5109+         DC    CL32'START of test: 68'
00036808  40A385A2 A37A40F6
00036810  F8404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00036818  40404040 40404040
00036820                     00036820              5110+          USING *,R5                base for test data and test routine
00036820                     00000000              5111+          USING PLOETEST,R5
00036820  0003684C                                 5112+T68       DC    A(CODE68)           address of test code
00036824  0044                                     5113+          DC    H'68'               test number
00036826  00                                       5114+          DC    X'00'
00036827  22                                       5115+          DC    HL1'34'             Function
00036828  02                                       5116+          DC    HL1'2'              expected unequal results (UR)
00036829  00                                       5117+          DC    HL1'00'             found UR
0003682A  02                                       5118+          DC    HL1'2'              cc
0003682B  0D                                       5119+          DC    HL1'13'             not expected CC mask
0003682C  00                                       5120+          DS    X                   extracted cc
00036830  00000000 00000000                        5121+          DS    2F                  extract PSW after test (has CC)
00036838  000371E0                                 5122+          DC    A(FMSG68)           address of test fail message
0003683C  00036F40                                 5123+          DC    A(OPSRC68)          address of operand source
00036840  00036E20                                 5124+          DC    A(OPDST68)          address of operand destination
00036844  00037060                                 5125+          DC    A(PL68)             address of parameter list
00036848  000041FC                                 5126+          DC    A(RANDX+68)         address of random source data
                                                   5127
                                                   5128           TSTCODE
0003684C                                           5129+CODE68    DS    0F
                                                   5130 *                                   load random source
0003684C  E310 501C 0014             0000001C      5131           LGF   R1,OPSRCA
00036852  E320 5028 0014             00000028      5132           LGF   R2,RANDXA
00036858  E330 5024 0014             00000024      5133           LGF   R3,PLA
0003685E  E340 5020 0014             00000020      5134           LGF   R4,OPREPA
                                                   5135 *
                                                   5136 * Quadruple Compare and Swap (QCS: 34)
                                                   5137 * Build Parambter List
                                                   5138 *
00036864  D20F 1000 2000  00000000   00000000      5139           MVC   0(16,R1),0(R2)      load Op1-Op4 data in source
0003686A  D203 3040 1000  00000040   00000000      5140           MVC   64(4,R3),0(R1)      copy to PL OP1-Compare
00036870  D203 3048 1004  00000048   00000004      5141           MVC   72(4,R3),4(R1)      copy to PL OP2-Compare
00036876  D203 3050 1008  00000050   00000008      5142           MVC   80(4,R3),8(R1)      copy to PL OP3-Compare
0003687C  D203 3058 100C  00000058   0000000C      5143           MVC   88(4,R3),12(R1)     copy to PL OP4-Compare
                                                   5144
00036882  D21F 4000 2080  00000000   00000080      5145           MVC   0(32,R4),128(R2)    load initial Op1-Op4 data
00036888  D203 3060 4000  00000060   00000000      5146           MVC   96(4,R3),0(R4)      copy to PL OP1-Replacement
0003688E  D203 3068 4004  00000068   00000004      5147           MVC   104(4,R3),4(R4)     copy to PL OP2-Replacement
00036894  D203 3070 4008  00000070   00000008      5148           MVC   112(4,R3),8(R4)     copy to PL OP3-Replacement
0003689A  D203 3078 400C  00000078   0000000C      5149           MVC   120(4,R3),12(R4)    copy to PL OP4-Replacement
                                                   5150 ****
                                                   5151 * force a failure by changing the 2nd operand
                                                   5152 ****
000368A0  D203 1004 4000  00000004   00000000      5153           MVC   4(4,R1),0(R4)
                                                   5154 ****
                                                   5155
000368A6  E313 0000 0024             00000000      5156           STG   R1,0(R3)            load OP1 address in PL
                                                   5157
000368AC  4111 0004                  00000004      5158           LA    R1,4(R1)
000368B0  E313 0008 0024             00000008      5159           STG   R1,8(R3)            load OP2 address in PL
                                                   5160
000368B6  4111 0004                  00000004      5161           LA    R1,4(R1)
000368BA  E313 0010 0024             00000010      5162           STG   R1,16(R3)           load OP3 address in PL
                                                   5163
000368C0  4111 0004                  00000004      5164           LA    R1,4(R1)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000368C4  E313 0018 0024                 00000018  5165           STG   R1,24(R3)           load OP4 address in PL
                                                   5166
000368CA  B982 0000                              5167           XGR   R0,R0
000368CE  4100 0022                      00000022  5168           LA    R0,34
000368D2  EE00 0000 3000      00000000   00000000  5169           PLO   0,0,0,0(R3)         Quadruple C/S (QCS: 34)
                                                   5170
000368D8  B98D 0020                              5171           EPSW  R2,R0               extract psw
000368DC  5020 5010                      00000010  5172           ST    R2,CCPSW              to save CC
000368E0  E310 500B 0076                 0000000B  5173           LB    R1,CCMASK           (failure CC mask)
000368E6  8910 0004                      00000004  5174           SLL   R1,4                (shift to BC instr CC position)
000368EA  4410 5128                      00036948  5175           EX    R1,F342TSTCC            fail if...
                                                   5176
                                000368EE  00000001  5177 F342CKUR EQU   *
000368EE  D200 5009 3030      00000009   00000030  5178           MVC   URFOUND,48(R3)      Extract UR
000368F4  D500 5008 3030      00000008   00000030  5179           CLC   UR,48(R3)
000368FA  4780 50E2                      00036902  5180           BE    F342CKRES
000368FE  45F0 82C8                      000022C8  5181           BAL   R15,URMSG
                                                   5182 *
                                                   5183 * compare results
                                                   5184 *
                                00036902  00000001  5185 F342CKRES EQU   *
                                                   5186 *                                  reset adresses
00036902  E310 501C 0014                 0000001C  5187           LGF   R1,OPSRCA
00036908  E320 5028 0014                 00000028  5188           LGF   R2,RANDXA
0003690E  E330 5024 0014                 00000024  5189           LGF   R3,PLA
00036914  E340 5020 0014                 00000020  5190           LGF   R4,OPREPA
                                                   5191
0003691A  D503 1000 3040      00000000   00000040  5192           CLC   0(4,R1),64(R3)
00036920  4770 5122                      00036942  5193           BNE   F342XX              OP1C Different, error
00036924  D503 1004 3048      00000004   00000048  5194           CLC   4(4,R1),72(R3)
0003692A  4770 5122                      00036942  5195           BNE   F342XX              OP2C Different, error
0003692E  D503 1008 3050      00000008   00000050  5196           CLC   8(4,R1),80(R3)
00036934  4770 5122                      00036942  5197           BNE   F342XX              OP3C Different, error
00036938  D503 100C 3058      0000000C   00000058  5198           CLC   12(4,R1),88(R3)
0003693E  478B 0000                      00000000  5199           BE    0(R11)              OP4Csame, return
                                                   5200
                                00036942  00000001  5201 F342XX   EQU   *
00036942  45F0 83E8                      000023E8  5202           BAL   R15,FAILMSG
00036946  07FB                                     5203           BR    R11
                                                   5204
00036948  4700 512C                      0003694C  5205 F342TSTCC BC    0,F342CCMSG         fail if unexpected condition code
                                0003694C  00000001  5206 F342CCMSG EQU   *
0003694C  45F0 8196                      00002196  5207           BAL   R15,CCMSG
00036950  47F0 50CE                      000368EE  5208           B     F342CKUR
                                                   5209
                                                   5210           TSTDATA
00036954                        00036954  00036E00  5211+          ORG   PLOETST+(128*K)+30*(3*K)+(K+512)
                                                   5212           TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00036E00  6E6E6ED6 D7C4E2E3                        5213+          DC    CL32'>>>OPDST68<<<' eye catcher
00036E08  F6F84C4C 4C404040
00036E10  40404040 40404040
00036E18  40404040 40404040
00036E20                                           5214+OPDST68   DS    0C
00036E20  00000000 00000000                        5215           DC    XL256'0'
00036E28  00000000 00000000
00036E30  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00036E38  00000000 00000000
00036E40  00000000 00000000
00036E48  00000000 00000000
00036E50  00000000 00000000
00036E58  00000000 00000000
00036E60  00000000 00000000
00036E68  00000000 00000000
00036E70  00000000 00000000
00036E78  00000000 00000000
00036E80  00000000 00000000
00036E88  00000000 00000000
00036E90  00000000 00000000
00036E98  00000000 00000000
00036EA0  00000000 00000000
00036EA8  00000000 00000000
00036EB0  00000000 00000000
00036EB8  00000000 00000000
00036EC0  00000000 00000000
00036EC8  00000000 00000000
00036ED0  00000000 00000000
00036ED8  00000000 00000000
00036EE0  00000000 00000000
00036EE8  00000000 00000000
00036EF0  00000000 00000000
00036EF8  00000000 00000000
00036F00  00000000 00000000
00036F08  00000000 00000000
00036F10  00000000 00000000
00036F18  00000000 00000000
                                      5216
                                      5217           TSTLBL OPSRC              Source: OP1-OP4
00036F20  6E6E6ED6 D7E2D9C3           5218+          DC    CL32'>>>OPSRC68<<<' eye catcher
00036F28  F6F84C4C 4C404040
00036F30  40404040 40404040
00036F38  40404040 40404040
00036F40                              5219+OPSRC68   DS    0C
00036F40  00000000 00000000           5220           DC    XL256'0'
00036F48  00000000 00000000
00036F50  00000000 00000000
00036F58  00000000 00000000
00036F60  00000000 00000000
00036F68  00000000 00000000
00036F70  00000000 00000000
00036F78  00000000 00000000
00036F80  00000000 00000000
00036F88  00000000 00000000
00036F90  00000000 00000000
00036F98  00000000 00000000
00036FA0  00000000 00000000
00036FA8  00000000 00000000
00036FB0  00000000 00000000
00036FB8  00000000 00000000
00036FC0  00000000 00000000
00036FC8  00000000 00000000
00036FD0  00000000 00000000
00036FD8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00036FE0  00000000 00000000
00036FE8  00000000 00000000
00036FF0  00000000 00000000
00036FF8  00000000 00000000
00037000  00000000 00000000
00037008  00000000 00000000
00037010  00000000 00000000
00037018  00000000 00000000
00037020  00000000 00000000
00037028  00000000 00000000
00037030  00000000 00000000
00037038  00000000 00000000
                                        5221
                                        5222           TSTLBL PL                  Parameter List
00037040  6E6E6ED7 D3F6F84C             5223+          DC     CL32'>>>PL68<<<'    eye catcher
00037048  4C4C4040 40404040
00037050  40404040 40404040
00037058  40404040 40404040
00037060                                5224+PL68      DS     0C
00037060  00000000 00000000             5225           DC     XL352'00'
00037068  00000000 00000000
00037070  00000000 00000000
00037078  00000000 00000000
00037080  00000000 00000000
00037088  00000000 00000000
00037090  00000000 00000000
00037098  00000000 00000000
000370A0  00000000 00000000
000370A8  00000000 00000000
000370B0  00000000 00000000
000370B8  00000000 00000000
000370C0  00000000 00000000
000370C8  00000000 00000000
000370D0  00000000 00000000
000370D8  00000000 00000000
000370E0  00000000 00000000
000370E8  00000000 00000000
000370F0  00000000 00000000
000370F8  00000000 00000000
00037100  00000000 00000000
00037108  00000000 00000000
00037110  00000000 00000000
00037118  00000000 00000000
00037120  00000000 00000000
00037128  00000000 00000000
00037130  00000000 00000000
00037138  00000000 00000000
00037140  00000000 00000000
00037148  00000000 00000000
00037150  00000000 00000000
00037158  00000000 00000000
00037160  00000000 00000000
00037168  00000000 00000000
00037170  00000000 00000000
00037178  00000000 00000000
00037180  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00037188  00000000 00000000
00037190  00000000 00000000
00037198  00000000 00000000
000371A0  00000000 00000000
000371A8  00000000 00000000
000371B0  00000000 00000000
000371B8  00000000 00000000
                                        5226
                                        5227          TSTLBL FMSG                fail message
000371C0  6E6E6EC6 D4E2C7F6             5228+         DC    CL32'>>>FMSG68<<<'   eye catcher
000371C8  F84C4C4C 40404040
000371D0  40404040 40404040
000371D8  40404040 40404040
000371E0                                5229+FMSG68   DS    0C
000371E0  86818993 858440D7             5230          DC    C'failed PLO-Extension '
000371E8  D3D660C5 A7A38595
000371F0  A2899695 40
000371F5  D8A48184 99A49793             5231          DC    CL96'Quadruple Compare and Swap (QCS: 34): 2nd NE'
000371FD  8540C396 94978199
00037205  85408195 8440E2A6
0003720D  8197404D D8C3E27A
00037215  40F3F45D 7A40F295
0003721D  8440D5C5 40404040
00037225  40404040 40404040
0003722D  40404040 40404040
00037235  40404040 40404040
0003723D  40404040 40404040
00037245  40404040 40404040
0003724D  40404040 40404040
                                        5232
                                        5233          TSTEND
00037255  C5D5C440 968640A3             5234+         DC    CL32'END of test: 68 <<<'
0003725D  85A2A37A 40F6F840
00037265  4C4C4C40 40404040
0003726D  40404040 40404040
                                        5235
                                        5236 * ------------------------------------------------------------------
                                        5237 * Test Quadruple Compare and Swap (QCS: 34) instruction : 3rd NE
                                        5238 * ------------------------------------------------------------------
                                        5239          TSTHDR 2,3,34              3rd Operand NOT Equal (cc=2, UR=3)
00037275                  00037275  00037400  5240+         ORG   PLOETST+(128*K)+31*(3*K)
00037400  E2E3C1D9 E3409686             5241+         DC    CL32'START of test: 69'
00037408  40A385A2 A37A40F6
00037410  F9404040 40404040
00037418  40404040 40404040
00037420                  00037420          5242+         USING *,R5                 base for test data and test routine
00037420                  00000000          5243+         USING PLOETEST,R5
00037420  0003744C                        5244+T69      DC    A(CODE69)            address of test code
00037424  0045                            5245+         DC    H'69'                test number
00037426  00                              5246+         DC    X'00'
00037427  22                              5247+         DC    HL1'34'              Function
00037428  03                              5248+         DC    HL1'3'               expected unequal results (UR)
00037429  00                              5249+         DC    HL1'00'              found UR
0003742A  02                              5250+         DC    HL1'2'               cc
0003742B  0D                              5251+         DC    HL1'13'              not expected CC mask
0003742C  00                              5252+         DS    X                    extracted cc
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00037430  00000000 00000000                        5253+          DS    2F                  extract PSW after test (has CC)
00037438  00037DE0                                 5254+          DC    A(FMSG69)           address of test fail message
0003743C  00037B40                                 5255+          DC    A(OPSRC69)          address of operand source
00037440  00037A20                                 5256+          DC    A(OPDST69)          address of operand destination
00037444  00037C60                                 5257+          DC    A(PL69)             address of parameter list
00037448  000041FD                                 5258+          DC    A(RANDX+69)         address of random source data
                                                   5259
                                                   5260           TSTCODE
0003744C                                           5261+CODE69    DS    0F
                                                   5262 *                                   load random source
0003744C  E310 501C 0014                0000001C   5263           LGF   R1,OPSRCA
00037452  E320 5028 0014                00000028   5264           LGF   R2,RANDXA
00037458  E330 5024 0014                00000024   5265           LGF   R3,PLA
0003745E  E340 5020 0014                00000020   5266           LGF   R4,OPREPA
                                                   5267 *
                                                   5268 * Quadruple Compare and Swap (QCS: 34)
                                                   5269 * Build Parambter List
                                                   5270 *
00037464  D20F 1000 2000      00000000  00000000   5271           MVC   0(16,R1),0(R2)      load Op1-Op4 data in source
0003746A  D203 3040 1000      00000040  00000000   5272           MVC   64(4,R3),0(R1)      copy to PL OP1-Compare
00037470  D203 3048 1004      00000048  00000004   5273           MVC   72(4,R3),4(R1)      copy to PL OP2-Compare
00037476  D203 3050 1008      00000050  00000008   5274           MVC   80(4,R3),8(R1)      copy to PL OP3-Compare
0003747C  D203 3058 100C      00000058  0000000C   5275           MVC   88(4,R3),12(R1)     copy to PL OP4-Compare
                                                   5276
00037482  D21F 4000 2080      00000000  00000080   5277           MVC   0(32,R4),128(R2)    load initial Op1-Op4 data
00037488  D203 3060 4000      00000060  00000000   5278           MVC   96(4,R3),0(R4)      copy to PL OP1-Replacement
0003748E  D203 3068 4004      00000068  00000004   5279           MVC   104(4,R3),4(R4)     copy to PL OP2-Replacement
00037494  D203 3070 4008      00000070  00000008   5280           MVC   112(4,R3),8(R4)     copy to PL OP3-Replacement
0003749A  D203 3078 400C      00000078  0000000C   5281           MVC   120(4,R3),12(R4)    copy to PL OP4-Replacement
                                                   5282 ****
                                                   5283 * force a failure by changing the 3rd operand
                                                   5284 ****
000374A0  D203 1008 4000      00000008  00000000   5285           MVC   8(4,R1),0(R4)
                                                   5286 ****
                                                   5287
000374A6  E313 0000 0024                00000000   5288           STG   R1,0(R3)            load OP1 address in PL
                                                   5289
000374AC  4111 0004                     00000004   5290           LA    R1,4(R1)
000374B0  E313 0008 0024                00000008   5291           STG   R1,8(R3)            load OP2 address in PL
                                                   5292
000374B6  4111 0004                     00000004   5293           LA    R1,4(R1)
000374BA  E313 0010 0024                00000010   5294           STG   R1,16(R3)           load OP3 address in PL
                                                   5295
000374C0  4111 0004                     00000004   5296           LA    R1,4(R1)
000374C4  E313 0018 0024                00000018   5297           STG   R1,24(R3)           load OP4 address in PL
                                                   5298
000374CA  B982 0000                                5299           XGR   R0,R0
000374CE  4100 0022                     00000022   5300           LA    R0,34
000374D2  EE00 0000 3000      00000000  00000000   5301           PLO   0,0,0,0(R3)         Quadruple C/S (QCS: 34)
                                                   5302
000374D8  B98D 0020                                5303           EPSW  R2,R0               extract psw
000374DC  5020 5010                     00000010   5304           ST    R2,CCPSW              to save CC
000374E0  E310 500B 0076                0000000B   5305           LB    R1,CCMASK           (failure CC mask)
000374E6  8910 0004                     00000004   5306           SLL   R1,4                (shift to BC instr CC position)
000374EA  4410 5128                     00037548   5307           EX    R1,F343TSTCC            fail if...
                                                   5308
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                               000374EE  00000001  5309 F343CKUR EQU   *
000374EE  D200 5009 3030       00000009  00000030  5310          MVC   URFOUND,48(R3)      Extract UR
000374F4  D500 5008 3030       00000008  00000030  5311          CLC   UR,48(R3)
000374FA  4780 50E2                      00037502  5312          BE    F343CKRES
000374FE  45F0 82C8                      000022C8  5313          BAL   R15,URMSG
                                                   5314 *
                                                   5315 * compare results
                                                   5316 *
                               00037502  00000001  5317 F343CKRES EQU   *
                                                   5318 *                               reset adresses
00037502  E310 501C 0014                 0000001C  5319          LGF   R1,OPSRCA
00037508  E320 5028 0014                 00000028  5320          LGF   R2,RANDXA
0003750E  E330 5024 0014                 00000024  5321          LGF   R3,PLA
00037514  E340 5020 0014                 00000020  5322          LGF   R4,OPREPA
                                                   5323
0003751A  D503 1000 3040       00000000  00000040  5324          CLC   0(4,R1),64(R3)
00037520  4770 5122                      00037542  5325          BNE   F343XX              OP1C Different, error
00037524  D503 1004 3048       00000004  00000048  5326          CLC   4(4,R1),72(R3)
0003752A  4770 5122                      00037542  5327          BNE   F343XX              OP2C Different, error
0003752E  D503 1008 3050       00000008  00000050  5328          CLC   8(4,R1),80(R3)
00037534  4770 5122                      00037542  5329          BNE   F343XX              OP3C Different, error
00037538  D503 100C 3058       0000000C  00000058  5330          CLC   12(4,R1),88(R3)
0003753E  478B 0000                      00000000  5331          BE    0(R11)              OP4Csame, return
                                                   5332
                               00037542  00000001  5333 F343XX   EQU   *
00037542  45F0 83E8                      000023E8  5334          BAL   R15,FAILMSG
00037546  07FB                                     5335          BR    R11
                                                   5336
00037548  4700 512C                      0003754C  5337 F343TSTCC BC    0,F343CCMSG         fail if unexpected condition code
                               0003754C  00000001  5338 F343CCMSG EQU   *
0003754C  45F0 8196                      00002196  5339          BAL   R15,CCMSG
00037550  47F0 50CE                      000374EE  5340          B     F343CKUR
                                                   5341
                                                   5342          TSTDATA
00037554                       00037554  00037A00  5343+         ORG   PLOETST+(128*K)+31*(3*K)+(K+512)
                                                   5344          TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00037A00  6E6E6ED6 D7C4E2E3                        5345+         DC    CL32'>>>OPDST69<<<' eye catcher
00037A08  F6F94C4C 4C404040
00037A10  40404040 40404040
00037A18  40404040 40404040
00037A20                                           5346+OPDST69  DS    0C
00037A20  00000000 00000000                        5347          DC    XL256'0'
00037A28  00000000 00000000
00037A30  00000000 00000000
00037A38  00000000 00000000
00037A40  00000000 00000000
00037A48  00000000 00000000
00037A50  00000000 00000000
00037A58  00000000 00000000
00037A60  00000000 00000000
00037A68  00000000 00000000
00037A70  00000000 00000000
00037A78  00000000 00000000
00037A80  00000000 00000000
00037A88  00000000 00000000
00037A90  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00037A98  00000000 00000000
00037AA0  00000000 00000000
00037AA8  00000000 00000000
00037AB0  00000000 00000000
00037AB8  00000000 00000000
00037AC0  00000000 00000000
00037AC8  00000000 00000000
00037AD0  00000000 00000000
00037AD8  00000000 00000000
00037AE0  00000000 00000000
00037AE8  00000000 00000000
00037AF0  00000000 00000000
00037AF8  00000000 00000000
00037B00  00000000 00000000
00037B08  00000000 00000000
00037B10  00000000 00000000
00037B18  00000000 00000000
                                        5348
                                        5349          TSTLBL OPSRC              Source: OP1-OP4
00037B20  6E6E6ED6 D7E2D9C3             5350+         DC    CL32'>>>OPSRC69<<<' eye catcher
00037B28  F6F94C4C 4C404040
00037B30  40404040 40404040
00037B38  40404040 40404040
00037B40                                5351+OPSRC69  DS    0C
00037B40  00000000 00000000             5352          DC    XL256'0'
00037B48  00000000 00000000
00037B50  00000000 00000000
00037B58  00000000 00000000
00037B60  00000000 00000000
00037B68  00000000 00000000
00037B70  00000000 00000000
00037B78  00000000 00000000
00037B80  00000000 00000000
00037B88  00000000 00000000
00037B90  00000000 00000000
00037B98  00000000 00000000
00037BA0  00000000 00000000
00037BA8  00000000 00000000
00037BB0  00000000 00000000
00037BB8  00000000 00000000
00037BC0  00000000 00000000
00037BC8  00000000 00000000
00037BD0  00000000 00000000
00037BD8  00000000 00000000
00037BE0  00000000 00000000
00037BE8  00000000 00000000
00037BF0  00000000 00000000
00037BF8  00000000 00000000
00037C00  00000000 00000000
00037C08  00000000 00000000
00037C10  00000000 00000000
00037C18  00000000 00000000
00037C20  00000000 00000000
00037C28  00000000 00000000
00037C30  00000000 00000000
00037C38  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    5353
                                                    5354          TSTLBL PL                 Parameter List
00037C40  6E6E6ED7 D3F6F94C                         5355+         DC    CL32'>>>PL69<<<'    eye catcher
00037C48  4C4C4040 40404040
00037C50  40404040 40404040
00037C58  40404040 40404040
00037C60                                            5356+PL69     DS    0C
00037C60  00000000 00000000                         5357          DC    XL352'00'
00037C68  00000000 00000000
00037C70  00000000 00000000
00037C78  00000000 00000000
00037C80  00000000 00000000
00037C88  00000000 00000000
00037C90  00000000 00000000
00037C98  00000000 00000000
00037CA0  00000000 00000000
00037CA8  00000000 00000000
00037CB0  00000000 00000000
00037CB8  00000000 00000000
00037CC0  00000000 00000000
00037CC8  00000000 00000000
00037CD0  00000000 00000000
00037CD8  00000000 00000000
00037CE0  00000000 00000000
00037CE8  00000000 00000000
00037CF0  00000000 00000000
00037CF8  00000000 00000000
00037D00  00000000 00000000
00037D08  00000000 00000000
00037D10  00000000 00000000
00037D18  00000000 00000000
00037D20  00000000 00000000
00037D28  00000000 00000000
00037D30  00000000 00000000
00037D38  00000000 00000000
00037D40  00000000 00000000
00037D48  00000000 00000000
00037D50  00000000 00000000
00037D58  00000000 00000000
00037D60  00000000 00000000
00037D68  00000000 00000000
00037D70  00000000 00000000
00037D78  00000000 00000000
00037D80  00000000 00000000
00037D88  00000000 00000000
00037D90  00000000 00000000
00037D98  00000000 00000000
00037DA0  00000000 00000000
00037DA8  00000000 00000000
00037DB0  00000000 00000000
00037DB8  00000000 00000000
                                                    5358
                                                    5359          TSTLBL FMSG               fail message
00037DC0  6E6E6EC6 D4E2C7F6                         5360+         DC    CL32'>>>FMSG69<<<'  eye catcher
00037DC8  F94C4C4C 40404040
00037DD0  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00037DD8  40404040 40404040
00037DE0                                         5361+FMSG69   DS    0C
00037DE0  86818993 858440D7                      5362          DC    C'failed PLO-Extension '
00037DE8  D3D660C5 A7A38595
00037DF0  A2899695 40
00037DF5  D8A48184 99A49793                      5363          DC    CL96'Quadruple Compare and Swap (QCS: 34): 3rd NE'
00037DFD  8540C396 94978199
00037E05  85408195 8440E2A6
00037E0D  8197404D D8C3E27A
00037E15  40F3F45D 7A40F399
00037E1D  8440D5C5 40404040
00037E25  40404040 40404040
00037E2D  40404040 40404040
00037E35  40404040 40404040
00037E3D  40404040 40404040
00037E45  40404040 40404040
00037E4D  40404040 40404040
                                                 5364
                                                 5365          TSTEND
00037E55  C5D5C440 968640A3                      5366+         DC    CL32'END of test: 69 <<<'
00037E5D  85A2A37A 40F6F940
00037E65  4C4C4C40 40404040
00037E6D  40404040 40404040
                                                 5367
                                                 5368 * ------------------------------------------------------------------
                                                 5369 * Test Quadruple Compare and Swap (QCS: 34) instruction : 4th NE
                                                 5370 * ------------------------------------------------------------------
                                                 5371          TSTHDR 2,4,34             4th Operand NOT Equal (cc=2, UR=4)
00037E75                        00037E75  00038000 5372+         ORG   PLOETST+(128*K)+32*(3*K)
00038000  E2E3C1D9 E3409686                      5373+         DC    CL32'START of test: 70'
00038008  40A385A2 A37A40F7
00038010  F0404040 40404040
00038018  40404040 40404040
00038020                        00038020         5374+         USING *,R5                base for test data and test routine
00038020                        00000000         5375+         USING PLOETEST,R5
00038020  0003804C                               5376+T70      DC    A(CODE70)           address of test code
00038024  0046                                   5377+         DC    H'70'               test number
00038026  00                                     5378+         DC    X'00'
00038027  22                                     5379+         DC    HL1'34'             Function
00038028  04                                     5380+         DC    HL1'4'              expected unequal results (UR)
00038029  00                                     5381+         DC    HL1'00'             found UR
0003802A  02                                     5382+         DC    HL1'2'              cc
0003802B  0D                                     5383+         DC    HL1'13'             not expected CC mask
0003802C  00                                     5384+         DS    X                   extracted cc
00038030  00000000 00000000                      5385+         DS    2F                  extract PSW after test (has CC)
00038038  000389E0                               5386+         DC    A(FMSG70)           address of test fail message
0003803C  00038740                               5387+         DC    A(OPSRC70)          address of operand source
00038040  00038620                               5388+         DC    A(OPDST70)          address of operand destination
00038044  00038860                               5389+         DC    A(PL70)             address of parameter list
00038048  000041FE                               5390+         DC    A(RANDX+70)         address of random source data
                                                 5391
                                                 5392          TSTCODE
0003804C                                         5393+CODE70   DS    0F
                                                 5394 *                                  load random source
0003804C  E310 501C 0014              0000001C   5395          LGF   R1,OPSRCA
00038052  E320 5028 0014              00000028   5396          LGF   R2,RANDXA
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00038058  E330 5024 0014                 00000024  5397          LGF   R3,PLA
0003805E  E340 5020 0014                 00000020  5398          LGF   R4,OPREPA
                                                   5399 *
                                                   5400 * Quadruple Compare and Swap (QCS: 34)
                                                   5401 * Build Parambter List
                                                   5402 *
00038064  D20F 1000 2000       00000000  00000000  5403          MVC   0(16,R1),0(R2)      load Op1-Op4 data in source
0003806A  D203 3040 1000       00000040  00000000  5404          MVC   64(4,R3),0(R1)      copy to PL OP1-Compare
00038070  D203 3048 1004       00000048  00000004  5405          MVC   72(4,R3),4(R1)      copy to PL OP2-Compare
00038076  D203 3050 1008       00000050  00000008  5406          MVC   80(4,R3),8(R1)      copy to PL OP3-Compare
0003807C  D203 3058 100C       00000058  0000000C  5407          MVC   88(4,R3),12(R1)     copy to PL OP4-Compare
                                                   5408
00038082  D21F 4000 2080       00000000  00000080  5409          MVC   0(32,R4),128(R2)    load initial Op1-Op4 data
00038088  D203 3060 4000       00000060  00000000  5410          MVC   96(4,R3),0(R4)      copy to PL OP1-Replacement
0003808E  D203 3068 4004       00000068  00000004  5411          MVC   104(4,R3),4(R4)     copy to PL OP2-Replacement
00038094  D203 3070 4008       00000070  00000008  5412          MVC   112(4,R3),8(R4)     copy to PL OP3-Replacement
0003809A  D203 3078 400C       00000078  0000000C  5413          MVC   120(4,R3),12(R4)    copy to PL OP4-Replacement
                                                   5414 ****
                                                   5415 * force a failure by changing the 4th operand
                                                   5416 ****
000380A0  D203 100C 4000       0000000C  00000000  5417          MVC   12(4,R1),0(R4)
                                                   5418 ****
                                                   5419
000380A6  E313 0000 0024                 00000000  5420          STG   R1,0(R3)            load OP1 address in PL
                                                   5421
000380AC  4111 0004                      00000004  5422          LA    R1,4(R1)
000380B0  E313 0008 0024                 00000008  5423          STG   R1,8(R3)            load OP2 address in PL
                                                   5424
000380B6  4111 0004                      00000004  5425          LA    R1,4(R1)
000380BA  E313 0010 0024                 00000010  5426          STG   R1,16(R3)           load OP3 address in PL
                                                   5427
000380C0  4111 0004                      00000004  5428          LA    R1,4(R1)
000380C4  E313 0018 0024                 00000018  5429          STG   R1,24(R3)           load OP4 address in PL
                                                   5430
000380CA  B982 0000                                5431          XGR   R0,R0
000380CE  4100 0022                      00000022  5432          LA    R0,34
000380D2  EE00 0000 3000       00000000  00000000  5433          PLO   0,0,0,0(R3)         Quadruple C/S (QCS: 34)
                                                   5434
000380D8  B98D 0020                                5435          EPSW  R2,R0               extract psw
000380DC  5020 5010                      00000010  5436          ST    R2,CCPSW              to save CC
000380E0  E310 500B 0076                 0000000B  5437          LB    R1,CCMASK           (failure CC mask)
000380E6  8910 0004                      00000004  5438          SLL   R1,4                (shift to BC instr CC position)
000380EA  4410 5128                      00038148  5439          EX    R1,F344TSTCC            fail if...
                                                   5440
                                 000380EE  00000001  5441 F344CKUR EQU   *
000380EE  D200 5009 3030       00000009  00000030  5442          MVC   URFOUND,48(R3)      Extract UR
000380F4  D500 5008 3030       00000008  00000030  5443          CLC   UR,48(R3)
000380FA  4780 50E2                      00038102  5444          BE    F344CKRES
000380FE  45F0 82C8                      000022C8  5445          BAL   R15,URMSG
                                                   5446 *
                                                   5447 * compare results
                                                   5448 *
                                 00038102  00000001  5449 F344CKRES EQU   *
                                                   5450 *                                reset adresses
00038102  E310 501C 0014                 0000001C  5451          LGF   R1,OPSRCA
00038108  E320 5028 0014                 00000028  5452          LGF   R2,RANDXA
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003810E  E330 5024 0014                 00000024  5453           LGF   R3,PLA
00038114  E340 5020 0014                 00000020  5454           LGF   R4,OPREPA
                                                   5455
0003811A  D503 1000 3040       00000000  00000040  5456           CLC   0(4,R1),64(R3)
00038120  4770 5122                      00038142  5457           BNE   F344XX                OP1C Different, error
00038124  D503 1004 3048       00000004  00000048  5458           CLC   4(4,R1),72(R3)
0003812A  4770 5122                      00038142  5459           BNE   F344XX                OP2C Different, error
0003812E  D503 1008 3050       00000008  00000050  5460           CLC   8(4,R1),80(R3)
00038134  4770 5122                      00038142  5461           BNE   F344XX                OP3C Different, error
00038138  D503 100C 3058       0000000C  00000058  5462           CLC   12(4,R1),88(R3)
0003813E  478B 0000                      00000000  5463           BE    0(R11)                OP4Csame, return
                                                   5464
                               00038142  00000001  5465 F344XX    EQU   *
00038142  45F0 83E8                      000023E8  5466           BAL   R15,FAILMSG
00038146  07FB                                     5467           BR    R11
                                                   5468
00038148  4700 512C                      0003814C  5469 F344TSTCC BC     0,F344CCMSG         fail if unexpected condition code
                               0003814C  00000001  5470 F344CCMSG EQU   *
0003814C  45F0 8196                      00002196  5471           BAL   R15,CCMSG
00038150  47F0 50CE                      000380EE  5472           B     F344CKUR
                                                   5473
                                                   5474           TSTDATA
00038154                       00038154  00038600  5475+          ORG   PLOETST+(128*K)+32*(3*K)+(K+512)
                                                   5476           TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00038600  6E6E6ED6 D7C4E2E3                       5477+          DC    CL32'>>>OPDST70<<<' eye catcher
00038608  F7F04C4C 4C404040
00038610  40404040 40404040
00038618  40404040 40404040
00038620                                                 5478+OPDST70  DS    0C
00038620  00000000 00000000                              5479          DC    XL256'0'
00038628  00000000 00000000
00038630  00000000 00000000
00038638  00000000 00000000
00038640  00000000 00000000
00038648  00000000 00000000
00038650  00000000 00000000
00038658  00000000 00000000
00038660  00000000 00000000
00038668  00000000 00000000
00038670  00000000 00000000
00038678  00000000 00000000
00038680  00000000 00000000
00038688  00000000 00000000
00038690  00000000 00000000
00038698  00000000 00000000
000386A0  00000000 00000000
000386A8  00000000 00000000
000386B0  00000000 00000000
000386B8  00000000 00000000
000386C0  00000000 00000000
000386C8  00000000 00000000
000386D0  00000000 00000000
000386D8  00000000 00000000
000386E0  00000000 00000000
000386E8  00000000 00000000
000386F0  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000386F8  00000000 00000000
00038700  00000000 00000000
00038708  00000000 00000000
00038710  00000000 00000000
00038718  00000000 00000000
                                        5480
                                        5481          TSTLBL OPSRC              Source: OP1-OP4
00038720  6E6E6ED6 D7E2D9C3             5482+         DC    CL32'>>>OPSRC70<<<' eye catcher
00038728  F7F04C4C 4C404040
00038730  40404040 40404040
00038738  40404040 40404040
00038740                                5483+OPSRC70  DS    0C
00038740  00000000 00000000             5484          DC    XL256'0'
00038748  00000000 00000000
00038750  00000000 00000000
00038758  00000000 00000000
00038760  00000000 00000000
00038768  00000000 00000000
00038770  00000000 00000000
00038778  00000000 00000000
00038780  00000000 00000000
00038788  00000000 00000000
00038790  00000000 00000000
00038798  00000000 00000000
000387A0  00000000 00000000
000387A8  00000000 00000000
000387B0  00000000 00000000
000387B8  00000000 00000000
000387C0  00000000 00000000
000387C8  00000000 00000000
000387D0  00000000 00000000
000387D8  00000000 00000000
000387E0  00000000 00000000
000387E8  00000000 00000000
000387F0  00000000 00000000
000387F8  00000000 00000000
00038800  00000000 00000000
00038808  00000000 00000000
00038810  00000000 00000000
00038818  00000000 00000000
00038820  00000000 00000000
00038828  00000000 00000000
00038830  00000000 00000000
00038838  00000000 00000000
                                        5485
                                        5486          TSTLBL PL                 Parameter List
00038840  6E6E6ED7 D3F7F04C             5487+         DC    CL32'>>>PL70<<<'    eye catcher
00038848  4C4C4040 40404040
00038850  40404040 40404040
00038858  40404040 40404040
00038860                                5488+PL70     DS    0C
00038860  00000000 00000000             5489          DC    XL352'00'
00038868  00000000 00000000
00038870  00000000 00000000
00038878  00000000 00000000
00038880  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00038888  00000000 00000000
00038890  00000000 00000000
00038898  00000000 00000000
000388A0  00000000 00000000
000388A8  00000000 00000000
000388B0  00000000 00000000
000388B8  00000000 00000000
000388C0  00000000 00000000
000388C8  00000000 00000000
000388D0  00000000 00000000
000388D8  00000000 00000000
000388E0  00000000 00000000
000388E8  00000000 00000000
000388F0  00000000 00000000
000388F8  00000000 00000000
00038900  00000000 00000000
00038908  00000000 00000000
00038910  00000000 00000000
00038918  00000000 00000000
00038920  00000000 00000000
00038928  00000000 00000000
00038930  00000000 00000000
00038938  00000000 00000000
00038940  00000000 00000000
00038948  00000000 00000000
00038950  00000000 00000000
00038958  00000000 00000000
00038960  00000000 00000000
00038968  00000000 00000000
00038970  00000000 00000000
00038978  00000000 00000000
00038980  00000000 00000000
00038988  00000000 00000000
00038990  00000000 00000000
00038998  00000000 00000000
000389A0  00000000 00000000
000389A8  00000000 00000000
000389B0  00000000 00000000
000389B8  00000000 00000000
                                      5490
                                      5491          TSTLBL FMSG                fail message
000389C0  6E6E6EC6 D4E2C7F7           5492+         DC     CL32'>>>FMSG70<<<'  eye catcher
000389C8  F04C4C4C 40404040
000389D0  40404040 40404040
000389D8  40404040 40404040
000389E0                              5493+FMSG70   DS     0C
000389E0  86818993 858440D7           5494          DC     C'failed PLO-Extension '
000389E8  D3D660C5 A7A38595
000389F0  A2899695 40
000389F5  D8A48184 99A49793           5495          DC     CL96'Quadruple Compare and Swap (QCS: 34): 4th NE'
000389FD  8540C396 94978199
00038A05  85408195 8440E2A6
00038A0D  8197404D D8C3E27A
00038A15  40F3F45D 7A40F4A3
00038A1D  8840D5C5 40404040
00038A25  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00038A2D  40404040 40404040                        5496
00038A35  40404040 40404040                        5497          TSTEND
00038A3D  40404040 40404040                        5498+         DC    CL32'END of test: 70 <<<'
00038A45  40404040 40404040
00038A4D  40404040 40404040
                                                   5499
                                                   5500
                                                   5501 * ------------------------------------------------------------------
                                                   5502 * Test Quadruple Compare and Swap (QCS: 34) instruction : All EQ
                                                   5503 * ------------------------------------------------------------------
                                                   5504          TSTHDR 0,99,34             All EQ (cc=0, UR=99)
00038A75                    00038A75  00038C00     5505+         ORG   PLOETST+(128*K)+33*(3*K)
00038C00  E2E3C1D9 E3409686                        5506+         DC    CL32'START of test: 71'
00038C08  40A385A2 A37A40F7
00038C10  F1404040 40404040
00038C18  40404040 40404040
00038C20                    00038C20              5507+         USING *,R5                 base for test data and test routine
00038C20                    00000000              5508+         USING PLOETEST,R5
00038C20  00038C4C                                 5509+T71      DC    A(CODE71)            address of test code
00038C24  0047                                     5510+         DC    H'71'                test number
00038C26  00                                       5511+         DC    X'00'
00038C27  22                                       5512+         DC    HL1'34'              Function
00038C28  63                                       5513+         DC    HL1'99'              expected unequal results (UR)
00038C29  00                                       5514+         DC    HL1'00'              found UR
00038C2A  00                                       5515+         DC    HL1'0'               cc
00038C2B  07                                       5516+         DC    HL1'7'               not expected CC mask
00038C2C  00                                       5517+         DS    X                    extracted cc
00038C30  00000000 00000000                        5518+         DS    2F                   extract PSW after test (has CC)
00038C38  000395E0                                 5519+         DC    A(FMSG71)            address of test fail message
00038C3C  00039340                                 5520+         DC    A(OPSRC71)           address of operand source
00038C40  00039220                                 5521+         DC    A(OPDST71)           address of operand destination
00038C44  00039460                                 5522+         DC    A(PL71)              address of parameter list
00038C48  000041FF                                 5523+         DC    A(RANDX+71)          address of random source data
                                                   5524
                                                   5525          TSTCODE
00038C4C                                           5526+CODE71   DS    0F
                                                   5527 *                                   load random source
00038C4C  E310 501C 0014              0000001C     5528          LGF   R1,OPSRCA
00038C52  E320 5028 0014              00000028     5529          LGF   R2,RANDXA
00038C58  E330 5024 0014              00000024     5530          LGF   R3,PLA
00038C5E  E340 5020 0014              00000020     5531          LGF   R4,OPREPA
                                                   5532 *
                                                   5533 * Quadruple Compare and Swap (QCS: 34)
                                                   5534 * Build Parambter List
                                                   5535 *
00038C64  D20F 1000 2000      00000000  00000000   5536          MVC   0(16,R1),0(R2)       load Op1-Op4 data in source
00038C6A  D203 3040 1000      00000040  00000000   5537          MVC   64(4,R3),0(R1)       copy to PL OP1-Compare
00038C70  D203 3048 1004      00000048  00000004   5538          MVC   72(4,R3),4(R1)       copy to PL OP2-Compare
00038C76  D203 3050 1008      00000050  00000008   5539          MVC   80(4,R3),8(R1)       copy to PL OP3-Compare
00038C7C  D203 3058 100C      00000058  0000000C   5540          MVC   88(4,R3),12(R1)      copy to PL OP4-Compare
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    5541
00038C82  D21F 4000 2080      00000000  00000080  5542          MVC   0(32,R4),128(R2)    load initial Op1-Op4 data
00038C88  D203 3060 4000      00000060  00000000  5543          MVC   96(4,R3),0(R4)      copy to PL OP1-Replacement
00038C8E  D203 3068 4004      00000068  00000004  5544          MVC   104(4,R3),4(R4)     copy to PL OP2-Replacement
00038C94  D203 3070 4008      00000070  00000008  5545          MVC   112(4,R3),8(R4)     copy to PL OP3-Replacement
00038C9A  D203 3078 400C      00000078  0000000C  5546          MVC   120(4,R3),12(R4)    copy to PL OP4-Replacement
                                                    5547 ****
                                                    5548 * ALL EQ: do not force a failure by changing an operand
                                                    5549 ****
                                                    5550 *         MVC   12(4,R1),0(R4)
                                                    5551 ****
                                                    5552
00038CA0  E313 0000 0024                  00000000  5553          STG   R1,0(R3)            load OP1 address in PL
                                                    5554
00038CA6  4111 0004                       00000004  5555          LA    R1,4(R1)
00038CAA  E313 0008 0024                  00000008  5556          STG   R1,8(R3)            load OP2 address in PL
                                                    5557
00038CB0  4111 0004                       00000004  5558          LA    R1,4(R1)
00038CB4  E313 0010 0024                  00000010  5559          STG   R1,16(R3)           load OP3 address in PL
                                                    5560
00038CBA  4111 0004                       00000004  5561          LA    R1,4(R1)
00038CBE  E313 0018 0024                  00000018  5562          STG   R1,24(R3)           load OP4 address in PL
                                                    5563
00038CC4  B982 0000                                 5564          XGR   R0,R0
00038CC8  4100 0022                       00000022  5565          LA    R0,34
00038CCC  EE00 0000 3000      00000000  00000000  5566          PLO   0,0,0,0(R3)         Quadruple C/S (QCS: 34)
                                                    5567
00038CD2  B98D 0020                                 5568          EPSW  R2,R0               extract psw
00038CD6  5020 5010                       00000010  5569          ST    R2,CCPSW              to save CC
00038CDA  E310 500B 0076                  0000000B  5570          LB    R1,CCMASK           (failure CC mask)
00038CE0  8910 0004                       00000004  5571          SLL   R1,4                (shift to BC instr CC position)
00038CE4  4410 510E                       00038D2E  5572          EX    R1,F345TSTCC            fail if...
                                                    5573 *
                                                    5574 * compare results
                                                    5575 *
                                  00038CE8  00000001  5576 F345CKRES EQU   *
                                                    5577 *                                  reset adresses
00038CE8  E310 501C 0014                  0000001C  5578          LGF   R1,OPSRCA
00038CEE  E320 5028 0014                  00000028  5579          LGF   R2,RANDXA
00038CF4  E330 5024 0014                  00000024  5580          LGF   R3,PLA
00038CFA  E340 5020 0014                  00000020  5581          LGF   R4,OPREPA
                                                    5582
                                                    5583 * All equal; replacement values copied to operand source
                                                    5584
00038D00  D503 1000 3060      00000000  00000060  5585          CLC   0(4,R1),96(R3)
00038D06  4770 5108                       00038D28  5586          BNE   F345XX              OP1R Different, error
00038D0A  D503 1004 3068      00000004  00000068  5587          CLC   4(4,R1),104(R3)
00038D10  4770 5108                       00038D28  5588          BNE   F345XX              OP2R Different, error
00038D14  D503 1008 3070      00000008  00000070  5589          CLC   8(4,R1),112(R3)
00038D1A  4770 5108                       00038D28  5590          BNE   F345XX              OP3R Different, error
00038D1E  D503 100C 3078      0000000C  00000078  5591          CLC   12(4,R1),120(R3)
00038D24  478B 0000                       00000000  5592          BE    0(R11)              OP4R same, return
                                                    5593
                                  00038D28  00000001  5594 F345XX   EQU   *
00038D28  45F0 83E8                       000023E8  5595          BAL   R15,FAILMSG
00038D2C  07FB                                      5596          BR    R11
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    5597
00038D2E  4700 5112                       00038D32  5598 F345TSTCC BC    0,F345CCMSG        fail if unexpected condition code
                                00038D32  00000001  5599 F345CCMSG EQU   *
00038D32  45F0 8196                       00002196  5600           BAL   R15,CCMSG
00038D36  47F0 50C8                       00038CE8  5601           B     F345CKRES
                                                    5602
                                                    5603           TSTDATA
00038D3A                        00038D3A  00039200  5604+          ORG   PLOETST+(128*K)+33*(3*K)+(K+512)
                                                    5605           TSTLBL OPDST               Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00039200  6E6E6ED6 D7C4E2E3                        5606+          DC    CL32'>>>OPDST71<<<' eye catcher
00039208  F7F14C4C 4C404040
00039210  40404040 40404040
00039218  40404040 40404040
00039220                                           5607+OPDST71   DS    0C
00039220  00000000 00000000                        5608           DC    XL256'0'
00039228  00000000 00000000
00039230  00000000 00000000
00039238  00000000 00000000
00039240  00000000 00000000
00039248  00000000 00000000
00039250  00000000 00000000
00039258  00000000 00000000
00039260  00000000 00000000
00039268  00000000 00000000
00039270  00000000 00000000
00039278  00000000 00000000
00039280  00000000 00000000
00039288  00000000 00000000
00039290  00000000 00000000
00039298  00000000 00000000
000392A0  00000000 00000000
000392A8  00000000 00000000
000392B0  00000000 00000000
000392B8  00000000 00000000
000392C0  00000000 00000000
000392C8  00000000 00000000
000392D0  00000000 00000000
000392D8  00000000 00000000
000392E0  00000000 00000000
000392E8  00000000 00000000
000392F0  00000000 00000000
000392F8  00000000 00000000
00039300  00000000 00000000
00039308  00000000 00000000
00039310  00000000 00000000
00039318  00000000 00000000
                                                    5609
                                                    5610           TSTLBL OPSRC               Source: OP1-OP4
00039320  6E6E6ED6 D7E2D9C3                        5611+          DC    CL32'>>>OPSRC71<<<' eye catcher
00039328  F7F14C4C 4C404040
00039330  40404040 40404040
00039338  40404040 40404040
00039340                                           5612+OPSRC71   DS    0C
00039340  00000000 00000000                        5613           DC    XL256'0'
00039348  00000000 00000000
00039350  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00039358  00000000 00000000
00039360  00000000 00000000
00039368  00000000 00000000
00039370  00000000 00000000
00039378  00000000 00000000
00039380  00000000 00000000
00039388  00000000 00000000
00039390  00000000 00000000
00039398  00000000 00000000
000393A0  00000000 00000000
000393A8  00000000 00000000
000393B0  00000000 00000000
000393B8  00000000 00000000
000393C0  00000000 00000000
000393C8  00000000 00000000
000393D0  00000000 00000000
000393D8  00000000 00000000
000393E0  00000000 00000000
000393E8  00000000 00000000
000393F0  00000000 00000000
000393F8  00000000 00000000
00039400  00000000 00000000
00039408  00000000 00000000
00039410  00000000 00000000
00039418  00000000 00000000
00039420  00000000 00000000
00039428  00000000 00000000
00039430  00000000 00000000
00039438  00000000 00000000
                                      5614
                                      5615          TSTLBL PL                  Parameter List
00039440  6E6E6ED7 D3F7F14C           5616+         DC    CL32'>>>PL71<<<'     eye catcher
00039448  4C4C4040 40404040
00039450  40404040 40404040
00039458  40404040 40404040
00039460                              5617+PL71     DS    0C
00039460  00000000 00000000           5618          DC    XL352'00'
00039468  00000000 00000000
00039470  00000000 00000000
00039478  00000000 00000000
00039480  00000000 00000000
00039488  00000000 00000000
00039490  00000000 00000000
00039498  00000000 00000000
000394A0  00000000 00000000
000394A8  00000000 00000000
000394B0  00000000 00000000
000394B8  00000000 00000000
000394C0  00000000 00000000
000394C8  00000000 00000000
000394D0  00000000 00000000
000394D8  00000000 00000000
000394E0  00000000 00000000
000394E8  00000000 00000000
000394F0  00000000 00000000
000394F8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00039500  00000000 00000000
00039508  00000000 00000000
00039510  00000000 00000000
00039518  00000000 00000000
00039520  00000000 00000000
00039528  00000000 00000000
00039530  00000000 00000000
00039538  00000000 00000000
00039540  00000000 00000000
00039548  00000000 00000000
00039550  00000000 00000000
00039558  00000000 00000000
00039560  00000000 00000000
00039568  00000000 00000000
00039570  00000000 00000000
00039578  00000000 00000000
00039580  00000000 00000000
00039588  00000000 00000000
00039590  00000000 00000000
00039598  00000000 00000000
000395A0  00000000 00000000
000395A8  00000000 00000000
000395B0  00000000 00000000
000395B8  00000000 00000000
                                          5619
                                          5620          TSTLBL FMSG                fail message
000395C0  6E6E6EC6 D4E2C7F7               5621+         DC    CL32'>>>FMSG71<<<'  eye catcher
000395C8  F14C4C4C 40404040
000395D0  40404040 40404040
000395D8  40404040 40404040
000395E0                                  5622+FMSG71   DS    0C
000395E0  86818993 858440D7               5623          DC    C'failed PLO-Extension '
000395E8  D3D660C5 A7A38595
000395F0  A2899695 40
000395F5  D8A48184 99A49793               5624          DC    CL96'Quadruple Compare and Swap (QCS: 34): All EQ'
000395FD  8540C396 94978199
00039605  85408195 8440E2A6
0003960D  8197404D D8C3E27A
00039615  40F3F45D 7A40C193
0003961D  9340C5D8 40404040
00039625  40404040 40404040
0003962D  40404040 40404040
00039635  40404040 40404040
0003963D  40404040 40404040
00039645  40404040 40404040
0003964D  40404040 40404040
                                          5625
                                          5626          TSTEND
00039655  C5D5C440 968640A3               5627+         DC    CL32'END of test: 71 <<<'
0003965D  85A2A37A 40F7F140
00039665  4C4C4C40 40404040
0003966D  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   5629 *=============================================================
                                                   5630 *       Triple Compare and Swap Tests
                                                   5631 *=============================================================
                                                   5632
                                                   5633 * -------------------------------------------------------------
                                                   5634 * Test Triple Compare and Swap (TCSO: 33) instruction : 1st NE
                                                   5635 * -------------------------------------------------------------
                                                   5636          TSTHDR 2,1,33             1st Operand NOT Equal (cc=2, UR=1)
00039675                          00039675  00039800 5637+         ORG   PLOETST+(128*K)+34*(3*K)
00039800  E2E3C1D9 E3409686                        5638+         DC    CL32'START of test: 72'
00039808  40A385A2 A37A40F7
00039810  F2404040 40404040
00039818  40404040 40404040
00039820                          00039820         5639+         USING *,R5                base for test data and test routine
00039820                          00000000         5640+         USING PLOETEST,R5
00039820  0003984C                                 5641+T72      DC    A(CODE72)           address of test code
00039824  0048                                     5642+         DC    H'72'               test number
00039826  00                                       5643+         DC    X'00'
00039827  21                                       5644+         DC    HL1'33'             Function
00039828  01                                       5645+         DC    HL1'1'              expected unequal results (UR)
00039829  00                                       5646+         DC    HL1'00'             found UR
0003982A  02                                       5647+         DC    HL1'2'              cc
0003982B  0D                                       5648+         DC    HL1'13'             not expected CC mask
0003982C  00                                       5649+         DS    X                   extracted cc
00039830  00000000 00000000                        5650+         DS    2F                  extract PSW after test (has CC)
00039838  0003A1E0                                 5651+         DC    A(FMSG72)           address of test fail message
0003983C  00039F40                                 5652+         DC    A(OPSRC72)          address of operand source
00039840  00039E20                                 5653+         DC    A(OPDST72)          address of operand destination
00039844  0003A060                                 5654+         DC    A(PL72)             address of parameter list
00039848  00004200                                 5655+         DC    A(RANDX+72)         address of random source data
                                                   5656
                                                   5657          TSTCODE
0003984C                                           5658+CODE72   DS    0F
                                                   5659 *                                  load random source
0003984C  E310 501C 0014                0000001C   5660          LGF   R1,OPSRCA
00039852  E320 5028 0014                00000028   5661          LGF   R2,RANDXA
00039858  E330 5024 0014                00000024   5662          LGF   R3,PLA
0003985E  E340 5020 0014                00000020   5663          LGF   R4,OPREPA
                                                   5664 *
                                                   5665 * Triple Compare and Swap (TCSO: 33)
                                                   5666 * Build Parambter List
                                                   5667 *
00039864  D25F 1000 2000      00000000  00000000   5668          MVC   0(96,R1),0(R2)      load Op1-Op3 data in source
0003986A  D25F 3040 1000      00000040  00000000   5669          MVC   64(96,R3),0(R1)     copy to PL OP1-Op3 Compare
                                                   5670
00039870  D25F 4000 2080      00000000  00000080   5671          MVC   0(96,R4),128(R2)    load initial Op1-Op3 data
00039876  D25F 30A0 4000      000000A0  00000000   5672          MVC   160(96,R3),0(R4)    copy to PL Op1-Op3 Replacement
                                                   5673 ****
                                                   5674 * force a failure by changing the first operand
                                                   5675 ****
0003987C  D207 1000 4000      00000000  00000000   5676          MVC   0(8,R1),0(R4)
                                                   5677 ****
                                                   5678
00039882  E313 0000 0024                00000000   5679          STG   R1,0(R3)            load OP1 address in PL
                                                   5680
00039888  4111 0020                     00000020   5681          LA    R1,32(R1)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003988C  E313 0008 0024                 00000008  5682           STG   R1,8(R3)            load OP2 address in PL
                                                   5683
00039892  4111 0020                      00000020  5684           LA    R1,32(R1)
00039896  E313 0010 0024                 00000010  5685           STG   R1,16(R3)           load OP3 address in PL
                                                   5686
0003989C  B982 0000                                5687           XGR   R0,R0
000398A0  4100 0021                      00000021  5688           LA    R0,33
000398A4  EE00 0000 3000       00000000  00000000  5689           PLO   0,0,0,0(R3)         Triple C/S (TCSO: 33)
                                                   5690
000398AA  B98D 0020                                5691           EPSW  R2,R0               extract psw
000398AE  5020 5010                      00000010  5692           ST    R2,CCPSW              to save CC
000398B2  E310 500B 0076                 0000000B  5693           LB    R1,CCMASK           (failure CC mask)
000398B8  8910 0004                      00000004  5694           SLL   R1,4                (shift to BC instr CC position)
000398BC  4410 50F0                      00039910  5695           EX    R1,F331TSTCC            fail if...
                                                   5696
                                 000398C0  00000001  5697 F331CKUR EQU   *
000398C0  D200 5009 3030       00000009  00000030  5698           MVC   URFOUND,48(R3)      Extract UR
000398C6  D500 5008 3030       00000008  00000030  5699           CLC   UR,48(R3)
000398CC  4780 50B4                      000398D4  5700           BE    F331CKRES
000398D0  45F0 82C8                      000022C8  5701           BAL   R15,URMSG
                                                   5702 *
                                                   5703 * compare results
                                                   5704 *
                                 000398D4  00000001  5705 F331CKRES EQU   *
                                                   5706 *                                  reset adresses
000398D4  E310 501C 0014                 0000001C  5707           LGF   R1,OPSRCA
000398DA  E320 5028 0014                 00000028  5708           LGF   R2,RANDXA
000398E0  E330 5024 0014                 00000024  5709           LGF   R3,PLA
000398E6  E340 5020 0014                 00000020  5710           LGF   R4,OPREPA
                                                   5711
000398EC  D51F 1000 3040       00000000  00000040  5712           CLC   0(32,R1),64(R3)
000398F2  4770 50EA                      0003990A  5713           BNE   F331XX              OP1C Different, error
000398F6  D51F 1020 3060       00000020  00000060  5714           CLC   32(32,R1),96(R3)
000398FC  4770 50EA                      0003990A  5715           BNE   F331XX              OP2C Different, error
00039900  D51F 1040 3080       00000040  00000080  5716           CLC   64(32,R1),128(R3)
00039906  478B 0000                      00000000  5717           BE    0(R11)              OP3C same, return
                                                   5718
                                 0003990A  00000001  5719 F331XX   EQU   *
0003990A  45F0 83E8                      000023E8  5720           BAL   R15,FAILMSG
0003990E  07FB                                     5721           BR    R11
                                                   5722
00039910  4700 50F4                      00039914  5723 F331TSTCC BC    0,F331CCMSG         fail if unexpected condition code
                                 00039914  00000001  5724 F331CCMSG EQU   *
00039914  45F0 8196                      00002196  5725           BAL   R15,CCMSG
00039918  47F0 50A0                      000398C0  5726           B     F331CKUR
                                                   5727
                                                   5728           TSTDATA
0003991C                         0003991C  00039E00  5729+          ORG   PLOETST+(128*K)+34*(3*K)+(K+512)
                                                   5730           TSTLBL OPDST              Dest/Data: Op1-R, Op2-R, Op3-R
00039E00  6E6E6ED6 D7C4E2E3                        5731+          DC    CL32'>>>OPDST72<<<' eye catcher
00039E08  F7F24C4C 4C404040
00039E10  40404040 40404040
00039E18  40404040 40404040
00039E20                                           5732+OPDST72   DS    0C
00039E20  00000000 00000000                        5733           DC    XL256'0'
00039E28  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00039E30  00000000 00000000
00039E38  00000000 00000000
00039E40  00000000 00000000
00039E48  00000000 00000000
00039E50  00000000 00000000
00039E58  00000000 00000000
00039E60  00000000 00000000
00039E68  00000000 00000000
00039E70  00000000 00000000
00039E78  00000000 00000000
00039E80  00000000 00000000
00039E88  00000000 00000000
00039E90  00000000 00000000
00039E98  00000000 00000000
00039EA0  00000000 00000000
00039EA8  00000000 00000000
00039EB0  00000000 00000000
00039EB8  00000000 00000000
00039EC0  00000000 00000000
00039EC8  00000000 00000000
00039ED0  00000000 00000000
00039ED8  00000000 00000000
00039EE0  00000000 00000000
00039EE8  00000000 00000000
00039EF0  00000000 00000000
00039EF8  00000000 00000000
00039F00  00000000 00000000
00039F08  00000000 00000000
00039F10  00000000 00000000
00039F18  00000000 00000000
                                      5734
                                      5735           TSTLBL OPSRC               Source: OP1-OP3
00039F20  6E6E6ED6 D7E2D9C3           5736+          DC    CL32'>>>OPSRC72<<<' eye catcher
00039F28  F7F24C4C 4C404040
00039F30  40404040 40404040
00039F38  40404040 40404040
00039F40                              5737+OPSRC72   DS    0C
00039F40  00000000 00000000           5738           DC    XL256'0'
00039F48  00000000 00000000
00039F50  00000000 00000000
00039F58  00000000 00000000
00039F60  00000000 00000000
00039F68  00000000 00000000
00039F70  00000000 00000000
00039F78  00000000 00000000
00039F80  00000000 00000000
00039F88  00000000 00000000
00039F90  00000000 00000000
00039F98  00000000 00000000
00039FA0  00000000 00000000
00039FA8  00000000 00000000
00039FB0  00000000 00000000
00039FB8  00000000 00000000
00039FC0  00000000 00000000
00039FC8  00000000 00000000
00039FD0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00039FD8  00000000 00000000
00039FE0  00000000 00000000
00039FE8  00000000 00000000
00039FF0  00000000 00000000
00039FF8  00000000 00000000
0003A000  00000000 00000000
0003A008  00000000 00000000
0003A010  00000000 00000000
0003A018  00000000 00000000
0003A020  00000000 00000000
0003A028  00000000 00000000
0003A030  00000000 00000000
0003A038  00000000 00000000
                                          5739
                                          5740           TSTLBL PL                    Parameter List
0003A040  6E6E6ED7 D3F7F24C               5741+          DC    CL32'>>>PL72<<<'       eye catcher
0003A048  4C4C4040 40404040
0003A050  40404040 40404040
0003A058  40404040 40404040
0003A060                                  5742+PL72      DS    0C
0003A060  00000000 00000000               5743           DC    XL352'00'
0003A068  00000000 00000000
0003A070  00000000 00000000
0003A078  00000000 00000000
0003A080  00000000 00000000
0003A088  00000000 00000000
0003A090  00000000 00000000
0003A098  00000000 00000000
0003A0A0  00000000 00000000
0003A0A8  00000000 00000000
0003A0B0  00000000 00000000
0003A0B8  00000000 00000000
0003A0C0  00000000 00000000
0003A0C8  00000000 00000000
0003A0D0  00000000 00000000
0003A0D8  00000000 00000000
0003A0E0  00000000 00000000
0003A0E8  00000000 00000000
0003A0F0  00000000 00000000
0003A0F8  00000000 00000000
0003A100  00000000 00000000
0003A108  00000000 00000000
0003A110  00000000 00000000
0003A118  00000000 00000000
0003A120  00000000 00000000
0003A128  00000000 00000000
0003A130  00000000 00000000
0003A138  00000000 00000000
0003A140  00000000 00000000
0003A148  00000000 00000000
0003A150  00000000 00000000
0003A158  00000000 00000000
0003A160  00000000 00000000
0003A168  00000000 00000000
0003A170  00000000 00000000
0003A178  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003A180  00000000 00000000
0003A188  00000000 00000000
0003A190  00000000 00000000
0003A198  00000000 00000000
0003A1A0  00000000 00000000
0003A1A8  00000000 00000000
0003A1B0  00000000 00000000
0003A1B8  00000000 00000000
                                        5744
                                        5745          TSTLBL FMSG                fail message
0003A1C0  6E6E6EC6 D4E2C7F7             5746+         DC    CL32'>>>FMSG72<<<'   eye catcher
0003A1C8  F24C4C4C 40404040
0003A1D0  40404040 40404040
0003A1D8  40404040 40404040
0003A1E0                                5747+FMSG72   DS    0C
0003A1E0  86818993 858440D7             5748          DC    C'failed PLO-Extension '
0003A1E8  D3D660C5 A7A38595
0003A1F0  A2899695 40
0003A1F5  E3998997 938540C3             5749          DC    CL96'Triple Compare and Swap (TCSO: 33): 1st NE'
0003A1FD  96949781 99854081
0003A205  958440E2 A6819740
0003A20D  4DE3C3E2 D67A40F3
0003A215  F35D7A40 F1A2A340
0003A21D  D5C54040 40404040
0003A225  40404040 40404040
0003A22D  40404040 40404040
0003A235  40404040 40404040
0003A23D  40404040 40404040
0003A245  40404040 40404040
0003A24D  40404040 40404040
                                        5750
                                        5751          TSTEND
0003A255  C5D5C440 968640A3             5752+         DC    CL32'END of test: 72 <<<'
0003A25D  85A2A37A 40F7F240
0003A265  4C4C4C40 40404040
0003A26D  40404040 40404040
                                        5753
                                        5754 * ------------------------------------------------------------------
                                        5755 * Test Triple Compare and Swap (TCSO: 33) instruction : 2nd NE
                                        5756 * ------------------------------------------------------------------
                                        5757          TSTHDR 2,2,33              2nd Operand NOT Equal (cc=2, UR=2)
0003A275                        0003A275  0003A400    5758+         ORG   PLOETST+(128*K)+35*(3*K)
0003A400  E2E3C1D9 E3409686             5759+         DC    CL32'START of test: 73'
0003A408  40A385A2 A37A40F7
0003A410  F3404040 40404040
0003A418  40404040 40404040
0003A420                        0003A420              5760+         USING *,R5                 base for test data and test routine
0003A420                        00000000              5761+         USING PLOETEST,R5
0003A420  0003A44C                                    5762+T73      DC    A(CODE73)            address of test code
0003A424  0049                                        5763+         DC    H'73'                test number
0003A426  00                                          5764+         DC    X'00'
0003A427  21                                          5765+         DC    HL1'33'              Function
0003A428  02                                          5766+         DC    HL1'2'               expected unequal results (UR)
0003A429  00                                          5767+         DC    HL1'00'              found UR
0003A42A  02                                          5768+         DC    HL1'2'               cc
0003A42B  0D                                          5769+         DC    HL1'13'              not expected CC mask
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003A42C  00                                       5770+          DS    X                   extracted cc
0003A430  00000000 00000000                        5771+          DS    2F                  extract PSW after test (has CC)
0003A438  0003ADE0                                 5772+          DC    A(FMSG73)           address of test fail message
0003A43C  0003AB40                                 5773+          DC    A(OPSRC73)          address of operand source
0003A440  0003AA20                                 5774+          DC    A(OPDST73)          address of operand destination
0003A444  0003AC60                                 5775+          DC    A(PL73)             address of parameter list
0003A448  00004201                                 5776+          DC    A(RANDX+73)         address of random source data
                                                   5777
                                                   5778           TSTCODE
0003A44C                                           5779+CODE73    DS    0F
                                                   5780 *                                   load random source
0003A44C  E310 501C 0014                0000001C   5781           LGF   R1,OPSRCA
0003A452  E320 5028 0014                00000028   5782           LGF   R2,RANDXA
0003A458  E330 5024 0014                00000024   5783           LGF   R3,PLA
0003A45E  E340 5020 0014                00000020   5784           LGF   R4,OPREPA
                                                   5785 *
                                                   5786 * Triple Compare and Swap (TCSO: 33)
                                                   5787 * Build Parambter List
                                                   5788 *
0003A464  D25F 1000 2000      00000000  00000000   5789           MVC   0(96,R1),0(R2)      load Op1-Op3 data in source
0003A46A  D25F 3040 1000      00000040  00000000   5790           MVC   64(96,R3),0(R1)     copy to PL OP1-Op3 Compare
                                                   5791
0003A470  D25F 4000 2080      00000000  00000080   5792           MVC   0(96,R4),128(R2)    load initial Op1-Op3 data
0003A476  D25F 30A0 4000      000000A0  00000000   5793           MVC   160(96,R3),0(R4)    copy to PL Op1-Op3 Replacement
                                                   5794 ****
                                                   5795 * force a failure by changing the 2nd operand
                                                   5796 ****
0003A47C  D207 1020 4000      00000020  00000000   5797           MVC   32(8,R1),0(R4)
                                                   5798 ****
                                                   5799
0003A482  E313 0000 0024                00000000   5800           STG   R1,0(R3)            load OP1 address in PL
                                                   5801
0003A488  4111 0020                     00000020   5802           LA    R1,32(R1)
0003A48C  E313 0008 0024                00000008   5803           STG   R1,8(R3)            load OP2 address in PL
                                                   5804
0003A492  4111 0020                     00000020   5805           LA    R1,32(R1)
0003A496  E313 0010 0024                00000010   5806           STG   R1,16(R3)           load OP3 address in PL
                                                   5807
0003A49C  B982 0000                                5808           XGR   R0,R0
0003A4A0  4100 0021                     00000021   5809           LA    R0,33
0003A4A4  EE00 0000 3000      00000000  00000000   5810           PLO   0,0,0,0(R3)         Triple C/S (TCSO: 33)
                                                   5811
0003A4AA  B98D 0020                                5812           EPSW  R2,R0               extract psw
0003A4AE  5020 5010                     00000010   5813           ST    R2,CCPSW              to save CC
0003A4B2  E310 500B 0076                0000000B   5814           LB    R1,CCMASK           (failure CC mask)
0003A4B8  8910 0004                     00000004   5815           SLL   R1,4                (shift to BC instr CC position)
0003A4BC  4410 50F0                     0003A510   5816           EX    R1,F332TSTCC            fail if...
                                                   5817
                              0003A4C0  00000001   5818 F332CKUR  EQU   *
0003A4C0  D200 5009 3030      00000009  00000030   5819           MVC   URFOUND,48(R3)      Extract UR
0003A4C6  D500 5008 3030      00000008  00000030   5820           CLC   UR,48(R3)
0003A4CC  4780 50B4                     0003A4D4   5821           BE    F332CKRES
0003A4D0  45F0 82C8                     000022C8   5822           BAL   R15,URMSG
                                                   5823 *
                                                   5824 * compare results
                                                   5825 *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                0003A4D4  00000001  5826 F332CKRES EQU   *
                                                    5827 *                                reset adresses
0003A4D4  E310 501C 0014                  0000001C  5828          LGF   R1,OPSRCA
0003A4DA  E320 5028 0014                  00000028  5829          LGF   R2,RANDXA
0003A4E0  E330 5024 0014                  00000024  5830          LGF   R3,PLA
0003A4E6  E340 5020 0014                  00000020  5831          LGF   R4,OPREPA
                                                    5832
0003A4EC  D51F 1000 3040  00000000        00000040  5833          CLC   0(32,R1),64(R3)
0003A4F2  4770 50EA                       0003A50A  5834          BNE   F332XX                OP1C Different, error
0003A4F6  D51F 1020 3060  00000020        00000060  5835          CLC   32(32,R1),96(R3)
0003A4FC  4770 50EA                       0003A50A  5836          BNE   F332XX                OP2C Different, error
0003A500  D51F 1040 3080  00000040        00000080  5837          CLC   64(32,R1),128(R3)
0003A506  478B 0000                       00000000  5838          BE    0(R11)                OP3C same, return
                                                    5839
                                0003A50A  00000001  5840 F332XX   EQU   *
0003A50A  45F0 83E8                       000023E8  5841          BAL   R15,FAILMSG
0003A50E  07FB                                      5842          BR    R11
                                                    5843
0003A510  4700 50F4                       0003A514  5844 F332TSTCC BC    0,F332CCMSG           fail if unexpected condition code
                                0003A514  00000001  5845 F332CCMSG EQU   *
0003A514  45F0 8196                       00002196  5846          BAL   R15,CCMSG
0003A518  47F0 50A0                       0003A4C0  5847          B     F332CKUR
                                                    5848
                                                    5849          TSTDATA
0003A51C                        0003A51C  0003AA00  5850+         ORG   PLOETST+(128*K)+35*(3*K)+(K+512)
                                                    5851          TSTLBL OPDST              Dest/Data: Op1-R, Op2-R, Op3-R
0003AA00  6E6E6ED6 D7C4E2E3               5852+         DC    CL32'>>>OPDST73<<<' eye catcher
0003AA08  F7F34C4C 4C404040
0003AA10  40404040 40404040
0003AA18  40404040 40404040
0003AA20                                  5853+OPDST73  DS    0C
0003AA20  00000000 00000000               5854          DC    XL256'0'
0003AA28  00000000 00000000
0003AA30  00000000 00000000
0003AA38  00000000 00000000
0003AA40  00000000 00000000
0003AA48  00000000 00000000
0003AA50  00000000 00000000
0003AA58  00000000 00000000
0003AA60  00000000 00000000
0003AA68  00000000 00000000
0003AA70  00000000 00000000
0003AA78  00000000 00000000
0003AA80  00000000 00000000
0003AA88  00000000 00000000
0003AA90  00000000 00000000
0003AA98  00000000 00000000
0003AAA0  00000000 00000000
0003AAA8  00000000 00000000
0003AAB0  00000000 00000000
0003AAB8  00000000 00000000
0003AAC0  00000000 00000000
0003AAC8  00000000 00000000
0003AAD0  00000000 00000000
0003AAD8  00000000 00000000
0003AAE0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0003AAE8  00000000 00000000
0003AAF0  00000000 00000000
0003AAF8  00000000 00000000
0003AB00  00000000 00000000
0003AB08  00000000 00000000
0003AB10  00000000 00000000
0003AB18  00000000 00000000
                                        5855
                                        5856          TSTLBL OPSRC               Source: OP1-OP3
0003AB20  6E6E6ED6 D7E2D9C3             5857+         DC    CL32'>>>OPSRC73<<<'  eye catcher
0003AB28  F7F34C4C 4C404040
0003AB30  40404040 40404040
0003AB38  40404040 40404040
0003AB40                                5858+OPSRC73  DS    0C
0003AB40  00000000 00000000             5859          DC    XL256'0'
0003AB48  00000000 00000000
0003AB50  00000000 00000000
0003AB58  00000000 00000000
0003AB60  00000000 00000000
0003AB68  00000000 00000000
0003AB70  00000000 00000000
0003AB78  00000000 00000000
0003AB80  00000000 00000000
0003AB88  00000000 00000000
0003AB90  00000000 00000000
0003AB98  00000000 00000000
0003ABA0  00000000 00000000
0003ABA8  00000000 00000000
0003ABB0  00000000 00000000
0003ABB8  00000000 00000000
0003ABC0  00000000 00000000
0003ABC8  00000000 00000000
0003ABD0  00000000 00000000
0003ABD8  00000000 00000000
0003ABE0  00000000 00000000
0003ABE8  00000000 00000000
0003ABF0  00000000 00000000
0003ABF8  00000000 00000000
0003AC00  00000000 00000000
0003AC08  00000000 00000000
0003AC10  00000000 00000000
0003AC18  00000000 00000000
0003AC20  00000000 00000000
0003AC28  00000000 00000000
0003AC30  00000000 00000000
0003AC38  00000000 00000000
                                        5860
                                        5861          TSTLBL PL                  Parameter List
0003AC40  6E6E6ED7 D3F7F34C             5862+         DC    CL32'>>>PL73<<<'     eye catcher
0003AC48  4C4C4040 40404040
0003AC50  40404040 40404040
0003AC58  40404040 40404040
0003AC60                                5863+PL73     DS    0C
0003AC60  00000000 00000000             5864          DC    XL352'00'
0003AC68  00000000 00000000
0003AC70  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003AC78  00000000 00000000
0003AC80  00000000 00000000
0003AC88  00000000 00000000
0003AC90  00000000 00000000
0003AC98  00000000 00000000
0003ACA0  00000000 00000000
0003ACA8  00000000 00000000
0003ACB0  00000000 00000000
0003ACB8  00000000 00000000
0003ACC0  00000000 00000000
0003ACC8  00000000 00000000
0003ACD0  00000000 00000000
0003ACD8  00000000 00000000
0003ACE0  00000000 00000000
0003ACE8  00000000 00000000
0003ACF0  00000000 00000000
0003ACF8  00000000 00000000
0003AD00  00000000 00000000
0003AD08  00000000 00000000
0003AD10  00000000 00000000
0003AD18  00000000 00000000
0003AD20  00000000 00000000
0003AD28  00000000 00000000
0003AD30  00000000 00000000
0003AD38  00000000 00000000
0003AD40  00000000 00000000
0003AD48  00000000 00000000
0003AD50  00000000 00000000
0003AD58  00000000 00000000
0003AD60  00000000 00000000
0003AD68  00000000 00000000
0003AD70  00000000 00000000
0003AD78  00000000 00000000
0003AD80  00000000 00000000
0003AD88  00000000 00000000
0003AD90  00000000 00000000
0003AD98  00000000 00000000
0003ADA0  00000000 00000000
0003ADA8  00000000 00000000
0003ADB0  00000000 00000000
0003ADB8  00000000 00000000
                                          5865
                                          5866          TSTLBL FMSG                fail message
0003ADC0  6E6E6EC6 D4E2C7F7               5867+         DC    CL32'>>>FMSG73<<<'   eye catcher
0003ADC8  F34C4C4C 40404040
0003ADD0  40404040 40404040
0003ADD8  40404040 40404040
0003ADE0                                  5868+FMSG73   DS    0C
0003ADE0  86818993 858440D7               5869          DC    C'failed PLO-Extension '
0003ADE8  D3D660C5 A7A38595
0003ADF0  A2899695 40
0003ADF5  E3998997 938540C3               5870          DC    CL96'Triple Compare and Swap (TCSO: 33): 2nd NE'
0003ADFD  96949781 99854081
0003AE05  958440E2 A6819740
0003AE0D  4DE3C3E2 D67A40F3
0003AE15  F35D7A40 F2958440
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003AE1D  D5C54040 40404040
0003AE25  40404040 40404040
0003AE2D  40404040 40404040
0003AE35  40404040 40404040
0003AE3D  40404040 40404040
0003AE45  40404040 40404040
0003AE4D  40404040 40404040
                                        5871
                                        5872          TSTEND
0003AE55  C5D5C440 968640A3             5873+         DC    CL32'END of test: 73 <<<'
0003AE5D  85A2A37A 40F7F340
0003AE65  4C4C4C40 40404040
0003AE6D  40404040 40404040
                                        5874
                                        5875 * ------------------------------------------------------------------
                                        5876 * Test Triple Compare and Swap (TCSO: 33) instruction : 3rd NE
                                        5877 * ------------------------------------------------------------------
                                        5878          TSTHDR 2,3,33             3rd Operand NOT Equal (cc=2, UR=3)
0003AE75                        0003AE75  0003B000  5879+         ORG   PLOETST+(128*K)+36*(3*K)
0003B000  E2E3C1D9 E3409686             5880+         DC    CL32'START of test: 74'
0003B008  40A385A2 A37A40F7
0003B010  F4404040 40404040
0003B018  40404040 40404040
0003B020                        0003B020            5881+         USING *,R5                base for test data and test routine
0003B020                        00000000            5882+         USING PLOETEST,R5
0003B020  0003B04C                      5883+T74    DC    A(CODE74)                 address of test code
0003B024  004A                          5884+         DC    H'74'                     test number
0003B026  00                            5885+         DC    X'00'
0003B027  21                            5886+         DC    HL1'33'                   Function
0003B028  03                            5887+         DC    HL1'3'                    expected unequal results (UR)
0003B029  00                            5888+         DC    HL1'00'                   found UR
0003B02A  02                            5889+         DC    HL1'2'                    cc
0003B02B  0D                            5890+         DC    HL1'13'                   not expected CC mask
0003B02C  00                            5891+         DS    X                         extracted cc
0003B030  00000000 00000000             5892+         DS    2F                        extract PSW after test (has CC)
0003B038  0003B9E0                      5893+         DC    A(FMSG74)                 address of test fail message
0003B03C  0003B740                      5894+         DC    A(OPSRC74)                address of operand source
0003B040  0003B620                      5895+         DC    A(OPDST74)                address of operand destination
0003B044  0003B860                      5896+         DC    A(PL74)                   address of parameter list
0003B048  00004202                      5897+         DC    A(RANDX+74)               address of random source data
                                        5898
                                        5899          TSTCODE
0003B04C                                5900+CODE74   DS    0F
                                        5901 *                                  load random source
0003B04C  E310 501C 0014                        0000001C  5902          LGF   R1,OPSRCA
0003B052  E320 5028 0014                        00000028  5903          LGF   R2,RANDXA
0003B058  E330 5024 0014                        00000024  5904          LGF   R3,PLA
0003B05E  E340 5020 0014                        00000020  5905          LGF   R4,OPREPA
                                        5906 *
                                        5907 * Triple Compare and Swap (TCSO: 33)
                                        5908 * Build Parambter List
                                        5909 *
0003B064  D25F 1000 2000        00000000  00000000  5910          MVC   0(96,R1),0(R2)      load Op1-Op3 data in source
0003B06A  D25F 3040 1000        00000040  00000000  5911          MVC   64(96,R3),0(R1)     copy to PL OP1-Op3 Compare
                                                    5912
0003B070  D25F 4000 2080        00000000  00000080  5913          MVC   0(96,R4),128(R2)    load initial Op1-Op3 data
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003B076  D25F 30A0 4000      000000A0  00000000  5914          MVC   160(96,R3),0(R4)    copy to PL Op1-Op3 Replacement
                                                  5915 ****
                                                  5916 * force a failure by changing the 3rd operand
                                                  5917 ****
0003B07C  D207 1040 4000      00000040  00000000  5918          MVC   64(8,R1),0(R4)
                                                  5919 ****
                                                  5920
0003B082  E313 0000 0024                00000000  5921          STG   R1,0(R3)            load OP1 address in PL
                                                  5922
0003B088  4111 0020                     00000020  5923          LA    R1,32(R1)
0003B08C  E313 0008 0024                00000008  5924          STG   R1,8(R3)            load OP2 address in PL
                                                  5925
0003B092  4111 0020                     00000020  5926          LA    R1,32(R1)
0003B096  E313 0010 0024                00000010  5927          STG   R1,16(R3)           load OP3 address in PL
                                                  5928
0003B09C  B982 0000                               5929          XGR   R0,R0
0003B0A0  4100 0021                     00000021  5930          LA    R0,33
0003B0A4  EE00 0000 3000      00000000  00000000  5931          PLO   0,0,0,0(R3)         Triple C/S (TCSO: 33)
                                                  5932
0003B0AA  B98D 0020                               5933          EPSW  R2,R0               extract psw
0003B0AE  5020 5010                     00000010  5934          ST    R2,CCPSW              to save CC
0003B0B2  E310 500B 0076                0000000B  5935          LB    R1,CCMASK           (failure CC mask)
0003B0B8  8910 0004                     00000004  5936          SLL   R1,4                (shift to BC instr CC position)
0003B0BC  4410 50F0                     0003B110  5937          EX    R1,F333TSTCC            fail if...
                                                  5938
                                0003B0C0  00000001  5939 F333CKUR EQU   *
0003B0C0  D200 5009 3030      00000009  00000030  5940          MVC   URFOUND,48(R3)      Extract UR
0003B0C6  D500 5008 3030      00000008  00000030  5941          CLC   UR,48(R3)
0003B0CC  4780 50B4                     0003B0D4  5942          BE    F333CKRES
0003B0D0  45F0 82C8                     000022C8  5943          BAL   R15,URMSG
                                                  5944 *
                                                  5945 * compare results
                                                  5946 *
                                0003B0D4  00000001  5947 F333CKRES EQU   *
                                                  5948 *                                reset adresses
0003B0D4  E310 501C 0014                0000001C  5949          LGF   R1,OPSRCA
0003B0DA  E320 5028 0014                00000028  5950          LGF   R2,RANDXA
0003B0E0  E330 5024 0014                00000024  5951          LGF   R3,PLA
0003B0E6  E340 5020 0014                00000020  5952          LGF   R4,OPREPA
                                                  5953
0003B0EC  D51F 1000 3040      00000000  00000040  5954          CLC   0(32,R1),64(R3)
0003B0F2  4770 50EA                     0003B10A  5955          BNE   F333XX              OP1C Different, error
0003B0F6  D51F 1020 3060      00000020  00000060  5956          CLC   32(32,R1),96(R3)
0003B0FC  4770 50EA                     0003B10A  5957          BNE   F333XX              OP2C Different, error
0003B100  D51F 1040 3080      00000040  00000080  5958          CLC   64(32,R1),128(R3)
0003B106  478B 0000                     00000000  5959          BE    0(R11)              OP3C same, return
                                                  5960
                                0003B10A  00000001  5961 F333XX   EQU   *
0003B10A  45F0 83E8                     000023E8  5962          BAL   R15,FAILMSG
0003B10E  07FB                                    5963          BR    R11
                                                  5964
0003B110  4700 50F4                     0003B114  5965 F333TSTCC BC    0,F333CCMSG        fail if unexpected condition code
                                0003B114  00000001  5966 F333CCMSG EQU  *
0003B114  45F0 8196                     00002196  5967          BAL   R15,CCMSG
0003B118  47F0 50A0                     0003B0C0  5968          B     F333CKUR
                                                  5969
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    5970           TSTDATA
0003B11C                      0003B11C  0003B600  5971+          ORG   PLOETST+(128*K)+36*(3*K)+(K+512)
                                                    5972           TSTLBL OPDST              Dest/Data: Op1-R, Op2-R, Op3-R
0003B600  6E6E6ED6 D7C4E2E3                         5973+          DC    CL32'>>>OPDST74<<<' eye catcher
0003B608  F7F44C4C 4C404040
0003B610  40404040 40404040
0003B618  40404040 40404040
0003B620                                            5974+OPDST74   DS    0C
0003B620  00000000 00000000                         5975           DC    XL256'0'
0003B628  00000000 00000000
0003B630  00000000 00000000
0003B638  00000000 00000000
0003B640  00000000 00000000
0003B648  00000000 00000000
0003B650  00000000 00000000
0003B658  00000000 00000000
0003B660  00000000 00000000
0003B668  00000000 00000000
0003B670  00000000 00000000
0003B678  00000000 00000000
0003B680  00000000 00000000
0003B688  00000000 00000000
0003B690  00000000 00000000
0003B698  00000000 00000000
0003B6A0  00000000 00000000
0003B6A8  00000000 00000000
0003B6B0  00000000 00000000
0003B6B8  00000000 00000000
0003B6C0  00000000 00000000
0003B6C8  00000000 00000000
0003B6D0  00000000 00000000
0003B6D8  00000000 00000000
0003B6E0  00000000 00000000
0003B6E8  00000000 00000000
0003B6F0  00000000 00000000
0003B6F8  00000000 00000000
0003B700  00000000 00000000
0003B708  00000000 00000000
0003B710  00000000 00000000
0003B718  00000000 00000000
                                                    5976
                                                    5977           TSTLBL OPSRC              Source: OP1-OP3
0003B720  6E6E6ED6 D7E2D9C3                         5978+          DC    CL32'>>>OPSRC74<<<' eye catcher
0003B728  F7F44C4C 4C404040
0003B730  40404040 40404040
0003B738  40404040 40404040
0003B740                                            5979+OPSRC74   DS    0C
0003B740  00000000 00000000                         5980           DC    XL256'0'
0003B748  00000000 00000000
0003B750  00000000 00000000
0003B758  00000000 00000000
0003B760  00000000 00000000
0003B768  00000000 00000000
0003B770  00000000 00000000
0003B778  00000000 00000000
0003B780  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0003B788  00000000 00000000
0003B790  00000000 00000000
0003B798  00000000 00000000
0003B7A0  00000000 00000000
0003B7A8  00000000 00000000
0003B7B0  00000000 00000000
0003B7B8  00000000 00000000
0003B7C0  00000000 00000000
0003B7C8  00000000 00000000
0003B7D0  00000000 00000000
0003B7D8  00000000 00000000
0003B7E0  00000000 00000000
0003B7E8  00000000 00000000
0003B7F0  00000000 00000000
0003B7F8  00000000 00000000
0003B800  00000000 00000000
0003B808  00000000 00000000
0003B810  00000000 00000000
0003B818  00000000 00000000
0003B820  00000000 00000000
0003B828  00000000 00000000
0003B830  00000000 00000000
0003B838  00000000 00000000
                                          5981
                                          5982          TSTLBL PL                   Parameter List
0003B840  6E6E6ED7 D3F7F44C               5983+         DC    CL32'>>>PL74<<<'      eye catcher
0003B848  4C4C4040 40404040
0003B850  40404040 40404040
0003B858  40404040 40404040
0003B860                                  5984+PL74     DS    0C
0003B860  00000000 00000000               5985          DC    XL352'00'
0003B868  00000000 00000000
0003B870  00000000 00000000
0003B878  00000000 00000000
0003B880  00000000 00000000
0003B888  00000000 00000000
0003B890  00000000 00000000
0003B898  00000000 00000000
0003B8A0  00000000 00000000
0003B8A8  00000000 00000000
0003B8B0  00000000 00000000
0003B8B8  00000000 00000000
0003B8C0  00000000 00000000
0003B8C8  00000000 00000000
0003B8D0  00000000 00000000
0003B8D8  00000000 00000000
0003B8E0  00000000 00000000
0003B8E8  00000000 00000000
0003B8F0  00000000 00000000
0003B8F8  00000000 00000000
0003B900  00000000 00000000
0003B908  00000000 00000000
0003B910  00000000 00000000
0003B918  00000000 00000000
0003B920  00000000 00000000
0003B928  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003B930  00000000 00000000
0003B938  00000000 00000000
0003B940  00000000 00000000
0003B948  00000000 00000000
0003B950  00000000 00000000
0003B958  00000000 00000000
0003B960  00000000 00000000
0003B968  00000000 00000000
0003B970  00000000 00000000
0003B978  00000000 00000000
0003B980  00000000 00000000
0003B988  00000000 00000000
0003B990  00000000 00000000
0003B998  00000000 00000000
0003B9A0  00000000 00000000
0003B9A8  00000000 00000000
0003B9B0  00000000 00000000
0003B9B8  00000000 00000000
                                      5986
                                      5987          TSTLBL FMSG               fail message
0003B9C0  6E6E6EC6 D4E2C7F7           5988+         DC    CL32'>>>FMSG74<<<'  eye catcher
0003B9C8  F44C4C4C 40404040
0003B9D0  40404040 40404040
0003B9D8  40404040 40404040
0003B9E0                              5989+FMSG74   DS    0C
0003B9E0  86818993 858440D7           5990          DC    C'failed PLO-Extension '
0003B9E8  D3D660C5 A7A38595
0003B9F0  A2899695 40
0003B9F5  E3998997 938540C3           5991          DC    CL96'Triple Compare and Swap (TCSO: 33): 3rd NE'
0003B9FD  96949781 99854081
0003BA05  958440E2 A6819740
0003BA0D  4DE3C3E2 D67A40F3
0003BA15  F35D7A40 F3998440
0003BA1D  D5C54040 40404040
0003BA25  40404040 40404040
0003BA2D  40404040 40404040
0003BA35  40404040 40404040
0003BA3D  40404040 40404040
0003BA45  40404040 40404040
0003BA4D  40404040 40404040
                                      5992
                                      5993          TSTEND
0003BA55  C5D5C440 968640A3           5994+         DC    CL32'END of test: 74 <<<'
0003BA5D  85A2A37A 40F7F440
0003BA65  4C4C4C40 40404040
0003BA6D  40404040 40404040
                                      5995
                                      5996 * ------------------------------------------------------------------
                                      5997 * Test Triple Compare and Swap (TCSO: 33) instruction : ALL EQ
                                      5998 * ------------------------------------------------------------------
                                      5999          TSTHDR 0,99,33            ALL EQ (cc=0, UR=99)
0003BA75                 0003BA75  0003BC00  6000+         ORG   PLOETST+(128*K)+37*(3*K)
0003BC00  E2E3C1D9 E3409686           6001+         DC    CL32'START of test: 75'
0003BC08  40A385A2 A37A40F7
0003BC10  F5404040 40404040
0003BC18  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003BC20                     0003BC20              6002+          USING *,R5                base for test data and test routine
0003BC20                     00000000              6003+          USING PLOETEST,R5
0003BC20  0003BC4C                                 6004+T75       DC    A(CODE75)           address of test code
0003BC24  004B                                     6005+          DC    H'75'               test number
0003BC26  00                                       6006+          DC    X'00'
0003BC27  21                                       6007+          DC    HL1'33'             Function
0003BC28  63                                       6008+          DC    HL1'99'             expected unequal results (UR)
0003BC29  00                                       6009+          DC    HL1'00'             found UR
0003BC2A  00                                       6010+          DC    HL1'0'              cc
0003BC2B  07                                       6011+          DC    HL1'7'              not expected CC mask
0003BC2C  00                                       6012+          DS    X                   extracted cc
0003BC30  00000000 00000000                        6013+          DS    2F                  extract PSW after test (has CC)
0003BC38  0003C5E0                                 6014+          DC    A(FMSG75)           address of test fail message
0003BC3C  0003C340                                 6015+          DC    A(OPSRC75)          address of operand source
0003BC40  0003C220                                 6016+          DC    A(OPDST75)          address of operand destination
0003BC44  0003C460                                 6017+          DC    A(PL75)             address of parameter list
0003BC48  00004203                                 6018+          DC    A(RANDX+75)         address of random source data
                                                   6019
                                                   6020           TSTCODE
0003BC4C                                           6021+CODE75    DS    0F
                                                   6022 *                                   load random source
0003BC4C  E310 501C 0014               0000001C    6023           LGF   R1,OPSRCA
0003BC52  E320 5028 0014               00000028    6024           LGF   R2,RANDXA
0003BC58  E330 5024 0014               00000024    6025           LGF   R3,PLA
0003BC5E  E340 5020 0014               00000020    6026           LGF   R4,OPREPA
                                                   6027 *
                                                   6028 * Triple Compare and Swap (TCSO: 33)
                                                   6029 * Build Parambter List
                                                   6030 *
0003BC64  D25F 1000 2000      00000000 00000000    6031           MVC   0(96,R1),0(R2)      load Op1-Op3 data in source
0003BC6A  D25F 3040 1000      00000040 00000000    6032           MVC   64(96,R3),0(R1)     copy to PL OP1-Op3 Compare
                                                   6033
0003BC70  D25F 4000 2080      00000000 00000080    6034           MVC   0(96,R4),128(R2)    load initial Op1-Op3 data
0003BC76  D25F 30A0 4000      000000A0 00000000    6035           MVC   160(96,R3),0(R4)    copy to PL Op1-Op3 Replacement
                                                   6036 ****
                                                   6037 * Do NOTforce a failure by changing an operand
                                                   6038 ****
                                                   6039 *         MVC   64(8,R1),0(R4)
                                                   6040 ****
                                                   6041
0003BC7C  E313 0000 0024               00000000    6042           STG   R1,0(R3)            load OP1 address in PL
                                                   6043
0003BC82  4111 0020                    00000020    6044           LA    R1,32(R1)
0003BC86  E313 0008 0024               00000008    6045           STG   R1,8(R3)            load OP2 address in PL
                                                   6046
0003BC8C  4111 0020                    00000020    6047           LA    R1,32(R1)
0003BC90  E313 0010 0024               00000010    6048           STG   R1,16(R3)           load OP3 address in PL
                                                   6049
0003BC96  B982 0000                                6050           XGR   R0,R0
0003BC9A  4100 0021                    00000021    6051           LA    R0,33
0003BC9E  EE00 0000 3000      00000000 00000000    6052           PLO   0,0,0,0(R3)         Triple C/S (TCSO: 33)
                                                   6053
0003BCA4  B98D 0020                                6054           EPSW  R2,R0               extract psw
0003BCA8  5020 5010                    00000010    6055           ST    R2,CCPSW              to save CC
0003BCAC  E310 500B 0076               0000000B    6056           LB    R1,CCMASK           (failure CC mask)
0003BCB2  8910 0004                    00000004    6057           SLL   R1,4                (shift to BC instr CC position)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003BCB6  4410 50D6                     0003BCF6  6058          EX    R1,F334TSTCC            fail if...
                                                  6059 *
                                                  6060 * compare results
                                                  6061 *
                          0003BCBA  00000001      6062 F334CKRES EQU   *
                                                  6063 *                                reset adresses
0003BCBA  E310 501C 0014                0000001C  6064          LGF   R1,OPSRCA
0003BCC0  E320 5028 0014                00000028  6065          LGF   R2,RANDXA
0003BCC6  E330 5024 0014                00000024  6066          LGF   R3,PLA
0003BCCC  E340 5020 0014                00000020  6067          LGF   R4,OPREPA
                                                  6068
                                                  6069 * All equal; replacement values copied to operand source
                                                  6070
0003BCD2  D51F 1000 30A0  00000000      000000A0  6071          CLC   0(32,R1),160(R3)
0003BCD8  4770 50D0                     0003BCF0  6072          BNE   F334XX                OP1R Different, error
0003BCDC  D51F 1020 30C0  00000020      000000C0  6073          CLC   32(32,R1),192(R3)
0003BCE2  4770 50D0                     0003BCF0  6074          BNE   F334XX                OP2R Different, error
0003BCE6  D51F 1040 30E0  00000040      000000E0  6075          CLC   64(32,R1),224(R3)
0003BCEC  478B 0000                     00000000  6076          BE    0(R11)
                                                  6077
                          0003BCF0  00000001      6078 F334XX   EQU   *
0003BCF0  45F0 83E8                     000023E8  6079          BAL   R15,FAILMSG
0003BCF4  07FB                                    6080          BR    R11
                                                  6081
0003BCF6  4700 50DA                     0003BCFA  6082 F334TSTCC BC    0,F334CCMSG          fail if unexpected condition code
                          0003BCFA  00000001      6083 F334CCMSG EQU   *
0003BCFA  45F0 8196                     00002196  6084          BAL   R15,CCMSG
0003BCFE  47F0 509A                     0003BCBA  6085          B     F334CKRES
                                                  6086
                                                  6087          TSTDATA
0003BD02                  0003BD02  0003C200      6088+         ORG   PLOETST+(128*K)+37*(3*K)+(K+512)
                                                  6089          TSTLBL OPDST              Dest/Data: Op1-R, Op2-R, Op3-R
0003C200  6E6E6ED6 D7C4E2E3                       6090+         DC    CL32'>>>OPDST75<<<' eye catcher
0003C208  F7F54C4C 4C404040
0003C210  40404040 40404040
0003C218  40404040 40404040
0003C220                                          6091+OPDST75  DS    0C
0003C220  00000000 00000000                       6092          DC    XL256'0'
0003C228  00000000 00000000
0003C230  00000000 00000000
0003C238  00000000 00000000
0003C240  00000000 00000000
0003C248  00000000 00000000
0003C250  00000000 00000000
0003C258  00000000 00000000
0003C260  00000000 00000000
0003C268  00000000 00000000
0003C270  00000000 00000000
0003C278  00000000 00000000
0003C280  00000000 00000000
0003C288  00000000 00000000
0003C290  00000000 00000000
0003C298  00000000 00000000
0003C2A0  00000000 00000000
0003C2A8  00000000 00000000
0003C2B0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003C2B8  00000000 00000000
0003C2C0  00000000 00000000
0003C2C8  00000000 00000000
0003C2D0  00000000 00000000
0003C2D8  00000000 00000000
0003C2E0  00000000 00000000
0003C2E8  00000000 00000000
0003C2F0  00000000 00000000
0003C2F8  00000000 00000000
0003C300  00000000 00000000
0003C308  00000000 00000000
0003C310  00000000 00000000
0003C318  00000000 00000000
                                          6093
                                          6094          TSTLBL OPSRC              Source: OP1-OP3
0003C320  6E6E6ED6 D7E2D9C3               6095+         DC    CL32'>>>OPSRC75<<<' eye catcher
0003C328  F7F54C4C 4C404040
0003C330  40404040 40404040
0003C338  40404040 40404040
0003C340                                  6096+OPSRC75  DS    0C
0003C340  00000000 00000000               6097          DC    XL256'0'
0003C348  00000000 00000000
0003C350  00000000 00000000
0003C358  00000000 00000000
0003C360  00000000 00000000
0003C368  00000000 00000000
0003C370  00000000 00000000
0003C378  00000000 00000000
0003C380  00000000 00000000
0003C388  00000000 00000000
0003C390  00000000 00000000
0003C398  00000000 00000000
0003C3A0  00000000 00000000
0003C3A8  00000000 00000000
0003C3B0  00000000 00000000
0003C3B8  00000000 00000000
0003C3C0  00000000 00000000
0003C3C8  00000000 00000000
0003C3D0  00000000 00000000
0003C3D8  00000000 00000000
0003C3E0  00000000 00000000
0003C3E8  00000000 00000000
0003C3F0  00000000 00000000
0003C3F8  00000000 00000000
0003C400  00000000 00000000
0003C408  00000000 00000000
0003C410  00000000 00000000
0003C418  00000000 00000000
0003C420  00000000 00000000
0003C428  00000000 00000000
0003C430  00000000 00000000
0003C438  00000000 00000000
                                          6098
                                          6099          TSTLBL PL                 Parameter List
0003C440  6E6E6ED7 D3F7F54C               6100+         DC    CL32'>>>PL75<<<'    eye catcher
0003C448  4C4C4040 40404040
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003C450  40404040 40404040
0003C458  40404040 40404040
0003C460                                             6101+PL75     DS    0C
0003C460  00000000 00000000                          6102          DC    XL352'00'
0003C468  00000000 00000000
0003C470  00000000 00000000
0003C478  00000000 00000000
0003C480  00000000 00000000
0003C488  00000000 00000000
0003C490  00000000 00000000
0003C498  00000000 00000000
0003C4A0  00000000 00000000
0003C4A8  00000000 00000000
0003C4B0  00000000 00000000
0003C4B8  00000000 00000000
0003C4C0  00000000 00000000
0003C4C8  00000000 00000000
0003C4D0  00000000 00000000
0003C4D8  00000000 00000000
0003C4E0  00000000 00000000
0003C4E8  00000000 00000000
0003C4F0  00000000 00000000
0003C4F8  00000000 00000000
0003C500  00000000 00000000
0003C508  00000000 00000000
0003C510  00000000 00000000
0003C518  00000000 00000000
0003C520  00000000 00000000
0003C528  00000000 00000000
0003C530  00000000 00000000
0003C538  00000000 00000000
0003C540  00000000 00000000
0003C548  00000000 00000000
0003C550  00000000 00000000
0003C558  00000000 00000000
0003C560  00000000 00000000
0003C568  00000000 00000000
0003C570  00000000 00000000
0003C578  00000000 00000000
0003C580  00000000 00000000
0003C588  00000000 00000000
0003C590  00000000 00000000
0003C598  00000000 00000000
0003C5A0  00000000 00000000
0003C5A8  00000000 00000000
0003C5B0  00000000 00000000
0003C5B8  00000000 00000000
                                                     6103
                                                     6104          TSTLBL FMSG                fail message
0003C5C0  6E6E6EC6 D4E2C7F7                          6105+         DC    CL32'>>>FMSG75<<<'   eye catcher
0003C5C8  F54C4C4C 40404040
0003C5D0  40404040 40404040
0003C5D8  40404040 40404040
0003C5E0                                             6106+FMSG75   DS    0C
0003C5E0  86818993 858440D7                          6107          DC    C'failed PLO-Extension '
0003C5E8  D3D660C5 A7A38595
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0003C5F0  A2899695 40
0003C5F5  E3998997 938540C3                        6108          DC    CL96'Triple Compare and Swap (TCSO: 33):  ALL EQ'
0003C5FD  96949781 99854081
0003C605  958440E2 A6819740
0003C60D  4DE3C3E2 D67A40F3
0003C615  F35D7A40 40C1D3D3
0003C61D  40C5D840 40404040
0003C625  40404040 40404040
0003C62D  40404040 40404040
0003C635  40404040 40404040
0003C63D  40404040 40404040
0003C645  40404040 40404040
0003C64D  40404040 40404040
                                                   6109
                                                   6110          TSTEND
0003C655  C5D5C440 968640A3                        6111+         DC    CL32'END of test: 75 <<<'
0003C65D  85A2A37A 40F7F540
0003C665  4C4C4C40 40404040
0003C66D  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6113 *=======================================================================
                                                    6114 *       Triple Compare and Swap Tests
                                                    6115 *=======================================================================
                                                    6116
                                                    6117 * ---------------------------------------------------------------------
                                                    6118 * Test Triple Compare and Swap (TCSX: 32) instruction : 1st NE
                                                    6119 * ---------------------------------------------------------------------
                                                    6120          TSTHDR 2,1,32             1st Operand NOT Equal (cc=2, UR=1)
0003C675                          0003C675  0003C800  6121+         ORG   PLOETST+(128*K)+38*(3*K)
0003C800  E2E3C1D9 E3409686                         6122+         DC    CL32'START of test: 76'
0003C808  40A385A2 A37A40F7
0003C810  F6404040 40404040
0003C818  40404040 40404040
0003C820                          0003C820          6123+         USING *,R5                base for test data and test routine
0003C820                          00000000          6124+         USING PLOETEST,R5
0003C820  0003C84C                                  6125+T76      DC    A(CODE76)           address of test code
0003C824  004C                                      6126+         DC    H'76'               test number
0003C826  00                                        6127+         DC    X'00'
0003C827  20                                        6128+         DC    HL1'32'             Function
0003C828  01                                        6129+         DC    HL1'1'              expected unequal results (UR)
0003C829  00                                        6130+         DC    HL1'00'             found UR
0003C82A  02                                        6131+         DC    HL1'2'              cc
0003C82B  0D                                        6132+         DC    HL1'13'             not expected CC mask
0003C82C  00                                        6133+         DS    X                   extracted cc
0003C830  00000000 00000000                         6134+         DS    2F                  extract PSW after test (has CC)
0003C838  0003D1E0                                  6135+         DC    A(FMSG76)           address of test fail message
0003C83C  0003CF40                                  6136+         DC    A(OPSRC76)          address of operand source
0003C840  0003CE20                                  6137+         DC    A(OPDST76)          address of operand destination
0003C844  0003D060                                  6138+         DC    A(PL76)             address of parameter list
0003C848  00004204                                  6139+         DC    A(RANDX+76)         address of random source data
                                                    6140
                                                    6141          TSTCODE
0003C84C                                            6142+CODE76   DS    0F
                                                    6143 *                                  load random source
0003C84C  E310 501C 0014                 0000001C   6144          LGF   R1,OPSRCA
0003C852  E320 5028 0014                 00000028   6145          LGF   R2,RANDXA
0003C858  E330 5024 0014                 00000024   6146          LGF   R3,PLA
0003C85E  E340 5020 0014                 00000020   6147          LGF   R4,OPREPA
                                                    6148 *
                                                    6149 * Triple Compare and Swap (TCSX: 32)
                                                    6150 * Build Parambter List
                                                    6151 *
0003C864  D22F 1000 2000  00000000       00000000   6152          MVC   0(48,R1),0(R2)      load Op1-Op3 data in source
0003C86A  D22F 3040 1000  00000040       00000000   6153          MVC   64(48,R3),0(R1)     copy to PL OP1-Op3 Compare
                                                    6154
0003C870  D22F 4000 2080  00000000       00000080   6155          MVC   0(48,R4),128(R2)    load initial Op1-Op3 data
0003C876  D22F 30A0 4000  000000A0       00000000   6156          MVC   160(48,R3),0(R4)    copy to PL Op1-Op3 Replacement
                                                    6157 ****
                                                    6158 * force a failure by changing the 1st operand
                                                    6159 ****
0003C87C  D203 1000 4000  00000000       00000000   6160          MVC   0(4,R1),0(R4)
                                                    6161 ****
                                                    6162
0003C882  E313 0000 0024                 00000000   6163          STG   R1,0(R3)            load OP1 address in PL
                                                    6164
0003C888  4111 0010                      00000010   6165          LA    R1,16(R1)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003C88C  E313 0008 0024                 00000008  6166           STG   R1,8(R3)            load OP2 address in PL
                                                   6167
0003C892  4111 0010                      00000010  6168           LA    R1,16(R1)
0003C896  E313 0010 0024                 00000010  6169           STG   R1,16(R3)           load OP3 address in PL
                                                   6170
0003C89C  B982 0000                                6171           XGR   R0,R0
0003C8A0  4100 0020                      00000020  6172           LA    R0,32
0003C8A4  EE00 0000 3000       00000000  00000000  6173           PLO   0,0,0,0(R3)         Triple C/S (TCSX: 32)
                                                   6174
0003C8AA  B98D 0020                                6175           EPSW  R2,R0               extract psw
0003C8AE  5020 5010                      00000010  6176           ST    R2,CCPSW              to save CC
0003C8B2  E310 500B 0076                 0000000B  6177           LB    R1,CCMASK           (failure CC mask)
0003C8B8  8910 0004                      00000004  6178           SLL   R1,4                (shift to BC instr CC position)
0003C8BC  4410 50F0                      0003C910  6179           EX    R1,F321TSTCC            fail if...
                                                   6180
                                0003C8C0  00000001  6181 F321CKUR EQU   *
0003C8C0  D200 5009 3030        00000009  00000030  6182           MVC   URFOUND,48(R3)      Extract UR
0003C8C6  D500 5008 3030        00000008  00000030  6183           CLC   UR,48(R3)
0003C8CC  4780 50B4                      0003C8D4  6184           BE    F321CKRES
0003C8D0  45F0 82C8                      000022C8  6185           BAL   R15,URMSG
                                                   6186 *
                                                   6187 * compare results
                                                   6188 *
                                0003C8D4  00000001  6189 F321CKRES EQU   *
                                                   6190 *                                  reset adresses
0003C8D4  E310 501C 0014                 0000001C  6191           LGF   R1,OPSRCA
0003C8DA  E320 5028 0014                 00000028  6192           LGF   R2,RANDXA
0003C8E0  E330 5024 0014                 00000024  6193           LGF   R3,PLA
0003C8E6  E340 5020 0014                 00000020  6194           LGF   R4,OPREPA
                                                   6195
0003C8EC  D50F 1000 3040       00000000  00000040  6196           CLC   0(16,R1),64(R3)
0003C8F2  4770 50EA                      0003C90A  6197           BNE   F321XX              OP1C Different, error
0003C8F6  D50F 1010 3050       00000010  00000050  6198           CLC   16(16,R1),80(R3)
0003C8FC  4770 50EA                      0003C90A  6199           BNE   F321XX              OP2C Different, error
0003C900  D50F 1020 3060       00000020  00000060  6200           CLC   32(16,R1),96(R3)
0003C906  478B 0000                      00000000  6201           BE    0(R11)              OP3C same, return
                                                   6202
                                0003C90A  00000001  6203 F321XX   EQU   *
0003C90A  45F0 83E8                      000023E8  6204           BAL   R15,FAILMSG
0003C90E  07FB                                     6205           BR    R11
                                                   6206
0003C910  4700 50F4                      0003C914  6207 F321TSTCC BC    0,F321CCMSG         fail if unexpected condition code
                                0003C914  00000001  6208 F321CCMSG EQU   *
0003C914  45F0 8196                      00002196  6209           BAL   R15,CCMSG
0003C918  47F0 50A0                      0003C8C0  6210           B     F321CKUR
                                                   6211
                                                   6212           TSTDATA
0003C91C                        0003C91C  0003CE00  6213+          ORG   PLOETST+(128*K)+38*(3*K)+(K+512)
                                                   6214           TSTLBL OPDST              Dest/Data: Op1-R, Op2-R, Op3-R
0003CE00  6E6E6ED6 D7C4E2E3                        6215+          DC    CL32'>>>OPDST76<<<' eye catcher
0003CE08  F7F64C4C 4C404040
0003CE10  40404040 40404040
0003CE18  40404040 40404040
0003CE20                                           6216+OPDST76   DS    0C
0003CE20  00000000 00000000                        6217           DC    XL256'0'
0003CE28  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003CE30  00000000 00000000
0003CE38  00000000 00000000
0003CE40  00000000 00000000
0003CE48  00000000 00000000
0003CE50  00000000 00000000
0003CE58  00000000 00000000
0003CE60  00000000 00000000
0003CE68  00000000 00000000
0003CE70  00000000 00000000
0003CE78  00000000 00000000
0003CE80  00000000 00000000
0003CE88  00000000 00000000
0003CE90  00000000 00000000
0003CE98  00000000 00000000
0003CEA0  00000000 00000000
0003CEA8  00000000 00000000
0003CEB0  00000000 00000000
0003CEB8  00000000 00000000
0003CEC0  00000000 00000000
0003CEC8  00000000 00000000
0003CED0  00000000 00000000
0003CED8  00000000 00000000
0003CEE0  00000000 00000000
0003CEE8  00000000 00000000
0003CEF0  00000000 00000000
0003CEF8  00000000 00000000
0003CF00  00000000 00000000
0003CF08  00000000 00000000
0003CF10  00000000 00000000
0003CF18  00000000 00000000
                                      6218
                                      6219           TSTLBL OPSRC               Source: OP1-OP3
0003CF20  6E6E6ED6 D7E2D9C3           6220+          DC     CL32'>>>OPSRC76<<<' eye catcher
0003CF28  F7F64C4C 4C404040
0003CF30  40404040 40404040
0003CF38  40404040 40404040
0003CF40                              6221+OPSRC76   DS     0C
0003CF40  00000000 00000000           6222           DC     XL256'0'
0003CF48  00000000 00000000
0003CF50  00000000 00000000
0003CF58  00000000 00000000
0003CF60  00000000 00000000
0003CF68  00000000 00000000
0003CF70  00000000 00000000
0003CF78  00000000 00000000
0003CF80  00000000 00000000
0003CF88  00000000 00000000
0003CF90  00000000 00000000
0003CF98  00000000 00000000
0003CFA0  00000000 00000000
0003CFA8  00000000 00000000
0003CFB0  00000000 00000000
0003CFB8  00000000 00000000
0003CFC0  00000000 00000000
0003CFC8  00000000 00000000
0003CFD0  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0003CFD8  00000000 00000000
0003CFE0  00000000 00000000
0003CFE8  00000000 00000000
0003CFF0  00000000 00000000
0003CFF8  00000000 00000000
0003D000  00000000 00000000
0003D008  00000000 00000000
0003D010  00000000 00000000
0003D018  00000000 00000000
0003D020  00000000 00000000
0003D028  00000000 00000000
0003D030  00000000 00000000
0003D038  00000000 00000000
                                          6223
                                          6224          TSTLBL PL                   Parameter List
0003D040  6E6E6ED7 D3F7F64C               6225+         DC    CL32'>>>PL76<<<'      eye catcher
0003D048  4C4C4040 40404040
0003D050  40404040 40404040
0003D058  40404040 40404040
0003D060                                  6226+PL76     DS    0C
0003D060  00000000 00000000               6227          DC    XL352'00'
0003D068  00000000 00000000
0003D070  00000000 00000000
0003D078  00000000 00000000
0003D080  00000000 00000000
0003D088  00000000 00000000
0003D090  00000000 00000000
0003D098  00000000 00000000
0003D0A0  00000000 00000000
0003D0A8  00000000 00000000
0003D0B0  00000000 00000000
0003D0B8  00000000 00000000
0003D0C0  00000000 00000000
0003D0C8  00000000 00000000
0003D0D0  00000000 00000000
0003D0D8  00000000 00000000
0003D0E0  00000000 00000000
0003D0E8  00000000 00000000
0003D0F0  00000000 00000000
0003D0F8  00000000 00000000
0003D100  00000000 00000000
0003D108  00000000 00000000
0003D110  00000000 00000000
0003D118  00000000 00000000
0003D120  00000000 00000000
0003D128  00000000 00000000
0003D130  00000000 00000000
0003D138  00000000 00000000
0003D140  00000000 00000000
0003D148  00000000 00000000
0003D150  00000000 00000000
0003D158  00000000 00000000
0003D160  00000000 00000000
0003D168  00000000 00000000
0003D170  00000000 00000000
0003D178  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003D180  00000000 00000000
0003D188  00000000 00000000
0003D190  00000000 00000000
0003D198  00000000 00000000
0003D1A0  00000000 00000000
0003D1A8  00000000 00000000
0003D1B0  00000000 00000000
0003D1B8  00000000 00000000
                                      6228
                                      6229          TSTLBL FMSG                fail message
0003D1C0  6E6E6EC6 D4E2C7F7           6230+         DC    CL32'>>>FMSG76<<<'   eye catcher
0003D1C8  F64C4C4C 40404040
0003D1D0  40404040 40404040
0003D1D8  40404040 40404040
0003D1E0                              6231+FMSG76   DS    0C
0003D1E0  86818993 858440D7           6232          DC    C'failed PLO-Extension '
0003D1E8  D3D660C5 A7A38595
0003D1F0  A2899695 40
0003D1F5  E3998997 938540C3           6233          DC    CL96'Triple Compare and Swap (TCSX: 32): 1st NE'
0003D1FD  96949781 99854081
0003D205  958440E2 A6819740
0003D20D  4DE3C3E2 E77A40F3
0003D215  F25D7A40 F1A2A340
0003D21D  D5C54040 40404040
0003D225  40404040 40404040
0003D22D  40404040 40404040
0003D235  40404040 40404040
0003D23D  40404040 40404040
0003D245  40404040 40404040
0003D24D  40404040 40404040
                                      6234
                                      6235          TSTEND
0003D255  C5D5C440 968640A3           6236+         DC    CL32'END of test: 76 <<<'
0003D25D  85A2A37A 40F7F640
0003D265  4C4C4C40 40404040
0003D26D  40404040 40404040
                                      6237
                                      6238 * ------------------------------------------------------------------------
                                      6239 * Test Triple Compare and Swap (TCSX: 32) instruction : 2nd NE
                                      6240 * ------------------------------------------------------------------------
                                      6241          TSTHDR 2,2,32              2nd Operand NOT Equal (cc=2, UR=2)
0003D275                  0003D275  0003D400  6242+         ORG   PLOETST+(128*K)+39*(3*K)
0003D400  E2E3C1D9 E3409686           6243+         DC    CL32'START of test: 77'
0003D408  40A385A2 A37A40F7
0003D410  F7404040 40404040
0003D418  40404040 40404040
0003D420                  0003D420    6244+         USING *,R5                 base for test data and test routine
0003D420                  00000000    6245+         USING PLOETEST,R5
0003D420  0003D44C                    6246+T77      DC    A(CODE77)            address of test code
0003D424  004D                        6247+         DC    H'77'                test number
0003D426  00                          6248+         DC    X'00'
0003D427  20                          6249+         DC    HL1'32'              Function
0003D428  02                          6250+         DC    HL1'2'               expected unequal results (UR)
0003D429  00                          6251+         DC    HL1'00'              found UR
0003D42A  02                          6252+         DC    HL1'2'               cc
0003D42B  0D                          6253+         DC    HL1'13'              not expected CC mask
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003D42C  00                                       6254+          DS    X                   extracted cc
0003D430  00000000 00000000                        6255+          DS    2F                  extract PSW after test (has CC)
0003D438  0003DDE0                                 6256+          DC    A(FMSG77)           address of test fail message
0003D43C  0003DB40                                 6257+          DC    A(OPSRC77)          address of operand source
0003D440  0003DA20                                 6258+          DC    A(OPDST77)          address of operand destination
0003D444  0003DC60                                 6259+          DC    A(PL77)             address of parameter list
0003D448  00004205                                 6260+          DC    A(RANDX+77)         address of random source data
                                                   6261
                                                   6262           TSTCODE
0003D44C                                           6263+CODE77    DS    0F
                                                   6264 *                                   load random source
0003D44C  E310 501C 0014                0000001C   6265           LGF   R1,OPSRCA
0003D452  E320 5028 0014                00000028   6266           LGF   R2,RANDXA
0003D458  E330 5024 0014                00000024   6267           LGF   R3,PLA
0003D45E  E340 5020 0014                00000020   6268           LGF   R4,OPREPA
                                                   6269 *
                                                   6270 * Triple Compare and Swap (TCSX: 32)
                                                   6271 * Build Parambter List
                                                   6272 *
0003D464  D22F 1000 2000      00000000  00000000   6273           MVC   0(48,R1),0(R2)      load Op1-Op3 data in source
0003D46A  D22F 3040 1000      00000040  00000000   6274           MVC   64(48,R3),0(R1)     copy to PL OP1-Op3 Compare
                                                   6275
0003D470  D22F 4000 2080      00000000  00000080   6276           MVC   0(48,R4),128(R2)    load initial Op1-Op3 data
0003D476  D22F 30A0 4000      000000A0  00000000   6277           MVC   160(48,R3),0(R4)    copy to PL Op1-Op3 Replacement
                                                   6278 ****
                                                   6279 * force a failure by changing the 2nd operand
                                                   6280 ****
0003D47C  D207 1010 4000      00000010  00000000   6281           MVC   16(8,R1),0(R4)
                                                   6282 ****
                                                   6283
0003D482  E313 0000 0024                00000000   6284           STG   R1,0(R3)            load OP1 address in PL
                                                   6285
0003D488  4111 0010                     00000010   6286           LA    R1,16(R1)
0003D48C  E313 0008 0024                00000008   6287           STG   R1,8(R3)            load OP2 address in PL
                                                   6288
0003D492  4111 0010                     00000010   6289           LA    R1,16(R1)
0003D496  E313 0010 0024                00000010   6290           STG   R1,16(R3)           load OP3 address in PL
                                                   6291
0003D49C  B982 0000                                6292           XGR   R0,R0
0003D4A0  4100 0020                     00000020   6293           LA    R0,32
0003D4A4  EE00 0000 3000      00000000  00000000   6294           PLO   0,0,0,0(R3)         Triple C/S (TCSX: 32)
                                                   6295
0003D4AA  B98D 0020                                6296           EPSW  R2,R0               extract psw
0003D4AE  5020 5010                     00000010   6297           ST    R2,CCPSW              to save CC
0003D4B2  E310 500B 0076                0000000B   6298           LB    R1,CCMASK           (failure CC mask)
0003D4B8  8910 0004                     00000004   6299           SLL   R1,4                (shift to BC instr CC position)
0003D4BC  4410 50F0                     0003D510   6300           EX    R1,F322TSTCC            fail if...
                                                   6301
                              0003D4C0  00000001   6302 F322CKUR  EQU   *
0003D4C0  D200 5009 3030      00000009  00000030   6303           MVC   URFOUND,48(R3)      Extract UR
0003D4C6  D500 5008 3030      00000008  00000030   6304           CLC   UR,48(R3)
0003D4CC  4780 50B4                     0003D4D4   6305           BE    F322CKRES
0003D4D0  45F0 82C8                     000022C8   6306           BAL   R15,URMSG
                                                   6307 *
                                                   6308 * compare results
                                                   6309 *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                               0003D4D4  00000001  6310 F322CKRES EQU   *
                                                   6311 *                                reset adresses
0003D4D4  E310 501C 0014                 0000001C  6312          LGF   R1,OPSRCA
0003D4DA  E320 5028 0014                 00000028  6313          LGF   R2,RANDXA
0003D4E0  E330 5024 0014                 00000024  6314          LGF   R3,PLA
0003D4E6  E340 5020 0014                 00000020  6315          LGF   R4,OPREPA
                                                   6316
0003D4EC  D50F 1000 3040  00000000       00000040  6317          CLC   0(16,R1),64(R3)
0003D4F2  4770 50EA                      0003D50A  6318          BNE   F322XX                OP1C Different, error
0003D4F6  D50F 1010 3050  00000010       00000050  6319          CLC   16(16,R1),80(R3)
0003D4FC  4770 50EA                      0003D50A  6320          BNE   F322XX                OP2C Different, error
0003D500  D50F 1020 3060  00000020       00000060  6321          CLC   32(16,R1),96(R3)
0003D506  478B 0000                      00000000  6322          BE    0(R11)                OP3C same, return
                                                   6323
                               0003D50A  00000001  6324 F322XX   EQU   *
0003D50A  45F0 83E8                      000023E8  6325          BAL   R15,FAILMSG
0003D50E  07FB                                     6326          BR    R11
                                                   6327
0003D510  4700 50F4                      0003D514  6328 F322TSTCC BC    0,F322CCMSG           fail if unexpected condition code
                               0003D514  00000001  6329 F322CCMSG EQU   *
0003D514  45F0 8196                      00002196  6330          BAL   R15,CCMSG
0003D518  47F0 50A0                      0003D4C0  6331          B     F322CKUR
                                                   6332
                                                   6333          TSTDATA
0003D51C                       0003D51C  0003DA00  6334+         ORG   PLOETST+(128*K)+39*(3*K)+(K+512)
                                                   6335          TSTLBL OPDST                Dest/Data: Op1-R, Op2-R, Op3-R
0003DA00  6E6E6ED6 D7C4E2E3              6336+         DC    CL32'>>>OPDST77<<<' eye catcher
0003DA08  F7F74C4C 4C404040
0003DA10  40404040 40404040
0003DA18  40404040 40404040
0003DA20                                 6337+OPDST77  DS    0C
0003DA20  00000000 00000000              6338          DC    XL256'0'
0003DA28  00000000 00000000
0003DA30  00000000 00000000
0003DA38  00000000 00000000
0003DA40  00000000 00000000
0003DA48  00000000 00000000
0003DA50  00000000 00000000
0003DA58  00000000 00000000
0003DA60  00000000 00000000
0003DA68  00000000 00000000
0003DA70  00000000 00000000
0003DA78  00000000 00000000
0003DA80  00000000 00000000
0003DA88  00000000 00000000
0003DA90  00000000 00000000
0003DA98  00000000 00000000
0003DAA0  00000000 00000000
0003DAA8  00000000 00000000
0003DAB0  00000000 00000000
0003DAB8  00000000 00000000
0003DAC0  00000000 00000000
0003DAC8  00000000 00000000
0003DAD0  00000000 00000000
0003DAD8  00000000 00000000
0003DAE0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0003DAE8  00000000 00000000
0003DAF0  00000000 00000000
0003DAF8  00000000 00000000
0003DB00  00000000 00000000
0003DB08  00000000 00000000
0003DB10  00000000 00000000
0003DB18  00000000 00000000
                                        6339
                                        6340          TSTLBL OPSRC               Source: OP1-OP3
0003DB20  6E6E6ED6 D7E2D9C3             6341+         DC    CL32'>>>OPSRC77<<<'  eye catcher
0003DB28  F7F74C4C 4C404040
0003DB30  40404040 40404040
0003DB38  40404040 40404040
0003DB40                                6342+OPSRC77  DS    0C
0003DB40  00000000 00000000             6343          DC    XL256'0'
0003DB48  00000000 00000000
0003DB50  00000000 00000000
0003DB58  00000000 00000000
0003DB60  00000000 00000000
0003DB68  00000000 00000000
0003DB70  00000000 00000000
0003DB78  00000000 00000000
0003DB80  00000000 00000000
0003DB88  00000000 00000000
0003DB90  00000000 00000000
0003DB98  00000000 00000000
0003DBA0  00000000 00000000
0003DBA8  00000000 00000000
0003DBB0  00000000 00000000
0003DBB8  00000000 00000000
0003DBC0  00000000 00000000
0003DBC8  00000000 00000000
0003DBD0  00000000 00000000
0003DBD8  00000000 00000000
0003DBE0  00000000 00000000
0003DBE8  00000000 00000000
0003DBF0  00000000 00000000
0003DBF8  00000000 00000000
0003DC00  00000000 00000000
0003DC08  00000000 00000000
0003DC10  00000000 00000000
0003DC18  00000000 00000000
0003DC20  00000000 00000000
0003DC28  00000000 00000000
0003DC30  00000000 00000000
0003DC38  00000000 00000000
                                        6344
                                        6345          TSTLBL PL                  Parameter List
0003DC40  6E6E6ED7 D3F7F74C             6346+         DC    CL32'>>>PL77<<<'     eye catcher
0003DC48  4C4C4040 40404040
0003DC50  40404040 40404040
0003DC58  40404040 40404040
0003DC60                                6347+PL77     DS    0C
0003DC60  00000000 00000000             6348          DC    XL352'00'
0003DC68  00000000 00000000
0003DC70  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003DC78  00000000 00000000
0003DC80  00000000 00000000
0003DC88  00000000 00000000
0003DC90  00000000 00000000
0003DC98  00000000 00000000
0003DCA0  00000000 00000000
0003DCA8  00000000 00000000
0003DCB0  00000000 00000000
0003DCB8  00000000 00000000
0003DCC0  00000000 00000000
0003DCC8  00000000 00000000
0003DCD0  00000000 00000000
0003DCD8  00000000 00000000
0003DCE0  00000000 00000000
0003DCE8  00000000 00000000
0003DCF0  00000000 00000000
0003DCF8  00000000 00000000
0003DD00  00000000 00000000
0003DD08  00000000 00000000
0003DD10  00000000 00000000
0003DD18  00000000 00000000
0003DD20  00000000 00000000
0003DD28  00000000 00000000
0003DD30  00000000 00000000
0003DD38  00000000 00000000
0003DD40  00000000 00000000
0003DD48  00000000 00000000
0003DD50  00000000 00000000
0003DD58  00000000 00000000
0003DD60  00000000 00000000
0003DD68  00000000 00000000
0003DD70  00000000 00000000
0003DD78  00000000 00000000
0003DD80  00000000 00000000
0003DD88  00000000 00000000
0003DD90  00000000 00000000
0003DD98  00000000 00000000
0003DDA0  00000000 00000000
0003DDA8  00000000 00000000
0003DDB0  00000000 00000000
0003DDB8  00000000 00000000
                             6349
                             6350          TSTLBL FMSG                fail message
0003DDC0  6E6E6EC6 D4E2C7F7  6351+         DC     CL32'>>>FMSG77<<<'  eye catcher
0003DDC8  F74C4C4C 40404040
0003DDD0  40404040 40404040
0003DDD8  40404040 40404040
0003DDE0                     6352+FMSG77   DS     0C
0003DDE0  86818993 858440D7  6353          DC     C'failed PLO-Extension '
0003DDE8  D3D660C5 A7A38595
0003DDF0  A2899695 40
0003DDF5  E3998997 938540C3  6354          DC     CL96'Triple Compare and Swap (TCSX: 32): 2nd NE'
0003DDFD  96949781 99854081
0003DE05  958440E2 A6819740
0003DE0D  4DE3C3E2 E77A40F3
0003DE15  F25D7A40 F2958440
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003DE1D  D5C54040 40404040
0003DE25  40404040 40404040
0003DE2D  40404040 40404040
0003DE35  40404040 40404040
0003DE3D  40404040 40404040
0003DE45  40404040 40404040
0003DE4D  40404040 40404040
                                       6355
                                       6356          TSTEND
0003DE55  C5D5C440 968640A3            6357+         DC    CL32'END of test: 77 <<<'
0003DE5D  85A2A37A 40F7F740
0003DE65  4C4C4C40 40404040
0003DE6D  40404040 40404040
                                       6358
                                       6359 * ------------------------------------------------------------------
                                       6360 * Test Triple Compare and Swap (TCSX: 32) instruction : 3nd NE
                                       6361 * ------------------------------------------------------------------
                                       6362          TSTHDR 2,3,32              3nd Operand NOT Equal (cc=2, UR=3)
0003DE75                  0003DE75  0003E000  6363+         ORG   PLOETST+(128*K)+40*(3*K)
0003E000  E2E3C1D9 E3409686            6364+         DC    CL32'START of test: 78'
0003E008  40A385A2 A37A40F7
0003E010  F8404040 40404040
0003E018  40404040 40404040
0003E020                  0003E020     6365+         USING *,R5                 base for test data and test routine
0003E020                  00000000     6366+         USING PLOETEST,R5
0003E020  0003E04C                     6367+T78      DC    A(CODE78)            address of test code
0003E024  004E                         6368+         DC    H'78'                test number
0003E026  00                           6369+         DC    X'00'
0003E027  20                           6370+         DC    HL1'32'              Function
0003E028  03                           6371+         DC    HL1'3'               expected unequal results (UR)
0003E029  00                           6372+         DC    HL1'00'              found UR
0003E02A  02                           6373+         DC    HL1'2'               cc
0003E02B  0D                           6374+         DC    HL1'13'              not expected CC mask
0003E02C  00                           6375+         DS    X                    extracted cc
0003E030  00000000 00000000            6376+         DS    2F                   extract PSW after test (has CC)
0003E038  0003E9E0                     6377+         DC    A(FMSG78)            address of test fail message
0003E03C  0003E740                     6378+         DC    A(OPSRC78)           address of operand source
0003E040  0003E620                     6379+         DC    A(OPDST78)           address of operand destination
0003E044  0003E860                     6380+         DC    A(PL78)              address of parameter list
0003E048  00004206                     6381+         DC    A(RANDX+78)          address of random source data
                                       6382
                                       6383          TSTCODE
0003E04C                               6384+CODE78   DS    0F
                                       6385 *                                   load random source
0003E04C  E310 501C 0014                   0000001C  6386          LGF   R1,OPSRCA
0003E052  E320 5028 0014                   00000028  6387          LGF   R2,RANDXA
0003E058  E330 5024 0014                   00000024  6388          LGF   R3,PLA
0003E05E  E340 5020 0014                   00000020  6389          LGF   R4,OPREPA
                                       6390 *
                                       6391 * Triple Compare and Swap (TCSX: 32)
                                       6392 * Build Parambter List
                                       6393 *
0003E064  D22F 1000 2000       00000000  00000000  6394          MVC   0(48,R1),0(R2)       load Op1-Op3 data in source
0003E06A  D22F 3040 1000       00000040  00000000  6395          MVC   64(48,R3),0(R1)      copy to PL OP1-Op3 Compare
                                                   6396
0003E070  D22F 4000 2080       00000000  00000080  6397          MVC   0(48,R4),128(R2)     load initial Op1-Op3 data
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003E076  D22F 30A0 4000      000000A0  00000000  6398          MVC   160(48,R3),0(R4)    copy to PL Op1-Op3 Replacement
                                                  6399 ****
                                                  6400 * force a failure by changing the 3nd operand
                                                  6401 ****
0003E07C  D207 1020 4000      00000020  00000000  6402          MVC   32(8,R1),0(R4)
                                                  6403 ****
                                                  6404
0003E082  E313 0000 0024                00000000  6405          STG   R1,0(R3)            load OP1 address in PL
                                                  6406
0003E088  4111 0010                     00000010  6407          LA    R1,16(R1)
0003E08C  E313 0008 0024                00000008  6408          STG   R1,8(R3)            load OP2 address in PL
                                                  6409
0003E092  4111 0010                     00000010  6410          LA    R1,16(R1)
0003E096  E313 0010 0024                00000010  6411          STG   R1,16(R3)           load OP3 address in PL
                                                  6412
0003E09C  B982 0000                               6413          XGR   R0,R0
0003E0A0  4100 0020                     00000020  6414          LA    R0,32
0003E0A4  EE00 0000 3000      00000000  00000000  6415          PLO   0,0,0,0(R3)         Triple C/S (TCSX: 32)
                                                  6416
0003E0AA  B98D 0020                               6417          EPSW  R2,R0               extract psw
0003E0AE  5020 5010                     00000010  6418          ST    R2,CCPSW              to save CC
0003E0B2  E310 500B 0076                0000000B  6419          LB    R1,CCMASK           (failure CC mask)
0003E0B8  8910 0004                     00000004  6420          SLL   R1,4                (shift to BC instr CC position)
0003E0BC  4410 50F0                     0003E110  6421          EX    R1,F323TSTCC            fail if...
                                                  6422
                                0003E0C0  00000001  6423 F323CKUR EQU   *
0003E0C0  D200 5009 3030      00000009  00000030  6424          MVC   URFOUND,48(R3)      Extract UR
0003E0C6  D500 5008 3030      00000008  00000030  6425          CLC   UR,48(R3)
0003E0CC  4780 50B4                     0003E0D4  6426          BE    F323CKRES
0003E0D0  45F0 82C8                     000022C8  6427          BAL   R15,URMSG
                                                  6428 *
                                                  6429 * compare results
                                                  6430 *
                                0003E0D4  00000001  6431 F323CKRES EQU   *
                                                  6432 *                              reset adresses
0003E0D4  E310 501C 0014                0000001C  6433          LGF   R1,OPSRCA
0003E0DA  E320 5028 0014                00000028  6434          LGF   R2,RANDXA
0003E0E0  E330 5024 0014                00000024  6435          LGF   R3,PLA
0003E0E6  E340 5020 0014                00000020  6436          LGF   R4,OPREPA
                                                  6437
0003E0EC  D50F 1000 3040      00000000  00000040  6438          CLC   0(16,R1),64(R3)
0003E0F2  4770 50EA                     0003E10A  6439          BNE   F323XX              OP1C Different, error
0003E0F6  D50F 1010 3050      00000010  00000050  6440          CLC   16(16,R1),80(R3)
0003E0FC  4770 50EA                     0003E10A  6441          BNE   F323XX              OP2C Different, error
0003E100  D50F 1020 3060      00000020  00000060  6442          CLC   32(16,R1),96(R3)
0003E106  478B 0000                     00000000  6443          BE    0(R11)              OP3C same, return
                                                  6444
                                0003E10A  00000001  6445 F323XX   EQU   *
0003E10A  45F0 83E8                     000023E8  6446          BAL   R15,FAILMSG
0003E10E  07FB                                    6447          BR    R11
                                                  6448
0003E110  4700 50F4                     0003E114  6449 F323TSTCC BC    0,F323CCMSG         fail if unexpected condition code
                                0003E114  00000001  6450 F323CCMSG EQU  *
0003E114  45F0 8196                     00002196  6451          BAL   R15,CCMSG
0003E118  47F0 50A0                     0003E0C0  6452          B     F323CKUR
                                                  6453
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6454           TSTDATA
0003E11C                      0003E11C  0003E600  6455+          ORG   PLOETST+(128*K)+40*(3*K)+(K+512)
                                                    6456           TSTLBL OPDST              Dest/Data: Op1-R, Op2-R, Op3-R
0003E600  6E6E6ED6 D7C4E2E3                         6457+          DC    CL32'>>>OPDST78<<<' eye catcher
0003E608  F7F84C4C 4C404040
0003E610  40404040 40404040
0003E618  40404040 40404040
0003E620                                            6458+OPDST78   DS    0C
0003E620  00000000 00000000                         6459           DC    XL256'0'
0003E628  00000000 00000000
0003E630  00000000 00000000
0003E638  00000000 00000000
0003E640  00000000 00000000
0003E648  00000000 00000000
0003E650  00000000 00000000
0003E658  00000000 00000000
0003E660  00000000 00000000
0003E668  00000000 00000000
0003E670  00000000 00000000
0003E678  00000000 00000000
0003E680  00000000 00000000
0003E688  00000000 00000000
0003E690  00000000 00000000
0003E698  00000000 00000000
0003E6A0  00000000 00000000
0003E6A8  00000000 00000000
0003E6B0  00000000 00000000
0003E6B8  00000000 00000000
0003E6C0  00000000 00000000
0003E6C8  00000000 00000000
0003E6D0  00000000 00000000
0003E6D8  00000000 00000000
0003E6E0  00000000 00000000
0003E6E8  00000000 00000000
0003E6F0  00000000 00000000
0003E6F8  00000000 00000000
0003E700  00000000 00000000
0003E708  00000000 00000000
0003E710  00000000 00000000
0003E718  00000000 00000000
                                                    6460
                                                    6461           TSTLBL OPSRC              Source: OP1-OP3
0003E720  6E6E6ED6 D7E2D9C3                         6462+          DC    CL32'>>>OPSRC78<<<' eye catcher
0003E728  F7F84C4C 4C404040
0003E730  40404040 40404040
0003E738  40404040 40404040
0003E740                                            6463+OPSRC78   DS    0C
0003E740  00000000 00000000                         6464           DC    XL256'0'
0003E748  00000000 00000000
0003E750  00000000 00000000
0003E758  00000000 00000000
0003E760  00000000 00000000
0003E768  00000000 00000000
0003E770  00000000 00000000
0003E778  00000000 00000000
0003E780  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0003E788  00000000 00000000
0003E790  00000000 00000000
0003E798  00000000 00000000
0003E7A0  00000000 00000000
0003E7A8  00000000 00000000
0003E7B0  00000000 00000000
0003E7B8  00000000 00000000
0003E7C0  00000000 00000000
0003E7C8  00000000 00000000
0003E7D0  00000000 00000000
0003E7D8  00000000 00000000
0003E7E0  00000000 00000000
0003E7E8  00000000 00000000
0003E7F0  00000000 00000000
0003E7F8  00000000 00000000
0003E800  00000000 00000000
0003E808  00000000 00000000
0003E810  00000000 00000000
0003E818  00000000 00000000
0003E820  00000000 00000000
0003E828  00000000 00000000
0003E830  00000000 00000000
0003E838  00000000 00000000
                                        6465
                                        6466          TSTLBL PL                  Parameter List
0003E840  6E6E6ED7 D3F7F84C             6467+         DC    CL32'>>>PL78<<<'     eye catcher
0003E848  4C4C4040 40404040
0003E850  40404040 40404040
0003E858  40404040 40404040
0003E860                                6468+PL78     DS    0C
0003E860  00000000 00000000             6469          DC    XL352'00'
0003E868  00000000 00000000
0003E870  00000000 00000000
0003E878  00000000 00000000
0003E880  00000000 00000000
0003E888  00000000 00000000
0003E890  00000000 00000000
0003E898  00000000 00000000
0003E8A0  00000000 00000000
0003E8A8  00000000 00000000
0003E8B0  00000000 00000000
0003E8B8  00000000 00000000
0003E8C0  00000000 00000000
0003E8C8  00000000 00000000
0003E8D0  00000000 00000000
0003E8D8  00000000 00000000
0003E8E0  00000000 00000000
0003E8E8  00000000 00000000
0003E8F0  00000000 00000000
0003E8F8  00000000 00000000
0003E900  00000000 00000000
0003E908  00000000 00000000
0003E910  00000000 00000000
0003E918  00000000 00000000
0003E920  00000000 00000000
0003E928  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003E930  00000000 00000000
0003E938  00000000 00000000
0003E940  00000000 00000000
0003E948  00000000 00000000
0003E950  00000000 00000000
0003E958  00000000 00000000
0003E960  00000000 00000000
0003E968  00000000 00000000
0003E970  00000000 00000000
0003E978  00000000 00000000
0003E980  00000000 00000000
0003E988  00000000 00000000
0003E990  00000000 00000000
0003E998  00000000 00000000
0003E9A0  00000000 00000000
0003E9A8  00000000 00000000
0003E9B0  00000000 00000000
0003E9B8  00000000 00000000
                                       6470
                                       6471          TSTLBL FMSG               fail message
0003E9C0  6E6E6EC6 D4E2C7F7            6472+         DC    CL32'>>>FMSG78<<<'  eye catcher
0003E9C8  F84C4C4C 40404040
0003E9D0  40404040 40404040
0003E9D8  40404040 40404040
0003E9E0                               6473+FMSG78   DS    0C
0003E9E0  86818993 858440D7            6474          DC    C'failed PLO-Extension '
0003E9E8  D3D660C5 A7A38595
0003E9F0  A2899695 40
0003E9F5  E3998997 938540C3            6475          DC    CL96'Triple Compare and Swap (TCSX: 32): 3nd NE'
0003E9FD  96949781 99854081
0003EA05  958440E2 A6819740
0003EA0D  4DE3C3E2 E77A40F3
0003EA15  F25D7A40 F3958440
0003EA1D  D5C54040 40404040
0003EA25  40404040 40404040
0003EA2D  40404040 40404040
0003EA35  40404040 40404040
0003EA3D  40404040 40404040
0003EA45  40404040 40404040
0003EA4D  40404040 40404040
                                       6476
                                       6477          TSTEND
0003EA55  C5D5C440 968640A3            6478+         DC    CL32'END of test: 78 <<<'
0003EA5D  85A2A37A 40F7F840
0003EA65  4C4C4C40 40404040
0003EA6D  40404040 40404040
                                       6479
                                       6480 * ------------------------------------------------------------------
                                       6481 * Test Triple Compare and Swap (TCSX: 32) instruction : All EQ
                                       6482 * ------------------------------------------------------------------
                                       6483          TSTHDR 0,99,32             All Equal (cc=0, UR=99)
0003EA75                  0003EA75  0003EC00   6484+         ORG   PLOETST+(128*K)+41*(3*K)
0003EC00  E2E3C1D9 E3409686            6485+         DC    CL32'START of test: 79'
0003EC08  40A385A2 A37A40F7
0003EC10  F9404040 40404040
0003EC18  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0003EC20                     0003EC20              6486+          USING *,R5                base for test data and test routine
0003EC20                     00000000              6487+          USING PLOETEST,R5
0003EC20  0003EC4C                                 6488+T79       DC    A(CODE79)           address of test code
0003EC24  004F                                     6489+          DC    H'79'               test number
0003EC26  00                                       6490+          DC    X'00'
0003EC27  20                                       6491+          DC    HL1'32'             Function
0003EC28  63                                       6492+          DC    HL1'99'             expected unequal results (UR)
0003EC29  00                                       6493+          DC    HL1'00'             found UR
0003EC2A  00                                       6494+          DC    HL1'0'              cc
0003EC2B  07                                       6495+          DC    HL1'7'              not expected CC mask
0003EC2C  00                                       6496+          DS    X                   extracted cc
0003EC30  00000000 00000000                        6497+          DS    2F                  extract PSW after test (has CC)
0003EC38  0003F5E0                                 6498+          DC    A(FMSG79)           address of test fail message
0003EC3C  0003F340                                 6499+          DC    A(OPSRC79)          address of operand source
0003EC40  0003F220                                 6500+          DC    A(OPDST79)          address of operand destination
0003EC44  0003F460                                 6501+          DC    A(PL79)             address of parameter list
0003EC48  00004207                                 6502+          DC    A(RANDX+79)         address of random source data
                                                   6503
                                                   6504           TSTCODE
0003EC4C                                           6505+CODE79    DS    0F
                                                   6506 *                                   load random source
0003EC4C  E310 501C 0014               0000001C    6507           LGF   R1,OPSRCA
0003EC52  E320 5028 0014               00000028    6508           LGF   R2,RANDXA
0003EC58  E330 5024 0014               00000024    6509           LGF   R3,PLA
0003EC5E  E340 5020 0014               00000020    6510           LGF   R4,OPREPA
                                                   6511 *
                                                   6512 * Triple Compare and Swap (TCSX: 32)
                                                   6513 * Build Parambter List
                                                   6514 *
0003EC64  D22F 1000 2000      00000000 00000000    6515           MVC   0(48,R1),0(R2)      load Op1-Op3 data in source
0003EC6A  D22F 3040 1000      00000040 00000000    6516           MVC   64(48,R3),0(R1)     copy to PL OP1-Op3 Compare
                                                   6517
0003EC70  D22F 4000 2080      00000000 00000080    6518           MVC   0(48,R4),128(R2)    load initial Op1-Op3 data
0003EC76  D22F 30A0 4000      000000A0 00000000    6519           MVC   160(48,R3),0(R4)    copy to PL Op1-Op3 Replacement
                                                   6520 ****
                                                   6521 * Do Not force a failure by changing the 3nd operand
                                                   6522 ****
                                                   6523 *          MVC   32(8,R1),0(R4)
                                                   6524 ****
                                                   6525
0003EC7C  E313 0000 0024               00000000    6526           STG   R1,0(R3)            load OP1 address in PL
                                                   6527
0003EC82  4111 0010                    00000010    6528           LA    R1,16(R1)
0003EC86  E313 0008 0024               00000008    6529           STG   R1,8(R3)            load OP2 address in PL
                                                   6530
0003EC8C  4111 0010                    00000010    6531           LA    R1,16(R1)
0003EC90  E313 0010 0024               00000010    6532           STG   R1,16(R3)           load OP3 address in PL
                                                   6533
0003EC96  B982 0000                                6534           XGR   R0,R0
0003EC9A  4100 0020                    00000020    6535           LA    R0,32
0003EC9E  EE00 0000 3000      00000000 00000000    6536           PLO   0,0,0,0(R3)         Triple C/S (TCSX: 32)
                                                   6537
0003ECA4  B98D 0020                                6538           EPSW  R2,R0               extract psw
0003ECA8  5020 5010                    00000010    6539           ST    R2,CCPSW              to save CC
0003ECAC  E310 500B 0076               0000000B    6540           LB    R1,CCMASK           (failure CC mask)
0003ECB2  8910 0004                    00000004    6541           SLL   R1,4                (shift to BC instr CC position)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003ECB6  4410 50D6                     0003ECF6  6542          EX    R1,F324TSTCC            fail if...
                                                  6543 *
                                                  6544 * compare results
                                                  6545 *
                          0003ECBA  00000001      6546 F324CKRES EQU   *
                                                  6547 *                                reset adresses
0003ECBA  E310 501C 0014                0000001C  6548          LGF   R1,OPSRCA
0003ECC0  E320 5028 0014                00000028  6549          LGF   R2,RANDXA
0003ECC6  E330 5024 0014                00000024  6550          LGF   R3,PLA
0003ECCC  E340 5020 0014                00000020  6551          LGF   R4,OPREPA
                                                  6552
                                                  6553 * All equal; replacement values copied to operand source
                                                  6554
0003ECD2  D50F 1000 3070  00000000      00000070  6555          CLC   0(16,R1),112(R3)
0003ECD8  4770 50D0                     0003ECF0  6556          BNE   F324XX                OP1R Different, error
0003ECDC  D50F 1010 3080  00000010      00000080  6557          CLC   16(16,R1),128(R3)
0003ECE2  4770 50D0                     0003ECF0  6558          BNE   F324XX                OP2R Different, error
0003ECE6  D50F 1020 3090  00000020      00000090  6559          CLC   32(16,R1),144(R3)
0003ECEC  478B 0000                     00000000  6560          BE    0(R11)
                                                  6561
                          0003ECF0  00000001      6562 F324XX   EQU   *
0003ECF0  45F0 83E8                     000023E8  6563          BAL   R15,FAILMSG
0003ECF4  07FB                                    6564          BR    R11
                                                  6565
0003ECF6  4700 50DA                     0003ECFA  6566 F324TSTCC BC    0,F324CCMSG           fail if unexpected condition code
                          0003ECFA  00000001      6567 F324CCMSG EQU   *
0003ECFA  45F0 8196                     00002196  6568          BAL   R15,CCMSG
0003ECFE  47F0 509A                     0003ECBA  6569          B     F324CKRES
                                                  6570
                                                  6571          TSTDATA
0003ED02                  0003ED02  0003F200      6572+         ORG   PLOETST+(128*K)+41*(3*K)+(K+512)
                                                  6573          TSTLBL OPDST                Dest/Data: Op1-R, Op2-R, Op3-R
0003F200  6E6E6ED6 D7C4E2E3                       6574+         DC    CL32'>>>OPDST79<<<' eye catcher
0003F208  F7F94C4C 4C404040
0003F210  40404040 40404040
0003F218  40404040 40404040
0003F220                                          6575+OPDST79  DS    0C
0003F220  00000000 00000000                       6576          DC    XL256'0'
0003F228  00000000 00000000
0003F230  00000000 00000000
0003F238  00000000 00000000
0003F240  00000000 00000000
0003F248  00000000 00000000
0003F250  00000000 00000000
0003F258  00000000 00000000
0003F260  00000000 00000000
0003F268  00000000 00000000
0003F270  00000000 00000000
0003F278  00000000 00000000
0003F280  00000000 00000000
0003F288  00000000 00000000
0003F290  00000000 00000000
0003F298  00000000 00000000
0003F2A0  00000000 00000000
0003F2A8  00000000 00000000
0003F2B0  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003F2B8  00000000 00000000
0003F2C0  00000000 00000000
0003F2C8  00000000 00000000
0003F2D0  00000000 00000000
0003F2D8  00000000 00000000
0003F2E0  00000000 00000000
0003F2E8  00000000 00000000
0003F2F0  00000000 00000000
0003F2F8  00000000 00000000
0003F300  00000000 00000000
0003F308  00000000 00000000
0003F310  00000000 00000000
0003F318  00000000 00000000
                                          6577
                                          6578          TSTLBL OPSRC              Source: OP1-OP3
0003F320  6E6E6ED6 D7E2D9C3               6579+         DC    CL32'>>>OPSRC79<<<' eye catcher
0003F328  F7F94C4C 4C404040
0003F330  40404040 40404040
0003F338  40404040 40404040
0003F340                                  6580+OPSRC79  DS    0C
0003F340  00000000 00000000               6581          DC    XL256'0'
0003F348  00000000 00000000
0003F350  00000000 00000000
0003F358  00000000 00000000
0003F360  00000000 00000000
0003F368  00000000 00000000
0003F370  00000000 00000000
0003F378  00000000 00000000
0003F380  00000000 00000000
0003F388  00000000 00000000
0003F390  00000000 00000000
0003F398  00000000 00000000
0003F3A0  00000000 00000000
0003F3A8  00000000 00000000
0003F3B0  00000000 00000000
0003F3B8  00000000 00000000
0003F3C0  00000000 00000000
0003F3C8  00000000 00000000
0003F3D0  00000000 00000000
0003F3D8  00000000 00000000
0003F3E0  00000000 00000000
0003F3E8  00000000 00000000
0003F3F0  00000000 00000000
0003F3F8  00000000 00000000
0003F400  00000000 00000000
0003F408  00000000 00000000
0003F410  00000000 00000000
0003F418  00000000 00000000
0003F420  00000000 00000000
0003F428  00000000 00000000
0003F430  00000000 00000000
0003F438  00000000 00000000
                                          6582
                                          6583          TSTLBL PL                 Parameter List
0003F440  6E6E6ED7 D3F7F94C               6584+         DC    CL32'>>>PL79<<<'    eye catcher
0003F448  4C4C4040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0003F450  40404040 40404040
0003F458  40404040 40404040
0003F460                                          6585+PL79     DS    0C
0003F460  00000000 00000000                       6586          DC    XL352'00'
0003F468  00000000 00000000
0003F470  00000000 00000000
0003F478  00000000 00000000
0003F480  00000000 00000000
0003F488  00000000 00000000
0003F490  00000000 00000000
0003F498  00000000 00000000
0003F4A0  00000000 00000000
0003F4A8  00000000 00000000
0003F4B0  00000000 00000000
0003F4B8  00000000 00000000
0003F4C0  00000000 00000000
0003F4C8  00000000 00000000
0003F4D0  00000000 00000000
0003F4D8  00000000 00000000
0003F4E0  00000000 00000000
0003F4E8  00000000 00000000
0003F4F0  00000000 00000000
0003F4F8  00000000 00000000
0003F500  00000000 00000000
0003F508  00000000 00000000
0003F510  00000000 00000000
0003F518  00000000 00000000
0003F520  00000000 00000000
0003F528  00000000 00000000
0003F530  00000000 00000000
0003F538  00000000 00000000
0003F540  00000000 00000000
0003F548  00000000 00000000
0003F550  00000000 00000000
0003F558  00000000 00000000
0003F560  00000000 00000000
0003F568  00000000 00000000
0003F570  00000000 00000000
0003F578  00000000 00000000
0003F580  00000000 00000000
0003F588  00000000 00000000
0003F590  00000000 00000000
0003F598  00000000 00000000
0003F5A0  00000000 00000000
0003F5A8  00000000 00000000
0003F5B0  00000000 00000000
0003F5B8  00000000 00000000
                                          6587
                                          6588          TSTLBL FMSG               fail message
0003F5C0  6E6E6EC6 D4E2C7F7               6589+         DC    CL32'>>>FMSG79<<<'  eye catcher
0003F5C8  F94C4C4C 40404040
0003F5D0  40404040 40404040
0003F5D8  40404040 40404040
0003F5E0                                  6590+FMSG79   DS    0C
0003F5E0  86818993 858440D7               6591          DC    C'failed PLO-Extension '
0003F5E8  D3D660C5 A7A38595
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0003F5F0  A2899695 40
0003F5F5  E3998997 938540C3                         6592          DC    CL96'Triple Compare and Swap (TCSX: 32): All EQ'
0003F5FD  96949781 99854081
0003F605  958440E2 A6819740
0003F60D  4DE3C3E2 E77A40F3
0003F615  F25D7A40 C1939340
0003F61D  C5D84040 40404040
0003F625  40404040 40404040
0003F62D  40404040 40404040
0003F635  40404040 40404040
0003F63D  40404040 40404040
0003F645  40404040 40404040
0003F64D  40404040 40404040
                                                    6593
                                                    6594          TSTEND
0003F655  C5D5C440 968640A3                         6595+         DC    CL32'END of test: 79 <<<'
0003F65D  85A2A37A 40F7F940
0003F665  4C4C4C40 40404040
0003F66D  40404040 40404040
                                                    6596
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6598 *=============================================================================
                                                    6599 *       Triple Compare and Swap Tests
                                                    6600 *=============================================================================
                                                    6601
                                                    6602 * -----------------------------------------------------------------------
                                                    6603 * Test Triple Compare and Swap (TCSG: 31) instruction : 1st NE
                                                    6604 * -----------------------------------------------------------------------
                                                    6605          TSTHDR 2,1,31             1st Operand NOT Equal (cc=2, UR=1)
0003F675                          0003F675  0003F800  6606+         ORG   PLOETST+(128*K)+42*(3*K)
0003F800  E2E3C1D9 E3409686                         6607+         DC    CL32'START of test: 80'
0003F808  40A385A2 A37A40F8
0003F810  F0404040 40404040
0003F818  40404040 40404040
0003F820                          0003F820          6608+         USING *,R5                base for test data and test routine
0003F820                          00000000          6609+         USING PLOETEST,R5
0003F820  0003F84C                                  6610+T80      DC    A(CODE80)           address of test code
0003F824  0050                                      6611+         DC    H'80'               test number
0003F826  00                                        6612+         DC    X'00'
0003F827  1F                                        6613+         DC    HL1'31'             Function
0003F828  01                                        6614+         DC    HL1'1'              expected unequal results (UR)
0003F829  00                                        6615+         DC    HL1'00'             found UR
0003F82A  02                                        6616+         DC    HL1'2'              cc
0003F82B  0D                                        6617+         DC    HL1'13'             not expected CC mask
0003F82C  00                                        6618+         DS    X                   extracted cc
0003F830  00000000 00000000                         6619+         DS    2F                  extract PSW after test (has CC)
0003F838  000401E0                                  6620+         DC    A(FMSG80)           address of test fail message
0003F83C  0003FF40                                  6621+         DC    A(OPSRC80)          address of operand source
0003F840  0003FE20                                  6622+         DC    A(OPDST80)          address of operand destination
0003F844  00040060                                  6623+         DC    A(PL80)             address of parameter list
0003F848  00004208                                  6624+         DC    A(RANDX+80)         address of random source data
                                                    6625
                                                    6626          TSTCODE
0003F84C                                            6627+CODE80   DS    0F
                                                    6628 *                                  load random source
0003F84C  E310 501C 0014                 0000001C   6629          LGF   R1,OPSRCA
0003F852  E320 5028 0014                 00000028   6630          LGF   R2,RANDXA
0003F858  E330 5024 0014                 00000024   6631          LGF   R3,PLA
0003F85E  E340 5020 0014                 00000020   6632          LGF   R4,OPREPA
                                                    6633 *
                                                    6634 * Triple Compare and Swap (TCSG: 31)
                                                    6635 * Build Parambter List
                                                    6636 *
0003F864  D217 1000 2000       00000000  00000000   6637          MVC   0(24,R1),0(R2)      load Op1-Op3 data in source
0003F86A  D217 3040 1000       00000040  00000000   6638          MVC   64(24,R3),0(R1)     copy to PL OP1-Op3 Compare
                                                    6639
0003F870  D217 4000 2080       00000000  00000080   6640          MVC   0(24,R4),128(R2)    load initial Op1-Op3 data
0003F876  D217 3058 4000       00000058  00000000   6641          MVC   88(24,R3),0(R4)     copy to PL Op1-Op3 Replacement
                                                    6642 ****
                                                    6643 * force a failure by changing the 1st operand
                                                    6644 ****
0003F87C  D203 1000 4000       00000000  00000000   6645          MVC   0(4,R1),0(R4)
                                                    6646 ****
                                                    6647
0003F882  E313 0000 0024                 00000000   6648          STG   R1,0(R3)            load OP1 address in PL
                                                    6649
0003F888  4111 0008                      00000008   6650          LA    R1,8(R1)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003F88C  E313 0008 0024                 00000008  6651           STG   R1,8(R3)            load OP2 address in PL
                                                   6652
0003F892  4111 0008                      00000008  6653           LA    R1,8(R1)
0003F896  E313 0010 0024                 00000010  6654           STG   R1,16(R3)           load OP3 address in PL
                                                   6655
0003F89C  B982 0000                                6656           XGR   R0,R0
0003F8A0  4100 001F                      0000001F  6657           LA    R0,31
0003F8A4  EE00 0000 3000       00000000  00000000  6658           PLO   0,0,0,0(R3)         Triple C/S (TCSG: 31)
                                                   6659
0003F8AA  B98D 0020                                6660           EPSW  R2,R0               extract psw
0003F8AE  5020 5010                      00000010  6661           ST    R2,CCPSW              to save CC
0003F8B2  E310 500B 0076                 0000000B  6662           LB    R1,CCMASK           (failure CC mask)
0003F8B8  8910 0004                      00000004  6663           SLL   R1,4                (shift to BC instr CC position)
0003F8BC  4410 50F0                      0003F910  6664           EX    R1,F311TSTCC            fail if...
                                                   6665
                                0003F8C0  00000001  6666 F311CKUR EQU   *
0003F8C0  D200 5009 3030        00000009  00000030  6667           MVC   URFOUND,48(R3)      Extract UR
0003F8C6  D500 5008 3030        00000008  00000030  6668           CLC   UR,48(R3)
0003F8CC  4780 50B4                      0003F8D4  6669           BE    F311CKRES
0003F8D0  45F0 82C8                      000022C8  6670           BAL   R15,URMSG
                                                   6671 *
                                                   6672 * compare results
                                                   6673 *
                                0003F8D4  00000001  6674 F311CKRES EQU   *
                                                   6675 *                                  reset adresses
0003F8D4  E310 501C 0014                 0000001C  6676           LGF   R1,OPSRCA
0003F8DA  E320 5028 0014                 00000028  6677           LGF   R2,RANDXA
0003F8E0  E330 5024 0014                 00000024  6678           LGF   R3,PLA
0003F8E6  E340 5020 0014                 00000020  6679           LGF   R4,OPREPA
                                                   6680
0003F8EC  D507 1000 3040       00000000  00000040  6681           CLC   0(8,R1),64(R3)
0003F8F2  4770 50EA                      0003F90A  6682           BNE   F311XX              OP1C Different, error
0003F8F6  D507 1008 3048       00000008  00000048  6683           CLC   8(8,R1),72(R3)
0003F8FC  4770 50EA                      0003F90A  6684           BNE   F311XX              OP2C Different, error
0003F900  D507 1010 3050       00000010  00000050  6685           CLC   16(8,R1),80(R3)
0003F906  478B 0000                      00000000  6686           BE    0(R11)              OP3C same, return
                                                   6687
                                0003F90A  00000001  6688 F311XX   EQU   *
0003F90A  45F0 83E8                      000023E8  6689           BAL   R15,FAILMSG
0003F90E  07FB                                     6690           BR    R11
                                                   6691
0003F910  4700 50F4                      0003F914  6692 F311TSTCC BC    0,F311CCMSG         fail if unexpected condition code
                                0003F914  00000001  6693 F311CCMSG EQU   *
0003F914  45F0 8196                      00002196  6694           BAL   R15,CCMSG
0003F918  47F0 50A0                      0003F8C0  6695           B     F311CKUR
                                                   6696
                                                   6697           TSTDATA
0003F91C                        0003F91C  0003FE00  6698+          ORG   PLOETST+(128*K)+42*(3*K)+(K+512)
                                                   6699           TSTLBL OPDST              Dest/Data: Op1-R, Op2-R, Op3-R
0003FE00  6E6E6ED6 D7C4E2E3                        6700+          DC    CL32'>>>OPDST80<<<' eye catcher
0003FE08  F8F04C4C 4C404040
0003FE10  40404040 40404040
0003FE18  40404040 40404040
0003FE20                                           6701+OPDST80   DS    0C
0003FE20  00000000 00000000                        6702           DC    XL256'0'
0003FE28  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003FE30  00000000 00000000
0003FE38  00000000 00000000
0003FE40  00000000 00000000
0003FE48  00000000 00000000
0003FE50  00000000 00000000
0003FE58  00000000 00000000
0003FE60  00000000 00000000
0003FE68  00000000 00000000
0003FE70  00000000 00000000
0003FE78  00000000 00000000
0003FE80  00000000 00000000
0003FE88  00000000 00000000
0003FE90  00000000 00000000
0003FE98  00000000 00000000
0003FEA0  00000000 00000000
0003FEA8  00000000 00000000
0003FEB0  00000000 00000000
0003FEB8  00000000 00000000
0003FEC0  00000000 00000000
0003FEC8  00000000 00000000
0003FED0  00000000 00000000
0003FED8  00000000 00000000
0003FEE0  00000000 00000000
0003FEE8  00000000 00000000
0003FEF0  00000000 00000000
0003FEF8  00000000 00000000
0003FF00  00000000 00000000
0003FF08  00000000 00000000
0003FF10  00000000 00000000
0003FF18  00000000 00000000
                                      6703
                                      6704          TSTLBL OPSRC              Source: OP1-OP3
0003FF20  6E6E6ED6 D7E2D9C3           6705+         DC    CL32'>>>OPSRC80<<<' eye catcher
0003FF28  F8F04C4C 4C404040
0003FF30  40404040 40404040
0003FF38  40404040 40404040
0003FF40                              6706+OPSRC80  DS    0C
0003FF40  00000000 00000000           6707          DC    XL256'0'
0003FF48  00000000 00000000
0003FF50  00000000 00000000
0003FF58  00000000 00000000
0003FF60  00000000 00000000
0003FF68  00000000 00000000
0003FF70  00000000 00000000
0003FF78  00000000 00000000
0003FF80  00000000 00000000
0003FF88  00000000 00000000
0003FF90  00000000 00000000
0003FF98  00000000 00000000
0003FFA0  00000000 00000000
0003FFA8  00000000 00000000
0003FFB0  00000000 00000000
0003FFB8  00000000 00000000
0003FFC0  00000000 00000000
0003FFC8  00000000 00000000
0003FFD0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003FFD8  00000000 00000000
0003FFE0  00000000 00000000
0003FFE8  00000000 00000000
0003FFF0  00000000 00000000
0003FFF8  00000000 00000000
00040000  00000000 00000000
00040008  00000000 00000000
00040010  00000000 00000000
00040018  00000000 00000000
00040020  00000000 00000000
00040028  00000000 00000000
00040030  00000000 00000000
00040038  00000000 00000000
                                          6708
                                          6709          TSTLBL PL                   Parameter List
00040040  6E6E6ED7 D3F8F04C               6710+         DC    CL32'>>>PL80<<<'      eye catcher
00040048  4C4C4040 40404040
00040050  40404040 40404040
00040058  40404040 40404040
00040060                                  6711+PL80     DS    0C
00040060  00000000 00000000               6712          DC    XL352'00'
00040068  00000000 00000000
00040070  00000000 00000000
00040078  00000000 00000000
00040080  00000000 00000000
00040088  00000000 00000000
00040090  00000000 00000000
00040098  00000000 00000000
000400A0  00000000 00000000
000400A8  00000000 00000000
000400B0  00000000 00000000
000400B8  00000000 00000000
000400C0  00000000 00000000
000400C8  00000000 00000000
000400D0  00000000 00000000
000400D8  00000000 00000000
000400E0  00000000 00000000
000400E8  00000000 00000000
000400F0  00000000 00000000
000400F8  00000000 00000000
00040100  00000000 00000000
00040108  00000000 00000000
00040110  00000000 00000000
00040118  00000000 00000000
00040120  00000000 00000000
00040128  00000000 00000000
00040130  00000000 00000000
00040138  00000000 00000000
00040140  00000000 00000000
00040148  00000000 00000000
00040150  00000000 00000000
00040158  00000000 00000000
00040160  00000000 00000000
00040168  00000000 00000000
00040170  00000000 00000000
00040178  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00040180  00000000 00000000
00040188  00000000 00000000
00040190  00000000 00000000
00040198  00000000 00000000
000401A0  00000000 00000000
000401A8  00000000 00000000
000401B0  00000000 00000000
000401B8  00000000 00000000
                                        6713
                                        6714          TSTLBL FMSG                fail message
000401C0  6E6E6EC6 D4E2C7F8             6715+         DC    CL32'>>>FMSG80<<<'   eye catcher
000401C8  F04C4C4C 40404040
000401D0  40404040 40404040
000401D8  40404040 40404040
000401E0                                6716+FMSG80   DS    0C
000401E0  86818993 858440D7             6717          DC    C'failed PLO-Extension '
000401E8  D3D660C5 A7A38595
000401F0  A2899695 40
000401F5  E3998997 938540C3             6718          DC    CL96'Triple Compare and Swap (TCSG: 31): 1st NE'
000401FD  96949781 99854081
00040205  958440E2 A6819740
0004020D  4DE3C3E2 C77A40F3
00040215  F15D7A40 F1A2A340
0004021D  D5C54040 40404040
00040225  40404040 40404040
0004022D  40404040 40404040
00040235  40404040 40404040
0004023D  40404040 40404040
00040245  40404040 40404040
0004024D  40404040 40404040
                                        6719
                                        6720          TSTEND
00040255  C5D5C440 968640A3             6721+         DC    CL32'END of test: 80 <<<'
0004025D  85A2A37A 40F8F040
00040265  4C4C4C40 40404040
0004026D  40404040 40404040
                                        6722
                                        6723 * ------------------------------------------------------------------------
                                        6724 * Test Triple Compare and Swap (TCSG: 31) instruction : 2nd NE
                                        6725 * ------------------------------------------------------------------------
                                        6726          TSTHDR 2,2,31              2nd Operand NOT Equal (cc=2, UR=2)
00040275                        00040275  00040400    6727+         ORG   PLOETST+(128*K)+43*(3*K)
00040400  E2E3C1D9 E3409686             6728+         DC    CL32'START of test: 81'
00040408  40A385A2 A37A40F8
00040410  F1404040 40404040
00040418  40404040 40404040
00040420                        00040420            6729+         USING *,R5                 base for test data and test routine
00040420                        00000000            6730+         USING PLOETEST,R5
00040420  0004044C                                  6731+T81      DC    A(CODE81)            address of test code
00040424  0051                                      6732+         DC    H'81'                test number
00040426  00                                        6733+         DC    X'00'
00040427  1F                                        6734+         DC    HL1'31'              Function
00040428  02                                        6735+         DC    HL1'2'               expected unequal results (UR)
00040429  00                                        6736+         DC    HL1'00'              found UR
0004042A  02                                        6737+         DC    HL1'2'               cc
0004042B  0D                                        6738+         DC    HL1'13'              not expected CC mask
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004042C  00                                      6739+          DS    X                   extracted cc
00040430  00000000 00000000                       6740+          DS    2F                  extract PSW after test (has CC)
00040438  00040DE0                                6741+          DC    A(FMSG81)           address of test fail message
0004043C  00040B40                                6742+          DC    A(OPSRC81)          address of operand source
00040440  00040A20                                6743+          DC    A(OPDST81)          address of operand destination
00040444  00040C60                                6744+          DC    A(PL81)             address of parameter list
00040448  00004209                                6745+          DC    A(RANDX+81)         address of random source data
                                                  6746
                                                  6747           TSTCODE
0004044C                                          6748+CODE81    DS    0F
                                                  6749 *                                   load random source
0004044C  E310 501C 0014                 0000001C 6750           LGF   R1,OPSRCA
00040452  E320 5028 0014                 00000028 6751           LGF   R2,RANDXA
00040458  E330 5024 0014                 00000024 6752           LGF   R3,PLA
0004045E  E340 5020 0014                 00000020 6753           LGF   R4,OPREPA
                                                  6754 *
                                                  6755 * Triple Compare and Swap (TCSG: 31)
                                                  6756 * Build Parambter List
                                                  6757 *
00040464  D217 1000 2000       00000000  00000000 6758           MVC   0(24,R1),0(R2)      load Op1-Op3 data in source
0004046A  D217 3040 1000       00000040  00000000 6759           MVC   64(24,R3),0(R1)     copy to PL OP1-Op3 Compare
                                                  6760
00040470  D217 4000 2080       00000000  00000080 6761           MVC   0(24,R4),128(R2)    load initial Op1-Op3 data
00040476  D217 3058 4000       00000058  00000000 6762           MVC   88(24,R3),0(R4)     copy to PL Op1-Op3 Replacement
                                                  6763 ****
                                                  6764 * force a failure by changing the 2nd operand
                                                  6765 ****
0004047C  D203 1008 4000       00000008  00000000 6766           MVC   8(4,R1),0(R4)
                                                  6767 ****
                                                  6768
00040482  E313 0000 0024                 00000000 6769           STG   R1,0(R3)            load OP1 address in PL
                                                  6770
00040488  4111 0008                      00000008 6771           LA    R1,8(R1)
0004048C  E313 0008 0024                 00000008 6772           STG   R1,8(R3)            load OP2 address in PL
                                                  6773
00040492  4111 0008                      00000008 6774           LA    R1,8(R1)
00040496  E313 0010 0024                 00000010 6775           STG   R1,16(R3)           load OP3 address in PL
                                                  6776
0004049C  B982 0000                               6777           XGR   R0,R0
000404A0  4100 001F                      0000001F 6778           LA    R0,31
000404A4  EE00 0000 3000       00000000  00000000 6779           PLO   0,0,0,0(R3)         Triple C/S (TCSG: 31)
                                                  6780
000404AA  B98D 0020                               6781           EPSW  R2,R0               extract psw
000404AE  5020 5010                      00000010 6782           ST    R2,CCPSW              to save CC
000404B2  E310 500B 0076                 0000000B 6783           LB    R1,CCMASK           (failure CC mask)
000404B8  8910 0004                      00000004 6784           SLL   R1,4                (shift to BC instr CC position)
000404BC  4410 50F0                      00040510 6785           EX    R1,F312TSTCC            fail if...
                                                  6786
                                 000404C0 00000001 6787 F312CKUR EQU   *
000404C0  D200 5009 3030       00000009  00000030 6788           MVC   URFOUND,48(R3)      Extract UR
000404C6  D500 5008 3030       00000008  00000030 6789           CLC   UR,48(R3)
000404CC  4780 50B4                      000404D4 6790           BE    F312CKRES
000404D0  45F0 82C8                      000022C8 6791           BAL   R15,URMSG
                                                  6792 *
                                                  6793 * compare results
                                                  6794 *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                               000404D4  00000001  6795 F312CKRES EQU   *
                                                   6796 *                                reset adresses
000404D4  E310 501C 0014                 0000001C  6797          LGF   R1,OPSRCA
000404DA  E320 5028 0014                 00000028  6798          LGF   R2,RANDXA
000404E0  E330 5024 0014                 00000024  6799          LGF   R3,PLA
000404E6  E340 5020 0014                 00000020  6800          LGF   R4,OPREPA
                                                   6801
000404EC  D507 1000 3040  00000000       00000040  6802          CLC   0(8,R1),64(R3)
000404F2  4770 50EA                      0004050A  6803          BNE   F312XX                OP1C Different, error
000404F6  D507 1008 3048  00000008       00000048  6804          CLC   8(8,R1),72(R3)
000404FC  4770 50EA                      0004050A  6805          BNE   F312XX                OP2C Different, error
00040500  D507 1010 3050  00000010       00000050  6806          CLC   16(8,R1),80(R3)
00040506  478B 0000                      00000000  6807          BE    0(R11)                OP3C same, return
                                                   6808
                               0004050A  00000001  6809 F312XX   EQU   *
0004050A  45F0 83E8                      000023E8  6810          BAL   R15,FAILMSG
0004050E  07FB                                     6811          BR    R11
                                                   6812
00040510  4700 50F4                      00040514  6813 F312TSTCC BC    0,F312CCMSG           fail if unexpected condition code
                               00040514  00000001  6814 F312CCMSG EQU   *
00040514  45F0 8196                      00002196  6815          BAL   R15,CCMSG
00040518  47F0 50A0                      000404C0  6816          B     F312CKUR
                                                   6817
                                                   6818          TSTDATA
0004051C                       0004051C  00040A00  6819+         ORG   PLOETST+(128*K)+43*(3*K)+(K+512)
                                                   6820          TSTLBL OPDST                Dest/Data: Op1-R, Op2-R, Op3-R
00040A00  6E6E6ED6 D7C4E2E3              00040A00  6821+         DC    CL32'>>>OPDST81<<<' eye catcher
00040A08  F8F14C4C 4C404040
00040A10  40404040 40404040
00040A18  40404040 40404040
00040A20                                 00040A20  6822+OPDST81  DS    0C
00040A20  00000000 00000000              00040A20  6823          DC    XL256'0'
00040A28  00000000 00000000
00040A30  00000000 00000000
00040A38  00000000 00000000
00040A40  00000000 00000000
00040A48  00000000 00000000
00040A50  00000000 00000000
00040A58  00000000 00000000
00040A60  00000000 00000000
00040A68  00000000 00000000
00040A70  00000000 00000000
00040A78  00000000 00000000
00040A80  00000000 00000000
00040A88  00000000 00000000
00040A90  00000000 00000000
00040A98  00000000 00000000
00040AA0  00000000 00000000
00040AA8  00000000 00000000
00040AB0  00000000 00000000
00040AB8  00000000 00000000
00040AC0  00000000 00000000
00040AC8  00000000 00000000
00040AD0  00000000 00000000
00040AD8  00000000 00000000
00040AE0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00040AE8  00000000 00000000
00040AF0  00000000 00000000
00040AF8  00000000 00000000
00040B00  00000000 00000000
00040B08  00000000 00000000
00040B10  00000000 00000000
00040B18  00000000 00000000
                                          6824
                                          6825           TSTLBL OPSRC               Source: OP1-OP3
00040B20  6E6E6ED6 D7E2D9C3               6826+          DC    CL32'>>>OPSRC81<<<'  eye catcher
00040B28  F8F14C4C 4C404040
00040B30  40404040 40404040
00040B38  40404040 40404040
00040B40                                  6827+OPSRC81   DS    0C
00040B40  00000000 00000000               6828           DC    XL256'0'
00040B48  00000000 00000000
00040B50  00000000 00000000
00040B58  00000000 00000000
00040B60  00000000 00000000
00040B68  00000000 00000000
00040B70  00000000 00000000
00040B78  00000000 00000000
00040B80  00000000 00000000
00040B88  00000000 00000000
00040B90  00000000 00000000
00040B98  00000000 00000000
00040BA0  00000000 00000000
00040BA8  00000000 00000000
00040BB0  00000000 00000000
00040BB8  00000000 00000000
00040BC0  00000000 00000000
00040BC8  00000000 00000000
00040BD0  00000000 00000000
00040BD8  00000000 00000000
00040BE0  00000000 00000000
00040BE8  00000000 00000000
00040BF0  00000000 00000000
00040BF8  00000000 00000000
00040C00  00000000 00000000
00040C08  00000000 00000000
00040C10  00000000 00000000
00040C18  00000000 00000000
00040C20  00000000 00000000
00040C28  00000000 00000000
00040C30  00000000 00000000
00040C38  00000000 00000000
                                          6829
                                          6830           TSTLBL PL                  Parameter List
00040C40  6E6E6ED7 D3F8F14C               6831+          DC    CL32'>>>PL81<<<'     eye catcher
00040C48  4C4C4040 40404040
00040C50  40404040 40404040
00040C58  40404040 40404040
00040C60                                  6832+PL81      DS    0C
00040C60  00000000 00000000               6833           DC    XL352'00'
00040C68  00000000 00000000
00040C70  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00040C78  00000000 00000000
00040C80  00000000 00000000
00040C88  00000000 00000000
00040C90  00000000 00000000
00040C98  00000000 00000000
00040CA0  00000000 00000000
00040CA8  00000000 00000000
00040CB0  00000000 00000000
00040CB8  00000000 00000000
00040CC0  00000000 00000000
00040CC8  00000000 00000000
00040CD0  00000000 00000000
00040CD8  00000000 00000000
00040CE0  00000000 00000000
00040CE8  00000000 00000000
00040CF0  00000000 00000000
00040CF8  00000000 00000000
00040D00  00000000 00000000
00040D08  00000000 00000000
00040D10  00000000 00000000
00040D18  00000000 00000000
00040D20  00000000 00000000
00040D28  00000000 00000000
00040D30  00000000 00000000
00040D38  00000000 00000000
00040D40  00000000 00000000
00040D48  00000000 00000000
00040D50  00000000 00000000
00040D58  00000000 00000000
00040D60  00000000 00000000
00040D68  00000000 00000000
00040D70  00000000 00000000
00040D78  00000000 00000000
00040D80  00000000 00000000
00040D88  00000000 00000000
00040D90  00000000 00000000
00040D98  00000000 00000000
00040DA0  00000000 00000000
00040DA8  00000000 00000000
00040DB0  00000000 00000000
00040DB8  00000000 00000000
                             6834
                             6835          TSTLBL FMSG                fail message
00040DC0  6E6E6EC6 D4E2C7F8  6836+         DC     CL32'>>>FMSG81<<<'  eye catcher
00040DC8  F14C4C4C 40404040
00040DD0  40404040 40404040
00040DD8  40404040 40404040
00040DE0                     6837+FMSG81   DS     0C
00040DE0  86818993 858440D7  6838          DC     C'failed PLO-Extension '
00040DE8  D3D660C5 A7A38595
00040DF0  A2899695 40
00040DF5  E3998997 938540C3  6839          DC     CL96'Triple Compare and Swap (TCSG: 31): 2nd NE'
00040DFD  96949781 99854081
00040E05  958440E2 A6819740
00040E0D  4DE3C3E2 C77A40F3
00040E15  F15D7A40 F2958440
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00040E1D  D5C54040 40404040
00040E25  40404040 40404040
00040E2D  40404040 40404040
00040E35  40404040 40404040
00040E3D  40404040 40404040
00040E45  40404040 40404040
00040E4D  40404040 40404040
                                        6840
                                        6841          TSTEND
00040E55  C5D5C440 968640A3             6842+         DC    CL32'END of test: 81 <<<'
00040E5D  85A2A37A 40F8F140
00040E65  4C4C4C40 40404040
00040E6D  40404040 40404040
                                        6843
                                        6844 * ------------------------------------------------------------------
                                        6845 * Test Triple Compare and Swap (TCSG: 31) instruction : 3rd NE
                                        6846 * ------------------------------------------------------------------
                                        6847          TSTHDR 2,3,31             3rd Operand NOT Equal (cc=2, UR=3)
00040E75                        00040E75  00041000    6848+         ORG   PLOETST+(128*K)+44*(3*K)
00041000  E2E3C1D9 E3409686             6849+         DC    CL32'START of test: 82'
00041008  40A385A2 A37A40F8
00041010  F2404040 40404040
00041018  40404040 40404040
00041020                        00041020              6850+         USING *,R5                base for test data and test routine
00041020                        00000000              6851+         USING PLOETEST,R5
00041020  0004104C                                    6852+T82      DC    A(CODE82)           address of test code
00041024  0052                                        6853+         DC    H'82'               test number
00041026  00                                          6854+         DC    X'00'
00041027  1F                                          6855+         DC    HL1'31'             Function
00041028  03                                          6856+         DC    HL1'3'              expected unequal results (UR)
00041029  00                                          6857+         DC    HL1'00'             found UR
0004102A  02                                          6858+         DC    HL1'2'              cc
0004102B  0D                                          6859+         DC    HL1'13'             not expected CC mask
0004102C  00                                          6860+         DS    X                   extracted cc
00041030  00000000 00000000                           6861+         DS    2F                  extract PSW after test (has CC)
00041038  000419E0                                    6862+         DC    A(FMSG82)           address of test fail message
0004103C  00041740                                    6863+         DC    A(OPSRC82)          address of operand source
00041040  00041620                                    6864+         DC    A(OPDST82)          address of operand destination
00041044  00041860                                    6865+         DC    A(PL82)             address of parameter list
00041048  0000420A                                    6866+         DC    A(RANDX+82)         address of random source data
                                                      6867
                                                      6868          TSTCODE
0004104C                                              6869+CODE82   DS    0F
                                                      6870 *                                  load random source
0004104C  E310 501C 0014                0000001C      6871          LGF   R1,OPSRCA
00041052  E320 5028 0014                00000028      6872          LGF   R2,RANDXA
00041058  E330 5024 0014                00000024      6873          LGF   R3,PLA
0004105E  E340 5020 0014                00000020      6874          LGF   R4,OPREPA
                                                      6875 *
                                                      6876 * Triple Compare and Swap (TCSG: 31)
                                                      6877 * Build Parambter List
                                                      6878 *
00041064  D217 1000 2000        00000000  00000000    6879          MVC   0(24,R1),0(R2)      load Op1-Op3 data in source
0004106A  D217 3040 1000        00000040  00000000    6880          MVC   64(24,R3),0(R1)     copy to PL OP1-Op3 Compare
                                                      6881
00041070  D217 4000 2080        00000000  00000080    6882          MVC   0(24,R4),128(R2)    load initial Op1-Op3 data
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00041076  D217 3058 4000      00000058  00000000  6883          MVC   88(24,R3),0(R4)     copy to PL Op1-Op3 Replacement
                                                   6884 ****
                                                   6885 * force a failure by changing the 3rd operand
                                                   6886 ****
0004107C  D203 1010 4000      00000010  00000000  6887          MVC   16(4,R1),0(R4)
                                                   6888 ****
                                                   6889
00041082  E313 0000 0024                00000000  6890          STG   R1,0(R3)           load OP1 address in PL
                                                   6891
00041088  4111 0008                     00000008  6892          LA    R1,8(R1)
0004108C  E313 0008 0024                00000008  6893          STG   R1,8(R3)           load OP2 address in PL
                                                   6894
00041092  4111 0008                     00000008  6895          LA    R1,8(R1)
00041096  E313 0010 0024                00000010  6896          STG   R1,16(R3)          load OP3 address in PL
                                                   6897
0004109C  B982 0000                               6898          XGR   R0,R0
000410A0  4100 001F                     0000001F  6899          LA    R0,31
000410A4  EE00 0000 3000      00000000  00000000  6900          PLO   0,0,0,0(R3)        Triple C/S (TCSG: 31)
                                                   6901
000410AA  B98D 0020                               6902          EPSW  R2,R0              extract psw
000410AE  5020 5010                     00000010  6903          ST    R2,CCPSW             to save CC
000410B2  E310 500B 0076                0000000B  6904          LB    R1,CCMASK          (failure CC mask)
000410B8  8910 0004                     00000004  6905          SLL   R1,4               (shift to BC instr CC position)
000410BC  4410 50F0                     00041110  6906          EX    R1,F313TSTCC           fail if...
                                                   6907
                                000410C0  00000001  6908 F313CKUR EQU   *
000410C0  D200 5009 3030      00000009  00000030  6909          MVC   URFOUND,48(R3)     Extract UR
000410C6  D500 5008 3030      00000008  00000030  6910          CLC   UR,48(R3)
000410CC  4780 50B4                     000410D4  6911          BE    F313CKRES
000410D0  45F0 82C8                     000022C8  6912          BAL   R15,URMSG
                                                   6913 *
                                                   6914 * compare results
                                                   6915 *
                                000410D4  00000001  6916 F313CKRES EQU   *
                                                   6917 *                           reset adresses
000410D4  E310 501C 0014                0000001C  6918          LGF   R1,OPSRCA
000410DA  E320 5028 0014                00000028  6919          LGF   R2,RANDXA
000410E0  E330 5024 0014                00000024  6920          LGF   R3,PLA
000410E6  E340 5020 0014                00000020  6921          LGF   R4,OPREPA
                                                   6922
000410EC  D507 1000 3040      00000000  00000040  6923          CLC   0(8,R1),64(R3)
000410F2  4770 50EA                     0004110A  6924          BNE   F313XX             OP1C Different, error
000410F6  D507 1008 3048      00000008  00000048  6925          CLC   8(8,R1),72(R3)
000410FC  4770 50EA                     0004110A  6926          BNE   F313XX             OP2C Different, error
00041100  D507 1010 3050      00000010  00000050  6927          CLC   16(8,R1),80(R3)
00041106  478B 0000                     00000000  6928          BE    0(R11)             OP3C same, return
                                                   6929
                                0004110A  00000001  6930 F313XX   EQU   *
0004110A  45F0 83E8                     000023E8  6931          BAL   R15,FAILMSG
0004110E  07FB                                    6932          BR    R11
                                                   6933
00041110  4700 50F4                     00041114  6934 F313TSTCC BC    0,F313CCMSG        fail if unexpected condition code
                                00041114  00000001  6935 F313CCMSG EQU  *
00041114  45F0 8196                     00002196  6936          BAL   R15,CCMSG
00041118  47F0 50A0                     000410C0  6937          B     F313CKUR
                                                   6938
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6939           TSTDATA
0004111C                      0004111C  00041600  6940+          ORG   PLOETST+(128*K)+44*(3*K)+(K+512)
                                                    6941           TSTLBL OPDST              Dest/Data: Op1-R, Op2-R, Op3-R
00041600  6E6E6ED6 D7C4E2E3                        6942+          DC    CL32'>>>OPDST82<<<' eye catcher
00041608  F8F24C4C 4C404040
00041610  40404040 40404040
00041618  40404040 40404040
00041620                                            6943+OPDST82  DS    0C
00041620  00000000 00000000                         6944          DC    XL256'0'
00041628  00000000 00000000
00041630  00000000 00000000
00041638  00000000 00000000
00041640  00000000 00000000
00041648  00000000 00000000
00041650  00000000 00000000
00041658  00000000 00000000
00041660  00000000 00000000
00041668  00000000 00000000
00041670  00000000 00000000
00041678  00000000 00000000
00041680  00000000 00000000
00041688  00000000 00000000
00041690  00000000 00000000
00041698  00000000 00000000
000416A0  00000000 00000000
000416A8  00000000 00000000
000416B0  00000000 00000000
000416B8  00000000 00000000
000416C0  00000000 00000000
000416C8  00000000 00000000
000416D0  00000000 00000000
000416D8  00000000 00000000
000416E0  00000000 00000000
000416E8  00000000 00000000
000416F0  00000000 00000000
000416F8  00000000 00000000
00041700  00000000 00000000
00041708  00000000 00000000
00041710  00000000 00000000
00041718  00000000 00000000
                                                    6945
                                                    6946           TSTLBL OPSRC              Source: OP1-OP3
00041720  6E6E6ED6 D7E2D9C3                        6947+          DC    CL32'>>>OPSRC82<<<' eye catcher
00041728  F8F24C4C 4C404040
00041730  40404040 40404040
00041738  40404040 40404040
00041740                                            6948+OPSRC82  DS    0C
00041740  00000000 00000000                         6949          DC    XL256'0'
00041748  00000000 00000000
00041750  00000000 00000000
00041758  00000000 00000000
00041760  00000000 00000000
00041768  00000000 00000000
00041770  00000000 00000000
00041778  00000000 00000000
00041780  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00041788  00000000 00000000
00041790  00000000 00000000
00041798  00000000 00000000
000417A0  00000000 00000000
000417A8  00000000 00000000
000417B0  00000000 00000000
000417B8  00000000 00000000
000417C0  00000000 00000000
000417C8  00000000 00000000
000417D0  00000000 00000000
000417D8  00000000 00000000
000417E0  00000000 00000000
000417E8  00000000 00000000
000417F0  00000000 00000000
000417F8  00000000 00000000
00041800  00000000 00000000
00041808  00000000 00000000
00041810  00000000 00000000
00041818  00000000 00000000
00041820  00000000 00000000
00041828  00000000 00000000
00041830  00000000 00000000
00041838  00000000 00000000
                                        6950
                                        6951          TSTLBL PL                   Parameter List
00041840  6E6E6ED7 D3F8F24C             6952+         DC    CL32'>>>PL82<<<'      eye catcher
00041848  4C4C4040 40404040
00041850  40404040 40404040
00041858  40404040 40404040
00041860                                6953+PL82     DS    0C
00041860  00000000 00000000             6954          DC    XL352'00'
00041868  00000000 00000000
00041870  00000000 00000000
00041878  00000000 00000000
00041880  00000000 00000000
00041888  00000000 00000000
00041890  00000000 00000000
00041898  00000000 00000000
000418A0  00000000 00000000
000418A8  00000000 00000000
000418B0  00000000 00000000
000418B8  00000000 00000000
000418C0  00000000 00000000
000418C8  00000000 00000000
000418D0  00000000 00000000
000418D8  00000000 00000000
000418E0  00000000 00000000
000418E8  00000000 00000000
000418F0  00000000 00000000
000418F8  00000000 00000000
00041900  00000000 00000000
00041908  00000000 00000000
00041910  00000000 00000000
00041918  00000000 00000000
00041920  00000000 00000000
00041928  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00041930  00000000 00000000
00041938  00000000 00000000
00041940  00000000 00000000
00041948  00000000 00000000
00041950  00000000 00000000
00041958  00000000 00000000
00041960  00000000 00000000
00041968  00000000 00000000
00041970  00000000 00000000
00041978  00000000 00000000
00041980  00000000 00000000
00041988  00000000 00000000
00041990  00000000 00000000
00041998  00000000 00000000
000419A0  00000000 00000000
000419A8  00000000 00000000
000419B0  00000000 00000000
000419B8  00000000 00000000
                                      6955
                                      6956          TSTLBL FMSG               fail message
000419C0  6E6E6EC6 D4E2C7F8           6957+         DC    CL32'>>>FMSG82<<<'  eye catcher
000419C8  F24C4C4C 40404040
000419D0  40404040 40404040
000419D8  40404040 40404040
000419E0                              6958+FMSG82   DS    0C
000419E0  86818993 858440D7           6959          DC    C'failed PLO-Extension '
000419E8  D3D660C5 A7A38595
000419F0  A2899695 40
000419F5  E3998997 938540C3           6960          DC    CL96'Triple Compare and Swap (TCSG: 31): 3rd NE'
000419FD  96949781 99854081
00041A05  958440E2 A6819740
00041A0D  4DE3C3E2 C77A40F3
00041A15  F15D7A40 F3998440
00041A1D  D5C54040 40404040
00041A25  40404040 40404040
00041A2D  40404040 40404040
00041A35  40404040 40404040
00041A3D  40404040 40404040
00041A45  40404040 40404040
00041A4D  40404040 40404040
                                      6961
                                      6962          TSTEND
00041A55  C5D5C440 968640A3           6963+         DC    CL32'END of test: 82 <<<'
00041A5D  85A2A37A 40F8F240
00041A65  4C4C4C40 40404040
00041A6D  40404040 40404040
                                      6964
                                      6965 * ------------------------------------------------------------------
                                      6966 * Test Triple Compare and Swap (TCSG: 31) instruction : All EQ
                                      6967 * ------------------------------------------------------------------
                                      6968          TSTHDR 0,99,31            All Operands Equal (cc=0, UR=99)
00041A75                 00041A75  00041C00  6969+         ORG   PLOETST+(128*K)+45*(3*K)
00041C00  E2E3C1D9 E3409686           6970+         DC    CL32'START of test: 83'
00041C08  40A385A2 A37A40F8
00041C10  F3404040 40404040
00041C18  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00041C20                     00041C20              6971+          USING *,R5                base for test data and test routine
00041C20                     00000000              6972+          USING PLOETEST,R5
00041C20  00041C4C                                 6973+T83       DC    A(CODE83)           address of test code
00041C24  0053                                     6974+          DC    H'83'               test number
00041C26  00                                       6975+          DC    X'00'
00041C27  1F                                       6976+          DC    HL1'31'             Function
00041C28  63                                       6977+          DC    HL1'99'             expected unequal results (UR)
00041C29  00                                       6978+          DC    HL1'00'             found UR
00041C2A  00                                       6979+          DC    HL1'0'              cc
00041C2B  07                                       6980+          DC    HL1'7'              not expected CC mask
00041C2C  00                                       6981+          DS    X                   extracted cc
00041C30  00000000 00000000                        6982+          DS    2F                  extract PSW after test (has CC)
00041C38  000425E0                                 6983+          DC    A(FMSG83)           address of test fail message
00041C3C  00042340                                 6984+          DC    A(OPSRC83)          address of operand source
00041C40  00042220                                 6985+          DC    A(OPDST83)          address of operand destination
00041C44  00042460                                 6986+          DC    A(PL83)             address of parameter list
00041C48  0000420B                                 6987+          DC    A(RANDX+83)         address of random source data
                                                   6988
                                                   6989           TSTCODE
00041C4C                                           6990+CODE83    DS    0F
                                                   6991 *                                   load random source
00041C4C  E310 501C 0014               0000001C    6992           LGF   R1,OPSRCA
00041C52  E320 5028 0014               00000028    6993           LGF   R2,RANDXA
00041C58  E330 5024 0014               00000024    6994           LGF   R3,PLA
00041C5E  E340 5020 0014               00000020    6995           LGF   R4,OPREPA
                                                   6996 *
                                                   6997 * Triple Compare and Swap (TCSG: 31)
                                                   6998 * Build Parambter List
                                                   6999 *
00041C64  D217 1000 2000      00000000 00000000    7000           MVC   0(24,R1),0(R2)      load Op1-Op3 data in source
00041C6A  D217 3040 1000      00000040 00000000    7001           MVC   64(24,R3),0(R1)     copy to PL OP1-Op3 Compare
                                                   7002
00041C70  D217 4000 2080      00000000 00000080    7003           MVC   0(24,R4),128(R2)    load initial Op1-Op3 data
00041C76  D217 3058 4000      00000058 00000000    7004           MVC   88(24,R3),0(R4)     copy to PL Op1-Op3 Replacement
                                                   7005 ****
                                                   7006 * do not force a failure by changing the 3rd operand
                                                   7007 ****
                                                   7008 *          MVC   16(4,R1),0(R4)
                                                   7009 ****
                                                   7010
00041C7C  E313 0000 0024               00000000    7011           STG   R1,0(R3)            load OP1 address in PL
                                                   7012
00041C82  4111 0008                    00000008    7013           LA    R1,8(R1)
00041C86  E313 0008 0024               00000008    7014           STG   R1,8(R3)            load OP2 address in PL
                                                   7015
00041C8C  4111 0008                    00000008    7016           LA    R1,8(R1)
00041C90  E313 0010 0024               00000010    7017           STG   R1,16(R3)           load OP3 address in PL
                                                   7018
00041C96  B982 0000                                7019           XGR   R0,R0
00041C9A  4100 001F                    0000001F    7020           LA    R0,31
00041C9E  EE00 0000 3000      00000000 00000000    7021           PLO   0,0,0,0(R3)         Triple C/S (TCSG: 31)
                                                   7022
00041CA4  B98D 0020                                7023           EPSW  R2,R0               extract psw
00041CA8  5020 5010                    00000010    7024           ST    R2,CCPSW              to save CC
00041CAC  E310 500B 0076               0000000B    7025           LB    R1,CCMASK           (failure CC mask)
00041CB2  8910 0004                    00000004    7026           SLL   R1,4                (shift to BC instr CC position)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00041CB6  4410 50D6                     00041CF6  7027          EX    R1,F314TSTCC            fail if...
                                                  7028 *
                                                  7029 * compare results
                                                  7030 *
                          00041CBA  00000001      7031 F314CKRES EQU   *
                                                  7032 *                                reset adresses
00041CBA  E310 501C 0014                0000001C  7033          LGF   R1,OPSRCA
00041CC0  E320 5028 0014                00000028  7034          LGF   R2,RANDXA
00041CC6  E330 5024 0014                00000024  7035          LGF   R3,PLA
00041CCC  E340 5020 0014                00000020  7036          LGF   R4,OPREPA
                                                  7037
                                                  7038 * All equal; replacement values copied to operand source
                                                  7039
00041CD2  D507 1000 3058  00000000      00000058  7040          CLC   0(8,R1),88(R3)
00041CD8  4770 50D0                     00041CF0  7041          BNE   F314XX              OP1R Different, error
00041CDC  D507 1008 3060  00000008      00000060  7042          CLC   8(8,R1),96(R3)
00041CE2  4770 50D0                     00041CF0  7043          BNE   F314XX              OP2R Different, error
00041CE6  D507 1010 3068  00000010      00000068  7044          CLC   16(8,R1),104(R3)
00041CEC  478B 0000                     00000000  7045          BE    0(R11)
                                                  7046
                          00041CF0      00000001  7047 F314XX   EQU   *
00041CF0  45F0 83E8                     000023E8  7048          BAL   R15,FAILMSG
00041CF4  07FB                                    7049          BR    R11
                                                  7050
00041CF6  4700 50DA                     00041CFA  7051 F314TSTCC BC    0,F314CCMSG         fail if unexpected condition code
                          00041CFA      00000001  7052 F314CCMSG EQU   *
00041CFA  45F0 8196                     00002196  7053          BAL   R15,CCMSG
00041CFE  47F0 509A                     00041CBA  7054          B     F314CKRES
                                                  7055
                                                  7056          TSTDATA
00041D02                  00041D02      00042200  7057+          ORG   PLOETST+(128*K)+45*(3*K)+(K+512)
                                                  7058          TSTLBL OPDST              Dest/Data: Op1-R, Op2-R, Op3-R
00042200  6E6E6ED6 D7C4E2E3                      7059+          DC    CL32'>>>OPDST83<<<' eye catcher
00042208  F8F34C4C 4C404040
00042210  40404040 40404040
00042218  40404040 40404040
00042220                                  7060+OPDST83  DS    0C
00042220  00000000 00000000               7061          DC    XL256'0'
00042228  00000000 00000000
00042230  00000000 00000000
00042238  00000000 00000000
00042240  00000000 00000000
00042248  00000000 00000000
00042250  00000000 00000000
00042258  00000000 00000000
00042260  00000000 00000000
00042268  00000000 00000000
00042270  00000000 00000000
00042278  00000000 00000000
00042280  00000000 00000000
00042288  00000000 00000000
00042290  00000000 00000000
00042298  00000000 00000000
000422A0  00000000 00000000
000422A8  00000000 00000000
000422B0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
000422B8  00000000 00000000
000422C0  00000000 00000000
000422C8  00000000 00000000
000422D0  00000000 00000000
000422D8  00000000 00000000
000422E0  00000000 00000000
000422E8  00000000 00000000
000422F0  00000000 00000000
000422F8  00000000 00000000
00042300  00000000 00000000
00042308  00000000 00000000
00042310  00000000 00000000
00042318  00000000 00000000
                                          7062
                                          7063          TSTLBL OPSRC              Source: OP1-OP3
00042320  6E6E6ED6 D7E2D9C3               7064+         DC    CL32'>>>OPSRC83<<<' eye catcher
00042328  F8F34C4C 4C404040
00042330  40404040 40404040
00042338  40404040 40404040
00042340                                  7065+OPSRC83  DS    0C
00042340  00000000 00000000               7066          DC    XL256'0'
00042348  00000000 00000000
00042350  00000000 00000000
00042358  00000000 00000000
00042360  00000000 00000000
00042368  00000000 00000000
00042370  00000000 00000000
00042378  00000000 00000000
00042380  00000000 00000000
00042388  00000000 00000000
00042390  00000000 00000000
00042398  00000000 00000000
000423A0  00000000 00000000
000423A8  00000000 00000000
000423B0  00000000 00000000
000423B8  00000000 00000000
000423C0  00000000 00000000
000423C8  00000000 00000000
000423D0  00000000 00000000
000423D8  00000000 00000000
000423E0  00000000 00000000
000423E8  00000000 00000000
000423F0  00000000 00000000
000423F8  00000000 00000000
00042400  00000000 00000000
00042408  00000000 00000000
00042410  00000000 00000000
00042418  00000000 00000000
00042420  00000000 00000000
00042428  00000000 00000000
00042430  00000000 00000000
00042438  00000000 00000000
                                          7067
                                          7068          TSTLBL PL                 Parameter List
00042440  6E6E6ED7 D3F8F34C               7069+         DC    CL32'>>>PL83<<<'    eye catcher
00042448  4C4C4040 40404040
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00042450  40404040 40404040
00042458  40404040 40404040
00042460                                    7070+PL83     DS    0C
00042460  00000000 00000000                 7071          DC    XL352'00'
00042468  00000000 00000000
00042470  00000000 00000000
00042478  00000000 00000000
00042480  00000000 00000000
00042488  00000000 00000000
00042490  00000000 00000000
00042498  00000000 00000000
000424A0  00000000 00000000
000424A8  00000000 00000000
000424B0  00000000 00000000
000424B8  00000000 00000000
000424C0  00000000 00000000
000424C8  00000000 00000000
000424D0  00000000 00000000
000424D8  00000000 00000000
000424E0  00000000 00000000
000424E8  00000000 00000000
000424F0  00000000 00000000
000424F8  00000000 00000000
00042500  00000000 00000000
00042508  00000000 00000000
00042510  00000000 00000000
00042518  00000000 00000000
00042520  00000000 00000000
00042528  00000000 00000000
00042530  00000000 00000000
00042538  00000000 00000000
00042540  00000000 00000000
00042548  00000000 00000000
00042550  00000000 00000000
00042558  00000000 00000000
00042560  00000000 00000000
00042568  00000000 00000000
00042570  00000000 00000000
00042578  00000000 00000000
00042580  00000000 00000000
00042588  00000000 00000000
00042590  00000000 00000000
00042598  00000000 00000000
000425A0  00000000 00000000
000425A8  00000000 00000000
000425B0  00000000 00000000
000425B8  00000000 00000000
                                            7072
                                            7073          TSTLBL FMSG               fail message
000425C0  6E6E6EC6 D4E2C7F8                 7074+         DC    CL32'>>>FMSG83<<<'  eye catcher
000425C8  F34C4C4C 40404040
000425D0  40404040 40404040
000425D8  40404040 40404040
000425E0                                    7075+FMSG83   DS    0C
000425E0  86818993 858440D7                 7076          DC    C'failed PLO-Extension '
000425E8  D3D660C5 A7A38595
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000425F0  A2899695 40
000425F5  E3998997 938540C3                         7077          DC    CL96'Triple Compare and Swap (TCSG: 31): All EQ'
000425FD  96949781 99854081
00042605  958440E2 A6819740
0004260D  4DE3C3E2 C77A40F3
00042615  F15D7A40 C1939340
0004261D  C5D84040 40404040
00042625  40404040 40404040
0004262D  40404040 40404040
00042635  40404040 40404040
0004263D  40404040 40404040
00042645  40404040 40404040
0004264D  40404040 40404040
                                                    7078
                                                    7079          TSTEND
00042655  C5D5C440 968640A3                         7080+         DC    CL32'END of test: 83 <<<'
0004265D  85A2A37A 40F8F340
00042665  4C4C4C40 40404040
0004266D  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   7082 *=====================================================================
                                                   7083 *       Triple Compare and Swap Tests
                                                   7084 *=====================================================================
                                                   7085
                                                   7086 * ---------------------------------------------------------------------
                                                   7087 * Test Triple Compare and Swap (TCS: 30) instruction : 1st NE
                                                   7088 * ---------------------------------------------------------------------
                                                   7089          TSTHDR 2,1,30             1st Operand NOT Equal (cc=2, UR=1)
00042675                          00042675  00042800 7090+         ORG   PLOETST+(128*K)+46*(3*K)
00042800  E2E3C1D9 E3409686                        7091+         DC    CL32'START of test: 84'
00042808  40A385A2 A37A40F8
00042810  F4404040 40404040
00042818  40404040 40404040
00042820                          00042820          7092+         USING *,R5                base for test data and test routine
00042820                          00000000          7093+         USING PLOETEST,R5
00042820  0004284C                                  7094+T84      DC    A(CODE84)           address of test code
00042824  0054                                      7095+         DC    H'84'               test number
00042826  00                                        7096+         DC    X'00'
00042827  1E                                        7097+         DC    HL1'30'             Function
00042828  01                                        7098+         DC    HL1'1'              expected unequal results (UR)
00042829  00                                        7099+         DC    HL1'00'             found UR
0004282A  02                                        7100+         DC    HL1'2'              cc
0004282B  0D                                        7101+         DC    HL1'13'             not expected CC mask
0004282C  00                                        7102+         DS    X                   extracted cc
00042830  00000000 00000000                         7103+         DS    2F                  extract PSW after test (has CC)
00042838  000431E0                                  7104+         DC    A(FMSG84)           address of test fail message
0004283C  00042F40                                  7105+         DC    A(OPSRC84)          address of operand source
00042840  00042E20                                  7106+         DC    A(OPDST84)          address of operand destination
00042844  00043060                                  7107+         DC    A(PL84)             address of parameter list
00042848  0000420C                                  7108+         DC    A(RANDX+84)         address of random source data
                                                    7109
                                                    7110          TSTCODE
0004284C                                            7111+CODE84   DS    0F
                                                    7112 *                                  load random source
0004284C  E310 501C 0014                 0000001C   7113          LGF   R1,OPSRCA
00042852  E320 5028 0014                 00000028   7114          LGF   R2,RANDXA
00042858  E330 5024 0014                 00000024   7115          LGF   R3,PLA
0004285E  E340 5020 0014                 00000020   7116          LGF   R4,OPREPA
                                                    7117 *
                                                    7118 * Triple Compare and Swap (TCS: 30)
                                                    7119 * Build Parambter List
                                                    7120 *
00042864  D20B 1000 2000       00000000  00000000   7121          MVC   0(12,R1),0(R2)      load Op1-Op3 data in source
0004286A  D203 3040 1000       00000040  00000000   7122          MVC   64(4,R3),0(R1)      copy to PL OP1-Compare
00042870  D203 3048 1004       00000048  00000004   7123          MVC   72(4,R3),4(R1)      copy to PL OP2-Compare
00042876  D203 3050 1008       00000050  00000008   7124          MVC   80(4,R3),8(R1)      copy to PL OP3-Compare
                                                    7125
0004287C  D20B 4000 2080       00000000  00000080   7126          MVC   0(12,R4),128(R2)    load initial Op1-Op3 data
00042882  D203 3058 4000       00000058  00000000   7127          MVC   88(4,R3),0(R4)      copy to PL OP1-Replacement
00042888  D203 3060 4004       00000060  00000004   7128          MVC   96(4,R3),4(R4)      copy to PL OP2-Replacement
0004288E  D203 3068 4008       00000068  00000008   7129          MVC   104(4,R3),8(R4)     copy to PL OP3-Replacement
                                                    7130
                                                    7131 ****
                                                    7132 * force a failure by changing the 1st operand
                                                    7133 ****
00042894  D203 1000 4000       00000000  00000000   7134          MVC   0(4,R1),0(R4)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   7135 ****
                                                   7136
0004289A  E313 0000 0024                 00000000  7137          STG   R1,0(R3)           load OP1 address in PL
                                                   7138
000428A0  4111 0004                      00000004  7139          LA    R1,4(R1)
000428A4  E313 0008 0024                 00000008  7140          STG   R1,8(R3)           load OP2 address in PL
                                                   7141
000428AA  4111 0004                      00000004  7142          LA    R1,4(R1)
000428AE  E313 0010 0024                 00000010  7143          STG   R1,16(R3)          load OP3 address in PL
                                                   7144
000428B4  B982 0000                                7145          XGR   R0,R0
000428B8  4100 001E                      0000001E  7146          LA    R0,30
000428BC  EE00 0000 3000       00000000  00000000  7147          PLO   0,0,0,0(R3)        Triple C/S (TCS: 30)
                                                   7148
000428C2  B98D 0020                                7149          EPSW  R2,R0              extract psw
000428C6  5020 5010                      00000010  7150          ST    R2,CCPSW             to save CC
000428CA  E310 500B 0076                 0000000B  7151          LB    R1,CCMASK          (failure CC mask)
000428D0  8910 0004                      00000004  7152          SLL   R1,4               (shift to BC instr CC position)
000428D4  4410 5108                      00042928  7153          EX    R1,F301TSTCC         fail if...
                                                   7154
                                000428D8  00000001  7155 F301CKUR EQU   *
000428D8  D200 5009 3030       00000009  00000030  7156          MVC   URFOUND,48(R3)     Extract UR
000428DE  D500 5008 3030       00000008  00000030  7157          CLC   UR,48(R3)
000428E4  4780 50CC                      000428EC  7158          BE    F301CKRES
000428E8  45F0 82C8                      000022C8  7159          BAL   R15,URMSG
                                                   7160 *
                                                   7161 * compare results
                                                   7162 *
                                000428EC  00000001  7163 F301CKRES EQU   *
                                                   7164 *                                reset adresses
000428EC  E310 501C 0014                 0000001C  7165          LGF   R1,OPSRCA
000428F2  E320 5028 0014                 00000028  7166          LGF   R2,RANDXA
000428F8  E330 5024 0014                 00000024  7167          LGF   R3,PLA
000428FE  E340 5020 0014                 00000020  7168          LGF   R4,OPREPA
                                                   7169
00042904  D503 1000 3040       00000000  00000040  7170          CLC   0(4,R1),64(R3)
0004290A  4770 5102                      00042922  7171          BNE   F301XX             OP1C Different, error
0004290E  D503 1004 3048       00000004  00000048  7172          CLC   4(4,R1),72(R3)
00042914  4770 5102                      00042922  7173          BNE   F301XX             OP2C Different, error
00042918  D503 1008 3050       00000008  00000050  7174          CLC   8(4,R1),80(R3)
0004291E  478B 0000                      00000000  7175          BE    0(R11)             OP3C same, return
                                                   7176
                                00042922  00000001  7177 F301XX   EQU   *
00042922  45F0 83E8                      000023E8  7178          BAL   R15,FAILMSG
00042926  07FB                                     7179          BR    R11
                                                   7180
00042928  4700 510C                      0004292C  7181 F301TSTCC BC    0,F301CCMSG        fail if unexpected condition code
                                0004292C  00000001  7182 F301CCMSG EQU   *
0004292C  45F0 8196                      00002196  7183          BAL   R15,CCMSG
00042930  47F0 50B8                      000428D8  7184          B     F301CKUR
                                                   7185
                                                   7186          TSTDATA
00042934                        00042934  00042E00  7187+         ORG   PLOETST+(128*K)+46*(3*K)+(K+512)
                                                   7188          TSTLBL OPDST             Dest/Data: Op1-R, Op2-R, Op3-R
00042E00  6E6E6ED6 D7C4E2E3                       7189+         DC    CL32'>>>OPDST84<<<' eye catcher
00042E08  F8F44C4C 4C404040
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00042E10  40404040 40404040
00042E18  40404040 40404040
00042E20                                             7190+OPDST84  DS    0C
00042E20  00000000 00000000                          7191          DC    XL256'0'
00042E28  00000000 00000000
00042E30  00000000 00000000
00042E38  00000000 00000000
00042E40  00000000 00000000
00042E48  00000000 00000000
00042E50  00000000 00000000
00042E58  00000000 00000000
00042E60  00000000 00000000
00042E68  00000000 00000000
00042E70  00000000 00000000
00042E78  00000000 00000000
00042E80  00000000 00000000
00042E88  00000000 00000000
00042E90  00000000 00000000
00042E98  00000000 00000000
00042EA0  00000000 00000000
00042EA8  00000000 00000000
00042EB0  00000000 00000000
00042EB8  00000000 00000000
00042EC0  00000000 00000000
00042EC8  00000000 00000000
00042ED0  00000000 00000000
00042ED8  00000000 00000000
00042EE0  00000000 00000000
00042EE8  00000000 00000000
00042EF0  00000000 00000000
00042EF8  00000000 00000000
00042F00  00000000 00000000
00042F08  00000000 00000000
00042F10  00000000 00000000
00042F18  00000000 00000000
                                                     7192
                                                     7193          TSTLBL OPSRC              Source: OP1-OP3
00042F20  6E6E6ED6 D7E2D9C3                          7194+         DC    CL32'>>>OPSRC84<<<' eye catcher
00042F28  F8F44C4C 4C404040
00042F30  40404040 40404040
00042F38  40404040 40404040
00042F40                                             7195+OPSRC84  DS    0C
00042F40  00000000 00000000                          7196          DC    XL256'0'
00042F48  00000000 00000000
00042F50  00000000 00000000
00042F58  00000000 00000000
00042F60  00000000 00000000
00042F68  00000000 00000000
00042F70  00000000 00000000
00042F78  00000000 00000000
00042F80  00000000 00000000
00042F88  00000000 00000000
00042F90  00000000 00000000
00042F98  00000000 00000000
00042FA0  00000000 00000000
00042FA8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00042FB0  00000000 00000000
00042FB8  00000000 00000000
00042FC0  00000000 00000000
00042FC8  00000000 00000000
00042FD0  00000000 00000000
00042FD8  00000000 00000000
00042FE0  00000000 00000000
00042FE8  00000000 00000000
00042FF0  00000000 00000000
00042FF8  00000000 00000000
00043000  00000000 00000000
00043008  00000000 00000000
00043010  00000000 00000000
00043018  00000000 00000000
00043020  00000000 00000000
00043028  00000000 00000000
00043030  00000000 00000000
00043038  00000000 00000000
                                        7197
                                        7198          TSTLBL PL                 Parameter List
00043040  6E6E6ED7 D3F8F44C             7199+         DC    CL32'>>>PL84<<<'    eye catcher
00043048  4C4C4040 40404040
00043050  40404040 40404040
00043058  40404040 40404040
00043060                                7200+PL84     DS    0C
00043060  00000000 00000000             7201          DC    XL352'00'
00043068  00000000 00000000
00043070  00000000 00000000
00043078  00000000 00000000
00043080  00000000 00000000
00043088  00000000 00000000
00043090  00000000 00000000
00043098  00000000 00000000
000430A0  00000000 00000000
000430A8  00000000 00000000
000430B0  00000000 00000000
000430B8  00000000 00000000
000430C0  00000000 00000000
000430C8  00000000 00000000
000430D0  00000000 00000000
000430D8  00000000 00000000
000430E0  00000000 00000000
000430E8  00000000 00000000
000430F0  00000000 00000000
000430F8  00000000 00000000
00043100  00000000 00000000
00043108  00000000 00000000
00043110  00000000 00000000
00043118  00000000 00000000
00043120  00000000 00000000
00043128  00000000 00000000
00043130  00000000 00000000
00043138  00000000 00000000
00043140  00000000 00000000
00043148  00000000 00000000
00043150  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00043158  00000000 00000000
00043160  00000000 00000000
00043168  00000000 00000000
00043170  00000000 00000000
00043178  00000000 00000000
00043180  00000000 00000000
00043188  00000000 00000000
00043190  00000000 00000000
00043198  00000000 00000000
000431A0  00000000 00000000
000431A8  00000000 00000000
000431B0  00000000 00000000
000431B8  00000000 00000000
                                        7202
                                        7203          TSTLBL FMSG                fail message
000431C0  6E6E6EC6 D4E2C7F8             7204+         DC    CL32'>>>FMSG84<<<'   eye catcher
000431C8  F44C4C4C 40404040
000431D0  40404040 40404040
000431D8  40404040 40404040
000431E0                                7205+FMSG84   DS    0C
000431E0  86818993 858440D7             7206          DC    C'failed PLO-Extension '
000431E8  D3D660C5 A7A38595
000431F0  A2899695 40
000431F5  E3998997 938540C3             7207          DC    CL96'Triple Compare and Swap (TCS: 30): 1st NE'
000431FD  96949781 99854081
00043205  958440E2 A6819740
0004320D  4DE3C3E2 7A40F3F0
00043215  5D7A40F1 A2A340D5
0004321D  C5404040 40404040
00043225  40404040 40404040
0004322D  40404040 40404040
00043235  40404040 40404040
0004323D  40404040 40404040
00043245  40404040 40404040
0004324D  40404040 40404040
                                        7208          TSTEND
00043255  C5D5C440 968640A3             7209+         DC    CL32'END of test: 84 <<<'
0004325D  85A2A37A 40F8F440
00043265  4C4C4C40 40404040
0004326D  40404040 40404040
                                        7210
                                        7211 * ------------------------------------------------------------------
                                        7212 * Test Triple Compare and Swap (TCS: 30) instruction : 2nd NE
                                        7213 * ------------------------------------------------------------------
                                        7214          TSTHDR 2,2,30              2nd Operand NOT Equal (cc=2, UR=2)
00043275                  00043275  00043400  7215+         ORG   PLOETST+(128*K)+47*(3*K)
00043400  E2E3C1D9 E3409686             7216+         DC    CL32'START of test: 85'
00043408  40A385A2 A37A40F8
00043410  F5404040 40404040
00043418  40404040 40404040
00043420                  00043420          7217+         USING *,R5                 base for test data and test routine
00043420                  00000000          7218+         USING PLOETEST,R5
00043420  0004344C                        7219+T85      DC    A(CODE85)            address of test code
00043424  0055                            7220+         DC    H'85'                test number
00043426  00                              7221+         DC    X'00'
00043427  1E                              7222+         DC    HL1'30'              Function
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00043428  02                                       7223+          DC    HL1'2'              expected unequal results (UR)
00043429  00                                       7224+          DC    HL1'00'             found UR
0004342A  02                                       7225+          DC    HL1'2'              cc
0004342B  0D                                       7226+          DC    HL1'13'             not expected CC mask
0004342C  00                                       7227+          DS    X                   extracted cc
00043430  00000000 00000000                        7228+          DS    2F                  extract PSW after test (has CC)
00043438  00043DE0                                 7229+          DC    A(FMSG85)           address of test fail message
0004343C  00043B40                                 7230+          DC    A(OPSRC85)          address of operand source
00043440  00043A20                                 7231+          DC    A(OPDST85)          address of operand destination
00043444  00043C60                                 7232+          DC    A(PL85)             address of parameter list
00043448  0000420D                                 7233+          DC    A(RANDX+85)         address of random source data
                                                   7234
                                                   7235           TSTCODE
0004344C                                           7236+CODE85    DS    0F
                                                   7237 *                                   load random source
0004344C  E310 501C 0014                0000001C   7238           LGF   R1,OPSRCA
00043452  E320 5028 0014                00000028   7239           LGF   R2,RANDXA
00043458  E330 5024 0014                00000024   7240           LGF   R3,PLA
0004345E  E340 5020 0014                00000020   7241           LGF   R4,OPREPA
                                                   7242 *
                                                   7243 * Triple Compare and Swap (TCS: 30)
                                                   7244 * Build Parambter List
                                                   7245 *
00043464  D20B 1000 2000       00000000 00000000   7246           MVC   0(12,R1),0(R2)      load Op1-Op3 data in source
0004346A  D203 3040 1000       00000040 00000000   7247           MVC   64(4,R3),0(R1)      copy to PL OP1-Compare
00043470  D203 3048 1004       00000048 00000004   7248           MVC   72(4,R3),4(R1)      copy to PL OP2-Compare
00043476  D203 3050 1008       00000050 00000008   7249           MVC   80(4,R3),8(R1)      copy to PL OP3-Compare
                                                   7250
0004347C  D20B 4000 2080       00000000 00000080   7251           MVC   0(12,R4),128(R2)    load initial Op1-Op3 data
00043482  D203 3058 4000       00000058 00000000   7252           MVC   88(4,R3),0(R4)      copy to PL OP1-Replacement
00043488  D203 3060 4004       00000060 00000004   7253           MVC   96(4,R3),4(R4)      copy to PL OP2-Replacement
0004348E  D203 3068 4008       00000068 00000008   7254           MVC   104(4,R3),8(R4)     copy to PL OP3-Replacement
                                                   7255
                                                   7256 ****
                                                   7257 * force a failure by changing the 2nd operand
                                                   7258 ****
00043494  D203 1004 4000       00000004 00000000   7259           MVC   4(4,R1),0(R4)
                                                   7260 ****
                                                   7261
0004349A  E313 0000 0024                00000000   7262           STG   R1,0(R3)            load OP1 address in PL
                                                   7263
000434A0  4111 0004                     00000004   7264           LA    R1,4(R1)
000434A4  E313 0008 0024                00000008   7265           STG   R1,8(R3)            load OP2 address in PL
                                                   7266
000434AA  4111 0004                     00000004   7267           LA    R1,4(R1)
000434AE  E313 0010 0024                00000010   7268           STG   R1,16(R3)           load OP3 address in PL
                                                   7269
000434B4  B982 0000                                7270           XGR   R0,R0
000434B8  4100 001E                     0000001E   7271           LA    R0,30
000434BC  EE00 0000 3000       00000000 00000000   7272           PLO   0,0,0,0(R3)         Triple C/S (TCS: 30)
                                                   7273
000434C2  B98D 0020                                7274           EPSW  R2,R0               extract psw
000434C6  5020 5010                     00000010   7275           ST    R2,CCPSW              to save CC
000434CA  E310 500B 0076                0000000B   7276           LB    R1,CCMASK           (failure CC mask)
000434D0  8910 0004                     00000004   7277           SLL   R1,4                (shift to BC instr CC position)
000434D4  4410 5108                     00043528   7278           EX    R1,F302TSTCC            fail if...
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    7279
                               000434D8  00000001  7280 F302CKUR EQU   *
000434D8  D200 5009 3030       00000009  00000030  7281          MVC   URFOUND,48(R3)      Extract UR
000434DE  D500 5008 3030       00000008  00000030  7282          CLC   UR,48(R3)
000434E4  4780 50CC                      000434EC  7283          BE    F302CKRES
000434E8  45F0 82C8                      000022C8  7284          BAL   R15,URMSG
                                                    7285 *
                                                    7286 * compare results
                                                    7287 *
                               000434EC  00000001  7288 F302CKRES EQU   *
                                                    7289 *                               reset adresses
000434EC  E310 501C 0014                 0000001C  7290          LGF   R1,OPSRCA
000434F2  E320 5028 0014                 00000028  7291          LGF   R2,RANDXA
000434F8  E330 5024 0014                 00000024  7292          LGF   R3,PLA
000434FE  E340 5020 0014                 00000020  7293          LGF   R4,OPREPA
                                                    7294
00043504  D503 1000 3040       00000000  00000040  7295          CLC   0(4,R1),64(R3)
0004350A  4770 5102                      00043522  7296          BNE   F302XX             OP1C Different, error
0004350E  D503 1004 3048       00000004  00000048  7297          CLC   4(4,R1),72(R3)
00043514  4770 5102                      00043522  7298          BNE   F302XX             OP2C Different, error
00043518  D503 1008 3050       00000008  00000050  7299          CLC   8(4,R1),80(R3)
0004351E  478B 0000                      00000000  7300          BE    0(R11)             OP3C same, return
                                                    7301
                               00043522  00000001  7302 F302XX   EQU   *
00043522  45F0 83E8                      000023E8  7303          BAL   R15,FAILMSG
00043526  07FB                                     7304          BR    R11
                                                    7305
00043528  4700 510C                      0004352C  7306 F302TSTCC BC    0,F302CCMSG        fail if unexpected condition code
                               0004352C  00000001  7307 F302CCMSG EQU   *
0004352C  45F0 8196                      00002196  7308          BAL   R15,CCMSG
00043530  47F0 50B8                      000434D8  7309          B     F302CKUR
                                                    7310
                                                    7311          TSTDATA
00043534                       00043534  00043A00  7312+         ORG   PLOETST+(128*K)+47*(3*K)+(K+512)
                                                    7313          TSTLBL OPDST             Dest/Data: Op1-R, Op2-R, Op3-R
00043A00  6E6E6ED6 D7C4E2E3                        7314+         DC    CL32'>>>OPDST85<<<' eye catcher
00043A08  F8F54C4C 4C404040
00043A10  40404040 40404040
00043A18  40404040 40404040
00043A20                                           7315+OPDST85  DS    0C
00043A20  00000000 00000000                        7316          DC    XL256'0'
00043A28  00000000 00000000
00043A30  00000000 00000000
00043A38  00000000 00000000
00043A40  00000000 00000000
00043A48  00000000 00000000
00043A50  00000000 00000000
00043A58  00000000 00000000
00043A60  00000000 00000000
00043A68  00000000 00000000
00043A70  00000000 00000000
00043A78  00000000 00000000
00043A80  00000000 00000000
00043A88  00000000 00000000
00043A90  00000000 00000000
00043A98  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00043AA0  00000000 00000000
00043AA8  00000000 00000000
00043AB0  00000000 00000000
00043AB8  00000000 00000000
00043AC0  00000000 00000000
00043AC8  00000000 00000000
00043AD0  00000000 00000000
00043AD8  00000000 00000000
00043AE0  00000000 00000000
00043AE8  00000000 00000000
00043AF0  00000000 00000000
00043AF8  00000000 00000000
00043B00  00000000 00000000
00043B08  00000000 00000000
00043B10  00000000 00000000
00043B18  00000000 00000000
                                          7317
                                          7318          TSTLBL OPSRC              Source: OP1-OP3
00043B20  6E6E6ED6 D7E2D9C3               7319+         DC    CL32'>>>OPSRC85<<<' eye catcher
00043B28  F8F54C4C 4C404040
00043B30  40404040 40404040
00043B38  40404040 40404040
00043B40                                  7320+OPSRC85  DS    0C
00043B40  00000000 00000000               7321          DC    XL256'0'
00043B48  00000000 00000000
00043B50  00000000 00000000
00043B58  00000000 00000000
00043B60  00000000 00000000
00043B68  00000000 00000000
00043B70  00000000 00000000
00043B78  00000000 00000000
00043B80  00000000 00000000
00043B88  00000000 00000000
00043B90  00000000 00000000
00043B98  00000000 00000000
00043BA0  00000000 00000000
00043BA8  00000000 00000000
00043BB0  00000000 00000000
00043BB8  00000000 00000000
00043BC0  00000000 00000000
00043BC8  00000000 00000000
00043BD0  00000000 00000000
00043BD8  00000000 00000000
00043BE0  00000000 00000000
00043BE8  00000000 00000000
00043BF0  00000000 00000000
00043BF8  00000000 00000000
00043C00  00000000 00000000
00043C08  00000000 00000000
00043C10  00000000 00000000
00043C18  00000000 00000000
00043C20  00000000 00000000
00043C28  00000000 00000000
00043C30  00000000 00000000
00043C38  00000000 00000000
                                          7322
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   7323           TSTLBL PL                  Parameter List
00043C40  6E6E6ED7 D3F8F54C                        7324+          DC    CL32'>>>PL85<<<'     eye catcher
00043C48  4C4C4040 40404040
00043C50  40404040 40404040
00043C58  40404040 40404040
00043C60                                           7325+PL85      DS    0C
00043C60  00000000 00000000                        7326           DC    XL352'00'
00043C68  00000000 00000000
00043C70  00000000 00000000
00043C78  00000000 00000000
00043C80  00000000 00000000
00043C88  00000000 00000000
00043C90  00000000 00000000
00043C98  00000000 00000000
00043CA0  00000000 00000000
00043CA8  00000000 00000000
00043CB0  00000000 00000000
00043CB8  00000000 00000000
00043CC0  00000000 00000000
00043CC8  00000000 00000000
00043CD0  00000000 00000000
00043CD8  00000000 00000000
00043CE0  00000000 00000000
00043CE8  00000000 00000000
00043CF0  00000000 00000000
00043CF8  00000000 00000000
00043D00  00000000 00000000
00043D08  00000000 00000000
00043D10  00000000 00000000
00043D18  00000000 00000000
00043D20  00000000 00000000
00043D28  00000000 00000000
00043D30  00000000 00000000
00043D38  00000000 00000000
00043D40  00000000 00000000
00043D48  00000000 00000000
00043D50  00000000 00000000
00043D58  00000000 00000000
00043D60  00000000 00000000
00043D68  00000000 00000000
00043D70  00000000 00000000
00043D78  00000000 00000000
00043D80  00000000 00000000
00043D88  00000000 00000000
00043D90  00000000 00000000
00043D98  00000000 00000000
00043DA0  00000000 00000000
00043DA8  00000000 00000000
00043DB0  00000000 00000000
00043DB8  00000000 00000000
                                                   7327
                                                   7328           TSTLBL FMSG                fail message
00043DC0  6E6E6EC6 D4E2C7F8                        7329+          DC    CL32'>>>FMSG85<<<'   eye catcher
00043DC8  F54C4C4C 40404040
00043DD0  40404040 40404040
00043DD8  40404040 40404040
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00043DE0                                            7330+FMSG85   DS    0C
00043DE0  86818993 858440D7                         7331          DC    C'failed PLO-Extension '
00043DE8  D3D660C5 A7A38595
00043DF0  A2899695 40
00043DF5  E3998997 938540C3                         7332          DC    CL96'Triple Compare and Swap (TCS: 30): 2nd NE'
00043DFD  96949781 99854081
00043E05  958440E2 A6819740
00043E0D  4DE3C3E2 7A40F3F0
00043E15  5D7A40F2 958440D5
00043E1D  C5404040 40404040
00043E25  40404040 40404040
00043E2D  40404040 40404040
00043E35  40404040 40404040
00043E3D  40404040 40404040
00043E45  40404040 40404040
00043E4D  40404040 40404040
                                                    7333          TSTEND
00043E55  C5D5C440 968640A3                         7334+         DC    CL32'END of test: 85 <<<'
00043E5D  85A2A37A 40F8F540
00043E65  4C4C4C40 40404040
00043E6D  40404040 40404040
                                                    7335
                                                    7336 * ------------------------------------------------------------------
                                                    7337 * Test Triple Compare and Swap (TCS: 30) instruction : 3rd NE
                                                    7338 * ------------------------------------------------------------------
                                                    7339          TSTHDR 2,3,30            3rd Operand NOT Equal (cc=2, UR=3)
00043E75                            00043E75  00044000  7340+         ORG   PLOETST+(128*K)+48*(3*K)
00044000  E2E3C1D9 E3409686                         7341+         DC    CL32'START of test: 86'
00044008  40A385A2 A37A40F8
00044010  F6404040 40404040
00044018  40404040 40404040
00044020                            00044020        7342+         USING *,R5               base for test data and test routine
00044020                            00000000        7343+         USING PLOETEST,R5
00044020  0004404C                                  7344+T86      DC    A(CODE86)          address of test code
00044024  0056                                      7345+         DC    H'86'              test number
00044026  00                                        7346+         DC    X'00'
00044027  1E                                        7347+         DC    HL1'30'            Function
00044028  03                                        7348+         DC    HL1'3'             expected unequal results (UR)
00044029  00                                        7349+         DC    HL1'00'            found UR
0004402A  02                                        7350+         DC    HL1'2'             cc
0004402B  0D                                        7351+         DC    HL1'13'            not expected CC mask
0004402C  00                                        7352+         DS    X                  extracted cc
00044030  00000000 00000000                         7353+         DS    2F                 extract PSW after test (has CC)
00044038  000449E0                                  7354+         DC    A(FMSG86)          address of test fail message
0004403C  00044740                                  7355+         DC    A(OPSRC86)         address of operand source
00044040  00044620                                  7356+         DC    A(OPDST86)         address of operand destination
00044044  00044860                                  7357+         DC    A(PL86)            address of parameter list
00044048  0000420E                                  7358+         DC    A(RANDX+86)        address of random source data
                                                    7359
                                                    7360          TSTCODE
0004404C                                            7361+CODE86   DS    0F
                                                    7362 *                                 load random source
0004404C  E310 501C 0014                0000001C    7363          LGF   R1,OPSRCA
00044052  E320 5028 0014                00000028    7364          LGF   R2,RANDXA
00044058  E330 5024 0014                00000024    7365          LGF   R3,PLA
0004405E  E340 5020 0014                00000020    7366          LGF   R4,OPREPA
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   7367 *
                                                   7368 * Triple Compare and Swap (TCS: 30)
                                                   7369 * Build Parambter List
                                                   7370 *
00044064  D20B 1000 2000      00000000  00000000  7371          MVC   0(12,R1),0(R2)      load Op1-Op3 data in source
0004406A  D203 3040 1000      00000040  00000000  7372          MVC   64(4,R3),0(R1)      copy to PL OP1-Compare
00044070  D203 3048 1004      00000048  00000004  7373          MVC   72(4,R3),4(R1)      copy to PL OP2-Compare
00044076  D203 3050 1008      00000050  00000008  7374          MVC   80(4,R3),8(R1)      copy to PL OP3-Compare
                                                   7375
0004407C  D20B 4000 2080      00000000  00000080  7376          MVC   0(12,R4),128(R2)    load initial Op1-Op3 data
00044082  D203 3058 4000      00000058  00000000  7377          MVC   88(4,R3),0(R4)      copy to PL OP1-Replacement
00044088  D203 3060 4004      00000060  00000004  7378          MVC   96(4,R3),4(R4)      copy to PL OP2-Replacement
0004408E  D203 3068 4008      00000068  00000008  7379          MVC   104(4,R3),8(R4)     copy to PL OP3-Replacement
                                                   7380
                                                   7381 ****
                                                   7382 * force a failure by changing the 3rd operand
                                                   7383 ****
00044094  D203 1008 4000      00000008  00000000  7384          MVC   8(4,R1),0(R4)
                                                   7385 ****
                                                   7386
0004409A  E313 0000 0024                00000000  7387          STG   R1,0(R3)            load OP1 address in PL
                                                   7388
000440A0  4111 0004                     00000004  7389          LA    R1,4(R1)
000440A4  E313 0008 0024                00000008  7390          STG   R1,8(R3)            load OP2 address in PL
                                                   7391
000440AA  4111 0004                     00000004  7392          LA    R1,4(R1)
000440AE  E313 0010 0024                00000010  7393          STG   R1,16(R3)           load OP3 address in PL
                                                   7394
000440B4  B982 0000                               7395          XGR   R0,R0
000440B8  4100 001E                     0000001E  7396          LA    R0,30
000440BC  EE00 0000 3000      00000000  00000000  7397          PLO   0,0,0,0(R3)         Triple C/S (TCS: 30)
                                                   7398
000440C2  B98D 0020                               7399          EPSW  R2,R0               extract psw
000440C6  5020 5010                     00000010  7400          ST    R2,CCPSW              to save CC
000440CA  E310 500B 0076                0000000B  7401          LB    R1,CCMASK           (failure CC mask)
000440D0  8910 0004                     00000004  7402          SLL   R1,4                (shift to BC instr CC position)
000440D4  4410 5108                     00044128  7403          EX    R1,F303TSTCC            fail if...
                                                   7404
                                000440D8  00000001  7405 F303CKUR EQU   *
000440D8  D200 5009 3030      00000009  00000030  7406          MVC   URFOUND,48(R3)      Extract UR
000440DE  D500 5008 3030      00000008  00000030  7407          CLC   UR,48(R3)
000440E4  4780 50CC                     000440EC  7408          BE    F303CKRES
000440E8  45F0 82C8                     000022C8  7409          BAL   R15,URMSG
                                                   7410 *
                                                   7411 * compare results
                                                   7412 *
                                000440EC  00000001  7413 F303CKRES EQU   *
                                                   7414 *                            reset adresses
000440EC  E310 501C 0014                0000001C  7415          LGF   R1,OPSRCA
000440F2  E320 5028 0014                00000028  7416          LGF   R2,RANDXA
000440F8  E330 5024 0014                00000024  7417          LGF   R3,PLA
000440FE  E340 5020 0014                00000020  7418          LGF   R4,OPREPA
                                                   7419
00044104  D503 1000 3040      00000000  00000040  7420          CLC   0(4,R1),64(R3)
0004410A  4770 5102                     00044122  7421          BNE   F303XX              OP1C Different, error
0004410E  D503 1004 3048      00000004  00000048  7422          CLC   4(4,R1),72(R3)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00044114  4770 5102                      00044122  7423           BNE    F303XX              OP2C Different, error
00044118  D503 1008 3050      00000008  00000050  7424           CLC    8(4,R1),80(R3)
0004411E  478B 0000                      00000000  7425           BE     0(R11)              OP3C same, return
                                                   7426
                              00044122  00000001  7427 F303XX    EQU    *
00044122  45F0 83E8                      000023E8  7428           BAL    R15,FAILMSG
00044126  07FB                                     7429           BR     R11
                                                   7430
00044128  4700 510C                      0004412C  7431 F303TSTCC BC     0,F303CCMSG         fail if unexpected condition code
                              0004412C  00000001  7432 F303CCMSG EQU    *
0004412C  45F0 8196                      00002196  7433           BAL    R15,CCMSG
00044130  47F0 50B8                      000440D8  7434           B      F303CKUR
                                                   7435
                                                   7436           TSTDATA
00044134                      00044134  00044600  7437+          ORG    PLOETST+(128*K)+48*(3*K)+(K+512)
                                                   7438           TSTLBL OPDST               Dest/Data: Op1-R, Op2-R, Op3-R
00044600  6E6E6ED6 D7C4E2E3                       7439+          DC     CL32'>>>OPDST86<<<' eye catcher
00044608  F8F64C4C 4C404040
00044610  40404040 40404040
00044618  40404040 40404040
00044620                                           7440+OPDST86   DS     0C
00044620  00000000 00000000                        7441           DC     XL256'0'
00044628  00000000 00000000
00044630  00000000 00000000
00044638  00000000 00000000
00044640  00000000 00000000
00044648  00000000 00000000
00044650  00000000 00000000
00044658  00000000 00000000
00044660  00000000 00000000
00044668  00000000 00000000
00044670  00000000 00000000
00044678  00000000 00000000
00044680  00000000 00000000
00044688  00000000 00000000
00044690  00000000 00000000
00044698  00000000 00000000
000446A0  00000000 00000000
000446A8  00000000 00000000
000446B0  00000000 00000000
000446B8  00000000 00000000
000446C0  00000000 00000000
000446C8  00000000 00000000
000446D0  00000000 00000000
000446D8  00000000 00000000
000446E0  00000000 00000000
000446E8  00000000 00000000
000446F0  00000000 00000000
000446F8  00000000 00000000
00044700  00000000 00000000
00044708  00000000 00000000
00044710  00000000 00000000
00044718  00000000 00000000
                                                   7442
                                                   7443           TSTLBL OPSRC               Source: OP1-OP3
00044720  6E6E6ED6 D7E2D9C3                       7444+          DC     CL32'>>>OPSRC86<<<' eye catcher
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00044728  F8F64C4C 4C404040
00044730  40404040 40404040
00044738  40404040 40404040
00044740                                           7445+OPSRC86  DS    0C
00044740  00000000 00000000                        7446          DC    XL256'0'
00044748  00000000 00000000
00044750  00000000 00000000
00044758  00000000 00000000
00044760  00000000 00000000
00044768  00000000 00000000
00044770  00000000 00000000
00044778  00000000 00000000
00044780  00000000 00000000
00044788  00000000 00000000
00044790  00000000 00000000
00044798  00000000 00000000
000447A0  00000000 00000000
000447A8  00000000 00000000
000447B0  00000000 00000000
000447B8  00000000 00000000
000447C0  00000000 00000000
000447C8  00000000 00000000
000447D0  00000000 00000000
000447D8  00000000 00000000
000447E0  00000000 00000000
000447E8  00000000 00000000
000447F0  00000000 00000000
000447F8  00000000 00000000
00044800  00000000 00000000
00044808  00000000 00000000
00044810  00000000 00000000
00044818  00000000 00000000
00044820  00000000 00000000
00044828  00000000 00000000
00044830  00000000 00000000
00044838  00000000 00000000
                                           7447
                                           7448          TSTLBL PL                  Parameter List
00044840  6E6E6ED7 D3F8F64C                7449+         DC    CL32'>>>PL86<<<'     eye catcher
00044848  4C4C4040 40404040
00044850  40404040 40404040
00044858  40404040 40404040
00044860                                           7450+PL86     DS    0C
00044860  00000000 00000000                        7451          DC    XL352'00'
00044868  00000000 00000000
00044870  00000000 00000000
00044878  00000000 00000000
00044880  00000000 00000000
00044888  00000000 00000000
00044890  00000000 00000000
00044898  00000000 00000000
000448A0  00000000 00000000
000448A8  00000000 00000000
000448B0  00000000 00000000
000448B8  00000000 00000000
000448C0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000448C8  00000000 00000000
000448D0  00000000 00000000
000448D8  00000000 00000000
000448E0  00000000 00000000
000448E8  00000000 00000000
000448F0  00000000 00000000
000448F8  00000000 00000000
00044900  00000000 00000000
00044908  00000000 00000000
00044910  00000000 00000000
00044918  00000000 00000000
00044920  00000000 00000000
00044928  00000000 00000000
00044930  00000000 00000000
00044938  00000000 00000000
00044940  00000000 00000000
00044948  00000000 00000000
00044950  00000000 00000000
00044958  00000000 00000000
00044960  00000000 00000000
00044968  00000000 00000000
00044970  00000000 00000000
00044978  00000000 00000000
00044980  00000000 00000000
00044988  00000000 00000000
00044990  00000000 00000000
00044998  00000000 00000000
000449A0  00000000 00000000
000449A8  00000000 00000000
000449B0  00000000 00000000
000449B8  00000000 00000000
                                        7452
                                        7453          TSTLBL FMSG                fail message
000449C0  6E6E6EC6 D4E2C7F8             7454+         DC    CL32'>>>FMSG86<<<'   eye catcher
000449C8  F64C4C4C 40404040
000449D0  40404040 40404040
000449D8  40404040 40404040
000449E0                                7455+FMSG86   DS    0C
000449E0  86818993 858440D7             7456          DC    C'failed PLO-Extension '
000449E8  D3D660C5 A7A38595
000449F0  A2899695 40
000449F5  E3998997 938540C3             7457          DC    CL96'Triple Compare and Swap (TCS: 30): 3rd NE'
000449FD  96949781 99854081
00044A05  958440E2 A6819740
00044A0D  4DE3C3E2 7A40F3F0
00044A15  5D7A40F3 998440D5
00044A1D  C5404040 40404040
00044A25  40404040 40404040
00044A2D  40404040 40404040
00044A35  40404040 40404040
00044A3D  40404040 40404040
00044A45  40404040 40404040
00044A4D  40404040 40404040
                                        7458          TSTEND
00044A55  C5D5C440 968640A3             7459+         DC    CL32'END of test: 86 <<<'
00044A5D  85A2A37A 40F8F640
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00044A65  4C4C4C40 40404040
00044A6D  40404040 40404040
                                          7460
                                          7461 * ------------------------------------------------------------------------
                                          7462 * Test Triple Compare and Swap (TCS: 30) instruction : All EQ
                                          7463 * ------------------------------------------------------------------------
                                          7464          TSTHDR 0,99,30            All Operand Equal (cc=0, UR=99)
00044A75                  00044A75  00044C00  7465+         ORG   PLOETST+(128*K)+49*(3*K)
00044C00  E2E3C1D9 E3409686                 7466+         DC    CL32'START of test: 87'
00044C08  40A385A2 A37A40F8
00044C10  F7404040 40404040
00044C18  40404040 40404040
00044C20                  00044C20          7467+         USING *,R5                base for test data and test routine
00044C20                  00000000          7468+         USING PLOETEST,R5
00044C20  00044C4C                          7469+T87      DC    A(CODE87)           address of test code
00044C24  0057                              7470+         DC    H'87'               test number
00044C26  00                                7471+         DC    X'00'
00044C27  1E                                7472+         DC    HL1'30'             Function
00044C28  63                                7473+         DC    HL1'99'             expected unequal results (UR)
00044C29  00                                7474+         DC    HL1'00'             found UR
00044C2A  00                                7475+         DC    HL1'0'              cc
00044C2B  07                                7476+         DC    HL1'7'              not expected CC mask
00044C2C  00                                7477+         DS    X                   extracted cc
00044C30  00000000 00000000                 7478+         DS    2F                  extract PSW after test (has CC)
00044C38  000455E0                          7479+         DC    A(FMSG87)           address of test fail message
00044C3C  00045340                          7480+         DC    A(OPSRC87)          address of operand source
00044C40  00045220                          7481+         DC    A(OPDST87)          address of operand destination
00044C44  00045460                          7482+         DC    A(PL87)             address of parameter list
00044C48  0000420F                          7483+         DC    A(RANDX+87)         address of random source data
                                          7484
                                          7485          TSTCODE
00044C4C                                  7486+CODE87   DS    0F
                                          7487 *                                    load random source
00044C4C  E310 501C 0014          0000001C  7488          LGF   R1,OPSRCA
00044C52  E320 5028 0014          00000028  7489          LGF   R2,RANDXA
00044C58  E330 5024 0014          00000024  7490          LGF   R3,PLA
00044C5E  E340 5020 0014          00000020  7491          LGF   R4,OPREPA
                                          7492 *
                                          7493 * Triple Compare and Swap (TCS: 30)
                                          7494 * Build Parambter List
                                          7495 *
00044C64  D20B 1000 2000  00000000  00000000  7496          MVC   0(12,R1),0(R2)      load Op1-Op3 data in source
00044C6A  D203 3040 1000  00000040  00000000  7497          MVC   64(4,R3),0(R1)      copy to PL OP1-Compare
00044C70  D203 3048 1004  00000048  00000004  7498          MVC   72(4,R3),4(R1)      copy to PL OP2-Compare
00044C76  D203 3050 1008  00000050  00000008  7499          MVC   80(4,R3),8(R1)      copy to PL OP3-Compare
                                          7500
00044C7C  D20B 4000 2080  00000000  00000080  7501          MVC   0(12,R4),128(R2)    load initial Op1-Op3 data
00044C82  D203 3058 4000  00000058  00000000  7502          MVC   88(4,R3),0(R4)      copy to PL OP1-Replacement
00044C88  D203 3060 4004  00000060  00000004  7503          MVC   96(4,R3),4(R4)      copy to PL OP2-Replacement
00044C8E  D203 3068 4008  00000068  00000008  7504          MVC   104(4,R3),8(R4)     copy to PL OP3-Replacement
                                          7505
                                          7506 ****
                                          7507 * Do NOT force a failure by changing the 3rd operand
                                          7508 ****
                                          7509 *          MVC   8(4,R1),0(R4)
                                          7510 ****
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    7511
00044C94  E313 0000 0024                00000000  7512           STG   R1,0(R3)            load OP1 address in PL
                                                    7513
00044C9A  4111 0004                     00000004  7514           LA    R1,4(R1)
00044C9E  E313 0008 0024                00000008  7515           STG   R1,8(R3)            load OP2 address in PL
                                                    7516
00044CA4  4111 0004                     00000004  7517           LA    R1,4(R1)
00044CA8  E313 0010 0024                00000010  7518           STG   R1,16(R3)           load OP3 address in PL
                                                    7519
00044CAE  B982 0000                               7520           XGR   R0,R0
00044CB2  4100 001E                     0000001E  7521           LA    R0,30
00044CB6  EE00 0000 3000  00000000      00000000  7522           PLO   0,0,0,0(R3)         Triple C/S (TCS: 30)
                                                    7523
00044CBC  B98D 0020                               7524           EPSW  R2,R0               extract psw
00044CC0  5020 5010                     00000010  7525           ST    R2,CCPSW              to save CC
00044CC4  E310 500B 0076                0000000B  7526           LB    R1,CCMASK           (failure CC mask)
00044CCA  8910 0004                     00000004  7527           SLL   R1,4                (shift to BC instr CC position)
00044CCE  4410 50EE                     00044D0E  7528           EX    R1,F304TSTCC            fail if...
                                                    7529 *
                                                    7530 * compare results
                                                    7531 *
                          00044CD2      00000001  7532 F304CKRES EQU   *
                                                    7533 *                                  reset adresses
00044CD2  E310 501C 0014                0000001C  7534           LGF   R1,OPSRCA
00044CD8  E320 5028 0014                00000028  7535           LGF   R2,RANDXA
00044CDE  E330 5024 0014                00000024  7536           LGF   R3,PLA
00044CE4  E340 5020 0014                00000020  7537           LGF   R4,OPREPA
                                                    7538
                                                    7539 * All equal; replacement values copied to operand source
                                                    7540
00044CEA  D503 1000 3058  00000000      00000058  7541           CLC   0(4,R1),88(R3)
00044CF0  4770 50E8                     00044D08  7542           BNE   F304XX              OP1R Different, error
00044CF4  D503 1004 3060  00000004      00000060  7543           CLC   4(4,R1),96(R3)
00044CFA  4770 50E8                     00044D08  7544           BNE   F304XX              OP2R Different, error
00044CFE  D503 1008 3068  00000008      00000068  7545           CLC   8(4,R1),104(R3)
00044D04  478B 0000                     00000000  7546           BE    0(R11)              OP4R same, return
                                                    7547
                          00044D08      00000001  7548 F304XX    EQU   *
00044D08  45F0 83E8                     000023E8  7549           BAL   R15,FAILMSG
00044D0C  07FB                                    7550           BR    R11
                                                    7551
00044D0E  4700 50F2                     00044D12  7552 F304TSTCC BC    0,F304CCMSG         fail if unexpected condition code
                          00044D12      00000001  7553 F304CCMSG EQU   *
00044D12  45F0 8196                     00002196  7554           BAL   R15,CCMSG
00044D16  47F0 50B2                     00044CD2  7555           B     F304CKRES
                                                    7556
                                                    7557           TSTDATA
00044D1A                  00044D1A      00045200  7558+          ORG   PLOETST+(128*K)+49*(3*K)+(K+512)
                                                    7559           TSTLBL OPDST              Dest/Data: Op1-R, Op2-R, Op3-R
00045200  6E6E6ED6 D7C4E2E3                        7560+          DC    CL32'>>>OPDST87<<<' eye catcher
00045208  F8F74C4C 4C404040
00045210  40404040 40404040
00045218  40404040 40404040
00045220                                  00045220  7561+OPDST87  DS    0C
00045220  00000000 00000000                       7562           DC    XL256'0'
00045228  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00045230  00000000 00000000
00045238  00000000 00000000
00045240  00000000 00000000
00045248  00000000 00000000
00045250  00000000 00000000
00045258  00000000 00000000
00045260  00000000 00000000
00045268  00000000 00000000
00045270  00000000 00000000
00045278  00000000 00000000
00045280  00000000 00000000
00045288  00000000 00000000
00045290  00000000 00000000
00045298  00000000 00000000
000452A0  00000000 00000000
000452A8  00000000 00000000
000452B0  00000000 00000000
000452B8  00000000 00000000
000452C0  00000000 00000000
000452C8  00000000 00000000
000452D0  00000000 00000000
000452D8  00000000 00000000
000452E0  00000000 00000000
000452E8  00000000 00000000
000452F0  00000000 00000000
000452F8  00000000 00000000
00045300  00000000 00000000
00045308  00000000 00000000
00045310  00000000 00000000
00045318  00000000 00000000
                                      7563
                                      7564           TSTLBL OPSRC               Source: OP1-OP3
00045320  6E6E6ED6 D7E2D9C3           7565+          DC     CL32'>>>OPSRC87<<<' eye catcher
00045328  F8F74C4C 4C404040
00045330  40404040 40404040
00045338  40404040 40404040
00045340                              7566+OPSRC87   DS     0C
00045340  00000000 00000000           7567           DC     XL256'0'
00045348  00000000 00000000
00045350  00000000 00000000
00045358  00000000 00000000
00045360  00000000 00000000
00045368  00000000 00000000
00045370  00000000 00000000
00045378  00000000 00000000
00045380  00000000 00000000
00045388  00000000 00000000
00045390  00000000 00000000
00045398  00000000 00000000
000453A0  00000000 00000000
000453A8  00000000 00000000
000453B0  00000000 00000000
000453B8  00000000 00000000
000453C0  00000000 00000000
000453C8  00000000 00000000
000453D0  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000453D8  00000000 00000000
000453E0  00000000 00000000
000453E8  00000000 00000000
000453F0  00000000 00000000
000453F8  00000000 00000000
00045400  00000000 00000000
00045408  00000000 00000000
00045410  00000000 00000000
00045418  00000000 00000000
00045420  00000000 00000000
00045428  00000000 00000000
00045430  00000000 00000000
00045438  00000000 00000000
                                          7568
                                          7569          TSTLBL PL                   Parameter List
00045440  6E6E6ED7 D3F8F74C               7570+         DC    CL32'>>>PL87<<<'      eye catcher
00045448  4C4C4040 40404040
00045450  40404040 40404040
00045458  40404040 40404040
00045460                                  7571+PL87     DS    0C
00045460  00000000 00000000               7572          DC    XL352'00'
00045468  00000000 00000000
00045470  00000000 00000000
00045478  00000000 00000000
00045480  00000000 00000000
00045488  00000000 00000000
00045490  00000000 00000000
00045498  00000000 00000000
000454A0  00000000 00000000
000454A8  00000000 00000000
000454B0  00000000 00000000
000454B8  00000000 00000000
000454C0  00000000 00000000
000454C8  00000000 00000000
000454D0  00000000 00000000
000454D8  00000000 00000000
000454E0  00000000 00000000
000454E8  00000000 00000000
000454F0  00000000 00000000
000454F8  00000000 00000000
00045500  00000000 00000000
00045508  00000000 00000000
00045510  00000000 00000000
00045518  00000000 00000000
00045520  00000000 00000000
00045528  00000000 00000000
00045530  00000000 00000000
00045538  00000000 00000000
00045540  00000000 00000000
00045548  00000000 00000000
00045550  00000000 00000000
00045558  00000000 00000000
00045560  00000000 00000000
00045568  00000000 00000000
00045570  00000000 00000000
00045578  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00045580  00000000 00000000
00045588  00000000 00000000
00045590  00000000 00000000
00045598  00000000 00000000
000455A0  00000000 00000000
000455A8  00000000 00000000
000455B0  00000000 00000000
000455B8  00000000 00000000
                                          7573
                                          7574          TSTLBL FMSG                fail message
000455C0  6E6E6EC6 D4E2C7F8               7575+         DC    CL32'>>>FMSG87<<<'   eye catcher
000455C8  F74C4C4C 40404040
000455D0  40404040 40404040
000455D8  40404040 40404040
000455E0                                  7576+FMSG87   DS    0C
000455E0  86818993 858440D7               7577          DC    C'failed PLO-Extension '
000455E8  D3D660C5 A7A38595
000455F0  A2899695 40
000455F5  E3998997 938540C3               7578          DC    CL96'Triple Compare and Swap (TCS: 30): All EQ'
000455FD  96949781 99854081
00045605  958440E2 A6819740
0004560D  4DE3C3E2 7A40F3F0
00045615  5D7A40C1 939340C5
0004561D  D8404040 40404040
00045625  40404040 40404040
0004562D  40404040 40404040
00045635  40404040 40404040
0004563D  40404040 40404040
00045645  40404040 40404040
0004564D  40404040 40404040
                                          7579          TSTEND
00045655  C5D5C440 968640A3               7580+         DC    CL32'END of test: 87 <<<'
0004565D  85A2A37A 40F8F740
00045665  4C4C4C40 40404040
0004566D  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    7582 *=====================================================================
                                                    7583 *       Compare and Swap and Triple Store Tests
                                                    7584 *=====================================================================
                                                    7585
                                                    7586 * ---------------------------------------------------------------------
                                                    7587 * Test Compare and Swap and Triple Store (CSTSTO: 29) instruction : NE
                                                    7588 * ---------------------------------------------------------------------
                                                    7589          TSTHDR 1,99,29           1st Operand NOT Equal (cc=1, UR=99)
00045675                          00045675  00045800  7590+         ORG   PLOETST+(128*K)+50*(3*K)
00045800  E2E3C1D9 E3409686                         7591+         DC    CL32'START of test: 88'
00045808  40A385A2 A37A40F8
00045810  F8404040 40404040
00045818  40404040 40404040
00045820                          00045820          7592+         USING *,R5               base for test data and test routine
00045820                          00000000          7593+         USING PLOETEST,R5
00045820  0004584C                                  7594+T88      DC    A(CODE88)          address of test code
00045824  0058                                      7595+         DC    H'88'              test number
00045826  00                                        7596+         DC    X'00'
00045827  1D                                        7597+         DC    HL1'29'            Function
00045828  63                                        7598+         DC    HL1'99'            expected unequal results (UR)
00045829  00                                        7599+         DC    HL1'00'            found UR
0004582A  01                                        7600+         DC    HL1'1'             cc
0004582B  0B                                        7601+         DC    HL1'11'            not expected CC mask
0004582C  00                                        7602+         DS    X                  extracted cc
00045830  00000000 00000000                         7603+         DS    2F                 extract PSW after test (has CC)
00045838  000461E0                                  7604+         DC    A(FMSG88)          address of test fail message
0004583C  00045F40                                  7605+         DC    A(OPSRC88)         address of operand source
00045840  00045E20                                  7606+         DC    A(OPDST88)         address of operand destination
00045844  00046060                                  7607+         DC    A(PL88)            address of parameter list
00045848  00004210                                  7608+         DC    A(RANDX+88)        address of random source data
                                                    7609
                                                    7610          TSTCODE
0004584C                                            7611+CODE88   DS    0F
                                                    7612 *                                 load random source
0004584C  E310 501C 0014                 0000001C   7613          LGF   R1,OPSRCA
00045852  E320 5028 0014                 00000028   7614          LGF   R2,RANDXA
00045858  E330 5024 0014                 00000024   7615          LGF   R3,PLA
0004585E  E340 5020 0014                 00000020   7616          LGF   R4,OPREPA
                                                    7617 *
                                                    7618 * Compare and Swap and Triple Store (CSTSTO: 29)
                                                    7619 * Build Parambter List
                                                    7620 *
00045864  D29F 1000 2000       00000000  00000000  7621          MVC   0(160,R1),0(R2)    load Op1-C, Op3, Op5, Op7, Op1-R
0004586A  D21F 3000 1000       00000000  00000000  7622          MVC   0(32,R3),0(R1)     copy to PL OP1-C
00045870  D21F 3090 1020       00000090  00000020  7623          MVC   144(32,R3),32(R1)  copy to PL OP3
00045876  D21F 30B0 1040       000000B0  00000040  7624          MVC   176(32,R3),64(R1)  copy to PL OP5
0004587C  D21F 30D0 1060       000000D0  00000060  7625          MVC   208(32,R3),96(R1)  copy to PL OP7
00045882  D21F 3020 1080       00000020  00000080  7626          MVC   32(32,R3),128(R1)  copy to PL OP1-R
                                                    7627
00045888  D27F 4000 2080       00000000  00000080  7628          MVC   0(128,R4),128(R2)  load initial Op2, Op4, Op6, Op8
                                                    7629
                                                    7630 ****
                                                    7631 * force a failure by changing the Op1-c
                                                    7632 ****
0004588E  D203 3000 2021       00000000  00000021  7633          MVC   0(4,R3),33(R2)
                                                    7634 ****
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00045894  B982 0000                       7635          XGR   R0,R0
00045898  4104 0020                 00000020  7636          LA    R0,32(R4)
0004589C  E303 0048 0024            00000048  7637          STG   R0,72(R3)          load OP4 address in PL
                                              7638
000458A2  4104 0040                 00000040  7639          LA    R0,64(R4)
000458A6  E303 0068 0024            00000068  7640          STG   R0,104(R3)         load OP6 address in PL
                                              7641
000458AC  4104 0060                 00000060  7642          LA    R0,96(R4)
000458B0  E303 0088 0024            00000088  7643          STG   R0,136(R3)         load OP8 address in PL
                                              7644
000458B6  B982 0000                           7645          XGR   R0,R0
000458BA  4100 001D                 0000001D  7646          LA    R0,29
000458BE  EE00 4000 3000  00000000  00000000  7647          PLO   0,0(R4),0,0(R3)    C/S Triple Store (CSTSTO: 29)
                                              7648
000458C4  B98D 0020                           7649          EPSW  R2,R0              extract psw
000458C8  5020 5010                 00000010  7650          ST    R2,CCPSW             to save CC
000458CC  E310 500B 0076            0000000B  7651          LB    R1,CCMASK          (failure CC mask)
000458D2  8910 0004                 00000004  7652          SLL   R1,4               (shift to BC instr CC position)
000458D6  4410 50E2                 00045902  7653          EX    R1,F291TSTCC           fail if...
                                              7654 *
                                              7655 * compare results
                                              7656 *
                            000458DA  00000001  7657 F291CKRES EQU   *
                                              7658 *                               reset adresses
000458DA  E310 501C 0014            0000001C  7659          LGF   R1,OPSRCA          Op1, Op3, Op5, Op7
000458E0  E320 5028 0014            00000028  7660          LGF   R2,RANDXA
000458E6  E330 5024 0014            00000024  7661          LGF   R3,PLA             PL
000458EC  E340 5020 0014            00000020  7662          LGF   R4,OPREPA          OP2, Op4, Op6, Op8
                                              7663
000458F2  D51F 4000 3000  00000000  00000000  7664          CLC   0(32,R4),0(R3)     Op2 == OP1C?
000458F8  478B 0000                 00000000  7665          BE    0(R11)             same, return
                                              7666
                            000458FC  00000001  7667 F291XX   EQU   *
000458FC  45F0 83E8                 000023E8  7668          BAL   R15,FAILMSG
00045900  07FB                                7669          BR    R11
                                              7670
00045902  4700 50E6                 00045906  7671 F291TSTCC BC    0,F291CCMSG        fail if unexpected condition code
                            00045906  00000001  7672 F291CCMSG EQU  *
00045906  45F0 8196                 00002196  7673          BAL   R15,CCMSG
0004590A  47F0 50BA                 000458DA  7674          B     F291CKRES
                                              7675
                                              7676          TSTDATA
0004590E                    0004590E  00045E00  7677+         ORG   PLOETST+(128*K)+50*(3*K)+(K+512)
                                              7678          TSTLBL OPDST             OP2, Op4, Op6, Op8
00045E00  6E6E6ED6 D7C4E2E3                   7679+         DC    CL32'>>>OPDST88<<<' eye catcher
00045E08  F8F84C4C 4C404040
00045E10  40404040 40404040
00045E18  40404040 40404040
00045E20                                      7680+OPDST88  DS    0C
00045E20  00000000 00000000                   7681          DC    XL256'0'
00045E28  00000000 00000000
00045E30  00000000 00000000
00045E38  00000000 00000000
00045E40  00000000 00000000
00045E48  00000000 00000000
00045E50  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00045E58  00000000 00000000
00045E60  00000000 00000000
00045E68  00000000 00000000
00045E70  00000000 00000000
00045E78  00000000 00000000
00045E80  00000000 00000000
00045E88  00000000 00000000
00045E90  00000000 00000000
00045E98  00000000 00000000
00045EA0  00000000 00000000
00045EA8  00000000 00000000
00045EB0  00000000 00000000
00045EB8  00000000 00000000
00045EC0  00000000 00000000
00045EC8  00000000 00000000
00045ED0  00000000 00000000
00045ED8  00000000 00000000
00045EE0  00000000 00000000
00045EE8  00000000 00000000
00045EF0  00000000 00000000
00045EF8  00000000 00000000
00045F00  00000000 00000000
00045F08  00000000 00000000
00045F10  00000000 00000000
00045F18  00000000 00000000
                                                7682
                                                7683           TSTLBL OPSRC                Op1-C, Op3, Op5, Op7, Op1-R
00045F20  6E6E6ED6 D7E2D9C3                     7684+          DC    CL32'>>>OPSRC88<<<' eye catcher
00045F28  F8F84C4C 4C404040
00045F30  40404040 40404040
00045F38  40404040 40404040
00045F40                                        7685+OPSRC88   DS    0C
00045F40  00000000 00000000                     7686           DC    XL256'0'
00045F48  00000000 00000000
00045F50  00000000 00000000
00045F58  00000000 00000000
00045F60  00000000 00000000
00045F68  00000000 00000000
00045F70  00000000 00000000
00045F78  00000000 00000000
00045F80  00000000 00000000
00045F88  00000000 00000000
00045F90  00000000 00000000
00045F98  00000000 00000000
00045FA0  00000000 00000000
00045FA8  00000000 00000000
00045FB0  00000000 00000000
00045FB8  00000000 00000000
00045FC0  00000000 00000000
00045FC8  00000000 00000000
00045FD0  00000000 00000000
00045FD8  00000000 00000000
00045FE0  00000000 00000000
00045FE8  00000000 00000000
00045FF0  00000000 00000000
00045FF8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00046000  00000000 00000000
00046008  00000000 00000000
00046010  00000000 00000000
00046018  00000000 00000000
00046020  00000000 00000000
00046028  00000000 00000000
00046030  00000000 00000000
00046038  00000000 00000000
                                        7687
                                        7688          TSTLBL PL                  Parameter List
00046040  6E6E6ED7 D3F8F84C             7689+         DC    CL32'>>>PL88<<<'     eye catcher
00046048  4C4C4040 40404040
00046050  40404040 40404040
00046058  40404040 40404040
00046060                                7690+PL88     DS    0C
00046060  00000000 00000000             7691          DC    XL352'00'
00046068  00000000 00000000
00046070  00000000 00000000
00046078  00000000 00000000
00046080  00000000 00000000
00046088  00000000 00000000
00046090  00000000 00000000
00046098  00000000 00000000
000460A0  00000000 00000000
000460A8  00000000 00000000
000460B0  00000000 00000000
000460B8  00000000 00000000
000460C0  00000000 00000000
000460C8  00000000 00000000
000460D0  00000000 00000000
000460D8  00000000 00000000
000460E0  00000000 00000000
000460E8  00000000 00000000
000460F0  00000000 00000000
000460F8  00000000 00000000
00046100  00000000 00000000
00046108  00000000 00000000
00046110  00000000 00000000
00046118  00000000 00000000
00046120  00000000 00000000
00046128  00000000 00000000
00046130  00000000 00000000
00046138  00000000 00000000
00046140  00000000 00000000
00046148  00000000 00000000
00046150  00000000 00000000
00046158  00000000 00000000
00046160  00000000 00000000
00046168  00000000 00000000
00046170  00000000 00000000
00046178  00000000 00000000
00046180  00000000 00000000
00046188  00000000 00000000
00046190  00000000 00000000
00046198  00000000 00000000
000461A0  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000461A8  00000000 00000000
000461B0  00000000 00000000
000461B8  00000000 00000000
                                        7692
                                        7693          TSTLBL FMSG               fail message
000461C0  6E6E6EC6 D4E2C7F8             7694+         DC    CL32'>>>FMSG88<<<'  eye catcher
000461C8  F84C4C4C 40404040
000461D0  40404040 40404040
000461D8  40404040 40404040
000461E0                                7695+FMSG88   DS    0C
000461E0  86818993 858440D7             7696          DC    C'failed PLO-Extension '
000461E8  D3D660C5 A7A38595
000461F0  A2899695 40
000461F5  C3969497 81998540             7697          DC    C'Compare and Swap and Triple Store (CSTSTO: 29) : NE'
000461FD  81958440 E2A68197
00046205  40819584 40E39989
0004620D  97938540 E2A39699
00046215  85404DC3 E2E3E2E3
0004621D  D67A40F2 F95D407A
00046225  40D5C5
00046228  40404040 40404040             7698          DC    CL64' '
00046230  40404040 40404040
00046238  40404040 40404040
00046240  40404040 40404040
00046248  40404040 40404040
00046250  40404040 40404040
00046258  40404040 40404040
00046260  40404040 40404040
                                        7699          TSTEND
00046268  C5D5C440 968640A3             7700+         DC    CL32'END of test: 88 <<<'
00046270  85A2A37A 40F8F840
00046278  4C4C4C40 40404040
00046280  40404040 40404040
                                        7701
                                        7702 * ------------------------------------------------------------------
                                        7703 * Test Compare and Swap and Triple Store (CSTSTO: 29) instruction : EQ
                                        7704 * ------------------------------------------------------------------
                                        7705          TSTHDR 0,99,29            1st Operand NOT Equal (cc=1, UR=99)
00046288                  00046288  00046400  7706+         ORG   PLOETST+(128*K)+51*(3*K)
00046400  E2E3C1D9 E3409686             7707+         DC    CL32'START of test: 89'
00046408  40A385A2 A37A40F8
00046410  F9404040 40404040
00046418  40404040 40404040
00046420                  00046420          7708+         USING *,R5                base for test data and test routine
00046420                  00000000          7709+         USING PLOETEST,R5
00046420  0004644C                        7710+T89      DC    A(CODE89)           address of test code
00046424  0059                            7711+         DC    H'89'               test number
00046426  00                              7712+         DC    X'00'
00046427  1D                              7713+         DC    HL1'29'             Function
00046428  63                              7714+         DC    HL1'99'             expected unequal results (UR)
00046429  00                              7715+         DC    HL1'00'             found UR
0004642A  00                              7716+         DC    HL1'0'              cc
0004642B  07                              7717+         DC    HL1'7'              not expected CC mask
0004642C  00                              7718+         DS    X                   extracted cc
00046430  00000000 00000000               7719+         DS    2F                  extract PSW after test (has CC)
00046438  00046DE0                        7720+         DC    A(FMSG89)           address of test fail message
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004643C  00046B40                                7721+          DC    A(OPSRC89)          address of operand source
00046440  00046A20                                7722+          DC    A(OPDST89)          address of operand destination
00046444  00046C60                                7723+          DC    A(PL89)             address of parameter list
00046448  00004211                                7724+          DC    A(RANDX+89)         address of random source data
                                                  7725
                                                  7726           TSTCODE
0004644C                                          7727+CODE89    DS    0F
                                                  7728 *                                   load random source
0004644C  E310 501C 0014                0000001C  7729           LGF   R1,OPSRCA           Op1-C, Op3, Op5, Op7, Op1-R
00046452  E320 5028 0014                00000028  7730           LGF   R2,RANDXA
00046458  E330 5024 0014                00000024  7731           LGF   R3,PLA
0004645E  E340 5020 0014                00000020  7732           LGF   R4,OPREPA           OP2, Op4, Op6, Op8
                                                  7733 *
                                                  7734 * Compare and Swap and Triple Store (CSTSTO: 29)
                                                  7735 * Build Parambter List
                                                  7736 *
00046464  D29F 1000 2000      00000000  00000000  7737           MVC   0(160,R1),0(R2)     load Op1-C, Op3, Op5, Op7, Op1-R
0004646A  D21F 3000 1000      00000000  00000000  7738           MVC   0(32,R3),0(R1)      copy to PL OP1-C
00046470  D21F 3090 1020      00000090  00000020  7739           MVC   144(32,R3),32(R1)   copy to PL OP3
00046476  D21F 30B0 1040      000000B0  00000040  7740           MVC   176(32,R3),64(R1)   copy to PL OP5
0004647C  D21F 30D0 1060      000000D0  00000060  7741           MVC   208(32,R3),96(R1)   copy to PL OP7
00046482  D21F 3020 1080      00000020  00000080  7742           MVC   32(32,R3),128(R1)   copy to PL OP1-R
                                                  7743
00046488  D27F 4000 2080      00000000  00000080  7744           MVC   0(128,R4),128(R2)   load initial Op2, Op4, Op6, Op8
                                                  7745
                                                  7746 ****
                                                  7747 * Force a EQ by changing the Op1-C == Op2
                                                  7748 ****
0004648E  D21F 3000 4000      00000000  00000000  7749           MVC   0(32,R3),0(R4)
                                                  7750 ****
00046494  B982 0000                               7751           XGR   R0,R0
00046498  4104 0020                     00000020  7752           LA    R0,32(R4)
0004649C  E303 0048 0024                00000048  7753           STG   R0,72(R3)           load OP4 address in PL
                                                  7754
000464A2  4104 0040                     00000040  7755           LA    R0,64(R4)
000464A6  E303 0068 0024                00000068  7756           STG   R0,104(R3)          load OP6 address in PL
                                                  7757
000464AC  4104 0060                     00000060  7758           LA    R0,96(R4)
000464B0  E303 0088 0024                00000088  7759           STG   R0,136(R3)          load OP8 address in PL
                                                  7760
000464B6  B982 0000                               7761           XGR   R0,R0
000464BA  4100 001D                     0000001D  7762           LA    R0,29
000464BE  EE00 4000 3000      00000000  00000000  7763           PLO   0,0(R4),0,0(R3)     C/S Triple Store (CSTSTO: 29)
                                                  7764
000464C4  B98D 0020                               7765           EPSW  R2,R0               extract psw
000464C8  5020 5010                     00000010  7766           ST    R2,CCPSW              to save CC
000464CC  E310 500B 0076                0000000B  7767           LB    R1,CCMASK           (failure CC mask)
000464D2  8910 0004                     00000004  7768           SLL   R1,4                (shift to BC instr CC position)
000464D6  4410 5100                     00046520  7769           EX    R1,F292TSTCC            fail if...
                                                  7770 *
                                                  7771 * compare results
                                                  7772 *
                          000464DA  00000001      7773 F292CKRES EQU   *
                                                  7774 *                                   reset adresses
000464DA  E310 501C 0014                0000001C  7775           LGF   R1,OPSRCA           Op1, Op3, Op5, Op7
000464E0  E320 5028 0014                00000028  7776           LGF   R2,RANDXA
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000464E6  E330 5024 0014                 00000024  7777           LGF   R3,PLA               PL
000464EC  E340 5020 0014                 00000020  7778           LGF   R4,OPREPA            OP2, Op4, Op6, Op8
                                                   7779
000464F2  D51F 4020 3090       00000020  00000090  7780           CLC   32(32,R4),144(R3)    OP4 ==  PL Op3
000464F8  4770 50FA                      0004651A  7781           BNE   F292XX                    different, error
000464FC  D51F 4040 30B0       00000040  000000B0  7782           CLC   64(32,R4),176(R3)    OP6 ==  PL Op5
00046502  4770 50FA                      0004651A  7783           BNE   F292XX                    different, error
00046506  D51F 4060 30D0       00000060  000000D0  7784           CLC   96(32,R4),208(R3)    OP8 ==  PL Op7
0004650C  4770 50FA                      0004651A  7785           BNE   F292XX                    different, error
00046510  D51F 4000 3020       00000000  00000020  7786           CLC   0(32,R4),32(R3)      OP2 ==  PL Op-R
00046516  478B 0000                      00000000  7787           BE    0(R11)                    same, return
                                                   7788
                               0004651A  00000001  7789 F292XX    EQU   *
0004651A  45F0 83E8                      000023E8  7790           BAL   R15,FAILMSG
0004651E  07FB                                     7791           BR    R11
                                                   7792
00046520  4700 5104                      00046524  7793 F292TSTCC BC    0,F292CCMSG          fail if unexpected condition code
                               00046524  00000001  7794 F292CCMSG EQU   *
00046524  45F0 8196                      00002196  7795           BAL   R15,CCMSG
00046528  47F0 50BA                      000464DA  7796           B     F292CKRES
                                                   7797
                                                   7798           TSTDATA
0004652C                       0004652C  00046A00  7799+          ORG   PLOETST+(128*K)+51*(3*K)+(K+512)
                                                   7800           TSTLBL OPDST               OP2, Op4, Op6, Op8
00046A00  6E6E6ED6 D7C4E2E3                        7801+          DC    CL32'>>>OPDST89<<<' eye catcher
00046A08  F8F94C4C 4C404040
00046A10  40404040 40404040
00046A18  40404040 40404040
00046A20                                           7802+OPDST89   DS    0C
00046A20  00000000 00000000                        7803           DC    XL256'0'
00046A28  00000000 00000000
00046A30  00000000 00000000
00046A38  00000000 00000000
00046A40  00000000 00000000
00046A48  00000000 00000000
00046A50  00000000 00000000
00046A58  00000000 00000000
00046A60  00000000 00000000
00046A68  00000000 00000000
00046A70  00000000 00000000
00046A78  00000000 00000000
00046A80  00000000 00000000
00046A88  00000000 00000000
00046A90  00000000 00000000
00046A98  00000000 00000000
00046AA0  00000000 00000000
00046AA8  00000000 00000000
00046AB0  00000000 00000000
00046AB8  00000000 00000000
00046AC0  00000000 00000000
00046AC8  00000000 00000000
00046AD0  00000000 00000000
00046AD8  00000000 00000000
00046AE0  00000000 00000000
00046AE8  00000000 00000000
00046AF0  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00046AF8  00000000 00000000
00046B00  00000000 00000000
00046B08  00000000 00000000
00046B10  00000000 00000000
00046B18  00000000 00000000
                                        7804
                                        7805          TSTLBL OPSRC              Op1-C, Op3, Op5, Op7, Op1-R
00046B20  6E6E6ED6 D7E2D9C3             7806+         DC    CL32'>>>OPSRC89<<<' eye catcher
00046B28  F8F94C4C 4C404040
00046B30  40404040 40404040
00046B38  40404040 40404040
00046B40                                7807+OPSRC89  DS    0C
00046B40  00000000 00000000             7808          DC    XL256'0'
00046B48  00000000 00000000
00046B50  00000000 00000000
00046B58  00000000 00000000
00046B60  00000000 00000000
00046B68  00000000 00000000
00046B70  00000000 00000000
00046B78  00000000 00000000
00046B80  00000000 00000000
00046B88  00000000 00000000
00046B90  00000000 00000000
00046B98  00000000 00000000
00046BA0  00000000 00000000
00046BA8  00000000 00000000
00046BB0  00000000 00000000
00046BB8  00000000 00000000
00046BC0  00000000 00000000
00046BC8  00000000 00000000
00046BD0  00000000 00000000
00046BD8  00000000 00000000
00046BE0  00000000 00000000
00046BE8  00000000 00000000
00046BF0  00000000 00000000
00046BF8  00000000 00000000
00046C00  00000000 00000000
00046C08  00000000 00000000
00046C10  00000000 00000000
00046C18  00000000 00000000
00046C20  00000000 00000000
00046C28  00000000 00000000
00046C30  00000000 00000000
00046C38  00000000 00000000
                                        7809
                                        7810          TSTLBL PL                 Parameter List
00046C40  6E6E6ED7 D3F8F94C             7811+         DC    CL32'>>>PL89<<<'    eye catcher
00046C48  4C4C4040 40404040
00046C50  40404040 40404040
00046C58  40404040 40404040
00046C60                                7812+PL89     DS    0C
00046C60  00000000 00000000             7813          DC    XL352'00'
00046C68  00000000 00000000
00046C70  00000000 00000000
00046C78  00000000 00000000
00046C80  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00046C88  00000000 00000000
00046C90  00000000 00000000
00046C98  00000000 00000000
00046CA0  00000000 00000000
00046CA8  00000000 00000000
00046CB0  00000000 00000000
00046CB8  00000000 00000000
00046CC0  00000000 00000000
00046CC8  00000000 00000000
00046CD0  00000000 00000000
00046CD8  00000000 00000000
00046CE0  00000000 00000000
00046CE8  00000000 00000000
00046CF0  00000000 00000000
00046CF8  00000000 00000000
00046D00  00000000 00000000
00046D08  00000000 00000000
00046D10  00000000 00000000
00046D18  00000000 00000000
00046D20  00000000 00000000
00046D28  00000000 00000000
00046D30  00000000 00000000
00046D38  00000000 00000000
00046D40  00000000 00000000
00046D48  00000000 00000000
00046D50  00000000 00000000
00046D58  00000000 00000000
00046D60  00000000 00000000
00046D68  00000000 00000000
00046D70  00000000 00000000
00046D78  00000000 00000000
00046D80  00000000 00000000
00046D88  00000000 00000000
00046D90  00000000 00000000
00046D98  00000000 00000000
00046DA0  00000000 00000000
00046DA8  00000000 00000000
00046DB0  00000000 00000000
00046DB8  00000000 00000000
                                      7814
                                      7815          TSTLBL FMSG                fail message
00046DC0  6E6E6EC6 D4E2C7F8           7816+         DC     CL32'>>>FMSG89<<<'  eye catcher
00046DC8  F94C4C4C 40404040
00046DD0  40404040 40404040
00046DD8  40404040 40404040
00046DE0                              7817+FMSG89   DS     0C
00046DE0  86818993 858440D7           7818          DC     C'failed PLO-Extension '
00046DE8  D3D660C5 A7A38595
00046DF0  A2899695 40
00046DF5  C3969497 81998540           7819          DC     C'Compare and Swap and Triple Store (CSTSTO: 29) : EQ'
00046DFD  81958440 E2A68197
00046E05  40819584 40E39989
00046E0D  97938540 E2A39699
00046E15  85404DC3 E2E3E2E3
00046E1D  D67A40F2 F95D407A
00046E25  40C5D8
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00046E28  40404040 40404040                         7820           DC    CL64' '
00046E30  40404040 40404040
00046E38  40404040 40404040
00046E40  40404040 40404040
00046E48  40404040 40404040
00046E50  40404040 40404040
00046E58  40404040 40404040
00046E60  40404040 40404040
                                                    7821           TSTEND
00046E68  C5D5C440 968640A3                         7822+          DC    CL32'END of test: 89 <<<'
00046E70  85A2A37A 40F8F940
00046E78  4C4C4C40 40404040
00046E80  40404040 40404040
                                                    7823
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   7825 *=====================================================================
                                                   7826 *       Compare and Swap and Double Store Tests
                                                   7827 *=====================================================================
                                                   7828
                                                   7829 * ---------------------------------------------------------------------
                                                   7830 * Test Compare and Swap and Double Store (CSDSTO: 28) instruction : NE
                                                   7831 * ---------------------------------------------------------------------
                                                   7832          TSTHDR 1,99,28           1Op1-C != OP2 (cc=1, UR=99)
00046E88                          00046E88  00047000 7833+         ORG   PLOETST+(128*K)+52*(3*K)
00047000  E2E3C1D9 E3409686                        7834+         DC    CL32'START of test: 90'
00047008  40A385A2 A37A40F9
00047010  F0404040 40404040
00047018  40404040 40404040
00047020                          00047020          7835+         USING *,R5               base for test data and test routine
00047020                          00000000          7836+         USING PLOETEST,R5
00047020  0004704C                                 7837+T90      DC    A(CODE90)          address of test code
00047024  005A                                     7838+         DC    H'90'              test number
00047026  00                                       7839+         DC    X'00'
00047027  1C                                       7840+         DC    HL1'28'            Function
00047028  63                                       7841+         DC    HL1'99'            expected unequal results (UR)
00047029  00                                       7842+         DC    HL1'00'            found UR
0004702A  01                                       7843+         DC    HL1'1'             cc
0004702B  0B                                       7844+         DC    HL1'11'            not expected CC mask
0004702C  00                                       7845+         DS    X                  extracted cc
00047030  00000000 00000000                        7846+         DS    2F                 extract PSW after test (has CC)
00047038  000479E0                                 7847+         DC    A(FMSG90)          address of test fail message
0004703C  00047740                                 7848+         DC    A(OPSRC90)         address of operand source
00047040  00047620                                 7849+         DC    A(OPDST90)         address of operand destination
00047044  00047860                                 7850+         DC    A(PL90)            address of parameter list
00047048  00004212                                 7851+         DC    A(RANDX+90)        address of random source data
                                                   7852
                                                   7853          TSTCODE
0004704C                                           7854+CODE90   DS    0F
                                                   7855 *                                 load random source
0004704C  E310 501C 0014                0000001C   7856          LGF   R1,OPSRCA          Op1-C, Op3, Op5, Op1-R
00047052  E320 5028 0014                00000028   7857          LGF   R2,RANDXA
00047058  E330 5024 0014                00000024   7858          LGF   R3,PLA
0004705E  E340 5020 0014                00000020   7859          LGF   R4,OPREPA          OP2, Op4, Op6
                                                   7860 *
                                                   7861 * Compare and Swap and Double Store (CSDSTO: 28)
                                                   7862 * Build Parambter List
                                                   7863 *
00047064  D27F 1000 2000  00000000      00000000   7864          MVC   0(128,R1),0(R2)    load Op1-C, Op3, Op5, Op1-R
0004706A  D21F 3000 1000  00000000      00000000   7865          MVC   0(32,R3),0(R1)     copy to PL OP1-C
00047070  D21F 3070 1020  00000070      00000020   7866          MVC   112(32,R3),32(R1)  copy to PL OP3
00047076  D21F 3090 1040  00000090      00000040   7867          MVC   144(32,R3),64(R1)  copy to PL OP5
0004707C  D21F 3020 1060  00000020      00000060   7868          MVC   32(32,R3),96(R1)   copy to PL OP1-R
                                                   7869
00047082  D25F 4000 2080  00000000      00000080   7870          MVC   0(96,R4),128(R2)   load initial Op2, Op4, Op6
                                                   7871
                                                   7872 ****
                                                   7873 * force a failure by changing the Op1-c
                                                   7874 ****
00047088  D203 3000 2021  00000000      00000021   7875          MVC   0(4,R3),33(R2)
                                                   7876 ****
0004708E  B982 0000                                7877          XGR   R0,R0
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00047092  4104 0020                     00000020  7878           LA    R0,32(R4)
00047096  E303 0048 0024                 00000048  7879           STG   R0,72(R3)           load OP4 address in PL
                                                   7880
0004709C  4104 0040                      00000040  7881           LA    R0,64(R4)
000470A0  E303 0068 0024                 00000068  7882           STG   R0,104(R3)          load OP6 address in PL
                                                   7883
000470A6  B982 0000                                7884           XGR   R0,R0
000470AA  4100 001C                      0000001C  7885           LA    R0,28
000470AE  EE00 4000 3000       00000000  00000000  7886           PLO   0,0(R4),0,0(R3)     C/S Double Store (CSDSTO: 28)
                                                   7887
000470B4  B98D 0020                                7888           EPSW  R2,R0               extract psw
000470B8  5020 5010                      00000010  7889           ST    R2,CCPSW              to save CC
000470BC  E310 500B 0076                 0000000B  7890           LB    R1,CCMASK           (failure CC mask)
000470C2  8910 0004                      00000004  7891           SLL   R1,4                (shift to BC instr CC position)
000470C6  4410 50D2                      000470F2  7892           EX    R1,F281TSTCC            fail if...
                                                   7893 *
                                                   7894 * compare results
                                                   7895 *
                         000470CA  00000001        7896 F281CKRES EQU   *
                                                   7897 *                                    reset adresses
000470CA  E310 501C 0014                 0000001C  7898           LGF   R1,OPSRCA           Op1, Op3, Op5
000470D0  E320 5028 0014                 00000028  7899           LGF   R2,RANDXA
000470D6  E330 5024 0014                 00000024  7900           LGF   R3,PLA              PL
000470DC  E340 5020 0014                 00000020  7901           LGF   R4,OPREPA           OP2, Op4, Op6
                                                   7902
000470E2  D51F 4000 3000       00000000  00000000  7903           CLC   0(32,R4),0(R3)      Op2 == OP1C?
000470E8  478B 0000                      00000000  7904           BE    0(R11)              same, return
                                                   7905
                         000470EC  00000001        7906 F281XX    EQU   *
000470EC  45F0 83E8                      000023E8  7907           BAL   R15,FAILMSG
000470F0  07FB                                     7908           BR    R11
                                                   7909
000470F2  4700 50D6                      000470F6  7910 F281TSTCC BC    0,F281CCMSG         fail if unexpected condition code
                         000470F6  00000001        7911 F281CCMSG EQU   *
000470F6  45F0 8196                      00002196  7912           BAL   R15,CCMSG
000470FA  47F0 50AA                      000470CA  7913           B     F281CKRES
                                                   7914
                                                   7915           TSTDATA
000470FE                 000470FE  00047600        7916+          ORG   PLOETST+(128*K)+52*(3*K)+(K+512)
                                                   7917           TSTLBL OPDST              OP2, Op4, Op6
00047600  6E6E6ED6 D7C4E2E3                       7918+          DC    CL32'>>>OPDST90<<<' eye catcher
00047608  F9F04C4C 4C404040
00047610  40404040 40404040
00047618  40404040 40404040
00047620                                          7919+OPDST90   DS    0C
00047620  00000000 00000000                       7920           DC    XL256'0'
00047628  00000000 00000000
00047630  00000000 00000000
00047638  00000000 00000000
00047640  00000000 00000000
00047648  00000000 00000000
00047650  00000000 00000000
00047658  00000000 00000000
00047660  00000000 00000000
00047668  00000000 00000000
00047670  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00047678  00000000 00000000
00047680  00000000 00000000
00047688  00000000 00000000
00047690  00000000 00000000
00047698  00000000 00000000
000476A0  00000000 00000000
000476A8  00000000 00000000
000476B0  00000000 00000000
000476B8  00000000 00000000
000476C0  00000000 00000000
000476C8  00000000 00000000
000476D0  00000000 00000000
000476D8  00000000 00000000
000476E0  00000000 00000000
000476E8  00000000 00000000
000476F0  00000000 00000000
000476F8  00000000 00000000
00047700  00000000 00000000
00047708  00000000 00000000
00047710  00000000 00000000
00047718  00000000 00000000
                                        7921
                                        7922          TSTLBL OPSRC              Op1-C, Op3, Op5, Op1-R
00047720  6E6E6ED6 D7E2D9C3             7923+         DC    CL32'>>>OPSRC90<<<' eye catcher
00047728  F9F04C4C 4C404040
00047730  40404040 40404040
00047738  40404040 40404040
00047740                                7924+OPSRC90  DS    0C
00047740  00000000 00000000             7925          DC    XL256'0'
00047748  00000000 00000000
00047750  00000000 00000000
00047758  00000000 00000000
00047760  00000000 00000000
00047768  00000000 00000000
00047770  00000000 00000000
00047778  00000000 00000000
00047780  00000000 00000000
00047788  00000000 00000000
00047790  00000000 00000000
00047798  00000000 00000000
000477A0  00000000 00000000
000477A8  00000000 00000000
000477B0  00000000 00000000
000477B8  00000000 00000000
000477C0  00000000 00000000
000477C8  00000000 00000000
000477D0  00000000 00000000
000477D8  00000000 00000000
000477E0  00000000 00000000
000477E8  00000000 00000000
000477F0  00000000 00000000
000477F8  00000000 00000000
00047800  00000000 00000000
00047808  00000000 00000000
00047810  00000000 00000000
00047818  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00047820  00000000 00000000
00047828  00000000 00000000
00047830  00000000 00000000
00047838  00000000 00000000
                                        7926
                                        7927          TSTLBL PL                   Parameter List
00047840  6E6E6ED7 D3F9F04C             7928+         DC    CL32'>>>PL90<<<'      eye catcher
00047848  4C4C4040 40404040
00047850  40404040 40404040
00047858  40404040 40404040
00047860                                7929+PL90     DS    0C
00047860  00000000 00000000             7930          DC    XL352'00'
00047868  00000000 00000000
00047870  00000000 00000000
00047878  00000000 00000000
00047880  00000000 00000000
00047888  00000000 00000000
00047890  00000000 00000000
00047898  00000000 00000000
000478A0  00000000 00000000
000478A8  00000000 00000000
000478B0  00000000 00000000
000478B8  00000000 00000000
000478C0  00000000 00000000
000478C8  00000000 00000000
000478D0  00000000 00000000
000478D8  00000000 00000000
000478E0  00000000 00000000
000478E8  00000000 00000000
000478F0  00000000 00000000
000478F8  00000000 00000000
00047900  00000000 00000000
00047908  00000000 00000000
00047910  00000000 00000000
00047918  00000000 00000000
00047920  00000000 00000000
00047928  00000000 00000000
00047930  00000000 00000000
00047938  00000000 00000000
00047940  00000000 00000000
00047948  00000000 00000000
00047950  00000000 00000000
00047958  00000000 00000000
00047960  00000000 00000000
00047968  00000000 00000000
00047970  00000000 00000000
00047978  00000000 00000000
00047980  00000000 00000000
00047988  00000000 00000000
00047990  00000000 00000000
00047998  00000000 00000000
000479A0  00000000 00000000
000479A8  00000000 00000000
000479B0  00000000 00000000
000479B8  00000000 00000000
                                        7931
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   7932           TSTLBL FMSG                fail message
000479C0  6E6E6EC6 D4E2C7F9                        7933+          DC     CL32'>>>FMSG90<<<'  eye catcher
000479C8  F04C4C4C 40404040
000479D0  40404040 40404040
000479D8  40404040 40404040
000479E0                                           7934+FMSG90    DS     0C
000479E0  86818993 858440D7                        7935           DC     C'failed PLO-Extension '
000479E8  D3D660C5 A7A38595
000479F0  A2899695 40
000479F5  C3969497 81998540                        7936           DC     C'Compare and Swap and Double Store (CSDSTO: 28) : NE'
000479FD  81958440 E2A68197
00047A05  40819584 40C496A4
00047A0D  82938540 E2A39699
00047A15  85404DC3 E2C4E2E3
00047A1D  D67A40F2 F85D407A
00047A25  40D5C5
00047A28  40404040 40404040                        7937           DC     CL64' '
00047A30  40404040 40404040
00047A38  40404040 40404040
00047A40  40404040 40404040
00047A48  40404040 40404040
00047A50  40404040 40404040
00047A58  40404040 40404040
00047A60  40404040 40404040
                                                   7938           TSTEND
00047A68  C5D5C440 968640A3                        7939+          DC     CL32'END of test: 90 <<<'
00047A70  85A2A37A 40F9F040
00047A78  4C4C4C40 40404040
00047A80  40404040 40404040
                                                   7940
                                                   7941 * ------------------------------------------------------------------------
                                                   7942 * Test Compare and Swap and Double Store (CSDSTO: 28) instruction : EQ
                                                   7943 * ------------------------------------------------------------------------
                                                   7944           TSTHDR 0,99,28             Op1-C == OP2 (cc=0, UR=99)
00047A88                          00047A88  00047C00 7945+          ORG    PLOETST+(128*K)+53*(3*K)
00047C00  E2E3C1D9 E3409686                        7946+          DC     CL32'START of test: 91'
00047C08  40A385A2 A37A40F9
00047C10  F1404040 40404040
00047C18  40404040 40404040
00047C20                          00047C20         7947+          USING  *,R5               base for test data and test routine
00047C20                          00000000         7948+          USING  PLOETEST,R5
00047C20  00047C4C                                 7949+T91       DC     A(CODE91)          address of test code
00047C24  005B                                     7950+          DC     H'91'              test number
00047C26  00                                       7951+          DC     X'00'
00047C27  1C                                       7952+          DC     HL1'28'            Function
00047C28  63                                       7953+          DC     HL1'99'            expected unequal results (UR)
00047C29  00                                       7954+          DC     HL1'00'            found UR
00047C2A  00                                       7955+          DC     HL1'0'             cc
00047C2B  07                                       7956+          DC     HL1'7'             not expected CC mask
00047C2C  00                                       7957+          DS     X                  extracted cc
00047C30  00000000 00000000                        7958+          DS     2F                 extract PSW after test (has CC)
00047C38  000485E0                                 7959+          DC     A(FMSG91)          address of test fail message
00047C3C  00048340                                 7960+          DC     A(OPSRC91)         address of operand source
00047C40  00048220                                 7961+          DC     A(OPDST91)         address of operand destination
00047C44  00048460                                 7962+          DC     A(PL91)            address of parameter list
00047C48  00004213                                 7963+          DC     A(RANDX+91)        address of random source data
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    7964
                                                    7965          TSTCODE
00047C4C                                            7966+CODE91   DS    0F
                                                    7967 *                                load random source
00047C4C  E310 501C 0014                 0000001C   7968          LGF   R1,OPSRCA           Op1-C, Op3, Op5, Op1-R
00047C52  E320 5028 0014                 00000028   7969          LGF   R2,RANDXA
00047C58  E330 5024 0014                 00000024   7970          LGF   R3,PLA
00047C5E  E340 5020 0014                 00000020   7971          LGF   R4,OPREPA           OP2, Op4, Op6
                                                    7972 *
                                                    7973 * Compare and Swap and Double Store (CSDSTO: 28)
                                                    7974 * Build Parambter List
                                                    7975 *
00047C64  D27F 1000 2000       00000000  00000000   7976          MVC   0(128,R1),0(R2)     load Op1-C, Op3, Op5, Op1-R
00047C6A  D21F 3000 1000       00000000  00000000   7977          MVC   0(32,R3),0(R1)      copy to PL OP1-C
00047C70  D21F 3070 1020       00000070  00000020   7978          MVC   112(32,R3),32(R1)   copy to PL OP3
00047C76  D21F 3090 1040       00000090  00000040   7979          MVC   144(32,R3),64(R1)   copy to PL OP5
00047C7C  D21F 3020 1060       00000020  00000060   7980          MVC   32(32,R3),96(R1)    copy to PL OP1-R
                                                    7981
00047C82  D25F 4000 2080       00000000  00000080   7982          MVC   0(96,R4),128(R2)    load initial Op2, Op4, Op6
                                                    7983
                                                    7984 ****
                                                    7985 * Force a EQ by changing the Op1-C == Op2
                                                    7986 ****
00047C88  D21F 3000 4000       00000000  00000000   7987          MVC   0(32,R3),0(R4)
                                                    7988 ****
00047C8E  B982 0000                                 7989          XGR   R0,R0
00047C92  4104 0020                      00000020   7990          LA    R0,32(R4)
00047C96  E303 0048 0024                 00000048   7991          STG   R0,72(R3)           load OP4 address in PL
                                                    7992
00047C9C  4104 0040                      00000040   7993          LA    R0,64(R4)
00047CA0  E303 0068 0024                 00000068   7994          STG   R0,104(R3)          load OP6 address in PL
                                                    7995
00047CA6  B982 0000                                 7996          XGR   R0,R0
00047CAA  4100 001C                      0000001C   7997          LA    R0,28
00047CAE  EE00 4000 3000       00000000  00000000   7998          PLO   0,0(R4),0,0(R3)     C/S Double Store (CSDSTO: 28)
                                                    7999
00047CB4  B98D 0020                                 8000          EPSW  R2,R0               extract psw
00047CB8  5020 5010                      00000010   8001          ST    R2,CCPSW              to save CC
00047CBC  E310 500B 0076                 0000000B   8002          LB    R1,CCMASK           (failure CC mask)
00047CC2  8910 0004                      00000004   8003          SLL   R1,4                (shift to BC instr CC position)
00047CC6  4410 50E6                      00047D06   8004          EX    R1,F282TSTCC          fail if...
                                                    8005 *
                                                    8006 * compare results
                                                    8007 *
                               00047CCA  00000001   8008 F282CKRES EQU   *
                                                    8009 *                                reset adresses
00047CCA  E310 501C 0014                 0000001C   8010          LGF   R1,OPSRCA           Op1, Op3, Op5
00047CD0  E320 5028 0014                 00000028   8011          LGF   R2,RANDXA
00047CD6  E330 5024 0014                 00000024   8012          LGF   R3,PLA              PL
00047CDC  E340 5020 0014                 00000020   8013          LGF   R4,OPREPA           OP2, Op4, Op6
                                                    8014
00047CE2  D51F 4020 3070       00000020  00000070   8015          CLC   32(32,R4),112(R3)   OP4 ==  PL Op3
00047CE8  4770 50E0                      00047D00   8016          BNE   F282XX                   different, error
00047CEC  D51F 4040 3090       00000040  00000090   8017          CLC   64(32,R4),144(R3)   OP6 ==  PL Op5
00047CF2  4770 50E0                      00047D00   8018          BNE   F282XX                   different, error
00047CF6  D51F 4000 3020       00000000  00000020   8019          CLC   0(32,R4),32(R3)     OP2 ==  PL Op-R
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00047CFC  478B 0000                    00000000  8020           BE    0(R11)                    same, return
                                                 8021
                                00047D00  00000001  8022 F282XX    EQU   *
00047D00  45F0 83E8                    000023E8  8023           BAL   R15,FAILMSG
00047D04  07FB                                   8024           BR    R11
                                                 8025
00047D06  4700 50EA                    00047D0A  8026 F282TSTCC BC    0,F282CCMSG          fail if unexpected condition code
                                00047D0A  00000001  8027 F282CCMSG EQU   *
00047D0A  45F0 8196                    00002196  8028           BAL   R15,CCMSG
00047D0E  47F0 50AA                    00047CCA  8029           B     F282CKRES
                                                 8030
                                                 8031           TSTDATA
00047D12                       00047D12  00048200  8032+          ORG   PLOETST+(128*K)+53*(3*K)+(K+512)
                                                 8033           TSTLBL OPDST               OP2, Op4, Op6
00048200  6E6E6ED6 D7C4E2E3                      8034+          DC    CL32'>>>OPDST91<<<' eye catcher
00048208  F9F14C4C 4C404040
00048210  40404040 40404040
00048218  40404040 40404040
00048220                                         8035+OPDST91   DS    0C
00048220  00000000 00000000                      8036           DC    XL256'0'
00048228  00000000 00000000
00048230  00000000 00000000
00048238  00000000 00000000
00048240  00000000 00000000
00048248  00000000 00000000
00048250  00000000 00000000
00048258  00000000 00000000
00048260  00000000 00000000
00048268  00000000 00000000
00048270  00000000 00000000
00048278  00000000 00000000
00048280  00000000 00000000
00048288  00000000 00000000
00048290  00000000 00000000
00048298  00000000 00000000
000482A0  00000000 00000000
000482A8  00000000 00000000
000482B0  00000000 00000000
000482B8  00000000 00000000
000482C0  00000000 00000000
000482C8  00000000 00000000
000482D0  00000000 00000000
000482D8  00000000 00000000
000482E0  00000000 00000000
000482E8  00000000 00000000
000482F0  00000000 00000000
000482F8  00000000 00000000
00048300  00000000 00000000
00048308  00000000 00000000
00048310  00000000 00000000
00048318  00000000 00000000
                                                 8037
                                                 8038           TSTLBL OPSRC               Op1-C, Op3, Op5, Op1-R
00048320  6E6E6ED6 D7E2D9C3                      8039+          DC    CL32'>>>OPSRC91<<<' eye catcher
00048328  F9F14C4C 4C404040
00048330  40404040 40404040
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00048338  40404040 40404040
00048340                                             8040+OPSRC91  DS    0C
00048340  00000000 00000000                          8041          DC    XL256'0'
00048348  00000000 00000000
00048350  00000000 00000000
00048358  00000000 00000000
00048360  00000000 00000000
00048368  00000000 00000000
00048370  00000000 00000000
00048378  00000000 00000000
00048380  00000000 00000000
00048388  00000000 00000000
00048390  00000000 00000000
00048398  00000000 00000000
000483A0  00000000 00000000
000483A8  00000000 00000000
000483B0  00000000 00000000
000483B8  00000000 00000000
000483C0  00000000 00000000
000483C8  00000000 00000000
000483D0  00000000 00000000
000483D8  00000000 00000000
000483E0  00000000 00000000
000483E8  00000000 00000000
000483F0  00000000 00000000
000483F8  00000000 00000000
00048400  00000000 00000000
00048408  00000000 00000000
00048410  00000000 00000000
00048418  00000000 00000000
00048420  00000000 00000000
00048428  00000000 00000000
00048430  00000000 00000000
00048438  00000000 00000000
                                                     8042
                                                     8043          TSTLBL PL                   Parameter List
00048440  6E6E6ED7 D3F9F14C                          8044+         DC    CL32'>>>PL91<<<'      eye catcher
00048448  4C4C4040 40404040
00048450  40404040 40404040
00048458  40404040 40404040
00048460                                             8045+PL91     DS    0C
00048460  00000000 00000000                          8046          DC    XL352'00'
00048468  00000000 00000000
00048470  00000000 00000000
00048478  00000000 00000000
00048480  00000000 00000000
00048488  00000000 00000000
00048490  00000000 00000000
00048498  00000000 00000000
000484A0  00000000 00000000
000484A8  00000000 00000000
000484B0  00000000 00000000
000484B8  00000000 00000000
000484C0  00000000 00000000
000484C8  00000000 00000000
000484D0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000484D8  00000000 00000000
000484E0  00000000 00000000
000484E8  00000000 00000000
000484F0  00000000 00000000
000484F8  00000000 00000000
00048500  00000000 00000000
00048508  00000000 00000000
00048510  00000000 00000000
00048518  00000000 00000000
00048520  00000000 00000000
00048528  00000000 00000000
00048530  00000000 00000000
00048538  00000000 00000000
00048540  00000000 00000000
00048548  00000000 00000000
00048550  00000000 00000000
00048558  00000000 00000000
00048560  00000000 00000000
00048568  00000000 00000000
00048570  00000000 00000000
00048578  00000000 00000000
00048580  00000000 00000000
00048588  00000000 00000000
00048590  00000000 00000000
00048598  00000000 00000000
000485A0  00000000 00000000
000485A8  00000000 00000000
000485B0  00000000 00000000
000485B8  00000000 00000000
                             8047
                             8048          TSTLBL FMSG               fail message
000485C0  6E6E6EC6 D4E2C7F9  8049+         DC    CL32'>>>FMSG91<<<'  eye catcher
000485C8  F14C4C4C 40404040
000485D0  40404040 40404040
000485D8  40404040 40404040
000485E0                     8050+FMSG91   DS    0C
000485E0  86818993 858440D7  8051          DC    C'failed PLO-Extension '
000485E8  D3D660C5 A7A38595
000485F0  A2899695 40
000485F5  C3969497 81998540  8052          DC    C'Compare and Swap and Double Store (CSDSTO: 28) : EQ'
000485FD  81958440 E2A68197
00048605  40819584 40C496A4
0004860D  82938540 E2A39699
00048615  85404DC3 E2C4E2E3
0004861D  D67A40F2 F85D407A
00048625  40C5D8
00048628  40404040 40404040  8053          DC    CL64' '
00048630  40404040 40404040
00048638  40404040 40404040
00048640  40404040 40404040
00048648  40404040 40404040
00048650  40404040 40404040
00048658  40404040 40404040
00048660  40404040 40404040
                             8054          TSTEND
00048668  C5D5C440 968640A3  8055+         DC    CL32'END of test: 91 <<<'
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00048670  85A2A37A 40F9F140
00048678  4C4C4C40 40404040
00048680  40404040 40404040
                                                     8056
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    8058 *=====================================================================
                                                    8059 *       Compare and Swap and Store Tests
                                                    8060 *=====================================================================
                                                    8061
                                                    8062 * ---------------------------------------------------------------------
                                                    8063 * Test Compare and Swap and Store (CSSTO: 27) instruction : NE
                                                    8064 * ---------------------------------------------------------------------
                                                    8065          TSTHDR 1,99,26           Op1-C != OP2 (cc=1, UR=99)
00048688                          00048688  00048800  8066+          ORG   PLOETST+(128*K)+54*(3*K)
00048800  E2E3C1D9 E3409686                         8067+          DC    CL32'START of test: 92'
00048808  40A385A2 A37A40F9
00048810  F2404040 40404040
00048818  40404040 40404040
00048820                          00048820          8068+          USING *,R5               base for test data and test routine
00048820                          00000000          8069+          USING PLOETEST,R5
00048820  0004884C                                  8070+T92       DC    A(CODE92)          address of test code
00048824  005C                                      8071+          DC    H'92'              test number
00048826  00                                        8072+          DC    X'00'
00048827  1A                                        8073+          DC    HL1'26'            Function
00048828  63                                        8074+          DC    HL1'99'            expected unequal results (UR)
00048829  00                                        8075+          DC    HL1'00'            found UR
0004882A  01                                        8076+          DC    HL1'1'             cc
0004882B  0B                                        8077+          DC    HL1'11'            not expected CC mask
0004882C  00                                        8078+          DS    X                  extracted cc
00048830  00000000 00000000                         8079+          DS    2F                 extract PSW after test (has CC)
00048838  000491E0                                  8080+          DC    A(FMSG92)          address of test fail message
0004883C  00048F40                                  8081+          DC    A(OPSRC92)         address of operand source
00048840  00048E20                                  8082+          DC    A(OPDST92)         address of operand destination
00048844  00049060                                  8083+          DC    A(PL92)            address of parameter list
00048848  00004214                                  8084+          DC    A(RANDX+92)        address of random source data
                                                    8085
                                                    8086          TSTCODE
0004884C                                            8087+CODE92    DS    0F
                                                    8088 *                                 load random source
0004884C  E310 501C 0014                 0000001C   8089          LGF   R1,OPSRCA          Op1-C, Op3, Op1-R
00048852  E320 5028 0014                 00000028   8090          LGF   R2,RANDXA
00048858  E330 5024 0014                 00000024   8091          LGF   R3,PLA
0004885E  E340 5020 0014                 00000020   8092          LGF   R4,OPREPA          OP2, Op4
                                                    8093 *
                                                    8094 * Compare and Swap and Store (CSSTO: 27)
                                                    8095 * Build Parambter List
                                                    8096 *
00048864  D25F 1000 2000       00000000  00000000  8097          MVC   0(96,R1),0(R2)     load Op1-C, Op3, Op1-R
0004886A  D21F 3000 1000       00000000  00000000  8098          MVC   0(32,R3),0(R1)     copy to PL OP1-C
00048870  D21F 3050 1020       00000050  00000020  8099          MVC   80(32,R3),32(R1)   copy to PL OP3
00048876  D21F 3020 1060       00000020  00000060  8100          MVC   32(32,R3),96(R1)   copy to PL OP1-R
                                                    8101
0004887C  D23F 4000 2080       00000000  00000080  8102          MVC   0(64,R4),128(R2)   load initial Op2, Op4
                                                    8103
                                                    8104 ****
                                                    8105 * force a failure by changing the Op1-c
                                                    8106 ****
00048882  D203 3000 2021       00000000  00000021  8107          MVC   0(4,R3),33(R2)
                                                    8108 ****
00048888  B982 0000                                 8109          XGR   R0,R0
0004888C  4104 0020                      00000020  8110          LA    R0,32(R4)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00048890  E303 0048 0024                 00000048  8111           STG   R0,72(R3)           load OP4 address in PL
                                                   8112
00048896  B982 0000                              8113           XGR   R0,R0
0004889A  4100 001B                 0000001B     8114           LA    R0,27
0004889E  EE00 4000 3000      00000000  00000000  8115           PLO   0,0(R4),0,0(R3)     C/S Store (CSSTO: 27)
                                                   8116
000488A4  B98D 0020                              8117           EPSW  R2,R0               extract psw
000488A8  5020 5010                 00000010     8118           ST    R2,CCPSW              to save CC
000488AC  E310 500B 0076            0000000B     8119           LB    R1,CCMASK           (failure CC mask)
000488B2  8910 0004                 00000004     8120           SLL   R1,4                (shift to BC instr CC position)
000488B6  4410 50C2                 000488E2     8121           EX    R1,F271TSTCC          fail if...
                                                   8122 *
                                                   8123 * compare results
                                                   8124 *
                            000488BA 00000001      8125 F271CKRES EQU   *
                                                   8126 *                                  reset adresses
000488BA  E310 501C 0014            0000001C     8127           LGF   R1,OPSRCA           Op1-C, Op3, Op1-R
000488C0  E320 5028 0014            00000028     8128           LGF   R2,RANDXA
000488C6  E330 5024 0014            00000024     8129           LGF   R3,PLA
000488CC  E340 5020 0014            00000020     8130           LGF   R4,OPREPA           OP2, Op4
                                                   8131
000488D2  D51F 4000 3000      00000000  00000000  8132           CLC   0(32,R4),0(R3)      Op2 == OP1C?
000488D8  478B 0000                 00000000     8133           BE    0(R11)              same, return
                                                   8134
                            000488DC 00000001      8135 F271XX   EQU   *
000488DC  45F0 83E8                 000023E8     8136           BAL   R15,FAILMSG
000488E0  07FB                                   8137           BR    R11
                                                   8138
000488E2  4700 50C6                 000488E6     8139 F271TSTCC BC    0,F271CCMSG         fail if unexpected condition code
                            000488E6 00000001      8140 F271CCMSG EQU   *
000488E6  45F0 8196                 00002196     8141           BAL   R15,CCMSG
000488EA  47F0 509A                 000488BA     8142           B     F271CKRES
                                                   8143
                                                   8144           TSTDATA
000488EE                    000488EE 00048E00      8145+          ORG   PLOETST+(128*K)+54*(3*K)+(K+512)
                                                   8146           TSTLBL OPDST              OP2, Op4
00048E00  6E6E6ED6 D7C4E2E3                        8147+          DC    CL32'>>>OPDST92<<<' eye catcher
00048E08  F9F24C4C 4C404040
00048E10  40404040 40404040
00048E18  40404040 40404040
00048E20                                         8148+OPDST92  DS    0C
00048E20  00000000 00000000                      8149           DC    XL256'0'
00048E28  00000000 00000000
00048E30  00000000 00000000
00048E38  00000000 00000000
00048E40  00000000 00000000
00048E48  00000000 00000000
00048E50  00000000 00000000
00048E58  00000000 00000000
00048E60  00000000 00000000
00048E68  00000000 00000000
00048E70  00000000 00000000
00048E78  00000000 00000000
00048E80  00000000 00000000
00048E88  00000000 00000000
00048E90  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00048E98  00000000 00000000
00048EA0  00000000 00000000
00048EA8  00000000 00000000
00048EB0  00000000 00000000
00048EB8  00000000 00000000
00048EC0  00000000 00000000
00048EC8  00000000 00000000
00048ED0  00000000 00000000
00048ED8  00000000 00000000
00048EE0  00000000 00000000
00048EE8  00000000 00000000
00048EF0  00000000 00000000
00048EF8  00000000 00000000
00048F00  00000000 00000000
00048F08  00000000 00000000
00048F10  00000000 00000000
00048F18  00000000 00000000
                                        8150
                                        8151          TSTLBL OPSRC              Op1-C, Op3, Op1-R
00048F20  6E6E6ED6 D7E2D9C3             8152+         DC    CL32'>>>OPSRC92<<<' eye catcher
00048F28  F9F24C4C 4C404040
00048F30  40404040 40404040
00048F38  40404040 40404040
00048F40                                8153+OPSRC92  DS    0C
00048F40  00000000 00000000             8154          DC    XL256'0'
00048F48  00000000 00000000
00048F50  00000000 00000000
00048F58  00000000 00000000
00048F60  00000000 00000000
00048F68  00000000 00000000
00048F70  00000000 00000000
00048F78  00000000 00000000
00048F80  00000000 00000000
00048F88  00000000 00000000
00048F90  00000000 00000000
00048F98  00000000 00000000
00048FA0  00000000 00000000
00048FA8  00000000 00000000
00048FB0  00000000 00000000
00048FB8  00000000 00000000
00048FC0  00000000 00000000
00048FC8  00000000 00000000
00048FD0  00000000 00000000
00048FD8  00000000 00000000
00048FE0  00000000 00000000
00048FE8  00000000 00000000
00048FF0  00000000 00000000
00048FF8  00000000 00000000
00049000  00000000 00000000
00049008  00000000 00000000
00049010  00000000 00000000
00049018  00000000 00000000
00049020  00000000 00000000
00049028  00000000 00000000
00049030  00000000 00000000
00049038  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   8155
                                                   8156          TSTLBL PL                 Parameter List
00049040  6E6E6ED7 D3F9F24C                        8157+         DC    CL32'>>>PL92<<<'    eye catcher
00049048  4C4C4040 40404040
00049050  40404040 40404040
00049058  40404040 40404040
00049060                                           8158+PL92     DS    0C
00049060  00000000 00000000                        8159          DC    XL352'00'
00049068  00000000 00000000
00049070  00000000 00000000
00049078  00000000 00000000
00049080  00000000 00000000
00049088  00000000 00000000
00049090  00000000 00000000
00049098  00000000 00000000
000490A0  00000000 00000000
000490A8  00000000 00000000
000490B0  00000000 00000000
000490B8  00000000 00000000
000490C0  00000000 00000000
000490C8  00000000 00000000
000490D0  00000000 00000000
000490D8  00000000 00000000
000490E0  00000000 00000000
000490E8  00000000 00000000
000490F0  00000000 00000000
000490F8  00000000 00000000
00049100  00000000 00000000
00049108  00000000 00000000
00049110  00000000 00000000
00049118  00000000 00000000
00049120  00000000 00000000
00049128  00000000 00000000
00049130  00000000 00000000
00049138  00000000 00000000
00049140  00000000 00000000
00049148  00000000 00000000
00049150  00000000 00000000
00049158  00000000 00000000
00049160  00000000 00000000
00049168  00000000 00000000
00049170  00000000 00000000
00049178  00000000 00000000
00049180  00000000 00000000
00049188  00000000 00000000
00049190  00000000 00000000
00049198  00000000 00000000
000491A0  00000000 00000000
000491A8  00000000 00000000
000491B0  00000000 00000000
000491B8  00000000 00000000
                                                   8160
                                                   8161          TSTLBL FMSG               fail message
000491C0  6E6E6EC6 D4E2C7F9                        8162+         DC    CL32'>>>FMSG92<<<'  eye catcher
000491C8  F24C4C4C 40404040
000491D0  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000491D8  40404040 40404040
000491E0                                          8163+FMSG92   DS    0C
000491E0  86818993 858440D7                       8164          DC    C'failed PLO-Extension '
000491E8  D3D660C5 A7A38595
000491F0  A2899695 40
000491F5  C3969497 81998540                       8165          DC    C'Compare and Swap and Store (CSSTO: 27) : NE'
000491FD  81958440 E2A68197
00049205  40819584 40E2A396
0004920D  9985404D C3E2E2E3
00049215  D67A40F2 F75D407A
0004921D  40D5C5
00049220  40404040 40404040                       8166          DC    CL64' '
00049228  40404040 40404040
00049230  40404040 40404040
00049238  40404040 40404040
00049240  40404040 40404040
00049248  40404040 40404040
00049250  40404040 40404040
00049258  40404040 40404040
                                                  8167          TSTEND
00049260  C5D5C440 968640A3                       8168+         DC    CL32'END of test: 92 <<<'
00049268  85A2A37A 40F9F240
00049270  4C4C4C40 40404040
00049278  40404040 40404040
                                                  8169
                                                  8170 * ------------------------------------------------------------------
                                                  8171 * Test Compare and Swap and Store (CSSTO: 27) instruction : EQ
                                                  8172 * ------------------------------------------------------------------
                                                  8173          TSTHDR 0,99,27            Op1-C == OP2 (cc=0, UR=99)
00049280                          00049280  00049400  8174+         ORG   PLOETST+(128*K)+55*(3*K)
00049400  E2E3C1D9 E3409686                       8175+         DC    CL32'START of test: 93'
00049408  40A385A2 A37A40F9
00049410  F3404040 40404040
00049418  40404040 40404040
00049420                          00049420          8176+         USING *,R5                base for test data and test routine
00049420                          00000000          8177+         USING PLOETEST,R5
00049420  0004944C                                8178+T93      DC    A(CODE93)           address of test code
00049424  005D                                    8179+         DC    H'93'               test number
00049426  00                                      8180+         DC    X'00'
00049427  1B                                      8181+         DC    HL1'27'             Function
00049428  63                                      8182+         DC    HL1'99'             expected unequal results (UR)
00049429  00                                      8183+         DC    HL1'00'             found UR
0004942A  00                                      8184+         DC    HL1'0'              cc
0004942B  07                                      8185+         DC    HL1'7'              not expected CC mask
0004942C  00                                      8186+         DS    X                   extracted cc
00049430  00000000 00000000                       8187+         DS    2F                  extract PSW after test (has CC)
00049438  00049DE0                                8188+         DC    A(FMSG93)           address of test fail message
0004943C  00049B40                                8189+         DC    A(OPSRC93)          address of operand source
00049440  00049A20                                8190+         DC    A(OPDST93)          address of operand destination
00049444  00049C60                                8191+         DC    A(PL93)             address of parameter list
00049448  00004215                                8192+         DC    A(RANDX+93)         address of random source data
                                                  8193
                                                  8194          TSTCODE
0004944C                                          8195+CODE93   DS    0F
                                                  8196 *                                  load random source
0004944C  E310 501C 0014                0000001C  8197          LGF   R1,OPSRCA           Op1-C, Op3, Op1-R
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00049452  E320 5028 0014               00000028  8198          LGF   R2,RANDXA
00049458  E330 5024 0014               00000024  8199          LGF   R3,PLA
0004945E  E340 5020 0014               00000020  8200          LGF   R4,OPREPA           OP2, Op4
                                                 8201 *
                                                 8202 * Compare and Swap and Store (CSSTO: 27)
                                                 8203 * Build Parambter List
                                                 8204 *
00049464  D25F 1000 2000      00000000  00000000  8205          MVC   0(96,R1),0(R2)      load Op1-C, Op3, Op1-R
0004946A  D21F 3000 1000      00000000  00000000  8206          MVC   0(32,R3),0(R1)      copy to PL OP1-C
00049470  D21F 3050 1020      00000050  00000020  8207          MVC   80(32,R3),32(R1)    copy to PL OP3
00049476  D21F 3020 1060      00000020  00000060  8208          MVC   32(32,R3),96(R1)    copy to PL OP1-R
                                                 8209
0004947C  D23F 4000 2080      00000000  00000080  8210          MVC   0(64,R4),128(R2)    load initial Op2, Op4
                                                 8211
                                                 8212 ****
                                                 8213 * Force a EQ by changing the Op1-C == Op2
                                                 8214 ****
00049482  D21F 3000 4000      00000000  00000000  8215          MVC   0(32,R3),0(R4)
                                                 8216 ****
00049488  B982 0000                             8217          XGR   R0,R0
0004948C  4104 0020                     00000020  8218          LA    R0,32(R4)
00049490  E303 0048 0024                00000048  8219          STG   R0,72(R3)           load OP4 address in PL
                                                 8220
00049496  4104 0040                     00000040  8221          LA    R0,64(R4)
0004949A  E303 0068 0024                00000068  8222          STG   R0,104(R3)          load OP6 address in PL
                                                 8223
000494A0  B982 0000                             8224          XGR   R0,R0
000494A4  4100 001B                     0000001B  8225          LA    R0,27
000494A8  EE00 4000 3000      00000000  00000000  8226          PLO   0,0(R4),0,0(R3)     C/S Store (CSSTO: 27)
                                                 8227
000494AE  B98D 0020                             8228          EPSW  R2,R0               extract psw
000494B2  5020 5010                     00000010  8229          ST    R2,CCPSW              to save CC
000494B6  E310 500B 0076                0000000B  8230          LB    R1,CCMASK           (failure CC mask)
000494BC  8910 0004                     00000004  8231          SLL   R1,4                (shift to BC instr CC position)
000494C0  4410 50D6                     000494F6  8232          EX    R1,F272TSTCC            fail if...
                                                 8233 *
                                                 8234 * compare results
                                                 8235 *
                                000494C4 00000001 8236 F272CKRES EQU   *
                                                 8237 *                                  reset adresses
000494C4  E310 501C 0014                0000001C  8238          LGF   R1,OPSRCA           Op1-C, Op3, Op1-R
000494CA  E320 5028 0014                00000028  8239          LGF   R2,RANDXA
000494D0  E330 5024 0014                00000024  8240          LGF   R3,PLA
000494D6  E340 5020 0014                00000020  8241          LGF   R4,OPREPA           OP2, Op4
                                                 8242
000494DC  D51F 4020 3050      00000020  00000050  8243          CLC   32(32,R4),80(R3)    OP4 ==  PL Op3
000494E2  4770 50D0                     000494F0  8244          BNE   F272XX                different, error
000494E6  D51F 4000 3020      00000000  00000020  8245          CLC   0(32,R4),32(R3)     OP2 ==  PL Op-R
000494EC  478B 0000                     00000000  8246          BE    0(R11)                same, return
                                                 8247
                                000494F0 00000001 8248 F272XX   EQU   *
000494F0  45F0 83E8                     000023E8  8249          BAL   R15,FAILMSG
000494F4  07FB                                   8250          BR    R11
                                                 8251
000494F6  4700 50DA                     000494FA  8252 F272TSTCC BC    0,F272CCMSG        fail if unexpected condition code
                                000494FA 00000001 8253 F272CCMSG EQU   *
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
000494FA  45F0 8196                       00002196  8254           BAL   R15,CCMSG
000494FE  47F0 50A4                       000494C4  8255           B     F272CKRES
                                                    8256
                                                    8257           TSTDATA
00049502                        00049502  00049A00  8258+          ORG   PLOETST+(128*K)+55*(3*K)+(K+512)
                                                    8259           TSTLBL OPDST               OP2, Op4
00049A00  6E6E6ED6 D7C4E2E3                         8260+          DC    CL32'>>>OPDST93<<<' eye catcher
00049A08  F9F34C4C 4C404040
00049A10  40404040 40404040
00049A18  40404040 40404040
00049A20                                            8261+OPDST93   DS    0C
00049A20  00000000 00000000                         8262           DC    XL256'0'
00049A28  00000000 00000000
00049A30  00000000 00000000
00049A38  00000000 00000000
00049A40  00000000 00000000
00049A48  00000000 00000000
00049A50  00000000 00000000
00049A58  00000000 00000000
00049A60  00000000 00000000
00049A68  00000000 00000000
00049A70  00000000 00000000
00049A78  00000000 00000000
00049A80  00000000 00000000
00049A88  00000000 00000000
00049A90  00000000 00000000
00049A98  00000000 00000000
00049AA0  00000000 00000000
00049AA8  00000000 00000000
00049AB0  00000000 00000000
00049AB8  00000000 00000000
00049AC0  00000000 00000000
00049AC8  00000000 00000000
00049AD0  00000000 00000000
00049AD8  00000000 00000000
00049AE0  00000000 00000000
00049AE8  00000000 00000000
00049AF0  00000000 00000000
00049AF8  00000000 00000000
00049B00  00000000 00000000
00049B08  00000000 00000000
00049B10  00000000 00000000
00049B18  00000000 00000000
                                                    8263
                                                    8264           TSTLBL OPSRC               Op1-C, Op3, Op1-R
00049B20  6E6E6ED6 D7E2D9C3                         8265+          DC    CL32'>>>OPSRC93<<<' eye catcher
00049B28  F9F34C4C 4C404040
00049B30  40404040 40404040
00049B38  40404040 40404040
00049B40                                            8266+OPSRC93   DS    0C
00049B40  00000000 00000000                         8267           DC    XL256'0'
00049B48  00000000 00000000
00049B50  00000000 00000000
00049B58  00000000 00000000
00049B60  00000000 00000000
00049B68  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00049B70  00000000 00000000
00049B78  00000000 00000000
00049B80  00000000 00000000
00049B88  00000000 00000000
00049B90  00000000 00000000
00049B98  00000000 00000000
00049BA0  00000000 00000000
00049BA8  00000000 00000000
00049BB0  00000000 00000000
00049BB8  00000000 00000000
00049BC0  00000000 00000000
00049BC8  00000000 00000000
00049BD0  00000000 00000000
00049BD8  00000000 00000000
00049BE0  00000000 00000000
00049BE8  00000000 00000000
00049BF0  00000000 00000000
00049BF8  00000000 00000000
00049C00  00000000 00000000
00049C08  00000000 00000000
00049C10  00000000 00000000
00049C18  00000000 00000000
00049C20  00000000 00000000
00049C28  00000000 00000000
00049C30  00000000 00000000
00049C38  00000000 00000000
                             8268
                             8269          TSTLBL PL                 Parameter List
00049C40  6E6E6ED7 D3F9F34C  8270+         DC    CL32'>>>PL93<<<'    eye catcher
00049C48  4C4C4040 40404040
00049C50  40404040 40404040
00049C58  40404040 40404040
00049C60                     8271+PL93     DS    0C
00049C60  00000000 00000000  8272          DC    XL352'00'
00049C68  00000000 00000000
00049C70  00000000 00000000
00049C78  00000000 00000000
00049C80  00000000 00000000
00049C88  00000000 00000000
00049C90  00000000 00000000
00049C98  00000000 00000000
00049CA0  00000000 00000000
00049CA8  00000000 00000000
00049CB0  00000000 00000000
00049CB8  00000000 00000000
00049CC0  00000000 00000000
00049CC8  00000000 00000000
00049CD0  00000000 00000000
00049CD8  00000000 00000000
00049CE0  00000000 00000000
00049CE8  00000000 00000000
00049CF0  00000000 00000000
00049CF8  00000000 00000000
00049D00  00000000 00000000
00049D08  00000000 00000000
00049D10  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00049D18  00000000 00000000
00049D20  00000000 00000000
00049D28  00000000 00000000
00049D30  00000000 00000000
00049D38  00000000 00000000
00049D40  00000000 00000000
00049D48  00000000 00000000
00049D50  00000000 00000000
00049D58  00000000 00000000
00049D60  00000000 00000000
00049D68  00000000 00000000
00049D70  00000000 00000000
00049D78  00000000 00000000
00049D80  00000000 00000000
00049D88  00000000 00000000
00049D90  00000000 00000000
00049D98  00000000 00000000
00049DA0  00000000 00000000
00049DA8  00000000 00000000
00049DB0  00000000 00000000
00049DB8  00000000 00000000
                                      8273
                                      8274          TSTLBL FMSG                fail message
00049DC0  6E6E6EC6 D4E2C7F9           8275+         DC    CL32'>>>FMSG93<<<'   eye catcher
00049DC8  F34C4C4C 40404040
00049DD0  40404040 40404040
00049DD8  40404040 40404040
00049DE0                              8276+FMSG93   DS    0C
00049DE0  86818993 858440D7           8277          DC    C'failed PLO-Extension '
00049DE8  D3D660C5 A7A38595
00049DF0  A2899695 40
00049DF5  C3969497 81998540           8278          DC    C'Compare and Swap and Store (CSSTO: 27) : EQ'
00049DFD  81958440 E2A68197
00049E05  40819584 40E2A396
00049E0D  9985404D C3E2E2E3
00049E15  D67A40F2 F75D407A
00049E1D  40C5D8
00049E20  40404040 40404040           8279          DC    CL64' '
00049E28  40404040 40404040
00049E30  40404040 40404040
00049E38  40404040 40404040
00049E40  40404040 40404040
00049E48  40404040 40404040
00049E50  40404040 40404040
00049E58  40404040 40404040
                                      8280          TSTEND
00049E60  C5D5C440 968640A3           8281+         DC    CL32'END of test: 93 <<<'
00049E68  85A2A37A 40F9F340
00049E70  4C4C4C40 40404040
00049E78  40404040 40404040
                                      8282
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   8284 *=====================================================================
                                                   8285 *       Double Compare and Swap Tests
                                                   8286 *=====================================================================
                                                   8287
                                                   8288 * -------------------------------------------------------------------
                                                   8289 * Test Double Compare and Swap (DCSO: 26) instruction : NE Op1!=Op2
                                                   8290 * -------------------------------------------------------------------
                                                   8291          TSTHDR 1,99,26           Op1-C != OP2 (cc=1, UR=99)
00049E80                        00049E80  0004A000  8292+         ORG   PLOETST+(128*K)+56*(3*K)
0004A000  E2E3C1D9 E3409686                        8293+         DC    CL32'START of test: 94'
0004A008  40A385A2 A37A40F9
0004A010  F4404040 40404040
0004A018  40404040 40404040
0004A020                        0004A020          8294+         USING *,R5               base for test data and test routine
0004A020                        00000000          8295+         USING PLOETEST,R5
0004A020  0004A04C                                8296+T94      DC    A(CODE94)          address of test code
0004A024  005E                                    8297+         DC    H'94'              test number
0004A026  00                                      8298+         DC    X'00'
0004A027  1A                                      8299+         DC    HL1'26'            Function
0004A028  63                                      8300+         DC    HL1'99'            expected unequal results (UR)
0004A029  00                                      8301+         DC    HL1'00'            found UR
0004A02A  01                                      8302+         DC    HL1'1'             cc
0004A02B  0B                                      8303+         DC    HL1'11'            not expected CC mask
0004A02C  00                                      8304+         DS    X                  extracted cc
0004A030  00000000 00000000                       8305+         DS    2F                 extract PSW after test (has CC)
0004A038  0004A9E0                                8306+         DC    A(FMSG94)          address of test fail message
0004A03C  0004A740                                8307+         DC    A(OPSRC94)         address of operand source
0004A040  0004A620                                8308+         DC    A(OPDST94)         address of operand destination
0004A044  0004A860                                8309+         DC    A(PL94)            address of parameter list
0004A048  00004216                                8310+         DC    A(RANDX+94)        address of random source data
                                                   8311
                                                   8312          TSTCODE
0004A04C                                          8313+CODE94   DS    0F
                                                   8314 *                                 load random source
0004A04C  E310 501C 0014                0000001C  8315          LGF   R1,OPSRCA          Op1-C, Op3, Op1-R, Op3-R
0004A052  E320 5028 0014                00000028  8316          LGF   R2,RANDXA
0004A058  E330 5024 0014                00000024  8317          LGF   R3,PLA
0004A05E  E340 5020 0014                00000020  8318          LGF   R4,OPREPA          OP2, Op4
                                                   8319 *
                                                   8320 * Double Compare and Swap (DCSO: 26)
                                                   8321 * Build Parambter List
                                                   8322 *
0004A064  D27F 1000 2000      00000000  00000000  8323          MVC   0(128,R1),0(R2)    load Op1-C, Op3, Op1-R, Op3-R
0004A06A  D21F 3000 1000      00000000  00000000  8324          MVC   0(32,R3),0(R1)     copy to PL OP1-C
0004A070  D21F 3050 1020      00000050  00000020  8325          MVC   80(32,R3),32(R1)   copy to PL OP3
0004A076  D21F 3020 1040      00000020  00000040  8326          MVC   32(32,R3),64(R1)   copy to PL OP1-R
0004A07C  D21F 3070 1060      00000070  00000060  8327          MVC   112(32,R3),96(R1)  copy to PL OP3-R
                                                   8328
0004A082  D23F 4000 2080      00000000  00000080  8329          MVC   0(64,R4),128(R2)   load initial Op2, Op4
                                                   8330
                                                   8331 ****
                                                   8332 * force a failure by changing the Op1-c
                                                   8333 ****
0004A088  D203 3000 2021      00000000  00000021  8334          MVC   0(4,R3),33(R2)
                                                   8335 ****
0004A08E  B982 0000                               8336          XGR   R0,R0
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004A092  4104 0020                      00000020  8337           LA    R0,32(R4)
0004A096  E303 0048 0024                 00000048  8338           STG   R0,72(R3)           load OP4 address in PL
                                                   8339
0004A09C  B982 0000                                8340           XGR   R0,R0
0004A0A0  4100 001A                      0000001A  8341           LA    R0,26
0004A0A4  EE00 4000 3000  00000000       00000000  8342           PLO   0,0(R4),0,0(R3)     Double C/S (DCSO: 26)
                                                   8343
0004A0AA  B98D 0020                                8344           EPSW  R2,R0               extract psw
0004A0AE  5020 5010                      00000010  8345           ST    R2,CCPSW              to save CC
0004A0B2  E310 500B 0076                 0000000B  8346           LB    R1,CCMASK           (failure CC mask)
0004A0B8  8910 0004                      00000004  8347           SLL   R1,4                (shift to BC instr CC position)
0004A0BC  4410 50C8                      0004A0E8  8348           EX    R1,F261TSTCC            fail if...
                                                   8349 *
                                                   8350 * compare results
                                                   8351 *
                         0004A0C0  00000001        8352 F261CKRES EQU   *
                                                   8353 *                                  reset adresses
0004A0C0  E310 501C 0014                 0000001C  8354           LGF   R1,OPSRCA           Op1-C, Op3, Op1-R, Op3-R
0004A0C6  E320 5028 0014                 00000028  8355           LGF   R2,RANDXA
0004A0CC  E330 5024 0014                 00000024  8356           LGF   R3,PLA
0004A0D2  E340 5020 0014                 00000020  8357           LGF   R4,OPREPA           OP2, Op4
                                                   8358
0004A0D8  D51F 4000 3000  00000000       00000000  8359           CLC   0(32,R4),0(R3)      Op2 == OP1C?
0004A0DE  478B 0000                      00000000  8360           BE    0(R11)              same, return
                                                   8361
                         0004A0E2  00000001        8362 F261XX   EQU   *
0004A0E2  45F0 83E8                      000023E8  8363           BAL   R15,FAILMSG
0004A0E6  07FB                                     8364           BR    R11
                                                   8365
0004A0E8  4700 50CC                      0004A0EC  8366 F261TSTCC BC    0,F261CCMSG         fail if unexpected condition code
                         0004A0EC  00000001        8367 F261CCMSG EQU   *
0004A0EC  45F0 8196                      00002196  8368           BAL   R15,CCMSG
0004A0F0  47F0 50A0                      0004A0C0  8369           B     F261CKRES
                                                   8370
                                                   8371           TSTDATA
0004A0F4                  0004A0F4       0004A600  8372+          ORG   PLOETST+(128*K)+56*(3*K)+(K+512)
                                                   8373           TSTLBL OPDST              OP2, Op4
0004A600  6E6E6ED6 D7C4E2E3                        8374+          DC    CL32'>>>OPDST94<<<' eye catcher
0004A608  F9F44C4C 4C404040
0004A610  40404040 40404040
0004A618  40404040 40404040
0004A620                                           8375+OPDST94   DS    0C
0004A620  00000000 00000000                        8376           DC    XL256'0'
0004A628  00000000 00000000
0004A630  00000000 00000000
0004A638  00000000 00000000
0004A640  00000000 00000000
0004A648  00000000 00000000
0004A650  00000000 00000000
0004A658  00000000 00000000
0004A660  00000000 00000000
0004A668  00000000 00000000
0004A670  00000000 00000000
0004A678  00000000 00000000
0004A680  00000000 00000000
0004A688  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0004A690  00000000 00000000
0004A698  00000000 00000000
0004A6A0  00000000 00000000
0004A6A8  00000000 00000000
0004A6B0  00000000 00000000
0004A6B8  00000000 00000000
0004A6C0  00000000 00000000
0004A6C8  00000000 00000000
0004A6D0  00000000 00000000
0004A6D8  00000000 00000000
0004A6E0  00000000 00000000
0004A6E8  00000000 00000000
0004A6F0  00000000 00000000
0004A6F8  00000000 00000000
0004A700  00000000 00000000
0004A708  00000000 00000000
0004A710  00000000 00000000
0004A718  00000000 00000000
                                        8377
                                        8378          TSTLBL OPSRC              Op1-C, Op3, Op1-R, Op3-R
0004A720  6E6E6ED6 D7E2D9C3             8379+         DC    CL32'>>>OPSRC94<<<' eye catcher
0004A728  F9F44C4C 4C404040
0004A730  40404040 40404040
0004A738  40404040 40404040
0004A740                                8380+OPSRC94  DS    0C
0004A740  00000000 00000000             8381          DC    XL256'0'
0004A748  00000000 00000000
0004A750  00000000 00000000
0004A758  00000000 00000000
0004A760  00000000 00000000
0004A768  00000000 00000000
0004A770  00000000 00000000
0004A778  00000000 00000000
0004A780  00000000 00000000
0004A788  00000000 00000000
0004A790  00000000 00000000
0004A798  00000000 00000000
0004A7A0  00000000 00000000
0004A7A8  00000000 00000000
0004A7B0  00000000 00000000
0004A7B8  00000000 00000000
0004A7C0  00000000 00000000
0004A7C8  00000000 00000000
0004A7D0  00000000 00000000
0004A7D8  00000000 00000000
0004A7E0  00000000 00000000
0004A7E8  00000000 00000000
0004A7F0  00000000 00000000
0004A7F8  00000000 00000000
0004A800  00000000 00000000
0004A808  00000000 00000000
0004A810  00000000 00000000
0004A818  00000000 00000000
0004A820  00000000 00000000
0004A828  00000000 00000000
0004A830  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0004A838  00000000 00000000                          8382
                                                     8383          TSTLBL PL                 Parameter List
0004A840  6E6E6ED7 D3F9F44C                          8384+         DC    CL32'>>>PL94<<<'    eye catcher
0004A848  4C4C4040 40404040
0004A850  40404040 40404040
0004A858  40404040 40404040
0004A860                                             8385+PL94     DS    0C
0004A860  00000000 00000000                          8386          DC    XL352'00'
0004A868  00000000 00000000
0004A870  00000000 00000000
0004A878  00000000 00000000
0004A880  00000000 00000000
0004A888  00000000 00000000
0004A890  00000000 00000000
0004A898  00000000 00000000
0004A8A0  00000000 00000000
0004A8A8  00000000 00000000
0004A8B0  00000000 00000000
0004A8B8  00000000 00000000
0004A8C0  00000000 00000000
0004A8C8  00000000 00000000
0004A8D0  00000000 00000000
0004A8D8  00000000 00000000
0004A8E0  00000000 00000000
0004A8E8  00000000 00000000
0004A8F0  00000000 00000000
0004A8F8  00000000 00000000
0004A900  00000000 00000000
0004A908  00000000 00000000
0004A910  00000000 00000000
0004A918  00000000 00000000
0004A920  00000000 00000000
0004A928  00000000 00000000
0004A930  00000000 00000000
0004A938  00000000 00000000
0004A940  00000000 00000000
0004A948  00000000 00000000
0004A950  00000000 00000000
0004A958  00000000 00000000
0004A960  00000000 00000000
0004A968  00000000 00000000
0004A970  00000000 00000000
0004A978  00000000 00000000
0004A980  00000000 00000000
0004A988  00000000 00000000
0004A990  00000000 00000000
0004A998  00000000 00000000
0004A9A0  00000000 00000000
0004A9A8  00000000 00000000
0004A9B0  00000000 00000000
0004A9B8  00000000 00000000
                                                     8387
                                                     8388          TSTLBL FMSG               fail message
0004A9C0  6E6E6EC6 D4E2C7F9                          8389+         DC    CL32'>>>FMSG94<<<'  eye catcher
0004A9C8  F44C4C4C 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004A9D0  40404040 40404040
0004A9D8  40404040 40404040
0004A9E0                                         8390+FMSG94   DS    0C
0004A9E0  86818993 858440D7                      8391          DC    C'failed PLO-Extension '
0004A9E8  D3D660C5 A7A38595
0004A9F0  A2899695 40
0004A9F5  C496A482 938540C3                      8392          DC    C'Double Compare and Swap (DCSO: 26) : NE Op1 != Op2'
0004A9FD  96949781 99854081
0004AA05  958440E2 A6819740
0004AA0D  4DC4C3E2 D67A40F2
0004AA15  F65D407A 40D5C540
0004AA1D  D697F140 5A7E40D6
0004AA25  97F2
0004AA27  40404040 40404040                      8393          DC    CL64' '
0004AA2F  40404040 40404040
0004AA37  40404040 40404040
0004AA3F  40404040 40404040
0004AA47  40404040 40404040
0004AA4F  40404040 40404040
0004AA57  40404040 40404040
0004AA5F  40404040 40404040
                                                 8394          TSTEND
0004AA67  C5D5C440 968640A3                      8395+         DC    CL32'END of test: 94 <<<'
0004AA6F  85A2A37A 40F9F440
0004AA77  4C4C4C40 40404040
0004AA7F  40404040 40404040
                                                 8396
                                                 8397 * ------------------------------------------------------------------
                                                 8398 * Test Double Compare and Swap (DCSO: 26) instruction : NE Op3!=Op4
                                                 8399 * ------------------------------------------------------------------
                                                 8400          TSTHDR 1,99,26            Op3 != OP4 (cc=2, UR=99)
0004AA87                    0004AA87  0004AC00   8401+         ORG   PLOETST+(128*K)+57*(3*K)
0004AC00  E2E3C1D9 E3409686                      8402+         DC    CL32'START of test: 95'
0004AC08  40A385A2 A37A40F9
0004AC10  F5404040 40404040
0004AC18  40404040 40404040
0004AC20                    0004AC20             8403+         USING *,R5                base for test data and test routine
0004AC20                    00000000             8404+         USING PLOETEST,R5
0004AC20  0004AC4C                               8405+T95      DC    A(CODE95)           address of test code
0004AC24  005F                                   8406+         DC    H'95'               test number
0004AC26  00                                     8407+         DC    X'00'
0004AC27  1A                                     8408+         DC    HL1'26'             Function
0004AC28  63                                     8409+         DC    HL1'99'             expected unequal results (UR)
0004AC29  00                                     8410+         DC    HL1'00'             found UR
0004AC2A  01                                     8411+         DC    HL1'1'              cc
0004AC2B  0B                                     8412+         DC    HL1'11'             not expected CC mask
0004AC2C  00                                     8413+         DS    X                   extracted cc
0004AC30  00000000 00000000                      8414+         DS    2F                  extract PSW after test (has CC)
0004AC38  0004B5E0                               8415+         DC    A(FMSG95)           address of test fail message
0004AC3C  0004B340                               8416+         DC    A(OPSRC95)          address of operand source
0004AC40  0004B220                               8417+         DC    A(OPDST95)          address of operand destination
0004AC44  0004B460                               8418+         DC    A(PL95)             address of parameter list
0004AC48  00004217                               8419+         DC    A(RANDX+95)         address of random source data
                                                 8420
                                                 8421          TSTCODE
0004AC4C                                         8422+CODE95   DS    0F
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    8423 *                                load random source
0004AC4C  E310 501C 0014                0000001C  8424          LGF   R1,OPSRCA           Op1-C, Op3, Op1-R, Op3-R
0004AC52  E320 5028 0014                00000028  8425          LGF   R2,RANDXA
0004AC58  E330 5024 0014                00000024  8426          LGF   R3,PLA
0004AC5E  E340 5020 0014                00000020  8427          LGF   R4,OPREPA           OP2, Op4
                                                    8428 *
                                                    8429 * Double Compare and Swap (DCSO: 26)
                                                    8430 * Build Parambter List
                                                    8431 *
0004AC64  D27F 1000 2000  00000000      00000000  8432          MVC   0(128,R1),0(R2)     load Op1-C, Op3, Op1-R, Op3-R
0004AC6A  D21F 3000 1000  00000000      00000000  8433          MVC   0(32,R3),0(R1)      copy to PL OP1-C
0004AC70  D21F 3050 1020  00000050      00000020  8434          MVC   80(32,R3),32(R1)    copy to PL OP3
0004AC76  D21F 3020 1040  00000020      00000040  8435          MVC   32(32,R3),64(R1)    copy to PL OP1-R
0004AC7C  D21F 3070 1060  00000070      00000060  8436          MVC   112(32,R3),96(R1)   copy to PL OP3-R
                                                    8437
0004AC82  D23F 4000 2080  00000000      00000080  8438          MVC   0(64,R4),128(R2)    load initial Op2, Op4
                                                    8439
                                                    8440 ****
                                                    8441 * force a failure by changing the Op3-C
                                                    8442 ****
0004AC88  D203 3050 2021  00000050      00000021  8443          MVC   80(4,R3),33(R2)
                                                    8444 ****
0004AC8E  B982 0000                                 8445          XGR   R0,R0
0004AC92  4104 0020                     00000020  8446          LA    R0,32(R4)
0004AC96  E303 0048 0024                00000048  8447          STG   R0,72(R3)           load OP4 address in PL
                                                    8448
0004AC9C  B982 0000                                 8449          XGR   R0,R0
0004ACA0  4100 001A                     0000001A  8450          LA    R0,26
0004ACA4  EE00 4000 3000  00000000      00000000  8451          PLO   0,0(R4),0,0(R3)     Double C/S (DCSO: 26)
                                                    8452
0004ACAA  B98D 0020                                 8453          EPSW  R2,R0               extract psw
0004ACAE  5020 5010                     00000010  8454          ST    R2,CCPSW              to save CC
0004ACB2  E310 500B 0076                0000000B  8455          LB    R1,CCMASK           (failure CC mask)
0004ACB8  8910 0004                     00000004  8456          SLL   R1,4                (shift to BC instr CC position)
0004ACBC  4410 50D2                     0004ACF2  8457          EX    R1,F262TSTCC            fail if...
                                                    8458 *
                                                    8459 * compare results
                                                    8460 *
                          0004ACC0      00000001  8461 F262CKRES EQU   *
                                                    8462 *                                reset adresses
0004ACC0  E310 501C 0014                0000001C  8463          LGF   R1,OPSRCA           Op1-C, Op3, Op1-R, Op3-R
0004ACC6  E320 5028 0014                00000028  8464          LGF   R2,RANDXA
0004ACCC  E330 5024 0014                00000024  8465          LGF   R3,PLA
0004ACD2  E340 5020 0014                00000020  8466          LGF   R4,OPREPA           OP2, Op4
                                                    8467
0004ACD8  D51F 4000 3000  00000000      00000000  8468          CLC   0(32,R4),0(R3)      Op2 == OP1C?
0004ACDE  4770 50CC                     0004ACEC  8469          BNE   F262XX
0004ACE2  D51F 3050 4020  00000050      00000020  8470          CLC   80(32,R3),32(R4)    Op3-C == OP4
0004ACE8  477B 0000                     00000000  8471          BNE    0(R11)               NE, matches test Op3 != OP4
                                                    8472
                          0004ACEC      00000001  8473 F262XX   EQU   *
0004ACEC  45F0 83E8                     000023E8  8474          BAL   R15,FAILMSG
0004ACF0  07FB                                    8475          BR    R11
                                                    8476
0004ACF2  4700 50D6                     0004ACF6  8477 F262TSTCC BC    0,F262CCMSG        fail if unexpected condition code
                          0004ACF6      00000001  8478 F262CCMSG EQU   *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004ACF6  45F0 8196                     00002196  8479           BAL   R15,CCMSG
0004ACFA  47F0 50A0                     0004ACC0  8480           B     F262CKRES
                                                  8481
                                                  8482           TSTDATA
0004ACFE                    0004ACFE    0004B200  8483+          ORG   PLOETST+(128*K)+57*(3*K)+(K+512)
                                                  8484           TSTLBL OPDST               OP2, Op4
0004B200  6E6E6ED6 D7C4E2E3                       8485+          DC    CL32'>>>OPDST95<<<' eye catcher
0004B208  F9F54C4C 4C404040
0004B210  40404040 40404040
0004B218  40404040 40404040
0004B220                                          8486+OPDST95   DS    0C
0004B220  00000000 00000000                       8487           DC    XL256'0'
0004B228  00000000 00000000
0004B230  00000000 00000000
0004B238  00000000 00000000
0004B240  00000000 00000000
0004B248  00000000 00000000
0004B250  00000000 00000000
0004B258  00000000 00000000
0004B260  00000000 00000000
0004B268  00000000 00000000
0004B270  00000000 00000000
0004B278  00000000 00000000
0004B280  00000000 00000000
0004B288  00000000 00000000
0004B290  00000000 00000000
0004B298  00000000 00000000
0004B2A0  00000000 00000000
0004B2A8  00000000 00000000
0004B2B0  00000000 00000000
0004B2B8  00000000 00000000
0004B2C0  00000000 00000000
0004B2C8  00000000 00000000
0004B2D0  00000000 00000000
0004B2D8  00000000 00000000
0004B2E0  00000000 00000000
0004B2E8  00000000 00000000
0004B2F0  00000000 00000000
0004B2F8  00000000 00000000
0004B300  00000000 00000000
0004B308  00000000 00000000
0004B310  00000000 00000000
0004B318  00000000 00000000
                                                  8488
                                                  8489           TSTLBL OPSRC               Op1-C, Op3, Op1-R, Op3-R
0004B320  6E6E6ED6 D7E2D9C3                       8490+          DC    CL32'>>>OPSRC95<<<' eye catcher
0004B328  F9F54C4C 4C404040
0004B330  40404040 40404040
0004B338  40404040 40404040
0004B340                                          8491+OPSRC95   DS    0C
0004B340  00000000 00000000                       8492           DC    XL256'0'
0004B348  00000000 00000000
0004B350  00000000 00000000
0004B358  00000000 00000000
0004B360  00000000 00000000
0004B368  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0004B370  00000000 00000000
0004B378  00000000 00000000
0004B380  00000000 00000000
0004B388  00000000 00000000
0004B390  00000000 00000000
0004B398  00000000 00000000
0004B3A0  00000000 00000000
0004B3A8  00000000 00000000
0004B3B0  00000000 00000000
0004B3B8  00000000 00000000
0004B3C0  00000000 00000000
0004B3C8  00000000 00000000
0004B3D0  00000000 00000000
0004B3D8  00000000 00000000
0004B3E0  00000000 00000000
0004B3E8  00000000 00000000
0004B3F0  00000000 00000000
0004B3F8  00000000 00000000
0004B400  00000000 00000000
0004B408  00000000 00000000
0004B410  00000000 00000000
0004B418  00000000 00000000
0004B420  00000000 00000000
0004B428  00000000 00000000
0004B430  00000000 00000000
0004B438  00000000 00000000
                                                8493
                                                8494           TSTLBL PL                  Parameter List
0004B440  6E6E6ED7 D3F9F54C                     8495+          DC    CL32'>>>PL95<<<'     eye catcher
0004B448  4C4C4040 40404040
0004B450  40404040 40404040
0004B458  40404040 40404040
0004B460                                        8496+PL95      DS    0C
0004B460  00000000 00000000                     8497           DC    XL352'00'
0004B468  00000000 00000000
0004B470  00000000 00000000
0004B478  00000000 00000000
0004B480  00000000 00000000
0004B488  00000000 00000000
0004B490  00000000 00000000
0004B498  00000000 00000000
0004B4A0  00000000 00000000
0004B4A8  00000000 00000000
0004B4B0  00000000 00000000
0004B4B8  00000000 00000000
0004B4C0  00000000 00000000
0004B4C8  00000000 00000000
0004B4D0  00000000 00000000
0004B4D8  00000000 00000000
0004B4E0  00000000 00000000
0004B4E8  00000000 00000000
0004B4F0  00000000 00000000
0004B4F8  00000000 00000000
0004B500  00000000 00000000
0004B508  00000000 00000000
0004B510  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0004B518  00000000 00000000
0004B520  00000000 00000000
0004B528  00000000 00000000
0004B530  00000000 00000000
0004B538  00000000 00000000
0004B540  00000000 00000000
0004B548  00000000 00000000
0004B550  00000000 00000000
0004B558  00000000 00000000
0004B560  00000000 00000000
0004B568  00000000 00000000
0004B570  00000000 00000000
0004B578  00000000 00000000
0004B580  00000000 00000000
0004B588  00000000 00000000
0004B590  00000000 00000000
0004B598  00000000 00000000
0004B5A0  00000000 00000000
0004B5A8  00000000 00000000
0004B5B0  00000000 00000000
0004B5B8  00000000 00000000
                                        8498
                                        8499          TSTLBL FMSG                fail message
0004B5C0  6E6E6EC6 D4E2C7F9             8500+         DC    CL32'>>>FMSG95<<<'   eye catcher
0004B5C8  F54C4C4C 40404040
0004B5D0  40404040 40404040
0004B5D8  40404040 40404040
0004B5E0                                8501+FMSG95   DS    0C
0004B5E0  86818993 858440D7             8502          DC    C'failed PLO-Extension '
0004B5E8  D3D660C5 A7A38595
0004B5F0  A2899695 40
0004B5F5  C496A482 938540C3             8503          DC    C'Double Compare and Swap (DCSO: 26) : Op3 != OP4'
0004B5FD  96949781 99854081
0004B605  958440E2 A6819740
0004B60D  4DC4C3E2 D67A40F2
0004B615  F65D407A 40D697F3
0004B61D  405A7E40 D6D7F4
0004B624  40404040 40404040             8504          DC    CL64' '
0004B62C  40404040 40404040
0004B634  40404040 40404040
0004B63C  40404040 40404040
0004B644  40404040 40404040
0004B64C  40404040 40404040
0004B654  40404040 40404040
0004B65C  40404040 40404040
                                        8505          TSTEND
0004B664  C5D5C440 968640A3             8506+         DC    CL32'END of test: 95 <<<'
0004B66C  85A2A37A 40F9F540
0004B674  4C4C4C40 40404040
0004B67C  40404040 40404040
                                        8507
                                        8508 * ------------------------------------------------------------------
                                        8509 * Test Double Compare and Swap (DCSO: 26) instruction : EQ
                                        8510 * ------------------------------------------------------------------
                                        8511          TSTHDR 0,99,28             Op1-C == OP2 && Op3 == OP4
0004B684                  0004B684  0004B800  8512+         ORG   PLOETST+(128*K)+58*(3*K)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004B800  E2E3C1D9 E3409686                        8513+         DC    CL32'START of test: 96'
0004B808  40A385A2 A37A40F9
0004B810  F6404040 40404040
0004B818  40404040 40404040
0004B820                    0004B820              8514+         USING *,R5                base for test data and test routine
0004B820                    00000000              8515+         USING PLOETEST,R5
0004B820  0004B84C                                8516+T96      DC    A(CODE96)           address of test code
0004B824  0060                                    8517+         DC    H'96'               test number
0004B826  00                                      8518+         DC    X'00'
0004B827  1C                                      8519+         DC    HL1'28'             Function
0004B828  63                                      8520+         DC    HL1'99'             expected unequal results (UR)
0004B829  00                                      8521+         DC    HL1'00'             found UR
0004B82A  00                                      8522+         DC    HL1'0'              cc
0004B82B  07                                      8523+         DC    HL1'7'              not expected CC mask
0004B82C  00                                      8524+         DS    X                   extracted cc
0004B830  00000000 00000000                       8525+         DS    2F                  extract PSW after test (has CC)
0004B838  0004C1E0                                8526+         DC    A(FMSG96)           address of test fail message
0004B83C  0004BF40                                8527+         DC    A(OPSRC96)          address of operand source
0004B840  0004BE20                                8528+         DC    A(OPDST96)          address of operand destination
0004B844  0004C060                                8529+         DC    A(PL96)             address of parameter list
0004B848  00004218                                8530+         DC    A(RANDX+96)         address of random source data
                                                  8531 *                                          (cc=0, UR=99)
                                                  8532          TSTCODE
0004B84C                                          8533+CODE96   DS    0F
                                                  8534 *                                    load random source
0004B84C  E310 501C 0014                0000001C  8535          LGF   R1,OPSRCA           Op1-C, Op3, Op1-R, Op3-R
0004B852  E320 5028 0014                00000028  8536          LGF   R2,RANDXA
0004B858  E330 5024 0014                00000024  8537          LGF   R3,PLA
0004B85E  E340 5020 0014                00000020  8538          LGF   R4,OPREPA           OP2, Op4
                                                  8539 *
                                                  8540 * Compare and Swap and Double Store (CSDSTO: 28)
                                                  8541 * Build Parambter List
                                                  8542 *
0004B864  D27F 1000 2000       00000000 00000000  8543          MVC   0(128,R1),0(R2)     load Op1-C, Op3, Op1-R, Op3-R
0004B86A  D21F 3000 1000       00000000 00000000  8544          MVC   0(32,R3),0(R1)      copy to PL OP1-C
0004B870  D21F 3050 1020       00000050 00000020  8545          MVC   80(32,R3),32(R1)    copy to PL OP3
0004B876  D21F 3020 1040       00000020 00000040  8546          MVC   32(32,R3),64(R1)    copy to PL OP1-R
0004B87C  D21F 3070 1060       00000070 00000060  8547          MVC   112(32,R3),96(R1)   copy to PL OP3-R
                                                  8548
0004B882  D23F 4000 2080       00000000 00000080  8549          MVC   0(64,R4),128(R2)    load initial Op2, Op4
                                                  8550
                                                  8551 ****
                                                  8552 * Force a EQ by changing Op1-C == OP2 && Op3 == OP4
                                                  8553 ****
0004B888  D21F 3000 4000       00000000 00000000  8554          MVC   0(32,R3),0(R4)      Op1-C <-- Op2
0004B88E  D21F 3050 4020       00000050 00000020  8555          MVC   80(32,R3),32(R4)    Op3-C <-- Op4
                                                  8556 ****
0004B894  B982 0000                               8557          XGR   R0,R0
0004B898  4104 0020                     00000020  8558          LA    R0,32(R4)
0004B89C  E303 0048 0024                00000048  8559          STG   R0,72(R3)           load OP4 address in PL
                                                  8560
0004B8A2  B982 0000                               8561          XGR   R0,R0
0004B8A6  4100 001A                     0000001A  8562          LA    R0,26
0004B8AA  EE00 4000 3000       00000000 00000000  8563          PLO   0,0(R4),0,0(R3)     C/S Double Store (CSDSTO: 28)
                                                  8564
0004B8B0  B98D 0020                               8565          EPSW  R2,R0               extract psw
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004B8B4  5020 5010                     00000010  8566          ST    R2,CCPSW              to save CC
0004B8B8  E310 500B 0076                0000000B  8567          LB    R1,CCMASK            (failure CC mask)
0004B8BE  8910 0004                     00000004  8568          SLL   R1,4                (shift to BC instr CC position)
0004B8C2  4410 50D8                     0004B8F8  8569          EX    R1,F263TSTCC           fail if...
                                                  8570 *
                                                  8571 * compare results
                                                  8572 *
                          0004B8C6      00000001  8573 F263CKRES EQU   *
                                                  8574 *                                    reset adresses
0004B8C6  E310 501C 0014                0000001C  8575          LGF   R1,OPSRCA            Op1-C, Op3, Op1-R, Op3-R
0004B8CC  E320 5028 0014                00000028  8576          LGF   R2,RANDXA
0004B8D2  E330 5024 0014                00000024  8577          LGF   R3,PLA
0004B8D8  E340 5020 0014                00000020  8578          LGF   R4,OPREPA            OP2, Op4
                                                  8579
0004B8DE  D51F 4020 3070  00000020      00000070  8580          CLC   32(32,R4),112(R3)    OP4 ==  PL Op3-R
0004B8E4  4770 50D2                     0004B8F2  8581          BNE   F263XX                    different, error
0004B8E8  D51F 4000 3020  00000000      00000020  8582          CLC   0(32,R4),32(R3)      OP2 ==  PL Op1-R
0004B8EE  478B 0000                     00000000  8583          BE    0(R11)                    same, return
                                                  8584
                          0004B8F2      00000001  8585 F263XX   EQU   *
0004B8F2  45F0 83E8                     000023E8  8586          BAL   R15,FAILMSG
0004B8F6  07FB                                    8587          BR    R11
                                                  8588
0004B8F8  4700 50DC                     0004B8FC  8589 F263TSTCC BC    0,F263CCMSG          fail if unexpected condition code
                          0004B8FC      00000001  8590 F263CCMSG EQU   *
0004B8FC  45F0 8196                     00002196  8591          BAL   R15,CCMSG
0004B900  47F0 50A6                     0004B8C6  8592          B     F263CKRES
                                                  8593
                                                  8594          TSTDATA
0004B904                  0004B904      0004BE00  8595+          ORG   PLOETST+(128*K)+58*(3*K)+(K+512)
                                                  8596          TSTLBL OPDST               OP2, Op4
0004BE00  6E6E6ED6 D7C4E2E3                       8597+          DC    CL32'>>>OPDST96<<<' eye catcher
0004BE08  F9F64C4C 4C404040
0004BE10  40404040 40404040
0004BE18  40404040 40404040
0004BE20                                          8598+OPDST96  DS    0C
0004BE20  00000000 00000000                       8599          DC    XL256'0'
0004BE28  00000000 00000000
0004BE30  00000000 00000000
0004BE38  00000000 00000000
0004BE40  00000000 00000000
0004BE48  00000000 00000000
0004BE50  00000000 00000000
0004BE58  00000000 00000000
0004BE60  00000000 00000000
0004BE68  00000000 00000000
0004BE70  00000000 00000000
0004BE78  00000000 00000000
0004BE80  00000000 00000000
0004BE88  00000000 00000000
0004BE90  00000000 00000000
0004BE98  00000000 00000000
0004BEA0  00000000 00000000
0004BEA8  00000000 00000000
0004BEB0  00000000 00000000
0004BEB8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0004BEC0  00000000 00000000
0004BEC8  00000000 00000000
0004BED0  00000000 00000000
0004BED8  00000000 00000000
0004BEE0  00000000 00000000
0004BEE8  00000000 00000000
0004BEF0  00000000 00000000
0004BEF8  00000000 00000000
0004BF00  00000000 00000000
0004BF08  00000000 00000000
0004BF10  00000000 00000000
0004BF18  00000000 00000000
                                          8600
                                          8601           TSTLBL OPSRC              Op1-C, Op3, Op1-R, Op3-R
0004BF20  6E6E6ED6 D7E2D9C3               8602+          DC    CL32'>>>OPSRC96<<<' eye catcher
0004BF28  F9F64C4C 4C404040
0004BF30  40404040 40404040
0004BF38  40404040 40404040
0004BF40                                  8603+OPSRC96   DS    0C
0004BF40  00000000 00000000               8604           DC    XL256'0'
0004BF48  00000000 00000000
0004BF50  00000000 00000000
0004BF58  00000000 00000000
0004BF60  00000000 00000000
0004BF68  00000000 00000000
0004BF70  00000000 00000000
0004BF78  00000000 00000000
0004BF80  00000000 00000000
0004BF88  00000000 00000000
0004BF90  00000000 00000000
0004BF98  00000000 00000000
0004BFA0  00000000 00000000
0004BFA8  00000000 00000000
0004BFB0  00000000 00000000
0004BFB8  00000000 00000000
0004BFC0  00000000 00000000
0004BFC8  00000000 00000000
0004BFD0  00000000 00000000
0004BFD8  00000000 00000000
0004BFE0  00000000 00000000
0004BFE8  00000000 00000000
0004BFF0  00000000 00000000
0004BFF8  00000000 00000000
0004C000  00000000 00000000
0004C008  00000000 00000000
0004C010  00000000 00000000
0004C018  00000000 00000000
0004C020  00000000 00000000
0004C028  00000000 00000000
0004C030  00000000 00000000
0004C038  00000000 00000000
                                          8605
                                          8606           TSTLBL PL                 Parameter List
0004C040  6E6E6ED7 D3F9F64C               8607+          DC    CL32'>>>PL96<<<'    eye catcher
0004C048  4C4C4040 40404040
0004C050  40404040 40404040
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0004C058  40404040 40404040
0004C060                                             8608+PL96     DS    0C
0004C060  00000000 00000000                          8609          DC    XL352'00'
0004C068  00000000 00000000
0004C070  00000000 00000000
0004C078  00000000 00000000
0004C080  00000000 00000000
0004C088  00000000 00000000
0004C090  00000000 00000000
0004C098  00000000 00000000
0004C0A0  00000000 00000000
0004C0A8  00000000 00000000
0004C0B0  00000000 00000000
0004C0B8  00000000 00000000
0004C0C0  00000000 00000000
0004C0C8  00000000 00000000
0004C0D0  00000000 00000000
0004C0D8  00000000 00000000
0004C0E0  00000000 00000000
0004C0E8  00000000 00000000
0004C0F0  00000000 00000000
0004C0F8  00000000 00000000
0004C100  00000000 00000000
0004C108  00000000 00000000
0004C110  00000000 00000000
0004C118  00000000 00000000
0004C120  00000000 00000000
0004C128  00000000 00000000
0004C130  00000000 00000000
0004C138  00000000 00000000
0004C140  00000000 00000000
0004C148  00000000 00000000
0004C150  00000000 00000000
0004C158  00000000 00000000
0004C160  00000000 00000000
0004C168  00000000 00000000
0004C170  00000000 00000000
0004C178  00000000 00000000
0004C180  00000000 00000000
0004C188  00000000 00000000
0004C190  00000000 00000000
0004C198  00000000 00000000
0004C1A0  00000000 00000000
0004C1A8  00000000 00000000
0004C1B0  00000000 00000000
0004C1B8  00000000 00000000
                                                     8610
                                                     8611          TSTLBL FMSG                fail message
0004C1C0  6E6E6EC6 D4E2C7F9                          8612+         DC    CL32'>>>FMSG96<<<'   eye catcher
0004C1C8  F64C4C4C 40404040
0004C1D0  40404040 40404040
0004C1D8  40404040 40404040
0004C1E0                                             8613+FMSG96   DS    0C
0004C1E0  86818993 858440D7                          8614          DC    C'failed PLO-Extension '
0004C1E8  D3D660C5 A7A38595
0004C1F0  A2899695 40
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0004C1F5  C496A482 938540C3                         8615          DC    C'Double Compare and Swap (DCSO: 26) : EQ'
0004C1FD  96949781 99854081
0004C205  958440E2 A6819740
0004C20D  4DC4C3E2 D67A40F2
0004C215  F65D407A 40C5D8
0004C21C  40404040 40404040                         8616          DC    CL64' '
0004C224  40404040 40404040
0004C22C  40404040 40404040
0004C234  40404040 40404040
0004C23C  40404040 40404040
0004C244  40404040 40404040
0004C24C  40404040 40404040
0004C254  40404040 40404040
                                                    8617          TSTEND
0004C25C  C5D5C440 968640A3                         8618+         DC    CL32'END of test: 96 <<<'
0004C264  85A2A37A 40F9F640
0004C26C  4C4C4C40 40404040
0004C274  40404040 40404040
                                                    8619
                                                    8620
```

```
 LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                  8622 *=======================================================================
                                                  8623 *       Compare and Swap Tests
                                                  8624 *=======================================================================
                                                  8625
                                                  8626 * ---------------------------------------------------------------------
                                                  8627 * Test Compare and Swap (CSO: 25) instruction : NE Op1-C!=Op2
                                                  8628 * ---------------------------------------------------------------------
                                                  8629          TSTHDR 1,99,26           Op1-C != OP2 (cc=1, UR=99)
0004C27C                          0004C27C  0004C400  8630+         ORG   PLOETST+(128*K)+59*(3*K)
0004C400  E2E3C1D9 E3409686                         8631+         DC    CL32'START of test: 97'
0004C408  40A385A2 A37A40F9
0004C410  F7404040 40404040
0004C418  40404040 40404040
0004C420                          0004C420          8632+         USING *,R5               base for test data and test routine
0004C420                          00000000          8633+         USING PLOETEST,R5
0004C420  0004C44C                                  8634+T97      DC    A(CODE97)          address of test code
0004C424  0061                                      8635+         DC    H'97'              test number
0004C426  00                                        8636+         DC    X'00'
0004C427  1A                                        8637+         DC    HL1'26'            Function
0004C428  63                                        8638+         DC    HL1'99'            expected unequal results (UR)
0004C429  00                                        8639+         DC    HL1'00'            found UR
0004C42A  01                                        8640+         DC    HL1'1'             cc
0004C42B  0B                                        8641+         DC    HL1'11'            not expected CC mask
0004C42C  00                                        8642+         DS    X                  extracted cc
0004C430  00000000 00000000                         8643+         DS    2F                 extract PSW after test (has CC)
0004C438  0004CDE0                                  8644+         DC    A(FMSG97)          address of test fail message
0004C43C  0004CB40                                  8645+         DC    A(OPSRC97)         address of operand source
0004C440  0004CA20                                  8646+         DC    A(OPDST97)         address of operand destination
0004C444  0004CC60                                  8647+         DC    A(PL97)            address of parameter list
0004C448  00004219                                  8648+         DC    A(RANDX+97)        address of random source data
                                                  8649
                                                  8650          TSTCODE
0004C44C                                            8651+CODE97   DS    0F
                                                  8652 *                                   load random source
0004C44C  E310 501C 0014                0000001C  8653          LGF   R1,OPSRCA          Op1-C, Op1-R
0004C452  E320 5028 0014                00000028  8654          LGF   R2,RANDXA
0004C458  E330 5024 0014                00000024  8655          LGF   R3,PLA
0004C45E  E340 5020 0014                00000020  8656          LGF   R4,OPREPA          OP2
                                                  8657 *
                                                  8658 * Compare and Swap (CSO: 25)
                                                  8659 * Build Parambter List
                                                  8660 *
0004C464  D23F 1000 2000  00000000  00000000      8661          MVC   0(64,R1),0(R2)     load Op1-C, Op1-R
0004C46A  D21F 3000 1000  00000000  00000000      8662          MVC   0(32,R3),0(R1)     copy to PL OP1-C
0004C470  D21F 3020 1020  00000020  00000020      8663          MVC   32(32,R3),32(R1)   copy to PL OP1-R
                                                  8664
0004C476  D21F 4000 2080  00000000  00000080      8665          MVC   0(32,R4),128(R2)   load initial Op2
                                                  8666
                                                  8667 ****
                                                  8668 * force a NE by changing the Op1-c
                                                  8669 ****
0004C47C  D203 3000 2021  00000000  00000021      8670          MVC   0(4,R3),33(R2)
                                                  8671 ****
0004C482  B982 0000                               8672          XGR   R0,R0
0004C486  4100 0019                     00000019  8673          LA    R0,25
0004C48A  EE00 4000 3000  00000000  00000000      8674          PLO   0,0(R4),0,0(R3)    C/S (CSO: 25)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    8675
0004C490  B98D 0020                                 8676          EPSW  R2,R0              extract psw
0004C494  5020 5010                       00000010  8677          ST    R2,CCPSW              to save CC
0004C498  E310 500B 0076                  0000000B  8678          LB    R1,CCMASK             (failure CC mask)
0004C49E  8910 0004                       00000004  8679          SLL   R1,4                  (shift to BC instr CC position)
0004C4A2  4410 50AE                       0004C4CE  8680          EX    R1,F251TSTCC          fail if...
                                                    8681 *
                                                    8682 * compare results
                                                    8683 *
                            0004C4A6      00000001  8684 F251CKRES EQU   *
                                                    8685 *                                    reset adresses
0004C4A6  E310 501C 0014                  0000001C  8686          LGF   R1,OPSRCA             Op1-C, Op1-R
0004C4AC  E320 5028 0014                  00000028  8687          LGF   R2,RANDXA
0004C4B2  E330 5024 0014                  00000024  8688          LGF   R3,PLA
0004C4B8  E340 5020 0014                  00000020  8689          LGF   R4,OPREPA             OP2
                                                    8690
0004C4BE  D51F 4000 3000  00000000        00000000  8691          CLC   0(32,R4),0(R3)        Op2 == OP1-C?
0004C4C4  478B 0000                       00000000  8692          BE    0(R11)                same, return
                                                    8693
                            0004C4C8      00000001  8694 F251XX   EQU   *
0004C4C8  45F0 83E8                       000023E8  8695          BAL   R15,FAILMSG
0004C4CC  07FB                                      8696          BR    R11
                                                    8697
0004C4CE  4700 50B2                       0004C4D2  8698 F251TSTCC BC    0,F251CCMSG           fail if unexpected condition code
                            0004C4D2      00000001  8699 F251CCMSG EQU   *
0004C4D2  45F0 8196                       00002196  8700          BAL   R15,CCMSG
0004C4D6  47F0 5086                       0004C4A6  8701          B     F251CKRES
                                                    8702
                                                    8703          TSTDATA
0004C4DA                    0004C4DA      0004CA00  8704+         ORG   PLOETST+(128*K)+59*(3*K)+(K+512)
                                                    8705          TSTLBL OPDST                OP2
0004CA00  6E6E6ED6 D7C4E2E3                         8706+         DC    CL32'>>>OPDST97<<<' eye catcher
0004CA08  F9F74C4C 4C404040
0004CA10  40404040 40404040
0004CA18  40404040 40404040
0004CA20                                            8707+OPDST97  DS    0C
0004CA20  00000000 00000000                         8708          DC    XL256'0'
0004CA28  00000000 00000000
0004CA30  00000000 00000000
0004CA38  00000000 00000000
0004CA40  00000000 00000000
0004CA48  00000000 00000000
0004CA50  00000000 00000000
0004CA58  00000000 00000000
0004CA60  00000000 00000000
0004CA68  00000000 00000000
0004CA70  00000000 00000000
0004CA78  00000000 00000000
0004CA80  00000000 00000000
0004CA88  00000000 00000000
0004CA90  00000000 00000000
0004CA98  00000000 00000000
0004CAA0  00000000 00000000
0004CAA8  00000000 00000000
0004CAB0  00000000 00000000
0004CAB8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0004CAC0  00000000 00000000
0004CAC8  00000000 00000000
0004CAD0  00000000 00000000
0004CAD8  00000000 00000000
0004CAE0  00000000 00000000
0004CAE8  00000000 00000000
0004CAF0  00000000 00000000
0004CAF8  00000000 00000000
0004CB00  00000000 00000000
0004CB08  00000000 00000000
0004CB10  00000000 00000000
0004CB18  00000000 00000000
                                        8709
                                        8710          TSTLBL OPSRC              Op1-C, Op1-R
0004CB20  6E6E6ED6 D7E2D9C3             8711+         DC    CL32'>>>OPSRC97<<<' eye catcher
0004CB28  F9F74C4C 4C404040
0004CB30  40404040 40404040
0004CB38  40404040 40404040
0004CB40                                8712+OPSRC97  DS    0C
0004CB40  00000000 00000000             8713          DC    XL256'0'
0004CB48  00000000 00000000
0004CB50  00000000 00000000
0004CB58  00000000 00000000
0004CB60  00000000 00000000
0004CB68  00000000 00000000
0004CB70  00000000 00000000
0004CB78  00000000 00000000
0004CB80  00000000 00000000
0004CB88  00000000 00000000
0004CB90  00000000 00000000
0004CB98  00000000 00000000
0004CBA0  00000000 00000000
0004CBA8  00000000 00000000
0004CBB0  00000000 00000000
0004CBB8  00000000 00000000
0004CBC0  00000000 00000000
0004CBC8  00000000 00000000
0004CBD0  00000000 00000000
0004CBD8  00000000 00000000
0004CBE0  00000000 00000000
0004CBE8  00000000 00000000
0004CBF0  00000000 00000000
0004CBF8  00000000 00000000
0004CC00  00000000 00000000
0004CC08  00000000 00000000
0004CC10  00000000 00000000
0004CC18  00000000 00000000
0004CC20  00000000 00000000
0004CC28  00000000 00000000
0004CC30  00000000 00000000
0004CC38  00000000 00000000
                                        8714
                                        8715          TSTLBL PL                 Parameter List
0004CC40  6E6E6ED7 D3F9F74C             8716+         DC    CL32'>>>PL97<<<'    eye catcher
0004CC48  4C4C4040 40404040
0004CC50  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0004CC58  40404040 40404040
0004CC60                                           8717+PL97     DS    0C
0004CC60  00000000 00000000                        8718          DC    XL352'00'
0004CC68  00000000 00000000
0004CC70  00000000 00000000
0004CC78  00000000 00000000
0004CC80  00000000 00000000
0004CC88  00000000 00000000
0004CC90  00000000 00000000
0004CC98  00000000 00000000
0004CCA0  00000000 00000000
0004CCA8  00000000 00000000
0004CCB0  00000000 00000000
0004CCB8  00000000 00000000
0004CCC0  00000000 00000000
0004CCC8  00000000 00000000
0004CCD0  00000000 00000000
0004CCD8  00000000 00000000
0004CCE0  00000000 00000000
0004CCE8  00000000 00000000
0004CCF0  00000000 00000000
0004CCF8  00000000 00000000
0004CD00  00000000 00000000
0004CD08  00000000 00000000
0004CD10  00000000 00000000
0004CD18  00000000 00000000
0004CD20  00000000 00000000
0004CD28  00000000 00000000
0004CD30  00000000 00000000
0004CD38  00000000 00000000
0004CD40  00000000 00000000
0004CD48  00000000 00000000
0004CD50  00000000 00000000
0004CD58  00000000 00000000
0004CD60  00000000 00000000
0004CD68  00000000 00000000
0004CD70  00000000 00000000
0004CD78  00000000 00000000
0004CD80  00000000 00000000
0004CD88  00000000 00000000
0004CD90  00000000 00000000
0004CD98  00000000 00000000
0004CDA0  00000000 00000000
0004CDA8  00000000 00000000
0004CDB0  00000000 00000000
0004CDB8  00000000 00000000
                                                   8719
                                                   8720          TSTLBL FMSG                fail message
0004CDC0  6E6E6EC6 D4E2C7F9                        8721+         DC    CL32'>>>FMSG97<<<'   eye catcher
0004CDC8  F74C4C4C 40404040
0004CDD0  40404040 40404040
0004CDD8  40404040 40404040
0004CDE0                                           8722+FMSG97   DS    0C
0004CDE0  86818993 858440D7                        8723          DC    C'failed PLO-Extension '
0004CDE8  D3D660C5 A7A38595
0004CDF0  A2899695 40
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004CDF5  C3969497 81998540                        8724          DC    C'Compare and Swap (CSO: 25) : NE Op1-C != Op2'
0004CDFD  81958440 E2A68197
0004CE05  404DC3E2 D67A40F2
0004CE0D  F55D407A 40D5C540
0004CE15  D697F160 C3405A7E
0004CE1D  40D697F2
0004CE21  40404040 40404040                        8725          DC    CL64' '
0004CE29  40404040 40404040
0004CE31  40404040 40404040
0004CE39  40404040 40404040
0004CE41  40404040 40404040
0004CE49  40404040 40404040
0004CE51  40404040 40404040
0004CE59  40404040 40404040
                                                   8726          TSTEND
0004CE61  C5D5C440 968640A3                        8727+         DC    CL32'END of test: 97 <<<'
0004CE69  85A2A37A 40F9F740
0004CE71  4C4C4C40 40404040
0004CE79  40404040 40404040
                                                   8728
                                                   8729 * ------------------------------------------------------------------
                                                   8730 * Test Compare and Swap (DCSO: 25) instruction : EQ Op1-C == OP2
                                                   8731 * ------------------------------------------------------------------
                                                   8732          TSTHDR 0,99,25              EQ Op1-C == OP2 (cc=0, UR=99)
0004CE81                        0004CE81  0004D000 8733+         ORG   PLOETST+(128*K)+60*(3*K)
0004D000  E2E3C1D9 E3409686                        8734+         DC    CL32'START of test: 98'
0004D008  40A385A2 A37A40F9
0004D010  F8404040 40404040
0004D018  40404040 40404040
0004D020                        0004D020           8735+         USING *,R5                  base for test data and test routine
0004D020                        00000000           8736+         USING PLOETEST,R5
0004D020  0004D04C                                 8737+T98      DC    A(CODE98)             address of test code
0004D024  0062                                     8738+         DC    H'98'                 test number
0004D026  00                                       8739+         DC    X'00'
0004D027  19                                       8740+         DC    HL1'25'               Function
0004D028  63                                       8741+         DC    HL1'99'               expected unequal results (UR)
0004D029  00                                       8742+         DC    HL1'00'               found UR
0004D02A  00                                       8743+         DC    HL1'0'                cc
0004D02B  07                                       8744+         DC    HL1'7'                not expected CC mask
0004D02C  00                                       8745+         DS    X                     extracted cc
0004D030  00000000 00000000                        8746+         DS    2F                    extract PSW after test (has CC)
0004D038  0004D9E0                                 8747+         DC    A(FMSG98)             address of test fail message
0004D03C  0004D740                                 8748+         DC    A(OPSRC98)            address of operand source
0004D040  0004D620                                 8749+         DC    A(OPDST98)            address of operand destination
0004D044  0004D860                                 8750+         DC    A(PL98)               address of parameter list
0004D048  0000421A                                 8751+         DC    A(RANDX+98)           address of random source data
                                                   8752
                                                   8753          TSTCODE
0004D04C                                           8754+CODE98   DS    0F
                                                   8755 *                                    load random source
0004D04C  E310 501C 0014                0000001C   8756          LGF   R1,OPSRCA             Op1-C, Op1-R
0004D052  E320 5028 0014                00000028   8757          LGF   R2,RANDXA
0004D058  E330 5024 0014                00000024   8758          LGF   R3,PLA
0004D05E  E340 5020 0014                00000020   8759          LGF   R4,OPREPA             OP2
                                                   8760 *
                                                   8761 * Compare and Swap (DCSO: 25)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   8762 * Build Parambter List
                                                   8763 *
0004D064  D23F 1000 2000      00000000  00000000  8764          MVC   0(64,R1),0(R2)      load Op1-C, Op1-R
0004D06A  D21F 3000 1000      00000000  00000000  8765          MVC   0(32,R3),0(R1)      copy to PL OP1-C
0004D070  D21F 3020 1020      00000020  00000020  8766          MVC   32(32,R3),32(R1)    copy to PL OP1-R
                                                   8767
0004D076  D21F 4000 2080      00000000  00000080  8768          MVC   0(32,R4),128(R2)    load initial Op2
                                                   8769
                                                   8770 ****
                                                   8771 * force an EQ by changing the Op1-C
                                                   8772 ****
0004D07C  D21F 3000 4000      00000000  00000000  8773          MVC   0(32,R3),0(R4)      Op1-C <-- Op2
                                                   8774 ****
0004D082  B982 0000                               8775          XGR   R0,R0
0004D086  4100 0019                     00000019  8776          LA    R0,25
0004D08A  EE00 4000 3000      00000000  00000000  8777          PLO   0,0(R4),0,0(R3)     C/S (CSO: 25)
                                                   8778
0004D090  B98D 0020                               8779          EPSW  R2,R0               extract psw
0004D094  5020 5010                     00000010  8780          ST    R2,CCPSW              to save CC
0004D098  E310 500B 0076                0000000B  8781          LB    R1,CCMASK           (failure CC mask)
0004D09E  8910 0004                     00000004  8782          SLL   R1,4                (shift to BC instr CC position)
0004D0A2  4410 50AE                     0004D0CE  8783          EX    R1,F252TSTCC            fail if...
                                                   8784 *
                                                   8785 * compare results
                                                   8786 *
                                0004D0A6  00000001 8787 F252CKRES EQU   *
                                                   8788 *                                  reset adresses
0004D0A6  E310 501C 0014                0000001C  8789          LGF   R1,OPSRCA           Op1-C, Op1-R
0004D0AC  E320 5028 0014                00000028  8790          LGF   R2,RANDXA
0004D0B2  E330 5024 0014                00000024  8791          LGF   R3,PLA
0004D0B8  E340 5020 0014                00000020  8792          LGF   R4,OPREPA           OP2
                                                   8793
0004D0BE  D51F 3020 4000      00000020  00000000  8794          CLC   32(32,R3),0(R4)     Op1-R == OP2
0004D0C4  478B 0000                     00000000  8795          BE    0(R11)              same, return
                                                   8796
                                0004D0C8  00000001 8797 F252XX   EQU   *
0004D0C8  45F0 83E8                     000023E8  8798          BAL   R15,FAILMSG
0004D0CC  07FB                                    8799          BR    R11
                                                   8800
0004D0CE  4700 50B2                     0004D0D2  8801 F252TSTCC BC    0,F252CCMSG        fail if unexpected condition code
                                0004D0D2  00000001 8802 F252CCMSG EQU  *
0004D0D2  45F0 8196                     00002196  8803          BAL   R15,CCMSG
0004D0D6  47F0 5086                     0004D0A6  8804          B     F252CKRES
                                                   8805
                                                   8806          TSTDATA
0004D0DA                        0004D0DA  0004D600 8807+         ORG   PLOETST+(128*K)+60*(3*K)+(K+512)
                                                   8808          TSTLBL OPDST              OP2
0004D600  6E6E6ED6 D7C4E2E3                        8809+         DC    CL32'>>>OPDST98<<<' eye catcher
0004D608  F9F84C4C 4C404040
0004D610  40404040 40404040
0004D618  40404040 40404040
0004D620                                           8810+OPDST98  DS    0C
0004D620  00000000 00000000                        8811          DC    XL256'0'
0004D628  00000000 00000000
0004D630  00000000 00000000
0004D638  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0004D640  00000000 00000000
0004D648  00000000 00000000
0004D650  00000000 00000000
0004D658  00000000 00000000
0004D660  00000000 00000000
0004D668  00000000 00000000
0004D670  00000000 00000000
0004D678  00000000 00000000
0004D680  00000000 00000000
0004D688  00000000 00000000
0004D690  00000000 00000000
0004D698  00000000 00000000
0004D6A0  00000000 00000000
0004D6A8  00000000 00000000
0004D6B0  00000000 00000000
0004D6B8  00000000 00000000
0004D6C0  00000000 00000000
0004D6C8  00000000 00000000
0004D6D0  00000000 00000000
0004D6D8  00000000 00000000
0004D6E0  00000000 00000000
0004D6E8  00000000 00000000
0004D6F0  00000000 00000000
0004D6F8  00000000 00000000
0004D700  00000000 00000000
0004D708  00000000 00000000
0004D710  00000000 00000000
0004D718  00000000 00000000
                                                8812
                                                8813           TSTLBL OPSRC               Op1-C, Op1-R
0004D720  6E6E6ED6 D7E2D9C3                     8814+          DC    CL32'>>>OPSRC98<<<' eye catcher
0004D728  F9F84C4C 4C404040
0004D730  40404040 40404040
0004D738  40404040 40404040
0004D740                                        8815+OPSRC98   DS    0C
0004D740  00000000 00000000                     8816           DC    XL256'0'
0004D748  00000000 00000000
0004D750  00000000 00000000
0004D758  00000000 00000000
0004D760  00000000 00000000
0004D768  00000000 00000000
0004D770  00000000 00000000
0004D778  00000000 00000000
0004D780  00000000 00000000
0004D788  00000000 00000000
0004D790  00000000 00000000
0004D798  00000000 00000000
0004D7A0  00000000 00000000
0004D7A8  00000000 00000000
0004D7B0  00000000 00000000
0004D7B8  00000000 00000000
0004D7C0  00000000 00000000
0004D7C8  00000000 00000000
0004D7D0  00000000 00000000
0004D7D8  00000000 00000000
0004D7E0  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0004D7E8  00000000 00000000
0004D7F0  00000000 00000000
0004D7F8  00000000 00000000
0004D800  00000000 00000000
0004D808  00000000 00000000
0004D810  00000000 00000000
0004D818  00000000 00000000
0004D820  00000000 00000000
0004D828  00000000 00000000
0004D830  00000000 00000000
0004D838  00000000 00000000
                                          8817
                                          8818           TSTLBL PL                  Parameter List
0004D840  6E6E6ED7 D3F9F84C               8819+          DC    CL32'>>>PL98<<<'     eye catcher
0004D848  4C4C4040 40404040
0004D850  40404040 40404040
0004D858  40404040 40404040
0004D860                                  8820+PL98      DS    0C
0004D860  00000000 00000000               8821           DC    XL352'00'
0004D868  00000000 00000000
0004D870  00000000 00000000
0004D878  00000000 00000000
0004D880  00000000 00000000
0004D888  00000000 00000000
0004D890  00000000 00000000
0004D898  00000000 00000000
0004D8A0  00000000 00000000
0004D8A8  00000000 00000000
0004D8B0  00000000 00000000
0004D8B8  00000000 00000000
0004D8C0  00000000 00000000
0004D8C8  00000000 00000000
0004D8D0  00000000 00000000
0004D8D8  00000000 00000000
0004D8E0  00000000 00000000
0004D8E8  00000000 00000000
0004D8F0  00000000 00000000
0004D8F8  00000000 00000000
0004D900  00000000 00000000
0004D908  00000000 00000000
0004D910  00000000 00000000
0004D918  00000000 00000000
0004D920  00000000 00000000
0004D928  00000000 00000000
0004D930  00000000 00000000
0004D938  00000000 00000000
0004D940  00000000 00000000
0004D948  00000000 00000000
0004D950  00000000 00000000
0004D958  00000000 00000000
0004D960  00000000 00000000
0004D968  00000000 00000000
0004D970  00000000 00000000
0004D978  00000000 00000000
0004D980  00000000 00000000
0004D988  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004D990  00000000 00000000
0004D998  00000000 00000000
0004D9A0  00000000 00000000
0004D9A8  00000000 00000000
0004D9B0  00000000 00000000
0004D9B8  00000000 00000000
                                        8822
                                        8823          TSTLBL FMSG               fail message
0004D9C0  6E6E6EC6 D4E2C7F9             8824+         DC    CL32'>>>FMSG98<<<'  eye catcher
0004D9C8  F84C4C4C 40404040
0004D9D0  40404040 40404040
0004D9D8  40404040 40404040
0004D9E0                                8825+FMSG98   DS    0C
0004D9E0  86818993 858440D7             8826          DC    C'failed PLO-Extension '
0004D9E8  D3D660C5 A7A38595
0004D9F0  A2899695 40
0004D9F5  C3969497 81998540             8827          DC    C'Compare and Swap (CSO: 25) : EQ Op1-C == OP2'
0004D9FD  81958440 E2A68197
0004DA05  404DC3E2 D67A40F2
0004DA0D  F55D407A 40C5D840
0004DA15  D697F160 C3407E7E
0004DA1D  40D6D7F2
0004DA21  40404040 40404040             8828          DC    CL64' '
0004DA29  40404040 40404040
0004DA31  40404040 40404040
0004DA39  40404040 40404040
0004DA41  40404040 40404040
0004DA49  40404040 40404040
0004DA51  40404040 40404040
0004DA59  40404040 40404040
                                        8829          TSTEND
0004DA61  C5D5C440 968640A3             8830+         DC    CL32'END of test: 98 <<<'
0004DA69  85A2A37A 40F9F840
0004DA71  4C4C4C40 40404040
0004DA79  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   8832 *=====================================================================
                                                   8833 *       Compare and Load Tests
                                                   8834 *=====================================================================
                                                   8835
                                                   8836 * -------------------------------------------------------------------
                                                   8837 * Test Compare and Load (CLO: 24) instruction : NE Op1-C!=Op2
                                                   8838 * -------------------------------------------------------------------
                                                   8839          TSTHDR 1,99,25           Op1-C != OP2 (cc=1, UR=99)
0004DA81                          0004DA81  0004DC00 8840+         ORG   PLOETST+(128*K)+61*(3*K)
0004DC00  E2E3C1D9 E3409686                        8841+         DC    CL32'START of test: 99'
0004DC08  40A385A2 A37A40F9
0004DC10  F9404040 40404040
0004DC18  40404040 40404040
0004DC20                          0004DC20         8842+         USING *,R5               base for test data and test routine
0004DC20                          00000000         8843+         USING PLOETEST,R5
0004DC20  0004DC4C                                 8844+T99      DC    A(CODE99)          address of test code
0004DC24  0063                                     8845+         DC    H'99'              test number
0004DC26  00                                       8846+         DC    X'00'
0004DC27  19                                       8847+         DC    HL1'25'            Function
0004DC28  63                                       8848+         DC    HL1'99'            expected unequal results (UR)
0004DC29  00                                       8849+         DC    HL1'00'            found UR
0004DC2A  01                                       8850+         DC    HL1'1'             cc
0004DC2B  0B                                       8851+         DC    HL1'11'            not expected CC mask
0004DC2C  00                                       8852+         DS    X                  extracted cc
0004DC30  00000000 00000000                        8853+         DS    2F                 extract PSW after test (has CC)
0004DC38  0004E5E0                                 8854+         DC    A(FMSG99)          address of test fail message
0004DC3C  0004E340                                 8855+         DC    A(OPSRC99)         address of operand source
0004DC40  0004E220                                 8856+         DC    A(OPDST99)         address of operand destination
0004DC44  0004E460                                 8857+         DC    A(PL99)            address of parameter list
0004DC48  0000421B                                 8858+         DC    A(RANDX+99)        address of random source data
                                                   8859
                                                   8860          TSTCODE
0004DC4C                                           8861+CODE99   DS    0F
                                                   8862 *                                 load random source
0004DC4C  E310 501C 0014                0000001C   8863          LGF   R1,OPSRCA          Op1-C, Op3
0004DC52  E320 5028 0014                00000028   8864          LGF   R2,RANDXA
0004DC58  E330 5024 0014                00000024   8865          LGF   R3,PLA
0004DC5E  E340 5020 0014                00000020   8866          LGF   R4,OPREPA          OP2, Op4
                                                   8867 *
                                                   8868 * Compare and Load (CLO: 24)
                                                   8869 * Build Parambter List
                                                   8870 *
0004DC64  D23F 1000 2000  00000000      00000000   8871          MVC   0(64,R1),0(R2)     load Op1-C, Op3
0004DC6A  D21F 3000 1000  00000000      00000000   8872          MVC   0(32,R3),0(R1)     copy to PL Op1-C
0004DC70  D21F 3020 1020  00000020      00000020   8873          MVC   32(32,R3),32(R1)   copy to PL Op3
                                                   8874
0004DC76  D23F 4000 2080  00000000      00000080   8875          MVC   0(64,R4),128(R2)   load initial Op2, Op4
                                                   8876
                                                   8877 ****
                                                   8878 * force a NE by changing the Op1-c
                                                   8879 ****
0004DC7C  D203 3000 2021  00000000      00000021   8880          MVC   0(4,R3),33(R2)
                                                   8881 ****
0004DC82  B982 0000                                8882          XGR   R0,R0
0004DC86  4104 0020                     00000020   8883          LA    R0,32(R4)
0004DC8A  E303 0048 0024                00000048   8884          STG   R0,72(R3)          load OP4 address in PL
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    8885
0004DC90  B982 0000                                 8886          XGR   R0,R0
0004DC94  4100 0018                       00000018  8887          LA    R0,24
0004DC98  EE00 4000 3000      00000000    00000000  8888          PLO   0,0(R4),0,0(R3)     C/L (CLO: 24)
                                                    8889
0004DC9E  B98D 0020                                 8890          EPSW  R2,R0              extract psw
0004DCA2  5020 5010                       00000010  8891          ST    R2,CCPSW             to save CC
0004DCA6  E310 500B 0076                  0000000B  8892          LB    R1,CCMASK          (failure CC mask)
0004DCAC  8910 0004                       00000004  8893          SLL   R1,4               (shift to BC instr CC position)
0004DCB0  4410 50BC                       0004DCDC  8894          EX    R1,F241TSTCC         fail if...
                                                    8895 *
                                                    8896 * compare results
                                                    8897 *
                            0004DCB4      00000001  8898 F241CKRES EQU   *
                                                    8899 *                                 reset adresses
0004DCB4  E310 501C 0014                  0000001C  8900          LGF   R1,OPSRCA          Op1-C, Op3
0004DCBA  E320 5028 0014                  00000028  8901          LGF   R2,RANDXA
0004DCC0  E330 5024 0014                  00000024  8902          LGF   R3,PLA
0004DCC6  E340 5020 0014                  00000020  8903          LGF   R4,OPREPA          OP2, Op4
                                                    8904
0004DCCC  D51F 4000 3000      00000000    00000000  8905          CLC   0(32,R4),0(R3)     Op2 == OP1-C?
0004DCD2  478B 0000                       00000000  8906          BE    0(R11)             same, return
                                                    8907
                            0004DCD6      00000001  8908 F241XX   EQU   *
0004DCD6  45F0 83E8                       000023E8  8909          BAL   R15,FAILMSG
0004DCDA  07FB                                      8910          BR    R11
                                                    8911
0004DCDC  4700 50C0                       0004DCE0  8912 F241TSTCC BC    0,F241CCMSG        fail if unexpected condition code
                            0004DCE0      00000001  8913 F241CCMSG EQU   *
0004DCE0  45F0 8196                       00002196  8914          BAL   R15,CCMSG
0004DCE4  47F0 5094                       0004DCB4  8915          B     F241CKRES
                                                    8916
                                                    8917          TSTDATA
0004DCE8                      0004DCE8    0004E200  8918+         ORG   PLOETST+(128*K)+61*(3*K)+(K+512)
                                                    8919          TSTLBL OPDST             OP2, Op4
0004E200  6E6E6ED6 D7C4E2E3                         8920+         DC    CL32'>>>OPDST99<<<' eye catcher
0004E208  F9F94C4C 4C404040
0004E210  40404040 40404040
0004E218  40404040 40404040
0004E220                                            8921+OPDST99  DS    0C
0004E220  00000000 00000000                         8922          DC    XL256'0'
0004E228  00000000 00000000
0004E230  00000000 00000000
0004E238  00000000 00000000
0004E240  00000000 00000000
0004E248  00000000 00000000
0004E250  00000000 00000000
0004E258  00000000 00000000
0004E260  00000000 00000000
0004E268  00000000 00000000
0004E270  00000000 00000000
0004E278  00000000 00000000
0004E280  00000000 00000000
0004E288  00000000 00000000
0004E290  00000000 00000000
0004E298  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0004E2A0  00000000 00000000
0004E2A8  00000000 00000000
0004E2B0  00000000 00000000
0004E2B8  00000000 00000000
0004E2C0  00000000 00000000
0004E2C8  00000000 00000000
0004E2D0  00000000 00000000
0004E2D8  00000000 00000000
0004E2E0  00000000 00000000
0004E2E8  00000000 00000000
0004E2F0  00000000 00000000
0004E2F8  00000000 00000000
0004E300  00000000 00000000
0004E308  00000000 00000000
0004E310  00000000 00000000
0004E318  00000000 00000000
                                          8923
                                          8924          TSTLBL OPSRC               Op1-C, Op3
0004E320  6E6E6ED6 D7E2D9C3               8925+         DC    CL32'>>>OPSRC99<<<' eye catcher
0004E328  F9F94C4C 4C404040
0004E330  40404040 40404040
0004E338  40404040 40404040
0004E340                                  8926+OPSRC99  DS    0C
0004E340  00000000 00000000               8927          DC    XL256'0'
0004E348  00000000 00000000
0004E350  00000000 00000000
0004E358  00000000 00000000
0004E360  00000000 00000000
0004E368  00000000 00000000
0004E370  00000000 00000000
0004E378  00000000 00000000
0004E380  00000000 00000000
0004E388  00000000 00000000
0004E390  00000000 00000000
0004E398  00000000 00000000
0004E3A0  00000000 00000000
0004E3A8  00000000 00000000
0004E3B0  00000000 00000000
0004E3B8  00000000 00000000
0004E3C0  00000000 00000000
0004E3C8  00000000 00000000
0004E3D0  00000000 00000000
0004E3D8  00000000 00000000
0004E3E0  00000000 00000000
0004E3E8  00000000 00000000
0004E3F0  00000000 00000000
0004E3F8  00000000 00000000
0004E400  00000000 00000000
0004E408  00000000 00000000
0004E410  00000000 00000000
0004E418  00000000 00000000
0004E420  00000000 00000000
0004E428  00000000 00000000
0004E430  00000000 00000000
0004E438  00000000 00000000
                                                                8928
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   8929           TSTLBL PL                  Parameter List
0004E440  6E6E6ED7 D3F9F94C                        8930+          DC    CL32'>>>PL99<<<'     eye catcher
0004E448  4C4C4040 40404040
0004E450  40404040 40404040
0004E458  40404040 40404040
0004E460                                           8931+PL99      DS    0C
0004E460  00000000 00000000                        8932           DC    XL352'00'
0004E468  00000000 00000000
0004E470  00000000 00000000
0004E478  00000000 00000000
0004E480  00000000 00000000
0004E488  00000000 00000000
0004E490  00000000 00000000
0004E498  00000000 00000000
0004E4A0  00000000 00000000
0004E4A8  00000000 00000000
0004E4B0  00000000 00000000
0004E4B8  00000000 00000000
0004E4C0  00000000 00000000
0004E4C8  00000000 00000000
0004E4D0  00000000 00000000
0004E4D8  00000000 00000000
0004E4E0  00000000 00000000
0004E4E8  00000000 00000000
0004E4F0  00000000 00000000
0004E4F8  00000000 00000000
0004E500  00000000 00000000
0004E508  00000000 00000000
0004E510  00000000 00000000
0004E518  00000000 00000000
0004E520  00000000 00000000
0004E528  00000000 00000000
0004E530  00000000 00000000
0004E538  00000000 00000000
0004E540  00000000 00000000
0004E548  00000000 00000000
0004E550  00000000 00000000
0004E558  00000000 00000000
0004E560  00000000 00000000
0004E568  00000000 00000000
0004E570  00000000 00000000
0004E578  00000000 00000000
0004E580  00000000 00000000
0004E588  00000000 00000000
0004E590  00000000 00000000
0004E598  00000000 00000000
0004E5A0  00000000 00000000
0004E5A8  00000000 00000000
0004E5B0  00000000 00000000
0004E5B8  00000000 00000000
                                                   8933
                                                   8934           TSTLBL FMSG                fail message
0004E5C0  6E6E6EC6 D4E2C7F9                        8935+          DC    CL32'>>>FMSG99<<<'   eye catcher
0004E5C8  F94C4C4C 40404040
0004E5D0  40404040 40404040
0004E5D8  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004E5E0                                          8936+FMSG99   DS    0C
0004E5E0  86818993 858440D7                       8937          DC    C'failed PLO-Extension '
0004E5E8  D3D660C5 A7A38595
0004E5F0  A2899695 40
0004E5F5  C3969497 81998540                       8938          DC    C'Compare and Load (CLO: 24) : NE Op1-C != Op2'
0004E5FD  81958440 D3968184
0004E605  404DC3D3 D67A40F2
0004E60D  F45D407A 40D5C540
0004E615  D697F160 C3405A7E
0004E61D  40D697F2
0004E621  40404040 40404040                       8939          DC    CL64' '
0004E629  40404040 40404040
0004E631  40404040 40404040
0004E639  40404040 40404040
0004E641  40404040 40404040
0004E649  40404040 40404040
0004E651  40404040 40404040
0004E659  40404040 40404040
                                                  8940          TSTEND
0004E661  C5D5C440 968640A3                       8941+         DC    CL32'END of test: 99 <<<'
0004E669  85A2A37A 40F9F940
0004E671  4C4C4C40 40404040
0004E679  40404040 40404040
                                                  8942
                                                  8943 * ------------------------------------------------------------------
                                                  8944 * Test Compare and Load (CLO: 24) instruction : EQ Op1-C == OP2
                                                  8945 * ------------------------------------------------------------------
                                                  8946          TSTHDR 0,99,25            EQ Op1-C == OP2 (cc=0, UR=99)
0004E681                    0004E681  0004E800    8947+         ORG   PLOETST+(128*K)+62*(3*K)
0004E800  E2E3C1D9 E3409686                       8948+         DC    CL32'START of test: 100'
0004E808  40A385A2 A37A40F1
0004E810  F0F04040 40404040
0004E818  40404040 40404040
0004E820                    0004E820              8949+         USING *,R5                base for test data and test routine
0004E820                    00000000              8950+         USING PLOETEST,R5
0004E820  0004E84C                                8951+T100     DC    A(CODE100)          address of test code
0004E824  0064                                    8952+         DC    H'100'              test number
0004E826  00                                      8953+         DC    X'00'
0004E827  19                                      8954+         DC    HL1'25'             Function
0004E828  63                                      8955+         DC    HL1'99'             expected unequal results (UR)
0004E829  00                                      8956+         DC    HL1'00'             found UR
0004E82A  00                                      8957+         DC    HL1'0'              cc
0004E82B  07                                      8958+         DC    HL1'7'              not expected CC mask
0004E82C  00                                      8959+         DS    X                   extracted cc
0004E830  00000000 00000000                       8960+         DS    2F                  extract PSW after test (has CC)
0004E838  0004F1E0                                8961+         DC    A(FMSG100)          address of test fail message
0004E83C  0004EF40                                8962+         DC    A(OPSRC100)         address of operand source
0004E840  0004EE20                                8963+         DC    A(OPDST100)         address of operand destination
0004E844  0004F060                                8964+         DC    A(PL100)            address of parameter list
0004E848  0000421C                                8965+         DC    A(RANDX+100)        address of random source data
                                                  8966
                                                  8967          TSTCODE
0004E84C                                          8968+CODE100  DS    0F
                                                  8969 *                                  load random source
0004E84C  E310 501C 0014              0000001C    8970          LGF   R1,OPSRCA           Op1-C, Op3
0004E852  E320 5028 0014              00000028    8971          LGF   R2,RANDXA
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004E858  E330 5024 0014                 00000024  8972          LGF   R3,PLA
0004E85E  E340 5020 0014                 00000020  8973          LGF   R4,OPREPA           OP2, Op4
                                                   8974 *
                                                   8975 * Compare and Swap (DCSO: 25)
                                                   8976 * Build Parambter List
                                                   8977 *
0004E864  D23F 1000 2000       00000000  00000000  8978          MVC   0(64,R1),0(R2)      load Op1-C, Op3
0004E86A  D21F 3000 1000       00000000  00000000  8979          MVC   0(32,R3),0(R1)      copy to PL Op1-C
0004E870  D21F 3020 1020       00000020  00000020  8980          MVC   32(32,R3),32(R1)    copy to PL Op3
                                                   8981
0004E876  D23F 4000 2080       00000000  00000080  8982          MVC   0(64,R4),128(R2)    load initial Op2, Op4
                                                   8983
                                                   8984 ****
                                                   8985 * force an EQ by changing the Op1-C
                                                   8986 ****
0004E87C  D21F 3000 4000       00000000  00000000  8987          MVC   0(32,R3),0(R4)      Op1-C <-- Op2
                                                   8988 ****
0004E882  B982 0000                                8989          XGR   R0,R0
0004E886  4104 0020                      00000020  8990          LA    R0,32(R4)
0004E88A  E303 0048 0024                 00000048  8991          STG   R0,72(R3)           load OP4 address in PL
                                                   8992
0004E890  B982 0000                                8993          XGR   R0,R0
0004E894  4100 0018                      00000018  8994          LA    R0,24
0004E898  EE00 4000 3000       00000000  00000000  8995          PLO   0,0(R4),0,0(R3)     C/S (CLO: 24)
                                                   8996
0004E89E  B98D 0020                                8997          EPSW  R2,R0               extract psw
0004E8A2  5020 5010                      00000010  8998          ST    R2,CCPSW              to save CC
0004E8A6  E310 500B 0076                 0000000B  8999          LB    R1,CCMASK           (failure CC mask)
0004E8AC  8910 0004                      00000004  9000          SLL   R1,4                (shift to BC instr CC position)
0004E8B0  4410 50BC                      0004E8DC  9001          EX    R1,F242TSTCC            fail if...
                                                   9002 *
                                                   9003 * compare results
                                                   9004 *
                                0004E8B4  00000001 9005 F242CKRES EQU   *
                                                   9006 *                                  reset adresses
0004E8B4  E310 501C 0014                 0000001C  9007          LGF   R1,OPSRCA           Op1-C, Op3
0004E8BA  E320 5028 0014                 00000028  9008          LGF   R2,RANDXA
0004E8C0  E330 5024 0014                 00000024  9009          LGF   R3,PLA
0004E8C6  E340 5020 0014                 00000020  9010          LGF   R4,OPREPA           OP2, Op4
                                                   9011
0004E8CC  D51F 3020 4020       00000020  00000020  9012          CLC   32(32,R3),32(R4)    Op3 == OP4
0004E8D2  478B 0000                      00000000  9013          BE    0(R11)              same, return
                                                   9014
                                0004E8D6  00000001 9015 F242XX   EQU   *
0004E8D6  45F0 83E8                      000023E8  9016          BAL   R15,FAILMSG
0004E8DA  07FB                                     9017          BR    R11
                                                   9018
0004E8DC  4700 50C0                      0004E8E0  9019 F242TSTCC BC    0,F242CCMSG        fail if unexpected condition code
                                0004E8E0  00000001 9020 F242CCMSG EQU   *
0004E8E0  45F0 8196                      00002196  9021          BAL   R15,CCMSG
0004E8E4  47F0 5094                      0004E8B4  9022          B     F242CKRES
                                                   9023
                                                   9024          TSTDATA
0004E8E8                        0004E8E8  0004EE00 9025+         ORG   PLOETST+(128*K)+62*(3*K)+(K+512)
                                                   9026          TSTLBL OPDST              OP2, Op4
                                                   9027+         DC    CL32'>>>OPDST100<<<' \
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0004EE00  6E6E6ED6 D7C4E2E3                          +                                   eye catcher
0004EE08  F1F0F04C 4C4C4040
0004EE10  40404040 40404040
0004EE18  40404040 40404040
0004EE20                                          9028+OPDST100 DS    0C
0004EE20  00000000 00000000                       9029          DC    XL256'0'
0004EE28  00000000 00000000
0004EE30  00000000 00000000
0004EE38  00000000 00000000
0004EE40  00000000 00000000
0004EE48  00000000 00000000
0004EE50  00000000 00000000
0004EE58  00000000 00000000
0004EE60  00000000 00000000
0004EE68  00000000 00000000
0004EE70  00000000 00000000
0004EE78  00000000 00000000
0004EE80  00000000 00000000
0004EE88  00000000 00000000
0004EE90  00000000 00000000
0004EE98  00000000 00000000
0004EEA0  00000000 00000000
0004EEA8  00000000 00000000
0004EEB0  00000000 00000000
0004EEB8  00000000 00000000
0004EEC0  00000000 00000000
0004EEC8  00000000 00000000
0004EED0  00000000 00000000
0004EED8  00000000 00000000
0004EEE0  00000000 00000000
0004EEE8  00000000 00000000
0004EEF0  00000000 00000000
0004EEF8  00000000 00000000
0004EF00  00000000 00000000
0004EF08  00000000 00000000
0004EF10  00000000 00000000
0004EF18  00000000 00000000
                                                  9030
                                                  9031          TSTLBL OPSRC              Op1-C, Op3
                                                  9032+         DC    CL32'>>>OPSRC100<<<' \
0004EF20  6E6E6ED6 D7E2D9C3                          +                                   eye catcher
0004EF28  F1F0F04C 4C4C4040
0004EF30  40404040 40404040
0004EF38  40404040 40404040
0004EF40                                          9033+OPSRC100 DS    0C
0004EF40  00000000 00000000                       9034          DC    XL256'0'
0004EF48  00000000 00000000
0004EF50  00000000 00000000
0004EF58  00000000 00000000
0004EF60  00000000 00000000
0004EF68  00000000 00000000
0004EF70  00000000 00000000
0004EF78  00000000 00000000
0004EF80  00000000 00000000
0004EF88  00000000 00000000
0004EF90  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0004EF98  00000000 00000000
0004EFA0  00000000 00000000
0004EFA8  00000000 00000000
0004EFB0  00000000 00000000
0004EFB8  00000000 00000000
0004EFC0  00000000 00000000
0004EFC8  00000000 00000000
0004EFD0  00000000 00000000
0004EFD8  00000000 00000000
0004EFE0  00000000 00000000
0004EFE8  00000000 00000000
0004EFF0  00000000 00000000
0004EFF8  00000000 00000000
0004F000  00000000 00000000
0004F008  00000000 00000000
0004F010  00000000 00000000
0004F018  00000000 00000000
0004F020  00000000 00000000
0004F028  00000000 00000000
0004F030  00000000 00000000
0004F038  00000000 00000000
                                          9035
                                          9036          TSTLBL PL                   Parameter List
0004F040  6E6E6ED7 D3F1F0F0               9037+         DC    CL32'>>>PL100<<<'     eye catcher
0004F048  4C4C4C40 40404040
0004F050  40404040 40404040
0004F058  40404040 40404040
0004F060                                  9038+PL100    DS    0C
0004F060  00000000 00000000               9039          DC    XL352'00'
0004F068  00000000 00000000
0004F070  00000000 00000000
0004F078  00000000 00000000
0004F080  00000000 00000000
0004F088  00000000 00000000
0004F090  00000000 00000000
0004F098  00000000 00000000
0004F0A0  00000000 00000000
0004F0A8  00000000 00000000
0004F0B0  00000000 00000000
0004F0B8  00000000 00000000
0004F0C0  00000000 00000000
0004F0C8  00000000 00000000
0004F0D0  00000000 00000000
0004F0D8  00000000 00000000
0004F0E0  00000000 00000000
0004F0E8  00000000 00000000
0004F0F0  00000000 00000000
0004F0F8  00000000 00000000
0004F100  00000000 00000000
0004F108  00000000 00000000
0004F110  00000000 00000000
0004F118  00000000 00000000
0004F120  00000000 00000000
0004F128  00000000 00000000
0004F130  00000000 00000000
0004F138  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004F140  00000000 00000000
0004F148  00000000 00000000
0004F150  00000000 00000000
0004F158  00000000 00000000
0004F160  00000000 00000000
0004F168  00000000 00000000
0004F170  00000000 00000000
0004F178  00000000 00000000
0004F180  00000000 00000000
0004F188  00000000 00000000
0004F190  00000000 00000000
0004F198  00000000 00000000
0004F1A0  00000000 00000000
0004F1A8  00000000 00000000
0004F1B0  00000000 00000000
0004F1B8  00000000 00000000
                                          9040
                                          9041          TSTLBL FMSG                fail message
0004F1C0  6E6E6EC6 D4E2C7F1               9042+         DC    CL32'>>>FMSG100<<<' eye catcher
0004F1C8  F0F04C4C 4C404040
0004F1D0  40404040 40404040
0004F1D8  40404040 40404040
0004F1E0                                  9043+FMSG100  DS    0C
0004F1E0  86818993 858440D7               9044          DC    C'failed PLO-Extension '
0004F1E8  D3D660C5 A7A38595
0004F1F0  A2899695 40
0004F1F5  C3969497 81998540               9045          DC    C'Compare and Load (CLO: 24) : EQ Op1-C == OP2'
0004F1FD  81958440 D3968184
0004F205  404DC3D3 D67A40F2
0004F20D  F45D407A 40C5D840
0004F215  D697F160 C3407E7E
0004F21D  40D6D7F2
0004F221  40404040 40404040               9046          DC    CL64' '
0004F229  40404040 40404040
0004F231  40404040 40404040
0004F239  40404040 40404040
0004F241  40404040 40404040
0004F249  40404040 40404040
0004F251  40404040 40404040
0004F259  40404040 40404040
                                          9047          TSTEND
0004F261  C5D5C440 968640A3               9048+         DC    CL32'END of test: 100 <<<'
0004F269  85A2A37A 40F1F0F0
0004F271  404C4C4C 40404040
0004F279  40404040 40404040
                                          9049
                                          9050 ********************************************************************************
                                          9051 * table of pointers to individual tests
                                          9052 ********************************************************************************
                                          9053
0004F281                    0004F281  000045B8  9054          ORG   PLOTESTS           (2k reserved for test pointers)
                                          9055          PTTABLE
000045B8                                  9056+TTABLE   DS    0F
000045B8  000049C0                        9057+         DC    A(T1)
000045BC  000049E0                        9058+         DC    A(T2)
000045C0  00004A00                        9059+         DC    A(T3)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000045C4  00004A20                             9060+          DC    A(T4)
000045C8  00004A40                             9061+          DC    A(T5)
000045CC  00004A60                             9062+          DC    A(T6)
000045D0  00004A80                             9063+          DC    A(T7)
000045D4  00004AA0                             9064+          DC    A(T8)
000045D8  00004AC0                             9065+          DC    A(T9)
000045DC  00004AE0                             9066+          DC    A(T10)
000045E0  00004B00                             9067+          DC    A(T11)
000045E4  00004B20                             9068+          DC    A(T12)
000045E8  00004B40                             9069+          DC    A(T13)
000045EC  00004B60                             9070+          DC    A(T14)
000045F0  00004B80                             9071+          DC    A(T15)
000045F4  00004BA0                             9072+          DC    A(T16)
000045F8  00004BC0                             9073+          DC    A(T17)
000045FC  00004BE0                             9074+          DC    A(T18)
00004600  00004C00                             9075+          DC    A(T19)
00004604  00004C20                             9076+          DC    A(T20)
00004608  00004C40                             9077+          DC    A(T21)
0000460C  00004C60                             9078+          DC    A(T22)
00004610  00004C80                             9079+          DC    A(T23)
00004614  00004CA0                             9080+          DC    A(T24)
00004618  00004CC0                             9081+          DC    A(T25)
0000461C  00004CE0                             9082+          DC    A(T26)
00004620  00004D00                             9083+          DC    A(T27)
00004624  00004D20                             9084+          DC    A(T28)
00004628  00004D40                             9085+          DC    A(T29)
0000462C  00004D60                             9086+          DC    A(T30)
00004630  00004D80                             9087+          DC    A(T31)
00004634  00004DA0                             9088+          DC    A(T32)
00004638  00004DC0                             9089+          DC    A(T33)
0000463C  00004DE0                             9090+          DC    A(T34)
00004640  00004E00                             9091+          DC    A(T35)
00004644  00004E20                             9092+          DC    A(T36)
00004648  00004E40                             9093+          DC    A(T37)
0000464C  00004E60                             9094+          DC    A(T38)
00004650  00020C20                             9095+          DC    A(T39)
00004654  00021820                             9096+          DC    A(T40)
00004658  00022420                             9097+          DC    A(T41)
0000465C  00023020                             9098+          DC    A(T42)
00004660  00023C20                             9099+          DC    A(T43)
00004664  00024820                             9100+          DC    A(T44)
00004668  00025420                             9101+          DC    A(T45)
0000466C  00026020                             9102+          DC    A(T46)
00004670  00026C20                             9103+          DC    A(T47)
00004674  00027820                             9104+          DC    A(T48)
00004678  00028420                             9105+          DC    A(T49)
0000467C  00029020                             9106+          DC    A(T50)
00004680  00029C20                             9107+          DC    A(T51)
00004684  0002A820                             9108+          DC    A(T52)
00004688  0002B420                             9109+          DC    A(T53)
0000468C  0002C020                             9110+          DC    A(T54)
00004690  0002CC20                             9111+          DC    A(T55)
00004694  0002D820                             9112+          DC    A(T56)
00004698  0002E420                             9113+          DC    A(T57)
0000469C  0002F020                             9114+          DC    A(T58)
000046A0  0002FC20                             9115+          DC    A(T59)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000046A4  00030820                          9116+          DC    A(T60)
000046A8  00031420                          9117+          DC    A(T61)
000046AC  00032020                          9118+          DC    A(T62)
000046B0  00032C20                          9119+          DC    A(T63)
000046B4  00033820                          9120+          DC    A(T64)
000046B8  00034420                          9121+          DC    A(T65)
000046BC  00035020                          9122+          DC    A(T66)
000046C0  00035C20                          9123+          DC    A(T67)
000046C4  00036820                          9124+          DC    A(T68)
000046C8  00037420                          9125+          DC    A(T69)
000046CC  00038020                          9126+          DC    A(T70)
000046D0  00038C20                          9127+          DC    A(T71)
000046D4  00039820                          9128+          DC    A(T72)
000046D8  0003A420                          9129+          DC    A(T73)
000046DC  0003B020                          9130+          DC    A(T74)
000046E0  0003BC20                          9131+          DC    A(T75)
000046E4  0003C820                          9132+          DC    A(T76)
000046E8  0003D420                          9133+          DC    A(T77)
000046EC  0003E020                          9134+          DC    A(T78)
000046F0  0003EC20                          9135+          DC    A(T79)
000046F4  0003F820                          9136+          DC    A(T80)
000046F8  00040420                          9137+          DC    A(T81)
000046FC  00041020                          9138+          DC    A(T82)
00004700  00041C20                          9139+          DC    A(T83)
00004704  00042820                          9140+          DC    A(T84)
00004708  00043420                          9141+          DC    A(T85)
0000470C  00044020                          9142+          DC    A(T86)
00004710  00044C20                          9143+          DC    A(T87)
00004714  00045820                          9144+          DC    A(T88)
00004718  00046420                          9145+          DC    A(T89)
0000471C  00047020                          9146+          DC    A(T90)
00004720  00047C20                          9147+          DC    A(T91)
00004724  00048820                          9148+          DC    A(T92)
00004728  00049420                          9149+          DC    A(T93)
0000472C  0004A020                          9150+          DC    A(T94)
00004730  0004AC20                          9151+          DC    A(T95)
00004734  0004B820                          9152+          DC    A(T96)
00004738  0004C420                          9153+          DC    A(T97)
0000473C  0004D020                          9154+          DC    A(T98)
00004740  0004DC20                          9155+          DC    A(T99)
00004744  0004E820                          9156+          DC    A(T100)
                                            9157+*
00004748  00000000                          9158+          DC    A(0)                END OF TABLE
0000474C  00000000                          9159+          DC    A(0)
```

```
 LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                        9161 ********************************************************************************
                                        9162 *        Register equates
                                        9163 ********************************************************************************

                            00000000  00000001  9165 R0       EQU   0
                            00000001  00000001  9166 R1       EQU   1
                            00000002  00000001  9167 R2       EQU   2
                            00000003  00000001  9168 R3       EQU   3
                            00000004  00000001  9169 R4       EQU   4
                            00000005  00000001  9170 R5       EQU   5
                            00000006  00000001  9171 R6       EQU   6
                            00000007  00000001  9172 R7       EQU   7
                            00000008  00000001  9173 R8       EQU   8
                            00000009  00000001  9174 R9       EQU   9
                            0000000A  00000001  9175 R10      EQU   10
                            0000000B  00000001  9176 R11      EQU   11
                            0000000C  00000001  9177 R12      EQU   12
                            0000000D  00000001  9178 R13      EQU   13
                            0000000E  00000001  9179 R14      EQU   14
                            0000000F  00000001  9180 R15      EQU   15

                                        9182 ********************************************************************************
                                        9183 *        Register equates
                                        9184 ********************************************************************************

                          00000000  00000001  9186 V0       EQU   0
                          00000001  00000001  9187 V1       EQU   1
                          00000002  00000001  9188 V2       EQU   2
                          00000003  00000001  9189 V3       EQU   3
                          00000004  00000001  9190 V4       EQU   4
                          00000005  00000001  9191 V5       EQU   5
                          00000006  00000001  9192 V6       EQU   6
                          00000007  00000001  9193 V7       EQU   7
                          00000008  00000001  9194 V8       EQU   8
                          00000009  00000001  9195 V9       EQU   9
                          0000000A  00000001  9196 V10      EQU   10
                          0000000B  00000001  9197 V11      EQU   11
                          0000000C  00000001  9198 V12      EQU   12
                          0000000D  00000001  9199 V13      EQU   13
                          0000000E  00000001  9200 V14      EQU   14
                          0000000F  00000001  9201 V15      EQU   15
                          00000010  00000001  9202 V16      EQU   16
                          00000011  00000001  9203 V17      EQU   17
                          00000012  00000001  9204 V18      EQU   18
                          00000013  00000001  9205 V19      EQU   19
                          00000014  00000001  9206 V20      EQU   20
                          00000015  00000001  9207 V21      EQU   21
```

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SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
A000LABL	U	000000	1	118	
AFUN	U	000007	1	724	422
ATNUM	H	000004	2	722	415
ATSUB	A	000000	4	721	
BEGIN	I	002000	2	156	125 152 153 154
CC	U	00000A	1	703	253 282
CCFOUND	X	00000C	1	706	264 289
CCMASK	U	00000B	1	704	3125 3251 3377 3503 3630 3762 3888 4014 4140 4267 4399 4525 4651 4777
					4903 5041 5173 5305 5437 5570 5693 5814 5935 6056 6177 6298 6419 6540
					6662 6783 6904 7025 7151 7276 7401 7526 7651 7767 7890 8002 8119 8230
					8346 8455 8567 8678 8781 8892 8999
CCMSG	U	002196	1	249	3159 3285 3411 3537 3666 3796 3922 4048 4174 4303 4433 4559 4685 4811
					4939 5075 5207 5339 5471 5600 5725 5846 5967 6084 6209 6330 6451 6568
					6694 6815 6936 7053 7183 7308 7433 7554 7673 7795 7912 8028 8141 8254
					8368 8479 8591 8700 8803 8914 9021
CCPRTEXP	C	0040CE	1	560	286
CCPRTFUN	C	0040BD	2	557	279
CCPRTGOT	C	0040DE	1	563	293
CCPRTLINE	C	004081	16	553	565 298
CCPRTLNG	U	00005F	1	565	297
CCPRTNUM	C	004091	3	554	272
CCPSW	F	000010	4	707	261 3124 3250 3376 3502 3629 3761 3887 4013 4139 4266 4398 4524 4650
					4776 4902 5040 5172 5304 5436 5569 5692 5813 5934 6055 6176 6297 6418
					6539 6661 6782 6903 7024 7150 7275 7400 7525 7650 7766 7889 8001 8118
					8229 8345 8454 8566 8677 8780 8891 8998
CODE1	F	0049C8	4	905	900
CODE10	F	004AE8	4	1085	1080
CODE100	F	04E84C	4	8968	8951
CODE11	F	004B08	4	1105	1100
CODE12	F	004B28	4	1125	1120
CODE13	F	004B48	4	1145	1140
CODE14	F	004B68	4	1165	1160
CODE15	F	004B88	4	1185	1180
CODE16	F	004BA8	4	1205	1200
CODE17	F	004BC8	4	1225	1220
CODE18	F	004BE8	4	1245	1240
CODE19	F	004C08	4	1265	1260
CODE2	F	0049E8	4	925	920
CODE20	F	004C28	4	1285	1280
CODE21	F	004C48	4	1305	1300
CODE22	F	004C68	4	1325	1320
CODE23	F	004C88	4	1345	1340
CODE24	F	004CA8	4	1365	1360
CODE25	F	004CC8	4	1385	1380
CODE26	F	004CE8	4	1405	1400
CODE27	F	004D08	4	1425	1420
CODE28	F	004D28	4	1445	1440
CODE29	F	004D48	4	1465	1460
CODE3	F	004A08	4	945	940
CODE30	F	004D68	4	1485	1480
CODE31	F	004D88	4	1505	1500
CODE32	F	004DA8	4	1525	1520
CODE33	F	004DC8	4	1545	1540
CODE34	F	004DE8	4	1565	1560
CODE35	F	004E08	4	1585	1580
CODE36	F	004E28	4	1605	1600

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
CODE37	F	004E48	4	1625	1620
CODE38	F	004E68	4	1645	1640
CODE39	F	020C4C	4	1709	1692
CODE4	F	004A28	4	965	960
CODE40	F	02184C	4	1823	1806
CODE41	F	02244C	4	1937	1920
CODE42	F	02304C	4	2051	2034
CODE43	F	023C4C	4	2169	2152
CODE44	F	02484C	4	2275	2258
CODE45	F	02544C	4	2381	2364
CODE46	F	02604C	4	2487	2470
CODE47	F	026C4C	4	2597	2580
CODE48	F	02784C	4	2695	2678
CODE49	F	02844C	4	2797	2780
CODE5	F	004A48	4	985	980
CODE50	F	02904C	4	2892	2875
CODE51	F	029C4C	4	2992	2975
CODE52	F	02A84C	4	3086	3069
CODE53	F	02B44C	4	3213	3196
CODE54	F	02C04C	4	3339	3322
CODE55	F	02CC4C	4	3465	3448
CODE56	F	02D84C	4	3592	3575
CODE57	F	02E44C	4	3724	3707
CODE58	F	02F04C	4	3850	3833
CODE59	F	02FC4C	4	3976	3959
CODE6	F	004A68	4	1005	1000
CODE60	F	03084C	4	4102	4085
CODE61	F	03144C	4	4229	4212
CODE62	F	03204C	4	4361	4344
CODE63	F	032C4C	4	4487	4470
CODE64	F	03384C	4	4613	4596
CODE65	F	03444C	4	4739	4722
CODE66	F	03504C	4	4865	4848
CODE67	F	035C4C	4	4997	4980
CODE68	F	03684C	4	5129	5112
CODE69	F	03744C	4	5261	5244
CODE7	F	004A88	4	1025	1020
CODE70	F	03804C	4	5393	5376
CODE71	F	038C4C	4	5526	5509
CODE72	F	03984C	4	5658	5641
CODE73	F	03A44C	4	5779	5762
CODE74	F	03B04C	4	5900	5883
CODE75	F	03BC4C	4	6021	6004
CODE76	F	03C84C	4	6142	6125
CODE77	F	03D44C	4	6263	6246
CODE78	F	03E04C	4	6384	6367
CODE79	F	03EC4C	4	6505	6488
CODE8	F	004AA8	4	1045	1040
CODE80	F	03F84C	4	6627	6610
CODE81	F	04044C	4	6748	6731
CODE82	F	04104C	4	6869	6852
CODE83	F	041C4C	4	6990	6973
CODE84	F	04284C	4	7111	7094
CODE85	F	04344C	4	7236	7219
CODE86	F	04404C	4	7361	7344
CODE87	F	044C4C	4	7486	7469

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
CODE88	F	04584C	4	7611	7594
CODE89	F	04644C	4	7727	7710
CODE9	F	004AC8	4	1065	1060
CODE90	F	04704C	4	7854	7837
CODE91	F	047C4C	4	7966	7949
CODE92	F	04884C	4	8087	8070
CODE93	F	04944C	4	8195	8178
CODE94	F	04A04C	4	8313	8296
CODE95	F	04AC4C	4	8422	8405
CODE96	F	04B84C	4	8533	8516
CODE97	F	04C44C	4	8651	8634
CODE98	F	04D04C	4	8754	8737
CODE99	F	04DC4C	4	8861	8844
CTLR0	F	00270C	4	511	166 167 168 169
DECNUM	C	0041A1	16	610	269 271 276 278 283 285 290 292 329 331 336 338 343 345 350 352 379 381 416 418 423 425
EDIT	X	004175	18	605	270 277 284 291 330 337 344 351 380 417 424
ENDTEST	U	002188	1	238	213
EOJ	I	002660	4	493	201 241
EOJPSW	D	002650	8	491	493
F241CCMSG	U	04DCE0	1	8913	8912
F241CKRES	U	04DCB4	1	8898	8915
F241TSTCC	I	04DCDC	4	8912	8894
F241XX	U	04DCD6	1	8908	
F242CCMSG	U	04E8E0	1	9020	9019
F242CKRES	U	04E8B4	1	9005	9022
F242TSTCC	I	04E8DC	4	9019	9001
F242XX	U	04E8D6	1	9015	
F251CCMSG	U	04C4D2	1	8699	8698
F251CKRES	U	04C4A6	1	8684	8701
F251TSTCC	I	04C4CE	4	8698	8680
F251XX	U	04C4C8	1	8694	
F252CCMSG	U	04D0D2	1	8802	8801
F252CKRES	U	04D0A6	1	8787	8804
F252TSTCC	I	04D0CE	4	8801	8783
F252XX	U	04D0C8	1	8797	
F261CCMSG	U	04A0EC	1	8367	8366
F261CKRES	U	04A0C0	1	8352	8369
F261TSTCC	I	04A0E8	4	8366	8348
F261XX	U	04A0E2	1	8362	
F262CCMSG	U	04ACF6	1	8478	8477
F262CKRES	U	04ACC0	1	8461	8480
F262TSTCC	I	04ACF2	4	8477	8457
F262XX	U	04ACEC	1	8473	8469
F263CCMSG	U	04B8FC	1	8590	8589
F263CKRES	U	04B8C6	1	8573	8592
F263TSTCC	I	04B8F8	4	8589	8569
F263XX	U	04B8F2	1	8585	8581
F271CCMSG	U	0488E6	1	8140	8139
F271CKRES	U	0488BA	1	8125	8142
F271TSTCC	I	0488E2	4	8139	8121
F271XX	U	0488DC	1	8135	
F272CCMSG	U	0494FA	1	8253	8252
F272CKRES	U	0494C4	1	8236	8255
F272TSTCC	I	0494F6	4	8252	8232
F272XX	U	0494F0	1	8248	8244

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
F281CCMSG	U	0470F6	1	7911	7910
F281CKRES	U	0470CA	1	7896	7913
F281TSTCC	I	0470F2	4	7910	7892
F281XX	U	0470EC	1	7906	
F282CCMSG	U	047D0A	1	8027	8026
F282CKRES	U	047CCA	1	8008	8029
F282TSTCC	I	047D06	4	8026	8004
F282XX	U	047D00	1	8022	8016 8018
F291CCMSG	U	045906	1	7672	7671
F291CKRES	U	0458DA	1	7657	7674
F291TSTCC	I	045902	4	7671	7653
F291XX	U	0458FC	1	7667	
F292CCMSG	U	046524	1	7794	7793
F292CKRES	U	0464DA	1	7773	7796
F292TSTCC	I	046520	4	7793	7769
F292XX	U	04651A	1	7789	7781 7783 7785
F301CCMSG	U	04292C	1	7182	7181
F301CKRES	U	0428EC	1	7163	7158
F301CKUR	U	0428D8	1	7155	7184
F301TSTCC	I	042928	4	7181	7153
F301XX	U	042922	1	7177	7171 7173
F302CCMSG	U	04352C	1	7307	7306
F302CKRES	U	0434EC	1	7288	7283
F302CKUR	U	0434D8	1	7280	7309
F302TSTCC	I	043528	4	7306	7278
F302XX	U	043522	1	7302	7296 7298
F303CCMSG	U	04412C	1	7432	7431
F303CKRES	U	0440EC	1	7413	7408
F303CKUR	U	0440D8	1	7405	7434
F303TSTCC	I	044128	4	7431	7403
F303XX	U	044122	1	7427	7421 7423
F304CCMSG	U	044D12	1	7553	7552
F304CKRES	U	044CD2	1	7532	7555
F304TSTCC	I	044D0E	4	7552	7528
F304XX	U	044D08	1	7548	7542 7544
F311CCMSG	U	03F914	1	6693	6692
F311CKRES	U	03F8D4	1	6674	6669
F311CKUR	U	03F8C0	1	6666	6695
F311TSTCC	I	03F910	4	6692	6664
F311XX	U	03F90A	1	6688	6682 6684
F312CCMSG	U	040514	1	6814	6813
F312CKRES	U	0404D4	1	6795	6790
F312CKUR	U	0404C0	1	6787	6816
F312TSTCC	I	040510	4	6813	6785
F312XX	U	04050A	1	6809	6803 6805
F313CCMSG	U	041114	1	6935	6934
F313CKRES	U	0410D4	1	6916	6911
F313CKUR	U	0410C0	1	6908	6937
F313TSTCC	I	041110	4	6934	6906
F313XX	U	04110A	1	6930	6924 6926
F314CCMSG	U	041CFA	1	7052	7051
F314CKRES	U	041CBA	1	7031	7054
F314TSTCC	I	041CF6	4	7051	7027
F314XX	U	041CF0	1	7047	7041 7043
F321CCMSG	U	03C914	1	6208	6207
F321CKRES	U	03C8D4	1	6189	6184

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES		
F321CKUR	U	03C8C0	1	6181	6210		
F321TSTCC	I	03C910	4	6207	6179		
F321XX	U	03C90A	1	6203	6197	6199	
F322CCMSG	U	03D514	1	6329	6328		
F322CKRES	U	03D4D4	1	6310	6305		
F322CKUR	U	03D4C0	1	6302	6331		
F322TSTCC	I	03D510	4	6328	6300		
F322XX	U	03D50A	1	6324	6318	6320	
F323CCMSG	U	03E114	1	6450	6449		
F323CKRES	U	03E0D4	1	6431	6426		
F323CKUR	U	03E0C0	1	6423	6452		
F323TSTCC	I	03E110	4	6449	6421		
F323XX	U	03E10A	1	6445	6439	6441	
F324CCMSG	U	03ECFA	1	6567	6566		
F324CKRES	U	03ECBA	1	6546	6569		
F324TSTCC	I	03ECF6	4	6566	6542		
F324XX	U	03ECF0	1	6562	6556	6558	
F331CCMSG	U	039914	1	5724	5723		
F331CKRES	U	0398D4	1	5705	5700		
F331CKUR	U	0398C0	1	5697	5726		
F331TSTCC	I	039910	4	5723	5695		
F331XX	U	03990A	1	5719	5713	5715	
F332CCMSG	U	03A514	1	5845	5844		
F332CKRES	U	03A4D4	1	5826	5821		
F332CKUR	U	03A4C0	1	5818	5847		
F332TSTCC	I	03A510	4	5844	5816		
F332XX	U	03A50A	1	5840	5834	5836	
F333CCMSG	U	03B114	1	5966	5965		
F333CKRES	U	03B0D4	1	5947	5942		
F333CKUR	U	03B0C0	1	5939	5968		
F333TSTCC	I	03B110	4	5965	5937		
F333XX	U	03B10A	1	5961	5955	5957	
F334CCMSG	U	03BCFA	1	6083	6082		
F334CKRES	U	03BCBA	1	6062	6085		
F334TSTCC	I	03BCF6	4	6082	6058		
F334XX	U	03BCF0	1	6078	6072	6074	
F341CCMSG	U	035D4C	1	5074	5073		
F341CKRES	U	035D02	1	5053	5048		
F341CKUR	U	035CEE	1	5045	5076		
F341TSTCC	I	035D48	4	5073	5043		
F341XX	U	035D42	1	5069	5061	5063	5065
F342CCMSG	U	03694C	1	5206	5205		
F342CKRES	U	036902	1	5185	5180		
F342CKUR	U	0368EE	1	5177	5208		
F342TSTCC	I	036948	4	5205	5175		
F342XX	U	036942	1	5201	5193	5195	5197
F343CCMSG	U	03754C	1	5338	5337		
F343CKRES	U	037502	1	5317	5312		
F343CKUR	U	0374EE	1	5309	5340		
F343TSTCC	I	037548	4	5337	5307		
F343XX	U	037542	1	5333	5325	5327	5329
F344CCMSG	U	03814C	1	5470	5469		
F344CKRES	U	038102	1	5449	5444		
F344CKUR	U	0380EE	1	5441	5472		
F344TSTCC	I	038148	4	5469	5439		
F344XX	U	038142	1	5465	5457	5459	5461

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES		
F345CCMSG	U	038D32	1	5599	5598		
F345CKRES	U	038CE8	1	5576	5601		
F345TSTCC	I	038D2E	4	5598	5572		
F345XX	U	038D28	1	5594	5586	5588	5590
F351CCMSG	U	032128	1	4432	4431		
F351CKRES	U	0320DE	1	4411	4406		
F351CKUR	U	0320CA	1	4403	4434		
F351TSTCC	I	032124	4	4431	4401		
F351XX	U	03211E	1	4427	4419	4421	4423
F352CCMSG	U	032D28	1	4558	4557		
F352CKRES	U	032CDE	1	4537	4532		
F352CKUR	U	032CCA	1	4529	4560		
F352TSTCC	I	032D24	4	4557	4527		
F352XX	U	032D1E	1	4553	4545	4547	4549
F353CCMSG	U	033928	1	4684	4683		
F353CKRES	U	0338DE	1	4663	4658		
F353CKUR	U	0338CA	1	4655	4686		
F353TSTCC	I	033924	4	4683	4653		
F353XX	U	03391E	1	4679	4671	4673	4675
F354CCMSG	U	034528	1	4810	4809		
F354CKRES	U	0344DE	1	4789	4784		
F354CKUR	U	0344CA	1	4781	4812		
F354TSTCC	I	034524	4	4809	4779		
F354XX	U	03451E	1	4805	4797	4799	4801
F355CCMSG	U	035122	1	4938	4937		
F355CKRES	U	0350D8	1	4915	4910		
F355CKUR	U	0350C4	1	4907	4940		
F355TSTCC	I	03511E	4	4937	4905		
F355XX	U	035118	1	4933	4925	4927	4929
F361CCMSG	U	02E528	1	3795	3794		
F361CKRES	U	02E4DE	1	3774	3769		
F361CKUR	U	02E4CA	1	3766	3797		
F361TSTCC	I	02E524	4	3794	3764		
F361XX	U	02E51E	1	3790	3782	3784	3786
F362CCMSG	U	02F128	1	3921	3920		
F362CKRES	U	02F0DE	1	3900	3895		
F362CKUR	U	02F0CA	1	3892	3923		
F362TSTCC	I	02F124	4	3920	3890		
F362XX	U	02F11E	1	3916	3908	3910	3912
F363CCMSG	U	02FD28	1	4047	4046		
F363CKRES	U	02FCDE	1	4026	4021		
F363CKUR	U	02FCCA	1	4018	4049		
F363TSTCC	I	02FD24	4	4046	4016		
F363XX	U	02FD1E	1	4042	4034	4036	4038
F364CCMSG	U	030928	1	4173	4172		
F364CKRES	U	0308DE	1	4152	4147		
F364CKUR	U	0308CA	1	4144	4175		
F364TSTCC	I	030924	4	4172	4142		
F365CCMSG	U	031522	1	4302	4301		
F365CKRES	U	0314D8	1	4279	4274		
F365CKUR	U	0314C4	1	4271	4304		
F365TSTCC	I	03151E	4	4301	4269		
F371CCMSG	U	02A928	1	3158	3157		
F371CKRES	U	02A8DE	1	3137	3132		
F371CKUR	U	02A8CA	1	3129	3160		
F371TSTCC	I	02A924	4	3157	3127		

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES													
F372CCMSG	U	02B528	1	3284	3283													
F372CKRES	U	02B4DE	1	3263	3258													
F372CKUR	U	02B4CA	1	3255	3286													
F372TSTCC	I	02B524	4	3283	3253													
F373CCMSG	U	02C128	1	3410	3409													
F373CKRES	U	02C0DE	1	3389	3384													
F373CKUR	U	02C0CA	1	3381	3412													
F373TSTCC	I	02C124	4	3409	3379													
F373XX	U	02C11E	1	3405	3397	3399	3401											
F374CCMSG	U	02CD28	1	3536	3535													
F374CKRES	U	02CCDE	1	3515	3510													
F374CKUR	U	02CCCA	1	3507	3538													
F374TSTCC	I	02CD24	4	3535	3505													
F374XX	U	02CD1E	1	3531	3523	3525	3527											
F375CCMSG	U	02D922	1	3665	3664													
F375CKRES	U	02D8D8	1	3642	3637													
F375CKUR	U	02D8C4	1	3634	3667													
F375TSTCC	I	02D91E	4	3664	3632													
F375XX	U	02D918	1	3660	3652	3654	3656											
F4053	U	026CCC	1	2640	2636													
F4154	U	0260EA	1	2538	2532	2534												
F4255	U	0254EA	1	2432	2426	2428												
F4352	U	0278CC	1	2738	2734													
F4356	U	0248EA	1	2326	2320	2322												
F4457	U	023CEA	1	2220	2214	2216												
F4558	U	023108	1	2110	2102	2104	2106											
F4659	U	022508	1	1996	1988	1990	1992											
F4760	U	021908	1	1882	1874	1876	1878											
F4861	U	020D08	1	1768	1760	1762	1764											
FAILED	F	004000	4	529	239	257	324	376	411									
FAILHLNG	U	000015	1	541														
FAILLINE	C	004008	16	538	541	544	393											
FAILLNG	U	000079	1	544	392													
FAILMSG	U	0023E8	1	374	1769	1883	1997	2111	2221	2327	2433	2539	2641	2739	2838	2933	3027	3154
					3280	3406	3532	3661	3791	3917	4043	4169	4298	4428	4554	4680	4806	4934
					5070	5202	5334	5466	5595	5720	5841	5962	6079	6204	6325	6446	6563	6689
					6810	6931	7048	7178	7303	7428	7549	7668	7790	7907	8023	8136	8249	8363
					8474	8586	8695	8798	8909	9016								
FAILPSW	D	002668	8	495	497													
FAILTEST	I	002678	4	497	242													
FAILTNUM	C	004018	3	539	382													
FAILTXT	C	00401D	100	543	384	387												
FAILTXTA	A	000018	4	709	386													
FB0001	F	002080	8	185	189	190	192											
FCCSAV	D	002248	8	307	300	304												
FLSAV	D	002438	8	402	395	399												
FMSG100	C	04F1E0	1	9043	8961													
FMSG39	C	021480	1	1791	1702													
FMSG40	C	022080	1	1905	1816													
FMSG41	C	022C80	1	2019	1930													
FMSG42	C	023880	1	2133	2044													
FMSG43	C	024480	1	2243	2162													
FMSG44	C	025080	1	2349	2268													
FMSG45	C	025C80	1	2455	2374													
FMSG46	C	026880	1	2561	2480													
FMSG47	C	027480	1	2663	2590													

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES		
FMSG48	C	028080	1	2761	2688		
FMSG49	C	028C80	1	2860	2790		
FMSG50	C	029880	1	2955	2885		
FMSG51	C	02A480	1	3049	2985		
FMSG52	C	02B1E0	1	3181	3079		
FMSG53	C	02BDE0	1	3307	3206		
FMSG54	C	02C9E0	1	3433	3332		
FMSG55	C	02D5E0	1	3559	3458		
FMSG56	C	02E1E0	1	3688	3585		
FMSG57	C	02EDE0	1	3818	3717		
FMSG58	C	02F9E0	1	3944	3843		
FMSG59	C	0305E0	1	4070	3969		
FMSG60	C	0311E0	1	4196	4095		
FMSG61	C	031DE0	1	4325	4222		
FMSG62	C	0329E0	1	4455	4354		
FMSG63	C	0335E0	1	4581	4480		
FMSG64	C	0341E0	1	4707	4606		
FMSG65	C	034DE0	1	4833	4732		
FMSG66	C	0359E0	1	4961	4858		
FMSG67	C	0365E0	1	5097	4990		
FMSG68	C	0371E0	1	5229	5122		
FMSG69	C	037DE0	1	5361	5254		
FMSG70	C	0389E0	1	5493	5386		
FMSG71	C	0395E0	1	5622	5519		
FMSG72	C	03A1E0	1	5747	5651		
FMSG73	C	03ADE0	1	5868	5772		
FMSG74	C	03B9E0	1	5989	5893		
FMSG75	C	03C5E0	1	6106	6014		
FMSG76	C	03D1E0	1	6231	6135		
FMSG77	C	03DDE0	1	6352	6256		
FMSG78	C	03E9E0	1	6473	6377		
FMSG79	C	03F5E0	1	6590	6498		
FMSG80	C	0401E0	1	6716	6620		
FMSG81	C	040DE0	1	6837	6741		
FMSG82	C	0419E0	1	6958	6862		
FMSG83	C	0425E0	1	7075	6983		
FMSG84	C	0431E0	1	7205	7104		
FMSG85	C	043DE0	1	7330	7229		
FMSG86	C	0449E0	1	7455	7354		
FMSG87	C	0455E0	1	7576	7479		
FMSG88	C	0461E0	1	7695	7604		
FMSG89	C	046DE0	1	7817	7720		
FMSG90	C	0479E0	1	7934	7847		
FMSG91	C	0485E0	1	8050	7959		
FMSG92	C	0491E0	1	8163	8080		
FMSG93	C	049DE0	1	8276	8188		
FMSG94	C	04A9E0	1	8390	8306		
FMSG95	C	04B5E0	1	8501	8415		
FMSG96	C	04C1E0	1	8613	8526		
FMSG97	C	04CDE0	1	8722	8644		
FMSG98	C	04D9E0	1	8825	8747		
FMSG99	C	04E5E0	1	8936	8854		
FUN	U	000007	1	699	275	335	
FXX137	U	02A91E	1	3153	3145	3147	3149
FXX237	U	02B51E	1	3279	3271	3273	3275
FXX437	U	03091E	1	4168	4160	4162	4164

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
FXX50	U	0290BE	1	2932	2928
FXX51	U	0284C6	1	2837	2833
FXX537	U	031518	1	4297	4289 4291 4293
IMAGE	1	000000	324225	0	
K	U	000400	1	516	517 518 519 1688 1773 1802 1887 1916 2001 2030 2115 2148 2225 2254 2331 2360 2437 2466 2543 2576 2645 2674 2743 2776 2842 2871 2937 2971 3031 3065 3163 3192 3289 3318 3415 3444 3541 3571 3670 3703 3800 3829 3926 3955 4052 4081 4178 4208 4307 4340 4437 4466 4563 4592 4689 4718 4815 4844 4943 4976 5079 5108 5211 5240 5343 5372 5475 5505 5604 5637 5729 5758 5850 5879 5971 6000 6088 6121 6213 6242 6334 6363 6455 6484 6572 6606 6698 6727 6819 6848 6940 6969 7057 7090 7187 7215 7312 7340 7437 7465 7558 7590 7677 7706 7799 7833 7916 7945 8032 8066 8145 8174 8258 8292 8372 8401 8483 8512 8595 8630 8704 8733 8807 8840 8918 8947 9025
K64	U	010000	1	518	
MB	U	100000	1	519	
MSG	I	002598	4	451	200 302 362 397 437
MSGCMD	C	0025E6	9	482	464 465
MSGMSG	C	0025EF	95	483	458 480 456
MSGMVC	I	0025E0	6	480	462
MSGOK	I	0025AE	2	460	457
MSGRET	I	0025CE	4	475	468 471
MSGSAVE	F	0025D4	4	478	454 475
NACCSAV	D	002518	8	442	435 439
NAMSG	U	0024B8	1	409	910 930 950 970 990 1010 1030 1050 1070 1090 1110 1130 1150 1170 1190 1210 1230 1250 1270 1290 1310 1330 1350 1370 1390 1410 1430 1450 1470 1490 1510 1530 1550 1570 1590 1610 1630 1650
NAPRTFUN	C	004161	2	596	426
NAPRTLINE	C	00413F	16	593	599 433
NAPRTLNG	U	000036	1	599	432
NAPRTNUM	C	00414F	3	594	419
NEXTTST	U	0020D4	1	210	231
OPDST100	C	04EE20	1	9028	8963
OPDST39	C	021220	1	1776	1704
OPDST40	C	021E20	1	1890	1818
OPDST41	C	022A20	1	2004	1932
OPDST42	C	023620	1	2118	2046
OPDST43	C	024220	1	2228	2164
OPDST44	C	024E20	1	2334	2270
OPDST45	C	025A20	1	2440	2376
OPDST46	C	026620	1	2546	2482
OPDST47	C	027220	1	2648	2592
OPDST48	C	027E20	1	2746	2690
OPDST49	C	028A20	1	2845	2792
OPDST50	C	029620	1	2940	2887
OPDST51	C	02A220	1	3034	2987
OPDST52	C	02AE20	1	3166	3081
OPDST53	C	02BA20	1	3292	3208
OPDST54	C	02C620	1	3418	3334
OPDST55	C	02D220	1	3544	3460
OPDST56	C	02DE20	1	3673	3587
OPDST57	C	02EA20	1	3803	3719
OPDST58	C	02F620	1	3929	3845
OPDST59	C	030220	1	4055	3971
OPDST60	C	030E20	1	4181	4097
OPDST61	C	031A20	1	4310	4224

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
OPDST62	C	032620	1	4440	4356
OPDST63	C	033220	1	4566	4482
OPDST64	C	033E20	1	4692	4608
OPDST65	C	034A20	1	4818	4734
OPDST66	C	035620	1	4946	4860
OPDST67	C	036220	1	5082	4992
OPDST68	C	036E20	1	5214	5124
OPDST69	C	037A20	1	5346	5256
OPDST70	C	038620	1	5478	5388
OPDST71	C	039220	1	5607	5521
OPDST72	C	039E20	1	5732	5653
OPDST73	C	03AA20	1	5853	5774
OPDST74	C	03B620	1	5974	5895
OPDST75	C	03C220	1	6091	6016
OPDST76	C	03CE20	1	6216	6137
OPDST77	C	03DA20	1	6337	6258
OPDST78	C	03E620	1	6458	6379
OPDST79	C	03F220	1	6575	6500
OPDST80	C	03FE20	1	6701	6622
OPDST81	C	040A20	1	6822	6743
OPDST82	C	041620	1	6943	6864
OPDST83	C	042220	1	7060	6985
OPDST84	C	042E20	1	7190	7106
OPDST85	C	043A20	1	7315	7231
OPDST86	C	044620	1	7440	7356
OPDST87	C	045220	1	7561	7481
OPDST88	C	045E20	1	7680	7606
OPDST89	C	046A20	1	7802	7722
OPDST90	C	047620	1	7919	7849
OPDST91	C	048220	1	8035	7961
OPDST92	C	048E20	1	8148	8082
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OPSRC40	C	021EC0	1	1895	1817
OPSRC41	C	022AC0	1	2009	1931
OPSRC42	C	0236C0	1	2123	2045
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OPSRC62	C	032740	1	4445	4355
OPSRC63	C	033340	1	4571	4481
OPSRC64	C	033F40	1	4697	4607
OPSRC65	C	034B40	1	4823	4733
OPSRC66	C	035740	1	4951	4859
OPSRC67	C	036340	1	5087	4991
OPSRC68	C	036F40	1	5219	5123
OPSRC69	C	037B40	1	5351	5255
OPSRC70	C	038740	1	5483	5387
OPSRC71	C	039340	1	5612	5520
OPSRC72	C	039F40	1	5737	5652
OPSRC73	C	03AB40	1	5858	5773
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PL56	C	02E060	1	3683	3588
PL57	C	02EC60	1	3813	3720
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PL59	C	030460	1	4065	3972
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PL61	C	031C60	1	4320	4225
PL62	C	032860	1	4450	4357
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PL65	C	034C60	1	4828	4735
PL66	C	035860	1	4956	4861
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PL73	C	03AC60	1	5863	5775
PL74	C	03B860	1	5984	5896
PL75	C	03C460	1	6101	6017
PL76	C	03D060	1	6226	6138
PL77	C	03DC60	1	6347	6259
PL78	C	03E860	1	6468	6380
PL79	C	03F460	1	6585	6501
PL80	C	040060	1	6711	6623
PL81	C	040C60	1	6832	6744
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PL86	C	044860	1	7450	7357

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PL92	C	049060	1	8158	8083
PL93	C	049C60	1	8271	8191
PL94	C	04A860	1	8385	8309
PL95	C	04B460	1	8496	8418
PL96	C	04C060	1	8608	8529
PL97	C	04CC60	1	8717	8647
PL98	C	04D860	1	8820	8750
PL99	C	04E460	1	8931	8857
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					8997	8998	9008											
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SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
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					4165 4233 4240 4243 4250 4253 4256 4259 4263 4272 4273 4283 4288 4290
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					4420 4422 4424 4491 4498 4501 4508 4511 4514 4517 4521 4530 4531 4541
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					5924 5927 5931 5940 5941 5951 5954 5956 5958 6025 6032 6035 6042 6045
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R4	U	000004	1	9169	3091 3100 3101 3105 3142 3218 3226 3227 3231 3268 3344 3352 3353 3357
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SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
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					8200 8210 8215 8218 8221 8226 8241 8243 8245 8318 8329 8337 8342 8357
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					1652 1690 1691 1804 1805 1918 1919 2032 2033 2150 2151 2256 2257 2362
					2363 2468 2469 2578 2579 2676 2677 2778 2779 2873 2874 2973 2974 3067
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R6	U	000006	1	9171	
R7	U	000007	1	9172	
R8	U	000008	1	9173	152 156 157 158 160
R9	U	000009	1	9174	153 160 161 163
RANDX	X	0041B8	16	616	1706 1820 1934 2048 2166 2272 2378 2484 2594 2692 2794 2889 2989 3083
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RANDXA	A	000028	4	714	1712 1826 1940 2054 2172 2278 2384 2490 2600 2698 2800 2895 2995 3089
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SKL0001	U	00004B	1	182	198
SKT0001	C	00202A	26	179	182 199
SVOLDPSW	U	000140	0	121	
SVTST	D	002108	8	233	218 228
T1	A	0049C0	4	900	9057
T10	A	004AE0	4	1080	9066
T100	A	04E820	4	8951	9156
T11	A	004B00	4	1100	9067
T12	A	004B20	4	1120	9068
T13	A	004B40	4	1140	9069
T14	A	004B60	4	1160	9070
T15	A	004B80	4	1180	9071

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
T16	A	004BA0	4	1200	9072
T17	A	004BC0	4	1220	9073
T18	A	004BE0	4	1240	9074
T19	A	004C00	4	1260	9075
T2	A	0049E0	4	920	9058
T20	A	004C20	4	1280	9076
T21	A	004C40	4	1300	9077
T22	A	004C60	4	1320	9078
T23	A	004C80	4	1340	9079
T24	A	004CA0	4	1360	9080
T25	A	004CC0	4	1380	9081
T26	A	004CE0	4	1400	9082
T27	A	004D00	4	1420	9083
T28	A	004D20	4	1440	9084
T29	A	004D40	4	1460	9085
T3	A	004A00	4	940	9059
T30	A	004D60	4	1480	9086
T31	A	004D80	4	1500	9087
T32	A	004DA0	4	1520	9088
T33	A	004DC0	4	1540	9089
T34	A	004DE0	4	1560	9090
T35	A	004E00	4	1580	9091
T36	A	004E20	4	1600	9092
T37	A	004E40	4	1620	9093
T38	A	004E60	4	1640	9094
T39	A	020C20	4	1692	9095
T4	A	004A20	4	960	9060
T40	A	021820	4	1806	9096
T41	A	022420	4	1920	9097
T42	A	023020	4	2034	9098
T43	A	023C20	4	2152	9099
T44	A	024820	4	2258	9100
T45	A	025420	4	2364	9101
T46	A	026020	4	2470	9102
T47	A	026C20	4	2580	9103
T48	A	027820	4	2678	9104
T49	A	028420	4	2780	9105
T5	A	004A40	4	980	9061
T50	A	029020	4	2875	9106
T51	A	029C20	4	2975	9107
T52	A	02A820	4	3069	9108
T53	A	02B420	4	3196	9109
T54	A	02C020	4	3322	9110
T55	A	02CC20	4	3448	9111
T56	A	02D820	4	3575	9112
T57	A	02E420	4	3707	9113
T58	A	02F020	4	3833	9114
T59	A	02FC20	4	3959	9115
T6	A	004A60	4	1000	9062
T60	A	030820	4	4085	9116
T61	A	031420	4	4212	9117
T62	A	032020	4	4344	9118
T63	A	032C20	4	4470	9119
T64	A	033820	4	4596	9120
T65	A	034420	4	4722	9121
T66	A	035020	4	4848	9122

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
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T68	A	036820	4	5112	9124
T69	A	037420	4	5244	9125
T7	A	004A80	4	1020	9063
T70	A	038020	4	5376	9126
T71	A	038C20	4	5509	9127
T72	A	039820	4	5641	9128
T73	A	03A420	4	5762	9129
T74	A	03B020	4	5883	9130
T75	A	03BC20	4	6004	9131
T76	A	03C820	4	6125	9132
T77	A	03D420	4	6246	9133
T78	A	03E020	4	6367	9134
T79	A	03EC20	4	6488	9135
T8	A	004AA0	4	1040	9064
T80	A	03F820	4	6610	9136
T81	A	040420	4	6731	9137
T82	A	041020	4	6852	9138
T83	A	041C20	4	6973	9139
T84	A	042820	4	7094	9140
T85	A	043420	4	7219	9141
T86	A	044020	4	7344	9142
T87	A	044C20	4	7469	9143
T88	A	045820	4	7594	9144
T89	A	046420	4	7710	9145
T9	A	004AC0	4	1060	9065
T90	A	047020	4	7837	9146
T91	A	047C20	4	7949	9147
T92	A	048820	4	8070	9148
T93	A	049420	4	8178	9149
T94	A	04A020	4	8296	9150
T95	A	04AC20	4	8405	9151
T96	A	04B820	4	8516	9152
T97	A	04C420	4	8634	9153
T98	A	04D020	4	8737	9154
T99	A	04DC20	4	8844	9155
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TNUM	H	000004	2	697	221 268 328 378
TSUB	A	000000	4	696	224
TTABLE	F	0045B8	4	9056	
UR	U	000008	1	700	320 342 3131 3257 3383 3509 3636 3768 3894 4020 4146 4273 4405 4531 4657 4783 4909 5047 5179 5311 5443 5699 5820 5941 6183 6304 6425 6668 6789 6910 7157 7282 7407
URFOUND	U	000009	1	701	349 3130 3256 3382 3508 3635 3767 3893 4019 4145 4272 4404 4530 4656 4782 4908 5046 5178 5310 5442 5698 5819 5940 6182 6303 6424 6667 6788 6909 7156 7281 7406
URMSG	U	0022C8	1	316	3133 3259 3385 3511 3638 3770 3896 4022 4148 4275 4407 4533 4659 4785 4911 5049 5181 5313 5445 5701 5822 5943 6185 6306 6427 6670 6791 6912 7159 7284 7409
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URPRTFUN	C	00411C	2	577	339
URPRTGOT	C	00413D	1	583	353
URPRTLINE	C	0040E0	16	573	585 358
URPRTLNG	U	00005F	1	585	357
URPRTNUM	C	0040F0	3	574	332
URSAV	D	002368	8	367	360 364

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
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V1	U	000001	1	9187	
V10	U	00000A	1	9196	
V11	U	00000B	1	9197	
V12	U	00000C	1	9198	
V13	U	00000D	1	9199	
V14	U	00000E	1	9200	
V15	U	00000F	1	9201	
V16	U	000010	1	9202	
V17	U	000011	1	9203	
V18	U	000012	1	9204	
V19	U	000013	1	9205	
V2	U	000002	1	9188	
V20	U	000014	1	9206	
V21	U	000015	1	9207	
V22	U	000016	1	9208	
V23	U	000017	1	9209	
V24	U	000018	1	9210	
V25	U	000019	1	9211	
V26	U	00001A	1	9212	
V27	U	00001B	1	9213	
V28	U	00001C	1	9214	
V29	U	00001D	1	9215	
V3	U	000003	1	9189	
V30	U	00001E	1	9216	
V31	U	00001F	1	9217	
V4	U	000004	1	9190	
V5	U	000005	1	9191	
V6	U	000006	1	9192	
V7	U	000007	1	9193	
V8	U	000008	1	9194	
V9	U	000009	1	9195	
X0001	U	0020A8	1	188	176 189
XC0001	U	0020D0	1	202	194
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=AL2(L'MSGMSG)	R	00268A	2	508	456
=CL128' '	C	00268C	128	509	384
=F'1'	F	00267C	4	504	193 256 323 375 410
=H'0'	H	002688	2	507	451
=XL4'3'	X	002684	4	506	263

```
 MACRO    DEFN   REFERENCES

FCHECK     68    175
PTTABLE   867   9055
TSTCODE   789   1708   1822   1936   2050   2168   2274   2380   2486   2596   2694   2796   2891   2991   3085   3212   3338   3464
                3591   3723   3849   3975   4101   4228   4360   4486   4612   4738   4864   4996   5128   5260   5392   5525   5657
                5778   5899   6020   6141   6262   6383   6504   6626   6747   6868   6989   7110   7235   7360   7485   7610   7726
                7853   7965   8086   8194   8312   8421   8532   8650   8753   8860   8967
TSTDATA   801   1772   1886   2000   2114   2224   2330   2436   2542   2644   2742   2841   2936   3030   3162   3288   3414   3540
                3669   3799   3925   4051   4177   4306   4436   4562   4688   4814   4942   5078   5210   5342   5474   5603   5728
                5849   5970   6087   6212   6333   6454   6571   6697   6818   6939   7056   7186   7311   7436   7557   7676   7798
                7915   8031   8144   8257   8371   8482   8594   8703   8806   8917   9024
TSTEND    780   1795   1909   2023   2137   2247   2353   2459   2565   2667   2765   2864   2959   3053   3185   3311   3437   3563
                3692   3822   3948   4074   4200   4329   4459   4585   4711   4837   4965   5101   5233   5365   5497   5626   5751
                5872   5993   6110   6235   6356   6477   6594   6720   6841   6962   7079   7208   7333   7458   7579   7699   7821
                7938   8054   8167   8280   8394   8505   8617   8726   8829   8940   9047
TSTHDR    814   1687   1801   1915   2029   2147   2253   2359   2465   2575   2673   2775   2870   2970   3064   3191   3317   3443
                3570   3702   3828   3954   4080   4207   4339   4465   4591   4717   4843   4975   5107   5239   5371   5504   5636
                5757   5878   5999   6120   6241   6362   6483   6605   6726   6847   6968   7089   7214   7339   7464   7589   7705
                7832   7944   8065   8173   8291   8400   8511   8629   8732   8839   8946
TSTLBL    768   1774   1779   1784   1789   1888   1893   1898   1903   2002   2007   2012   2017   2116   2121   2126   2131   2226
                2231   2236   2241   2332   2337   2342   2347   2438   2443   2448   2453   2544   2549   2554   2559   2646   2651
                2656   2661   2744   2749   2754   2759   2843   2848   2853   2858   2938   2943   2948   2953   3032   3037   3042
                3047   3164   3169   3174   3179   3290   3295   3300   3305   3416   3421   3426   3431   3542   3547   3552   3557
                3671   3676   3681   3686   3801   3806   3811   3816   3927   3932   3937   3942   4053   4058   4063   4068   4179
                4184   4189   4194   4308   4313   4318   4323   4438   4443   4448   4453   4564   4569   4574   4579   4690   4695
                4700   4705   4816   4821   4826   4831   4944   4949   4954   4959   5080   5085   5090   5095   5212   5217   5222
                5227   5344   5349   5354   5359   5476   5481   5486   5491   5605   5610   5615   5620   5730   5735   5740   5745
                5851   5856   5861   5866   5972   5977   5982   5987   6089   6094   6099   6104   6214   6219   6224   6229   6335
                6340   6345   6350   6456   6461   6466   6471   6573   6578   6583   6588   6699   6704   6709   6714   6820   6825
                6830   6835   6941   6946   6951   6956   7058   7063   7068   7073   7188   7193   7198   7203   7313   7318   7323
                7328   7438   7443   7448   7453   7559   7564   7569   7574   7678   7683   7688   7693   7800   7805   7810   7815
                7917   7922   7927   7932   8033   8038   8043   8048   8146   8151   8156   8161   8259   8264   8269   8274   8373
                8378   8383   8388   8484   8489   8494   8499   8596   8601   8606   8611   8705   8710   8715   8720   8808   8813
                8818   8823   8919   8924   8929   8934   9026   9031   9036   9041
TSTPLO    737    894    914    934    954    974    994   1014   1034   1054   1074   1094   1114   1134   1154   1174   1194   1214
                1234   1254   1274   1294   1314   1334   1354   1374   1394   1414   1434   1454   1474   1494   1514   1534   1554
                1574   1594   1614   1634
```

```
   DESC      SYMBOL    SIZE        POS           ADDR

Entry: 0

Image       IMAGE     324225   00000-4F280   00000-4F280
  Region              324225   00000-4F280   00000-4F280
    CSECT   PLOETST   324225   00000-4F280   00000-4F280
```

```
   STMT                     FILE NAME
1     /devstor/sharedvfp/tests/plo-01-extension.asm

** NO ERRORS FOUND **
```